



WORK ORDER NO. 1783264

CONTRACT NO.

## **P1360 Joint Maritime Facility**

At the  
**VARLOCS**

Final Design Submittal - Volume 3 of 5

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Submitted By:

Date:

APPROVED BY:

Specifications:

For Commander, NAVFAC:  
Date: 01/28/2026

*Wm B forbes*

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PROJECT NUMBER: 1838720

CONTRACT NUMBER:

TASK ORDER NUMBER:

**P1360 JOINT MAIRITIME FACILITY**  
**NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND**  
**ATLANTIC**

HAVING PARTICIPATED IN THE DESIGN OF PROJECT NO. 1838720, P1360 JOINT MARITIME FACILITY AND HAVING THOROUGHLY REVIEWED THE COMPLETED PROJECT DOCUMENTS, I DECLARE THAT THE FACILITY DESIGN INCLUDED HEREIN COMPLIES WITH ALL APPLICABLE CANADIAN CODES AND LAWS.

DATE

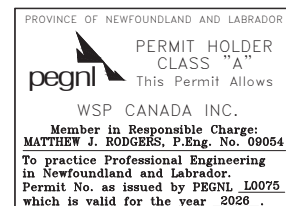
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(PROFESSIONAL SEAL)  
REGISTERED ENGINEER

January 23, 2026

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| 32 13 13.06 | 05/20               | PORTLAND CEMENT CONCRETE PAVEMENT FOR ROADS AND SITE FACILITIES |
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## SECTION 07 52 00

## MODIFIED BITUMINOUS MEMBRANE ROOFING

**05/12, CHG 5: 11/19**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

## AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.24 (2022) Roofing - Safety Requirements for Low-Sloped Roofs

## ASPHALT ROOFING MANUFACTURER'S ASSOCIATION (ARMA)

ARMA 410BUR88 (2001) Manual of Roof Maintenance and Repair

ARMA PMBRG98 (1998) Quality Control Guideline for the Application of Polymer Modified Bitumen Roofing

## ASTM INTERNATIONAL (ASTM)

ASTM C1153 (2023) Standard Practice for Location of Wet Insulation in Roofing Systems Using Infrared Imaging

ASTM C1289 (2023a) Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board

ASTM D41/D41M (2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing

ASTM D1970/D1970M (2019) Standard Specification for Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection

ASTM D4073/D4073M (2006; E 2019; R 2019) Standard Test Method for Tensile-Tear Strength of Bituminous Roofing Membranes

ASTM D4586/D4586M (2007; R 2018) Asphalt Roof Cement, Asbestos-Free



|                                                                             |                                                                                                                                                                       |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D4637/D4637M                                                           | (2015) EPDM Sheet Used in Single-Ply Roof Membrane                                                                                                                    |
| ASTM D5147/D5147M                                                           | (2014) Standard Test Methods for Sampling and Testing Modified Bituminous Sheet Material                                                                              |
| ASTM D6162/D6162M                                                           | (2016) Standard Specification for Styrene Butadiene Styrene (SBS) Modified Bituminous Sheet Materials Using a Combination of Polyester and Glass Fiber Reinforcements |
| ASTM D6164/D6164M                                                           | (2016) Standard Specification for Styrene Butadiene Styrene (SBS) Modified Bituminous Sheet Materials Using Polyester Reinforcements                                  |
| ASTM E108                                                                   | (2020a) Standard Test Methods for Fire Tests of Roof Coverings                                                                                                        |
| CSA Group (CSA)                                                             |                                                                                                                                                                       |
| CSA A123.21                                                                 | (2010) Wind Uplift Standards                                                                                                                                          |
| FM GLOBAL (FM)                                                              |                                                                                                                                                                       |
| FM APP GUIDE                                                                | (updated on-line) Approval Guide<br><a href="https://www.approvalguide.com/">https://www.approvalguide.com/</a>                                                       |
| INTERNATIONAL CODE COUNCIL (ICC)                                            |                                                                                                                                                                       |
| ICC IBC                                                                     | (2024) International Building Code                                                                                                                                    |
| NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)                             |                                                                                                                                                                       |
| NRCA C3701                                                                  | (1997) Repair Manual for Low Slope Membrane Roof Systems                                                                                                              |
| NRCA CONDET                                                                 | (2014) Construction Details Manual                                                                                                                                    |
| NRCA RoofMan                                                                | (2020) The NRCA Roofing Manual                                                                                                                                        |
| SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA) |                                                                                                                                                                       |
| SMACNA 1793                                                                 | (2012) Architectural Sheet Metal Manual, 7th Edition                                                                                                                  |
| SINGLE PLY ROOFING INDUSTRY (SPRI)                                          |                                                                                                                                                                       |
| ANSI/SPRI/FM 4435/ES-1                                                      | (2017) Test Standard for Edge Systems Used with Low Slope Roofing Systems                                                                                             |
| U.S. DEPARTMENT OF ENERGY (DOE)                                             |                                                                                                                                                                       |
| Energy Star                                                                 | (1992; R 2006) Energy Star Energy Efficiency Labeling System (FEMP)                                                                                                   |

## UL SOLUTIONS (UL)

|         |                                                                             |
|---------|-----------------------------------------------------------------------------|
| UL 790  | (2022) UL Standard for Safety Test Methods for Fire Tests of Roof Coverings |
| UL RMSD | (2012) Roofing Materials and Systems Directory                              |

## 1.2 DESCRIPTION OF ROOF MEMBRANE SYSTEM

Minimum three-ply SBS modified bitumen roof membrane consisting of modified bitumen base sheet, base sheet, interply sheet and cap sheet. Modified bitumen roof membrane must be set in cold-applied adhesive.

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All work must follow the NRCA RoofMan guidelines and standards stated within this Section.

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Roof Plan; G drawing depicting wind loads and boundaries of enhanced perimeter and corner attachments of roof system components, as applicable

## Field Inspection and Existing Conditions Report

Identify all fire safety issues including exposed or concealed combustible materials, which may require additional protection during roof installation.

## SD-03 Product Data

Modified Bitumen Sheets; G

Heat Island Reduction; S

Energy Star Label for Top Coating; S

Cold-Applied Membrane Adhesive; G

Primer; G

Modified Bitumen Roof Cement; G

Pre-Manufactured Accessories

Fasteners And Plates; G

Sample Warranty certificate; G

Submit all data required by Section 07 22 00 ROOF AND DECK INSULATION, together with requirements of this section. Include in data written acceptance by the roof membrane manufacturer of the products and accessories provided. Provide products as listed in the applicable wind uplift and fire rating classification listings, unless approved otherwise by the Contracting Officer.

#### SD-05 Design Data

Wind Uplift Calculations; G

Provide Engineering calculations, signed, sealed, and dated by a qualified Engineer validating the wind resistance per ASCE 7-16, ASTM D4073/D4073M, and ANSI/SPRI/FM 4435/ES-1 of non-rated roof system.

#### SD-07 Certificates

Provide evidence that products used within this specification are manufactured in the United States.

Qualification of Manufacturer

Certify that the manufacturer of the modified bitumen membrane meets requirements specified under paragraph QUALIFICATION OF MANUFACTURER.

Qualification of Applicator

Certify that the applicator meets requirements specified under paragraph QUALIFICATION OF APPLICATOR.

Qualification of Engineer of Record

Certify that the Engineer of Record is fully qualified, competent, and currently licensed to practice in the project jurisdiction.

Wind Uplift Resistance; G classification, as applicable

Fire Resistance classification; G

Submit the roof system assembly wind uplift and fire rating classification listings.

#### SD-08 Manufacturer's Instructions

Modified Bitumen Membrane Application; G

Flashing; G

Cold Adhesive Applied Modified Bitumen Membrane; G

Base Sheet attachment, including pattern and frequency of mechanical attachments required in field of roof, corners, and perimeters to provide for the specified wind resistance.

Primer

Fasteners

Cold Weather Installation; G

Include detailed application instructions and standard manufacturer drawings altered as required by these specifications. Explicitly identify in writing, differences between manufacturer's instructions and the specified requirements.

#### SD-11 Closeout Submittals

Warranty

Information Card

Instructions To Government Personnel

Include copies of Safety Data Sheets for maintenance/repair materials.

Submit 20 year "No-Dollar-Limit" warranty for labor and materials.

### 1.4 QUALITY ASSURANCE

#### 1.4.1 Qualification of Manufacturer

Modified bitumen sheet roofing system manufacturer must have a minimum of 5 five years experience in manufacturing modified bitumen roofing products.

#### 1.4.2 Qualification of Applicator

Roofing system applicator must be approved, authorized, or licensed in writing by the modified bitumen sheet roofing system manufacturer and have a minimum of 5 five years experience as an approved, authorized, or licensed applicator with that manufacturer and be approved at a level capable of providing the specified warranty. The applicator must supply the names, locations and client contact information of five projects of similar size and scope that the applicator has constructed using the manufacturer's roofing products submitted for this project within the previous three years.

#### 1.4.3 Qualification of Engineer of Record

Engineer of Record must be currently licensed within the jurisdiction of the project.

Wind uplift requirements in accordance with Local and Newfoundland, Canada ASCE 7-16, in accordance with International Building Code.

Snow load requirements per ICC IBC Chapter 16 Section 1608.3 and Section 7 of ASCE 7-16

#### 1.4.4 Fire Resistance

Complete roof covering assembly must:

- a. Be Class A rated in accordance with ASTM E108, CSA A123.21, or UL 790; and
- b. Be listed as part of Fire-Classified roof deck construction in UL RMSD, or Class I roof deck construction in FM APP GUIDE.

FM or UL approved components of the roof covering assembly must bear the appropriate FM or UL label.

#### 1.4.5 Wind Uplift Resistance

Provide a complete roof system assembly that is rated and installed to resist wind loads calculated in accordance with ASCE 7-16, CSA A123.21 and validated by uplift resistance testing in accordance with Factory Mutual (FM) test procedures. Do not install non-rated systems, except as approved by the Contracting Officer. Submit licensed engineer's Wind uplift calculations and substantiating data to validate any non-rated roof system. Base wind uplift measurements on a design wind speed as indicated on structural drawings and in accordance with ASCE 7-16 and other applicable building code requirements.

#### 1.4.6 Preroofing Conference

After approval of submittals and before performing roofing and insulation system installation work, hold a preroofing conference to review the following:

- a. Drawings, including Roof Plan, specifications and submittals related to the roof work
- b. Roof system components installation
- c. Procedure for the roof manufacturer's technical representative's onsite inspection and acceptance of the roof structure, and roofing substrate, the name of the manufacturer's technical representatives, the frequency of the onsite visits, distribution of copies of the inspection reports from the manufacturer's technical representatives to roof manufacturer
- d. Contractor's plan for coordination of the work of the various trades involved in providing the roofing system and other components secured to the roofing
- e. Quality control, (ARMA PMBRG98) plan for the roof system installation
- f. Safety requirements

Coordinate preroofing conference scheduling with the Contracting Officer. The conference must be attended by the Contractor, the Contracting Officer's designated personnel, and personnel directly responsible for the installation of roofing and insulation, flashing and sheet metal work, mechanicalelectrical work, other trades interfacing with the roof work, designated safety personnel trained to enforce and comply with ASSP A10.24,

Fire Marshall, and a representative of the roofing materials manufacturer. Before beginning roofing work, provide a copy of meeting notes and action items to all attending parties. Note action items requiring resolution prior to start of roof work.

## 1.5 DELIVERY, STORAGE, AND HANDLING

### 1.5.1 Delivery

Deliver materials in manufacturers' original unopened containers and rolls with labels intact and legible. Mark and remove wet or damaged materials from the site. Where materials are covered by a referenced specification, the container must bear the specification number, type, and class, as applicable. Deliver materials in sufficient quantity to allow work to proceed without interruption.

### 1.5.2 Storage

Protect materials against moisture absorption and contamination or other damage. Avoid crushing or crinkling of roll materials. Store roll materials on end on clean raised platforms or pallets one level high in dry locations with adequate ventilation, such as an enclosed building or closed trailer. Do not store roll materials in buildings under construction until concrete, mortar, and plaster work is finished and dry. Maintain roll materials at temperatures above 10 degrees C for 24 hours immediately before application. Do not store materials outdoors unless approved by the Contracting Officer. Completely cover felts stored outdoors, on and off roof, with waterproof canvas protective covering. Do not use polyethylene sheet as a covering. Tie covering securely to pallets to make completely weatherproof. Provide sufficient ventilation to prevent condensation. Do not store more materials on roof than can be installed the same day and remove unused materials at end of each days work. Distribute materials temporarily stored on roof to stay within live load limits of the roof construction.

Maintain a minimum distance of 10.67 meters for all stored flammable materials, including materials covered with shrink wraps, craft paper or tarps from all torch/welding applications.

Immediately remove wet, contaminated or otherwise damaged or unsuitable materials from the site. Damaged materials may be marked by the Contracting Officer.

### 1.5.3 Handling

Prevent damage to edges and ends of roll materials. Do not install damaged materials in the work. Select and operate material handling equipment to prevent damage to materials or applied roofing.

## 1.6 ENVIRONMENTAL REQUIREMENTS

Do not install roofing system when air temperature is below 4.44 degrees C, during any form of precipitation, including fog, or when there is ice, frost, moisture, or any other visible dampness on the roof deck. Follow manufacturer's printed instructions for Cold Weather Installation.

## 1.7 SEQUENCING

Coordinate the work with other trades to ensure that components which are

to be secured to or stripped into the roofing system are available and that permanent flashing and counter flashing, per NRCA CONDET, and are installed as the work progresses. Ensure temporary protection measures are in place to preclude moisture intrusion or damage to installed materials. Apply roofing immediately following application of insulation as a continuous operation. Coordinate roofing operations with insulation work so that all roof insulation applied each day is covered with roof membrane installation the same day.

## 1.8 WARRANTY

Provide roof system material and workmanship warranties meeting specified requirements. Provide revision or amendment to standard membrane manufacturer warranty as required to comply with the specified requirements. Provide a manufacturer's warranty that has no dollar limit, covers full system water-tightness, and has a minimum duration of 20 years.

### 1.8.1 Roof Membrane Manufacturer Warranty

Furnish the roof membrane manufacturer's 20-year no dollar limit roof system materials and installation workmanship warranty, including flashing, insulation in compliance with ASTM C1289, and accessories necessary for a watertight roof system construction. Provide warranty directly to the Government and commence warranty effective date at time of Government's acceptance of the roof work. The warranty must state that:

- a. If within the warranty period the roof system, as installed for its intended use in the normal climatic and environmental conditions of the facility, becomes non-watertight, shows evidence of moisture intrusion within the assembly, blisters, splits, tears, delaminates, separates at the seams, or shows evidence of excessive weathering due to defective materials or installation workmanship, the repair or replacement of the defective and damaged materials of the roof system assembly and correction of defective workmanship are the responsibility of the roof membrane manufacturer. All costs associated with the repair or replacement work are the responsibility of the roof membrane manufacturer.
- b. When the manufacturer or his approved applicator fail to perform the repairs within 72 hours of notification, emergency temporary repairs performed by others does not void the warranty.
- c. Upon completion of installation, and acceptance by the Contracting Officer and Roofing System Engineer of Record, the manufacturer must supply the appropriate warranty to the Owner.
- d. Installer must submit a minimum two year warranty to the membrane manufacturer from the date of acceptance, with a copy to the Contracting Officer and Roofing System Engineer of Record.

### 1.8.2 Roofing System Installer Warranty

The roof system installer must warrant for a period of two years that the roof system, as installed, is free from defects in installation workmanship, to include the roof membrane, flashing, insulation, accessories, attachments, and sheet metal installation integral to a complete watertight roof system assembly. Write the warranty directly to the Government. The roof system installer is responsible for correction of defective workmanship and replacement of damaged or affected

materials. The roof system installer is responsible for all costs associated with the repair or replacement work.

### 1.8.3 Continuance of Warranty

Repair or replacement work, ARMA 410BUR88, NRCA C3701 that becomes necessary within the warranty period and accomplished in a manner so as to restore the integrity of the roof system assembly and validity of the roof membrane manufacturer warranty for the remainder of the manufacturer warranty period.

## 1.9 CONFORMANCE AND COMPATIBILITY

Provide the entire roofing and flashing system in accordance with specified and indicated requirements, including fire and wind resistance (ANSI/SPRI/FM 4435/ES-1) requirements. Work not specifically addressed and any deviation from specified requirements must be in general accordance with recommendations of the NRCA Roofing and Waterproofing Manual, membrane manufacturer published recommendations and details, and compatible with surrounding components and construction. Submit any deviation from specified or indicated requirements to the Contracting Officer for approval prior to installation.

### 1.10 ELIMINATION, PREVENTION OF FALL HAZARDS

#### 1.10.1 Fall Protection

## PART 2 PRODUCTS

### 2.1 MATERIALS

Coordinate with other specification sections related to the roof work. Furnish a combination of specified materials that comprise a roof system acceptable to the roof membrane manufacturer and meeting specified requirements. Protect materials provided from defects and make suitable for the service and climatic conditions of the installation.

#### 2.1.1 Energy Performance

Install a roof system that meets an overall performance as specified on the drawings or by insulation specified in other sections.

### 2.2 MODIFIED BITUMEN SHEETS MATERIALS

Furnish a combination of specified materials that comprise the modified bitumen manufacturer's standard system of the number and type of plies specified. Provide materials suitable for the service and climatic conditions of the installation. Modified bitumen sheets must be watertight and visually free of pinholes, particles of foreign matter, non-dispersed raw material, factory splices, or other conditions that might affect serviceability. Polymer modifier must comply with ARMA PMBRG98 and be uniformly dispersed throughout the sheet. Edges of sheet must be straight and flat.

- c. SBS Base Sheet: ASTM D6162/D6162M or ASTM D6164/D6164M  
, Type II, Grade S, minimum 2.0 mm thick.



- d. SBS Interply Sheet: ASTM D6162/D6162M or ASTM D6164/D6164M, Type II, Grade S, minimum 2.0 mm thick.
- e. SBS Cap Sheet: ASTM D6162/D6162M or ASTM D6164/D6164M; Type II, Grade G, minimum 3.7 mm thick, and as required to provide specified fire safety rating.

## 2.3 BASE FLASHING MEMBRANE

Membrane manufacturer's standard, minimum two-ply modified bitumen membrane flashing system compatible with the roof membrane specified and as recommended in membrane manufacturer's published literature. Provide flashing membranes that meet or exceed the properties of the material standards specified for the modified bitumen base, interply and cap sheet, except that flashing membrane thickness must be as recommended by the membrane manufacturer.

## 2.4 COLD-APPLIED MEMBRANE ADHESIVE

Membrane manufacturer's recommended low volatile organic compound (VOC) cold process adhesive for application of the membrane plies.

## 2.5 MEMBRANE SURFACING

Provide modified bitumen roof membrane cap sheet with factory-applied granule surfacing of dark color as selected from membrane manufacturer's standard colors.

## 2.6 PRIMER

ASTM D41/D41M, or other primer compatible with the application and as approved in writing by the modified bitumen membrane manufacturer.

## 2.7 MODIFIED BITUMEN ROOF CEMENT

ASTM D4586/D4586M, Type II for vertical surfaces, Type I for horizontal surfaces, compatible with the modified bitumen roof membrane and as recommended by the modified bitumen membrane manufacturer.

## 2.8 CANT AND TAPERED EDGE STRIPS

Provide standard cants and tapered edge strips of the same material as the roof insulation treated with bituminous impregnation, sizing, or waxing and fabricated to provide maximum 45 degree change in direction of membrane. Cant strips must be minimum 38.1 mm thick and provide for minimum 127 mm face and 88.9 mm vertical height when installed at 45 degree face angle, except where clearance restricts height to lesser dimension. Taper edge strips at a rate of 25.4 mm to 38.1 mm per 304.8 mm to a minimum of 3.175 mm of thickness. Provide kiln-dried preservative-treated wood cants, in compliance with requirements of Section 06 10 00 ROUGH CARPENTRY at base of wood nailers set on edge and wood curbing and where otherwise indicated.

## 2.9 FASTENERS AND PLATES

Provide coated, corrosion-resistant fasteners as recommended by the modified bitumen sheet manufacturer's printed instructions and meeting the requirements of CSA A123.21 and FM APP GUIDE for Class I roof deck construction and the wind uplift resistance specified. For fastening of

membrane or felts to wood materials, provide fasteners driven through 25.4 mm diameter metal discs, or one piece composite fasteners with heads not less than 25.4 mm in diameter or 25.4 mm square with rounded or 45 degree tapered corners.

#### 2.9.1 Masonry or Concrete Walls and Vertical Surfaces

Use hardened steel nails or screws with flat heads, diamond shaped points, and mechanically deformed shanks not less than 25.4 mm long for securing felts, modified bitumen sheets, metal items, and accessories to masonry or concrete walls and vertical surfaces. Use power-driven fasteners only when approved in writing by the Contracting Officer.

#### 2.10 PRE-MANUFACTURED ACCESSORIES

Pre-manufactured accessories must be manufacturer's standard for intended purpose, compatible with the membrane roof system and approved for use by the modified bitumen membrane manufacturer.

#### 2.11 WALKPADS

Provide roof walkpads that are polyester reinforced, granule-surfaced modified bitumen membrane material, minimum 5 mm thick, compatible with the modified bitumen sheet roofing and as recommended by the modified bitumen sheet roofing manufacturer. Panels must not exceed 1.219 meters in length. Other walkpad materials require approval of the Contracting Officer prior to installation.

#### 2.12 ROOF INSULATION BELOW MODIFIED BITUMEN MEMBRANE SYSTEM

Provide insulation compatible with the roof membrane, approved by the membrane manufacturer and meeting all the requirements as specified in Section 07 22 00 ROOF AND DECK INSULATION.

#### 2.13 MEMBRANE LINER

Provide self-adhering modified bitumen underlayment conforming to ASTM D1970/D1970M, EPDM membrane liner conforming to ASTM D4637/D4637M, or other waterproof membrane liner material as approved by the Contracting Officer.

#### 2.14 TOP COATING

Provide a top coating product that is Energy Star labeled and is produced and compatible with the roof material of this specification. Provide data identifying Energy Star label for top coating product. Install to the manufacturer's written installation methods. Provide written confirmation that installation of a top coat will not modify or void the required roof warranty.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

Ensure that the following conditions exist prior to application of the roofing materials:

- a. Do not install items that show visual evidence of biological growth.

- b. Drains, curbs, cants, control joints, expansion joints, perimeter walls, roof penetrating components, and equipment supports are in place.
- c. Surfaces are rigid, clean, dry, smooth, and free from cracks, holes, and sharp changes in elevation. Joints in the substrate are sealed to prevent dripping of bitumen into building or down exterior walls.
- d. The plane of the substrate does not vary more than 6.35 mm within an area 3.048 by 3.048 meters when checked with a 3.048 meter straight edge placed anywhere on the substrate.
- e. Substrate is sloped as indicated to provide positive drainage.
- f. Walls and vertical surfaces are constructed to receive counter flashing, and will permit mechanical fastening of the base flashing materials.
- g. Treated wood nailers are in place on non-nailable surfaces, to permit nailing of base flashing at minimum height of 203.2 mm above finished roofing surface.
- h. Protect all combustible materials and surfaces which may contain concealed combustible or flammable materials. All fire extinguishing equipment has been placed as specified.
- i. Verify all Fire Watch personnel assignments.
- j. Treated wood nailers are fastened in place at eaves, gable ends, openings, and intersections with vertical surfaces for securing of membrane, edging strips, attachment flanges of sheet metal, and roof fixtures. Surface-applied nailers are the same thickness as the roof insulation.
- k. Cants are securely fastened in place in the angles formed by walls and other vertical surfaces. The angle of the cant is 45 degrees and the height of the vertical leg is not less than 88.9 mm.
- l. Insulation boards are installed smoothly and evenly, and are not broken, cracked, or curled. There are no gaps in insulation board joints exceeding 6.35 mm in width. Insulation is being roofed over on the same day the insulation is installed.
- m. Cast-in-place substrates have been allowed to cure and the surface dryness requirements specified under paragraph FIELD QUALITY CONTROL have been met.
- n. Roof deck and framing are sloped as indicated to provide positive drainage.

### 3.2 PREPARATION

#### 3.2.1 Protection of Property

##### 3.2.1.1 Protective Coverings

Install protective coverings at paving and building walls adjacent to hoists prior to starting the work. Lap protective coverings not less than

15.24 cm , secure against wind, and vent to prevent collection of moisture on covered surfaces. Keep protective coverings in place for the duration of the roofing work.

### 3.2.2 Equipment

#### 3.2.2.1 Mechanical Application Devices

Mount mechanical application devices on pneumatic-tired wheels. Use devices designed and maintained to operate without damaging the insulation, roofing membrane, or structural components.

#### 3.2.2.2 Electric-Heated Equipment

Provide adequate electrical service as required by manufacturer of electrical equipment to ensure against damage to equipment and property and to ensure proper application of roofing materials.

### 3.2.3 Priming of Surfaces

Prime all surfaces to be in contact with adhered membrane materials. Apply primer at the rate of 3 liters per 10 sq. meters or as recommended by modified bitumen sheet manufacturer's printed instructions to promote adhesion of membrane materials. Allow primer to dry prior to application of membrane materials to primed surface. Avoid flammable primer material conditions in torch applied membrane applications.

#### 3.2.3.1 Priming of Metal Surfaces

Prime flanges of metal components to be embedded into the roof system prior to setting in bituminous materials or stripping into roofing system.

### 3.2.4 Membrane Preparation

Unroll modified bitumen membrane materials and allow to relax a minimum of 30 minutes prior to installation. In cold weather, adhere to membrane manufacturer's additional recommendations for pre-installation membrane handling and preparation. Inspect for damage, pinholes, particles of foreign matter, non-dispersed raw material, factory splices, or other conditions that might affect serviceability. Edges of seams must be straight and flat so that they may be seamed to one another without forming fish mouths or wrinkles. Discard damaged or defective materials.

### 3.2.5 Substrate Preparation

Apply membrane to clean, dry surfaces only. Do not apply membrane to surfaces that have been wet by rain or frozen precipitation within the previous 12 hours. Provide cleaning and artificial drying with heated blowers or torches as necessary to ensure clean, dry surface prior to membrane application. Torches may not be used to ensure clean, dry surfaces prior to membrane applications if the roof deck or materials used in the installation of the roofing system are combustible.

## 3.3 APPLICATION

Apply roofing materials as specified herein unless approved otherwise by the Contracting Officer. Keep roofing materials dry before and during application. Complete application of roofing in a continuous operation. Begin and apply only as much roofing in one day as can be completed that

same day. Maintain specified temperatures for asphalt.

### 3.3.1 Phased Membrane Construction

Phased application of membrane plies is prohibited unless otherwise approved by the Contracting Officer and supported by the membrane manufacturer's written application instructions. If cap sheet installation is delayed, thoroughly clean the applied membrane material surface and dry immediately prior to cap sheet installation. Priming of the applied membrane surface may be required at the discretion of the Contracting Officer prior to cap sheet installation.

### 3.3.2 Temporary Roofing and Flashing

Provide watertight temporary roofing and flashing where considerable work by other trades, such as installing antennas, and pipes, is to be performed on the roof or where construction scheduling or weather conditions require protection of the building's interior before permanent roofing system can be installed. Do not install temporary roofing over permanently installed insulation. Provide rigid pads for traffic over temporary roofing.

#### 3.3.2.1 Removal

Completely remove temporary roofing and flashing before continuing with application of the permanent roofing system.

### 3.3.3 Application Method

#### 3.3.3.1 Cold Adhesive Applied Modified Bitumen Membrane

Apply cold adhesive with airless sprayer or 6.35 mm saw-toothed rubber squeegee to prepared surfaces in accordance with membrane manufacturer's application instructions. Fully cover substrate with adhesive. Roll or lay membrane in adhesive in accordance with manufacturer's recommendations and within the time limitations of adhesive application. Broom the membrane to ensure full contact with adhesive. Seal laps with adhesive or by heat fusing with hot air welder as required by membrane manufacturer. Minimize traffic on installed membrane during the adhesive cure and set time.

#### 3.3.4 Modified Bitumen Base Sheet

Fully adhere base sheets in accordance with membrane manufacturer's printed instructions. Apply cold adhesive with airless sprayer or a 6.35 mm saw-toothed rubber squeegee and at application rate recommended by the membrane manufacturer. Fully cover substrate with cold adhesive. Ensure laps areas of base sheet are fully sealed. Roll and broom in the base sheet to ensure full contact with the adhesive application. Apply sheets in a continuous operation. Apply sheets with side laps at a minimum of 50.8 mm unless greater side lap is recommended by the manufacturer's standard written application instructions. Provide end laps of not less than 152.4 mm and staggered a minimum of 914.4 mm. Apply sheets at right angles to the roof slope so that the direction of water flow is over and not against the laps. Extend base sheets approximately 50.8 mm above the top of cant strips at vertical surfaces and to the top of cant strips elsewhere. Trim base sheet to a neat fit around vent pipes, roof drains, and other projections through the roof. Application must be free of ridges, wrinkles, and buckles.

### 3.3.5 Modified Bitumen Membrane Application

Ensure proper sheet alignment prior to installation. Apply membrane layers perpendicular to slope of roof in shingle fashion to shed water, including application on areas of tapered insulation that change slope direction. Bucking or backwater laps are prohibited. Fully adhere membrane sheets to underlying substrate materials. Provide minimum 76.2 mm side laps and minimum 152.4 mm end laps and as otherwise required by membrane manufacturer. Stagger end laps minimum 914.4 mm. Offset side laps between membrane layers a minimum of 304.8 mm. Offset end laps between membrane layers a minimum of 914.4 mm. Install all membrane layers the same workday, unless supported otherwise by roof membrane manufacturer application instructions and approved by the Contracting Officer. Provide tight smooth laminations of each membrane layer without wrinkles, ridges, buckles, kinks, fishmouths, or voids. Ensure full membrane adhesion and full lap seals. Rework to seal any open laps prior to application of subsequent membrane layers. The completed membrane application must be free of surface abrasions, air pockets, blisters, ridges, wrinkles, buckles, kinks, fishmouths, voids, or open seams.

#### 3.3.5.1 Cap Sheet Installation

Underlying applied membrane must be inspected and repaired free of damage, holes, puncture, gouges, abrasions, and any other defects, and free of moisture, loose materials, debris, sediments, dust, and any other conditions required by the membrane manufacturer prior to cap sheet installation. Do not apply cap sheet if rain or frozen precipitation has occurred within the previous 24 hours. Align cap membrane and apply by the specified method with the proper side and end lap widths. Cut at a 45 degree angle across selvage edge of cap membrane to be overlapped in end lap areas prior to applying overlapping cap membrane. Apply matching granules in any areas of adhesive bleed out while the adhesive is still tacky. Minimize traffic on newly installed cap sheet membrane.

### 3.3.6 Membrane Flashing

Apply two-ply modified bitumen strip flashing and sheet flashing in the angles formed where the roof deck abuts walls, curbs, ventilators, pipes, and other vertical surfaces, and where necessary to make the work watertight. Apply membrane flashing in accordance with the roof membrane manufacturers printed instructions and as specified. Cut at a 45 degree angle across terminating end lap area of cap membrane prior to applying adjacent overlapping cap membrane. Press flashing into place to ensure full adhesion and avoid bridging. Ensure full lap seal in all lap areas. Mechanically fasten top edge of modified bituminous base flashing 150 mm (6 inches) on center through minimum 25.4 mm diameter tin caps with fasteners of sufficient length to embed minimum one inch into attachment substrate. Apply matching granules in any areas of adhesive bleed out while the adhesive is still tacky. Apply membrane liner over top of exposed nailers and blocking and to overlap top edge of base flashing installation at curbs, parapet walls, expansion joints and as otherwise indicated to serve as waterproof lining under sheet metal flashing components. Metal flashing per SMACNA 1793 guidelines and standards is specified under Section 07 60 00 FLASHING AND SHEET METAL. Do not set metal flashing in hot asphalt.

#### 3.3.6.1 Membrane Strip Flashing

Set primed flanges of metal flashing in full bed of modified bituminous

cement material and securely fasten through to attachment substrate. Strip-in with membrane flashing so that strip extends not less than 101.6 mm beyond outer edge of flange. Where multiple membrane stripping plies are installed, extend each additional stripping ply minimum 101.6 mm beyond edge of previous ply.

#### 3.3.6.2 Membrane Flashing at Roof Drain

Roof drains are specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE. Flashing for roof drains, is specified in Section 07 60 00 FLASHING AND SHEET METAL. Extend membrane sheets to edge of drain bowl opening at the roof drain deck flange in accordance with membrane manufacturer's printed application instructions. Securely clamp membrane sheets and metal roof drain flashing and strip flashing in the flashing clamping ring. Secure clamps so that sheets and metal flashing are free from wrinkles and folds. Trim stripping must be flush with inside of clamping ring.

#### 3.3.6.3 Pre-fabricated Curbs

Securely anchor prefabricated curbs to nailer or other base substrate and flash with modified bitumen membrane.

#### 3.3.6.4 Set-On Accessories

Where pipe or conduit blocking, supports and similar roof accessories are set on the membrane, adhere walkpad material to bottom of accessories prior to setting on roofing membrane. Install set-on accessories to permit normal movement due to expansion, contraction, vibration, and similar occurrences without damaging roofing membrane. Do not mechanically secure set-on accessories through roofing membrane into roof deck substrate.

#### 3.3.6.5 Lightning Protection

Flash and attach lightning protection system components to the roof membrane in a manner acceptable to the roof membrane manufacturer.

#### 3.3.7 Roof Walkpads

Install walkpads at roof access points and where otherwise indicated for traffic areas and for access to mechanical equipment, in accordance with the modified bitumen sheet roofing manufacturer's printed instructions. Provide minimum 152.4 mm separation between adjacent walkpads to accommodate drainage. Provide walkpad or an additional layer of cap sheet under precast concrete paver blocks to protect the roofing.

#### 3.3.8 Correction of Deficiencies

Where any form of deficiency is found, take additional measures as deemed necessary by the Contracting Officer to determine the extent of the deficiency and perform corrective actions as directed by the Contracting Officer.

#### 3.3.9 Clean Up

Remove debris, scraps, containers and other rubbish and trash resulting from installation of the roofing system from job site each day.

### 3.4 CORRECTION OF DEFICIENCIES

Where any form of deficiency is found, take additional measures as deemed necessary by the Contracting Officer to determine the extent of the deficiency and perform corrective actions as directed by the Contracting Officer.

### 3.5 PROTECTION OF APPLIED ROOFING

At the end of the day's work and when precipitation is imminent, protect applied modified bitumen roofing system from water intrusion.

#### 3.5.1 Water Cutoffs

Straighten insulation line using loose-laid cut insulation sheets and seal the terminated edge of modified bitumen roofing system in an effective manner. Seal off flutes in metal decking along the cutoff edge. Remove the water cut-offs to expose the insulation when resuming work, and remove the insulation sheets used for fill-in.

#### 3.5.2 Temporary Flashing for Permanent Roofing

Provide temporary flashing at drains, curbs, walls and other penetrations and terminations of roofing sheets until permanent flashing can be applied. Remove temporary flashing before applying permanent flashing.

#### 3.5.3 Temporary Walkways, Runways, and Platforms

Do not permit storing, walking, wheeling, and trucking directly on applied roofing materials. Provide temporary walkways, runways, and platforms of smooth clean boards, mats or planks as necessary to avoid damage to applied roofing materials, and to distribute weight to conform to live load limits of roof construction. Use rubber-tired equipment for roofing work.

### 3.6 FIELD QUALITY CONTROL

Perform field tests in the presence of the Contracting Officer. Notify the Contracting Officer one day before performing tests.

#### 3.6.1 Construction Monitoring

During progress of the roof work, make visual inspections as necessary to ensure compliance with specified parameters. Additionally, verify the following:

a. Materials comply with the specified requirements.

b. Materials are not installed in adverse weather conditions.

All materials are properly stored, handled and protected from moisture or other damages.

c. Equipment is in working order. Metering devices are accurate.

d. Substrates are in acceptable condition, in compliance with specification, prior to application of subsequent materials.

(1) Nailers and blocking are provided where and as needed.



Insulation substrate is smooth, properly secured to its substrate, and without excessive gaps prior to membrane application.

- (2) The proper number, type, and spacing of fasteners are installed.

Membrane heating, hot mopping, or adhesive application is provided uniformly and as necessary to ensure full adhesion of roll materials. Asphalt is heated and applied within the specified temperature range.

The proper number and types of plies are installed, with the specified overlaps.

Applied membrane surface is inspected, cleaned, dry, and repaired as necessary prior to cap sheet installation.

- (3) Lap areas of all plies are completely sealed.

Membrane is fully adhered without ridges, wrinkles, kinks, fishmouths, or other voids or delaminations.

Installer adheres to specified and detailed application parameters.

Associated flashing and sheet metal are installed in a timely manner in accord with the specified requirements.

Temporary protection measures are in place at the end of each work shift.

#### 3.6.1.1 Manufacturer's Inspection

Manufacturer's technical representative must visit the site a minimum of once per week during the installation for purposes of reviewing materials installation practices and adequacy of work in place. Inspections must occur during the first 20 squares of membrane installation, at mid-point of the installation, and at substantial completion, at a minimum. Additional inspections must not exceed one for each 100 squares of total roof area with the exception that follow-up inspections of previously noted deficiencies or application errors must be performed as requested by the Contracting Officer. After each inspection, submit a report, signed by the manufacturer's technical representative to the Contracting Officer within 3 working days. Note in the report overall quality of work, deficiencies and any other concerns, and recommended corrective action.

#### 3.6.2 Samples of Roofing

Take samples per ASTM D5147/D5147M, sized 101.6 mm by 1016 mm cut across width of modified bitumen sheets as directed by the Contracting Officer. Cut samples will be examined by the Contracting Officer for specified number of plies, proper lap width, complete lap seal, full uniform adhesive compound application and adhesion, full bond between plies, harmful foreign materials, presence of moisture, and wet insulation. Where cuts are not retained by the Contracting Officer or disposed, set cut strip back in cut area in bed of modified bitumen cement. Repair area of cut with new minimum two-ply modified bitumen membrane patch.

### 3.6.3 Roof Drain Test

After completing roofing, but prior to Government acceptance, perform the following test for watertight integrity. Plug roof drains and fill with water to edge of drain sump for 8 hours. Do not plug secondary overflow drains at the same time as adjacent primary drain. To ensure some drainage from roof, do not test all drains at same time. Measure water at beginning and end of the test period. When precipitation occurs during test period, repeat test. When water level falls, remove water, thoroughly dry, and inspect installation; repair or replace roofing at drain to provide for a properly installed watertight flashing seal. Repeat test until there is no water leakage.

### 3.7 INFRARED INSPECTION

Eight 8 months after completion of the roofing system, the Contractor must inspect the roof surface using infrared (IR) scanning as specified in ASTM C1153. Where the IR inspection indicates moisture intrusion, replace wet insulation and damaged or deficient materials or construction in a manner to provide watertight construction and maintain the specified roof system warranties. Coordinate infrared inspections with building envelope commissioning activities.

### ]3.8 INSTRUCTIONS TO GOVERNMENT PERSONNEL

Furnish written and verbal instructions on proper maintenance procedures to designated Government personnel. Furnish instructions by a competent representative of the modified bitumen membrane manufacturer and include a minimum of 4 hours on maintenance and emergency repair of the membrane. Include a demonstration of membrane repair, and give sources of required special tools. Furnish information on safety requirements during maintenance and emergency repair operations.

### 3.9 INFORMATION CARD

For each roof, furnish a typewritten information card for facility Records and a card laminated in plastic and framed for interior display at roof access point, or a photoengraved 1 mm thick aluminum card for exterior display. Card must be 215 mm by 275 mm minimum, identifying facility name and number; location; contract number; approximate roof area; detailed roof system description, including deck type, membrane, number of plies, method of application, manufacturer, insulation and cover board system and thickness; presence of tapered insulation for primary drainage, presence of vapor retarder; date of completion; installing contractor identification and contact information; membrane manufacturer warranty expiration, warranty reference number, and contact information. Install card at roof top or access location as directed by the Contracting Officer and provide a paper copy to the Contracting Officer.

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## SECTION 07 60 00

## FLASHING AND SHEET METAL

**08/23**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN WELDING SOCIETY (AWS)

AWS D1.2/D1.2M (2014; Errata 1 2014; Errata 2 2020)  
Structural Welding Code - Aluminum

## ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A308/A308M | (2010) Standard Specification for Steel Sheet, Terne (Lead-Tin Alloy) Coated by the Hot Dip Process                                     |
| ASTM A480/A480M | (2024) Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip       |
| ASTM A653/A653M | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process |
| ASTM A792/A792M | (2022) Standard Specification for Steel Sheet, 55% Aluminum-Zinc Alloy-Coated by the Hot-Dip Process                                    |
| ASTM B32        | (2020) Standard Specification for Solder Metal                                                                                          |
| ASTM B69        | (2021) Standard Specification for Rolled Zinc                                                                                           |
| ASTM B101       | (2022) Standard Specification for Lead-Coated Copper Sheet and Strip for Building Construction                                          |
| ASTM B209/B209M | (2021a) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                                          |
| ASTM B221M      | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)                   |
| ASTM B370       | (2022) Standard Specification for Copper Sheet and Strip for Building Construction                                                      |

|                                                                             |                                                                                                                                    |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D226/D226M                                                             | (2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing                                 |
| ASTM D1784                                                                  | (2020) Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds |
| ASTM D2244                                                                  | (2021) Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates  |
| ASTM D4214                                                                  | (2007; R 2015) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films                                  |
| ASTM D4586/D4586M                                                           | (2007; R 2018) Asphalt Roof Cement, Asbestos-Free                                                                                  |
| ASTM E2112                                                                  | (2023) Standard Practice for Installation of Exterior Windows, Doors and Skylights                                                 |
| NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)                             |                                                                                                                                    |
| NRCA 0429                                                                   | (2022) The NRCA Roofing Manual: Architectural Metal Flashing, Condensation and Air Leakage Control and Reroofing                   |
| SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA) |                                                                                                                                    |
| SMACNA 1793                                                                 | (2012) Architectural Sheet Metal Manual, 7th Edition                                                                               |
| SINGLE PLY ROOFING INDUSTRY (SPRI)                                          |                                                                                                                                    |
| ANSI/SPRI RD-1                                                              | (2019) Performance Standard for Retrofit Drains                                                                                    |
| ANSI/SPRI/FM 4435/ES-1                                                      | (2017) Test Standard for Edge Systems Used with Low Slope Roofing Systems                                                          |

## 1.2 GENERAL REQUIREMENTS

Finished sheet metal assemblies must form a weathertight enclosure without waves, warps, buckles, fastening stresses or distortion, while allowing for expansion and contraction without damage to the system. The sheet metal installer is responsible for cutting, fitting, drilling, and other operations in connection with sheet metal modifications required to accommodate the work of other trades. Coordinate installation of sheet metal items used in conjunction with roofing with roofing work to permit continuous, uninterrupted roofing operations.

### 1.2.1 General Material Requirements

All materials specified in this Section installed in conjunction with the roofing system must be provided by the roofing system manufacturer, or by

a manufacturer approved by the roofing system manufacturer for use in the roofing system, and must form a part of the Warranty as required by the applicable roofing system Section.

### 1.2.2 Additional Wind Uplift Requirements

All flashing and edge materials specified in this Section must be installed in conjunction with the requirements of the roofing system to meet the requirements of ES-1 wind uplift. Details must include but not limited to ANSI/SPRI/FM 4435/ES-1 (ES-1), "Wind Test Design Standard for Edge Systems" used with Low Slope Roofing Systems, Test Methods: RE-1, RE-2 and RE-3 using uplift roofing details of the NRCA 0429.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

- Exposed Sheet Metal Coverings; G
- Gutters; G
- Downspouts; G
- Expansion Joints; G
- Gravel Stops and fascia; G
- Splash Pans; G
- Flashing for Roof Drains; G
- Base Flashing; G
- Counterflashing; G
- Flashing at Roof Penetrations and Equipment Supports; G
- Reglets; G
- Scuppers; G
- Copings; G
- Drip Edges; G
- Conductor Heads; G
- Open Valley Flashing; G
- Eave Flashing; G
- Recycled Content; S

## SD-04 Samples

Finish Samples; G

## SD-07 Certificates

Warranty on Finishes; G

## SD-08 Manufacturer's Instructions

Instructions for Installation; G

Quality Control Plan; G

## SD-10 Operation and Maintenance Data

Cleaning and Maintenance; G

## 1.4 MISCELLANEOUS REQUIREMENTS

## 1.4.1 Product Data

Indicate thicknesses, dimensions, fastenings, anchoring methods, expansion joints, and other provisions necessary for thermal expansion and contraction. Scaled manufacturer's catalog data may be submitted for factory fabricated items.

## 1.4.2 Finish Samples

Submit two color charts and two finish sample chips from manufacturer's standard color and finish options for each type of finish indicated.

## 1.4.3 Operation and Maintenance Data

Submit detailed instructions for installation and quality control during installation, cleaning and maintenance, for each type of assembly indicated.

## 1.5 DELIVERY, HANDLING, AND STORAGE

Package and protect materials during shipment. Uncrate and inspect materials for damage, dampness, and wet-storage stains upon delivery to the job site. Remove from the site and replace damaged materials that cannot be restored to like-new condition. Handle sheet metal items to avoid damage to surfaces, edges, and ends. Store materials in dry, weather-tight, ventilated areas until installation.

## PART 2 PRODUCTS

## 2.1 RECYCLED CONTENT

Provide products with recycled content. Provide data for each product with recycled content, identifying percentage of recycled content.

## 2.2 MATERIALS

Use any metal listed by NRCA 0429 or SMACNA 1793 for a particular item, unless otherwise indicated. Provide materials and configurations in accordance with NRCA 0429 or SMACNA 1793 for each material, while also

meeting the minimum thickness requirements specified in this Section. Different items need not be of the same metal, except that if copper is selected for any exposed item, all exposed items must be copper, and that contact between dissimilar metals must be avoided.

Provide sheet metal items in 2400 to 3000 mm lengths. Single pieces less than 2400 mm long may be used to connect to factory-fabricated inside and outside corners, and at ends of runs. Factory fabricate corner pieces with minimum 300 mm legs. Provide accessories and other items essential to complete the sheet metal installation. Provide accessories made of the same or compatible materials as the items to which they are applied. Fabricate sheet metal items of the materials specified below and to the gage, thickness, or weight shown in Table I at the end of this section. Provide sheet metal items with mill finish unless specified otherwise. Where more than one material is listed for a particular item in Table I, each is acceptable and may be used, except as follows:

#### 2.2.1 Exposed Sheet Metal Items

Provide exposed sheet metal items of the same material. Consider the following as exposed sheet metal: gutters, including hangers; downspouts; gravel stops and fascia; cap, valley, steeped, base, and eave flashings and related accessories.

#### 2.2.2 Drainage

Do not use copper for an exposed item if drainage from that item will pass over exposed masonry, stonework or other metal surfaces. In addition to the metals listed in Table I, lead-coated copper may be used for such items.

#### 2.2.3 Copper, Sheet and Strip

Provide in accordance with ASTM B370, cold-rolled temper, H 00 (standard).

#### 2.2.4 Lead-Coated Copper Sheet

Provide in accordance with ASTM B101.

#### 2.2.5 Lead Sheet

Provide in a minimum weight of 19.6 kilograms per square meter.

#### 2.2.6 Steel Sheet, Zinc-Coated (Galvanized)

Provide in accordance with ASTM A653/A653M, a minimum of 0.70 mm, and a minimum zinc coat weighting of 275 grams per square meter.

#### 2.2.7 Zinc Sheet and Strip

Provide in accordance with ASTM B69, Type I, a minimum of 0.61 mm thick.

#### 2.2.8 Steel Sheet, Aluminum Zinc-Coated

Provide in accordance with ASTM A792/A792M, a minimum of 0.70 mm.

#### 2.2.9 Stainless Steel

Provide in accordance with ASTM A480/A480M, Type 302 or 304, 2D Finish,



fully annealed, dead-soft temper, a minimum of 0.64 mm.

#### 2.2.10 Terne-Coated Steel

Provide in accordance with ASTM A308/A308M, a minimum of 350 by 500 mm with minimum of 18 kilogram coating per double base box. ASTM A308/A308M.

#### 2.2.11 Aluminum Alloy Sheet and Plate

Provide in accordance with ASTM B209/B209M anodized color form alloy, and temper appropriate for use. Provide material not less than 0.813 mm in thickness.

##### 2.2.11.1 Alclad

When fabricated of aluminum, fabricate the following items with Alclad 3003, Alclad 3004, or Alclad 3005, clad on both sides unless otherwise indicated.

- a. Gutters, downspouts, and hangers
- b. Gravel stops and fascia
- c. Flashing

#### 2.2.12 Finishes

Provide exposed exterior sheet metal and aluminum with a baked on, factory applied coil fluoropolymer color coating or approved equal siliconized polyester coating. Dry film thickness of coatings must be 0.020 to 0.033 mm. Color to be selected from manufacturer's standard range of color choices. Field applications of color coatings are prohibited and will be rejected.

##### 2.2.12.1 Warranty on Finishes

Provide a manufacturer's warranty to repair, or replace, sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within 20 years from date of project substantial completion. Deterioration includes the following:

- a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
- b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
- c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

#### 2.2.13 Aluminum Alloy, Extruded Bars, Rods, Shapes, and Tubes

ASTM B221M.

#### 2.2.14 Solder

Provide in accordance with ASTM B32, 95-5 tin-antimony.

## 2.2.15 Reglets

### 2.2.15.1 Polyvinyl Chloride Reglets

Provide in accordance with ASTM D1784, Type II, Grade 1, Class 14333-D, 1.9 mm minimum thickness.

### 2.2.15.2 Metal Reglets

Provide factory fabricated caulked type or friction type reglets with a minimum opening of 6 mm and a depth of 30 mm, as approved.

#### 2.2.15.2.1 Caulked Reglets

Provide with rounded edges, temporary reinforcing cores, and accessories as required for securing to adjacent construction. Provide built-up mitered corner pieces for inside and outside corners.

#### 2.2.15.2.2 Friction Reglets

Provide with flashing receiving slots not less than 16 mm deep, 25 mm jointing tongues, and upper and lower anchoring flanges installed at 600 mm maximum snap-lock type receiver.

## 2.2.16 Scuppers

Line interiors of scupper openings with sheet metal. Provide a drip edge at bottom edges with returns of not less than 25 mm against the face of the outside wall at the top and sides. Provide the perimeter of the lining approximately 13 mm less than the perimeter of the scupper.

## 2.2.17 Conductor Heads

Provide conductor heads and screens in the same material as downspouts. Provide outlet tubes not less than 100 mm long.

## 2.2.18 Splash Pans

Provide splash pans where downspouts discharge onto roof surfaces and at locations indicated. Unless otherwise indicated, provide pans not less than 600 mm long by 450 mm wide with metal ribs across bottoms of pans. Provide sides of pans with vertical baffles not less than 25 mm high in the front, and 100 mm high in the back.

## 2.2.19 Copings

Unless otherwise indicated, provide copings in copper sheets, 2400 or 3000 mm long, joined by a 20 mm locked and soldered seam.

## 2.2.20 Bituminous Plastic Cement

Provide in accordance with ASTM D4586/D4586M, Type I.

## 2.2.21 Roofing Felt

Provide in accordance with ASTM D226/D226M Type II.

### 2.2.22 Fasteners

Use the same metal as, or a metal compatible with the item fastened. Use stainless steel fasteners to fasten. Confirm compatibility of fasteners and items to be fastened to avoid galvanic corrosion due to dissimilar materials.

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 Metal Roofing

##### 3.1.1.1 Standing-seam Method

Make standing seams parallel with slope of roof. Fabricate sheets into long lengths at shop by locking short dimensions together and thoroughly soldering joints thus formed. In applying metal, turn up one edge of course at each side seam at right angles 40 mm. Then install 50 by 75 mm cleats spaced 300 mm apart by fastening one end of each cleat to roof with two 25 mm long nails and folding roof end back over nail heads. Turn end adjoining turned-up side seam up over upstanding edge of course. Turn up adjoining edge of next course 45 mm and abutting upstanding edges locked, turned over, and flattened against one side of standing seam. Make standing seams straight, rounded neatly at the top edges, and stand about 25 mm above roof deck. All sheets must be same length, except as required to complete run or maintain pattern. Locate transverse joints of each panel half way between joints in adjacent sheets. Align joints of alternate sheets horizontally to produce uniform pattern, as shown in SMACNA 1793.

##### 3.1.2 Workmanship

Make lines and angles sharp and true. Free exposed surfaces from visible wave, warp, buckle, and tool marks. Fold back exposed edges neatly to form a 13 mm hem on the concealed side. Make sheet metal exposed to the weather watertight with provisions for expansion and contraction.

Make surfaces to receive sheet metal plumb and true, clean, even, smooth, dry, and free of defects and projections. For installation of items not shown in detail or not covered by specifications conform to the applicable requirements of SMACNA 1793, Architectural Sheet Metal Manual. Provide sheet metal flashing in the angles formed where roof decks abut walls, curbs, ventilators, pipes, or other vertical surfaces and wherever indicated and necessary to make the work watertight. Join sheet metal items together as shown in Table II.

##### 3.1.3 Nailing

Confine nailing of sheet metal generally to sheet metal having a maximum width of 450 mm. Confine nailing of flashing to one edge only. Space nails evenly not over 75 mm on center and approximately 13 mm from edge unless otherwise specified or indicated. Face nailing is not permitted. Where sheet metal is applied to other than wood surfaces, include in shop drawings, the locations for sleepers and nailing strips required to secure the work. Secure flashing at one-half the normal interval to ensure a wind-resistant installation.

### 3.1.4 Cleats

Provide cleats for sheet metal 450 mm and over in width. Space cleats evenly not over 300 mm on center unless otherwise specified or indicated. Unless otherwise specified, provide cleats of 50 mm wide by 75 mm long and of the same material and thickness as the sheet metal being installed. Secure one end of the cleat with two nails and the cleat folded back over the nailheads. Lock the other end into the seam. Where the fastening is to be made to concrete or masonry, use screws and drive in expansion shields set in concrete or masonry. Pre-tin cleats for soldered seams.

### 3.1.5 Bolts, Rivets, and Screws

Install bolts, rivets, and screws where indicated or required. Provide compatible washers where required to protect surface of sheet metal and to provide a watertight connection. Provide mechanically formed joints in aluminum sheets 1.0 mm or less in thickness.

### 3.1.6 Seams

Straight and uniform in width and height with no solder showing on the face.

#### 3.1.6.1 Flat-lock Seams

Finish not less than 20 mm wide.

#### 3.1.6.2 Lap Seams

Finish soldered seams not less than 25 mm wide. Overlap seams not soldered, not less than 75 mm.

#### 3.1.6.3 Loose-Lock Expansion Seams

Not less than 75 mm wide; provide minimum 25 mm movement within the joint. Completely fill the joints with the specified sealant, applied at not less than 3 mm thick bed.

#### 3.1.6.4 Standing Seams

Not less than 60 mm high, double locked without solder.

#### 3.1.6.5 Flat Seams

Make seams in the direction of the flow.

### 3.1.7 Soldering

Where soldering is specified, apply to copper, terne-coated stainless steel, zinc-coated steel, and stainless steel items. Pre-tin edges of sheet metal before soldering is begun. Seal the joints in aluminum sheets of 1.0 mm or less in thickness with specified sealants. Do not solder aluminum.

#### 3.1.7.1 Edges

Scrape or wire-brush the edges of lead-coated material to be soldered to produce a bright surface. Flux brush the seams in before soldering. Treat with soldering acid flux the edges of stainless steel to be

pre-tinned. Seal the joints in aluminum sheets of 1.0 mm or less in thickness with specified sealants. Do not solder aluminum.

### 3.1.8 Welding and Mechanical Fastening

Use welding for aluminum of thickness greater than 1.0 mm. Aluminum 1.0 mm or less in thickness must be butted and the space backed with formed flashing plate; or lock joined, mechanically fastened, and filled with sealant as recommended by the aluminum manufacturer.

#### 3.1.8.1 Welding of Aluminum

Use welding of the inert gas, shield-arc type. For procedures, appearance and quality of welds, and the methods used in correcting welding work, conform to AWS D1.2/D1.2M.

#### 3.1.8.2 Mechanical Fastening of Aluminum

Use No. 12, aluminum alloy, sheet metal screws or other suitable aluminum alloy or stainless steel fasteners. Drive fasteners in holes made with a No. 26 drill in securing side laps, end laps, and flashings. Space fasteners 300 mm maximum on center. Where end lap fasteners are required to improve closure, locate the end lap fasteners not more than 50 mm from the end of the overlapping sheet.

### 3.1.9 Protection from Contact with Dissimilar Materials

#### 3.1.9.1 Copper or Copper-bearing Alloys

Paint with heavy-bodied bituminous paint surfaces in contact with dissimilar metal, or separate the surfaces by means of moistureproof building felts.

#### 3.1.9.2 Aluminum

Do not allow aluminum surfaces in direct contact with other metals except stainless steel, zinc, or zinc coating. Where aluminum contacts another metal, paint the dissimilar metal with a primer followed by two coats of aluminum paint. Where drainage from a dissimilar metal passes over aluminum, paint the dissimilar metal with a non-lead pigmented paint.

#### 3.1.9.3 Metal Surfaces

Paint surfaces in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

#### 3.1.9.4 Wood or Other Absorptive Materials

Paint surfaces that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.

### 3.1.10 Expansion and Contraction

Provide expansion and contraction joints at not more than 9750 mm intervals for aluminum and at not more than 12 meter intervals for other metals. Provide an additional joint where the distance between the last expansion joint and the end of the continuous run is more than half the

required interval. Space joints evenly. Join extruded aluminum gravel stops and fascia by expansion and contraction joints spaced not more than 3600 mm apart.

#### 3.1.11 Base Flashing

Extend up vertical surfaces of the flashing not less than 200 mm and not less than 100 mm under the roof covering. Where finish wall coverings form a counterflashing, extend the vertical leg of the flashing up behind the applied wall covering not less than 150 mm. Overlap the flashing strips with the previously laid flashing not less than 75 mm. Fasten the strips at their upper edge to the deck. Horizontal flashing at vertical surfaces must extend vertically above the roof surface and fastened at their upper edge to the deck a minimum of 150 mm on center with hex headed, galvanized shielded screws a minimum of 150 mm lap of any surface. Solder end laps and provide for expansion and contraction. Extend the metal flashing over crickets at the up-slope side of chimneys, curbs, and similar vertical surfaces extending through sloping roofs, the metal flashings. Extend the metal flashings onto the roof covering not less than 115 mm at the lower side of dormer walls, chimneys, and similar vertical surfaces extending through the roof decks. Install and fit the flashings so as to be completely weathertight. Provide factory-fabricated base flashing for interior and exterior corners. Do not use metal base flashing on built-up roofing.

#### 3.1.12 Counterflashing

Except where indicated or specified otherwise, insert counterflashing in reglets located from 230 to 250 mm above roof decks, extend down vertical surfaces over upturned vertical leg of base flashings not less than 75 mm. Fold the exposed edges of counterflashings 13 mm. Where stepped counterflashings are required, they may be installed in short lengths a minimum 200 mm by 200 mm or may be of the preformed single piece type. Provide end laps in counterflashings not less than 75 mm and make it weathertight with plastic cement. Do not make lengths of metal counterflashings exceed 3000 mm. Form flashings to the required shapes before installation. Factory form corners not less than 300 mm from the angle. Secure the flashings in the reglets with lead wedges and space not more than 450 mm apart; on chimneys and stair/elevator towers short runs, place wedges closer together. Fill caulked-type reglets or raked joints which receive counterflashing with caulking compound. Turn up the concealed edge of counterflashings built into masonry or concrete walls not less than 6 mm and extend not less than 50 mm into the walls. Install counterflashing to provide a spring action against base flashing. Where bituminous base flashings are provided, extend down the counter flashing as close as practicable to the top of the cant strip. Factory form counter flashing to provide spring action against the base flashing. Utilize two piece counterflashing components to minimize damage of counterflashing and reglets during future reroofing activities.

#### 3.1.13 Metal Reglets

Keep temporary cores in place during installation. Ensure factory fabricated caulked type or friction type, reglets have a minimum opening of 6 mm and a minimum depth of 30 mm, when installed.

##### 3.1.13.1 Caulked Reglets

Wedge flashing in reglets with lead wedges every 450 mm, caulked full and

solid with an approved compound.

#### 3.1.13.2 Friction Reglets

Install flashing snap lock receivers at 600 mm by 600 mm on center maximum. When flashing has been inserted the full depth of the slot, caulk the slot, lock with wedges, and fill with sealant.

#### 3.1.14 Polyvinyl Chloride Reglets for Temporary Construction

Rigid polyvinyl chloride reglets may be provided in lieu of metal reglets for temporary construction.

#### 3.1.15 Gravel Stops and fascia

Prefabricate in the shapes and sizes indicated and in lengths not less than 2400 mm. Extend flange at least 100 mm onto roofing. Provide prefabricated, mitered corners internal and external corners. Install gravel stops and fascia after all plies of the roofing membrane have been applied, but before the flood coat of bitumen is applied. Prime roof flange of gravel stops and fascia on both sides with an asphalt primer. After primer has dried, set flange on roofing membrane and strip-in. Nail flange securely to wood nailer with large-head, barbed-shank roofing nails 38 mm long spaced not more than 75 mm on center, in two staggered rows.

##### 3.1.15.1 Edge Strip

Hook the lower edge of fascia at least 20 mm over a continuous strip of the same material bent outward at an angle not more than 45 degrees to form a drip. Nail hook strip to a wood nailer at 150 mm maximum on center. Where fastening is made to concrete or masonry, use screws spaced 300 mm on center driven in expansion shields set in the concrete or masonry. Where horizontal wood nailers are slotted to provide for insulation venting, install strips to prevent obstruction of vent slots. Where necessary, install strips over 2 mm thick compatible spacer or washers.

##### 3.1.15.2 Joints

Leave open the section ends of gravel stops and fascia 6 mm and backed with a formed flashing plate, mechanically fastened in place and lapping each section end a minimum of 100 mm set laps in plastic cement. Face nailing is not permitted. Install prefabricated aluminum gravel stops and fascia in accordance with the manufacturer's printed instructions and details.

#### 3.1.16 Metal Drip Edges

Provide a metal drip edge, designed to allow water run-off to drip free of underlying construction, at eaves and rakes prior to the application of roofing shingles. Apply directly on the wood deck at the eaves and over the underlay along the rakes. Extend back from the edge of the deck not more than 75 mm and secure with compatible nails spaced not more than 250 mm on center along upper edge.

#### 3.1.17 Gutters

The hung type of shape indicated and supported on underside by brackets that permit free thermal movement of the gutter. Provide gutters in sizes

indicated complete with mitered corners, end caps, outlets, brackets, and other accessories necessary for installation. Bead with hemmed edge or reinforce the outer edge of gutter with a stiffening bar not less than 20 by 5 mm of material compatible with gutter. Fabricate gutters in sections not less than 2400 mm. Lap the sections a minimum of 25 mm in the direction of flow or provide with concealed splice plate 150 mm minimum. Join the gutters, other than aluminum, by riveted and soldered joints. Join aluminum gutters with riveted sealed joints. Provide expansion-type slip joints midway between outlets. Install gutters below slope line of the roof so that snow and ice can slide clear. Support gutters on adjustable hangers spaced not more than 750 mm on center by cleats spaced not less than 900 mm apart. Adjust gutters to slope uniformly to outlets, with high points occurring midway between outlets. Fabricate hangers and fastenings from compatible metals.

### 3.1.18 Downspouts

Space supports for downspouts according to the manufacturer's recommendation for the masonry or substrate. Types, shapes and sizes are indicated. Provide complete including elbows and offsets. Provide downspouts in approximately 3000 mm lengths. Provide end joints to telescope not less than 13 mm and lock longitudinal joints. Provide gutter outlets with wire ball strainers for each outlet. Provide strainers to fit tightly into outlets and be of the same material used for gutters. Keep downspouts not less than 25 mm away from walls. Fasten to the walls at top, bottom, and at an intermediate point not to exceed 1500 mm on center with leader straps or concealed rack-and-pin type fasteners. Form straps and fasteners of metal compatible with the downspouts.

#### 3.1.18.1 Terminations

Neatly fit into the drainage connection the downspouts terminating in drainage lines and fill the joints with a portland cement mortar cap sloped away from the downspout. Provide downspouts terminating in splash blocks with elbow-type fittings. Provide splash pans as specified.

### 3.1.19 Flashing for Roof Drains

Provide a 750 mm square sheet indicated. Taper insulation to drain from 600 mm out. Set flashing on finished felts in a full bed of asphalt roof cement, ASTM D4586/D4586M. Heavily coat the drain flashing ring with asphalt roof cement. Clamp the roof membrane, flashing sheet, and stripping felt in the drain clamping ring. Secure clamps so that felts and drain flashing are free of wrinkles and folds. Retrofit roof drains must conform to ANSI/SPRI RD-1.

### 3.1.20 Scuppers

Extend the scupper liner through and project outside of, the wall it penetrates to form a bottom drip edge against the face of the wall. Fold outside edges under 13 mm on all sides. Join the top and sides of the lining on the roof deck side to a closure flange by a locked and soldered joint. Join the bottom edge by a locked and soldered joint to the closure flange, where required, form with a ridge to act as a gravel stop around the scupper inlet. Provide surfaces to receive the scupper lining and coat with bituminous plastic cement.



### 3.1.21 Conductor Heads

Set the depth of the top opening equal to two-thirds of the width or the conductor head. Flat-lock solder seams. Where conductor heads are used in conjunction with scuppers, set the conductor a minimum of 50 mm wider than the scupper. Attach conductor heads to the wall with masonry fasteners. Securely fasten screens to heads.

### 3.1.22 Splash Pans

Install splash pans lapped with horizontal roof flanges not less than 100 mm wide to form a continuous surface. Bend the rear flange of the pan to contour of can't strip and extend up 150 mm under the side wall covering or to height of base flashing under counterflashing. Bed the pans and roof flanges in plastic bituminous cement and strip-flash as specified.

### 3.1.23 Eave Flashing

One piece in width, applied in 2400 to 3000 mm lengths with expansion joints spaced as specified in paragraph EXPANSION AND CONTRACTION. Provide a 20 mm continuous fold in the upper edge of the sheet to engage cleats spaced not more than 250 mm on center. Locate the upper edge of flashing not less than 450 mm from the outside face of the building, measured along the roof slope. Fold lower edge of the flashing over and loose-lock into a continuous edge strip on the fascia. Where eave flashing intersects metal valley flashing, secure with 25 mm flat locked joints with cleats that are 250 mm on center.

### 3.1.24 Flashing at Wall Openings

Install pan flashing in the rough opening sill at all penetrations in the exterior wall assemblies, such as windows, louvers, storefronts and curtain walls. Pan sill flashing must have end dams at both jambs a minimum of 50 mm high and a rear dam of 50 mm high. Flashing must comply with ASTM E2112 and SMACNA 1793.

### 3.1.25 Sheet Metal Covering on Flat, Sloped, or Curved Surfaces

Except as specified or indicated otherwise, cover and flash all minor flat, sloped, or curved surfaces such as crickets, bulkheads, dormers and small decks with metal sheets of the material used for flashing; maximum size of sheets, 375 by 455 mm. Fasten sheets to sheathing with metal cleats. Lock seams and solder. Lock aluminum seams as recommended by aluminum manufacturer. Provide an underlayment of roofing felt for all sheet metal covering.

### 3.1.26 Flashing at Roof Penetrations and Equipment Supports

Provide metal flashing for all pipes, ducts, and conduits projecting through the roof surface and for equipment supports, guy wire anchors, and similar items supported by or attached to the roof deck. Goose-necks, rain hoods, power roof ventilators, and other similar items supported by or attached to the roof deck.

### 3.1.27 Single Pipe Vents

See Table I, footnote (d). Set flange of sleeve in bituminous plastic cement and nail 75 mm on center. Bend the top of sleeve over and extend

down into the vent pipe a minimum of 50 mm. For long runs or long rises above the deck, where it is impractical to cover the vent pipe with lead, use a two-piece formed metal housing. Set metal housing with a metal sleeve having a 100 mm roof flange in bituminous plastic cement and nailed 75 mm on center. Extend sleeve a minimum of 200 mm above the roof deck and lapped a minimum of 75 mm by a metal hood secured to the vent pipe by a draw band. Seal the area of hood in contact with vent pipe with an approved sealant.

#### 3.1.28 Stepped Flashing

Provide stepped flashing where sloping roofs surfaced with shingles abut vertical surfaces. Place separate pieces of base flashing in alternate shingle courses.

#### 3.1.29 Copings

Provide coping with locked and soldered seam. Terminate outer edges in edge strips. Install with sealed lap joints as indicated.

#### 3.1.30 Through Wall Flashing

Provide through wall flashing as required in the applicable wall system Sections.

### 3.2 PAINTING

Touch ups in the field may be applied only after metal substrates have been cleaned and pretreated in accordance with manufacturer's written instructions and products.

Field-paint dissimilar sheet metals in contact to separate and deter galvanic interactions.

#### 3.2.1 Aluminum Surfaces

Clean with solvent and apply one coat of zinc-molybdate primer and one coat of aluminum paint.

### 3.3 CLEANING

Clean exposed sheet metal work at completion of installation. Remove grease and oil films, handling marks, contamination from steel wool, fittings and drilling debris, and scrub-clean. Free the exposed metal surfaces of dents, creases, waves, scratch marks, and solder or weld marks.

### 3.4 REPAIRS TO FINISH

Scratches, abrasions, and minor surface defects of finish may be repaired in accordance with the manufacturer's printed instructions and as approved. Repair damaged surfaces caused by scratches, blemishes, and variations of color and surface texture. Replace items which cannot be repaired.

### 3.5 FIELD QUALITY CONTROL

Establish and maintain a Quality Control Plan for sheet metal used in conjunction with roofing to assure compliance of the installed sheet metalwork with the Contract requirements. Remove work that is not in

compliance with the Contract and replace or correct. Include quality control, but not be limited to, the following:

- a. Observation of environmental conditions; number and skill level of sheet metal workers; condition of substrate.
- b. Verification that specified material is provided and installed.
- c. Inspection of sheet metalwork, for proper size(s) and thickness(es), fastening and joining, and proper installation.

### 3.5.1 Procedure

Submit for approval prior to start of roofing work. Include a checklist of points to be observed. Document the actual quality control observations and inspections. Provide a copy of the documentation to the Contracting Officer at the end of each day.

| TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES |                                   |                           |                     |                                  |   |
|------------------------------------------------------|-----------------------------------|---------------------------|---------------------|----------------------------------|---|
| Sheet Metal Items                                    | Copper kilograms per square meter | Aluminum, mm              | Stainless Steel, mm | Terne-Coated Stainless Steel, mm |   |
| Cover                                                | 4.9                               | 1.60                      | 0.635               | 0.635                            |   |
| Covering on minor flat, pitched or curved surfaces   | 6.125                             | 1.60                      | 0.635               | 0.635                            |   |
| Downspouts and leaders                               | 4.9                               | 1.60                      | 0.635               | 0.635                            |   |
| Downspout clips and anchors                          | -                                 | 1.02 clip<br>3.175 anchor | -                   | -                                | - |
| Downspout straps, 50 mm                              | 14.7 (a)                          | 1.52                      | 1.27                | -                                |   |
| Conductor heads                                      | 4.9                               | 1.60                      | 0.635               | 0.635                            |   |
| Scupper lining                                       | 6.125                             | 1.60                      | 0.635               | 0.635                            |   |
| Strainers, wire diameter or gage                     | 4.0 gage                          | 3.66 diameter             | 2.77 diameter       | -                                |   |
| Flashings:                                           |                                   |                           |                     |                                  |   |
| Base                                                 | 6.125                             | 1.60                      | 0.635               | 0.635                            |   |
| Cap (Counter-flashing)                               | 4.9                               | 1.60                      | 0.635               | 0.635                            |   |
| Eave                                                 | 4.9                               | -                         | 0.635               | 0.635                            |   |

| TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES      |                                   |                   |                     |                                  |  |
|-----------------------------------------------------------|-----------------------------------|-------------------|---------------------|----------------------------------|--|
| Sheet Metal Items                                         | Copper kilograms per square meter | Aluminum, mm      | Stainless Steel, mm | Terne-Coated Stainless Steel, mm |  |
| Spandrel beam                                             | 3.1                               | -                 | 0.635               | 0.635                            |  |
| Bond barrier                                              | 4.9                               | -                 | 0.635               | 0.635                            |  |
| Stepped                                                   | 4.9                               | 1.60              | 0.635               | 0.635                            |  |
| Valley                                                    | 4.9                               | 1.60              | 0.635               | 0.635                            |  |
| Roof drain                                                | 4.9 (b)                           |                   |                     |                                  |  |
| Pipe vent sleeve (d)                                      |                                   |                   |                     |                                  |  |
| Coping                                                    | 4.9                               | -                 | -                   | -                                |  |
| Gravel stops and fascia:                                  |                                   |                   |                     |                                  |  |
| Extrusions                                                | -                                 | 1.91              | -                   | -                                |  |
| Sheets, corrugated                                        | 4.9                               | 1.60              | 0.635               | 0.635                            |  |
| Sheets, smooth                                            | 6.125                             | 1.60              | 0.635               | 0.635                            |  |
| Edge strip                                                | 7.35                              | 1.60              | 0.635               | -                                |  |
| Gutters:                                                  |                                   |                   |                     |                                  |  |
| Gutter section                                            | 4.9                               | 0.81              | 0.38                | 0.38                             |  |
| Continuous cleat                                          | 4.9                               | 0.81              | 0.38                | 0.38                             |  |
| Hangers, dimensions                                       | 25 mm by 3 mm (a)                 | 25 mm by 2 mm (c) | 25 mm by 1 mm       | -                                |  |
| Joint Cover plates (See Table II)                         | 4.9                               | 1.60              | 0.635               | 0.635                            |  |
| Reglets (c)                                               | 3.1                               | -                 | 0.635               | 0.635                            |  |
| Splash pans                                               | 4.9                               | 1.60              | 0.635               | 0.635                            |  |
| (a) Brass.                                                |                                   |                   |                     |                                  |  |
| (b) May be lead weighing 19.6 kilograms per square meter. |                                   |                   |                     |                                  |  |
| (c) May be polyvinyl chloride.                            |                                   |                   |                     |                                  |  |

| TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES                                                                                                     |                                   |              |                     |                                  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------|---------------------|----------------------------------|--|
| Sheet Metal Items                                                                                                                                        | Copper kilograms per square meter | Aluminum, mm | Stainless Steel, mm | Terne-Coated Stainless Steel, mm |  |
| (d) 12.25 kilogram minimum lead sleeve with 100 mm flange. Where lead sleeve is impractical, refer to paragraph SINGLE PIPE VENTS for optional material. |                                   |              |                     |                                  |  |

| TABLE II. SHEET METAL JOINTS                                 |                                                                             |                                                                    |                                                                                                                                                     |
|--------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| TYPE OF JOINT                                                |                                                                             |                                                                    |                                                                                                                                                     |
| Item Designation                                             | Copper, Terne-Coated Stainless Steel, Zinc-Coated Steel and Stainless Steel | Aluminum                                                           | Remarks                                                                                                                                             |
| Joint cap for building expansion seam, cleated joint at roof | 30 mm single lock, standing seam, cleated                                   | 30 mm single lock, standing                                        | --                                                                                                                                                  |
| Flashings                                                    |                                                                             |                                                                    |                                                                                                                                                     |
| Base                                                         | 25 mm<br>75 mm lap for expansion joint                                      | 25 mm flat locked, soldered; sealed; 75 mm lap for expansion joint | Aluminum manufacturer's recommended hard setting sealant for locked aluminum joints. Fill each metal expansion joint with a joint sealing compound. |

| TABLE II. SHEET METAL JOINTS                                                          |                                                                                  |                                                                                         |                                                                                 |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| TYPE OF JOINT                                                                         |                                                                                  |                                                                                         |                                                                                 |
| Item Designation                                                                      | Copper, Terne-Coated Stainless Steel, Zinc-Coated Steel and Stainless Steel      | Aluminum                                                                                | Remarks                                                                         |
| Cap-in reglet                                                                         | 75 mm lap                                                                        | 75 mm lap                                                                               | Seal groove with joint sealing compound.                                        |
| Reglets                                                                               | Butt joint                                                                       | --                                                                                      | Seal reglet groove with joint sealing compound.                                 |
| Eave                                                                                  | 25 mm flat locked, cleated. 25 mm loose locked, sealed expansion joint, cleated. | 25 mm flat locked, locked, cleated 25 mm loose locked, sealed expansion joints, cleated | Same as base flashing.                                                          |
| Stepped                                                                               | 75 mm lap                                                                        | 75 mm lap                                                                               | --                                                                              |
| Valley                                                                                | 150 mm lap cleated                                                               | 150 mm lap cleated                                                                      | --                                                                              |
| Edge strip                                                                            | Butt                                                                             | Butt                                                                                    | --                                                                              |
| Gravel stops:                                                                         |                                                                                  |                                                                                         |                                                                                 |
| Extrusions                                                                            | --                                                                               | Butt with 13 mm space                                                                   | Use sheet flashing beneath and a cover plate                                    |
| Sheet, smooth                                                                         | Butt with 6 mm space                                                             | Butt with 6 mm space                                                                    | Use sheet flashing backup plate.                                                |
| Sheet, corrugated                                                                     | Butt with 6 mm space                                                             | Butt with 6 mm space                                                                    | Use sheet flashing beneath and a cover plate or a combination unit              |
| Gutters                                                                               | 40 mm lap, riveted and soldered                                                  | 25 mm flat locked riveted and sealed                                                    | Aluminum producers recommended hard setting sealant for locked aluminum joints. |
| (a) Provide a 75 mm lap elastomeric flashing with manufacturer's recommended sealant. |                                                                                  |                                                                                         |                                                                                 |

| TABLE II. SHEET METAL JOINTS                                                |                                                                                      |          |         |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------|---------|
| TYPE OF JOINT                                                               |                                                                                      |          |         |
| Item Designation                                                            | Copper, Terne-Coated<br>Stainless Steel,<br>Zinc-Coated Steel and<br>Stainless Steel | Aluminum | Remarks |
| (b) Seal Polyvinyl chloride reglet with manufacturer's recommended sealant. |                                                                                      |          |         |

-- End of Section --

## SECTION 07 61 15.00 20

## ALUMINUM STANDING SEAM ROOFING

**08/16, CHG 2: 11/18**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ALUMINUM ASSOCIATION (AA)

- AA ADM (2020) Aluminum Design Manual
- AA ASM-35 (2000) Specifications for Aluminum Sheet Metal Work in Building Construction, Construction Manual Series Section 5

## AMERICAN IRON AND STEEL INSTITUTE (AISI)

- AISI SG03-3 (2002; Suppl 2001-2004; R 2008)  
Cold-Formed Steel Design Manual Set

## AMERICAN WOOD COUNCIL (AWC)

- AWC NDS (2018) National Design Specification (NDS)  
for Wood Construction

## ASTM INTERNATIONAL (ASTM)

- ASTM B117 (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
- ASTM B209M (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
- ASTM D226/D226M (2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing
- ASTM D522/D522M (2017) Mandrel Bend Test of Attached Organic Coatings
- ASTM D523 (2014; R 2018) Standard Test Method for Specular Gloss
- ASTM D714 (2002; R 2017) Standard Test Method for Evaluating Degree of Blistering of Paints
- ASTM D968 (2022) Standard Test Methods for Abrasion Resistance of Organic Coatings by Falling Abrasive
- ASTM D1654 (2008; R 2016; E 2017) Standard Test Method for Evaluation of Painted or Coated



## Specimens Subjected to Corrosive Environments

|                 |                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D2247      | (2015; R 2020) Standard Practice for Testing Water Resistance of Coatings in 100% Relative Humidity                                                              |
| ASTM D2565      | (2016) Standard Practice for Xenon Arc Exposure of Plastics Intended for Outdoor Applications                                                                    |
| ASTM D4214      | (2007; R 2015) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films                                                                |
| ASTM E84        | (2024) Standard Test Method for Surface Burning Characteristics of Building Materials                                                                            |
| ASTM E330/E330M | (2014; R 2021) Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference |
| ASTM E1592      | (2017) Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Difference                          |
| ASTM G152       | (2013; R 2021) Standard Practice for Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials                                       |
| ASTM G153       | (2013; R 2021) Standard Practice for Operating Enclosed Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials                                         |

## NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)

|             |                                    |
|-------------|------------------------------------|
| NRCA CONDET | (2014) Construction Details Manual |
|-------------|------------------------------------|

## SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

|             |                                                      |
|-------------|------------------------------------------------------|
| SMACNA 1793 | (2012) Architectural Sheet Metal Manual, 7th Edition |
|-------------|------------------------------------------------------|

## U.S. DEPARTMENT OF ENERGY (DOE)

|             |                                                                     |
|-------------|---------------------------------------------------------------------|
| Energy Star | (1992; R 2006) Energy Star Energy Efficiency Labeling System (FEMP) |
|-------------|---------------------------------------------------------------------|

## 1.2 DEFINITIONS

### 1.2.1 Field-Formed Seam

Seams of panels so configured that when adjacent sheets are installed the seam is sealed utilizing mechanical or hand seamers. Crimped (45 degree bend), roll formed (180 degree bend), double roll formed (2 - 180 degree bend), and roll and lock systems are types of field-formed seam systems.

### 1.2.2 Snap Together Seam

Panels so configured that the male and female portions of the seam interlock through the application of foot pressure or tamping with a mallet. Snap-on cap configurations are a type of snap together system.

### 1.2.3 Pre-Formed

Formed to the final, less field-formed seam, profile and configuration in the factory.

### 1.2.4 Field-Formed

Formed to the final, less field-formed seam, profile and configuration at the site of work prior to installation.

### 1.2.5 Roofing System

The roofing system is defined as the assembly of roofing components, including roofing panels, flashing, fasteners, and accessories which, when assembled properly result in a watertight installation.

### 1.2.6 SSMRS

Standing Seam Metal Roof System (SSMRS) is abbreviation of the entire roof system specified herein with all components and parts coming from a single manufacturer's system.

## 1.3 SYSTEM DESCRIPTION

### 1.3.1 Design Requirements

- a. Provide continuous length panels with no joints or seams, except where indicated. Individual panels must be removable for replacement of damaged material.
- b. There must be no exposed or penetrating fasteners except where shown on the approved shop drawings. Fasteners into wood must be stainless steel sheet metal screws with full length threads. Fasteners into steel must be stainless steel or cadmium plated stainless steel screws inserted into predrilled holes. Length and diameter of screws must be sufficient to meet the design loads with a suitable factor of safety for the material to which the roofing components are attached. Calculate fastener capacity in accordance with AISI S903-3, AA ADM or AWC NDS as applicable.
- c. Roof panel standing seam must include a capillary break and be mechanically locked closed by the manufacturer's locking tool. The seam must include a continuous sealant when required by the manufacturer to withstand the rainfall and wind specified in paragraph

## MANUFACTURER'S REQUIREMENTS.

- d. Roof panel anchor clips must be concealed and designed to allow for thermal movement of the panels, except where specific fixed points are indicated.
- e. The system must resist the positive and negative loads specified herein in accordance with "Sheet Building Sheathing Design Guide" of the AA ADM. Determine capacity in accordance with principles of ASTM E330/E330M modified as follows:
  - (1) Test panels must be production material of the type proposed for use. Use either full length or partial length panels with attachment representative of the main part of the roof.
  - (2) Test specimens must be five panels wide, span one or more supports, and must have no end or edge attachment or seals that will restrict crosswise movement of the panels under load. Do not bridge longitudinal seams with tape or film that can restrict separation.
  - (3) Test panels to failure. Report load at failure.
- f. Panels must support walking loads without excessive distortion or telegraphing of the structural supports. Panels must support a 115 kilogram load concentrated on a 2500 square millimeter (mm) area at the center of the panel without buckling or permanent distortion.

## 1.3.2 Performance Requirements

## 1.3.2.1 Wind Loads

Resistance to wind uplift generated by winds of 235 km/h. Roof system and attachment must resist the components and cladding wind pressure provided on sheet S-003 of the contract documents

## 1.3.2.2 Resistance to Water Infiltration

Roofing system must show no infiltration at seams, edges, flashings, counterflashings and penetrations when subjected to a rainfall of 125 mm per hour with 200124 km/h wind.

## 1.3.2.3 Thermal Movement

The system must be capable of withstanding thermal movement based on a temperature range of 5 degrees C below design low air temperature and 82 degrees C for dark colors.

## 1.3.2.4 Deflection

Panel deflection must not exceed  $L/140$ .

## 1.3.2.5 Structural Performance

The structural performance test methods and requirements must be in accordance with ASTM E1592.

#### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

###### Roofing Panels; G

Submit drawings as necessary to supplement the instructions and diagrams. Include design and erection drawings containing an isometric view of the roof showing the design uplift pressures and dimensions of edge, ridge and corner zones. Show typical and special conditions including flashings, accessory installation, materials and thicknesses, all dimensions, anchoring methods, sealant locations, sealant tape locations, fastener layout, sizes, spacing, provisions for thermal movement, terminations, penetrations, and attachments. Details of installation must be in accordance with the manufacturer's Standard Instructions and details or the SMACNA 1793. The manufacturer's technical engineering department must approve the drawings before they are submitted.

##### SD-03 Product Data

###### Roofing Panels; G

Energy Star Label for Aluminum Roofing Product; S

Recycled Content of Aluminum Roofing Products; S

Attachment Clips

Closures

Accessories

Underlayment

Sample Warranty Certificate; G

Submit for all materials to be provided. Submit data sufficient to indicate conformance to specified requirements.

##### SD-04 Samples

###### Roofing Panels

Submit a 300 mm long section of typical panel in color specified.

When colors are not indicated, submit samples of not less than six different manufacturer's standard colors for selection.

###### Accessories

Submit each type of accessory item used in the project including, but not limited to: each type of anchor clip,

closures, fasteners and leg clamps.

#### SD-05 Design Data

Load Calculations; G

#### SD-06 Test Reports

Structural Performance; G

Panel Finish; G

Submit reports of the tests required by this section.

Manufacturer's Field Inspection; G

Submit manufacturer's technical representative's inspection reports as required in paragraph MANUFACTURER'S FIELD INSPECTION.

#### SD-07 Certificates

Technical Representative

Qualification of Installer

Submit documentation proving the installer is factory-trained, has the specified experience and is authorized by the manufacturer to install the products specified.

Coil Stock Compatibility; G

Provide certification of coil compatibility with roll forming machinery to be used for forming panels without warping, waviness, and rippling not part of panel profile; to be done without damage, abrasion or marking of finish coating.

#### SD-08 Manufacturer's Instructions

Sealant

Submit manufacturer's sealant requirements for roofing.

Installation Manual; G

Submit manufacturer's printed installation manual/instructions and standard details.

#### SD-11 Closeout Submittals

Information Card

For each roofing installation, submit a typewritten card or photoengraved aluminum card containing the information listed on Form 1 located at the end of this section.

Warranty

### 1.5 LOAD CALCULATIONS

Submit load calculations for the following by a structural engineer registered as a Professional Engineer in any jurisdiction verifying that the system supplied meets the design loads indicated. Coordinate calculations with manufacturer's test results.

- a. Wind load uplift design pressure at roof locations specified in paragraph WIND LOADS.
- b. Clip spacing and allowable load per clip calculations.
- c. The fastening of clips to structure or intermediate support spacing.
- d. Intermediate support spacing and fastening to structure when required.
- e. Allowable panel span at anchorage spacing indicated.
- f. Safety factor used in determining loading.

### 1.6 QUALITY ASSURANCE

#### 1.6.1 Preroofing Conference

After submittals are received and approved but before roofing and insulation work, including associated work, is preformed, the Contractor must hold a preroofing conference to review the following:

- a. The drawings and specifications
- b. Procedure for on site inspection and acceptance of the roofing substrate and pertinent structural details relating to the roofing system
- c. Contractor's plan for coordination of the work of the various trades involved in providing the roofing system and other components secured to the roofing
- d. Safety requirements.

The preroofing conference must be attended by the Contractor and personnel directly responsible for the roofing and insulation installation, mechanical and electrical work, and the roofing manufacturer's technical representative. Conflicts among those attending the preroofing conference must be resolved and confirmed in writing before roofing work, including associated work, is begun. Prepare written minutes of the preroofing conference and submit to the Contracting Officer.

#### 1.6.2 Manufacturer's Technical Representative

The representative must have authorization from manufacturer to approve field changes and be thoroughly familiar with the products and with installations in the geographical area where construction will take place. The manufacturer's representative must be an employee of the manufacturer with at least 5 years experience in installing the roof system. The representative must be available to perform field inspections and attend meetings as required herein, and as requested by the Contracting Officer.

### 1.6.3 Qualification of Installer

The roofing system installer must be factory-trained, approved by the aluminum roofing system manufacturer to install the system, and must have a minimum of three years experience as an approved applicator with that manufacturer. The applicator must have applied five installations of similar size and scope to this project within the previous 3 years.

### 1.6.4 Single Source

Provide roofing panels, clips, closures and other accessories from a single manufacturer.

### 1.6.5 Manufacturer

The SSMRS must be the product of an aluminum roofing industry recognized SSMRS manufacturer who has been in the practice of manufacturing SSMRS for a period of not less than 5 years and who has been involved in at least 5 projects similar in size and complexity to this project.

### 1.6.6 Laboratory Tests For Panel Finish

Previously manufactured panels of the same type and finish as proposed for the project must have been tested by an approved testing laboratory to ensure conformance to specifications. The term "appearance of base metal" refers to the aluminum base metal. Panels must meet the following test requirements.

#### 1.6.6.1 Salt Spray Test

Panels must withstand a salt spray test for a minimum of 2000 hours in accordance with ASTM B117, including the scribe requirement in the test. Immediately upon removal of the panel from the test, coating must receive a rating of 10, no blistering, as determined by ASTM D714; and a rating of 7, 2 mm failure at scribe, as determined by ASTM D1654, Rating Schedule No. 1.

#### 1.6.6.2 Formability Test

For formability test, when subjected to a 180 degree bend over a 3 mm diameter mandrel in accordance with ASTM D522/D522M, exterior coating film must show only microchecking of the exterior film and there must be no loss of adhesion.

#### 1.6.6.3 Accelerated Weathering Test

Panels must withstand an accelerated weathering test for a minimum of 2000 hours in accordance with ASTM G152, ASTM G153 or ASTM D2565 without cracking, peeling, blistering, loss of adhesion of the protective coating, or corrosion of the base metal. Protective coating that can be readily removed from the base metal with a penknife blade or similar instrument will be considered to indicate loss of adhesion.

#### 1.6.6.4 Chalking Resistance

After the 2000-hour weatherometer test, exterior coating may not chalk greater than No. 8 rating when measured in accordance with ASTM D4214 test procedures.

#### 1.6.6.5 Abrasion Resistance Test for Color Coating

When subjected to the falling sand test in accordance with ASTM D968, coating system must withstand a minimum of 100 liters of sand per 0.025 mm (mil) of coating thickness before appearance of base metal.

#### 1.6.6.6 Humidity Test

When subjected to a humidity cabinet test in accordance with ASTM D2247 for 1000 hours, a scored panel must show no signs of blistering, cracking, creepage, or corrosion.

#### 1.6.6.7 Fire Hazard

The finish on factory-fabricated panels must have a flame spread rating of not more than 25 when tested in accordance with ASTM E84.

#### 1.6.6.8 Gloss

The gloss of the finish must be 30 plus or minus 5 at an angle of 60 degrees, when measured in accordance with ASTM D523.

#### 1.6.6.9 Glare Resistance

Surfaces of panels that will be exposed to the exterior must have a specular reflectance of not more than 10 when measured in accordance with ASTM D523 at an angle of 85 degrees. Requirements specified under FORMABILITY TEST will be waived if necessary to conform to this requirement.

### 1.7 DELIVERY, STORAGE, AND HANDLING

Deliver, store, and handle preformed panels, bulk roofing products and other manufactured items in a manner to prevent damage or deformation.

#### 1.7.1 Delivery

Provide adequate packaging to protect materials during shipment. Do not uncrate materials until ready for use except for inspection. Immediately upon arrival of materials at jobsite, inspect materials for damage, dampness, and staining. Replace damaged or permanently stained materials that cannot be restored to like-new condition with new material. If materials are wet, remove moisture, restack and protect panels until used.

#### 1.7.2 Handling

Handle material carefully to avoid damage to surfaces, edges and ends.

#### 1.7.3 Storage

Stack materials stored on the site on platforms or pallets and cover with tarpaulins or other suitable weathertight coverings which prevent water trapping or condensation. Store panels so that water which might have accumulated during transit or storage will drain off. Do not store the panels in contact with materials that might cause staining, such as mud, lime, cement, fresh concrete or chemicals. Protect stored panels from wind damage.



## 1.8 WARRANTY

Furnish manufacturer's no dollar limit materials and workmanship warranty for the roofing system. The warranty period must be not less than 20 years from the date of Government acceptance of the work. Issue the warranty directly to the Government. The warranty must provide that if within the warranty period the aluminum roofing system becomes non-watertight or shows evidence of corrosion, perforation, peeling paint, rupture or excess weathering due to deterioration of the roofing system resulting from defective materials or workmanship the repair or replacement of the defective materials and correction of the defective workmanship must be the responsibility of the roofing system manufacturer. Repairs that become necessary because of defective materials and workmanship while roofing is under warranty must be performed within 7 days after notification, unless additional time is approved by the Contracting Officer. Failure to perform repairs within the specified period of time will constitute grounds for having the repairs performed by others and the cost billed to the manufacturer. In addition, provide a 2 year contractor installation warranty.

## PART 2 PRODUCTS

### 2.1 ROOFING PANELS

#### 2.1.1 Material

3004 aluminum, ASTM B209M. Aluminum roofing materials must contain a minimum of 30 percent total recycled content. Provide data identifying percentage of recycled content of aluminum roofing products.

Provide aluminum roofing product that is Energy Star labeled. Provide data identifying Energy Star label for aluminum roofing product.

##### 2.1.1.1 Thickness

1.0 mm minimum.

##### 2.1.1.2 Finish

Alclad fluorocarbon baked enamel exterior and neutral washcoat interior.

##### 2.1.1.3 Texture

Smooth.

##### 2.1.1.4 Color

Exterior as selected from the manufacturer's standard colors.

##### 2.1.1.5 Configuration

Provide panels of continuous lengths from ridge to eaves or from top to eaves on shed roof designs. Panels must be 450 mm wide with a minimum 50 mm high vertical legs and two 9.4 mm stiffening ribs at 100 mm on center between the legs to minimize oil-canning and telegraphing of structural members. Leading vertical leg must have a continuous groove in the rib top for anti-siphon protection when hook-rib top of next panel is locked over leading vertical leg to form the standing seam. Panels from coil stock must be formed without warping, waviness or ripples not a part of

the panel profile, and must be free of damage to the finish coating system.

#### 2.1.1.6 Prefinished Coating System

Fluorocarbon baked enamel, factory-applied, minimum total dry film nominal thickness of 0.050 mm, and conforming to test requirements specified herein. Provide prefinished coating system on both faces. Interior face must receive same coating system, or, at the manufacturer's option, receive a coat of acrylic wash coat applied to a minimum total dry film nominal thickness of 0.005 mm.

### 2.2 ATTACHMENT CLIPS

Series 300 non-magnetic stainless steel. Size, shape, thickness and capacity as required to meet the load, insulation thickness and deflection criteria specified.

### 2.3 ACCESSORIES

Sheet metal flashings, trim, moldings, closure strips, caps, preformed crickets, equipment curbs, gutters, down spouts, and other similar sheet aluminum accessories provided in conjunction with preformed aluminum panels must be of the same material and finish as panels, except that such items which will be concealed after installation may be provided without the finish if they are aluminum or stainless steel. Provide ridge and rib closures, as specified. Aluminum must be of thickness not less than that of panels. Molded closure strips must be closed-cell synthetic rubber, neoprene, or polyvinyl chloride premolded to match configurations of preformed aluminum panels. Thermal spacer blocks and other thermal barriers at concealed fasteners must be as recommended by the roofing panel manufacturer.

#### 2.3.1 Closures

##### 2.3.1.1 Ridge Closure

Aluminum-clad foam or aluminum closure with foam secondary closure matching panel configuration for installation on surface of roof panel between panel ribs at ridge and headwall roof panel flashing conditions and terminations. Foam material must not absorb water.

##### 2.3.1.2 Rib Closure

Aluminum, closed-cell or solid-cell synthetic rubber, neoprene or polyvinyl chloride pre-molded to match configuration of rib opening. Material for closures must not absorb water.

#### 2.3.2 Fasteners

Series 300 stainless steel with composite metal and neoprene composition washers. Fasteners for attachment to structural supports and fasteners for attachment of panels must be as approved and in accordance with manufacturer's recommendation. Unless specified otherwise herein, fasteners must be either self-tapping screws, bolts and nuts, or self-locking bolts. Design fastening system to withstand design loads indicated. Fasteners must not be over-torqued and must develop full capacity of attachment clips.

#### 2.3.2.1 Screws

Provide not less than 0.242 mm diameter for self-tapping type and not less than 0.216 mm diameter for self-drilling and self-tapping.

#### 2.3.2.2 Bolts

Provide not less than 6 mm diameter, shouldered or plain shank as required, with proper nuts.

#### 2.3.2.3 Automatic End-Welded Studs

Provide shouldered type with a shank diameter of not less than 5 mm and cap or nut for holding covering against the shoulder.

#### 2.3.2.4 Explosive Driven Fasteners

Provide fasteners to be driven with explosive actuated tools and with a shank diameter of not less than 13 mm for fastening to steel and not less than 25 mm for fastening to concrete.

#### 2.3.2.5 Rivets

Blind rivets must be aluminum with 5 mm nominal diameter shank or stainless steel with 3 mm nominal diameter shank. Rivets must be threaded stem type if used for other than fastening trim. Rivets with hollow stems must have closed ends.

#### 2.3.3 Sealant

Elastomeric type containing no oil or asphalt. Exposed sealant must cure to a rubberlike consistency. Concealed sealant must be the non-hardening type. Seam sealant must be factory-applied, non-skinning, non-drying, and must conform to the roofing manufacturer's recommendations. Do not use silicone-based sealants in contact with finished metal panels and components unless approved otherwise by the Contracting Officer.

#### 2.3.4 Sealant Tape

Polyvinyl chloride closed cell foam tape or composed of 99 percent solids in a base of butyl polyisobutylene rubber with the following properties and characteristics:

- a. Webbing and Elongation: 100 percent minimum at 25 degrees C
- b. Adhesion: Excellent to surfaces used
- c. U-V light exposure: No effect
- d. Ozone: No effect
- e. Weathering: 1000 hours in QUV Test Apparatus - Excellent, no cracking, bleeding, or significant changes.
- f. Moisture Transmission: 0.05 to 0.15 grams per 62500 square mm in 24 hours.
- g. Service Temperature Tests: Bending over 13 mm mandrel at minus 50 degrees C with no cracking. Expose sealed typical metal lap joint to

plus 176 degrees C for 24 hours with no significant loss of original properties.

- h. Reaction to Metals: Non-corrosive to aluminum

## 2.4 UNDERLAYMENT FOR WOOD SUBSTRATES

Provide underlayment ASTM D226/D226M, Type I perforated, covered by water-resistant rosin sized building paper.

## 2.5 LINER PANELS

Fabricate liner panels of the same material as roof panels, and formed or patterned to prevent waviness and distortion. Liner panels must have a factory applied, 0.025 mm thick minimum painted coating on the inside face, and a prime coat on the liner side.

## 2.6 UNDERLAYMENT

### 2.6.1 Self-Adhering High Temperature Ice and Water Shield

Provide high-temperature self-adhering underlayment designed to resist temperatures up to 121 C. Provide a 40 mil membrane comprised of skid resistant polyolefin film laminated to highly adhesive rubberized asphalt. Refer to drawings for the extent of application.

- a. Membrane Tensile Strength: MD 33 lbf/in, CD 31 lbf/in in compliance with ASTM D-412 or UNE-EN 12311-1:2000

- b. Water Vapor Permeance: Not more than 0.05 perms in compliance with ASTM E-96, Method B, or UNE-EN 12086:2013

## PART 3 EXECUTION

### 3.1 EXAMINATION

Do not use building construction materials that show visible evidence of biological growth.

Examine surfaces to receive standing seam aluminum roofing and flashing. Provide plumb and true surfaces, clean, even, smooth and as dry as possible. Ensure that surfaces are free from defects and projections which might affect the installation. Report unsuitable conditions to Contracting Officer.

### 3.2 PROTECTION OF DISSIMILAR METALS

Where an aluminum component is in contact with, fastened to, or contacted by drainage from dissimilar metals other than stainless steel, give such dissimilar metals one of the following treatments:

- a. A heavy brush coat of primer followed by two coats of aluminum metal and masonry paint.
- b. A heavy coat of alkali-resistant bituminous paint.
- c. Separate contact surfaces with non-absorptive tape or gasket.

### 3.2.1 Contact with Masonry

Where aluminum is in contact with masonry, concrete, or plaster, apply a heavy coat of alkali-resistant bituminous paint.

### 3.2.2 Contact with Wood

Where aluminum is in contact with wood or other absorptive material subject to wetting, or with wood treated with a preservative not compatible with aluminum, seal joints with sealing compound and apply one heavy brush coat of aluminum pigmented bituminous paint.

## 3.3 INSTALLATION

Install in accordance with approved manufacturer's erection instructions shop drawings, and diagrams, except as specified otherwise herein. Provide panels in full and firm contact with clips. Obtain approval prior to installation on prefinished panels cut in the field, and factory applied coverings or coatings that were repaired after being abraded or damaged during handling or installation. Make repairs with material of same color as weather coating. Completely seal openings through panels. Correct defects or errors in materials in an approved manner. Replace materials which cannot be corrected in an approved manner with new materials. Provide molded closure strips where indicated and where necessary for weathertight construction. Use shims as required to ensure clip line is true. Use a spacing gage at each row of panels to ensure that panel width is not stretched or shortened.

### 3.3.1 Roof Panels

Apply roofing panels with standing seams parallel to slope of roof. Provide roofing panels in full lengths from ridge to eaves (top to eaves on shed roofs), with no transverse joints except at the junction of ventilators, curbs, skylights, chimneys, and similar openings. Form interlocking rib type panel seams in the field with an automatic mechanical seamer approved by the manufacturer. Attach panels to structure with concealed clips which are incorporated into the panel seams. Clip attachment must allow roof to move freely and independently of the structure, except at fixed points as indicated.

### 3.3.2 Flashings

Provide flashing and related closures and accessories in connection with preformed metal panels as indicated and as necessary to provide a weathertight installation. Install flashing to ensure positive water drainage away from roof penetrations. Flash and seal roof at ridge, eaves and rakes, at projections through roof, and elsewhere as necessary. Accomplish placement of closure strips, flashing, and sealing material in an approved manner that will ensure complete weathertightness. Details of installation which are not indicated must be in accordance with the NRCA CONDET, SMACNA 1793, AA ASM-35, panel manufacturer's printed instructions and details of the approved shop drawings. Installation must allow for expansion and contraction of flashing.

### 3.3.3 Flashing Fasteners

Fastener spacings must be in accordance with the panel manufacturer's recommendations and as necessary to withstand the indicated design loads.

Install fasteners in roof valleys as recommended by the manufacturer of the panels. Install fasteners in straight lines within a tolerance of 13 mm in the length of a bay. Drive exposed penetrating type fasteners normal to the surface and to a uniform depth to seat gasketed washers properly and drive so as not to damage factory applied coating. Exercise extreme care in drilling pilot holes for fastenings to keep drills perpendicular and centered. Do not drill through sealant tapes. After drilling, remove metal filings and burrs from holes prior to installing fasteners and washers. Torque used in applying fasteners must not exceed that recommended by the manufacturer. Remove panels deformed or otherwise damaged by over-torqued fastenings, and provide new panels.

#### 3.3.4 Closure/Closure Strips

Set closure/closure strips in joint sealant material.

#### 3.4 CLEANING

Clean exposed sheet metal work at completion of installation. Remove metal shavings, filings, nails, bolts, and wires from roofs on completion to prevent discoloration and harm to the panels and flashing. Remove grease and oil films, excess sealants handling marks, contamination from steel wool, fittings and drilling debris and scrub the work clean. Exposed metal surfaces must be free of dents, creases, waves, scratch marks, and solder or weld marks.

#### 3.5 MANUFACTURER'S FIELD INSPECTION

Manufacturer's technical representative must visit the site as necessary during the installation process to assure panels, flashings, and other components are being installed in a satisfactory manner. Manufacturer's technical representative must perform a field inspection during the first 20 squares of roof panel installation and at substantial completion prior to issuance of warranty, as a minimum, and as otherwise requested by the Contracting Officer. Additional inspections must not exceed one for each 100 squares of total roof area with the exception that follow-up inspections of previously noted deficiencies or application errors must be performed as requested by the Contracting Officer. Each inspection visit must include a review of the entire installation to date. After each inspection, submit a report, signed by the manufacturer's technical representative, to the Contracting Officer noting the overall quality of work, deficiencies and any other concerns, and recommended corrective actions in detail. Notify Contracting Officer a minimum of 2 working days prior to site visit by manufacturer's technical representative.

#### 3.6 COMPLETED WORK

Completed work must be plumb and true without oil canning, dents, ripples, abrasion, rust, staining, or other damage detrimental to the performance or aesthetics of the completed roof assembly.

#### 3.7 INFORMATION CARD

For each roof, provide a typewritten card, laminated in plastic and framed for interior display or a photoengraved 0.8 mm thick aluminum card for exterior display. Card to be 220 by 280 mm minimum and contain the information listed on Form 1 at end of this section. Install card near point of access to roof, or where indicated. Send a photostatic paper copy to LANTNAVFACENGCOM, Code 1613, 1510 Gilbert Street, Norfolk, VA

23511-2699.

### 3.8 SCHEDULE

Some metric measurements in this section are based on mathematical conversion of English unit measurements, and not on metric measurement commonly agreed to by the manufacturers or other parties. The English and metric units for the measurements shown are as follows:

| PRODUCTS          | ENGLISH UNITS | METRIC UNITS |
|-------------------|---------------|--------------|
| a. Sheet Aluminum | 0.040 inch    | 1.0 mm       |
| b. Panels         | 12 inches     | 300 mm       |
| - vertical legs   | 2 inches      | 50 mm        |
| - stiffening ribs | 4 inches      | 100 mm       |
| c. Screws         | No. 14        | 0.242 mm     |
|                   | No. 12        | 0.216 mm     |
| d. Bolts          | 1/4 inch      | 6 mm         |
| e. Studs          | 3/16 inch     | 5 mm         |
| f. Fasteners      | 1/2 inch      | 13 mm        |
|                   | One inch      | 25 mm        |
| g. Rivets         | 1/16 inch     | 5 mm         |
|                   | 1/8 inch      | 3 mm         |

### 3.9 FORM ONE

## FORM 1 - PREFORMED STEEL STANDING SEAM ROOFING SYSTEM COMPONENTS

1. Contract Number:
2. Building Number & Location:
3. NAVFAC Specification Number:
4. Deck/Substrate Type:
5. Slopes of Deck/Roof Structure:
6. Insulation Type & Thickness:
7. Insulation Manufacturer:
8. Vapor Retarder:     ( )Yes     ( )No
9. Vapor Retarder Type:
10. Preformed Steel Standing Seam Roofing Description:
  - a. Manufacturer (Name, Address, & Phone No.):
  - b. Product Name:
  - c. Width:
  - d. Gage:
  - e. Base Metal:
  - f. Method of Attachment:
11. Repair of Color Coating:
  - a. Coating Manufacturer (Name, Address & Phone No.):
  - b. Product Name:
  - c. Surface Preparation:
  - d. Recoating Formula:
  - e. Application Method:
12. Statement of Compliance or Exception: \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_
13. Date Roof Completed:
14. Warranty Period: From \_\_\_\_\_ To \_\_\_\_\_
15. Roofing Contractor (Name & Address):
16. Prime Contractor (Name & Address):

Contractor's Signature \_\_\_\_\_ Date:

Inspector's Signature \_\_\_\_\_ Date:

-- End of Section --



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## SECTION 07 84 00

## FIRESTOPPING

05/10, CHG 1: 08/13

## PART 1 GENERAL

## 1.1 SUMMARY

Furnish and install tested and listed firestopping systems, combination of materials, or devices to form an effective barrier against the spread of flame, smoke and gases, and maintain the integrity of fire resistance rated walls, partitions, floors, and ceiling-floor assemblies, including through-penetrations and construction joints and gaps.

- a. Through-penetrations include the annular space around pipes, tubes, conduit, wires, cables and vents.
- b. Construction joints include those used to accommodate expansion, contraction, wind, or seismic movement; do not allow firestopping material to interfere with the required movement of the joint.

Gaps requiring firestopping include gaps between the curtain wall and the floor slab and between the top of the fire-rated walls and the roof or floor deck above and at the intersection of shaft assemblies and adjoining fire resistance rated assemblies.

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E84          | (2024) Standard Test Method for Surface Burning Characteristics of Building Materials                                                     |
| ASTM E119         | (2024) Standard Test Methods for Fire Tests of Building Construction and Materials                                                        |
| ASTM E814         | (2023a) Standard Test Method for Fire Tests of Penetration Firestop Systems                                                               |
| ASTM E1399/E1399M | (1997; R 2022) Standard Test Method for Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems |
| ASTM E1966        | (2015; R 2019) Standard Test Method for Fire-Resistive Joint Systems                                                                      |
| ASTM E2174        | (2020a) Standard Practice for On-Site Inspection of Installed Firestop Systems                                                            |
| ASTM E2307        | (2023b) Standard Test Method for                                                                                                          |

Determining Fire Resistance of Perimeter  
Fire Barrier Systems Using  
Intermediate-Scale, Multi-story Test  
Apparatus

ASTM E2393

(2020a) Standard Practice for On-Site  
Inspection of Installed Fire Resistive  
Joint Systems and Perimeter Fire Barriers

FM GLOBAL (FM)

FM 4991

(2013) Approval of Firestop Contractors

FM APP GUIDE

(updated on-line) Approval Guide  
<https://www.approvalguide.com/>

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC

(2024) International Building Code

UNDERWRITERS LABORATORIES (UL)

UL 723

(2020) UL Standard for Safety Test for  
Surface Burning Characteristics of  
Building Materials

UL 1479

(2015; Reprint May 2021) Fire Tests of  
Through-Penetration Firestops

UL 2079

(2015; Reprint Jul 2020) Tests for Fire  
Resistance of Building Joint Systems

UL Fire Resistance

(2014) Fire Resistance Directory

ULC 115

Standard Method of Fire Tests of Firestop  
Systems

### 1.3 SEQUENCING

Coordinate the specified work with other trades. Apply firestopping materials, at penetrations of pipes and ducts, prior to insulating, unless insulation meets requirements specified for firestopping. Apply firestopping materials. at building joints and construction gaps, prior to completion of enclosing walls or assemblies. Locate cast-in-place firestop devices and install in place before concrete placement. Install pipe, conduit or cable bundles through cast-in-place device after concrete placement but before area is concealed or made inaccessible. Firestop material must be inspected and approved prior to final completion and enclosing of any assemblies that may conceal installed firestop.

### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Firestopping System; G

SD-03 Product Data

Firestopping Materials; G

SD-06 Test Reports

Inspection; G

SD-07 Certificates

Inspector Qualifications

Firestopping Materials

Installer Qualifications; G

## 1.5 QUALITY ASSURANCE

### 1.5.1 Installer

The installation of firestopping systems must be performed by a dedicated and qualified firestop contractor. Engage an experienced Installer who is:

- a. FM Research approved in accordance with FM 4991, or operating as a UL Certified Firestop Contractor, or
- b. Certified, licensed, or otherwise qualified by the firestopping manufacturer as having the necessary staff, training, and a minimum of 3 years experience in the installation of manufacturer's products in accordance with specified requirements. Submit documentation of this experience. A manufacturer's willingness to sell its firestopping products to the Contractor or to an installer engaged by the Contractor does not in itself confer installer qualifications on the buyer. The Installer must be a trained representative of the manufacturer (not distributor or agent) in the proper selection and installation procedures. Obtain and submit installer's written certification of training, and retain proof of certification for duration of firestop installation.

### 1.5.2 Inspector Qualifications

The inspector must have a minimum of two years experience in construction field inspections of firestopping systems, products, and assemblies. The inspector must be completely independent of, and divested from, the installer, the manufacturer, and the supplier of any material or item being inspected. The inspector must not be a competitor of the installer, the contractor, the manufacturer, or supplier of any material or item being inspected. Include in the qualifications submittal a notarized statement assuring compliance with the requirements stated herein.

## 1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in the original unopened packages or containers showing name of the manufacturer and the brand name. Store materials off the ground, protected from damage and exposure to elements and temperatures in accordance with manufacturer requirements. Remove damaged or deteriorated materials from the site. Use materials within their indicated shelf life.

## PART 2 PRODUCTS

### 2.1 FIRESTOPPING SYSTEM

Submit detail drawings including manufacturer's descriptive data, typical details conforming to UL Fire Resistance or other details certified by another nationally recognized testing laboratory, installation instructions or UL listing details for a firestopping assembly in lieu of fire-test data or report. For those firestop applications for which no UL tested system is available through a manufacturer, submit a manufacturer's engineering judgment, derived from similar UL system designs or other tests for review and approval prior to installation. Submittal must indicate the firestopping material to be provided for each type of application. When more than a total of 5 penetrations and/or construction joints are to receive firestopping, provide drawings that indicate location, "F" "T" and "L" ratings, and type of application.

Also, submit a written report indicating locations of and types of penetrations and types of firestopping used at each location; record type by UL list printed numbers.

### 2.2 FIRESTOPPING MATERIALS

Provide firestopping materials, supplied from a single domestic manufacturer, consisting of commercially manufactured, asbestos-free, nontoxic products FM APP GUIDE approved, or UL listed, for use with applicable construction and penetrating items, complying with the following minimum requirements:

#### 2.2.1 Fire Hazard Classification

Provide material that has a flame spread of 25 or less, and a smoke developed rating of 50 or less, when tested in accordance with ASTM E84 or UL 723. Provide an approved firestopping material as listed in UL Fire Resistance or by a nationally recognized testing laboratory.

#### 2.2.2 Toxicity

Provide material that is nontoxic and carcinogen free to humans at all stages of application or during fire conditions and does not contain hazardous chemicals or require harmful chemicals to clean material or equipment.

#### 2.2.3 Fire Resistance Rating

Firestop systems must be UL Fire Resistance listed or FM APP GUIDE approved with "F" rating at least equal to fire-rating of fire wall or floor in which penetrated openings are to be protected. Where required, firestop systems must also have "T" rating at least equal to the fire-rated floor in which the openings are to be protected.

##### 2.2.3.1 Through-Penetrations

Firestopping materials for through-penetrations, as described in paragraph SUMMARY, must provide "F", "T" and "L" fire resistance ratings in accordance with ASTM E814, UL 1479 or ULC 115. Provide fire resistance ratings as follows:

#### 2.2.3.1.1 Penetrations of Fire Resistance Rated Walls and Partitions

F Rating = Rating of wall or partition being penetrated.

#### 2.2.3.1.2 Penetrations of Fire Resistance Rated Floors, Floor-Ceiling Assemblies and the Ceiling Membrane of Roof-Ceiling Assemblies

F Rating = 3 hour, T Rating = 3 hour. Where the penetrating item is outside of a wall cavity the F rating must be equal to the fire resistance rating of the floor penetrated, and the T rating must be in accordance with the requirements of ICC IBC.

#### 2.2.3.2 Construction Joints and Gaps

Fire resistance ratings of construction joints, as described in paragraph SUMMARY, and gaps such as those between floor slabs and curtain walls must be the same as the construction in which they occur. Provide construction joints and gaps with firestopping materials and systems that have been tested in accordance with ASTM E119, ASTM E1966 or UL 2079 to meet the required fire resistance rating. Provide curtain wall joints with firestopping materials and systems that have been tested in accordance with ASTM E2307 to meet the required fire resistance rating. Systems installed at construction joints must meet the cycling requirements of ASTM E1399/E1399M or UL 2079. Provide a minimum class II movement capability for all joints at the intersection of the top of a fire resistance rated wall and the underside of a fire-rated floor, floor ceiling, or roof ceiling assembly.

#### 2.2.4 Material Certification

Submit certificates attesting that firestopping material complies with the specified requirements. Provide certification of compliance with UL 1479 for all intumescent firestop materials used in through penetration systems.

### PART 3 EXECUTION

#### 3.1 PREPARATION

Areas to receive firestopping must be free of dirt, grease, oil, or loose materials which may affect the fitting or fire resistance of the firestopping system. For cast-in-place firestop devices, formwork or metal deck to receive device prior to concrete placement must be sound and capable of supporting device. Prepare surfaces as recommended by the manufacturer.

#### 3.2 INSTALLATION

Completely fill void spaces with firestopping material regardless of geometric configuration, subject to tolerance established by the manufacturer. Firestopping systems for filling floor voids 100 mm or more in any direction must be capable of supporting the same load as the floor is designed to support or be protected by a permanent barrier to prevent loading or traffic in the firestopped area. Install firestopping in accordance with manufacturer's written instructions. Provide tested and listed firestop systems in the following locations, except in floor slabs on grade:

- a. Penetrations of duct, conduit, tubing, cable and pipe through floors and through fire-resistance rated walls, partitions, and ceiling-floor

assemblies.

- b. Penetrations of vertical shafts such as pipe chases, elevator shafts, and utility chutes.
- c. Gaps at the intersection of floor slabs and curtain walls, including inside of hollow curtain walls at the floor slab.
- d. Gaps at perimeter of fire-resistance rated walls and partitions, such as between the top of the walls and the bottom of roof decks.
- e. Construction joints in floors and fire rated walls and partitions.
- f. Other locations where required to maintain fire resistance rating of the construction.

### 3.2.1 Insulated Pipes and Ducts

Cut and remove thermal insulation where pipes or ducts pass through firestopping, unless insulation meets requirements specified for firestopping. Replace thermal insulation with a material having equal thermal insulating and firestopping characteristics.

### 3.2.2 Fire Dampers

Install and firestop fire dampers in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION. Firestop installed with fire damper must be tested and approved for use in fire damper system.

### 3.2.3 Data and Communication Cabling

Seal cabling for data and communication applications with re-enterable firestopping devices.

#### 3.2.3.1 Re-Enterable Devices

Provide firestopping devices that are pre-manufactured modular devices, containing built-in self-sealing intumescent inserts. Allow for cable moves, additions or changes without the need to remove or replace any firestop materials. Devices must be capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants; while maintaining "L" rating of <10 cfm/sf measured at ambient temperature and 205 degrees C at 0 percent to 100 percent visual fill.

### 3.3 INSPECTION

For Navy projects, install one of each type of penetration and have it inspected and accepted by the NAVFAC Atlantic Division, Naval Facilities Engineering Systems Command, Fire Protection Engineer prior to the installation of the remainder of the penetrations. At this inspection, the manufacturer's technical representative of the firestopping material must be present. For all projects, do not cover or enclose the firestopped areas until inspection is complete and approved by the Contracting Officer. The inspector must inspect the applications initially to ensure adequate preparations (clean surfaces suitable for application, etc.) and periodically during the work to assure that the completed work has been accomplished according to the manufacturer's written instructions and the specified requirements. Submit written reports indicating locations of

and types of penetrations and types of firestopping used at each location; record type by UL listed printed numbers.

### 3.3.1 Inspection Standards

Inspect all firestopping in accordance with ASTM E2393 and ASTM E2174 for firestop inspection, and document inspection results to be submitted.

### 3.3.2 Inspection Reports

Submit inspection report stating that firestopping work has been inspected and found to be applied according to the manufacturer's recommendations and the specified requirements.

-- End of Section --



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## SECTION 07 92 00

## JOINT SEALANTS

**08/23**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|            |                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------|
| ASTM C509  | (2006; R 2021) Standard Specification for Elastomeric Cellular Preformed Gasket and Sealing Material              |
| ASTM C734  | (2015; R 2019) Standard Test Method for Low-Temperature Flexibility of Latex Sealants After Artificial Weathering |
| ASTM C919  | (2024) Standard Practice for Use of Sealants in Acoustical Applications                                           |
| ASTM C920  | (2018; R 2024) Standard Specification for Elastomeric Joint Sealants                                              |
| ASTM C1193 | (2016) Standard Guide for Use of Joint Sealants                                                                   |
| ASTM C1311 | (2022) Standard Specification for Solvent Release Sealants                                                        |
| ASTM C1521 | (2019; R 2020) Standard Practice for Evaluating Adhesion of Installed Weatherproofing Sealant Joints              |
| ASTM D217  | (2021a) Standard Test Methods for Cone Penetration of Lubricating Grease                                          |
| ASTM D1056 | (2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber                         |
| ASTM E84   | (2024) Standard Test Method for Surface Burning Characteristics of Building Materials                             |

## CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

## SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

## SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

## UL SOLUTIONS (UL)

UL 2818 (2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-03 Product Data

Sealants; G

Primers; G

Bond Breakers; G

Backstops; G

Caulking; G

Cleaning Solvents; G

## SD-06 Test Reports

Field Adhesion; G

## SD-07 Certificates

Indoor Air Quality For Interior Sealants; S

Indoor Air Quality For Interior Floor Joint Sealants; S

Indoor Air Quality For Interior Acoustical Sealants; S

Indoor Air Quality For Interior Caulking; S

## 1.3 PRODUCT DATA

Include storage requirements, shelf life, curing time, instructions for mixing and application, and accessories. Provide manufacturer's Safety Data Sheets (SDS) for each solvent, primer and sealant material proposed.

## 1.4 CERTIFICATIONS

### 1.4.1 Indoor Air Quality Certifications

Submit required indoor air quality certifications in one submittal package.

#### 1.4.1.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

## 1.5 ENVIRONMENTAL CONDITIONS

Apply sealant when the ambient temperature is between 4 and 32 degrees C.

## 1.6 DELIVERY AND STORAGE

Deliver materials to the jobsite in unopened manufacturers' sealed shipping containers, with brand name, date of manufacture, and material designation clearly marked thereon. Label elastomeric sealant containers to identify type, class, grade, and use. Handle and store materials in accordance with manufacturer's printed instructions. Prevent exposure to foreign materials or subjection to sustained temperatures exceeding 32 degrees C or lower than 4 degrees C. Keep materials and containers closed and separated from absorptive materials such as wood and insulation.

## 1.7 QUALITY ASSURANCE

### 1.7.1 Compatibility with Substrate

Verify that each sealant is compatible for use with each joint substrate in accordance with sealant manufacturer's printed recommendations for each application.

### 1.7.2 Joint Tolerance

Provide joint tolerances in accordance with manufacturer's printed instructions.

### 1.7.3 Mock-Up

Provide a mock-up of each type of sealant using materials, colors, and techniques approved for use on the project. Approved mock-ups may be incorporated into the Work.

### 1.7.4 Adhesion

Provide in accordance with ASTM C1193 or ASTM C1521.

## PART 2 PRODUCTS

### 2.1 SEALANTS

Provide sealant products that have been tested, found suitable, and

documented as such by the manufacturer for the particular substrates to which they will be applied.

In areas with ambient temperatures that exceed 43.33 degrees C, do not use polybutene, bituminous, acrylic-latex, polyvinyl acetate latex sealants, polychloroprene (neoprene), polyvinyl chloride (PVC), and polyurethane foams, and neoprene, PVC, and styrene butadiene rubber extruded seals and closure strips due to these materials having maximum recommended surface temperature ranges from 54.44 degrees C to 82.22 degrees C.

#### 2.1.1 Interior Sealants

Provide ASTM C920, Type S or M, Grade NS, Class 12.5, Use NT. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior sealants. Location(s) and color(s) of sealant for the following (Note, color "as selected" refers to manufacturer's full range of color options):

| LOCATION                                                                                                                                                                 | COLOR                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| a. Small voids between walls or partitions and adjacent lockers, casework, shelving, door frames, built-in or surface mounted equipment and fixtures, and similar items. | Match adjacent wall color |
| b. Perimeter of frames at doors, windows, and access panels which adjoin exposed interior concrete and masonry surfaces.                                                 | Match adjacent wall color |
| c. Joints of interior masonry walls and partitions which adjoin columns, pilasters, concrete walls, and exterior walls unless otherwise detailed.                        | Match adjacent wall color |
| d. Joints between edge members for acoustical tile and adjoining vertical surfaces.                                                                                      | Match adjacent wall color |
| e. Interior locations, not otherwise indicated or specified, where small voids exist between materials specified to be painted.                                          | As selected               |
| f. Joints between bathtubs and ceramic tile; joints between shower receptors and ceramic tile; joints formed where non-planar tile surfaces meet.                        | Match Grout               |
| g. Joints formed between tile floors and tile base cove; joints between tile and dissimilar materials; joints occurring where substrates change.                         | Match Grout               |
| h. Behind escutcheon plates at valve pipe penetrations and showerheads in showers.                                                                                       | Match Grout               |
| i.                                                                                                                                                                       |                           |

### 2.1.2 Exterior Sealants

For joints in vertical surfaces, provide ASTM C920, Type S or M, Grade NS, Class 25, Use NT. For joints in horizontal surfaces, provide ASTM C920, Type S or M, Grade P, Class 25, Use T. Provide location(s) and color(s) of sealant as follows (Note, color "as selected" refers to manufacturer's full range of color options):

| LOCATION                                                                                                                                                                                                                   | COLOR                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| a. Joints and recesses formed where frames and subsills of windows, doors, louvers, and vents adjoin masonry, concrete, or metal frames. Use sealant at both exterior and interior surfaces of exterior wall penetrations. | Match adjacent surface color |
|                                                                                                                                                                                                                            |                              |
|                                                                                                                                                                                                                            |                              |
|                                                                                                                                                                                                                            |                              |
| e. Expansion and control joints.                                                                                                                                                                                           | Match adjacent surface color |
| f. Interior face of expansion joints in exterior concrete or masonry walls where metal expansion joint covers are not required.                                                                                            | Match adjacent surface color |
| g. Voids where items pass through exterior walls.                                                                                                                                                                          | Match adjacent surface color |
| h. Metal reglets, where flashing is inserted into masonry joints, and where flashing is penetrated by coping dowels.                                                                                                       | Match adjacent surface color |
| i. Metal-to-metal joints where sealant is indicated or specified.                                                                                                                                                          | Match adjacent surface color |
| j. Joints between ends of gravel stops, fascia, copings, and adjacent walls.                                                                                                                                               | Match adjacent surface color |
|                                                                                                                                                                                                                            |                              |

### 2.1.3 Floor Joint Sealants

ASTM C920, Type S or M, Grade P, Class 25, Use T. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom

spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior floor joint sealants. Provide location(s) and color(s) of sealant as follows (Note, color "as selected" refers to manufacturer's full range of color options):

| LOCATION                                                                      | COLOR                        |
|-------------------------------------------------------------------------------|------------------------------|
| a. Seats of metal thresholds for exterior doors.                              | As selected                  |
| b. Control and expansion joints in floors, slabs, ceramic tile, and walkways. | Match adjacent surface color |

#### 2.1.4 Acoustical Sealants

Rubber or polymer based acoustical sealant in accordance with ASTM C919 to have a flame spread of 25 or less and a smoke developed rating of 50 or less when tested in accordance with ASTM E84. Provide non-staining acoustical sealant with a consistency of 250 to 310 when tested in accordance with ASTM D217. Acoustical sealant must remain flexible and adhesive after 500 hours of accelerated weathering as specified in ASTM C734. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior acoustical sealants.

#### 2.2 PRIMERS

Non-staining, quick drying type and consistency as recommended by the sealant manufacturer for the particular application. Provide primers for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

#### 2.3 BOND BREAKERS

Type and consistency as recommended by the sealant manufacturer to prevent adhesion of the sealant to the backing or to the bottom of the joint. Provide bond breakers for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

#### 2.4 BACKSTOPS

Provide glass fiber roving, neoprene, butyl, polyurethane, or polyethylene foams free from oil or other staining elements as recommended by sealant manufacturer. Provide 25 to 33 percent oversized backing for closed cell and 40 to 50 percent oversized backing for open cell material, unless otherwise indicated. Provide backstop material that is compatible with sealant. Do not use oakum or other types of absorptive materials as backstops.

##### 2.4.1 Rubber

Provide in accordance with ASTM D1056, Type 2, closed cell, Class D, Grade round cross section for cellular rubber sponge backing.

#### 2.4.2 Synthetic Rubber

Provide in accordance with ASTM C509, Option I, Type II preformed rods or tubes for synthetic rubber backing.

#### 2.4.3 Neoprene

Provide in accordance with ASTM D1056, closed cell expanded neoprene cord Type 2, Class C, Grade 2C2] for neoprene backing.

#### 2.4.4 Butyl Rubber Based

Provide in accordance with ASTM C1311, from a single component, with solvent release. color as selected from manufacturer's full range of color choices.

#### 2.4.5 Silicone Rubber Base

Provide in accordance with ASTM C920, from a single component, with solvent release, Non-sag. Color as selected from manufacturer's full range of color choices.

### 2.5 CLEANING SOLVENTS

Provide type(s) recommended by the sealant manufacturer and in accordance with environmental requirements herein. Provide solvents for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

## PART 3 EXECUTION

### 3.1 FIELD QUALITY CONTROL

Perform a field adhesion test in accordance with manufacturer's instructions and ASTM C1193, Method A or ASTM C1521, Method A, Tail Procedure. Remove sealants that fail adhesion testing; clean substrates, reapply sealants, and re-test. Test sealants adjacent to failed sealants. Submit field adhesion test report indicating tests, locations, dates, results, and remedial actions taken.

### 3.2 SURFACE PREPARATION

Prepare surfaces according to manufacturer's printed installation instructions. Clean surfaces from dirt, frost, moisture, grease, oil, wax, lacquer, paint, or other foreign matter that would destroy or impair adhesion. Remove oil and grease with solvent; thoroughly remove solvents prior to sealant installation. Wipe surfaces dry with clean cloths. When resealing an existing joint, remove existing caulk or sealant prior to applying new sealant. For surface types not listed below, provide in accordance with sealant manufacturer's printed instructions for each specific surface.

#### 3.2.1 Steel Surfaces

Remove loose mill scale by sandblasting or, if sandblasting is impractical or would damage finished work, scraping and wire brushing. Remove protective coatings by sandblasting or using a residue free solvent. Remove resulting debris and solvent residue prior to sealant installation.



### 3.2.2 Aluminum or Bronze Surfaces

Remove temporary protective coatings from surfaces that will be in contact with sealant. When masking tape is used as a protective coating, remove tape and residual adhesive prior to sealant application. For removing protective coatings and final cleaning, use non-staining solvents recommended by the manufacturer of the item(s) containing aluminum or bronze surfaces.

### 3.2.3 Concrete and Masonry Surfaces

Where surfaces have been treated with curing compounds, oil, or other such materials, remove materials by sandblasting or wire brushing. Remove laitance, efflorescence and loose mortar from the joint cavity. Remove resulting debris prior to sealant installation.

### 3.2.4 Wood Surfaces

Ensure wood surfaces that will be in contact with sealants are free of splinters, sawdust and other loose particles.

## 3.3 SEALANT PREPARATION

Do not add liquids, solvents, or powders to sealants. Mix multicomponent elastomeric sealants in accordance with manufacturer's printed instructions.

## 3.4 APPLICATION

### 3.4.1 Joint Width-To-Depth Ratios

Acceptable Ratios:

| JOINT WIDTH                                    | JOINT DEPTH  |                |
|------------------------------------------------|--------------|----------------|
|                                                | Minimum      | Maximum        |
| For metal, glass, or other nonporous surfaces: |              |                |
| 6 mm (minimum)                                 | 6 mm         | 6 mm           |
| over 6 mm                                      | 1/2 of width | Equal to width |
| For wood, concrete, masonry, stone:            |              |                |
| 6 mm (minimum)                                 | 6 mm         | 6 mm           |
| over 6 mm to 13 mm                             | 6 mm         | Equal to width |
| over 13 mm to 25 mm                            | 50 mm        | 16 mm          |
| Over 25 mm                                     | prohibited   |                |

Unacceptable Ratios: Where joints of acceptable width-to-depth ratios have not been provided, clean out joints to acceptable depths and grind or

cut to acceptable widths without damage to the adjoining work. Grinding is prohibited at metal surfaces.

#### 3.4.2 Unacceptable Sealant Use

Do not install sealants in lieu of other required building enclosure weatherproofing components such as flashing, drainage components, and joint closure accessories, or to close gaps between walls, floors, roofs, windows, and doors, that exceed acceptable installation tolerances. Remove sealants that have been used in an unacceptable manner and correct building enclosure deficiencies to comply with contract documents requirements.

#### 3.4.3 Masking Tape

Place masking tape on the finished surface on one or both sides of joint cavities to protect adjacent finished surfaces from primer or sealant smears. Remove masking tape within 10 minutes of joint filling and tooling.

#### 3.4.4 Backstops

Provide backstops dry and free of tears or holes. Tightly pack the back or bottom of joint cavities with backstop material to provide joints in specified depths. Provide backstops where indicated and where backstops are not indicated but joint cavities exceed the acceptable maximum depths specified in JOINT WIDTH-TO-DEPTH RATIOS Table.

#### 3.4.5 Primer

Clean out loose particles from joints immediately prior to application of. Apply primer to joints in concrete masonry units, wood, and other porous surfaces in accordance with sealant manufacturer's printed instructions. Do not apply primer to exposed finished surfaces.

#### 3.4.6 Bond Breaker

Provide bond breakers to surfaces not intended to bond in accordance with, sealant manufacturer's printed instructions for each type of surface and sealant combination specified.

#### 3.4.7 Sealants

Provide sealants compatible with the material(s) to which they are applied. Do not use a sealant that has exceeded its shelf life or has jelled and cannot be discharged in a continuous flow from the sealant gun. Apply sealants in accordance with the manufacturer's printed instructions with a gun having a nozzle that fits the joint width. Work sealant into joints so as to fill the joints solidly without air pockets. Tool sealant after application to ensure adhesion. Apply sealant uniformly smooth and free of wrinkles. Upon completion of sealant application, roughen partially filled or unfilled joints, apply additional sealant, and tool smooth as specified. Apply sealer over sealants in accordance with the sealant manufacturer's printed instructions.

### 3.5 PROTECTION AND CLEANING

#### 3.5.1 Protection

Protect areas adjacent to joints from sealant smears. Masking tape may be used for this purpose if removed 5 to 10 minutes after the joint is filled and no residual tape marks remain.

#### 3.5.2 Final Cleaning

Upon completion of sealant application, remove remaining smears and stains and leave the work in a clean and neat condition.

- a. Masonry and Other Porous Surfaces: Immediately remove fresh sealant that has been smeared on adjacent masonry, rub clean with a solvent, and remove solvent residue, in accordance with sealant manufacturer's printed instructions. Allow excess sealant to cure for 24 hours then remove by wire brushing or sanding. Remove resulting debris.
- b. Metal and Other Non-Porous Surfaces: Remove excess sealant with a solvent moistened cloth. Remove solvent residue in accordance with solvent manufacturer's printed instructions.

-- End of Section --

## SECTION 08 11 13

## STEEL DOORS AND FRAMES

**08/20**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2025) Structural Welding Code - Steel

## ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A879/A879M (2012; R 2017) Standard Specification for Steel Sheet, zinc Coated by the Electrolytic Process for Applications Requiring Designation of the Coating Mass on Each Surface

ASTM A924/A924M (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process

ASTM C578 (2023) Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation

ASTM C591 (2022) Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation

ASTM C612 (2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation

ASTM D2863 (2019) Standard Test Method for Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)

ASTM E330/E330M (2014; R 2021) Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference

ASTM E1300 (2016) Standard Practice for Determining Load Resistance of Glass in Buildings

|            |                                                                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E1886 | (2019) Standard Test Method for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Missile(s) and Exposed to Cyclic Pressure Differentials |
| ASTM E1996 | (2017) Standard Specification for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Windborne Debris in Hurricanes                        |
| ASTM F2247 | (2018) Standard Test Method for Metal Doors Used in Blast Resistant Applications (Equivalent Static Load Method)                                                                         |
| ASTM F2248 | (2012) Standard Practice for Specifying an Equivalent 3-Second Duration Design Loading for Blast Resistant Glazing Fabricated with Laminated Glass                                       |
| ASTM F2927 | (2012) Standard Test Method for Door Systems Subject to Airblast Loadings                                                                                                                |

BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)

|                    |                                                             |
|--------------------|-------------------------------------------------------------|
| ANSI/BHMA A156.115 | (2016) Hardware Preparation in Steel Doors and Steel Frames |
|--------------------|-------------------------------------------------------------|

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

|                |                           |
|----------------|---------------------------|
| NAAMM HMMA 810 | (2009) Hollow Metal Doors |
|----------------|---------------------------|

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                         |
|----------|-------------------------------------------------------------------------|
| NFPA 80  | (2025) Standard for Fire Doors and Other Opening Protectives            |
| NFPA 105 | (2022) Standard for Smoke Door Assemblies and Other Opening Protectives |
| NFPA 252 | (2022) Standard Methods of Fire Tests of Door Assemblies                |

STEEL DOOR INSTITUTE (SDI/DOOR)

|                 |                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------|
| SDI/DOOR 111    | (2009) Recommended Details for Standard Steel Doors, Frames, and Accessories and Related Components                |
| SDI/DOOR 113    | (2023) Standard Practice for Determining the Steady-State Thermal Transmittance of Steel Door and Frame Assemblies |
| SDI/DOOR A250.3 | (2019) Test Procedure and Acceptance Criteria for Factory Applied Finish Coatings for Steel Doors and Frames       |

|                  |                                                                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| SDI/DOOR A250.6  | (2020) Recommended Practice for Hardware Reinforcing on Standard Steel Doors and Frames                                                       |
| SDI/DOOR A250.8  | (2023) Specifications for Standard Steel Doors and Frames                                                                                     |
| SDI/DOOR A250.11 | (2022) Recommended Erection Instructions for Steel Frames                                                                                     |
| SDI/DOOR A250.13 | (2008, R 2018) Testing and Ratings of Severe Windstorm Resistant Components for Swinging Door Assemblies for Protection of Building Envelopes |

U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                                               |
|--------------|-------------------------------------------------------------------------------|
| UFC 4-010-01 | (2018; with Change 2, 2022) DoD Minimum Antiterrorism Standards for Buildings |
|--------------|-------------------------------------------------------------------------------|

UNDERWRITERS LABORATORIES (UL)

|        |                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------|
| UL 10C | (2016; Reprint May 2021) UL Standard for Safety Positive Pressure Fire Tests of Door Assemblies |
|--------|-------------------------------------------------------------------------------------------------|

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

- Doors; G
- Frames; G
- Accessories
- Schedule of Doors; G
- Schedule of Frames; G

### SD-03 Product Data

- Doors; G
- Recycled Content for Steel Door Product; S
- Frames; G
- Recycled Content for Steel Frame Product; S
- Accessories

### SD-04 Samples

## Factory-applied Enamel Finish; G

## 1.3 DELIVERY, STORAGE, AND HANDLING

Deliver doors, frames, and accessories undamaged and with protective wrappings or packaging. Provide temporary steel spreaders securely fastened to the bottom of each welded frame. Store doors and frames on platforms under cover in clean, dry, ventilated, and accessible locations, with 6 mm airspace between doors. Remove damp or wet packaging immediately and wipe affected surfaces dry. Replace damaged materials with new.

## PART 2 PRODUCTS

## 2.1 STANDARD STEEL DOORS

SDI/DOOR A250.8, except as specified otherwise. Prepare doors to receive door hardware as specified in Section 08 71 00 DOOR HARDWARE. Undercut where indicated. Provide exterior doors with top edge closed flush and sealed to prevent water intrusion. Provide doors at 44.5 mm thick, unless otherwise indicated. Provide door material that uses a minimum of 25 percent recycled content. Provide data indicating percentage of recycled content for steel door product. Provide exterior glazing in accordance with ASTM F2248 and ASTM E1300. Exterior doors must be tested in accordance with ASTM F2247 or ASTM F2927 to meet requirements of UFC 4-010-01.

## 2.1.1 Classification - Level, Performance, Model

## 2.1.1.1 Extra Heavy Duty Doors

SDI/DOOR A250.8, Level 3, physical performance Level A, Model 1 with core construction as required by the manufacturer for interior doors of size(s) and design(s) indicated. Where vertical stiffener cores are required, the space between the stiffeners must be filled with mineral board insulation.

## 2.1.1.2 Maximum Duty Doors

SDI/DOOR A250.8, Level 4, physical performance Level A, Model 1 with core construction as required by the manufacturer for all exterior of size(s) and design(s) indicated. Where vertical stiffener cores are required, the space between the stiffeners must be filled with mineral board insulation.

## 2.2 CUSTOM HOLLOW METAL DOORS

Provide custom hollow metal doors where nonstandard steel doors are indicated. Provide custom steel doors in the door size(s), design(s), materials, construction, gages, and finish as specified for custom steel doors and complying with the requirements of NAAMM HMMA 810. Fill all spaces in exterior doors with insulation. Close top and bottom edges with steel channels not lighter than 1.5 mm thick. Close tops of exterior doors flush with an additional channel and seal to prevent water intrusion.

Prepare doors to receive hardware specified in Section 08 71 00 DOOR HARDWARE. Undercut doors where indicated. Provide doors at 44.5 mm thick, unless otherwise indicated. Provide exterior glazing in accordance with ASTM F2248 and ASTM E1300. Exterior doors must be tested in accordance with ASTM F2247 and ASTM F2927 to meet the requirements of UFC 4-010-01.

## 2.3 ACCESSORIES

### 2.3.1 Astragals

For pairs of exterior steel doors which will not have aluminum astragals or removable mullions, as specified in Section 08 71 00 DOOR HARDWARE provide overlapping steel astragals with the doors.

## 2.4 INSULATION CORES

Provide insulating cores at all exterior doors and other specific doors noted in the door schedule, and provide an apparent U-factor of .48 in accordance with SDI/DOOR 113 and conforming to:

- a. Rigid Cellular Polyisocyanurate Foam: ASTM C591, Type I or II, foamed-in-place or in board form, with oxygen index of not less than 22 percent when tested in accordance with ASTM D2863; or
- b. Rigid Polystyrene Foam Board: ASTM C578, Type I or II; or
- c. Mineral board: ASTM C612, Type I.

## 2.5 STANDARD STEEL FRAMES

SDI/DOOR A250.8, Level 2, except as otherwise specified. Form frames to sizes and shapes indicated, with welded corners. Provide steel frames for doors, , unless otherwise indicated. Provide frame product that uses a minimum of 25 percent recycled content. Provide data indicating percentage of recycled content for steel frame product.

### 2.5.1 Welded Frames

Continuously weld frame faces at corner joints. Mechanically interlock or continuously weld stops and rabbets. Grind welds smooth.

Weld frames in accordance with the recommended practice of the Structural Welding Code Sections 1 through 6, AWS D1.1/D1.1M and in accordance with the practice specified by the producer of the metal being welded.

### 2.5.2 Anchors

Provide anchors to secure the frame to adjoining construction. Provide steel anchors, zinc-coated not lighter than 1.2 mm thick. Provide stainless steel anchors in exterior door frames.

#### 2.5.2.1 Wall Anchors

Provide at least three anchors for each jamb. For frames which are more than 2285 mm in height, provide one additional anchor for each jamb for each additional 760 mm or fraction thereof.

- a. Concrete: Provide anchors of corrugated or perforated steel straps or 5 mm diameter steel wire, adjustable or T-shaped;
- b. Stud partitions: Weld or otherwise securely fasten anchors to backs of frames. Design anchors to be fastened to closed steel studs with sheet metal screws, and to open steel studs by wiring or welding;



- c. Completed openings: Secure frames to previously placed concrete or masonry with expansion bolts in accordance with SDI/DOOR 111; and

#### 2.5.2.2 Floor Anchors

Provide floor anchors drilled for 10 mm anchor bolts at bottom of each jamb member. Where floor fill occurs, terminate bottom of frames at the indicated finished floor levels and support by adjustable extension clips resting on and anchored to the structural slabs.

### 2.6 FIRE AND SMOKE DOORS AND FRAMES

Provide fire and smoke doors and frames in accordance with NFPA 80 and NFPA 105 and this specification. Include insulated core materials in fire doors where indicated in the door schedule.

#### 2.6.1 Labels

Provide fire doors and frames bearing the label of Underwriters Laboratories (UL), Factory Mutual Engineering and Research (FM), or Warnock Hersey International (WHI) attesting to the rating required. Testing must be in accordance with NFPA 252 or UL 10C. Provide labels that are metal with raised letters, bearing the name or file number of the door and frame manufacturer. Labels must be permanently affixed at the factory to frames and to the hinge edge of the door. Do not paint door and labels.

#### 2.6.2 Oversized Doors

For fire doors and frames which exceed the size for which testing and labeling are available, furnish certificates stating that the doors and frames are identical in design, materials, and construction to a door which has been tested and meets the requirements for the class indicated.

#### 2.6.3 Astragal on Fire and Smoke Doors

On pairs of labeled fire doors, conform to NFPA 80 and UL requirements. On smoke control doors, conform to NFPA 105.

### 2.7 EXTERIOR HURRICANE-RESISTANT DOOR ASSEMBLIES

Provide factory-assembled door system, including but not limited to: frames, panels, and hardware. Provide assembly in accordance with SDI/DOOR A250.13 and ASTM E330/E330M with impact resistance of missile level D in accordance with ASTM E1886 and ASTM E1996. Provide a minimum design positive and negative pressure of 2.79 kPa, 58 psf. Provide thermal insulation in all exterior frames. Provide frames of a minimum Level 4, with frames of a minimum thickness of 1.7 mm, 14 gage.

Provide testing label which identifies the manufacturer, product designation, performance characteristics, and approved inspection agency.

### 2.8 HARDWARE PREPARATION

Drill and tap doors and frames to receive finish hardware. Prepare doors and frames for hardware in accordance with the applicable requirements of

SDI/DOOR A250.8 and SDI/DOOR A250.6. For additional requirements refer to ANSI/BHMA A156.115. Drill and tap for surface-applied hardware at the project site. Build additional reinforcing for surface-applied hardware into the door at the factory. Punch door frames, with the exception of frames that will have weatherstripping or lightproof or soundproof gasketing, to receive a minimum of two rubber or vinyl door silencers on lock side of single doors and one silencer for each leaf at heads of double doors. Set lock strikes out to provide clearance for silencers.

## 2.9 FINISHES

### 2.9.1 Factory-Primed Finish

Thoroughly clean all surfaces of doors and frames then chemically treat and factory prime with a rust inhibiting coating as specified in SDI/DOOR A250.8.

### 2.9.2 Hot-Dip Zinc-Coated and Factory-Primed Finish

Fabricate exterior doors and frames from hot dipped zinc coated steel, alloyed type, that complies with ASTM A924/A924M and ASTM A653/A653M (on all surfaces of frame) The coating weight must meet or exceed the minimum requirements for coatings having 122 grams per square meter, total both sides, i.e., ZF120. Repair damaged zinc-coated surfaces by the application of zinc dust paint. Thoroughly clean and chemically treat to insure maximum paint adhesion. Factory prime as specified in SDI/DOOR A250.8. Provide for exterior doors .

### 2.9.3 Electrolytic Zinc-Coated Anchors and Accessories

Provide electrolytically deposited zinc-coated steel in accordance with ASTM A879/A879M, Commercial Quality, Coating Class A. Phosphate treat and factory prime zinc-coated surfaces as specified in SDI/DOOR A250.8.

### 2.9.4 Factory-Applied Enamel Finish

Provide coatings that meet test procedures and acceptance criteria in accordance with SDI/DOOR A250.3. After factory priming, apply two coats of low-gloss enamel to exposed surfaces. Separately bake or oven dry each coat. Drying time and temperature requirements must be in accordance with the coating manufacturer's recommendations. Provide finish coat color(s) as indicated to match approved color sample(s).

## 2.10 FABRICATION AND WORKMANSHIP

Provide finished doors and frames that are strong and rigid, neat in appearance, and free from defects, waves, scratches, cuts, dents, ridges, holes, warp, and buckle. Provide molded members that are clean cut, straight, and true, with joints coped or mitered, well formed, and in true alignment. Dress exposed welded and soldered joints smooth. Design door frame sections for use with the wall construction indicated. Corner joints must be well formed and in true alignment. Conceal fastenings where practicable. Design frames in exposed concrete walls or partitions to allow sufficient space between the inside back of trim and masonry to receive caulking compound.

## 2.11 PROVISIONS FOR GLAZING

Materials are specified in Section 08 81 00, GLAZING.

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 Frames

Set frames in accordance with SDI/DOOR A250.11. Plumb, align, and brace securely until permanent anchors are set. Anchor bottoms of frames with expansion bolts or powder-actuated fasteners. Build in or secure wall anchors to adjoining construction. Where frames require ceiling struts or overhead bracing, anchor frames to the struts or bracing.

#### 3.1.2 Doors

Hang doors in accordance with clearances specified in SDI/DOOR A250.8. After erection and glazing, clean and adjust hardware.

#### 3.1.3 Fire and Smoke Doors and Frames

Install fire doors and frames, including hardware, in accordance with NFPA 80. Install fire rated smoke doors and frames in accordance with NFPA 80 and NFPA 105].

### 3.2 PROTECTION

Protect doors and frames from damage. Repair damaged doors and frames prior to completion and acceptance of the project or replace with new, as directed. Wire brush rusted frames until rust is removed. Clean thoroughly. Apply an all-over coat of rust-inhibitive paint of the same type used for shop coat.

### 3.3 CLEANING

Upon completion, clean exposed surfaces of doors and frames thoroughly. Remove mastic smears and other unsightly marks.

-- End of Section --

## SECTION 08 34 73

## SOUND CONTROL DOOR ASSEMBLIES

**11/19, CHG 1: 02/21**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN WELDING SOCIETY (AWS)

AWS D1.3/D1.3M (2018) Structural Welding Code - Sheet Steel

## ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A108 (2024) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished

ASTM A568/A568M (2019a) Standard Specification for Steel, Sheet, Carbon, Structural, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, General Requirements for

ASTM A1008/A1008M (2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable

ASTM A1011/A1011M (2023) Standard Specification for Steel Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength

ASTM C143/C143M (2020) Standard Test Method for Slump of Hydraulic-Cement Concrete

ASTM C476 (2020) Standard Specification for Grout for Masonry

ASTM D1056 (2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber

ASTM D6386 (2016a) Standard Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron and Steel Product and Hardware Surfaces for Painting

|            |                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E90   | (2009; R2016) Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements |
| ASTM E336  | (2023) Standard Test Method for Measurement of Airborne Sound Attenuation between Rooms in Buildings                                  |
| ASTM E413  | (2022) Classification for Rating Sound Insulation                                                                                     |
| ASTM E1289 | (2008; R 2016) Standard Specification for Reference Specimen for Sound Transmission Loss                                              |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                          |
|----------|----------------------------------------------------------|
| NFPA 101 | (2024) Life Safety Code                                  |
| NFPA 252 | (2022) Standard Methods of Fire Tests of Door Assemblies |

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|             |                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 36 CFR 1191 | Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

#### Fabrication Drawings

### SD-03 Product Data

Hollow Metal Sound Retardant Doors; G]

Door Frames; G

Door Hardware; G

Door Frame Sound Infill; G

Intumescent Seals and Gasketing; G

Thresholds; G]

### SD-06 Test Reports

Acoustical Tests; G

#### SD-07 Certificates

Hollow Metal Sound Retardant Doors; G

Door Frames; G

Door Hardware; G

Intumescent Seals, Gasketing and Door Bottoms; G

Thresholds; G

Assembly Test Reports

### 1.3 QUALITY CONTROL

Ensure work within this section is designed and furnished by one manufacturer, who has been engaged in the manufacture of Sound Retardant Hollow Metal Door systems for at least five years prior to the start of this work.

Provide acoustic assemblies manufactured by a single source specializing in the production of this type work for a minimum of five years.

#### 1.3.1 Compliance and Labeling

##### 1.3.1.1 Compliance with Accessibility Requirements

Americans with Disabilities Act/Architectural Barriers Act (ADA/ABA)  
36 CFR 1191

##### 1.3.1.2 Category A Positive Pressure Fire Door Construction

Where requirements for positive pressure are met, include for doors all requirements as part of the door construction per Category A guidelines as published by ITS/Warnock-Hersey. Intumescent is not allowed on the frame. Applying smoke gasketing around the perimeter of the frame to meet the "S" smoke rating is permissible in instances where smoke control is required.

##### 1.3.1.3 Category B Positive Pressure Fire Door Construction

Conform all door openings to the applicable portions of NFPA 101 and NFPA 252. Incorporate field applied intumescent materials, applied by a licensed installer according to the manufacturers' instructions. Keep instructions on file. Additional gasketing may be required to meet the 'S' smoke rating. Submit Certificate for intumescent seals, gasketing and

door bottoms.

#### 1.3.1.4 Labeling

Ensure all positive pressure door assemblies carry the fire label for the complete opening, clearly identifying the:

- a. Manufacturer
- b. Third party testing and certification agency
- c. Fire door rating
- d. Installation limitations
- e. Compatible frame, hardware component ratings
- f. Required building code information, including temperature and smoke rating
- g. STC rating if required.

Indicate fire-ratings of applicable components. Provide documentation of ABA/ADA accessibility compliance of applicable components, as required by 36 CFR 1191 Appendix D - Technical.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Ship all doors in the manufacturer's undamaged individual cartons, securely bundled and wrapped with moisture-resistant covers and stored in accordance with the manufacturer's printed instructions in a dry, clean, and ventilated area.

Deliver and store wood doors in the building following the installation of concrete, terrazzo, plaster, or other wet materials, and only after the building has dried out and has a roof.

Store all materials on planks in a dry location. Store doors and frames vertically with minimum airspace between. Store doors on the edge to eliminate any potential damage to the door bottom seal. Cover all material to protect from damage but in a manner to allow proper circulation.

Maintain relative humidity in the building between 30 and 65 percent. Maintain the ambient temperature at 16 degrees C minimum at the time of installation of wood doors.

Perform final adjustment of seals when temperatures and humidity conditions replicate the interior conditions that will exist when the building is occupied.

#### 1.5 WARRANTY

Manufacturer's warranty for 5 years from date of supply, covering material and workmanship. Failures include, but are not limited to, the following:

- a. Failure to meet sound rating requirements

- b. Faulty operation of sound seals
- c. Deterioration of metals, metal finishes, and other materials beyond normal use or weathering.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

Provide sound retardant door assemblies of the thickness, width, and height indicated, complete with perimeter seals, seal housings, gasketing, thresholds, door frames, and astragals as required to conform to the specified STC per ASTM E90 and ASTM E1289.

Submit fabrication drawings for Hollow Metal Sound Retardant Doors, Door Frames and Door Frame Sound Infill.

Submit certificates showing conformance with the referenced standards in this section, and manufacturer's catalog data including STC ratings and UL fire rating, where applicable, for the following items: Hollow metal sound retardant doors; door frames; door hardware; intumescent seals and gasketing]; thresholds.

Provide assemblies that are complete with metal frame, sealing system, and Cam-lift hinges (when required).

#### 2.1.1 Design Requirements

##### 2.1.1.1 Door Design

No visible seams are permitted on door faces. Provide face gauges, internal sound retardant core and perimeter door edge construction per manufacturer's standard for the specified STC rating. No lead or asbestos is permitted in door construction to achieve STC performance. No lead or asbestos is permitted in door construction to achieve performance requirements.

##### 2.1.1.2 Frame Design

Provide sound Retardant Metal Frames conforming to ASTM A1008/A1008M, not less than 1.90 millimeter thick, and free from pitting, scale, stretcher strains, fluting, and surface defects with integral trim and shipped with temporary spreader. Knockdown frames are not acceptable.

Provide frames with 50 millimeter faces, profiles and dimensions as indicated, with mitered reinforced corners, welded the full depth of frame and trim, with exposed surfaces ground smooth and flush. Close contact edges to hairline joints.

#### 2.1.2 Performance Requirements

##### 2.1.2.1 STC (Sound Transmission Classification) Rating

Provide doors with an STC per the door schedule.

### 2.2 FABRICATION

Provide doors that are minimum 16 gauge, 4.445 centimeter thick with



welded, seamless construction. No visible joints are permitted on the exposed faces or edges. Join door skins at vertical edges by continuous welds, ground and dressed smooth to provide a flush finish. Reinforce top and bottom with 1.52 millimeter continuous inverted steel channels spot welded to both faces. Finish both top and bottom to provide a smooth flush condition. Bevel both vertical edges .3175 centimeter in 5.08 centimeter.

Clean and sand to smooth finish all doors to remove handling and storage marks, raised grain, minor surface marks and abrasions which are to receive a job site finish.

## 2.2.1 Hollow Metal Sound Retardant Doors

### 2.2.1.1 Construction

Conform to ASTM A1008/A1008M for door construction utilizing steel facing sheets. Conform stretcher level flatness to ASTM A568/A568M; not less than 1.52 millimeter thick; free from pitting, scale, and surface defects; separated by a core construction designed to meet the required STC; and tested and rated in accordance with ASTM E90.

Provide doors that have flush seamless face sheets and vertical edges, with continuous welded and smooth joints. Provide edges that are flush or rabbeted as required for perimeter seals.

Provide door surfaces that are visually flat and free from warp, waviness, and other surface irregularities and defects. Maximum allowable warp or twist-can not exceed 3 millimeter when measured with a 2100 millimeter straightedge along the diagonal and not exceed 1.5 millimeter when measured with a 2100 millimeter straightedge in the width or in any position along the length of the door.

Provide hardware reinforcement that is steel drilled, tapped to template requirements and welded in place. Provide minimum thicknesses as follows:

- a. Butts, 4.7 millimeter
- b. Lock strike, 3.04 millimeter
- c. Surface applied hardware 1.90 millimeter

Provide doors, including sound retardant type, to bear the UL 1-hour label fire rating and the specified STC.

### 2.2.1.2 Coating

Thoroughly clean all mill scale, rust, oil, grease, dirt, and other foreign materials from surfaces before the application of the shop coat of paint.

After cleaning, provide galvanized surfaces free of paint in accordance with ASTM D6386, Method A, B, C, or D.

Apply to clean prepared dry surfaces one shop coat of rust inhibitive metallic oxide or synthetic resin primer by brush, dipping, or other approved method to provide a continuous minimum dry film thickness (dft) of 0.023 millimeter.

Shop paint the exposed door surfaces, including surfaces that are galvanized.

## 2.3 COMPONENTS

### 2.3.1 Frames

Construct frames for Sound Retardant Doors from formed sheet steel or structural shapes and bars. Provide sheet steel that is commercial quality, level, cold rolled steel conforming to ASTM A1008/A1008M or hot rolled, pickled and oiled steel conforming to ASTM A1011/A1011M. Comply steel shapes with ASTM A36/A36M and steel bars with ASTM A108, Grade 1018.

### 2.3.2 Door Frame Sound Infill

Grout: Comply with ASTM C476, with a slump of not more than 102 mm as measured according to ASTM C143/C143M.

Corrosion-Resistant Coating: Cold-applied asphalt mastic, compounded for 0.381 millimeter dry film thickness per coat. Provide inert-type noncorrosive compound free of asbestos fibers, sulfur components, and other deleterious impurities.

### 2.3.3 Hardware Reinforcements

Factory mortise, reinforce, drill and tap frames for all mortise hardware as required by hardware manufacturer's template. Provide necessary reinforcement plates as required for surface mounted hardware; installer to perform all field drilling and tapping. Provide dust cover boxes on all frame mortises. Provide minimum thicknesses as follows:

- a. Butts, 4.7 millimeter
- b. Lock strike, 3.04 millimeter
- c. Surface applied hardware 1.90 millimeter

### 2.3.4 Jamb Anchors

Provide number and spacing of anchors as follows:

#### 2.3.4.1 Stud-Wall Type

Locate anchors not more than 457 mm from top and bottom of frame. Space anchors not more than 813 mm o.c. and as follows:

- a) Three anchors per jamb up to 1524 mm in height.
- b) Four anchors per jamb from 1524 to 2286 mm in height.
- c) Five anchors per jamb from 2286 to 2438 mm in height.
- d) Five anchors per jamb plus one additional anchor per jamb for each 610 mm, or fraction thereof, more than 2438 mm in height.

- e) Two anchors per head for frames more than 1066 mm wide and mounted in metal-stud partitions.

#### 2.3.5 Door Hardware

Provide the following STC related hardware with the door; cam-lift hinges, perimeter seals, door bottoms, thresholds, hardware standoff brackets.

Include on Fabrication drawings a finish hardware schedule for each door and a hollow metal door frame schedule for each door indicating profile, dimensions, hardware reinforcement, and frame anchorage. Also indicate perimeter seals, door-bottom devices and other hardware items that are assembled in the shop.

Refer to Section 08 71 00 DOOR HARDWARE for remaining hardware requirements.

#### 2.3.6 Head and Jamb Seals

Provide a closed-cell, expanded cellular rubber Seal material conforming to ASTM D1056, Type S, Grade SBE-42 or SCE-42 for heads, jambs, and door bottoms.

Install seals in formed steel or extruded aluminum shapes designed to receive and hold seals and to provide concealed adjustable attachment to door frames. Provide concealed adjustment screws that are not more than 300 millimeter on center and provide at least 10 millimeter adjustment.

#### 2.3.7 Door Bottoms

Neoprene or silicone gasket held in place by metal housing; mortised into bottom edge of door.

#### 2.3.8 Thresholds

Provide metal thresholds where indicated. Provide thresholds that are extruded aluminum, 6063-T5 alloy, mill finish, not less than 3 millimeter thick, with integral seal grooves formed to the indicated section.

Provide flat, smooth, unfluted thresholds as recommended by manufacturer; fabricated from aluminum or stainless steel.

- a. Finish: Color to match interior design schedule anodic finish.
- b. Color: As selected by Architect from full range of industry colors and color densities.

Provide hardwood thresholds where indicated made of clear, all-heartwood, free of streaks, pin or worm holes, uniform in color, free of defects, finish sanded, and ready for job site, transparent or paint finish.

### 2.4 TESTS, INSPECTIONS, AND VERIFICATIONS

#### 2.4.1 Sound Transmission Classification

Provide test reports prepared by a nationally recognized, independent laboratory for Acoustical Tests, Air Infiltration Tests, Wind Loading Tests, and Water Leakage Tests indicating that the sound transmission classification (STC) of the proposed door, based on tests at 16

third-octave band frequencies from 125 to 4,000 hertz, is no less than the specified STC when tested in accordance with ASTM E90, and that the door tested is hung in substantially the type of wall and frame as indicated and is fully operable with hardware and perimeter seals installed.

#### 2.4.2 Positive Pressure

Provide test reports, prepared by a nationally recognized, independent laboratory for Positive Pressure Tests, for all fire rated door assemblies, including Intumescent Seals, Gasketing, and Door Bottoms.

#### 2.4.3 Cam Lift Hinges

When required to achieve STC, manufacturer to furnish laboratory test data certifying hinges have been cycled a minimum of 1,000,000 while supporting a minimum door weight of 159 kg.

Full-mortise template type that raises the door 13 mm when door is fully open; with hardened pin; fabricated from stainless steel.

#### 2.4.4 Guarantee

Provide written guarantee that each door delivered to the project is equal in construction, sound transmission classification (STC), and positive pressure test rating where applicable, with appropriate labeling and markings, to that of the sample door tested. Clearly state in written guarantee that each door assembly, when installed in accordance with the manufacturer's printed instructions, has an in-place STC within 3 decibels of the specimen tested. Submit the following test data and Certificates with the written Guarantee:

##### a. Acoustical Tests

### PART 3 EXECUTION

#### 3.1 PREPARATION

Upon receipt of material, thoroughly inspect all frames, doors and accessories. Verify quantities and tag numbers according to the packing list provided. Report all discrepancies, deficiencies and/or damages immediately to Contracting Officer.

Prior to installation check all doors and frames for correct size and swing. Verify that frames are plumb, square and aligned without twist in accordance with tolerances published by NAAMM/HMMA and SDI.

##### 3.1.1 Frame Painting and Cleaning

Clean thoroughly all surfaces of all mill scale, rust, oil, grease, dirt, and other foreign materials before the application of the shop coat of paint.

Apply one shop coat of rust inhibitive metallic oxide or synthetic resin

primer applied to clean, dry, and prepared surfaces by brush, dipping, or other approved method to provide a continuous minimum dry film thickness of 0.023 millimeter.

### 3.2 INSTALLATION

#### 3.2.1 Frame

Install frames plumb and true with not more than 0.8 millimeter deviation in vertical alignment in 2440 millimeter. Anchor to the wall in accordance with the manufacturer's instructions. Grout frames solid with mortar in masonry, concrete, and plaster wall construction. Spot grout frames in dry wall partitions with mortar at the jamb anchor clips; fill the space between metal frame and stud partition solidly with fiberglass or mineral wool insulation.

Field splices may be required after installation because of shipping limitations. Field weld splices by certified welders per manufacturer's instructions and in accordance with AWS D1.3/D1.3M.

#### 3.2.2 Door

Install and adjust all doors, hardware, and seals in accordance with the approved drawings, hardware schedules, and the printed instructions of the door manufacturer.

Install and adjust perimeter seals to provide positive compression contact with the entire sealing surface with no gaps, openings, or breaks. Hinges or hardware which distort or pinch the perimeter seal during operation of the door will be rejected.

Install door bottom devices to seal the space between the door bottoms and the finished floor and the space between the seal and seal housing.

Field apply perimeter seal housings with mitered corners and with flush, aligned hairline joints.

Install components to manufacturer's written instructions. Coordinate with gypsum board wall construction for anchor placement. Set frames plumb, square, level and at correct elevation. Adjust operable parts for correct clearances and function. Install and adjust perimeter and bottom acoustic seals.

### 3.3 FIELD QUALITY CONTROL

Provide third party testing in accordance with ASTM E336. Verify in writing that installed product performs no less than five (5) ASTC or NIC rating points below the specified laboratory STC rating. Examine, adjust, and retest any installation not meeting that criteria until compliance is obtained.

#### 3.3.1 Testing and Performance

Provide assemblies that are identical to those tested at an independent acoustical laboratory qualified under the National Voluntary Laboratory

Accreditation Program (NVLAP) by the National Institute for Science and Technology (NIST) in accordance with ASTM E90 and ASTM E413. For the assembly test reports include the laboratory name, test report number and date of test.

-- End of Section --

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## SECTION 08 71 00

DOOR HARDWARE  
02/16, CHG 4: 02/22

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|           |                                                                                                                                                                                 |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E283 | (2019) Standard Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)

|                   |                                                                             |
|-------------------|-----------------------------------------------------------------------------|
| ANSI/BHMA A156.1  | (2021) Butts and Hinges                                                     |
| ANSI/BHMA A156.3  | (2020) Exit Devices                                                         |
| ANSI/BHMA A156.4  | (2013) Door Controls - Closers                                              |
| ANSI/BHMA A156.5  | (2020) Cylinder and Input Devices for Locks                                 |
| ANSI/BHMA A156.6  | (2021) Architectural Door Trim                                              |
| ANSI/BHMA A156.7  | (2016) Template Hinge Dimensions                                            |
| ANSI/BHMA A156.8  | (2021) Door Controls - Overhead Stops and Holders                           |
| ANSI/BHMA A156.13 | (2022) Mortise Locks & Latches Series 1000                                  |
| ANSI/BHMA A156.15 | (2021) Release Devices Closer Holder, Electromagnetic and Electromechanical |
| ANSI/BHMA A156.16 | (2023) Auxiliary Hardware                                                   |
| ANSI/BHMA A156.18 | (2020) Materials and Finishes                                               |
| ANSI/BHMA A156.21 | (2019) Thresholds                                                           |
| ANSI/BHMA A156.22 | (2021) Gasketing                                                            |
| ANSI/BHMA A156.24 | (2012) Delayed Egress Locking Systems                                       |
| ANSI/BHMA A156.25 | (2013) Electrified Locking Devices                                          |
| ANSI/BHMA A156.26 | (2012) Continuous Hinges                                                    |
| ANSI/BHMA A156.27 | (2011) Power and Manual Operated Revolving                                  |



## Pedestrian Doors

ANSI/BHMA A156.29 (2012) Exit Locks, Exit Alarms, Alarms for Exit Devices

ANSI/BHMA A156.30 (2014) High Security Cylinders

## U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FF-L-2890C (2019) Federal Specification Lock Extensions (Pedestrian Door Lock Assembly Preassembled Panic and Auxilliary Deadbolt)

FF-L-2740B (2011) Federal Specification Locks, Combinations, Electromechanical

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NFPA 72 (2022; ERTA 22-1) National Fire Alarm and Signaling Code

NFPA 80 (2025) Standard for Fire Doors and Other Opening Protectives

NFPA 101 (2024) Life Safety Code

NFPA 252 (2022) Standard Methods of Fire Tests of Door Assemblies

## STEEL DOOR INSTITUTE (SDI/DOOR)

SDI/DOOR A250.8 (2023) Specifications for Standard Steel Doors and Frames

## U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

## UNDERWRITERS LABORATORIES (UL)

UL Bld Mat Dir (updated continuously online) Building Materials Directory

UL 1784 (2015) UL Standard for Safety for Air Leakage Tests of Door Assemblies and Other Opening Protectives

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Manufacturer's Detail Drawings; G

Verification of Existing Conditions; G

Hardware Schedule; G

Keying System; G

## SD-03 Product Data

Hardware Items; G

## SD-08 Manufacturer's Instructions

Installation

## SD-10 Operation and Maintenance Data

Hardware Schedule Items, Data Package 1; G

## SD-11 Closeout Submittals

Key Bitting

## 1.3 SHOP DRAWINGS

Submit manufacturer's detail drawings indicating all hardware assembly components and interface with adjacent construction. Indicate power components and wiring coordination for electrified hardware. Base shop drawings on verified field measurements and include verification of existing conditions.

## 1.4 PRODUCT DATA

Indicate fire-ratings at applicable components. Provide documentation of ABA/ADA accessibility compliance of applicable components, as required by 36 CFR 1191 Appendix D - Technical.

## 1.5 HARDWARE SCHEDULE

Provide Hardware Item List and Hardware Schedule containing the following information, and additional information as needed to identify the complete make up of each hardware set and its application to each opening:

## 1.5.1 Hardware Item List:

- a. Hardware Type
- b. Item Number
- c. Quantity
- d. Size(s)
- e. Reference Publication / Type Number

- f. Manufacturer's Name / Catalog Number
- g. Key Control Symbols
- h. UL Mark (If fire rated and listed)
- i. BHMA Finish(es)
- j. Remarks

#### 1.5.2 Hardware Schedule

- a. Hardware Set Number
- b. Opening Number(s)
- c. Opening Description (single/double leaf, hand, size, door/frame material)
- d. Fire Rating
- e. Sound Rating
- f. Hardware Items
- g. Quantity
- h. Size
- i. BHMA Finish
- j. Remarks

In addition, submit hardware schedule data package 1 in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

#### 1.6 KEY BITTING CHART REQUIREMENTS

##### 1.6.1 Requirements

Submit key bitting charts to the Contracting Officer prior to completion of the work. Include:

- a. Complete listing of all keys (e.g. AA1 and AA2).
- b. Complete listing of all key cuts (AA1-123456, AA2-123458).
- c. Tabulation showing which key fits which door.
- d. Copy of floor plan showing doors and door numbers.
- e. Listing of 20 percent more key cuts than are presently required in each master system.

## 1.7 QUALITY ASSURANCE

### 1.7.1 Hardware Manufacturers and Modifications

Provide, as far as feasible, locks, hinges, and closers of one lock, hinge, or closer manufacturer's make. Modify hardware as necessary to provide features indicated or specified.

### 1.7.2 Key Shop Drawings Coordination Meeting

Prior to the submission of the key shop drawing, the Contracting Officer, Contractor, Door Hardware Subcontractor, using Activity and Base Locksmith must meet to discuss and coordinate key requirements for the facility.

### 1.7.3 Qualifications

Provide qualification of GSA door hardware installer for personnel who will install FF-L-2890C and FF-L-2740B. Qualification must demonstrate completion of GSA Authorized Safe and Vault Technician course.

## 1.8 DELIVERY, STORAGE, AND HANDLING

Deliver hardware in original individual containers, complete with necessary appurtenances including fasteners and instructions. Mark each individual container with item number as shown on hardware schedule.

## PART 2 PRODUCTS

### 2.1 TEMPLATE HARDWARE

Hardware applied to metal or to prefinished doors must be manufactured using a template. Provide templates to door and frame manufacturers in accordance with ANSI/BHMA A156.7 for template hinges. Coordinate hardware items to prevent interference with other hardware.

### 2.2 HARDWARE FOR FIRE DOORS AND EXIT DOORS

Provide all hardware necessary to meet the requirements of NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, NFPA 252 for fire tests of door assemblies, ABA/ADA accessibility requirements, and all other requirements indicated, even if such hardware is not specifically mentioned in paragraph HARDWARE SCHEDULE. Provide Underwriters Laboratories, Inc. labels for such hardware in accordance with UL Bld Mat Dir or equivalent labels in accordance with another testing laboratory approved in writing by the Contracting Officer.

### 2.3 HARDWARE ITEMS

Clearly and permanently mark with the manufacturer's name or trademark, hinges, pivots, locks, latches, exit devices, bolts and closers where the identifying mark is visible after the item is installed. For closers with covers, the name or trademark may be beneath the cover. Coordinate electrified door hardware components with corresponding components specified in Division 28 ELECTRONIC SECURITY SYSTEMS (ESS).

#### 2.3.1 Hinges

Provide in accordance with ANSI/BHMA A156.1. Provide hinges that are 114 by 114 mm unless otherwise indicated. Construct loose pin hinges for

interior doors and reverse-bevel exterior doors so that pins are non-removable when door is closed. Other anti-friction bearing hinges may be provided in lieu of ball bearing hinges.

### 2.3.2 Continuous Hinges

Where continuous hinges are required, provide in accordance with ANSI/BHMA A156.26.

### 2.3.3 Locks and Latches

- a. At exterior locations provide locksets of full stainless steel type 302 or 304 construction including fronts, strike, escutcheons, knobs, bolts and all interior working parts. Marine Grade I, fully non-ferrous.
- b. In non-air-conditioned interior environments or humid interior environments, provide interior locksets on the same Marine Grade I, fully non-ferrous as exterior locksets.

#### 2.3.3.1 Mortise Locks and Latches

Provide in accordance with ANSI/BHMA A156.13, Series 1000, Operational Grade 1, Security Grade 2.

### 2.3.4 Lock Extension (Pedestrian Door Lock Assembly Presassembled, Panic and Auxiliary Deadbolt

Provide in accordance with FF-L-2890C. For assemblies which contain electromechanical combination lock, provide in accordance with FF-L-2740B and include as a single pre-assembled system. Provide a minimum 10 gauge reinforcement at door for door locks.

### 2.3.5 Exit Devices

Provide in accordance with ANSI/BHMA A156.3, Grade 1. Provide adjustable strikes for rim type and vertical rod devices. Provide open back strikes for pairs of doors with mortise and vertical rod devices. Provide touch bars in lieu of conventional crossbars and arms.

### 2.3.6 Exit Locks With Alarm

Provide in accordance with ANSI/BHMA A156.3 and ANSI/BHMA A156.29, Type E0431 (with full width horizontal actuating bar) for single doors; Type E0431 (with actuating bar) or E0471 (with actuating bar and top and bottom bolts, both leaves active) for pairs of doors, unless otherwise specified.

Provide terminals for connection to remote indicating panel. Provide outside control key. Provide door alarms integrated with the fire alarm system in accordance with NFPA 72.

### 2.3.7 Cylinders and Cores

Provide cylinders and cores for new locks, including locks provided under other sections of this specification. Provide cylinders and cores with seven pin tumblers. Provide cylinders from the products of one manufacturer, and provide cores from the products of one manufacturer.

Provide cylinders for new locks, including locks provided under other sections of this specification. Provide fully compatible cylinders of

Grade 1 products from products of one manufacturer with interchangeable cores that are removable by a special control key. Factory set the cores with seven pin tumblers using the A4 system and F keyway. Submit a core code sheet with the cores. Provide master keyed cores in one system for this project. Provide construction interchangeable cores.

#### 2.3.7.1 High Security Cylinders

Provide in accordance with ANSI/BHMA A156.30, security level A for all high security cylinder components.

#### 2.3.8 Push Button Mechanisms

Provide in accordance with ANSI/BHMA A156.5, Grade 1.

#### 2.3.9 Electrified Hardware

Comply with the requirements of NFPA 70 for wiring of electrified hardware.

##### 2.3.9.1 Electrified Mortise Locks

Provide in accordance with ANSI/BHMA A156.25, Grade 1. Provide electrified mortise locks that remain secure during power failure. Provide facility interface devices that use dc power to energize solenoids. Provide solenoids, resistors, and signal switches in accordance with paragraph ELECTRIC STRIKES AND FRAME MOUNTED ACTUATORS.

##### 2.3.9.1.1 Power Transfer Units

Provide power transfer units with each electrified lock that route power and monitoring signals from the lockset to the door frame. Coordinate power transfer units with door frames.

##### 2.3.9.2 Card Readers and Keypad Access Control Hardware

Provide in accordance with ANSI/BHMA A156.5 and ANSI/BHMA A156.25, Grade 1 components. Provide devices that are tamper alarmed, tamper and vandal resistant, solid state, and do not contain electronics which could compromise the access control subsystem should the subsystem be attacked. Provide surface, semi-flush, pedestal, or weatherproof mountable devices as specified for each individual location. Each device to contain a visual display, either mounted on the face, or on an integral part of the device, to indicate access or exit request processing, request approval, and request denial. Provide insertion type card readers capable of reading Smart Card type access control cards. Provide keypads that contain an integral 12-digit tactile keyboard with digits. Provide keypads that are integrated into the card reader. Coordinate access control hardware with corresponding devices and systems specified in Division 28 ELECTRONIC SECURITY SYSTEMS (ESS).

##### 2.3.9.3 Release Devices

In accordance with ANSI/BHMA A156.15, Grade 1.

##### 2.3.9.3.1 Closer Holders

Provide header mounted closer holder devices connected by integral releasing to fire and smoke detecting devices.

#### 2.3.9.3.2 Release Devices

Provide wall/floor mounted Electromagnetic release devices connected to fire and smoke detecting devices.

#### 2.3.9.4 Delayed Egress Locking System

Provide in accordance with ANSI/BHMA A156.24, Grade 1.

#### 2.3.9.5 Power and Manual Operated Revolving Pedestrian Doors

Provide in accordance with ANSI/BHMA A156.27, Grade 1.

#### 2.3.10 Keying System

Provide a master keying system. Provide construction interchangeable cores.

#### 2.3.11 Lock Trim

Provide cast, forged, or heavy wrought construction and commercial plain design for lock trim.

##### 2.3.11.1 Lever Handles

Provide lever handles where indicated in the Hardware Schedule. Provide in accordance with ANSI/BHMA A156.3 for mortise locks of lever handles for exit devices. Provide lever handle locks with a breakaway feature (such as a weakened spindle or a shear key) to prevent irreparable damage to the lock when force in excess of that specified in ANSI/BHMA A156.13 is applied to the lever handle. Provide lever handles return to within 13 mm of the door face.

##### 2.3.12 Keys

Provide one file key, one duplicate key, and one working key for each key change and for each master keying system. Provide one additional working key for each lock of each keyed-alike group. Provide a quantity of key blanks equal to 20 percent of the total number of file keys. Stamp each key with appropriate key control symbol and "U.S. property - do not duplicate." Do not place room number on keys.

##### 2.3.13 Door Bolts

Provide in accordance with ANSI/BHMA A156.16. Provide dustproof strikes for bottom bolts, except at doors having metal thresholds. Provide automatic latching flush bolts in accordance with ANSI/BHMA A156.3, Type 25.

##### 2.3.14 Closers

Provide in accordance with ANSI/BHMA A156.4, Series C02000, Grade 1, with PT 4C. Provide with brackets, arms, mounting devices, fasteners, full size covers, except at storefront mounting, and other features necessary for the particular application. Size closers in accordance with manufacturer's printed recommendations, or provide multi-size closers, Sizes 1 through 6, and list sizes in the Hardware Schedule. Provide manufacturer's 10 year warranty.

#### 2.3.14.1 Identification Marking

Engrave each closer with manufacturer's name or trademark, date of manufacture, and manufacturer's size designation in locations that will be visible after installation.

#### 2.3.15 Overhead Holders

Provide in accordance with ANSI/BHMA A156.8.

#### 2.3.16 Door Protection Plates

Provide in accordance with ANSI/BHMA A156.6.

##### 2.3.16.1 Sizes of Mop and Kick Plates

50 mm less than door width for single doors; 25 mm less than door width for pairs of doors. Provide 200mm kick plates for flush doors . Provide 150 mm mop plates.

#### 2.3.17 Door Stops and Silencers

Provide in accordance with ANSI/BHMA A156.16.

#### 2.3.18 Thresholds

Provide in accordance with ANSI/BHMA A156.21.

#### 2.3.19 Weatherstripping Gasketing

Provide in accordance with ANSI/BHMA A156.22. Provide the type and function designation where specified in paragraph HARDWARE SCHEDULE. Provide a set to include head and jamb seals, sweep strips, and, for pairs of doors, astragals. Air leakage of weatherstripped doors not to exceed 2.19 by 10-5 cms per minute of air per square meter of door area when tested in accordance with ASTM E283. Provide weatherstripping with one of the following:

##### 2.3.19.1 Extruded Aluminum Retainers

Extruded aluminum retainers not less than 1.25 mm wall thickness with vinyl, neoprene, silicone rubber, or polyurethane inserts. Provide match adjacent anodized aluminum.

##### 2.3.19.2 Interlocking Type

Zinc or bronze not less than 0.45 mm thick.

##### 2.3.19.3 Spring Tension Type

Spring bronze or stainless steel not less than 0.20 mm thick.

#### 2.3.20 Soundproofing Gasketing

Provide in accordance with ANSI/BHMA A156.22. Provide adjustable doorstops at heads, jams and automatic door bottoms in accordance with the hardware set, of extruded aluminum, anodized, surface applied, with vinyl



fin seals between plunger and housing. Provide doorstops with solid neoprene tube, silicone rubber, or closed cell sponge gasket. Provide door bottoms with adjustable operating rod and silicone rubber or closed cell sponge neoprene gasket. Provide doorstops that are mitered at corners. Provide type and function designation where specified in paragraph HARDWARE SETS.

#### 2.3.21 Smoke and Draft Gasketing

Provide in accordance with ANSI/BHMA A156.22. Provide Category H in accordance with UL 1784.

#### 2.3.22 Rain Drips

Provide in accordance with ANSI/BHMA A156.22. Provide extruded aluminum rain drips, not less than 2.03 mm thick, factory painted finish. Provide the manufacturer's full range of color choices to the Contracting Officer for color selection. Provide rain drips with a 102 mm overlap on each side of each exterior door that is not protected by an awning, roof, eave or other horizontal projection. Set drips in sealant and fasten with stainless steel screws.

##### 2.3.22.1 Door Rain Drips

Approximately 38 mm high by 16 mm projection. Align bottom with bottom edge of door.

#### 2.3.23 Auxiliary Hardware (Other than locks)

Provide in accordance with ANSI/BHMA A156.16, Grade 1.

#### 2.3.24 Special Tools

Provide special tools, such as spanner and socket wrenches and dogging keys, as required to service and adjust hardware items.

### 2.4 FASTENERS

Provide fasteners of type, quality, size, and quantity appropriate to the specific application. Fastener finish to match hardware. Provide stainless steel or nonferrous metal fasteners in locations exposed to weather. Verify metals in contact with one another are compatible and will avoid galvanic corrosion when exposed to weather.

### 2.5 FINISHES

Provide in accordance with ANSI/BHMA A156.18. Provide hardware in BHMA 630 finish (satin stainless steel), unless specified otherwise. Provide items not manufactured in stainless steel in BHMA 626 finish (satin chromium plated) over brass or bronze, except aluminum paint finish for surface door closers, and except BHMA 652 finish (satin chromium plated) for steel hinges. Provide hinges for exterior doors in stainless steel with BHMA 630 finish. Furnish exit devices in BHMA 626 finish in lieu of BHMA 630 finish except where BHMA 630 is specified under paragraph HARDWARE SETS. Match exposed parts of concealed closers to lock and door trim. Match hardware finish for aluminum doors to the doors.

## 2.6 KEY CABINET AND CONTROL SYSTEM

Provide in accordance with ANSI/BHMA A156.5, Type required to yield a capacity (number of hooks) 50 percent greater than the number of key changes used for door locks.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Provide hardware in accordance with manufacturers' printed installation instructions. Fasten hardware to wood surfaces with full-threaded wood screws or sheet metal screws. Provide machine screws set in expansion shields for fastening hardware to solid concrete and masonry surfaces. Provide toggle bolts where required for fastening to hollow core construction. Provide through bolts where necessary for satisfactory installation.

#### 3.1.1 Weatherstripping Installation

Provide full contact, weathertight seals that allow operation of doors without binding the weatherstripping.

##### 3.1.1.1 Stop Applied Weatherstripping

Fasten in place with color matched sheet metal screws not more than 225 mm on center after doors and frames have been finish painted.

##### 3.1.1.2 Interlocking Type Weatherstripping

Provide interlocking, self adjusting type on heads and jambs and flexible hook type at sills. Nail weatherstripping to door 25 mm on center and to heads and jambs at 100 mm on center.

##### 3.1.1.3 Spring Tension Type Weatherstripping

Provide spring tension type on heads and jambs. Provide bronze nails with bronze. Provide stainless steel nails with stainless steel. Space nails not more than 38 mm on center.

#### 3.1.2 Soundproofing Installation

Provide as specified for stop applied weatherstripping.

#### 3.1.3 Threshold Installation

Extend thresholds the full width of the opening and notch end for jamb stops. Set thresholds in a full bed of sealant and anchor to floor with cadmium-plated, countersunk, steel screws. For aluminum thresholds placed on top of concrete surfaces, coat the underside surfaces that are in contact with the concrete with fluid applied waterproofing as a separation measure prior to placement.

### 3.2 FIRE DOORS AND EXIT DOORS

Provide hardware in accordance with NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, and NFPA 252 for fire tests of door assemblies. .

### 3.3 HARDWARE LOCATIONS

Provide in accordance with SDI/DOOR A250.8, unless indicated or specified otherwise.

- a. Kick and Armor Plates: Push side of single-acting doors. Both sides of double-acting doors.
- b. Mop Plates: Bottom flush with bottom of door.

### 3.4 KEY CABINET AND CONTROL SYSTEM

Locate where directed. Tag one set of file keys and one set of duplicate keys. Place other keys in appropriately marked envelopes, or tag each key. Provide complete instructions for setup and use of key control system. On tags and envelopes, indicate door and room numbers or master or grand master key.

### 3.5 FIELD QUALITY CONTROL

After installation, protect hardware from paint, stains, blemishes, and other damage until acceptance of work. Submit notice of testing 15 days before scheduled, so that testing can be witnessed by the Contracting Officer. Adjust hinges, locks, latches, bolts, holders, closers, and other items to operate properly. Demonstrate that permanent keys operate respective locks, and give keys to the Contracting Officer. Correct, repair, and finish, errors in cutting and fitting and damage to adjoining work.

### 3.6 HARDWARE SETS

## Provide

| Quantity                                                     | Description                                                                                                                              | BHMA Type / Function Numbers                                          | Finish |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------|
| SET 1.0 - Single Ext x Hurricane x Push Button x UNI x Seals |                                                                                                                                          |                                                                       |        |
| Doors:<br>100A,<br>111B                                      |                                                                                                                                          |                                                                       |        |
| 4                                                            | Hinge (heavy weight)                                                                                                                     | A8111 NRP                                                             |        |
| 1                                                            | Fail-secure electromechanical mortise lock exit device, door lever and cylinder key override on exterior side; push-pad on interior side | Type 3 Exit Device, L2-A156.13 F4 G1 - E01,E06 Electromechanical Lock | 630    |
| 1                                                            | High security interchangeable Core cylinder                                                                                              | E09241A                                                               | 26     |
| 1                                                            | Surface Closer                                                                                                                           | C02021                                                                |        |
| 1                                                            | Threshold                                                                                                                                | J32190                                                                |        |
| 1                                                            | Gasketing                                                                                                                                | R0E155                                                                |        |
| 1                                                            | Door Shoe with Rain Drip                                                                                                                 | R3D514                                                                |        |
| 1                                                            | Electric Power Transfer                                                                                                                  |                                                                       |        |
| 1                                                            | ElectroLynx Harness (in door)                                                                                                            |                                                                       |        |
| 1                                                            | ElectroLynx Harness (in frame)                                                                                                           |                                                                       |        |
| 1                                                            | Balance Magnetic Switch/ Contact                                                                                                         | BY OTHERS                                                             |        |
| 1                                                            | Card Reader with Keypad                                                                                                                  | BY OTHERS                                                             |        |
| 1                                                            | Power Supply                                                                                                                             |                                                                       |        |
| 1                                                            | Wiring Diagrams                                                                                                                          | BY OTHERS                                                             |        |
| 1                                                            | PIR REX Sensor                                                                                                                           | BY OTHERS                                                             |        |
| 1                                                            | Balance Magnetic Switch / Contact                                                                                                        | BY OTHERS                                                             |        |

| Quantity                                                                                                                                                                                                                                                                                                                                                                                                                                   | Description                                                                                       | BHMA Type / Function Numbers | Finish |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------|--------|
| Operational description: Door is normally closed and locked. Presenting a valid card and code at key pad will momentarily unlock outside level, allowing entry. Egress at all times by proximity to passive infrared (PIR) sensor which will deactivate the lock and allowing operation of inside panic exit push-pad. High security switch at door head monitors the position of the door and reports this status to the security system. |                                                                                                   |                              |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                   |                              |        |
| SET 1.1 - Single Ext x Hurricane x Exit Only x UNI x Seals                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                   |                              |        |
| Doors: 102                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                   |                              |        |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                          | Hinge (heavy weight)                                                                              | A8111 NRP                    |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                          | Surface Vertical Rod Exit Device, No Trim on Exterior Side, Push-Pad on Interior Side             | Type 02, Function 01         | 630    |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                          | Surface Closer                                                                                    | C02021                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                          | Threshold                                                                                         | J32190                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                          | Gasketing                                                                                         | R0Y165                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                          | Door Shoe with Rain Drip                                                                          | R3D514                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                          | Balance Magnetic Switch / Contact                                                                 | BY OTHERS                    |        |
| Operational description: Egress only door. Door is normally closed. Egress at all times with pushpad exit device and exiting. No door hardware trim on exterior side. High security switch at door head monitors the position of the door and reports this status to the security system.                                                                                                                                                  |                                                                                                   |                              |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                   |                              |        |
| Set 2.0 - Pair Ext x Hurricane x Exit Only x Fire x UNI x Seals                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                   |                              |        |
| Doors: 112A, 113A, 114A, 114B, 118B                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                   |                              |        |
| 8                                                                                                                                                                                                                                                                                                                                                                                                                                          | Hinge (heavy weight)                                                                              | A8111 NRP                    | 630    |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                          | Surface Vertical Rod Exit Device, Fire Rated, No Trim on Exterior Side, Push-Pad on Interior Side | Type 02, Function 01         |        |

| Quantity                                                                                                                                                                                                                                                                                                                                                                                | Description                                                                                                                                                                                                                                                        | BHMA Type / Function Numbers                                                    | Finish |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------|
| 2                                                                                                                                                                                                                                                                                                                                                                                       | Surface Closer                                                                                                                                                                                                                                                     | C02021                                                                          | 689    |
| 2                                                                                                                                                                                                                                                                                                                                                                                       | Kick Plate                                                                                                                                                                                                                                                         | J102                                                                            | US32D  |
| 1                                                                                                                                                                                                                                                                                                                                                                                       | Threshold                                                                                                                                                                                                                                                          | J32190                                                                          |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                       | Smoke & Draft Gasketing                                                                                                                                                                                                                                            | R0E145                                                                          |        |
| 2                                                                                                                                                                                                                                                                                                                                                                                       | Smoke & Draft Door Shoe with Rain Drip                                                                                                                                                                                                                             | R3D154                                                                          |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                       | Meeting Stile Astragal                                                                                                                                                                                                                                             | R0Y635                                                                          |        |
| 2                                                                                                                                                                                                                                                                                                                                                                                       | Balance Magnetic Switch / Contact                                                                                                                                                                                                                                  | BY OTHERS                                                                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                       | Wiring Diagrams                                                                                                                                                                                                                                                    | BY OTHERS                                                                       |        |
| Operational description: Double door, egress only door. Door is normally closed, with surface multipoint lock on interior side of door panels top. Egress at all times by operating panic push pad exit device and exiting. No door hardware trim on exterior side. High security switch at door head monitors the position of the door and reports this status to the security system. |                                                                                                                                                                                                                                                                    |                                                                                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                    |                                                                                 |        |
| Set 2.1 - Pair Ext x Hurricane x Exit Only x UNI x Seals x Alarm                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                    |                                                                                 |        |
| Doors 109A                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                    |                                                                                 |        |
| 8                                                                                                                                                                                                                                                                                                                                                                                       | Hinge (heavy weight)                                                                                                                                                                                                                                               | A8111 NRP                                                                       | 630    |
| 2                                                                                                                                                                                                                                                                                                                                                                                       | Flush Bolt                                                                                                                                                                                                                                                         | L04251/61                                                                       | US26D  |
| 1                                                                                                                                                                                                                                                                                                                                                                                       | Dust Proof Strike                                                                                                                                                                                                                                                  | L04021                                                                          | US26D  |
| 1                                                                                                                                                                                                                                                                                                                                                                                       | Pedestrian door preassembled lock (PDPL), with electromechanical lock, meeting federal specification FF-L-2890 complete with electric release capability. Fire Rated. Provide statement on pull-side of door "EMERGENCY EXIT ONLY, PULL TO EXIT, ALARM WILL SOUND" | Federal Specification FF-L-2890, type X, No hardware trim on push side of door. |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                       | Surface Closer                                                                                                                                                                                                                                                     | C02021                                                                          | 689    |
| 2                                                                                                                                                                                                                                                                                                                                                                                       | Kick Plate                                                                                                                                                                                                                                                         | J102                                                                            | US32D  |

| Quantity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Description                       | BHMA Type / Function Numbers | Finish |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------|--------|
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Threshold                         | J32190                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Gasketing                         | R0E154                       |        |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Door Shoe with Rain Drip          | R3D154                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Removable Mullion                 | Type 22                      |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Mullion Cylinder                  |                              |        |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Balance Magnetic Switch / Contact | BY OTHERS                    |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Emergency Exit Audible Door Alarm | BY OTHERS                    |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Wiring Diagrams                   | BY OTHERS                    |        |
| Operational description: Alarmed exit only from occupied area to exterior. Doors are normally closed. Manual egress at all times at active leaf by pressing paddle of exit device and exiting. In-active door leaf locked at all times with flush bolts at door sill and door head. Door contact monitors the position of the active and inactive door panels and signals to hard-wired exit alarm panel for 100db sound actuation in addition to reporting this status to the security system. Deactivation of alarm manually at PCU when equipment receiving / movement is necessary. |                                   |                              |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |                              |        |
| Set 3.0 - Single x Passage x UNI x Seals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                   |                              |        |
| Doors: 106                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                   |                              |        |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Hinge (heavy weight)              | A8111                        | 630    |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Passage Lock                      | F01                          | 626    |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Surface Closer                    | C02011                       | 689    |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Kick Plate                        | J102                         | US32D  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Wall Stop                         | L02251                       | US32D  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Gasketing                         | R0Y154                       |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |                              |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |                              |        |
| Set 3.1 - Fire Rated Single x Storeroom x UNI x Seals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                   |                              |        |
| Doors: 113B, 115A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                   |                              |        |

| Quantity                                                    | Description                         | BHMA Type / Function Numbers | Finish |
|-------------------------------------------------------------|-------------------------------------|------------------------------|--------|
| 3                                                           | Hinge (heavy weight)                | A8111 NRP                    | 630    |
| 1                                                           | Storeroom Lock, Fire Rated          | F07                          | 626    |
| 1                                                           | Small Format Interchangeable Core   | E09241                       | 26     |
| 1                                                           | Surface Closer                      | C02021                       | 689    |
| 1                                                           | Kick Plate                          | J102                         | US32D  |
| 1                                                           | Wall Stop                           | L02251                       | US32D  |
| 1                                                           | Smoke & Draft Gasketing             | R0E154                       |        |
| 1                                                           | Smoke & Draft Automatic Door Bottom | R3G324                       |        |
|                                                             |                                     |                              |        |
| Set 3.2 - Fire Rated Single x Storeroom x UNI x Seals       |                                     |                              |        |
| Doors: 117, 118A, 119                                       |                                     |                              |        |
| 3                                                           | Hinge (heavy weight)                | A8111 NRP                    | 630    |
| 1                                                           | Storeroom Lock, Fire Rated          | F07                          | 626    |
| 1                                                           | Small Format Interchangeable Core   | E09241                       | 26     |
| 1                                                           | Surface Closer                      | C02011                       | 689    |
| 1                                                           | Kick Plate                          | J102                         | US32D  |
| 1                                                           | Wall Stop                           | L02251                       | US32D  |
| 1                                                           | Smoke & Draft Gasketing             | R0E154                       |        |
| 1                                                           | Smoke & Draft Sweep                 | R3G324                       |        |
|                                                             |                                     |                              |        |
| Set 3.3 - Fire Rated, STC, Single x Storeroom x UNI x Seals |                                     |                              |        |
| Doors: 115B                                                 |                                     |                              |        |
| 3                                                           | Hinge (heavy weight)                | A8111 NRP                    | 630    |
| 1                                                           | Storeroom Lock, Fire Rated          | F07                          | 626    |
| 1                                                           | Small Format Interchangeable Core   | E09241                       | 26     |
| 1                                                           | Surface Closer                      | C02011                       | 689    |



| Quantity                                                                                                                                      | Description                         | BHMA Type / Function Numbers | Finish |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------|--------|
| 1                                                                                                                                             | Kick Plate                          | J102                         | US32D  |
| 1                                                                                                                                             | Wall Stop                           | L02251                       | US32D  |
| 1                                                                                                                                             | Smoke & Draft Gasketing             | R0E154                       |        |
| 1                                                                                                                                             | Smoke & Draft Automatic Door Bottom | R3G324                       |        |
| Operational Description: Provide hardware set components within factory assembled sound control assembly in accordance with section 08 34 73. |                                     |                              |        |
|                                                                                                                                               |                                     |                              |        |
| Set 3.4 - Single x Smoke x Store x UNI x Seals                                                                                                |                                     |                              |        |
| Doors: 110                                                                                                                                    |                                     |                              |        |
| 3                                                                                                                                             | Hinge (heavy weight)                | A8111                        | 630    |
| 1                                                                                                                                             | Store / Utility Lock                | F14                          | 626    |
| 1                                                                                                                                             | Surface Closer                      | C02011                       | 689    |
| 1                                                                                                                                             | Kick Plate                          | J102                         | US32D  |
| 1                                                                                                                                             | Wall Stop                           | L02251                       | US32D  |
| 1                                                                                                                                             | Smoke Gasketing                     | R0E154                       |        |
| 1                                                                                                                                             | Smoke Automatic Door Bottom         | R3G324                       |        |
|                                                                                                                                               |                                     |                              |        |
| Set 3.5 - Single x Smoke x Storeroom x UNI x Seals                                                                                            |                                     |                              |        |
| Doors: 112B                                                                                                                                   |                                     |                              |        |
| 3                                                                                                                                             | Hinge (heavy weight)                | A8111 NRP                    | 630    |
| 1                                                                                                                                             | Storeroom Lock                      | F07                          | 626    |
| 1                                                                                                                                             | Small Format Interchangeable Core   | E09241                       | 26     |
| 1                                                                                                                                             | Surface Closer                      | C02011                       | 689    |
| 1                                                                                                                                             | Kick Plate                          | J102                         | US32D  |
| 1                                                                                                                                             | Wall Stop                           | L02251                       | US32D  |
| 1                                                                                                                                             | Smoke Gasketing                     | R0E154                       |        |
| 1                                                                                                                                             | Smoke Automatic Door Bottom         | R3G324                       |        |

| Quantity                                                                                                                                                                                                             | Description                 | BHMA Type / Function Numbers | Finish |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------|--------|
| Set 4.0 - Single x Privacy x CL                                                                                                                                                                                      |                             |                              |        |
| Doors 103, 104                                                                                                                                                                                                       |                             |                              |        |
| 3                                                                                                                                                                                                                    | Hinge (heavy weight)        | A8111                        | 630    |
| 1                                                                                                                                                                                                                    | Privacy Lock w/indicator    | F19                          | 630    |
| 1                                                                                                                                                                                                                    | Surface Closer              | C02011                       | 689    |
| 1                                                                                                                                                                                                                    | Kick Plate                  | J102                         | US32D  |
| 1                                                                                                                                                                                                                    | Wall Stop                   | L02251                       | US32D  |
| Operational description: Single door with mortise lock with interior thumbturn to lock door with graphic indicator on exterior side with red background color when locked, and green background color when unlocked. |                             |                              |        |
| Set 4.1 - Single x Smoke x Privacy x STC                                                                                                                                                                             |                             |                              |        |
| Doors 107, 108                                                                                                                                                                                                       |                             |                              |        |
| 3                                                                                                                                                                                                                    | Hinge (heavy weight)        | A8111                        | 630    |
| 1                                                                                                                                                                                                                    | Privacy Lock                | F22                          | 626    |
| 1                                                                                                                                                                                                                    | Closer                      | C02011                       | 689    |
| 1                                                                                                                                                                                                                    | Wall Stop                   | L02251                       | US32D  |
| 1                                                                                                                                                                                                                    | Smoke Gasketing             | R0E154                       |        |
| 1                                                                                                                                                                                                                    | Smoke Automatic Door Bottom | R3G324                       |        |
| Set 5.0 - Single x Passage x UNI x Seals                                                                                                                                                                             |                             |                              |        |
| Doors 105                                                                                                                                                                                                            |                             |                              |        |
| 3                                                                                                                                                                                                                    | Hinge (heavy weight)        | A8111                        | 630    |
| 1                                                                                                                                                                                                                    | Flush Bolt                  | L04251/61                    | US26D  |
| 1                                                                                                                                                                                                                    | Dust Proof Strike           | L04021                       | US26D  |
| 1                                                                                                                                                                                                                    | Passage Latch               | F01                          | 626    |
| 1                                                                                                                                                                                                                    | Surface Closer              | C02011                       | 689    |

| Quantity                                     | Description                                 | BHMA Type / Function Numbers | Finish |
|----------------------------------------------|---------------------------------------------|------------------------------|--------|
| 1                                            | Kick Plate                                  | J102                         | US32D  |
| 1                                            | Gasketing                                   | R0E154                       |        |
|                                              |                                             |                              |        |
| Set 6.0 - Single x Fire x Classroom x CL     |                                             |                              |        |
| Doors 100B                                   |                                             |                              |        |
| 3                                            | Hinge (heavy weight)                        | A8111                        | 630    |
| 1                                            | Classroom Lock, Fire Rated                  | F05                          | 626    |
| 1                                            | High security interchangeable core cylinder | E09241A                      | 26     |
| 1                                            | Wall Stop                                   | L02251                       | US32D  |
| 1                                            | Surface Closer                              | C02021                       |        |
| 1                                            | Smoke & Draft Gasketing                     | R0E154                       |        |
| 1                                            | Smoke & Draft Automatic Door Bottom         | R3G324                       |        |
|                                              |                                             |                              |        |
| Set 6.1 - Single x Smoke x Classroom x CL    |                                             |                              |        |
| Doors 109B                                   |                                             |                              |        |
| 3                                            | Hinge (heavy weight)                        | A8111                        | 630    |
| 1                                            | Classroom Lock                              | F05                          | 626    |
| 1                                            | High security interchangeable core cylinder | E09241A                      | 26     |
| 1                                            | Wall Stop                                   | L02251                       | US32D  |
| 1                                            | Surface Closer                              | C02021                       |        |
| 1                                            | Smoke Gasketing                             | R0E154                       |        |
| 1                                            | Smoke Automatic Door Bottom                 | R3G324                       |        |
|                                              |                                             |                              |        |
| Set 7.0 - Pair x Fire Rated x Exit Only x CL |                                             |                              |        |
| Door: 109C                                   |                                             |                              |        |
| 2                                            | Continuous Hinge                            | A85331B                      |        |

| Quantity                                                                                                                                                                                                                                                                              | Description                                                                                                  | BHMA Type / Function Numbers | Finish |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------|--------|
| 2                                                                                                                                                                                                                                                                                     | Flush Bolt                                                                                                   | L04251/61                    | US26D  |
| 1                                                                                                                                                                                                                                                                                     | Dust Proof Strike                                                                                            | L04021                       | US26D  |
| 1                                                                                                                                                                                                                                                                                     | Mortise exit device with pushpad, no door trim on pull-side, strike mounted on removable mullion, Fire Rated | Type 3, Function 01          |        |
| 1                                                                                                                                                                                                                                                                                     | Surface Closer                                                                                               | C02021                       | 689    |
| 2                                                                                                                                                                                                                                                                                     | Threshold                                                                                                    | J12190                       | US32D  |
| 2                                                                                                                                                                                                                                                                                     | Wall Stop                                                                                                    | L02251                       |        |
| 2                                                                                                                                                                                                                                                                                     | Smoke Gasketing                                                                                              | R0E154                       |        |
| 2                                                                                                                                                                                                                                                                                     | Smoke Automatic Door Bottom                                                                                  | R3G324                       |        |
| 1                                                                                                                                                                                                                                                                                     | ElectroLynx Harness (in door)                                                                                |                              |        |
| 1                                                                                                                                                                                                                                                                                     | ElectroLynx Harness (in frame)                                                                               |                              |        |
| 2                                                                                                                                                                                                                                                                                     | Balance Magnetic Switch/Contact                                                                              | BY OTHERS                    |        |
| 1                                                                                                                                                                                                                                                                                     | Wiring Diagrams                                                                                              | BY OTHERS                    |        |
| Operational description: Door is normally closed. No hardware trim on pull side of door. Egress at all times by operating inside panic exit push pad and exiting. High security switch at door head monitors the position of the door and reports this status to the security system. |                                                                                                              |                              |        |
|                                                                                                                                                                                                                                                                                       |                                                                                                              |                              |        |
| Set 8.0 - Single x Fire x Fail Secure x CL x STC                                                                                                                                                                                                                                      |                                                                                                              |                              |        |
| Doors: 100C                                                                                                                                                                                                                                                                           |                                                                                                              |                              |        |
| 3                                                                                                                                                                                                                                                                                     | Hinge (heavy weight)                                                                                         | A8111 NRP                    | 630    |

| Quantity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Description                                                                                                                                                                                                      | BHMA Type / Function Numbers                                                   | Finish |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------|
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Pedestrian door lock assembly panic (PDLAP), with electromechanical lock, meeting federal specification FF-L-2890 with integrated request to exit. Fire Rated. Provide top sliding exterior environmental cover. | Federal Specification FF-L-2890, type II, with elctromechanical lock FF-L-2740 |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | High security interchangeable core cylinder                                                                                                                                                                      | E09241A                                                                        |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Surface Closer                                                                                                                                                                                                   | C02021                                                                         |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Wall Stop                                                                                                                                                                                                        | L02251                                                                         |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Smoke & Draft Gasketing                                                                                                                                                                                          | R0E154                                                                         |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Smoke & Draft Automatic Door Bottom                                                                                                                                                                              | R3G334                                                                         |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Electric Power Transfer                                                                                                                                                                                          |                                                                                |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ElectroLynx Harness (in door)                                                                                                                                                                                    |                                                                                |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ElectroLynx Harness (in frame)                                                                                                                                                                                   |                                                                                |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Balance Magnetic Switch/Contact                                                                                                                                                                                  | BY OTHERS                                                                      |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Card Reader with Keypad                                                                                                                                                                                          | BY OTHERS                                                                      |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Power Supply                                                                                                                                                                                                     |                                                                                |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Wiring Diagrams                                                                                                                                                                                                  | BY OTHERS                                                                      |        |
| Operational description: Door is normally closed and locked. Presenting a valid card to card reader and code at electromechanical lock will momentarily unlock outside lever, allowing entry. Egress at all times by operating inside panic exit push pad and exiting with integrated REX and keyed bypass. High security switch at door head monitors the position of the door and reports this status to the security system. Provide hardware set components within factory assembled sound control assembly in accordance with section 08 34 73. |                                                                                                                                                                                                                  |                                                                                |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                  |                                                                                |        |
| Set 8.1 - Single x Fire Rated x Fail Secure x CL x STC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                  |                                                                                |        |
| Doors: 116A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                  |                                                                                |        |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Hinge (heavy weight) (A8111 NRP)                                                                                                                                                                                 | A8111 NRP                                                                      | US26D  |

| Quantity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Description                                                                                                                                        | BHMA Type / Function Numbers | Finish |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------|
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Fail-secure electromechanical mortise lock with levers on both sides with integrated REX on pull-side. Provide cylinder key override on push-side. | U2-A156.3 F05 G1 - E01, E06  | 630    |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | High security interchangeable core cylinder                                                                                                        | E09241A                      |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Surface Closer                                                                                                                                     | C02011                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Threshold                                                                                                                                          | J12190                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Wall Stop                                                                                                                                          | L02251                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Smoke & Draft Gasketing                                                                                                                            | R0E154                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Smoke & Draft Automatic Door Bottom                                                                                                                | R3G324                       |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Electric Power Transfer                                                                                                                            |                              |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ElectroLynx Harness (in door)                                                                                                                      |                              |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ElectroLynx Harness (in frame)                                                                                                                     |                              |        |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Balance Magnetic Switch/Contact                                                                                                                    | BY OTHERS                    |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Card Reader with Keypad                                                                                                                            | BY OTHERS                    |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PIR REX Sensor                                                                                                                                     | BY OTHERS                    |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Power Supply                                                                                                                                       |                              |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Wiring Diagrams                                                                                                                                    | BY OTHERS                    |        |
| Operational description: Door is normally closed and locked. Presenting a valid card to card reader and code at key pad will momentarily unlock outside lever, allowing entry. Egress at all times by proximity to passive infrared (PIR) sensor which will deactivate the lock and allowing operation of inside lever. High security switch at door head monitors the position of the door and reports this status to the security system. Provide hardware set components within factory assembled sound control assembly in accordance with section 08 34 73. |                                                                                                                                                    |                              |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                    |                              |        |
| Set 8 .2 - Single x Fire Rated x Fail Secure x CL x Audible Alarm x Exit Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                    |                              |        |
| Doors: 116B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                    |                              |        |

| Quantity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Description                                                                                                                                                                                                                                                        | BHMA Type / Function Numbers                                                     | Finish |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Hinge (heavy weight) (A8111 NRP)                                                                                                                                                                                                                                   | A8111 NRP                                                                        | US26D  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Pedestrian door preassembled lock (PDPL), with electromechanical lock, meeting federal specification FF-L-2890 complete with electric release capability. Fire Rated. Provide statement on pull-side of door "EMERGENCY EXIT ONLY, PULL TO EXIT, ALARM WILL SOUND" | Federal Specification FF-L-2890, type IX, No hardware trim on push side of door. |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Surface Closer                                                                                                                                                                                                                                                     | C02011                                                                           |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Threshold                                                                                                                                                                                                                                                          | J12190                                                                           |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Wall Stop                                                                                                                                                                                                                                                          | L02251                                                                           |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Smoke & Draft Gasketing                                                                                                                                                                                                                                            | R0E154                                                                           |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Smoke & Draft Automatic Door Bottom                                                                                                                                                                                                                                | R3G324                                                                           |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Balance Magnetic Switch/Contact                                                                                                                                                                                                                                    | BY OTHERS                                                                        |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Emergency Exit Audible Door Alarm                                                                                                                                                                                                                                  | BY OTHERS                                                                        |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Power Unit                                                                                                                                                                                                                                                         |                                                                                  |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Wiring Diagrams                                                                                                                                                                                                                                                    | BY OTHERS                                                                        |        |
| Operational description: Door is normally closed and locked. No hardware trim on pull side of door. Egress at all times by operating inside lever and exiting. High security switch at door head monitors the position of the door panels and signals to hard-wired exit alarm panel for 100db sound actuation in addition to reporting this status to the security system. Provide signage on pull-side of the door. Provide hardware set components within factory assembled sound control assembly in accordance with section 08 34 73. |                                                                                                                                                                                                                                                                    |                                                                                  |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                    |                                                                                  |        |
| Set 8.3 - Single x Fail Secure x CL X STC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                    |                                                                                  |        |
| Doors:<br>111A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                    |                                                                                  |        |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Hinge (heavy weight)                                                                                                                                                                                                                                               | A8111 NRP                                                                        | 630    |

| Quantity | Description                                                                                                                                                         | BHMA Type / Function Numbers                                                   | Finish |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------|
| 1        | Pedestrian door lock assembly panic (PDLAP), with electromechanical lock, meeting federal specification FF-L-2890 with integrated request to exit and keyed bypass. | Federal Specification FF-L-2890, type II, with elctromechanical lock FF-L-2740 |        |
| 1        | High security interchangeable core cylinder                                                                                                                         | E09241A                                                                        |        |
| 1        | Surface Closer                                                                                                                                                      | C02011                                                                         |        |
| 1        | Wall Stop                                                                                                                                                           | L02251                                                                         |        |
| 1        | Acoustical Gasketing                                                                                                                                                | R0E154                                                                         |        |
| 1        | Acoustical Automatic Door Bottom                                                                                                                                    | R3G324                                                                         |        |
| 1        | Electric Power Transfer                                                                                                                                             |                                                                                |        |
| 1        | ElectroLynx Harness (in door)                                                                                                                                       |                                                                                |        |
| 1        | ElectroLynx Harness (in frame)                                                                                                                                      |                                                                                |        |
| 1        | Balance Magnetic Switch/Contact                                                                                                                                     | BY OTHERS                                                                      |        |
| 1        | Card Reader with Keypad                                                                                                                                             | BY OTHERS                                                                      |        |
| 1        | Power Supply                                                                                                                                                        |                                                                                |        |
| 1        | Wiring Diagrams                                                                                                                                                     | BY OTHERS                                                                      |        |



| Quantity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | BHMA Type / Function Numbers | Finish |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------|
|          | <p>Operational description: Door is normally closed and locked. Presenting a valid card to card reader and code at electromechanical lock will momentarily unlock outside lever, allowing entry. Electromechanical combination lock located on Room #111 (Corridor) side of the door leaf. Egress at all times by operating inside panic exit push pad and exiting with integrated REX. High security switch at door head monitors the position of the door and reports this status to the security system. Provide hardware set components within factory assembled sound control assembly in accordance with section 08 34 73.</p> |                              |        |

-- End of Section --

## SECTION 08 81 00

GLAZING  
05/19

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## Canadian General Standards Board (CGSB)

CGSB 12.1 2017, Safety Glazing

CGSB 12.8 2017, Insulating Glass Units

## AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 800 (2016) Voluntary Specifications and Test Methods for Sealants

## AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z97.1 (2015) Safety Glazing Materials Used in Buildings - Safety Performance Specifications and Methods of Test

## ASTM INTERNATIONAL (ASTM)

ASTM C509 (2006; R 2021) Standard Specification for Elastomeric Cellular Preformed Gasket and Sealing Material

ASTM C864 (2005; R 2015) Dense Elastomeric Compression Seal Gaskets, Setting Blocks, and Spacers

ASTM C920 (2018; R 2024) Standard Specification for Elastomeric Joint Sealants

ASTM C1021 (2008; R 2014) Standard Practice for Laboratories Engaged in Testing of Building Sealants

ASTM C1036 (2021) Standard Specification for Flat Glass

ASTM C1048 (2018) Standard Specification for Heat-Strengthened and Fully Tempered Flat Glass

ASTM C1087 (2016) Standard Test Method for Determining Compatibility of Liquid-Applied Sealants with Accessories Used in Structural Glazing Systems

|            |                                                                                      |
|------------|--------------------------------------------------------------------------------------|
| ASTM C1172 | (2019) Standard Specification for Laminated Architectural Flat Glass                 |
| ASTM C1184 | (2014) Standard Specification for Structural Silicone Sealants                       |
| ASTM C1281 | (2016) Standard Specification for Preformed Tape Sealants for Glazing Applications   |
| ASTM D395  | (2016; E 2017) Standard Test Methods for Rubber Property - Compression Set           |
| ASTM D2287 | (2019) Nonrigid Vinyl Chloride Polymer and Copolymer Molding and Extrusion Compounds |
| ASTM E1300 | (2016) Standard Practice for Determining Load Resistance of Glass in Buildings       |
| ASTM E2190 | (2010) Standard Specification for Insulating Glass Unit Performance and Evaluation   |

#### GLASS ASSOCIATION OF NORTH AMERICA (GANA)

|                     |                       |
|---------------------|-----------------------|
| GANA Glazing Manual | (2008) Glazing Manual |
| GANA Sealant Manual | (2008) Sealant Manual |

#### INSULATING GLASS MANUFACTURERS ALLIANCE (IGMA)

|              |                                                                                                                     |
|--------------|---------------------------------------------------------------------------------------------------------------------|
| IGMA TB-1200 | (1983; R 2016) Guidelines for Insulating Glass Dimensional Tolerances                                               |
| IGMA TB-3001 | (2001) Guidelines for Sloped Glazing                                                                                |
| IGMA TM-3000 | (1990; R 2016) North American Glazing Guidelines for Sealed Insulating Glass Units for Commercial & Residential Use |

#### U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|             |                                                     |
|-------------|-----------------------------------------------------|
| 16 CFR 1201 | Safety Standard for Architectural Glazing Materials |
|-------------|-----------------------------------------------------|

### 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Insulating Glass

Glazing Accessories

Sealants

Joint Backer

SD-04 Samples

Insulating Glass

Glazing Compound

Glazing Tape

Sealing Tapes

SD-07 Certificates

Insulating Glass

SD-08 Manufacturer's Instructions

Setting and Sealing Materials

Glass Setting

SD-11 Closeout Submittals

Warranty for Insulated Glass Units

### 1.3 SYSTEM DESCRIPTION

Fabricate and install watertight and airtight glazing systems to withstand thermal movement and wind loading without glass breakage, gasket failure, deterioration of glazing accessories, or defects in the work. Glazed panels must comply with the safety standards, in accordance with ANSI Z97.1, and comply with indicated wind/snow loading in accordance with ASTM E1300.

#### 1.3.1 Glazing for Passive Solar and Dynamic Control Fenestration

Identify glazing for Passive Solar and Dynamic Control Fenestration noted as part of a passive solar heating system and/or chromogenic fenestration and evaluate separately from other fenestration. Glazing for use in Passive Solar systems are exempt from SHGC requirements. Area-weighted averaging of chromogenic fenestration with other non-chromogenic fenestration is not permitted. For chromogenic fenestration systems, the lower-rated labeled SHGC must be used with automatic controls to modulate the amount of heat flow into the space in multiple steps in response to daylight levels or solar intensity.

### 1.4 QUALITY CONTROL

Submit two 203 by 254 mm samples of each of the following: insulating glass units.

### 1.5 DELIVERY, STORAGE, AND HANDLING

Deliver products to the site in unopened containers, labeled plainly with manufacturers' names and brands. Store glass and setting materials in safe, enclosed dry locations and do not unpack until needed for installation. Handle and install materials in a manner that will protect them from damage.

### 1.6 ENVIRONMENTAL REQUIREMENTS

Do not start glazing work until the outdoor temperature is above 4 degrees C and rising, unless procedures recommended by the glass manufacturer and approved by the Contracting Officer are made to warm the glass and rabbet surfaces. Provide ventilation to prevent condensation of moisture on glazing work during installation. Do not perform glazing work during damp or rainy weather.

### 1.7 WARRANTY

#### 1.7.1 Warranty for Insulated Glass Units

Warranty insulating glass units against development of material obstruction to vision (such as dust, fogging, or film formation on the inner glass surfaces) caused by failure of the hermetic seal, other than through glass breakage, for a 10-year period following acceptance of the work. Provide new units for any units failing to comply with terms of this warranty within 45 working days after receipt of notice from the Government.

## PART 2 PRODUCTS

### 2.1 PRODUCT SUSTAINABILITY CRITERIA

### 2.2 GLASS

ASTM C1036, unless specified otherwise. In doors and sidelights, provide safety glazing material conforming to 16 CFR 1201 and CAN/CGSB 12.1.

#### 2.2.1 Clear Glass

Type I, Class 1 (clear), Quality q4 (A). Provide for glazing openings not indicated or specified otherwise. Use double-strength sheet glass or 3 mm float glass for openings up to and including 1.39 square meters, 4.5 mm for glazing openings over 1.39 square meters but not over 2.79 square meters, and 6 mm for glazing openings over 2.79 square meters but not over 4.18 square meters.

#### 2.2.2 Laminated Glass

ASTM C1172, Laminated glass fabricated from two nominal 3 mm pieces of Type I, Class 1, Quality Q3, flat heat strengthened ultra clear glass conforming to ASTM C1048. Flat glass to be laminated together with a minimum of 0.75 mm thick, clear polyvinyl butyral laminate, conforming to requirements of 16 CFR 1201 and ASTM C1172. The total thickness of nominally 7 mm. Color to be clear.

### 2.2.3 Mirrors

#### 2.2.3.1 Glass Mirrors

Glass for mirrors must be Type I transparent flat type, Class 1-clear, Glazing Quality q1 6 mm thick conforming to ASTM C1036. Glass must be coated on one surface with silver coating, copper protective coating, and mirror backing paint. Silver coating must be highly adhesive pure silver coating of a thickness which must provide reflectivity of 83 percent or more of incident light when viewed through 6 mm thick glass, and must be free of pinholes or other defects. Copper protective coating must be pure bright reflective copper, homogeneous without sludge, pinholes or other defects, and must be of proper thickness to prevent "adhesion pull" by mirror backing paint. Mirror backing paint must consist of two coats of special scratch and abrasion-resistant paint, and must be baked in uniform thickness to provide a protection for silver and copper coatings which will permit normal cutting and edge fabrication.

### 2.3 INSULATING GLASS UNITS

TwoThree panes of glass separated by a dehydrated airspace, filled with argon gas and hermetically sealed, conforming to ASTM E2190 and CAN/CGSB 12.8. Submit performance and compliance documentation for each type of insulating glass.

Dimensional tolerances must be as specified in IGMA TB-1200. Spacer must be black, roll-formed, thermally broken aluminum, with bent or tightly welded or keyed and sealed joints to completely seal the spacer periphery and eliminate moisture and hydrocarbon vapor transmission into airspace through the corners. Primary seal must be compressed polyisobutylene and the secondary seal must be a specially formulated silicone.

### 2.4 SETTING AND SEALING MATERIALS

Provide as specified in the GANA Glazing Manual, IGMA TM-3000, IGMA TB-3001, and manufacturer's recommendations, unless specified otherwise herein. Do not use metal sash putty, nonskinning compounds, nonresilient preformed sealers, or impregnated preformed gaskets. Materials exposed to view and unpainted must be gray or neutral color. Sealant testing must be performed by a testing agency qualified according to ASTM C1021.

Submit glass manufacturer's recommendations for setting and sealing materials and for installation of each type of glazing material specified.

#### 2.4.1 Putty and Glazing Compound

Provide glazing compound as recommended by manufacturer for face-glazing metal sash. Putty must be linseed oil type. Do not use putty and glazing compounds with insulating glass or laminated glass.

#### 2.4.2 Glazing Compound

Use for face glazing metal sash. Do not use with insulating glass units or laminated glass.

### 2.4.3 Sealants

Provide elastomeric and structural sealants.

#### 2.4.3.1 Elastomeric Sealant

ASTM C920, Type S, Grade NS, Class 12.5, Use G. Use for channel or stop glazing sash. Sealants must be chemically compatible with setting blocks, edge blocks, and sealing tapes, with sealants used in manufacture of insulating glass units. Color of sealant must be white.

#### 2.4.3.2 Structural Sealant

ASTM C1184, Type S.

### 2.4.4 Joint Backer

Joint backer must have a diameter size at least 25 percent larger than joint width; type and material as recommended in writing by glass and sealant manufacturer.

### 2.4.5 Glazing Tapes

#### 2.4.5.1 Back-Bedding Mastic Glazing Tapes

Preformed, butyl-based, 100 percent solids elastomeric tape; nonstaining and nonmigrating in contact with nonporous surfaces; with or without spacer rod as recommended in writing by tape and glass manufacturers for application indicated; and complying with ASTM C1281 and AAMA 800 for products indicated below:

- a. AAMA 804.3 tape, where indicated.
- b. AAMA 806.3 tape, for glazing applications in which tape is subject to continuous pressure.
- c. AAMA 807.3 tape, for glazing applications in which tape is not subject to continuous pressure.

#### 2.4.5.2 Expanded Cellular Glazing Tapes

Closed-cell, PVC foam tapes; factory coated with adhesive on both surfaces; and complying with AAMA 800 for the following types:

- a. AAMA 810.1, Type 1, for glazing applications in which tape acts as the primary sealant.
- b. AAMA 810.1, Type 2, for glazing applications in which tape is used in combination with a full bead of liquid sealant.

### 2.4.6 Sealing Tapes

Preformed, semisolid, PVC-based material of proper size and compressibility for the particular condition, complying with ASTM D2287. Use only where glazing rabbet is designed for tape and tape is recommended by the glass or sealant manufacturer. Provide spacer shims for use with compressible tapes. Tapes must be chemically compatible with the product being set.

#### 2.4.7 Setting Blocks and Edge Blocks

Closed-cell neoprene setting blocks must be dense extruded type conforming to ASTM C509 and ASTM D395, Method B, Shore A durometer between 70 and 90. Edge blocking must be Shore A durometer of 50 (plus or minus 5). Provide silicone setting blocks when blocks are in contact with silicone sealant. Profiles, lengths and locations must be as required and recommended in writing by glass manufacturer. Block color must be black.

#### 2.4.8 Glazing Gaskets

Glazing gaskets must be extruded with continuous integral locking projection designed to engage into metal glass holding members to provide a watertight seal during dynamic loading, building movements and thermal movements. Glazing gaskets for a single glazed opening must be continuous one-piece units with factory-fabricated injection-molded corners free of flashing and burrs. Glazing gaskets must be in lengths or units recommended by manufacturer to ensure against pull-back at corners. Provide glazing gasket profiles as recommended by the manufacturer for the intended application.

##### 2.4.8.1 Fixed Glazing Gaskets

Fixed glazing gaskets must be closed-cell (sponge) smooth extruded compression gaskets of cured elastomeric virgin neoprene compounds conforming to ASTM C509, Type 2, Option 1.

##### 2.4.8.2 Wedge Glazing Gaskets

Wedge glazing gaskets must be high-quality extrusions of cured elastomeric virgin neoprene compounds, ozone resistant, conforming to ASTM C864, Option 1, Shore A durometer between 65 and 75.

##### 2.4.8.3 Aluminum Framing Glazing Gaskets

Glazing gaskets for aluminum framing must be permanent, elastic, non-shrinking, non-migrating, watertight and weathertight.

#### 2.4.9 Accessories

Provide as required for a complete installation, including glazing points, clips, shims, angles, beads, and spacer strips. Provide noncorroding metal accessories. Provide primer-sealers and cleaners as recommended by the glass and sealant manufacturers. Use ASTM C1087 to determine whether priming and other specific joint preparation techniques are required to obtain rapid, optimum adhesion of glazing sealants to surface.

### 2.5 MIRROR ACCESSORIES

#### 2.5.1 Mastic

Mastic for setting mirrors must be a polymer type mirror mastic resistant to water, shock, cracking, vibration and thermal expansion. Provide mastic compatible with mirror backing paint, and as approved by mirror manufacturer.

#### 2.5.2 Mirror Frames

Provide mirrors with mirror frames (J-mold channels) fabricated of



one-piece roll-formed Type 304 stainless steel with No. 4 brushed satin finish and concealed fasteners which will keep mirrors snug to wall. Frames must be 32 by 6 by 6 mm continuous at top and bottom of mirrors. Concealed fasteners of type to suit wall construction material must be provided with mirror frames.

### 2.5.3 Mirror Clips

Provide clips with concealed fasteners of type to suit wall construction material.

## PART 3 EXECUTION

Any materials that show visual evidence of biological growth due to the presence of moisture must not be installed on the building project.

### 3.1 PREPARATION

Preparation, unless otherwise specified or approved, must conform to applicable recommendations in the GANA Glazing Manual, GANA Sealant Manual, IGMA TB-3001, IGMA TM-3000, and manufacturer's recommendations. Determine the sizes to provide the required edge clearances by measuring the actual opening to receive the glass. Grind smooth in the shop glass edges that will be exposed in finish work. Leave labels in place until the installation is approved, except remove applied labels on heat-absorbing glass and on insulating glass units as soon as glass is installed. Securely fix movable items or keep in a closed and locked position until glazing compound has thoroughly set.

### 3.2 GLASS SETTING

Shop glaze or field glaze items to be glazed using glass of the quality and thickness specified or indicated. Glazing, unless otherwise specified or approved, must conform to applicable recommendations in the GANA Glazing Manual, GANA Sealant Manual, IGMA TB-3001, IGMA TM-3000, and manufacturer's recommendations. Aluminum windows, wood doors, and wood windows may be glazed in conformance with one of the glazing methods described in the standards under which they are produced, except that face puttying with no bedding will not be permitted. Handle and install glazing materials in accordance with manufacturer's instructions. Use beads or stops which are furnished with items to be glazed to secure the glass in place. Verify products are properly installed, connected, and adjusted.

#### 3.2.1 Insulating Glass Units

Do not grind, nip, or cut edges or corners of units after the units have left the factory. Springing, forcing, or twisting of units during setting will not be permitted. Handle units so as not to strike frames or other objects. Installation must conform to applicable recommendations of IGMA TB-3001 and IGMA TM-3000.

#### 3.2.2 Installation of Laminated Glass

Sashes which are to receive laminated glass must be weeped to the outside to allow water drainage into the channel.

### 3.3 CLEANING

Clean glass surfaces and remove labels, paint spots, putty, and other defacement as required to prevent staining. Glass must be clean at the time the work is accepted.

### 3.4 PROTECTION

Protect glass work immediately after installation. Identify glazed openings with suitable warning tapes, cloth or paper flags, attached with non-staining adhesives. Protect reflective glass with a protective material to eliminate any contamination of the reflective coating. Place protective material far enough away from the coated glass to allow air to circulate to reduce heat buildup and moisture accumulation on the glass. Upon removal, separate protective materials for reuse or recycling. Remove and replace glass units which are broken, chipped, cracked, abraded, or otherwise damaged during construction activities with new units.

### 3.5 SCHEDULE

Some metric measurements in this section are based on mathematical conversion of inch-pound measurements, and not on metric measurement commonly agreed to by the manufacturers or other parties. The inch-pound and metric measurements are as follows:

| <u>PRODUCTS</u>  | <u>INCH-POUND</u> | <u>METRIC</u> |
|------------------|-------------------|---------------|
| Glass            | 1/8 inch          | 3 mm          |
|                  | 3/16 inch         | 4.5 mm        |
|                  | 7/32 inch         | 6 mm          |
|                  | 1/4 inch          | 6 mm          |
|                  | 3/8 inch          | 10 mm         |
| Interlayer       | 0.015 inch        | 0.38 mm       |
| Glazing Channels | 1/4 inch          | 6 mm          |

-- End of Section --

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## SECTION 09 22 00

SUPPORTS FOR PLASTER AND GYPSUM BOARD  
08/24

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI S220 (2020) North American Standard for Cold-Formed Steel Nonstructural Framing

## ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM C635/C635M (2022) Standard Specification for Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-In Panel Ceilings

ASTM C754 (2020) Standard Specification for Installation of Steel Framing Members to Receive Screw-Attached Gypsum Panel Products

ASTM C841 (2023) Installation of Interior Lathing and Furring

ASTM C847 (2014a) Standard Specification for Metal Lath

## ICC EVALUATION SERVICE, INC. (ICC-ES)

ICC-ES AC98 (2021) Accreditation Criteria for Inspection Agencies

## UL SOLUTIONS (UL)

UL Fire Resistance (2014) Fire Resistance Directory

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Metal Support Systems; G

SD-03 Product Data

Metal Support Systems

Recycled Content for Metal Support Systems; S

#### 1.2.1 Sustainable Design Submittals

- a. Product Data: For recycled content, indicating postconsumer and preconsumer recycled content and cost.
- b. Environmental Product Declaration (EPD): For each product.
- c. Construction and Demolition Waste Management: For each product.
- d. Health Product Declaration (HPD): Provide documentation confirming product compliance with one of the following:
  - (1) Option 1: Material Ingredient Reporting: Use at least 20 different permanently installed products from at least five different manufacturers that use one of the qualifying programs to demonstrate the chemical inventory of the product to at least 0.1 percent (1000 ppm).
  - (2) Option 2: Material Ingredient Optimization: Use products that have a compliant material ingredient optimization report or action plan. Use at least five permanently installed products sourced from at least three different manufacturers.

SD-04 Evaluation Reports: Submit evaluation reports certified under an independent third-party inspection program administered by an agency accredited by IAS to ICC-ES AC98 accreditation criteria for inspection agencies.

#### 1.3 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the job site and store in ventilated dry locations permitting easy access for inspection and handling. If materials are stored outdoors, stack materials off the ground, supported on a level platform, and fully protected from the weather. Handle materials carefully to prevent damage. Remove damaged items and provide new items.

### PART 2 PRODUCTS

#### 2.1 MATERIALS

Provide steel materials for metal support systems with either a EQ (Equivalent Coating) coating or a galvanized coating AISI S220G-60 or equivalent; aluminum coating T1-25; or a 55-percent aluminum-zinc coating AZ50.

Provide metal support systems containing a minimum of 20 percent recycled content. Provide data identifying percentage of recycled content for metal support systems.

### 2.1.1 Materials for Attachment of Lath

#### 2.1.1.1 Suspended and Furred Ceiling Systems and Wall Furring

ASTM C841 and ASTM C847.

#### 2.1.1.2 Non-load Bearing Wall Framing

AISI S220.

### 2.1.2 Materials for Attachment of Gypsum Wallboard

#### 2.1.2.1 Suspended and Furred Ceiling Systems

AISI S220.

#### 2.1.2.2 Non-load Bearing Wall Framing and Furring

- a. Framing Members, General: Comply with AISI S220 for conditions indicated.
  - (1) Steel Sheet Components: Comply with AISI S220 requirements for metal unless otherwise indicated.
  - (2) Protective Coating: AISI S220, provide EQ (Equivalent Coating) coating, galvanized coating G-60 or equivalent. G40 or coating with equivalent corrosion resistance of G40 . Provide G-90 coating in high-humidity environments.
- b. Studs and Runners: AISI S220. Use either steel studs and runners or embossed steel studs and runners.

AISI S220, but not thinner than thickness.
- c. Slip-Type Head Joints: Where indicated, provide one of the following:
  - (1) Slotted Deflection Track: Steel sheet top runner manufactured to prevent cracking of finishes applied to interior partition framing resulting from deflection of structure above; in thickness not less than indicated for studs and in width to accommodate depth of studs.
  - (2) Clip System: Clips designed for use in head-of-wall deflection conditions that provide a positive attachment of studs to tracks while allowing minimum vertical movement.
  - 3) Single Long-Leg Track System: Top track with 50 mm deep flanges in thickness not less than indicated for studs, installed with studs friction fit into top track and with continuous bridging and spacer bar located within 300 mm of the top of studs to provide lateral bracing.
- d. Firestop Tracks: Top runner manufactured to allow partition heads to expand and contract with movement of the structure while maintaining continuity of fire-resistance-rated assembly indicated; in thickness not less than indicated for studs and in width to accommodate depth of studs.
- e. Backing Plate: Fire-retardant-treated wood blocking used for fixture

attachment.

- f. Sound Isolation Clips: Used with 0457 mm, (25 gauge), drywall furring channels to provide acoustical separation.
- g. Framed Openings: Galvanized-steel, one-piece header and jamb studs in lieu of built-up members.
- h. Partial Wall Framing Connection: Stud connector designed to support out-of-plane loading of cantilevered partial wall systems that are unsupported at the top track.
- i. Drywall Penetration Barrier Mesh: Supply and install security mesh, as a penetration barrier behind gypsum wallboard walls and/or ceilings, where indicated on Drawings.

#### 2.1.2.3 Furring Structural Steel Columns

AISI S220. Steel (furring) clips and support angles listed in UL Fire Resistance may be provided in lieu of steel studs for erection of gypsum wallboard around structural steel columns.

#### 2.1.2.4 Z-Furring Channels with Wall Insulation

Not lighter than 0.45 mm thick galvanized steel, Z-shaped, with 30 mm and 20 mm flanges and depth as required by the insulation thickness provided.

### 2.2 SUSPENSION SYSTEMS

#### 2.2.1 Structural Classification

- a. Main Beam/ Tee is to be Fire-Rated Heavy Duty classification in accordance with ASTM C635/C635M.
- b. Classification can require wires to be closer together for additional loading when used to support double layer gypsum, verticals, slopes, circles, soffits, canopies, and step conditions which call for loading or unusual designs and shapes in drywall construction.
- c. Deflection of fastening suspension system supporting light fixtures, ceiling grilles, access doors, verticals and horizontal loads are to have a maximum deflection of 1/360 of the span.

#### 2.2.2 Components

- a. Main Beam/ Tee: Is to be commercial quality, cold-rolled steel, double-web construction (minimum 0.45 mm prior to protective coating, AISI S220), hot dipped galvanized (per ASTM A653/A653M) and 40 mm high minimum with knurled face.
- b. Drywall Grid Cross Tees: Are to be commercial quality, cold-rolled steel, double-web steel construction (minimum 0.45 mm prior to protective coating, AISI S220), hot dipped galvanized (minimum G40 or G90 in accordance with ASTM A653/A653M) and 40 mm high minimum with knurled face.
- c. Wall Molding/ Channel:
  - (1) 50 mm by 50 mm, knurled surface, pre-punched holes in top flange

100 mm on center, 33 mil (20 gauge)

- (2) Locking Angle Molding, 30 mm by 30 mm with pre-engineered locking tabs punched 202 mm on center, knurled surface, screw stop hem, pre-punched holes in top flange, 101 mm on center, 18 mil. (25 gauge)
- (3) Knurled Angle molding, 30 mm by 30 mm, knurled surface, screw stop hem, pre-punched holes in top flange, 100 mm on center, 18 mil. (25 gauge)
- (4) Knurled Angle molding, 40 mm by 40 mm, knurled surface, screw stop hem, pre-punched holes in top flange 100 mm on center, 18 mil. (25 gauge)
- (5) Simple Curve Knurled Angle Molding, 40 mm by 40 mm, knurled face radius 940 mm, screw stop hem, pre-punched holes in top flange, 100 on center, 28 mil. (22 gauge)

d. Support Hanger:

- (1) Strong Back Support Hanger
- (2) Channel beam splice 150 mm used to suspend main beams to 40 mm black iron carrying channels.
- (3) Hanger Wire 12 gauge, galvanized or as noted on drawings.

e. Accessories

- (1) Radius Clip
- (2) 90 Degree Drywall Angle/ Transition Clip
- (3) 45 Degree Drywall Angle/ Transition Clip
- (4) 30 Degree Drywall Angle/ Transition Clip
- (5) Double Drywall Clip
- (6) Wall Attachment Clip
- (7) Splice Plate
- (8) Dome Hub
- (9) Close Mount Attachment clip

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Systems for Attachment of Lath

3.1.1.1 Suspended and Furred Ceiling Systems and Wall Furring

ASTM C841, except as indicated otherwise.



### 3.1.1.2 Non-load Bearing Wall Framing

ASTM C754, Installation Standard.

#### 3.1.1.2.1 Firestop Track

Where indicated, install to maintain continuity of fire-resistance-rated assembly indicated.

#### 3.1.1.2.2 Sound-Rated Partitions

Install framing to comply with sound-rated assembly indicated.

### 3.1.2 Systems for Attachment of Gypsum Wallboard

#### 3.1.2.1 Suspended and Furred Ceiling Systems

ASTM C754, except provide framing members 400 mm on center unless indicated otherwise.

#### 3.1.2.2 Non-load Bearing Wall Framing and Furring

ASTM C754, except as indicated otherwise. Provide a sealer gasket under bottom of track on concrete slab or foundation.

#### 3.1.2.3 Furring Structural Steel Columns

Install studs or galvanized steel clips and support angles for erection of gypsum wallboard around structural steel columns in accordance with the UL Fire Resistance, design number(s) of the fire resistance rating indicated.

#### 3.1.2.4 Z-Furring Channels with Wall Insulation

Install Z-furring channels vertically spaced not more than 600 mm on center. Locate Z-furring channels at interior and exterior corners in accordance with manufacturer's printed erection instructions. Fasten furring channels to concrete walls with powder-driven fasteners or hardened concrete steel nails through narrow flange of channel. Space fasteners not more than 600 mm on center.

#### 3.1.2.5 Firestop Tracks

In accordance with manufacturer's installation instructions, applicable building code and UL assembly.

#### 3.1.2.6 Backing Plates

In accordance with manufacturer's installation instructions.

#### 3.1.2.7 Framed Openings

In accordance with manufacturer's installation instructions and approved design.

#### 3.1.2.8 Partial Wall Framing Connection

In accordance with manufacturer's installation instructions. Anchor to floor as designed by Engineer of Record. Provide a sealer gasket under

bottom of track on concrete slab or foundation.

#### 3.1.2.9 Drywall Penetration Barrier Mesh

Per manufacturer's instructions. Installation and layout of the panels to be approved by General Contractor prior to installation, with the intent to attach panels on vertical framing members. Panels may be installed in either direction. Attachment with approved clips and threaded fasteners.

#### 3.1.2.10 Suspension Systems

- a. Install suspension system and panels in accordance with the manufacturer's instructions, in compliance with ASTM C635/C635M installation standard, and with applicable codes as required by the authorities having jurisdiction.
- b. Install according to approved Pre-construction drawings recommended for complex projects. Contact your local manufacturer representative for support.
- c. Drywall Grid System can be installed in interior or exterior applications.
- d. Install hanger wire as required with necessary on center spacing to support expected ceiling load requirements, following local practices, codes and regulations. Provide additional wires at light fixtures, grilles, and access doors where necessary. A pigtail knot is to be used with three tight wraps at top and bottom fastening locations.
- e. Main tees/ beams are to be suspended from the overhead construction with hanger wire, spaced as required for expected ceiling loads, along the length of the main beams.
- f. Install cross tees at on center spacing as specified by the drywall manufacturer. Typical drywall cross tee spacing:
  - (1) 400 mm on center with 15 mm or 13 mm gypsum board.
  - (2) 600 mm on center with 15 mm gypsum board.
- g. Use channel molding or angle molding to interface with Drywall Grid System to provide perimeter attachment or to obtain drop soffits, verticals, slopes.

#### 3.2 ERECTION TOLERANCES

Provide framing members which will be covered by finish materials such as wallboard, plaster, or ceramic tile set in a mortar setting bed, within the following limits:

- a. Layout of walls and partitions: 6 mm from intended position;
- b. Plates and runners: 6 mm in 2.4 meters from a straight line;
- c. Studs: 6 mm in 2.4 meters out of plumb, not cumulative; and
- d. Face of framing members: 6 mm in 2.4 meters from a true plane.

Provide framing members which will be covered by ceramic tile set in dry-set mortar, latex-portland cement mortar, or organic adhesive within the following limits:

- a. Layout of walls and partitions: 6 mm;
- b. Plates and runners: 3 mm in 2.4 meters from a straight line;
- c. Studs: 3 mm in 2.4 meters out of plumb, not cumulative and must not exceed minimum fastening requirements of sheathing or other finishing materials; and
- d. Face of framing members: 3 mm in 2.4 meters from a true plane.

-- End of Section --

## SECTION 09 29 00

GYPSUM BOARD  
08/16, CHG 4: 02/20

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A108.11 (1992; Reaffirmed 2005) Specifications for Interior Installation of Cementitious Backer Units

## ASTM INTERNATIONAL (ASTM)

ASTM C475/C475M (2017; R 2022) Standard Specification for Joint Compound and Joint Tape for Finishing Gypsum Board

ASTM C840 (2023) Standard Specification for Application and Finishing of Gypsum Board

ASTM C954 (2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.033 in. (0.84 mm) to 0.112 in. (2.84 mm) in Thickness

ASTM C1002 (2022) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs

ASTM C1047 (2019) Standard Specification for Accessories for Gypsum Wallboard and Gypsum Veneer Base

ASTM C1177/C1177M (2017) Standard Specification for Glass Mat Gypsum Substrate for Use as Sheathing

ASTM C1178/C1178M (2013) Standard Specification for Glass Mat Water-Resistant Gypsum Backing Panel

ASTM C1396/C1396M (2017) Standard Specification for Gypsum Board

ASTM C1629/C1629M (2023) Standard Classification for Abuse-Resistant Nondecorated Interior Gypsum Panel Products and Fiber-Reinforced Cement Panels

|                 |                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D226/D226M | (2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing                                                   |
| ASTM D412       | (2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension                                                    |
| ASTM D624       | (2000; R 2020) Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers                                 |
| ASTM D1037      | (2012; R 2020) Standard Test Methods for Evaluating Properties of Wood-Base Fiber and Particle Panel Materials                                       |
| ASTM D1149      | (2018) Standard Test Method for Rubber Deterioration - Cracking in an Ozone Controlled Environment                                                   |
| ASTM D2394      | (2017) Standard Test Methods for Simulated Service Testing of Wood and Wood-Base Finish Flooring                                                     |
| ASTM D3273      | (2021) Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber                         |
| ASTM D5420      | (2016) Standard Test Method for Impact Resistance of Flat, Rigid Plastic Specimen by Means of a Strike Impacted by a Falling Weight (Gardner Impact) |
| ASTM E84        | (2024) Standard Test Method for Surface Burning Characteristics of Building Materials                                                                |
| ASTM E336       | (2023) Standard Test Method for Measurement of Airborne Sound Attenuation between Rooms in Buildings                                                 |
| ASTM E695       | (2022) Standard Test Method of Measuring Relative Resistance of Wall, Floor, and Roof Construction to Impact Loading                                 |

## CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

## FM GLOBAL (FM)

|              |                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------|
| FM APP GUIDE | (updated on-line) Approval Guide<br><a href="https://www.approvalguide.com/">https://www.approvalguide.com/</a> |
|--------------|-----------------------------------------------------------------------------------------------------------------|

## GREEN SEAL (GS)

GS-36 (2013) Adhesives for Commercial Use

## GYPSUM ASSOCIATION (GA)

GA 214 (2010) Recommended Levels of Gypsum Board Finish

GA 216 (2016) Application and Finishing of Gypsum Panel Products

GA 253 (2012) Application of Gypsum Sheathing

## SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

## SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

## UNDERWRITERS LABORATORIES (UL)

UL 2818 (2022) GREENGUARD Certification Program For Chemical Emissions For Building Materials, Finishes And Furnishings

UL Fire Resistance (2014) Fire Resistance Directory

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-03 Product Data

Cementitious Backer Units

Glass Mat Water-Resistant Gypsum Tile Backing Board

Water-Resistant Gypsum Backing Board

Glass Mat Covered or Reinforced Gypsum Sheathing

Glass Mat Covered or Reinforced Gypsum Sheathing Sealant

Abuse Resistant Gypsum Board

Accessories

Submit for each type of gypsum board and for cementitious backer units.

Gypsum Board

Recycled Content for Gypsum Board; S

Recycled Content for Paper Facing and Gypsum Cores; S

VOC Content of Joint Compound; S

#### SD-04 Samples

Predecorated Gypsum Board; G

Submit for each color and pattern of predecorated gypsum board. Where colors are not indicated, submit color selection samples of not less than eight of the manufacturer's standard colors.

#### SD-06 Test Reports

ASTM E90 Factory Test Report; G

ASTM E336 Field Test Report; G

#### SD-07 Certificates

Asbestos Free Materials; G

Certify that gypsum board types, gypsum backing board types, cementitious backer units, and joint treating materials do not contain asbestos.

Indoor Air Quality for Gypsum Board; S

Indoor Air Quality for Non-aerosol Adhesives; S

Indoor Air Quality for Aerosol Adhesives; S

#### SD-08 Manufacturer's Instructions

Safety Data Sheets

#### SD-10 Operation and Maintenance Data

Manufacturer Maintenance Instructions

### 1.3 CERTIFICATIONS

#### 1.3.1 Indoor Air Quality Certifications

Submit required indoor air quality certifications in one submittal package.

##### 1.3.1.1 Ceiling and Wall Systems

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

#### 1.3.1.2 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

### 1.4 DELIVERY, STORAGE, AND HANDLING

#### 1.4.1 Delivery

Deliver materials in the original packages, containers, or bundles with each bearing the brand name, applicable standard designation, and name of manufacturer, or supplier.

#### 1.4.2 Storage

Keep materials dry by storing inside a sheltered building. Where necessary to store gypsum board and cementitious backer units outside, store off the ground, properly supported on a level platform, and protected from direct exposure to rain, snow, sunlight, and other extreme weather conditions. Provide adequate ventilation to prevent condensation. Store per manufacturer's recommendations for allowable temperature and humidity range. Do not store gypsum wallboard with materials which have high emissions of volatile organic compounds (VOCs) or other contaminants. Do not store panels near materials that may offgas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives. Do not use materials that have visible moisture or biological growth.

#### 1.4.3 Handling

Neatly stack gypsum board and cementitious backer units flat to prevent sagging or damage to the edges, ends, and surfaces.

### 1.5 QUALIFICATIONS

Furnish type of gypsum board work specialized by the installer with a minimum of 3 years of documented successful experience.

### 1.6 SCHEDULING

The gypsum wallboard must be taped, finished and primed before the installation of the highly-emitting materials.

Commence application only after the area scheduled for gypsum board work is completely weathertight. The heating, ventilating, and air-conditioning systems must be complete and in operation prior to application of the gypsum board. If the mechanical system cannot be activated before gypsum board is begun, the gypsum board work may proceed in accordance with an approved plan to maintain the environmental conditions specified below. Apply gypsum board prior to the installation of finish flooring and acoustic ceiling.



## 1.7 ENVIRONMENTAL REQUIREMENTS

Do not expose the gypsum board to excessive sunlight prior to gypsum board application. Maintain a continuous uniform temperature of not less than 10 degrees C and not more than 27 degrees C for at least one week prior to the application of gypsum board work, while the gypsum board application is being done, and for at least one week after the gypsum board is set. Shield air supply and distribution devices to prevent any uneven flow of air across the plastered surfaces. Provide ventilation to exhaust moist air to the outside during gypsum board application, set, and until gypsum board jointing is dry. In glazed areas, keep windows open top and bottom or side to side 75 to 100 mm. Reduce openings in cold weather to prevent freezing of joint compound when applied. For enclosed areas lacking natural ventilation, provide temporary mechanical means for ventilation. In unglazed areas subjected to hot, dry winds or temperature differentials from day to night of 10 degrees C or more, screen openings with cheesecloth or similar materials. Avoid rapid drying. During periods of low indoor humidity, provide minimum air circulation following gypsum boarding and until gypsum board jointing complete and is dry.

## 1.8 FIRE RESISTIVE CONSTRUCTION

Comply with specified fire-rated assemblies for design numbers indicated per UL Fire Resistance or FM APP GUIDE.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Conform to specifications, standards and requirements specified. Provide gypsum board types, gypsum backing board types, cementitious backing units, and joint treating materials manufactured from asbestos free materials only. Submit Safety Data Sheets and manufacturer maintenance instructions for gypsum materials including adhesives.

#### 2.1.1 Gypsum Board

ASTM C1396/C1396M. Gypsum board must contain a minimum of 10 percent post-consumer recycled content, or a minimum of 20 percent post-industrial recycled content. Provide data identifying percentage of recycled content for gypsum board. Provide gypsum wall board and panels meeting the emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type). Provide certification or validation of indoor air quality for gypsum board.

##### 2.1.1.1 Regular

1200 mm wide, 15.9 mm thick, tapered edges.

##### 2.1.1.2 Type X (Special Fire-Resistant)

1200 mm wide, 15.9 mm thick, tapered edges.

##### 2.1.1.3 Mold Resistant / Anti-Microbial Gypsum

ASTM D3273. 1200 mm wide, 15.9 mm thick, tapered edges.

## 2.1.2 Gypsum Backing Board

ASTM C1396/C1396M, gypsum backing board must be used as a base in a multilayer system.

### 2.1.2.1 Regular

1200 mm wide, 15.9 mm thick, square edges.

### 2.1.2.2 Type X (Special Fire-Resistant)

1200 mm wide, 15.9 mm thick, square edges.

## 2.1.3 Regular Water-Resistant Gypsum Backing Board

ASTM C1396/C1396M

### 2.1.3.1 Regular

1200 mm wide, 15.9 mm thick, tapered edges.

### 2.1.3.2 Type X (Special Fire-Resistant)

1200 mm wide, 15.9 mm thick, tapered edges.

## 2.1.4 Glass Mat Water-Resistant Gypsum Tile Backing Board

ASTM C1178/C1178M

### 2.1.4.1 Regular

1200 mm wide, 15.9 mm thick, square edges.

### 2.1.4.2 Type X (Special Fire-Resistant)

1200 mm wide, 15.9 mm thick, square edges.

## 2.1.5 Glass Mat Covered or Reinforced Gypsum Sheathing

Exceeds physical properties of ASTM C1396/C1396M and ASTM C1177/C1177M. Provide 15.9 mm, gypsum sheathing. Provide gypsum board of with a noncombustible water-resistant core, with glass mat surfaces embedded to the gypsum core or reinforcing embedded throughout the gypsum core. Warrant gypsum sheathing board for at least twelve months against delamination due to direct weather exposure. Provide continuous, asphalt impregnated, building felt to cover exterior face of sheathing. Seal all joints, seams, and penetrations with compatible sealant.

### 2.1.5.1 Glass Mat Covered or Reinforced Gypsum Sheathing Sealant

Provide sealant compatible with glass mat covered or reinforced gypsum sheathing, rubber washers for masonry veneer anchors, and other associated cavity wall components such as anchors and through wall flashing. Provide sealants for glass mat covered or reinforced gypsum sheathing board edge seams and veneer anchor penetrations recommended by the glass mat covered or reinforced gypsum sheathing manufacturer and have the following performance requirements:

- a. ASTM D412: Tensile Strength, 551 kilopascals

- b. ASTM D412: Ultimate Tensile Strength (maximum elongation), 1172 kilopascals
- c. ASTM D624: Tear Strength, dieB, 4.7 kN/m
- d. ASTM D1149: Joint Movement Capability after 14 Days cure, plus or minus 50 percent.

#### 2.1.6 Abuse Resistant Gypsum Board

1200 mm wide, 15.9 mm thick, tapered edges.  
Reinforced gypsum panel with imbedded fiber mesh or lexan backing tested in accordance with the following tests. Hard body impact test must attain a Level 2 performance in accordance with ASTM C1629/C1629M. Provide fasteners that meet manufacturer requirements and specifications stated within this section. Abuse resistant gypsum board, when tested in accordance with ASTM E84, have a flame spread rating of 25 or less and a smoke developed rating of 50 or less.

##### 2.1.6.1 Soft Body Impact Test

ASTM E695 or ASTM D2394 for impact penetration and deformation. ASTM E695 using a 27.2 kg leather bag filled with steel pellets, resisting no less than 407 N-m cumulative impact energy before failure or ASTM D2394 using 139.7 mm hemispherical projectile resisting no less than 357 N-m before failure. Provide test specimen stud spacing a minimum 406 mm on center.

##### 2.1.6.2 Hard Body Impact Test

Comply with hard body impact test in accordance with ASTM C1629/C1629M Classification Level 2.

##### 2.1.6.3 Surface Abrasion Test

Comply with test surface abrasion test in accordance with ASTM C1629/C1629M.

##### 2.1.6.4 Indentation Test

ASTM D5420 or ASTM D1037 for indentation resistance. ASTM D5420 using a .907 kg weight with a 16 mm hemispherical impacting head dropped once 915 mm creating not more than 3.5 mm indentation or ASTM D1037 using no less than 213 kg weight applied to the 11.13 mm diameter ball to create not more than a 0.5 mm indentation depth.

#### 2.1.7 Cementitious Backer Units

In accordance with the Tile Council of America (TCA) Handbook.

#### 2.1.8 Joint Treatment Materials

ASTM C475/C475M. Product must be low emitting VOC types with VOC limits not exceeding 50 g/L. Provide data identifying VOC content of joint compound. Use all purpose joint and texturing compound containing inert fillers and natural binders, including lime compound. Pre-mixed compounds must be free of antifreeze, vinyl adhesives, preservatives, biocides and other slow releasing compounds.

##### 2.1.8.1 Embedding Compound

Specifically formulated and manufactured for use in embedding tape at gypsum board joints and compatible with tape, substrate and fasteners.

#### 2.1.8.2 Finishing or Topping Compound

Specifically formulated and manufactured for use as a finishing compound.

#### 2.1.8.3 All-Purpose Compound

Specifically formulated and manufactured to serve as both a taping and a finishing compound and compatible with tape, substrate and fasteners.

#### 2.1.8.4 Setting or Hardening Type Compound

Specifically formulated and manufactured for use with fiber glass mesh tape.

#### 2.1.8.5 Joint Tape

Use cross-laminated, tapered edge, reinforced paper, or fiber glass mesh tape recommended by the manufacturer.

#### 2.1.9 Fasteners

##### 2.1.9.1 Screws

ASTM C1002, Type "G", Type "S" or Type "W" steel drill screws for fastening gypsum board to gypsum board, wood framing members and steel framing members less than 0.84 mm thick. ASTM C954 steel drill screws for fastening gypsum board to steel framing members 0.84 to 2.84 mm thick. Provide cementitious backer unit screws with a polymer coating.

##### 2.1.9.2 Staples

1.5 mm thick flattened galvanized wire staples with 11.1 mm wide crown outside measurement and divergent point for base ply of two-ply gypsum board application. Use as follows:

| <u>Length of Legs</u> | <u>Thickness of Gypsum Board</u> |
|-----------------------|----------------------------------|
| 28.6 mm               | 12.7 mm                          |
| 31.8 mm               | 15.9 mm                          |

##### 2.1.10 Adhesives

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for non-aerosol adhesives applied on the interior of the building (inside of the weatherproofing system). Provide certification or validation of indoor air quality for aerosol adhesives used on the interior of the building (inside of the weatherproofing system).

#### 2.1.10.1 Adhesive for Fastening Gypsum Board to Metal Framing

Not permitted.

#### 2.1.10.2 Adhesive for Laminating

Not permitted.

#### 2.1.11 Gypsum Studs

Provide 25 mm minimum thickness and 150 mm minimum width. Studs may be of 25 mm thick gypsum board or multilayers fastened to required thickness. Conform to ASTM C1396/C1396M for material and GA 216 for installation.

#### 2.1.12 Shaftwall Liner Panel

ASTM C1396/C1396M. Conform to the UL Fire Resistance for the Design Numbers(s) indicated for shaftwall liner panels. Manufacture liner panel for cavity shaftwall system, with water-resistant paper faces, bevel edges, single lengths to fit required conditions, 25.4 mm thick, by 610 mm wide or as required by UL Listed Assembly.

#### 2.1.13 Accessories

ASTM C1047. Fabricate from corrosion protected steel designed for intended use. Accessories manufactured with paper flanges are not acceptable. Flanges must be free of dirt, grease, and other materials that may adversely affect bond of joint treatment. Provide prefinished or job decorated materials.

#### 2.1.14 Asphalt Impregnated Building Felt

Provide a 6.7 kg asphalt moisture barrier over glass mat covered or reinforced gypsum sheathing. Conforming to ASTM D226/D226M Type 1 (No. 15) for asphalt impregnated building felt.

#### 2.1.15 Water

Provide clean, fresh, and potable water.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

##### 3.1.1 Framing and Furring

Verify that framing and furring are securely attached and of sizes and spacing to provide a suitable substrate to receive gypsum board and cementitious backer units. Verify that all blocking, headers and supports are in place to support plumbing fixtures and to receive soap dishes, grab bars, towel racks, and similar items. Do not proceed with work until framing and furring are acceptable for application of gypsum board and cementitious backer units.

##### 3.1.2 Gypsum Board and Framing

Verify that surfaces of gypsum board and framing to be bonded with an adhesive are free of dust, dirt, grease, and any other foreign matter. Do not proceed with work until surfaces are acceptable for application of

### gypsum board with adhesive.3.1.3 Concrete Walls

Verify that surfaces of concrete walls to receive gypsum board applied with adhesive are dry, free of dust, oil, form release agents, protrusions and voids, and any other foreign matter. Do not proceed with work until surfaces are acceptable for application of gypsum board with adhesive.

### 3.1.4 Building Construction Materials

Do not install building construction materials that show visual evidence of biological growth.

## 3.2 APPLICATION OF GYPSUM BOARD

Apply gypsum board to framing and furring members in accordance with ASTM C840 or GA 216 and the requirements specified. Apply gypsum board with separate panels in moderate contact; do not force in place. Stagger end joints of adjoining panels. Neatly fit abutting end and edge joints. Use gypsum board of maximum practical length; select panel sizes to minimize waste. Cut out gypsum board to make neat, close, and tight joints around openings. In vertical application of gypsum board, provide panels in lengths required to reach full height of vertical surfaces in one continuous piece. Lay out panels to minimize waste; reuse cutoffs whenever feasible. Surfaces of gypsum board and substrate members may not be bonded together with an adhesive. Treat edges of cutouts for plumbing pipes, screwheads, and joints with water-resistant compound as recommended by the gypsum board manufacturer. Provide type of gypsum board for use in each system specified herein as indicated.

### 3.2.1 Application of Two-Ply Gypsum Board to Wood Framing

Apply in accordance with ASTM C840, System II or GA 216.

### 3.2.2 Semi-Solid Gypsum Board Partitions

Provide in accordance with ASTM C840, System IV or GA 216 .

### 3.2.3 Solid Gypsum Board Partitions

Provide in accordance with ASTM C840, System V or GA 216.

### 3.2.4 Adhesive Application to Interior Masonry or Concrete Walls

Apply in accordance with ASTM C840, System VI or GA 216.

### 3.2.5 Application of Gypsum Board to Steel Framing and Furring

Apply in accordance with ASTM C840, System VIII or GA 216.

### 3.2.6 Arches and Bending Radii

Apply gypsum board in accordance with ASTM C840, System IX or GA 216.

### 3.2.7 Gypsum Board for Wall Tile or Tile Base Applied with Adhesive

In dry areas (areas other than tubs, shower enclosures, saunas, steam rooms, gang shower rooms), apply glass mat water-resistant gypsum tile backing board in accordance with ASTM C840, System X or GA 216.

### 3.2.8 Exterior Application

Apply exterior gypsum board (such as at soffits) in accordance with ASTM C840, System XI or GA 216.

### 3.2.9 Glass Mat Covered or Fiber Reinforced Gypsum Sheathing

Apply glass mat covered or fiber reinforced gypsum sheathing in accordance to gypsum association publications GA 253. Follow gypsum sheathing manufacturer's requirements of design details for joints and fasteners and be properly installed to protect the substrate from moisture intrusion. Do not leave exposed surfaces of the glass mat covered or fiber reinforced gypsum sheathing beyond the manufacturer's recommendation without a weather barrier cladding. Provide continuous asphalt impregnated building felt over sheathing surface in shingle fashion with edges and ends lapped a minimum of 150 mm. Properly flash the openings. Seal all joints, seams, and penetrations with a compatible silicone sealant.

### 3.2.10 Floating Interior Angles

Minimize framing by floating corners with single studs and drywall clips. Locate the attachment fasteners adjacent to ceiling and wall intersections in accordance with ASTM C840, System XII or GA 216, for single-ply and two-ply applications of gypsum board to wood framing.

### 3.2.11 Control Joints

Install expansion and contraction joints in ceilings and walls in accordance with ASTM C840, System XIII or GA 216. Fill control joints between studs in fire-rated construction with firesafing insulation to match the fire-rating of construction.

### 3.2.12 Application of Abuse Resistant Gypsum Board

Apply in accordance with applicable system of ASTM C840 as specified or GA 216. Follow manufacturers written instructions on how to cut, drill and attach board.

## 3.3 APPLICATION OF CEMENTITIOUS BACKER UNITS

### 3.3.1 Application

In wet areas (tubs, shower enclosures, saunas, steam rooms, gang shower rooms), apply cementitious backer units in accordance with ANSI A108.11. Place a 7.6 kg asphalt impregnated, continuous felt paper membrane behind cementitious backer units, between backer units and studs or base layer of gypsum board. Place membrane with a minimum 150 mm overlap of sheets laid shingle style.

### 3.3.2 Joint Treatment

ANSI A108.11.

## 3.4 FINISHING OF GYPSUM BOARD

Tape and finish gypsum board in accordance with ASTM C840, GA 214 and GA 216. Finish plenum areas above ceilings to Level 1 in accordance with GA 214. Finish water resistant gypsum backing board, ASTM C1396/C1396M, to receive ceramic tile to Level 2 in accordance with GA 214. Finish

walls and ceilings to receive a heavy-grade wall covering or heavy textured finish before painting to Level 3 in accordance with GA 214. Finish walls and ceilings without critical lighting to receive flat paints, light textures, or wall coverings to Level 4 in accordance with GA 214. Unless otherwise specified, finish all gypsum board walls, partitions and ceilings to Level 5 in accordance with GA 214. Provide joint, fastener depression, and corner treatment. Tool joints as smoothly as possible to minimize sanding and dust. Do not use self-adhering fiber glass mesh tape with conventional drying type joint compounds; use setting or hardening type compounds only. Provide treatment for water-resistant gypsum board as recommended by the gypsum board manufacturer. Protect workers, building occupants, and HVAC systems from gypsum dust.

#### 3.4.1 Uniform Surface

Wherever gypsum board is to receive eggshell, semigloss or gloss paint finish, or where severe, up or down lighting conditions occur, finish gypsum wall surface in accordance to GA 214 Level 5. In accordance with GA 214 Level 5, apply a thin skim coat of joint compound to the entire gypsum board surface, after the two-coat joint and fastener treatment is complete and dry.

#### 3.5 SEALING

Seal openings around pipes, fixtures, and other items projecting through gypsum board and cementitious backer units as specified in Section 07 92 00 JOINT SEALANTS. Apply material with exposed surface flush with gypsum board or cementitious backer units.

##### 3.5.1 Sealing for Glass Mat or Reinforced Gypsum Board Sheathing

Apply silicone sealant in a 9.5 mm bead to all joints and trowel flat. Apply enough of the same sealant to all fasteners penetrating through the glass mat gypsum board surface to completely cover the penetration when troweled flat. Do not place construction materials behind sheathing until a visual inspection of sealed joints during daylight hours has been completed by Contracting Officer.

#### 3.6 FIRE-RESISTANT ASSEMBLIES

Wherever fire-rated construction is indicated, provide materials and application methods, including types and spacing of fasteners, wall and ceiling framing in accordance with the specifications contained in UL Fire Resistance for the Design Number(s) indicated. Joints of fire-rated gypsum board enclosures must be closed and sealed in accordance with UL test requirements or GA requirements. Seal penetrations through rated partitions and ceilings tight in accordance with tested systems.

#### 3.7 SOUND RATED ASSEMBLIES

When sound rated assemblies are required, provide materials and application methods, including panels, insulation, types and spacing of fasteners, wall and ceiling framing in accordance with the contract document and the description of the assembly in the ASTM E90 Factory Test Report. Seal partitions continuously with acoustical foam or sealant (both sides) and finished to match wall wherever it abuts another element such as the floor, ceiling, wall, column, mullion, or another system or assembly.



### 3.8 PATCHING

Patch surface defects in gypsum board to a smooth, uniform appearance, ready to receive finishes.

### 3.9 SOUND RATED ASSEMBLY FIELD TESTING

Provide third party testing of sound rated assemblies tested in accordance with ASTM E336. Provide the ASTM E336 Field Test Report verifying that the installed assemblies perform no less than five ASTC rating points below the ASTM E90 Factory Test Report. Examine, modify adjust, and retest any installation not meeting the STC Rating until compliance is obtained.

-- End of Section --

## SECTION 09 30 10

CERAMIC, QUARRY, AND GLASS TILING  
08/20

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A108/A118/A136.1 (2024) American National Standard  
Specifications for the Installation of  
Ceramic Tile

ANSI A137.1 (2019) American National Standards  
Specifications for Ceramic Tile

## ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M (2024a) Standard Specification for  
Concrete Aggregates

ASTM C144 (2018) Standard Specification for  
Aggregate for Masonry Mortar

ASTM C150/C150M (2024) Standard Specification for Portland  
Cement

ASTM C373 (2018; R 2023) Standard Test Methods for  
Determination of Water Absorption and  
Associated Properties by Vacuum Method for  
Pressed Ceramic Tiles and Glass Tiles and  
Boil Method for Extruded Ceramic Tiles and  
Non-tile Fired Ceramic Whiteware Products

ASTM C648 (2020) Standard Test Method for Breaking  
Strength of Ceramic Tile

ASTM C1027 (2009; R 2017) Standard Test Method for  
Determining Visible Abrasion Resistance of  
Glazed Ceramic Tile

## CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for  
the Testing and Evaluation of Volatile  
Organic Chemical Emissions from Indoor  
Sources using Environmental Chambers

## GREEN SEAL (GS)

GS-36 (2013) Adhesives for Commercial Use

## SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

## SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

## TILE COUNCIL OF NORTH AMERICA (TCNA)

TCNA Hdbk (2017) Handbook for Ceramic, Glass, and Stone Tile Installation

TTMAC Terrazzo, Tile and Marble Association of Canada

## U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

## UL SOLUTIONS (UL)

UL 2818 (2022) GREENGUARD Certification Program For Chemical Emissions For Building Materials, Finishes And Furnishings

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Detail Drawings; G

## SD-03 Product Data

Porcelain Tile; G

Recycled Content for Porcelain Tile; S

Mosaic Tile; G

Transition Strips; G

Metal Strips; G

Setting-Bed; G

Mortar, Grout, and Adhesive; G

Reinforcing Wire Fabric

Cementitious Backer Units; G

Glass-Mat Gypsum Water-Resistant Backing Board; G

Waterproof Membrane; G  
SD-04 Samples

Tile; G

Accessories; G

Transition Strips; G

Metal Strips; G

Grout; G

SD-07 Certificates

Indoor Air Quality for Adhesives; S

Indoor Air Quality for Sealants; S

Water Absorption Rates

SD-08 Manufacturer's Instructions

Manufacturer's Approved Cleaning Instructions

SD-10 Operation and Maintenance Data  
Porcelain Tile, Data Package 1; G  
Mosaic Tile, Data Package 1; G

Transition Strips, Data Package 1; G

Metal Strips, Data Package 1; G

### 1.3 CERTIFICATIONS

#### 1.3.1 Indoor Air Quality Certifications

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited in this Section.

#### 1.3.2 Water Absorption Rates Certification

Provide certification for each tile type indicating compliance with the following water absorption (wa) rates per ANSI A137.1 criteria as tested per ASTM C373 requirements.

- a. Porcelain and Mosaic Tile (Impervious): Provide water absorption (wa) of 0.5 percent or less.

#### 1.4 QUALITY ASSURANCE

Provide installers having a minimum of two years of experience with a company specializing in performing the type of work described. Each type and color of tile to be provided from a single source. Each type and color of mortar, adhesive, and grout to be provided from the same source.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Ship tiles in sealed packages and clearly marked with the grade, type of tile, producer identification, and country of origin. Deliver materials to the project site in manufacturer's original unopened containers with seals unbroken and labels and hallmarks intact. Protect materials from weather, and store them under cover in accordance with manufacturer's printed instructions. Store and handle tiles per manufacturer's instructions for gauged porcelain tile and gauged porcelain tile panels/slabs.

#### 1.6 ENVIRONMENTAL REQUIREMENTS

Do not perform ceramic tile work unless the substrate and ambient temperature is at least 10 degrees C and rising. Maintain temperature above 10 degrees C while the work is being performed and for at least 7 days after completion of the work. When temporary heaters are used, ventilate the area to the outside to avoid carbon dioxide damage to new tilework.

#### 1.7 WARRANTY

Provide manufacturer's warranty to repair or replace defective tiling materials and workmanship for a period of one year from date of final acceptance of the work.

#### 1.8 EXTRA MATERIALS

Supply an extra 2 percent of each type tile used in clean and marked cartons.

### PART 2 PRODUCTS

#### 2.1 TILE

Provide tiles that comply with ANSI A137.1 and are standard grade tiles. Provide a minimum breaking strength of 57 kg for wall tile and 113 kg for floor tile in accordance with ASTM C648. Provide floor tiles with a minimum wet dynamic coefficient of friction (DCOF) value of 0.42 when tested in accordance with ANSI A137.1 requirements. Provide glazed floor tile with a Class IV-Commercial classification as rated by the manufacturer when tested in accordance with ASTM C1027 for visible abrasion resistance as related to foot traffic. For materials like tile, accessories, and transition strips submit samples of sufficient size to show color range, pattern, type and joints.

Submit manufacturers' descriptive product data for tiling indicated. Include manufacturers' literature, finishes, profiles and thicknesses of materials.

Submit manufacturers' operations and maintenance data for tiling indicated in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

### 2.1.1 Porcelain Tile

Provide through body (surface color and pattern go all the way through the tile body) porcelain tile and cove base. Provide tile with a V2 aesthetic classification. Blend tiles in factory and in a packages to have same color range and continuous blend for installation. Provide nominal tile size(s) of 150 by 150, 300 by 600 mm and 9.5 mm thick.

Provide porcelain tiling materials that contain a minimum of 10 percent recycled content. Provide data identifying percentage of recycled content for porcelain tile.

## ]2.2 SETTING-BED

Submit manufacturer's catalog data. Compose the setting-bed of the following materials:

### 2.2.1 Aggregate for Concrete Fill

Conform to ASTM C33/C33M for aggregate fill. Do not exceed one-half the thickness of concrete fill for maximum size of coarse aggregate.

### 2.2.2 Portland Cement

Conform to ASTM C150/C150M for cement, Type I, white for wall mortar and gray for other uses.

### 2.2.3 Sand

Conform to ASTM C144 for sand.

## 2.3 WATER

Provide potable water.

## 2.4 MORTAR, GROUT, AND ADHESIVE

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesives used on the interior of the building meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of GS-36. For products located on the interior of the building (inside of the weatherproofing system, provide certification or validation of indoor air quality for adhesives. Provide bond coat, mortar, and grout supplied from the same manufacturer.

### 2.4.1 Dry-Set Portland Cement Mortar

TCNA Hdbk and/or TTMAC Standards.

### 2.4.2 Latex-Portland Cement Mortar

TCNA Hdbk and/or TTMAC Standards.

#### 2.4.3 Organic Adhesive

TCNA Hdbk, Type I. Water-resistant. Comply with ANSI A108/A118/A136.1 and TTMAC.

#### 2.4.4 Epoxy Resin Grout

TCNA Hdbk. Water cleanable epoxy conforming to ANSI A108/A118/A136.1 and TTMAC; provide manufacturer proportioned and packaged kit having hardener, resin and colored filler and horizontal and vertical grade products as applicable.

#### 2.4.5 Sealants

Comply with applicable regulations regarding toxic and hazardous materials and as specified. Provide sealant that does not change the color or alter the appearance of the grout. Refer to Section 07 92 00 JOINT SEALANTS.

Provide sealants used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. For products located on the interior of the building (inside of the weatherproofing system), provide certification or validation of indoor air quality for sealants.

### 2.5 SUBSTRATES

Refer to Section 09 29 00 GYPSUM BOARD.

### 2.6 MISCELLANEOUS TRIMS

#### 2.6.1 Transition Strips

Provide clear anodized aluminum transitions between tile and other flooring. Provide types as recommended by flooring manufacturer for both edges and transitions of flooring materials specified. Provide transition strips that comply with 36 CFR 1191 requirements.

#### 2.6.2 Metal Strips

Provide trim shapes, height to match tile and setting thickness, designed specifically for flooring, and wall applications. Provide extruded square, clear anodized aluminum edging at tile surfaces with exposed outside and inside corners. Provide profiles appropriate for finished floor and wall materials as indicated.

### 2.7 WATERPROOF MEMBRANE

#### 2.7.1 General

Manufacturer's standard product that complies with ANSI A108/A118/A136.1 and is recommended by the manufacturer for the application indicated. Include reinforcement and accessories recommended by manufacturer.

#### 2.7.2 Chlorinated-Polyethylene Shower Waterproof Membrane

Nonplasticized, chlorinated polyethylene faced on both sides with nonwoven polyester fabric; 1 mm nominal thickness.

## 2.8 CRACK ISOLATION MEMBRANE

### 2.8.1 General

Manufacturer's standard product that complies with ANSI A108/A118/A136.1 and is recommended by the manufacturer for the application indicated. Include reinforcement and accessories recommended by manufacturer.

### 2.8.2 Chlorinated-Polyethylene Crack Isolation Membrane

Nonplasticized, chlorinated polyethylene faced on both sides with nonwoven polyester fabric; 0.75 mm nominal thickness.

## 2.9 COLOR, TEXTURE, AND PATTERN

Provide color, pattern and texture as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.

## PART 3 EXECUTION

### 3.1 PREPARATORY WORK AND WORKMANSHIP

Inspect surface to receive tile in conformance to the requirements of TTMAC and TCNA Hdbk for surface conditions for the type setting bed specified and for workmanship. Provide variations of tiled surfaces that fall within maximum values shown below:

| TYPE                         | WALLS             | FLOORS            |
|------------------------------|-------------------|-------------------|
| Dry-Set Mortar               | 3 mm in 2.4 meter | 3.0 mm in 3 meter |
| Organic Adhesives            | 3 mm in 2.4 meter | 1.5 mm in 1 meter |
| Latex-Portland Cement Mortar | 3 mm in 2.4 meter | 3.0 mm in 3 meter |
| Epoxy                        | 3 mm in 2.4 meter | 3.0 mm in 3 meter |

### 3.2 GENERAL INSTALLATION REQUIREMENTS

Do not start tile work until roughing in for mechanical and electrical work has been completed and tested, and built-in items requiring membrane waterproofing have been installed and tested. Close space, in which tile is being set, to traffic and other work. Keep closed until tile is firmly set. Do not start floor tile installation in spaces requiring wall tile until after wall tile has been installed. Apply tile in colors and patterns indicated in the area shown on the drawings. Install tile with the respective surfaces in true even planes to the elevations and grades shown. Provide special shapes as required for sills, jambs, recesses, offsets, external corners, and other conditions to provide a complete and neatly finished installation. Solidly back tile bases and coves with mortar. Do not walk or work on newly tiled floors without using kneeling boards or equivalent protection of the tiled surface. Keep traffic off horizontal portland cement mortar installations for at least 72 hours. Keep all traffic off epoxy installed floors for at least 40 hours after grouting, and heavy traffic off for at least 7 days, unless otherwise specifically authorized by manufacturer. Dimension and draw detail



drawings at a minimum scale of 1:50. Include drawings of pattern at inside corners, outside corners, termination points and location of all equipment items such as thermostats, switch plates, mirrors and toilet accessories mounted on surface. Submit drawings showing ceramic tile pattern elevations and floor plans. Submit manufacturer's preprinted installation instructions.

Do not install building construction materials that show visual evidence of biological growth.

### 3.3 INSTALLATION OF SUBSTRATES

#### 3.3.1 Cementitious Backer Units and Glass-Mat Water-Resistant Backing Board

Install as specified in Section 09 29 00 GYPSUM BOARD.

### 3.4 INSTALLATION OF WALL TILE

Install wall tile in accordance with the TCNA Hdbk and TTMAC, method W244C-20 and TTMAC and with grout joints as recommended by the manufacturer for the type of tile. Install thinner wall tile flush with thicker wall tile applied on same wall and provide installation materials as recommended by the tile and setting materials manufacturer's to achieve flush installation.

#### 3.4.1 Epoxy Resin Grout

Prepare and install epoxy resin grout in accordance with TCNA Hdbk method W244C-20 and/or TTMAC Standards.

### 3.5 INSTALLATION OF FLOOR TILE

Install floor tile in accordance with TCNA Hdbk method F205-20 and with grout joints as recommended by the manufacturer for the type of tile and/or TTMAC Standards.

#### 3.5.1 Workable or Cured Mortar Bed

Install floor tile over a workable mortar bed or a cured mortar bed at the option of the Contractor. Conform to TCNA Hdbk method F114-20 for workable mortar bed materials and installation. Conform to TCNA Hdbk method F114-20 for cured mortar bed materials and installation. Provide minimum 6 mm to maximum 10 mm joints in uniform width.

#### 3.5.2 Dry-Set and Latex-Portland Cement

Use latex-portland cement mortar to install tile directly over properly cured, plane, clean concrete slabs in accordance with TCNA Hdbk method F205-20. Use latex-portland cement when installing porcelain ceramic tile.

#### 3.5.3 Waterproof and Crack Isolation Membranes

Install as indicated in accordance with manufacturer's written instructions.

### 3.6 INSTALLATION OF MISCELLANEOUS TRIMS

#### 3.6.1 Transition Strips

Install transition strips where indicated, in a manner similar to that of the ceramic tile floor and as recommended by the manufacturer. Provide thresholds full width of the opening. Install head joints at ends not exceeding 6 mm in width and grouted full.

#### 3.6.2 Metal Trims

Install trim where indicated. Embed anchoring leg in setting mortar in accordance with manufacturer's instructions. During grouting of tile joints, immediately wipe grout from finish surface.

### 3.7 EXPANSION JOINTS

Form and seal joints as specified in Section 07 92 00 JOINT SEALANTS.

#### 3.7.1 Walls

Provide expansion joints at control joints in backing material. Wherever backing material changes, install an expansion joint to separate the different materials.

#### 3.7.2 Floors

Provide expansion joints over construction joints, control joints, and expansion joints in concrete slabs in accordance with TCNA Hdbk method EJ171 type and/or TTMACStandards to suit conditions. Provide expansion joints where tile abuts restraining surfaces such as perimeter walls, curbs and columns and at intervals of 6.1 to 7.6 m each way in large interior floor areas. Extend expansion joints through setting-beds and fill.

### 3.8 CLEANING AND PROTECTING

Upon completion, thoroughly clean tile surfaces in accordance with manufacturer's approved cleaning instructions. Do not use acid for cleaning glazed tile. Clean floor tile with resinous grout or with factory mixed grout in accordance with printed instructions of the grout manufacturer. After the grout has set, provide a protective coat of a noncorrosive soap or other approved method of protection for tile wall surfaces. Cover tiled floor areas with building paper before foot traffic is permitted over the finished tile floors. Provide board walkways on tiled floors that are to be continuously used as passageways by workmen. Replace damaged or defective tiles.

-- End of Section --

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## SECTION 09 51 00

## ACOUSTICAL CEILINGS

08/20

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A489         | (2018; E 2018) Standard Specification for Carbon Steel Eyebolts                                                                                              |
| ASTM A580/A580M   | (2023) Standard Specification for Stainless Steel Wire                                                                                                       |
| ASTM C423         | (2023; E 2024) Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method                                                           |
| ASTM C635/C635M   | (2022) Standard Specification for Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-In Panel Ceilings            |
| ASTM C636/C636M   | (2013) Standard Practice for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels                                          |
| ASTM C834         | (2017; R 2023) Standard Specification for Latex Sealants                                                                                                     |
| ASTM E413         | (2022) Classification for Rating Sound Insulation                                                                                                            |
| ASTM E580/E580M   | (2024a) Standard Practice for Installation of Ceiling Suspension Systems for Acoustical Tile and Lay-in Panels in Areas Subject to Earthquake Ground Motions |
| ASTM E795         | (2023) Standard Practices for Mounting Test Specimens During Sound Absorption Tests                                                                          |
| ASTM E1111/E1111M | (2014; R 2022) Standard Test Method for Measuring the Interzone Attenuation of Open Office Components                                                        |
| ASTM E1264        | (2023) Standard Classification for Acoustical Ceiling Products                                                                                               |
| ASTM E1414/E1414M | (2021a) Standard Test Method for Airborne Sound Attenuation Between Rooms Sharing a                                                                          |

## Common Ceiling Plenum

ASTM E1477 (1998; R 2022a) Standard Test Method for Luminous Reflectance Factor of Acoustical Materials by Use of Integrating-Sphere Reflectometers

## CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers

## GREEN SEAL (GS)

GS-36 (2013) Adhesives for Commercial Use

## SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

## U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 3, 2025) Structural Engineering

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Drawings;

## SD-03 Product Data

Acoustical Ceiling Systems;

Fire Resistive Ceilings;

Recycled Content for Type III Ceiling Tiles; S

Recycled Content for Suspension Systems; S

Acoustical Performance;

## SD-04 Samples

Acoustical Units;

Acoustical Ceiling Tiles;

## SD-06 Test Reports

## SD-07 Certificates

Indoor Air Quality for Type III Ceiling Tiles; S

Indoor Air Quality for Adhesives; S

Indoor Air Quality for Sealants; S

### 1.3 DELIVERY, STORAGE. AND HANDLING

Deliver materials to the site in the manufacturer's original unopened containers with brand name and type clearly marked. Carefully handle and store materials in dry, watertight enclosures. Immediately before installation, store acoustical units for not less than 24 hours at the same temperature and relative humidity as the space where they will be installed in order to assure proper temperature and moisture acclimation.

### 1.4 ENVIRONMENTAL REQUIREMENTS

Maintain a uniform temperature of not less than 16 degrees C nor more than 29 degrees C and a relative humidity of not more than 70 percent for 24 hours before, during, and 24 hours after installation of acoustical units.

### 1.5 SCHEDULING

Complete and dry interior finish work such as plastering, concrete and terrazzo work before ceiling installation. Complete mechanical, electrical, and other work above the ceiling line; install and start operating heating, ventilating, and air conditioning systems in order to maintain temperature and humidity requirements.

### 1.6 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials and workmanship including but not limited to, sagging and warping of panels and rusting and of grid systems, for a period of ten years from date of final acceptance of the work.

### 1.7 EXTRA MATERIALS

Furnish spare tiles, from the same lot as those installed, of each color at the rate of 5 tiles for each 1000 tiles installed.

## ]PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

Provide sound controlling units mechanically mounted on a ceiling suspension system for acoustical treatment. Provide the unit size, texture, finish, and color as specified. The Contractor has the option to substitute inch-pound (I-P) Recessed Light Fixtures (RLF) for metric RLF. If the Contractor opts to provide I-P RLF, then provide I-P products for other ceiling elements like acoustical ceiling tiles, air diffusers, air registers and grills. Coordinate the entire ceiling system with other details, like the location of access panels and ceiling penetrations, for instance, shown on the drawings. Submit drawings showing the location, extent and details of acoustical treatment including suspension system, method of anchoring and fastening, and reflected ceiling plan. Coordinate with paragraph RECLAMATION PROCEDURES for reclamation of mineral fiber acoustical ceiling panels to be removed from the job site.

## 2.1.1 Acoustical Performance

### 2.1.1.1 Ceiling Sound Transmission

Provide ceiling systems with the specified Ceiling Attenuation Class (CAC) ratings as determined in accordance with ASTM E1414/E1414M and ASTM E413. Provide sound attenuators over light fixtures, air terminals and other ceiling penetrations, provide acoustical blanket insulation on top of the ceiling or adjacent to partitions to provide lightweight acoustical plenum barriers above partitions as required to achieve the specified CAC ratings. Provide test ceiling continuous at the partition and assembled in the suspension system in the same manner that the ceiling will be installed on the project.

### 2.1.1.2 Ceiling Sound Absorption

Determine the Noise Reduction Coefficient (NRC) in accordance with ASTM C423. Determine Articulation Class (AC) in accordance with ASTM E1111/E1111M.

## 2.1.2 Light Reflectance

Determine light reflectance factor in accordance with ASTM E1477 test method.

## 2.2 ACOUSTICAL UNITS

Submit samples of each type of acoustical unit and each type of suspension grid tee section showing texture, finish, and color. Conform acoustical units to ASTM E1264, Class A, and the following requirements:

### 2.2.1 Units for Exposed-Grid System ACT1

#### 2.2.1.1 Type ACT1

III (non-asbestos mineral fiber with painted finish). Provide Type III Acoustical Ceiling Tiles containing a minimum of 30 percent recycled content. Provide data identifying percentage of recycled content for Type III ceiling tiles. Provide certification of indoor air quality for Type III Ceiling Tiles.

#### 2.2.1.2 Flame Spread

Class A, 25 or less

#### 2.2.1.3 Pattern

G

#### 2.2.1.4 Minimum NRC

0.90 when tested on mounting Type E-400 of ASTM E795.

#### 2.2.1.5 Minimum Light Reflectance Coefficient

0.85

#### 2.2.1.6 Nominal Size

600 by 600 mm

## 2.2.1.7 Edge Detail

Tegular

## 2.2.1.8 Finish

Factory-applied standard finish. See paragraph COLORS AND STANDARDS.

## 2.2.1.9 Minimum CAC

35

## 2.2.2 Units for Exposed-Grid System ACT22.2.2.1 Type ACT2

III (non-asbestos mineral fiber with painted finish). Provide data identifying percentage of recycled content for Type III ceiling tiles. Provide certification of indoor air quality for Type III Ceiling Tiles.

## 2.2.2.2 Flame Spread

Class A, 25 or less

## 2.2.2.3 Pattern

G

## 2.2.2.4 Minimum NRC

0.85 when tested on Mounting Type E-400 of ASTM E795.

## 2.2.2.5 Minimum Light Reflectance Coefficient

0.85

## 2.2.2.6 Nominal Size

600 by 600 mm

## 2.2.2.7 Edge Detail

Square

## 2.2.2.8 Finish

Factory-applied standard finish. See paragraph COLORS AND PATTERNS.

## 2.2.2.9 Minimum CAC

35

## 2.2.3 Unit Acoustical Absorbers

Provide individually mounted sound absorbing plaques composed of glass fibers or non-asbestos mineral fibers and having a NRC range of not less than 0.60 - 0.70 when tested in accordance with ASTM C423 and reported as a 4 frequency average.



### 2.3 SUSPENSION SYSTEM ACG1

Provide standard width flange suspension system conforming to ASTM C635/C635M for heavy-duty systems. Provide surfaces exposed to view of aluminum or steel with a factory-applied white baked-enamel finish. Provide wall molding having a flange of not less than 24 mm. Provide standard corners. Provide a suspension system with a maximum deflection of 1/360 of the span length capable of supporting the finished ceiling, light fixtures, air diffusers, and accessories, as shown. Conform seismic details to the guidance in UFC 3-301-01 and ASTM E580/E580M.

Provide Suspension System containing a minimum of 15 percent recycled content. Provide data identifying percentage of recycled content for suspension systems.

### 2.4 SUSPENSION SYSTEM ACG2

Provide structural suspension ceiling system conforming to ASTM C635/C635M for heavy-duty systems to directly support cable trays, utilities, light fixtures, HVAC registers and other accessories. Capable of a maximum static point load of 544 kg. Provide surfaces exposed to view of aluminum or steel with a factory-applied white baked-enamel finish. Provide wall molding having a flange of not less than 24 mm. Provide standard corners. Provide a suspension system with a maximum deflection of 1/360 of the span length capable of supporting the finished ceiling, light fixtures, air diffusers, and accessories, as shown. Conform seismic details to the guidance in UFC 3-301-01 and ASTM E580/E580M.

Provide Suspension System containing a minimum of 15 percent recycled content. Provide data identifying percentage of recycled content for suspension systems.

### 2.5 HANGERS

Provide hangers and attachment capable of supporting a minimum 1334 N ultimate vertical load without failure of supporting material or attachment for ACG1. Provide hangers and attachments including 9.5 mm threaded rod from structure for ACG2.

#### 2.5.1 Wires

Conform wires to ASTM A580/A580M, composition 302 or 304, condition annealed stainless steel, 2.0 mm in diameter.

#### 2.5.2 Rods

Provide 5 mm diameter threaded steel rods, zinc or cadmium coated for ACG2.

#### 2.5.3 Eyebolts

Provide eyebolts of weldless, forged-carbon-steel, with a straight-shank in accordance with ASTM A489. Provide minimum 6 mm, zinc coated eyebolts.

#### 2.5.4 Masonry Anchorage Devices

Comply with ASTM C636/C636M for anchorage devices for eyebolts.

## 2.6 ACCESS PANELS

Provide access panels that match adjacent acoustical units, designed and equipped with suitable framing and fastenings for removal and replacement without damage. Size panel to be not less than 300 by 300 mm or more than 300 by 600 mm.

- a. Attach an identification plate of 0.8 mm thick aluminum, 19 mm in diameter, stamped with the letters "AP" and finished the same as the unit, near one corner on the face of each access panel.
- b. Identify ceiling access panel by a number utilizing white identification plates or plastic buttons with contrasting numerals. Provide plates or buttons of minimum 25 mm diameter and securely attached to one corner of each access unit. Provide a typewritten card framed under glass listing the code identification numbers and corresponding system descriptions listed above. Mount the framed card where directed and furnish a duplicate card to the Contracting Officer. Code identification system is as follows:

- (1) Fire detection/alarm system
- (2) Air conditioning controls
- (3) Plumbing system
- (4) Heating and steam systems
- (5) Air conditioning duct system
- (6) Sprinkler system
- (7) Intercommunication system

## 2.7 ADHESIVE

Use adhesive as recommended by tile manufacturer. Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of GS-36. For products located on the interior of the building (inside of the weatherproofing system), provide certification or validation of indoor air quality for adhesives.

## 2.8 FINISHES

Use manufacturer's standard textures, patterns and finishes as specified for acoustical units and suspension system members. Treat ceiling suspension system components to inhibit corrosion.

## 2.9 COLORS AND PATTERNS

Use colors and patterns for acoustical units and suspension system

components as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.

## 2.10 ACOUSTICAL SEALANT

Conform acoustical sealant to ASTM C834, nonstaining. Provide sealants used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit the requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. For products located on the interior of the building (inside of the weatherproofing system), provide certification of indoor air quality for Sealants.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Examine surfaces to receive directly attached acoustical units for unevenness, irregularities, and dampness that would affect quality and execution of the work. Rid areas, where acoustical units will be cemented, of oils, form residue, or other materials that reduce bonding capabilities of the adhesive. Complete and dry interior finish work such as plastering, concrete, and terrazzo work before installation. Complete and approve mechanical, electrical, and other work above the ceiling line prior to the start of acoustical ceiling installation. Provide acoustical work complete with necessary fastenings, clips, and other accessories required for a complete installation. Do not expose mechanical fastenings in the finished work. Lay out hangers for each individual room or space. Provide hangers to support framing around beams, ducts, columns, grilles, and other penetrations through ceilings. Keep main runners and carrying channels clear of abutting walls and partitions. Provide at least two main runners for each ceiling span. Wherever required to bypass an object with the hanger wires, install a subsuspension system so that all hanger wires will be plumb.

#### 3.1.1 Suspension System

Install suspension system in accordance with ASTM C636/C636M and as specified herein. Do not suspend hanger wires or other loads from underside of steel decking.

##### 3.1.1.1 Plumb Hangers

Install hangers plumb and not pressing against insulation covering ducts and pipes. Where lighting fixtures are supported from the suspended ceiling system, provide hangers at a minimum of four hangers per fixture and located not more than 152 mm from each corner of each fixture.

##### 3.1.1.2 Splayed Hangers

Splay (slope or slant) hangers around obstructions, offsetting the resulting horizontal force by bracing, countersplaying, or other acceptable means.

### 3.1.2 Wall Molding

Provide wall molding where ceilings abut vertical surfaces. Miter corners where wall moldings intersect or install corner caps. Secure wall molding not more than 76 mm from ends of each length and not more than 406 mm on centers between end fastenings. Provide wall molding springs at each acoustical unit in semi-exposed or concealed systems.

### 3.1.3 Acoustical Units

Install acoustical units in accordance with the approved installation instructions of the manufacturer. Ensure that edges of acoustical units are in close contact with metal supports, with each other, and in true alignment. Arrange acoustical units so that units less than one-half width are minimized. Hold units in exposed-grid system in place with manufacturer's standard hold-down clips, if units weigh less than 4.9 kg/square meter or if required for fire resistance rating.

### 3.1.4 Acoustical Sealant

Seal all joints around pipes, ducts or electrical outlets penetrating the ceiling. Apply a continuous ribbon of acoustical sealant on vertical web of wall or edge moldings.

## 3.2 CEILING ACCESS PANELS

Locate ceiling access panels directly under the items which require access.

## 3.3 CLEANING

Following installation, clean dirty or discolored surfaces of acoustical units and leave them free from defects. Remove units that are damaged or improperly installed and provide new units as directed.

-- End of Section --

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## SECTION 09 62 38

## STATIC-CONTROL FLOORING

**08/17, CHG 1: 08/18**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|            |                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------|
| ASTM E648  | (2023) Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source           |
| ASTM F150  | (2006; R 2013) Standard Test Method for Electrical Resistance of Conductive and Static Dissipative Resilient Flooring        |
| ASTM F1700 | (2020) Standard Specification for Solid Vinyl Floor Tile                                                                     |
| ASTM F1861 | (2021) Standard Specification for Resilient Wall Base                                                                        |
| ASTM F1869 | (2023) Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride |
| ASTM F2170 | (2019a) Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes                  |

## CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

## ELECTROSTATIC DISCHARGE ASSOCIATION (ESD)

|          |                                                                                             |
|----------|---------------------------------------------------------------------------------------------|
| ESD S6.1 | (2019) Standard for the Protection of Electrostatic Discharge Susceptible Items - Grounding |
|----------|---------------------------------------------------------------------------------------------|

## GREEN SEAL (GS)

|       |                                     |
|-------|-------------------------------------|
| GS-36 | (2013) Adhesives for Commercial Use |
|-------|-------------------------------------|

## RESILIENT FLOOR COVERING INSTITUTE (RFCI)

|            |                              |
|------------|------------------------------|
| FLOORSCORE | FLOORSCORE IAQ Certification |
|------------|------------------------------|

## SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

## SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

## UL SOLUTIONS (UL)

UL 2818 (2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

## 1.2 SCHEDULING

Schedule static-control flooring work after any other work which would damage the finished surface of the flooring.

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-03 Product Data

Static-Control Resilient Flooring; G

Recycled content for Static-Dissipative Vinyl Tile; S  
Accessories; G

Adhesives; G

Warranty

## SD-04 Samples

Moldings; G  
Accessories; G

## SD-06 Test Reports

Fire Resistance

Moisture, Alkalinity and Bond

Testing

## SD-07 Certificates

Indoor Air Quality for Static-Dissipative Vinyl Tile; S]

Indoor Air Quality for Adhesives; S

Qualifications of Applicator

## SD-08 Manufacturer's Instructions

Static-Control Resilient Flooring; G

Accessories; G

## SD-10 Operation and Maintenance Data

Static-Control Resilient Flooring; G

Accessories; G

## 1.3.1 Samples

## 1.3.1.1 Static-Control Resilient Flooring

Submit three samples of each indicated color and type of flooring, base, moldings, and accessories sized a minimum 60 by 100 mm.

## 1.3.1.2 Moldings

Submit three pieces of each type at least 300 mm long.

## 1.3.1.3 Operations and Maintenance Data

- a. Submit Data Package 1 in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.
- b. Submit three copies of manufacturer's maintenance instructions for each type of flooring material describing recommended type of cleaning equipment and materials, spotting and cleaning methods, and cleaning cycles.

## 1.4 CERTIFICATIONS

## 1.4.1 Indoor Air Quality Certifications

## 1.4.1.1 Floor Covering Materials

Provide Static-Dissipative Vinyl Tile and wall base products certified to meet indoor air quality requirements by FLOORSCORE, UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body.

## 1.4.1.2 Adhesives

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body.

## 1.5 EXTRA MATERIALS

Provide extra material from same dye lot for future maintenance. Provide



a minimum of 2 percent of total square meters of each flooring and base type, pattern, and color.

#### 1.6 QUALITY ASSURANCE

The flooring manufacturer will approve the Qualifications of Applicator and certify that he/she has a minimum of 3 years of experience in the application of the materials to be used.

#### 1.7 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the building site in original unopened containers bearing the manufacturer's name, style name, pattern color name and number, size, production run, project identification, handling instructions and related information. Observe ventilation and safety procedures specified in the Safety Data Sheets (SDS). Do not store flooring near materials that may off-gas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives.

##### 1.7.1 Static-Control Resilient Flooring

Store materials in a clean, dry, secure, and well-ventilated area free from strong contaminant sources and residues with ambient air temperature range as recommended by the manufacturer but not less than 20 degrees C or more than 30 degrees C. Stack materials according to manufacturer's recommendations. Protect materials from the direct flow of heat from hot-air registers, radiators and other heating fixtures and appliances.

#### 1.8 ENVIRONMENTAL CONDITIONS

Provide temporary ventilation during work of this section.

##### 1.8.1 Static-Control Resilient Flooring

Maintain areas in which resilient flooring is to be installed at a temperature range as recommended by the manufacturer but not less than 20 degrees C or more than 30 degrees C for 3 days before application, during application and 2 days after application, unless otherwise directed by the flooring manufacturer for the flooring being installed. Maintain a minimum temperature range as recommended by the manufacturer but not less than 13 degrees C thereafter for the duration of the contract. Provide adequate ventilation to remove moisture from area and to comply with regulations limiting concentrations of hazardous vapors.

#### 1.9 WARRANTY

##### 1.9.1 Static-Control Resilient Flooring

Provide manufacturer's standard performance guarantees or warranties including a five year wear warranty and ten year conductivity warranty.

## PART 2 PRODUCTS

### 2.1 STATIC-CONTROL RESILIENT FLOORING

#### 2.1.1 Conductive Resilient Flooring

#### 2.1.2 Static-Dissipative Resilient Flooring

##### 2.1.2.1 Static-Dissipative Vinyl Tile

Static-dissipative vinyl tile must be a homogeneous vinyl product and conform to ASTM F1700. Provide electrical resistance from surface to surface and surface to ground between 1,000,000 ohms ( $1.0 \times 10$  to the 6th) and 1,000,000,000 ohms ( $1.0 \times 10$  to the 9th) when tested in accordance with ASTM F150. Tile must be 600 mm square and 3.2 mm thick.

Provide Static-Dissipative Vinyl Tile containing a minimum of 10 percent recycled content. Provide data identifying percentage of recycled content for Static-Dissipative Vinyl Tile.

### 2.2 WALL BASE

#### 2.2.1 Resilient Base

Resilient base must conform to ASTM F1861, Type TS (vulcanized thermoset rubber), Style B (coved - installed with resilient flooring). Provide 150 mm high and a minimum 3 mm thick wall base. Provide job formed corners in matching height, shape, and color.

### 2.3 ADHESIVES

Provide conductive adhesive as recommended by the manufacturer of the static-control flooring. Provide adhesive for wall base as recommended by the wall base manufacturer.

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for adhesives.

### 2.4 MOLDINGS

Provide heavy duty tapered moldings of clear anodized aluminum and types as recommended by flooring manufacturer for both edges and transitions of flooring materials specified. Provide vertical lip on molding of maximum 6 mm. Provide bevel change in level between 6 and 13 mm with a slope no greater than 1:2. .

### 2.5 ACCESSORIES

Use accessories recommended by the manufacturer of the flooring.

### 2.6 ELECTRICAL GROUND CONNECTION

Provide an electrical ground connection that meets the requirements of

ESD S6.1. Connection between the static-control floor system and the external grounding system must be provided. Contact with the static-control floor system must be with conductive grounding strip and must have the greater of the following: a minimum contact area of 5800 square mm or the dimensions recommended by the manufacturer. Provide the grounding conductor recommended by the manufacturer of the flooring. Connect and install the grounding conductor as recommend by the flooring manufacturer.

## 2.7 MANUFACTURER'S COLOR, PATTERN AND TEXTURE

Provide color, pattern and texture as indicated. Provide flooring in any one continuous area or replacement of damaged flooring in continuous area from same production run with same shade and pattern.

## 2.8 FIRE RESISTANCE TESTING REQUIREMENTS

Provide a minimum average critical radiant flux of 0.22 watts per square centimeter for flooring in corridors and exits when tested in accordance with ASTM E648.

# PART 3 EXECUTION

## 3.1 SURFACE PREPARATION

Before any work under this section is begun, defects such as rough or scaling concrete, low spots, high spots, and uneven surfaces must be corrected, and damaged portions of concrete slabs must be repaired in accordance with flooring manufacturer's recommended instructions. Floor must be in a level plane with a maximum variation of 3 mm every 3 m, except where indicated as sloped. Repair cracks and irregularities and prepare the subfloor in accordance with flooring manufacturer's recommended instructions. Curing and sealing compounds should not be used on concrete surfaces to receive flooring unless they have been tested and approved by the flooring manufacturer. In addition, remove paint, varnish, oils, release agents, sealers, waxes, and adhesives, as required by the flooring product in accordance with manufacturer's printed installation instructions. If a curing compound is required, it must be coordinated for compatibility with the flooring adhesive.

## 3.2 MOISTURE, ALKALINITY AND BOND TESTS

Determine the suitability of the concrete subfloor for receiving the flooring with regard to moisture content and pH level by moisture and alkalinity tests. Conduct moisture testing in accordance with ASTM F1869 or ASTM F2170, unless otherwise recommended by the flooring manufacturer. Conduct alkalinity testing as recommended by the flooring manufacturer. Determine the compatibility of the flooring adhesives to the concrete floors by a bond test in accordance with the flooring manufacturer's recommendations.

## 3.3 GENERAL INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

## 3.4 INSTALLATION OF STATIC-CONTROL RESILIENT TILE FLOORING

Install static-control resilient flooring, ground connections and

accessories in accordance with the approved manufacturer's installation instructions. Tile lines and joints must be kept square, symmetrical, tight, and even. Tile at the perimeter of the area to be finished may vary as necessary to maintain full-size tiles in the field, but no perimeter tile may be less than one-half the field tile size, except where irregular shaped rooms make it impossible. Tile must be cut, fitted, and scribed to walls, partitions, and projections after field flooring has been applied. Install grounding strips in accordance with manufacturer's installation instructions. Protect edges of flooring material meeting hard surface flooring with molding and install in accordance with the molding manufacturer's printed instructions.

### 3.5 INSTALLATION OF WALL BASE

#### 3.5.1 Resilient Base

Install wall base in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions. Tighten base joints and make even with adjacent resilient flooring. Fill voids along the top edge of base at masonry walls with caulk. Roll entire vertical surface of base with hand roller, and press toe of base with a straight piece of wood to ensure proper alignment. Avoid excess adhesive in corners.

### 3.6 CLEANING AND PROTECTION

The flooring must be cleaned in accordance with the manufacturer's recommendations. Flooring must be protected by a covering of heavy-duty building paper before foot traffic is permitted. Lap and secure edges of kraft paper protection to provide a continuous cover. Boardwalks must be placed over flooring in areas where subsequent building operations might damage the floor. Remove and replace flooring that becomes loose, broken, or curled prior to acceptance, or flooring that does not conform to resistance requirements of ASTM F150.

### 3.7 TESTING

Test the flooring in accordance with and conform to the requirements of ESD S6.1.

-- End of Section --

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## SECTION 09 67 23.15

FUEL RESISTIVE RESINOUS FLOORING, 3-COAT SYSTEM  
02/21

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

|            |                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------|
| ASTM C679  | (2015; R 2022) Standard Test Method for Tack-Free Time of Elastomeric Sealants                    |
| ASTM D412  | (2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension |
| ASTM D638  | (2014) Standard Test Method for Tensile Properties of Plastics                                    |
| ASTM D1308 | (2002; R 2013) Effect of Household Chemicals on Clear and Pigmented Organic Finishes              |
| ASTM D2240 | (2015; R 2021) Standard Test Method for Rubber Property - Durometer Hardness                      |
| ASTM D3925 | (2002; R 2015) Sampling Liquid Paints and Related Pigmented Coatings                              |
| ASTM D4541 | (2022) Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers     |
| ASTM D6237 | (2019) Standard Guide for Painting Inspectors (Concrete and Masonry Substrates)                   |
| ASTM E11   | (2024) Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves                     |

## INTERNATIONAL CONCRETE REPAIR INSTITUTE (ICRI)

|             |                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------|
| ICRI 310.2R | (2013) Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, Polymer Overlays, and Concrete Repair |
|-------------|---------------------------------------------------------------------------------------------------------------------------|

## INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

|          |                                                |
|----------|------------------------------------------------|
| ISO 9001 | (2015) Quality Management Systems-Requirements |
|----------|------------------------------------------------|

## MASTER PAINTERS INSTITUTE (MPI)

|         |                                                                              |
|---------|------------------------------------------------------------------------------|
| MPI 211 | (2018) Floor Coating, Primer, Thin Film, for Aircraft Maintenance Facilities |
| MPI 212 | (2018) Floor Coating, Thin Film, for Aircraft Maintenance Facilities         |

## SOCIETY FOR PROTECTIVE COATINGS (SSPC)

|                      |                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SSPC QP 5            | (2022) Standard Procedure for Evaluating the Qualifications of Coating and Lining Inspection Companies                                                                                             |
| SSPC QP 8            | (2015) Standard Procedure for Evaluating the Qualifications of Contracting Firms that Install Polymer Coatings, Surfacing, Linings or FRP Composites on Concrete and Other Cementitious Substrates |
| SSPC QS 1            | (2015) Standard Procedure for Evaluating a Contractor's Advanced Quality Management System                                                                                                         |
| SSPC-TU 2/NACE 6G197 | (1997) Design, Installation, and Maintenance of Coating Systems for Concrete Used in Secondary Containment                                                                                         |

## U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|                  |                        |
|------------------|------------------------|
| 29 CFR 1910.134  | Respiratory Protection |
| 29 CFR 1910.1000 | Air Contaminants       |
| 29 CFR 1926.59   | Hazard Communication   |

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-03 Product Data

Joint Sealant; G

Thin Film Flooring System; G

White Aluminum Oxide Non-Skid Grit; G

## SD-05, Design Data

Environmental Control System

## SD-06 Test Reports

Joint Sealant Test Report; G

Primer Coat; G

Urethane Topcoat; G

White Aluminum Oxide Non-Skid Grit; G

Patch Test Demonstration; G

Daily Inspection Report; G

Adhesion Testing; G

#### SD-07 Certificates

Coating Work Plan; G

Joint Sealant Certificates; G

Thin Film Flooring System Certificates; G

Qualifications of Certified Industrial Hygienist (CIH)

Qualifications of Certified Protective Coatings Specialist (PCS)

Qualifications of Coating Inspection Company

Qualifications of QC Specialist Coating Inspector

Qualifications of Coating Contractors; G

Warranty; G

#### SD-08 Manufacturer's Instructions

Joint Sealant Manufacturer's Instructions; G

Thin Film Flooring System Manufacturer's Instructions; G

Water-Based Alkaline Degreaser; G

#### SD-11 Closeout Submittals

Inspection Logbook; G

### 1.3 QUALITY ASSURANCE

#### 1.3.1 Qualifications of Certified Protective Coatings Specialist (PCS)

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party PCS. Submit documentation that the specialist is certified by SSPC: The Society for Protective Coatings (SSPC) as a PCS, including certification number and date of certification/recertification. If the PCS is employed by the same coating inspection company to which the coating inspector is employed, this does not violate the independent third-party requirements. The PCS must remain certified during the entire project, and the Contracting Officer must be notified of any change in certification status within 10 days of the change. The PCS must not be the designated coating inspector.



### 1.3.2 Coating Work Plan

- a. Include Coating Work Plan in Quality Control Plan.
- b. Provide procedures for reviewing Contract Documents immediately after award to identify errors, omissions, and discrepancies so that any such issues can be resolved prior to project planning and development of detailed procedures.
- c. Provide procedures for verification of key processes during Initial Phase to ensure that Contract requirements can be met. Key processes must include surface preparation, coating application and curing, inspection, and documentation, and any other process that might adversely impact orderly progression of work.
- d. Provide procedures for all phases of coating operations, including planned work, rework, repair, inspection, and documentation. Address mobilization and setup, surface preparation, coating application, coating initial cure, tracking and correction of non-compliant work, and demobilization. Coordinate work processes with health and safety plans and confined space entry plans. For each process, provide procedures that include appropriate work instructions, material and equipment requirements, personnel qualifications, controls, and process verification procedures. Provide procedures for inspecting work to verify and document compliance with Contract requirements, including inspection forms and checklists, and acceptance and rejection criteria.
- e. Provide procedures for determining the existing surface profile under paint, and procedures for ensuring that the profile is not increased beyond the maximum profile specified herein.
- f. Provide procedures for correcting non-compliant work. Detailed procedures are required in advance to avoid delays in meeting overcoat windows as well as to avoid delays in production. Provide procedures for repairing defects in the coating film, such as runs, drips, sags, holidays, overspray, as well as how to correct coating thickness non-compliance, any other areas of repair or rework that might be adversely affected by delays in preparing and approving new procedures.
- g. If a procedure is based on a proposed or approved request for deviation, the deviation must be referenced. Changes to procedures must be noted by submittal number and date approved, clearly delineating old requirements and new requirements, so that the records provide a continuous log of requirements and procedures.

### 1.3.3 Design Data

#### 1.3.3.1 Environmental Control System

Submit design details of the proposed environmental control system to include ventilation, humidity control, and temperature regulation. Provide calculations for humidity control during separate surface preparation and coating application procedures, ventilation requirements during coating application, and maximum allowable coating application rates to coincide with ventilation. Include basis of design data on local conditions. Provide equipment layout sketches and procedures showing function of each piece of equipment and fail-safe measures. A Certified Industrial Hygienist must approve calculations, work procedures and

personal protective equipment.

#### 1.3.4 Test Reports

##### 1.3.4.1 Joint Sealant Test Report

Submit test results that confirm sealant complies with the requirements of Table Ia. Samples must have been tested within the last three years.

##### 1.3.4.2 Daily Inspection Report

Submit one copy of the daily inspection report to the Contracting Officer within 24 hours of the date recorded.

#### 1.3.5 Certificates

##### 1.3.5.1 Qualifications of Certified Industrial Hygienist (CIH)

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party CIH. Submit documentation that hygienist is certified by the American Board of Industrial Hygiene in comprehensive practice, including certification number and date of certification/recertification. Provide evidence of experience with hazards involved in industrial coating application work.

##### 1.3.5.2 Qualifications of Coating Inspection Company

Submit documentation that the coating inspection company performing all coating inspection functions is certified by SSPC to the requirements of SSPC QP 5 prior to Contract award. The coating inspection company submitted and approved must remain and not changed through completion of the Contract. The coating inspection company must remain so certified for the duration of the project. If a coating inspection company's certification expires, the firm will not be allowed to perform any inspection functions, and all surface preparation and coating application work must stop, until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered. Notify the Contracting Officer of any change in coating inspection company certification status. Notify the Contracting Officer of all scheduled and unannounced on-site inspections from SSPC and furnish a copy of all inspection reports.

##### 1.3.5.3 Qualifications of QC Specialist Coating Inspector

Submit documentation that each coating inspector is employed, and qualified to SSPC QP 5, Level II, by the selected coating inspection company. Each inspector must remain employed by the coating inspection company while performing any coating inspection functions. In addition to the handwritten records, the inspector must employ an electronic reporting program with functionality as outlined in Table II. The Administrator must be the designated Government Representative for the project.

##### 1.3.5.4 Qualifications of Coating Contractors

All Contractors that perform surface preparation or coating application must be certified to SSPC QP 8 and should also be SSPC QS 1 certified prior to Contract award and must remain certified while accomplishing any surface preparation or coating application. The painting Contractors must remain so certified for the duration of the project. If a Contractor's

certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered. Notify the Contracting Officer of any change in Contractor certification status. Notify the Contracting Officer of all scheduled and unannounced on-site audits from SSPC and furnish a copy of all audit reports.

For OCONUS, non-US territories where documentation is provided that certified SSPC QP 8 with or without SSPC QS 1 Contractors did not bid and are not available, all Contractors that perform surface preparation or coating application must be certified to ISO 9001 prior to Contract award, and must remain certified while accomplishing any surface preparation or coating application. If a Contractor's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered. Notify the Contracting Officer of any change in Contractor certification status. Notify the Contracting Officer of all scheduled and unannounced on-site inspections from the ISO certifying organization and furnish a copy of all inspection reports.

Minimum requirements for the installation Contractor are as follows: Completed three or more jobs within the past two years applying the specified materials to concrete surfaces in which the total area exceeds 18,587 square meters. Submit documentation listing location of work, point of contact at job site, total square footage of applied materials, listing of both materials and equipment used, and validation from coating manufacturer documenting quality of materials purchased per job for work totaling 18,587 square meters within the past two years. In addition to the above requirements, be certified by the material manufacturer(s) to install the submitted coatings and sealant. Submit copy of certificates.

#### 1.3.5.5 Joint Sealant Certificates

Submit literature documenting the past performance of the sealant's use in automotive or aircraft maintenance shops. Minimum requirements are two or more maintenance shops with joint work totaling 3,048 linear meters where the sealant has performed for two years with less than 1 percent combined sealant failures and defects. List location of shops, total linear feet of sealant applied per shop, shop point of contact, date sealant was applied, and the name of the installed sealant material.

#### 1.3.5.6 Thin Film Flooring System Certificates

Provide manufacturer's certification of conformance to Contract requirements.

#### 1.3.6 Product Data

##### 1.3.6.1 Joint Sealant Manufacturer's Instructions

Submit manufacturer's printed instructions to include detailed application procedures, minimum and maximum application temperatures, and curing procedures. In accordance with 29 CFR 1926.59, include Safety Data Sheets (SDS) for the sealant to be used at the job site.

#### 1.3.6.2 Thin Film Flooring System Manufacturer's Instructions

Submit manufacturer's printed instructions to include detailed mixing, minimum and maximum application temperatures, acceptable atmospheric and interior climatic conditions, application procedures, curing procedures, and procedures for maintenance cleaning of flooring system. Provide explicit instructions detailing surface preparation, recoat windows and remedial actions in case recoat windows are missed, and, if applicable, solvent-wiping between coats with acceptable types and grades of solvents. In accordance with 29 CFR 1926.59, include SDSs for the coatings to be used at the job site.

#### 1.3.6.3 Water-Based Alkaline Degreaser

Submit manufacturer's printed instructions to include detailed mixing, rate of dilution, application procedures, and rinsing procedures. In accordance with 29 CFR 1926.59, include SDSs for the water-based alkaline degreaser to be used at this job site.

### 1.4 DELIVERY, STORAGE, AND HANDLING

Store coatings and sealant in spaces with temperatures from 5 degrees C to 24 degrees C. Inspect materials on site for damage prior to use. Return to manufacturer packaged materials in dented, rusty, or leaking containers. Conduct testing by manufacture of returned materials with an expired shelf life and if compliant, reissue a shelf life extension.

### 1.5 COATING HAZARDS

Ensure that employees are trained in all aspects of the safety plan. Follow the coating manufacturer's written safety precautions throughout mixing, application, and curing of coatings. Comply with respiratory protection requirements in 29 CFR 1910.134 and safe levels of airborne contaminants in 29 CFR 1910.1000.

### 1.6 JOB SITE REFERENCES

Make available to the Contracting Officer at least one copy each of ASTM D4541, ASTM D6237, SSPC-TU 2/NACE 6G197, and ICRI 310.2R, including replica standards ICRI 310.2R CSP 1 through CSP 9, at the job site.

### 1.7 PATCH TEST DEMONSTRATION

Prior to the submitted flooring system's approval, apply the complete coating system to a 3 meter by 3 meter square section of concrete as prepared in accordance with PART 3 EXECUTION. Within this area, perform three adhesion tests as described in the paragraph ADHESION TESTING. If adhesion testing produces cohesive failures within the concrete, no less than 1 mm concrete removed over 95 percent of each pull-off coupon, or adhesion more than 2.75 MPa, patch test adhesion is acceptable. If concrete surface preparation was insufficient, apply an additional coating system patch to properly prepared concrete followed by the above adhesion testing. If adhesion results are unacceptable for both the topcoats and the primer, submit a new coating system manufactured by a different coating vendor. Apply a patch of the new coating system and subject patch to the above requirements for adhesion prior to approval. If customer is not satisfied with the non-skid grit application, adjustments to the specifications can be made. Grit coarser than No. 60 aluminum oxide is not recommended.

## 1.8 WARRANTY

Warranty materials and workmanship for a minimum period of one year following coating and sealant application. The following terms and conditions form a part of the warranty: If the applied coating system develops either blisters (chemical), checks, softening, or lifting within one year following application, rework each area at Contractor's expense. The following conditions are excluded from the warranty: A) concrete cracking, flooring system mirrors cracks in concrete; B) cosmetic imperfections due to scratching and gouging; C) application to metallic concrete finishes; and D) application to concrete with a rate of Moisture Vapor Emission (MVE) greater than 197.0 micrograms moisture per second, square meter. If the coating system's adhesion is in question, perform one adhesion test per 9.3 square meters as described in the paragraph ADHESION TESTING. To satisfy the warranty, each adhesion test must produce cohesive failures, concrete removal over 95 percent of each pull-off coupon, or adhesion no less than 2.8 MPa. Require two additional adhesion tests to confirm results for each area failing to meet adhesion requirements. Within the warranty period, remove to sound material and rework all areas unable to meet adhesion requirements. There must be zero percent sealant failures within one year. Within the warranty period, remove and rework all sealant material with chemically attacked surfaces or lifting from joint walls. Topcoat cracking over sealant is excluded from warranty.

## PART 2 PRODUCTS

### 2.1 JOINT SEALANT

Formulate the joint sealant to exhibit the properties as listed in Table Ia.

### 2.2 THIN FILM FLOORING SYSTEM

A three-coat industrial flooring system consisting of primer and two urethane topcoats. Apply the coating system at a Dry Film Thickness (DFT) ranging from 325 to 500 microns and contain a broadcast of aluminum oxide non-skid grit.

#### 2.2.1 Primer Coat

In addition to the requirements of the thin film flooring system, use MPI 211 primer coat.

#### 2.2.2 Urethane Topcoat

In addition to the requirements of the thin film flooring system, use MPI 212 top coat.

### 2.3 WHITE ALUMINUM OXIDE NON-SKID GRIT

Size No. 60, dust free (washed and dry), minimum 99 percent pure, having the following sieve analysis when tested using a 1000 gram sample (ASTM E11 ):

|              |                        |
|--------------|------------------------|
| Sieve No. 40 | 100 percent passing    |
| Sieve No. 50 | 15-30 percent retained |
| Sieve No. 60 | 70-85 percent retained |
| Sieve No. 70 | 0-15 percent retained  |

### PART 3 EXECUTION

#### 3.1 COATING SAMPLE COLLECTION

The Contracting Officer and QC Manager must witness all material sampling. Notify the Contracting Officer a minimum of three days in advance of sampling. Obtain liquid samples of each component of primer and topcoat by random selection from sealed containers and in accordance with ASTM D3925. Samples may be either individual cans of liquid material or 1.0 liter quantities of properly mixed, extracted, and sealed liquid material. Identify samples by designated name, specification number, batch number, project Contract number, sample date, intended use, and quantity involved. When the applied coating system has met the requirements defined in the paragraph ADHESION TESTING, return coating to the installation Contractor for proper disposal.

#### 3.2 JOINT MATERIAL REMOVAL, RE-SAW CUTTING, CRACK CHASING

Remove 100 percent of the existing material in all joints including material bonded to joint walls and base. Rigid material may require saw cutting equipment to remove. Joints may be widened up to 3.2 mm when re-saw cutting. Chase concrete cracks identified for repair and open to a minimum depth of 12.5 mm below crack surface resulting in crack(s) with smooth vertical walls. Cracks greater than 18.75 mm width can be repaired using either epoxy mortar or epoxy concrete.

#### 3.3 DEGREASING

On both coated and uncoated concrete, degrease entire floor by scrubbing using a solution of hot potable water, 49 degrees C to 77 degrees C, and a concentrated water-based alkaline degreaser. Perform two complete degreasing cycles on the entire floor surface. Allow solution to soak into surfaces prior to scrubbing and remove using hot potable water under a minimum of 27.6 MPa. Rinsing is complete when the rinse water appears clear. If the industrial detergent is not biodegradable, collect all rinse water and dispose as hazardous waste. Squeegees and shop vacuums may be used to collect pooling rinse water. Fans may be used to aid drying of floor surfaces.

#### 3.4 SURFACE PREPARATION

Shot blast entire floor to produce a level of coarseness equal to ICRI 310.2R CSP 3. Overlap each pass of shot blasting by 6.25 mm to 12.5 mm. Add new shot to shot blasting equipment prior to blasting. Prepare surfaces inaccessible to shot blasting, base of perimeter walls and under secured equipment, using diamond disk grinding or light scarification to produce a level of coarseness equal to ICRI 310.2R CSP 2 or ICRI 310.2R CSP 4, respectively. Resulting surfaces must appear clean and contain the appropriate level of surface coarseness. If the resulting level of cleanliness cannot be determined, place numerous drops of water on surfaces that appear contaminated. If the water drops soak into concrete,

the surfaces are free of hydrocarbon contamination (oils, grease, skydrol). If the water drops bead up and do not flatten out, surfaces require additional degreasing as detailed in the Article DEGREASING. Shot blasting coarse concrete or broom finished concrete can produce a level of coarseness equal to ICRI 310.2R CSP 5: employ a best-effort attempt to minimize over-shot-blasting of coarse concrete. If coarse concrete is encountered, shot blasting to a level of coarseness equal to ICRI 310.2R CSP 5 is acceptable; however, extremely coarse concrete can require resurfacing prior to the flooring system's installation. Sweep, vacuum, and run a high powered magnet over all surfaces to be coated, including joints.

### 3.5 JOINT TREATMENT

Use the "Conventional Sealed Joint" as detailed in Figure 1 of SSPC-TU 2/NACE 6G197 to seal joints. Employ measures to reduce contamination from equipment and foot traffic. Limit floor access to essential Contractor personnel. Confirm joint surfaces are sufficiently clean.

#### 3.5.1 Install Backer Rod

Install a continuous length of round, closed-cell polyethylene backer rod into each joint using a backer rod tool. For 12.5 mm, 9.4 mm, and 6.25 mm wide joints, place backer rod to a depth of 9.4 mm (depth equals the distance from the concrete's surface to the highest point on the backer rod). For joints greater than or equal to 18.75 mm width, place backer to a depth of 15.6 mm below the concrete's surface. Fit backer rod tight between joint walls (30 percent compression). Remove and reinstall all backer rod that is installed using either the incorrect size (loose fit) or at the incorrect depth. Following backer rod installation, apply painter's tape to surfaces adjacent joints to protect from sealant.

#### 3.5.2 Joint Sealant Application

Apply sealant directly into joints using a bulk-caulking gun. At room temperature, the resulting sealant application must exhibit a concave recess between 3.2 mm to 1.6 mm below the concrete's surface. Remove and reapply cured sealant remaining either flush or greater. Following sealant application, remove painter's tape and sealant drips on concrete surfaces. Cure sealant a minimum of 24 hours, prior to the application of coatings.

### 3.6 PRE-APPLICATION TESTING FOR CONTAMINATION

Spot check surfaces for oil/grease contamination using the water break test. At a rate of 5 tests per 95 square meters, place one to two drops of water onto surfaces and observe for beading. Test all other surfaces that show visible signs of potential contamination. Apply additional degreasing techniques to surfaces displaying water beading in accordance with the Article DEGREASING.

### 3.7 COATING APPLICATION

Vacuum flooring space one additional time prior to coating application.

#### 3.7.1 Primer Application

Apply MPI 211 epoxy primer to flooring space at 175 microns to 375 microns

Dry Film Thickness (DFT). If the prepared concrete resembles an ICRI 310.2R CSP 3 surface, apply the primer at a minimum of 175 microns DFT. If the prepared concrete resembles an ICRI 310.2R CSP 5 surface, apply the primer at a maximum of 375 microns DFT. The previously applied sealant may be lightly coated.

### 3.7.2 Non-Skid Grit Broadcast

Broadcast non-skid grit at a rate of 450 grams per 9.3 square meters into the second urethane top coat and backroll. Map floor into 55.8 square meter sections where 4080 grams of non-skid grit is pre-weighed, placed into clean buckets and used in its entirety per marked 55.8 square meter section.

### 3.7.3 Application of Topcoats

Apply two coats of MPI 211 epoxy urethane topcoat to the epoxy primer and broadcast white aluminum oxide non-skid grit directly into the second urethane topcoat.

#### 3.7.3.1 First Topcoat

Apply a full coat of the urethane topcoat at a spreading rate from 62.5 to 80 microns Dry Film Thickness (DFT). Stripe coat perimeter edges and around equipment footings. Monitor and record a minimum of one Wet Film Thickness (WFT) reading per 55.8 square meter of floor surface. Sealant is to be lightly coated.

#### 3.7.3.2 Second Topcoat

Apply a second coat of the urethane topcoat at a spreading rate of 62.5 to 80 microns DFT. Stripe coat perimeter edges and around equipment footings. Monitor and record a minimum of one WFT reading per 55.8 square meters of floor surface prior to broadcasting non-skid grit. When the correct WFT has been applied per 55.8 square meters of area, immediately and evenly broadcast non-skid grit directly into the second topcoat of urethane and backroll in two directions. Test the adhesion of the thin film flooring system in accordance with the paragraph ADHESION TESTING.

#### 3.7.3.3 Walkway Stripe and Grounding Rod Markings

Place the walkway stripe and grounding rod markings according to Government drawings, if applicable. When the second topcoat is within its recoat window, apply a walkway stripe of the red/orange urethane topcoat at 75 microns DFT, completely hiding the top coat, in one coat. If insufficient hiding occurs, apply one additional coat of the walkway stripe. Lightly broadcast non-skid grit into the wet walkway stripe. Use solvent-resistant tape to protect the floor coating against stripe coat bleed. A thin clear coat of either epoxy or urethane may be required to prevent stripe coat bleed prior to the full application of the colored stripe coat. Apply grounding rod markings using similar procedures, urethane top coat, and colors and size according to Government drawings.

### 3.8 CURING

Cure installed materials to display performance equal to manufacturer's product literature. Remove and reapply improperly cured material.



### 3.9 FIELD QUALITY CONTROL

#### 3.9.1 Coating Inspector

Consider the Coating Inspector as a QC Specialist, who works for the QC Manager, and qualified in accordance with Section 01 45 00 QUALITY CONTROL. The Coating Inspector must be present during all field tests, surface preparation, flooring application, initial cure of the flooring system, and during all flooring repair work. The Coating Inspector must provide all tools/equipment necessary to perform field tests and inspection. The Coating Inspector is responsible for field tests and specified level of inspection.

#### 3.9.2 Inspection

Document weather conditions, job site occurrences, and report conditions and occurrences potentially detrimental to the flooring system. The listed inspection requirements are in addition to the QC inspection and reporting requirements defined in Section 01 45 00 QUALITY CONTROL. The Coating Inspector must prepare a project reference sheet outlining all requirements, tests, test methods, and evaluation criteria, and hold regular meetings with Contractor personnel, including shot blasting operators and applicators, to review requirements/evaluation criteria for upcoming work prior to execution. At the start of coating operations and every hour following until daily work is complete, record air temperature, substrate temperature, and relative humidity. Following the application of each coat, inspect surfaces for improperly cured material, blisters, inadequate or excessive coating thickness, and other defects. Document each inspection, test, non-compliant area, and location of each non-compliant area. List method of evaluation, evaluation criteria, areas requiring rework, and all other pertinent observations.

##### 3.9.2.1 Daily Inspection Report

Submit to the Contracting Officer one copy of the daily inspection report completed each day when performing work under this Section. Use Appendix X1 "Inspection Checklist" of ASTM D6237 to monitor daily activity and to assist in preparing the daily inspection report. Note each non-compliant issue and each issue identified for rework in accordance with the QC documentation procedures of Section 01 45 00 QUALITY CONTROL. Use of forms containing entry blocks for all required data is encouraged. The data must be legible and presented in a professional format. Submit report within 24 hours of the report date.

##### 3.9.2.2 Inspection Logbook

Maintain a continuous record of all activity related to this Section on a daily basis. A computer / software package as outlined in Table II is preferred to record all information provided in the Daily Inspection Reports, as well as other pertinent observations and information including photo documentation where appropriate. The designated Government Representative for the project is assigned the highest level Administrator privileges and only the Administrator must be able to modify reports. In areas where photography is not allowed, the computer must come with verification that the camera / photo capability has been removed.

Alternatively, a continuous record of all activity related to this Section must be maintained in an Inspection Logbook on a daily basis. The logbook must be hard or spiral bound with consecutively numbered pages, and must

be used to record all information provided in the Daily Inspection Reports, as well as other pertinent observations and information. Submit the original Inspection Logbook to the Contracting Officer upon completion of the project and prior to final payment.

#### 3.9.2.3 Inspection Equipment

Use equipment in good condition, operational within its design range, and calibrated as required by the specified standard for use with each device.

#### 3.9.3 Adhesion Testing

Perform a minimum of three adhesion tests in accordance with ASTM D4541 to the thin film flooring system. Select three random flooring locations spaced a minimum of 6 meters between each location. Prior to attaching pull-off coupons, lightly sand flooring surface and attach pull-off coupons containing a grit-blasted anchor profile. Adhere directly to the center of each sanded surface a 18.75 mm diameter pull-off coupon. When pull-off coupon adhesive has sufficiently cured, score circumference of each pull-off coupon to concrete substrate. Test adhesion and evaluate results. If testing produces cohesive failures within the concrete, no less than 1 mm concrete removed over 95 percent of each pull-off coupon, or adhesion more than 2.75 MPa coating system's adhesion is acceptable. If the above requirements are not satisfied, then perform one adhesion test per 9.3 square meters using the above procedures. Perform two additional tests per non-compliant area to confirm results. Remove to sound material and rework all areas unable to meet adhesion requirements. Repair each adhesion test using a combination of primer, sand-filled epoxy mortar (for deep cohesive failures, if applicable), and two urethane topcoats. Make repairs flush with adjacent coatings and display an equivalent appearance.

#### 3.10 FINAL CLEANUP

Following completion of the work, remove debris, equipment, and materials from the site. Remove temporary connections to Government or Contractor furnished water and electrical services. Restore existing facilities in and around the work areas to their original condition.

| TABLE I - MATERIALS REQUIREMENTS                                                                      |                                                                                                                         |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| TABLE Ia                                                                                              |                                                                                                                         |
| Test                                                                                                  | Minimum Requirement (maximum where indicated)                                                                           |
| Sealant System (two-pack: self-leveling)                                                              | Polysulfide (Manganese Cure; MnO <sub>2</sub> ) or Urethane                                                             |
| Percent Volume Solids                                                                                 | 100 percent                                                                                                             |
| Chemical Resistance to JP-8 plus 100 Fuel at 21 degrees C (ASTM D1308) (see note 1)                   | 48 hours immersion: 2.0 percent (max) weight increase, 5.0 percent (max) volume increase, 2.0 percent (max) weight loss |
| Chemical Resistance to Motor Oils at 21 degrees C (ASTM D1308) (see note 1)                           | 48 hours immersion: 2.0 percent (max) weight increase, 5.0 percent (max) volume increase, 2.0 percent (max) weight loss |
| Chemical Resistance to Skydrols at 21 degrees C (ASTM D1308) (see note 1)                             | 48 hours immersion: 2.0 percent (max) weight increase, 5.0 percent (max) volume increase, 2.0 percent (max) weight loss |
| Hardness (ASTM D2240:Shore A)                                                                         | 20                                                                                                                      |
| Tensile Strength (ASTM D412) (or ASTM D638)                                                           | 1.0 MPa                                                                                                                 |
| Percent Elongation (ASTM D412) (or ASTM D638)                                                         | 500 percent                                                                                                             |
| Tack Free at 18.3 degrees C (ASTM C679)                                                               | 12 hours maximum                                                                                                        |
| Adhesion to Concrete                                                                                  | 0.96 MPa                                                                                                                |
| Adhesion to Urethane Topcoats (paintable sealant)                                                     | 0.96 MPa                                                                                                                |
| NOTES: (1) Immerse and test a minimum of three 50 mm by 12.5 mm by 12.5 mm sections of cured sealant. |                                                                                                                         |



| TABLE II                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reporting Program Requirements QA/QC                                                                                                                                                                                                       |
| Administrative Controls:                                                                                                                                                                                                                   |
| Administrators must be able to turn on and off unique access to specific jobs and Contracts.                                                                                                                                               |
| Administrators must be able to remotely enable and disable access for users.                                                                                                                                                               |
| All enabled users must view the same active report in real time. There must be no opportunity for multiple versions of the same report to exist.                                                                                           |
| Administrators must be able to setup unique approval processes for each project and promote or remove unique people from this process at any time.                                                                                         |
| Administrators must be able to associate Contract specific documents and specification limits quickly and easily.                                                                                                                          |
| Administrators must be able to associate PDS, SDS, blueprints, scope of work and Contracts uniquely to each job.                                                                                                                           |
| Objectivity Controls:                                                                                                                                                                                                                      |
| Data Entry fields must be by multi-selectable choices, numeric keypads, pickers and skip logic to ensure repeatable data entry in a way that makes running analytics and metrics easy and objective.                                       |
| The program / hardware package must be able to communicate with inspection devices that provide (batch) data export capability such as Elcometer and Defelsko gages.                                                                       |
| The program / hardware package must automatically time, date and GPS stamp all reports without input or interference from the inspector.                                                                                                   |
| Real Time Syncing:                                                                                                                                                                                                                         |
| Forms must be available for approved associates to view at all times.                                                                                                                                                                      |
| Retrievable storage must be provided for all job related reports and documents for a minimum time of 5 years from completion of the job or project. Archiving of the documents after 5 years will be the responsibility of the Government. |
| Document Library:                                                                                                                                                                                                                          |
| All reports must be in searchable and annotatable PDF format.                                                                                                                                                                              |
| The Administrator must be able to upload and annotate job specific reports in real time. Examples include but not limited to Safety Data Sheets, Product Data Sheets and Blueprints.                                                       |

| TABLE II                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annotations and modifications must be locked and associated with the document. Only the Administrator has rights to modify or delete annotations or allow modifications to the document library especially all related inspection reports. |
| Customization:                                                                                                                                                                                                                             |
| The program must be capable of being customized to specific jobs, Contracts or specifications.                                                                                                                                             |

-- End of Section --

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## SECTION 09 69 13

## RIGID GRID ACCESS FLOORING

**11/24**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-22 (2022; Supp 1 2023; Supp 2 2023) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

## AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A1264.1 (2017) Safety Requirements for Workplace Walking/Working Surfaces and Their Access; Workplace, Floor, Wall and Roof Openings; Stairs and Guardrail/Handrail Systems

## ASTM INTERNATIONAL (ASTM)

ASTM A780/A780M (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings

ASTM E488/E488M (2022) Standard Test Methods for Strength of Anchors in Concrete Elements

ASTM E648 (2023) Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source

ASTM F150 (2006; R 2013) Standard Test Method for Electrical Resistance of Conductive and Static Dissipative Resilient Flooring

ASTM F1700 (2020) Standard Specification for Solid Vinyl Floor Tile

ASTM F1861 (2021) Standard Specification for Resilient Wall Base

## CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers



## CEILINGS AND INTERIOR SYSTEMS CONSTRUCTION ASSOCIATION (CISCA)

CISCA Access Floors (2007) Recommended Test Procedures for Access Floors

## GREEN SEAL (GS)

GS-36 (2013) Adhesives for Commercial Use

## ICC EVALUATION SERVICE, INC. (ICC-ES)

ICC-ES AC308 (2014) Acceptance Criteria for Access Floors

## INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2024) International Building Code

## NATIONAL RESEARCH COUNCIL CANADA (NRC)

NBC (2020) National Building Code of Canada, Part 8, Construction Safety Measures at Construction and Demolition Sites

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 75 (2024) Standard for the Protection of Information Technology Equipment

NFPA 99 (2024) Health Care Facilities Code

NFPA 253 (2011) Standard Method of Test for Critical Radiant Flux of Floor Covering Systems Using a Radiant Heat Energy Source

## SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

## U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 3, 2025) Structural Engineering

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Detailed Installation Drawings; G

Fabrication Drawings; G

## SD-03 Product Data

Access Flooring System; G

Recycled Content of Access Flooring System; S

Indoor Air Quality For Pedestal Adhesive; S

Indoor Air Quality For Concrete Sealer; S

Indoor Air Quality For Adhesives; S

SD-04 Samples

Floor Panels

Floor Covering; G

Panel Support System

Accessories; G

Fascia; G

Exposed Step and Ramp Structure; G

Railings; G

Perforated Directional Air Supply Panels; G

Cut Outs; G

SD-05 Design Data

Seismic Calculations

SD-06 Test Reports

Factory Tests

Concentrated Load

Uniform Live Load

Rolling Load

Impact Load

Ultimate Load

Stringer Load

Pedestal Axial Load

Bonding Strength of Pedestal Adhesive

Electrical Resistance

Field Tests

SD-07 Certificates

Compliance with ICC-ES AC308

Compliance with ICC IBC

Certificate of Compliance

Qualification of Manufacturer

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

Lifting Device

Warranty; G

### 1.3 SPARE PARTS

Provide four floor panels complete with specified floor covering for future use.

### 1.4 QUALITY CONTROL

#### 1.4.1 Qualification of Manufacturer

Access flooring manufacturer must have at least 5 years experience in manufacturing access flooring systems. Certify that the manufacturer of the access flooring system meets requirements specified under paragraph entitled QUALIFICATION OF MANUFACTURER.

### 1.5 DELIVERY, STORAGE, AND HANDLING

#### 1.5.1 Delivery

Deliver materials to site in undamaged condition, in original containers or packages, complete with accessories and instructions. Label packages with manufacturer's name and brand designations. Package materials covered by specific references bearing specification number, type and class as applicable.

#### 1.5.2 Storage

Store all materials in original protective packaging in a safe, dry, and clean location. Store panels at temperatures between 4 and 32 degrees C, and between 20 and 70 percent humidity. Replace defective or damaged materials.

#### 1.5.3 Handling

Handle and protect materials in a manner to prevent damage during the entire construction period.

### 1.6 WARRANTY

Minimum manufacturer warranty must have no dollar limit, cover full system, and must have a minimum duration of 1 years. Include an agreement

to repair or replace floor panels, pedestals or stringers that fail within the warranty period in the standard performance guarantee or warranty. Failures include, but are not limited to, sagging and warping of panels; rusting; and manufacturing defects of panels or support system. For static-dissipative vinyl tile provide manufacturer's standard performance guarantees or warranties that extend beyond one year, standard warranty must not be less than a five year wear warranty and ten year conductivity warranty.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

- a. Provide for self-alignment of floor panels, adjustable pedestals and readily removable floor panels covered as specified.
- b. Lateral stability of floor support system must be independent of panels. Provide a finished assembly that is rigid and free of vibration, noises, and rocking panels. Provide bolted stringer system with equipotential plane grounding.
- c. Submit certificate of compliance attesting that the installed access floor system meets specification requirements, including all special equipment loads and specific electrical and or cable requirements for the complete access flooring system including, but not limited to the following:
  - (1) Compliance with ICC-ES AC308 and Compliance with ICC IBC Acceptance Criteria for Access Floors and with NBC.
  - (2) Load-bearing capabilities of pedestals, floor panels, and pedestal adhesive resisting force.
  - (3) Supporting independent laboratory test reports. For panel, stringer and pedestal load test results include concentrated loads at center of panel, panel edge midpoint, ultimate loads and uniform loads.
  - (4) Floor electrical characteristics.
  - (5) Material requirements.
  - (6) An elevated floor system free of defects in materials, fabrication, finish, and installation, that will remain so for a period of not less than 1 years after completion.
- d. Submit manufacturer's product data for access flooring system consisting of descriptive data, catalog cuts, and installation instructions. Include in the data information about any design and production techniques, total system including all accessories and finish coatings of under-floor components, procedures and policies used to conserve energy, reduce material, improve waste management or incorporate green building/recycled products into the manufacture of their components or products. Include cleaning and maintenance instructions. Systems which contain zinc electroplated anti-corrosion coatings are prohibited.

### 2.1.1 Design Requirements

Conduct floor panel testing in accordance with CISCA Access Floors. When tested as specified, make all deflection and deformation measurements at the point of load application on the top surface of the panel. Floor panels must be capable of supporting the following loads:

- a. Concentrated load of 11120 N on 645 square mm, at any point on panel, without a top-surface deflection more than 2.54 mm, and a permanent set not to exceed 0.25 mm in any of the specified tests. Testing must be in accordance with CISCA Access Floors, Section 1 Concentrated Loads with test panels being supported by understructure to be used with installed system instead of steel support blocks.
- b. Uniform live load of 23.94 kPa, without a top-surface deflection more than 1.0 mm, and a permanent set not to exceed 0.25 mm in any of the specified tests, when tested in accordance with CISCA Access Floors, Section 7 Uniform Load Test with test panels being supported by understructure to be used with installed system instead of steel support blocks.
- c. A rolling load of 8900 N applied through hard rubber surfaced wheel 152 mm diameter by 51 mm wide for 10,000 cycles over the same path. Permanent set at conclusion of test must not exceed 1.0 mm when tested in accordance with CISCA Access Floors, Section 3 Rolling Loads.
- d. A rolling load of 8890 N applied through a 75 mm diameter by 30 mm wide caster for 10 cycles over the same path, without developing a local overall surface deformation greater than 1 mm. In accordance with CISCA Access Floors, Section 3 Rolling Loads, the permanent deformation limit under rolling load must be satisfied in all of the specified tests.
- e. An impact load of 670 N anywhere on the panel dropped from a height of 914 mm onto a 645 square mm area without failure of the system, according to CISCA Access Floors, Section 8 Drop Impact Load Test.
- f. Ultimate Load. Panels must meet manufactures published Ultimate Load rating of 22250 N when tested in accordance with CISCA Access Floors, Section 2 Ultimate Loading.
- g. Safety Factor. Panels must provide a minimum Safety Factor of 5 times the uniform load specified above in accordance with ICC-ES AC308.
- h. Recycled Content. Provide Access Flooring System (panels, stringers and pedestals) containing a minimum of 20 percent recycled content. Provide data identifying percentage of recycled content of access flooring system.

### 2.1.2 Allowable Tolerances

#### 2.1.2.1 Floor Panel Flatness

Plus or minus 0.89 mm on diagonal on top of panel or underneath edge.

#### 2.1.2.2 Floor Panel Length

Plus or minus 0.4 mm.

#### 2.1.2.3 Floor Panel Squareness

Plus or minus 0.5 mm in panel length.

#### 2.1.3 Stringers

Provide stringers capable of supporting a 2000 N concentrated load at midspan without permanent deformation in excess of 0.25 mm, when tested in accordance with Cisca Access Floors, Section 4 Stringer Load Testing.

#### 2.1.4 Pedestals

Pedestals must be capable of supporting a 26.69 kN axial load without permanent deformation, when tested in accordance with Cisca Access Floors, Section 5 Pedestal Axial Load Test.

#### 2.1.5 Bonding Strength of Pedestal Adhesive

Adhesive for anchoring pedestal bases must have a bonding strength capable of resisting an overturning moment of 113 Nm when a force is applied to the top of the pedestal in any direction, when tested in accordance with Cisca Access Floors, Section 6 Pedestal Overturning Moment Test. Pedestal adhesive must meet emissions requirement of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type). Provide validation of indoor air quality for pedestal adhesive.

#### 2.1.6 Bond Strength of Factory Installed Covering

Bond strength of floor covering must be sufficient to permit handling of the panels by use of the panel lifting device, and to withstand moving caster loads up to 8890 N, without separation of the covering from the panel.

#### 2.1.7 Seismic Calculations

##### 2.1.7.1 Navy Project Specific Requirements

Submit seismic calculations for lateral bracing, sealed by a Professional Engineer. Document that access flooring system complies with seismic requirements of ICC IBC and ASCE 7-22 for Occupancy Importance Factor ( $I_p$ ) of 1.0, and seismic horizontal force ( $F_p$ ) determined in accordance with UFC 3-301-01 and Section 1615 of the ICC IBC and ASCE 7-22, Minimum Design Loads for buildings and other structures.

### 2.2 FLOOR PANELS

#### 2.2.1 Floor System Drawings And Planer Quality

- a. Submit Fabrication Drawings for elevated floor systems consisting of fabrication and assembly details to be performed in the factory.
- b. Indicate on Location Drawings exact location of pedestals, ventilation openings, cable cutouts, and the panel installation pattern.
- c. Provide Detail Drawings showing details of the pedestals, pedestal-floor interlocks, floor panels, panel edging, floor openings, floor opening edging, floor registers, floor grilles, cable cutout treatment, perimeter base, expansion, and peripheral support facilities.

- d. Design and workmanship of the floor, as installed, must be completely planar within plus or minus 1.5 mm in 3050 mm, 2.5 mm for the entire floor, and 0.7 mm across panel joints.
- e. Floor-panel joint-width tolerances must not exceed 0.43 mm as measured with a feeler gage at any point in any joint when the panels are installed and as long as the air leakage requirements specified in this section are met.
- f. Submit three complete samples of floor panels.

#### 2.2.2 Detailed Installation Drawings

Submit Detailed Installation Drawings that as a minimum indicate the following:

- a. Location of panels
- b. Layout of supports, panels, and cutout locations
- c. Stair, handrail, and ramp framing
- d. Sizes and details of components
- e. Details at floor perimeter and height above structural floor
- f. Method of anchorage to structural subfloor
- g. Lateral bracing
- h. Typical cutout details
- i. Gasketing, return air grilles, supply air registers, and perforated panels. Include air transfer capacity of grilles, registers and panels
- j. Description of factory coating
- k. Floor finishes
- l. Location of connection to building grounding electrode

#### 2.2.3 Panel Construction

- a. Base access floor system on a 600 by 600 mm square module providing minimum of 600 mm clearance between structural floor and underside of panel and stringer. Fabricate so accurate job cutting and fitting may be done using standard sizes for perimeters and around columns.
- b. Do not expose metal on finished top surface of panels. Provide cutouts and cutout closures to accommodate utility systems and equipment intercabling. Reinforce cutouts to meet design load requirements. Provide extra support pedestals at each corner of cutout for cutout panels that do not meet specified design load requirements.
- c. Panel design must provide for convenient panel removal for underfloor servicing and for openings for new equipment. Use panels of uniform dimensions within specified tolerances. Permanently mark panels to

indicate load rating and model number.

- d. Machine square floor panels to within plus or minus 0.38 mm with edge straightness plus or minus 0.064 mm. If plastic edging is applied to the panel, the tolerances apply to the panel before the plastic edging is applied.

#### 2.2.3.1 Hollow Formed Steel

Steel panels must be of die-formed construction, consisting of a flat steel top sheet welded to one or more formed steel stiffener sheets or components. Panels must be chemically cleaned, bonderized, and painted with the manufacturer's standard finish.

#### 2.2.4 Floor Covering

Surface floor panels with factory applied finish materials firmly bonded in place with waterproof adhesive. Provide finish flooring materials in corridors and exits with a critical radiant flux of not less than 0.22 watts per square centimeter (Class 2) when tested in accordance with ASTM E648 or NFPA 253. The electrical resistance must remain stable over the life expectancy of the floor covering. Any anti-static agent used in the manufacturing process must be an integral part of the material, not surface applied. Bolt heads or similar attachments must not rise above the traffic surface. Submit three separate samples of each specified floor covering finish and color.

##### 2.2.4.1 Static-Dissipative Vinyl Tile

Provide factory applied static-dissipative vinyl tile that is a homogeneous vinyl product and conforms to ASTM F1700, Class I monolithic, Type A smooth surface. Provide electrical resistance from surface to surface and surface to ground between 1,000,000 ohms ( $1.0 \times 10^6$ ) and 1,000,000,000 ohms ( $1.0 \times 10^9$ ) when tested in accordance with ASTM F150. Material must consist of one piece to cover the face of the panel. Provide edge detail that is integral to the finish material.

#### 2.2.5 Accessories

Provide the manufacturer's standard registers, grilles, perforated panels, and plenum dividers type where indicated. Provide registers, grilles, and perforated panels designed to support the same static loads as floor panels without structural failure, and capable of delivering the air volumes indicated. Registers and perforated panels must be 25 percent open area and equipped with adjustable dampers. Submit three samples and colors of each accessory. Provide round perforations.

#### 2.2.6 Resilient Base

Conform to ASTM F1861, Type TS (vulcanized thermoset rubber) Style B (coved - installed with resilient flooring). Provide 150 mm high and a minimum 3.175 mm thick wall base. Provide job formed corners in matching height, shape, and color.

#### 2.2.7 Adhesives

Provide adhesives as recommended by the manufacturer. Provide non-aerosol adhesive products that meet either emissions requirements of CDPH SECTION 01350 (use the requirements for either office or classroom,



regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesives that meet either emissions requirements of CDPH SECTION 01350 (use the requirements for office or classroom, regardless of space type) or VOC content requirements of GS-36. Provide validation of indoor air quality for adhesives. Provide conductive adhesive as recommended by the manufacturer of the static-control flooring.

#### 2.2.8 Lifting Device

At turn over provide one floor panel lifting device standard with the floor manufacturer, for each individual floor area (room or corridor). Furnish a minimum of two devices.

### 2.3 PANEL SUPPORT SYSTEM

Design support system to allow for 360 degree clearance in laying out cable and cutouts for service to machines and so that panel and stringer together take up maximum of 50 mm. Submit one sample of suspension system proposed for use.

#### 2.3.1 Pedestals

Provide pedestals made of steel or aluminum or a combination thereof. Ferrous materials must have a factory-applied corrosion-resistant finish. Provide pedestal base plates with a minimum of 10,300 square mm of bearing surface and a minimum of 3 mm thickness. Pedestal shafts must be threaded to permit height adjustment within a range of approximately 50 mm, to permit overall floor adjustment within plus or minus 2.5 mm of the required elevation, and to permit leveling of the finished floor surface within 1.56 mm in 3000 mm in all directions. Provide locking devices to positively lock the final pedestal vertical adjustments in place. Pedestal caps must interlock with stringers to preclude tilting or rocking of the panels.

#### 2.3.2 Stringers

Provide stringers of rolled steel or extruded aluminum, to interlock with the pedestal heads to prevent lateral movement. Provide stringers that can be added or removed after floor is in place.

#### 2.3.3 Gaskets

Provide continuous gasketing at contact surfaces between panel and stringers to deaden sound and seal off the underfloor cavity from above for air tightness, and to maintain panel alignment.

### 2.4 FASCIA

Provide aluminum or steel fascia plates at open ends of floor, at sides of ramps and steps, and elsewhere as required to enclose the free area under the raised floor. Steel plates must have a factory applied baked enamel finish. Finish on aluminum plates must be standard with the floor system manufacturer. Fascia plates must be reinforced on the back, and supported using the manufacturer's standard lateral bracing at maximum 1200 mm on center. Provide trim, angles, and fasteners as required. Submit one color samples for fascia.

### 2.5 FACTORY TESTS

Factory test access flooring, using an independent laboratory, at the same position and maximum design elevation and in the same arrangement as shown

on the drawings for installation so as to duplicate service conditions as much as possible.

#### 2.5.1 Load Tests

Conduct floor panel, stringer, and pedestal testing in accordance with Cisca Access Floors to determine deformation and permanent set of panels and systems due to concentrated, Uniform, rolling, impact and ultimate loading when panels are supported by actual understructure.

#### 2.5.2 Bond Strength of Covering

Conduct test for bond strength of covering in accordance with Cisca Access Floors for rolling loads, except as specified. Panels must be tested with specified hard surface flooring and on the pedestals and stringers as specified for the installed floor. Brace the supports as necessary to prevent sideways movement during the test. Impose a test load of 8890 N on the test assembly through a 75 mm in diameter and 25 mm wide hard plastic caster. Roll the caster completely across the center of the panel. The panel must withstand 20 passes of the caster with no delamination or separation of the covering.

### 2.6 REGISTERS AND GRILLES

Registers and grilles must be 600 mm by 600 mm long with a minimum free area of 60 percent, made from extruded aluminum, in anti-static powder-coat finish, to sustain point loads of 2669 newton per vane without failure or permanent deformation. No part of a grille may project more than 3 mm above the floor. Registers and grills are not permitted in a laminate floor tile system.

### 2.7 PERFORATED AIR SUPPLY PANELS

Provide air supply floor panels that meet the design criteria specified for standard panels, to sustain point loads of 4448 newton without failure or permanent deformation, are fabricated of 2 mm perforated steel sheet welded to minimum 1.6 mm side channels, are covered with high pressure laminate to match standard panels, and have a uniform perforated pattern to allow even air distribution.

### 2.8 PERFORATED DIRECTIONAL AIR SUPPLY PANELS

Provide directional air supply floor panels that meet or exceed the design criteria specified for standard panels, are fabricated of perforated steel sheet welded to a formed steel pan with powder coat finish. Submit one color samples for perforated directional air supply panels.

### 2.9 CUT OUTS

Provide cable cutouts finished with rigid polyvinylchloride or molded polypropylene edging to conform to the appearance level of the floor surface and to cover raw edges of the cutout panel. Extrusion must be of a configuration to permit its effective and convenient use when new cable openings are required. Provide at least 7300 mm of additional extrusion for future use. Submit one color samples for cut outs.

- a. Provide non-metallic adapter for openings less than 100 mm wide. Secure adapter adhesively in cutout to preclude removal from panel. Provide at least two adapters per 10 square meter for future use.

- b. Openings larger than 100 mm wide must use rigid polyvinylchloride or molded polypropylene edging. Perform cutting of panels, including cutouts, outside of the building.
- c. When size of cutout reduces the performance requirement of panel, provide intermediate stringers adjacent to cutouts.

## 2.10 PEDESTAL BASE FASTENERS

Provide four fasteners at each pedestal support base sized per ASTM E488/E488M to support loads without failure up to 1.5 times loads due to pedestal overturning moment.

Provide post-installed fasteners for securing pedestal bases to supporting concrete floor slabs.

## 2.11 COLOR

Color must be as indicated. Color listed is not intended to limit the selection of equal colors from other manufacturers.

# PART 3 EXECUTION

## 3.1 INSTALLATION

Install access flooring at the location and elevation and in the arrangement shown on the approved detailed installation drawings. The floor system must be of the rigid grid stringer type, complete with all supplemental items, and be the standard product of a manufacturer specializing in access flooring systems.

Install the floor system in accordance with the manufacturer's instructions. Open ends of the floor, where the floor system does not abut wall or other construction, must have positive anchorage and rigid support. Maintain areas to receive access flooring between 10 and 32 degrees C, and between 20 and 70 percent humidity for 24 hours prior to and during installation.

### 3.1.1 Preparation for Installation

Clear out all debris in the area in which the floor system is to be installed. Thoroughly clean structural floor surfaces and remove all dust. Install floor coatings, required for dust or vapor control, prior to installation of pedestals, only if the pedestal adhesive will not damage the coating. If the coating and adhesive are not compatible, apply the coating after the pedestals have been installed and the adhesive has cured.

### 3.1.2 Pedestals

Pedestals must be accurately spaced, and set plumb and in true alignment. Set base plates in full and firm contact with the structural floor, and secured to the structural floor with adhesive or steel expansion anchors in accordance with manufacturer's instructions, and as required by seismic design.

### 3.1.3 Stringers

Interlock stringers with the pedestal caps to preclude lateral movement, spaced uniformly in parallel lines at the indicated elevation.

### 3.1.4 Auxiliary Framing

Provide auxiliary framing or pedestals around columns and other permanent construction, at sides of ramps, at open ends of the floor, and beneath panels that are substantially cut to accommodate utility systems. Use special framing for additional lateral support as shown on the approved detailed installation drawings. Provide additional pedestals and stringers designed to specific heights and lengths to meet structural irregularities and design loads. Connect auxiliary framing to main framing.

### 3.1.5 Panels

Interlock panels with supports in a manner that will preclude lateral movement. Fasten perimeter panels, cutout panels, and panels adjoining columns, stairs, and ramps to the supporting components to form a rigid boundary for the interior panels. Level floors within the specified tolerances. Cut edges of steel and wood-core panels must be finished as recommended by the panel manufacturer. Secure extruded vinyl edging in place at all cut edges of all panel cut-outs to prevent abrasion of cables.

### 3.1.6 Resilient Base

Provide base at vertical wall intersections as indicated in the drawings. Apply the base after the floor system has been completely installed. Install wall base in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions. Tighten base joints and make even with adjacent flooring. Fill voids along the top edge of base at masonry walls with caulk. Roll entire vertical surface of base with hand roller, and press toe of base with a straight piece of wood to ensure proper alignment. Avoid excess adhesive in corners.

### 3.1.7 Repair of Zinc Coating

Repair zinc coating that has been damaged, and cut edges of zinc-coated components and accessories, by the application of a galvanizing repair paint conforming to ASTM A780/A780M. Areas to be repaired must be thoroughly cleaned prior to application of the paint.

## 3.2 FIELD TESTS

Submit certified copies of test reports from an approved testing laboratory, attesting that the proposed floor system components meet the performance requirements specified.

### 3.2.1 Acceptance Tests

Conduct acceptance tests after installation of floor system. Make at least one test for each 40 square meters of floor area. Conduct tests in presence of Contracting Officer and representatives of manufacturer and installer. Submit certified copies of test reports from an approved testing laboratory, attesting that the proposed floor system components meet the performance requirements specified.

### 3.2.2 Air Leakage

When the space below the finished floor is an air plenum, air leakage through the joints between panels and around the perimeter of the floor system must not exceed 0.15 L/s of air per linear meter of joint subjected to 2.5 mm, water gauge, positive pressure in the plenum, when tested in accordance with CISCA Access Floors, Section 10 Air Leakage Test. Measure the leakage rate on the finished raised floor system, which may include carpet.

### 3.2.3 Grounding

Ground the access flooring system for safety hazard and static suppression. Provide positive contact between components for safe, continuous electrical grounding of entire floor system. Total system resistance from wearing surface of floor to building grounding electrode must be within range of 0.5 to 20,000 megohms.

#### 3.2.3.1 Metal Grilles

Exposed metal is not allowed at wearing surface of access floor system, except at metal grilles and registers. When grilles and metal registers are provided, insulate as required to provide same grounding resistance as wearing surface.

#### 3.2.3.2 Joint Resistance

Electrical joint resistance between individual stringer and pedestal junctions must be less than 10 ohms. Electrical resistance between stringers and floor panels, as mounted in normal use, must be less than 10 ohms when tested in accordance with ASTM F150.

### 3.2.4 Electrical Resistance

Conduct testing of electrical resistance, in the completed installation, in the presence of the Contracting Officer in accordance with NFPA 99, modified by placing one electrode on the center of the panel surface and connecting the other electrode to the metal flooring support. Take measurements at five or more locations. Each measurement must be the average of five readings of 15 seconds duration at each location. During the tests, relative humidity must be 45 to 55 percent and temperature set at 21 to 24 degrees C. Select panels used in the testing at random and include two panels most distant from the ground connection. Measure electrical resistance with instruments that are accurate within 2 percent and that have been calibrated within 60 days prior to the performance of the resistance tests. The metal-to-metal resistance from panel to supporting pedestal must not exceed 10 ohms. The resistance between the wearing surface of the floor covering and the ground connection, as measured on the completed installation, must be in accordance with paragraph FLOOR COVERING.

### 3.2.5 SEISMIC SPECIAL INSPECTION AND TESTING

Perform special inspections and testing for seismic-resisting systems and components in accordance with UFC 3-301-01 and Section 01 45 35 SPECIAL INSPECTIONS.

### 3.3 CLEANING AND PROTECTION

#### 3.3.1 Cleaning

Keep the space below the completed floor free of all debris. Before any traffic or other work on the completed raised floor is started, clean the completed floor in accordance with the floor covering manufacturer's instructions. Do not permit seepage of cleaner between individual panels.

#### 3.3.2 Protection

Protect traffic areas of raised floor systems with a covering of building paper, fiberboard, or other suitable material to prevent damage to the surface. Cover cutouts with material of sufficient strength to support the loads to be encountered. Place plywood or similar material on the floor to serve as runways for installation of heavy equipment not in excess of design load capacity. Maintain protection until the raised floor system is accepted.

#### 3.3.3 Surplus Material Removal

Clean surfaces of the work, and adjacent surfaces soiled as a result of the work. Remove all installation equipment, surplus materials, and rubbish from the work site.

### 3.4 FIRE SAFETY

Install an automatic detection system below the raised floor meeting the requirements of NFPA 75 paragraph 5-2.1 to sound an audible and visual alarm. Air space below the raised floor must be subdivided into areas not exceeding 929 square meters by tight, noncombustible bulkheads. Seal all penetrations for piping and cables to maintain bulkhead properties.

### 3.5 OPERATION AND MAINTENANCE MANUALS

Submit maintenance instructions for proper care of the floor panel surface. When conductive flooring is specified, also submit maintenance instructions to identify special cleaning and maintenance requirements to maintain "conductivity" properties of the panel finish. Include requirements for maintenance of required dry and wet flooring slip resistance for level and ramped surfaces complying with ASSP A1264.1.

-- End of Section --

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## SECTION 09 90 00

## PAINTS AND COATINGS

02/21

## PART 1 GENERAL

## 1.1 RELATED REQUIREMENTS

## 1.1.1 Painting Included

Where a space or surface is indicated to be painted, include the following unless indicated otherwise.

- a. Surfaces behind portable objects and surface mounted articles readily detachable by removal of fasteners, such as screws and bolts.
- b. New factory finished surfaces that require identification or color coding and factory finished surfaces that are damaged during performance of the work.
- c. Existing coated surfaces that are damaged during performance of the work.

## 1.1.1.1 Interior Painting

Includes new surfaces of the building and appurtenances as indicated and existing coated surfaces made bare by cleaning operations. Where a space or surface is indicated to be painted, include the following items, unless indicated otherwise.

- a. Exposed columns, girders, beams, joists, and metal deck; and
- b. Other contiguous surfaces.

## 1.1.2 Painting Excluded

Do not paint the following unless indicated otherwise.

- a. Surfaces concealed and made inaccessible by panelboards, fixed ductwork, machinery, and equipment fixed in place.
- b. Surfaces in concealed spaces. Concealed spaces are defined as enclosed spaces above suspended ceilings, furred spaces, attic spaces, crawl spaces, elevator shafts and chases.
- c. Steel to be embedded in concrete.
- d. Copper, stainless steel, aluminum, anodized aluminum, brass, and lead except existing coated surfaces.
- e. Hardware, fittings, and other factory finished items.

## 1.1.3 Mechanical and Electrical Painting

Includes field coating of interior new surfaces.

- a. Where a space or surface is indicated to be painted, include the



following items unless indicated otherwise.

- (1) Exposed piping, conduit, and ductwork;
- (2) Supports, hangers, air grilles, and registers;
- (3) Miscellaneous metalwork and insulation coverings.

#### 1.1.3.1 Fire Extinguishing Sprinkler Systems

Clean, pretreat, prime, and paint new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metalwork, and accessories. Apply coatings to clean, dry surfaces, using clean brushes.

#### 1.1.4 Miscellaneous Painting

##### 1.1.4.1 Lettering Building Number(s)

Provide lettering block type, black enamel. Samples must be approved before application.

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

|            |                                                                                                |
|------------|------------------------------------------------------------------------------------------------|
| ACGIH 0100 | (2017; Suppl 2020) Documentation of the Threshold Limit Values and Biological Exposure Indices |
|------------|------------------------------------------------------------------------------------------------|

#### ASTM INTERNATIONAL (ASTM)

|            |                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D235  | (2002; R 2012) Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent)                                                   |
| ASTM D523  | (2014; R 2018) Standard Test Method for Specular Gloss                                                                                  |
| ASTM D4263 | (1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method                                     |
| ASTM D4444 | (2013; R 2018) Standard Test Method for Laboratory Standardization and Calibration of Hand-Held Moisture Meters                         |
| ASTM D6386 | (2016a) Standard Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron and Steel Product and Hardware Surfaces for Painting |
| ASTM F1869 | (2023) Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride            |

## CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC)

Intelligence Bulletin 65                      (2013) Occupational Exposure to Carbon  
Nanotubes and Nanofibers

## MASTER PAINTERS INSTITUTE (MPI)

|         |                                                                                        |
|---------|----------------------------------------------------------------------------------------|
| MPI 3   | (2016) Primer, Alkali Resistant, Water Based                                           |
| MPI 9   | (2016) Alkyd, Exterior Gloss (MPI Gloss Level 6)                                       |
| MPI 47  | (2016) Alkyd, Interior, Semi-Gloss (MPI Gloss Level 5)                                 |
| MPI 48  | (2016) Alkyd, Interior, Gloss (MPI Gloss Level 6-7)                                    |
| MPI 49  | (2015) Alkyd, Interior, Flat (MPI Gloss Level 1)                                       |
| MPI 50  | (2015) Primer Sealer, Latex, Interior                                                  |
| MPI 51  | (2016) Alkyd, Interior, (MPI Gloss Level 3)2                                           |
| MPI 72  | (2016) Polyurethane, Two-Component, Pigmented, Gloss (MPI Gloss Level 6-7)             |
| MPI 76  | (2016) Primer, Alkyd, Quick Dry, for Metal                                             |
| MPI 77  | (2015) Epoxy, Gloss                                                                    |
| MPI 101 | (2016) Primer, Epoxy, Anti-Corrosive, for Metal                                        |
| MPI 107 | (2016) Primer, Rust-Inhibitive, Water Based                                            |
| MPI 138 | (2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 2)            |
| MPI 139 | (2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 3)            |
| MPI 141 | (2016) Latex, Interior, High Performance Architectural, Semi-Gloss (MPI Gloss Level 5) |
| MPI 144 | (2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 2)                |
| MPI 145 | (2016) Latex, Interior, Institutional Low Odor/VOC, ( MPI Gloss Level 3)               |
| MPI 147 | (May 2016) Latex, Interior, Institutional Low Odor/VOC, Semi-Gloss (MPI Gloss Level 5) |

|              |                                                                                           |
|--------------|-------------------------------------------------------------------------------------------|
| MPI 149      | (2016) Primer Sealer, Interior,<br>Institutional Low Odor/VOC                             |
| MPI 151      | (2016) Light Industrial Coating, Interior,<br>Water Based (MPI Gloss Level 3)             |
| MPI 153      | (2016) Light Industrial Coating, Interior,<br>Water Based, Semi-Gloss (MPI Gloss Level 5) |
| MPI 154      | (2016) Light Industrial Coating, Interior,<br>Water Based, Gloss (MPI Gloss Level 6)      |
| MPI ASM      | (2019) Architectural Painting<br>Specification Manual                                     |
| MPI GPS-1-14 | (2014) Green Performance Standard GPS-1-14                                                |
| MPI GPS-2-14 | (2014) Green Performance Standard GPS-2-14                                                |
| MPI MRM      | (2015) Maintenance Repainting Manual                                                      |

## SOCIETY FOR PROTECTIVE COATINGS (SSPC)

|                        |                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------|
| SSPC 7/NACE No.4       | (2007) Brush-Off Blast Cleaning                                                                                 |
| SSPC Glossary          | (2011) SSPC Protective Coatings Glossary                                                                        |
| SSPC PA 1              | (2024) Shop, Field, and Maintenance<br>Coating of Metals                                                        |
| SSPC SP 1              | (2015) Solvent Cleaning                                                                                         |
| SSPC SP 2              | (2018) Hand Tool Cleaning                                                                                       |
| SSPC SP 3              | (2018) Power Tool Cleaning                                                                                      |
| SSPC SP 6/NACE No.3    | (2007) Commercial Blast Cleaning                                                                                |
| SSPC SP 10/NACE No. 2  | (2015) Near-White Blast Cleaning                                                                                |
| SSPC VIS 1             | (2002; E 2004) Guide and Reference<br>Photographs for Steel Surfaces Prepared by<br>Dry Abrasive Blast Cleaning |
| SSPC VIS 3             | (2004) Guide and Reference Photographs for<br>Steel Surfaces Prepared by Hand and Power<br>Tool Cleaning        |
| SSPC VIS 4/NACE VIS 7  | (1998; E 2000; E 2004) Guide and Reference<br>Photographs for Steel Surfaces Prepared by<br>Waterjetting        |
| SSPC-SP WJ-1/NACE WJ-1 | (2012) Clean to Bare Substrate, Waterjet<br>Cleaning of Metals                                                  |
| SSPC-SP WJ-2/NACE WJ-2 | (2012) Very Thorough Cleaning, Waterjet<br>Cleaning of Metals                                                   |
| SSPC-SP WJ-3/NACE WJ-3 | (2012) Thorough Cleaning, Waterjet                                                                              |

## Cleaning of Metals

SSPC-SP WJ-4/NACE WJ-4

(2012) Light Cleaning, Waterjet Cleaning of Metals

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

(2024) Safety -- Safety and Health Requirements Manual

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FED-STD-313

(2018) Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000

Air Contaminants

## 1.3 DEFINITIONS

## 1.3.1 Qualification Testing

Qualification testing is the performance of all test requirements listed in the product specification. This testing is accomplished by MPI to qualify each product for the MPI Approved Product List, and may also be accomplished by Contractor's third-party testing lab if an alternative to Batch Quality Conformance Testing by MPI is desired.

## 1.3.2 Batch Quality Conformance Testing

Batch quality conformance testing determines that the product provided is the same as the product qualified to the appropriate product specification. This testing must be accomplished by an MPI testing lab.

## 1.3.3 Coating

SSPC Glossary; (1) A liquid, liquefiable, or mastic composition that is converted to a solid protective, decorative, or functional adherent film after application as a thin layer; (2) Generic term for paint, lacquer, enamel.

## 1.3.4 DFT or dft

Dry film thickness, the film thickness of the fully cured, dry paint or coating.

## 1.3.5 DSD

Degree of Surface Degradation, the MPI system of defining degree of surface degradation. Five levels are generically defined under the Assessment sections in the MPI MRM, MPI Maintenance Repainting Manual.

## 1.3.6 INT

MPI short term designation for an interior coating system.

### 1.3.7 Loose Paint

Paint or coating that can be removed with a dull putty knife.

### 1.3.8 micron / microns

The metric measurement for 0.001 mm or one one-thousandth of a millimeter.

### 1.3.9 mm

The metric measurement for millimeter, 0.001 meter or one one-thousandth of a meter.

### 1.3.10 MPI Gloss Levels

MPI system of defining gloss. Seven gloss levels (G1 to G7) are generically defined under the Evaluation sections of the MPI Manuals. Traditionally, Flat refers to G1/G2, Eggshell refers to G3, Semigloss refers to G5, and Gloss refers to G6.

Gloss levels are defined by MPI as follows:

| Gloss Level | Description   | Units at 60 degree angle | Units at 80 degree angle |
|-------------|---------------|--------------------------|--------------------------|
| G1          | Matte or Flat | 0 to 5                   | 10 max                   |
| G2          | Velvet        | 0 to 10                  | 10 to 35                 |
| G3          | Eggshell      | 10 to 25                 | 10 to 35                 |
| G4          | Satin         | 20 to 35                 | 35 min                   |
| G5          | Semi-Gloss    | 35 to 70                 |                          |
| G6          | Gloss         | 70 to 85                 |                          |
| G7          | High Gloss    |                          |                          |

Gloss is tested in accordance with ASTM D523. Historically, the Government has used Flat (G1 / G2), Eggshell (G3), Semi-Gloss (G5), and Gloss (G6).

### 1.3.11 MPI System Number

The MPI coating system number in each MPI Division found in either the MPI Architectural Painting Specification Manual or the Maintenance Repainting Manual and defined as an exterior (EXT/REX) or interior system (INT/RIN).

### 1.3.12 Paint

SSPC Glossary; (1) Any pigmented liquid, liquefiable, or mastic composition designed for application to a substrate in a thin layer that is converted to an opaque solid film after application. Used for protection, decoration, identification, or to serve some other functional purposes; (2) Application of a coating material.

### 1.3.13 RIN

MPI short term designation for an interior coating system used in repainting projects or over existing coating systems.

### 1.4 SCHEDULING

Allow paint, polyurethane, varnish, and wood stain installations to cure prior to the installation of materials that adsorb VOCs, including unprimed gypsum wallboard, and acoustical ceiling panels.

### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Samples of specified materials may be taken and tested for compliance with specification requirements.

#### SD-02 Shop Drawings

Piping Identification

#### SD-03 Product Data

Coating; G

Product Data Sheets

#### SD-04 Samples

Color; G

Textured Wall Coating System; G

#### SD-07 Certificates

Qualification Testing laboratory for coatings; G

Indoor Air Quality for Paints and Primers

#### SD-08 Manufacturer's Instructions

Application Instructions

Mixing

Manufacturer's Safety Data Sheets

#### SD-10 Operation and Maintenance Data

Coatings, Data Package 1; G

## 1.6 QUALITY ASSURANCE

### 1.6.1 Regulatory Requirements

#### 1.6.1.1 Environmental Protection

In addition to requirements specified elsewhere for environmental protection, provide coating materials that conform to the restrictions of the local Air Pollution Control District and regional jurisdiction. Notify Contracting Officer of any paint specified herein which fails to conform.

#### 1.6.1.2 Lead Content

Do not use coatings having a lead content over 0.06 percent by weight of nonvolatile content.

#### 1.6.1.3 Chromate Content

Do not use coatings containing zinc-chromate or strontium-chromate.

#### 1.6.1.4 Asbestos Content

Provide asbestos-free materials.

#### 1.6.1.5 Mercury Content

Provide materials free of mercury or mercury compounds.

#### 1.6.1.6 Silica

Provide abrasive blast media containing no free crystalline silica.

#### 1.6.1.7 Human Carcinogens

Provide materials that do not contain ACGIH 0100 confirmed human carcinogens (A1) or suspected human carcinogens (A2).

#### 1.6.1.8 Carbon Based Fibers / Tubes

Materials must not contain carbon based fibers such as carbon nanotubes or carbon nanofibers. Intelligence Bulletin 65 ranks toxicity of carbon nanotubes on a par with asbestos.

### 1.6.2 Coating Contractor's Qualification

Submit the name, address, telephone number, and e-mail address of the Contractor that will be performing all surface preparation and coating application. Submit evidence that key personnel have successfully performed surface preparation and application of coatings on concrete on a minimum of three similar projects within the past three years. List information by individual and include the following:

- a. Name of individual and proposed position for this work.
- b. Information about each previous assignment including:

Position or responsibility

Employer (if other than the Contractor)

Name of facility owner

Mailing address and telephone number of facility owner

Name of individual in facility owner's organization who can be contacted as a reference

Location, size and description of structure

Dates work was carried out

Description of work carried out on structure

#### 1.6.3 Approved Products List

The current MPI, "Approved Product List" which lists paint by brand, label, product name and product code as of the date of Contract award, will be used to determine compliance with the submittal requirements of this specification. The Contractor may choose to use a subsequent MPI "Approved Product List", however, only one list may be used for the entire Contract and each coating system is to be from a single manufacturer. Provide all coats on a particular substrate from a single manufacturer. No variation from the MPI Approved Products List is acceptable.

#### 1.6.4 Paints and Coatings Indoor Air Quality Certifications

Provide paint and coating products certified to meet indoor air quality requirements by MPI GPS-1-14, MPI GPS-2-14 or provide certification by other third-party programs. Provide current product certification documentation from certification body.

Provide certification of Indoor Air Quality for Paints and Primers. Submit required indoor air quality certifications in one submittal package.

#### 1.6.5 Field Samples and Tests

The Contracting Officer may choose up to two coatings that have been delivered to the site to be tested at no cost to the Government. Take samples of each chosen product as specified in the paragraph SAMPLING PROCEDURE. Test each chosen product as specified in the paragraph TESTING PROCEDURE. Remove products from the job site which do not conform, and replace with new products that conform to the referenced specification. Test replacement products that failed initial testing as specified in the paragraph TESTING PROCEDURE at no cost to the Government.

##### 1.6.5.1 Sampling Procedure

Select paint at random from the products that have been delivered to the job site for sample testing. The Contractor must provide one liter samples of the selected paint materials. Take samples in the presence of the Contracting Officer, and label, and identify each sample. Provide labels in accordance with the paragraph PACKAGING, LABELING, AND STORAGE.

##### 1.6.5.2 Testing Procedure

Provide Batch Quality Conformance Testing for specified products, as



defined by and performed by MPI. As an alternative to Batch Quality Conformance Testing, the Contractor may provide Qualification Testing for specified products above to the appropriate MPI product specification, using the third-party laboratory approved under the paragraph QUALIFICATION TESTING laboratory for coatings. Include the backup data and summary of the test results within the qualification testing lab report. Provide a summary listing of all the reference specification requirements and the result of each test. Clearly indicate in the summary whether the tested paint meets each test requirement. Note that Qualification Testing may take 4 to 6 weeks to perform, due to the extent of testing required.

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party laboratory selected to perform testing of coating samples for compliance with specification requirements. Submit documentation that laboratory is regularly engaged in testing of paint samples for conformance with specifications, and that employees performing testing are qualified. If MPI is chosen to perform the Batch Quality Conformance testing, the above submittal information is not required, only a letter is required from the Contractor stating that MPI will perform the testing.

#### 1.7 PACKAGING, LABELING, AND STORAGE

Provide paints in sealed containers that legibly show the Contract specification number, designation name, formula or specification number, batch number, color, quantity, date of manufacture, manufacturer's formulation number, manufacturer's directions including any warnings and special precautions, and name and address of manufacturer. Furnish pigmented paints in containers not larger than 20 liters. Store paints and thinners in accordance with the manufacturer's written directions, and as a minimum, stored off the ground, under cover, with sufficient ventilation to prevent the buildup of flammable vapors, and at temperatures between 4 to 35 degrees C. Do not store paint, polyurethane, varnish, or wood stain products with materials that have a high capacity to absorb VOC emissions. Do not store paint, polyurethane, varnish, or wood stain products in occupied spaces.

#### 1.8 SAFETY AND HEALTH

Comply with applicable Federal, State, and local laws and regulations, and with the ACCIDENT PREVENTION PLAN, including the Activity Hazard Analysis as specified in Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and in Appendix A of EM 385-1-1. Include in the Activity Hazard Analysis the potential impact of painting operations on painting personnel and on others involved in and adjacent to the work zone.

##### 1.8.1 Toxic Materials

To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

- a. The applicable manufacturer's Safety Data Sheets (SDS) or local regulation.
- b. 29 CFR 1910.1000.
- c. ACGIH 0100, threshold limit values.

Submit manufacturer's Safety Data Sheets for coatings, solvents, and other potentially hazardous materials, as defined in FED-STD-313.

## 1.9 ENVIRONMENTAL REQUIREMENTS

Comply, at minimum, with manufacturer recommendations for space ventilation during and after installation. Isolate area of application from rest of building when applying high-emission paints or coatings.

### 1.9.1 Coatings

Do not apply coating when air or substrate conditions are:

- a. Less than 3 degrees C above dew point;
- b. Below 10 degrees C or over 35 degrees C, unless specifically pre-approved by the Contracting Officer and the product manufacturer. Do not, under any circumstances, violate the manufacturer's application recommendations.

### 1.9.2 Post-Application

Vacate space for as long as possible after application. Wait a minimum of 48 hours before occupying freshly painted rooms. Maintain one of the following ventilation conditions during the curing period, or for 72 hours after application:

- a. Supply 100 percent outside air 24 hours a day.
- b. Supply airflow at a rate of 6 air changes per hour, when outside temperatures are between 13 degrees C and 29 degrees C and humidity is between 30 percent and 60 percent.
- c. Supply airflow at a rate of 1.5 air changes per hour, when outside air conditions are not within the range stipulated above.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Conform to the coating specifications and standards referenced in PART 3. Submit Product Data Sheets for specified coatings and solvents. Provide preprinted cleaning and maintenance instructions for all coating systems. Submit Manufacturer's Instructions on Mixing: Detailed mixing instructions, minimum and maximum application temperature and humidity, pot life, and curing and drying times between coats.

### 2.2 COLOR SELECTION OF FINISH COATS

Provide colors of finish coats as indicated or specified. Allow Contracting Officer to select colors not indicated or specified. Manufacturers' names and color identification are used for the purpose of color identification only. Named products are acceptable for use only if they conform to specified requirements. Products of other manufacturers are acceptable if the colors are approximately the colors indicated and the product conforms to specified requirements.

Provide color, texture, and pattern of wall coating systems as indicated. Submit manufacturer's samples of paint colors. Cross reference color

samples to color scheme as indicated. Submit color stencil codes. Tint each coat progressively darker to enable confirmation of the number of coats.

### PART 3 EXECUTION

#### 3.1 PROTECTION OF AREAS AND SPACES NOT TO BE PAINTED

Prior to surface preparation and coating applications, remove, mask, or otherwise protect hardware, hardware accessories, machined surfaces, radiator covers, plates, lighting fixtures, public and private property, and other such items not to be coated that are in contact with surfaces to be coated. Following completion of painting, reinstall removed items by workmen skilled in the trades. Restore surfaces contaminated by coating materials, to original condition and repair damaged items.

#### 3.2 SURFACE PREPARATION

Remove dirt, splinters, loose particles, grease, oil, and other foreign matter and substances deleterious to coating performance as specified for each substrate before application of paint or surface treatments. Remove oil and grease prior to mechanical cleaning. Schedule cleaning so that dust and other contaminants will not fall on wet, newly painted surfaces. Spot-prime exposed ferrous metals such as nail heads on or in contact with surfaces to be painted with water-thinned paints, with a suitable corrosion-inhibitive primer capable of preventing flash rusting and compatible with the coating specified for the adjacent areas. Refer to MPI ASM and MPI MRM for additional more specific substrate preparation requirements.

##### 3.2.1 Substrate Repair

- a. Repair substrate surface damaged during coating removal;
- b. Sand edges of adjacent soundly-adhered existing coatings so they are tapered as smooth as practical to areas involved with coating removal; and
- c. Clean and prime the substrate as specified.

#### 3.3 PREPARATION OF METAL SURFACES

##### 3.3.1 Existing and New Ferrous Surfaces

- a. Ferrous Surfaces including Shop-coated Surfaces and Small Areas That Contain Rust, Mill Scale and Other Foreign Substances: Solvent clean or detergent wash in accordance with SSPC SP 1 to remove oil and grease. Where shop coat is missing or damaged, clean according to SSPC SP 2, SSPC SP 3, SSPC SP 6/NACE No.3, or SSPC SP 10/NACE No. 2. Brush-off blast remaining surface in accordance with SSPC 7/NACE No.4; Water jetting to SSPC-SP WJ-4/NACE WJ-4 may be used to remove loose coating and other loose materials. Use inhibitor as recommended by coating manufacturer to prevent premature rusting. Protect shop-coated ferrous surfaces from corrosion by treating and touching up corroded areas immediately upon detection.

### 3.3.2 Final Ferrous Surface Condition:

#### 3.3.2.1 Tool Cleaned Surfaces

Comply with SSPC SP 2 and SSPC SP 3. Use as a visual reference, photographs in SSPC VIS 3 for the appearance of cleaned surfaces.

#### 3.3.2.2 Abrasive Blast Cleaned Surfaces

Comply with SSPC 7/NACE No.4, SSPC SP 6/NACE No.3, and SSPC SP 10/NACE No. 2. Use as a visual reference, photographs in SSPC VIS 1 for the appearance of cleaned surfaces.

#### 3.3.2.3 Waterjet Cleaned Surfaces

Comply with SSPC-SP WJ-1/NACE WJ-1, SSPC-SP WJ-2/NACE WJ-2, SSPC-SP WJ-3/NACE WJ-3 or SSPC-SP WJ-4/NACE WJ-4. Use as a visual reference, photographs in SSPC VIS 4/NACE VIS 7 for the appearance of cleaned surfaces.

### 3.3.3 Galvanized Surfaces

- a. New or Existing Galvanized Surfaces With Only Dirt and Zinc Oxidation Products: Clean with solvent, steam, or non-alkaline detergent solution in accordance with SSPC SP 1. Completely remove coating by brush-off abrasive blast if the galvanized metal has been passivated or stabilized. Do not "passivate" or "stabilize" new galvanized steel to be coated. If the absence of hexavalent stain inhibitors is not documented, test as described in ASTM D6386, Appendix X2, and remove by one of the methods described therein.

### 3.3.4 Non-Ferrous Metallic Surfaces

Aluminum and aluminum-alloy, lead, copper, and other nonferrous metal surfaces.

Surface Cleaning: Solvent clean in accordance with SSPC SP 1 and wash with mild non-alkaline detergent to remove dirt and water soluble contaminants.

### 3.3.5 Terne-Coated Metal Surfaces

Solvent clean surfaces with mineral spirits, ASTM D235. Wipe dry with clean, dry cloths.

## 3.4 PREPARATION OF CONCRETE AND CEMENTITIOUS SURFACE

### 3.4.1 Concrete and Masonry

- a. Curing: Allow concrete, stucco and masonry surfaces to cure at least 30 days before painting, and concrete slab on grade to cure at least 90 days before painting.
- b. Surface Cleaning: Remove the following deleterious substances.

- (1) Dirt, Grease, and Oil: Wash new surfaces with a solution composed of 0.2 liter trisodium phosphate, 0.1 liter household detergent, and 6.4 liters of warm water. Then rinse thoroughly with fresh water. For large areas, water blasting may be used.
  - (2) Fungus and Mold: Wash new surfaces with a solution composed of 0.2 liter trisodium phosphate, 0.1 liter household detergent, 1.6 liters 5 percent sodium hypochlorite solution and 4.8 liters of warm water. Rinse thoroughly with fresh water.
  - (3) Paint and Loose Particles: Remove by wire brushing.
  - (4) Efflorescence: Remove by scraping or wire brushing followed by washing with a 5 to 10 percent by weight aqueous solution of hydrochloric (muriatic) acid. Do not allow acid to remain on the surface for more than five minutes before rinsing with fresh water. Do not acid clean more than 0.4 square meter of surface, per workman, at one time.
- c. Cosmetic Repair of Minor Defects: Repair or fill mortar joints and minor defects, including but not limited to spalls, in accordance with manufacturer's recommendations and prior to coating application.
- d. Allowable Moisture Content: Latex coatings may be applied to damp surfaces, but not to surfaces with droplets of water. Do not apply epoxies to damp vertical surfaces as determined by ASTM D4263 or horizontal surfaces that exceed 169.0 micrograms moisture per second, square meter as determined by ASTM F1869. In all cases follow manufacturer's recommendations. Allow surfaces to cure a minimum of 30 days before painting.

### 3.4.2 Gypsum Board, Plaster, and Stucco

#### 3.4.2.1 Surface Cleaning

Verify that plaster and stucco surfaces are free from loose matter and that gypsum board is dry. Remove loose dirt and dust by brushing with a soft brush, rubbing with a dry cloth, or vacuum-cleaning prior to application of the first coat material. A damp cloth or sponge may be used if paint is water-based.

#### 3.4.2.2 Repair of Minor Defects

Prior to painting, repair joints, cracks, holes, surface irregularities, and other minor defects with patching plaster or spackling compound and sand smooth.

#### 3.4.2.3 Allowable Moisture Content

Latex coatings may be applied to damp surfaces, but not surfaces with droplets of water. Do not apply epoxies to damp surfaces as determined by ASTM D4263. Verify that new plaster to be coated has a maximum moisture content of 8 percent, when measured in accordance with ASTM D4444, Method A, unless otherwise authorized. In addition to moisture content requirements, allow new plaster to age a minimum of 30 days before preparation for painting.

### 3.5 APPLICATION

#### 3.5.1 Coating Application

- a. Comply with applicable federal, state and local laws enacted to ensure compliance with Federal Clean Air Standards. Apply coating materials in accordance with SSPC PA 1. SSPC PA 1 methods are applicable to all substrates, except as modified herein.
- b. At the time of application, paint must show no signs of deterioration. Maintain uniform suspension of pigments during application.
- c. Unless otherwise specified or recommended by the paint manufacturer, paint may be applied by brush, roller, or spray. Use trigger operated spray nozzles for water hoses. Use rollers for applying paints and enamels of a type designed for the coating to be applied and the surface to be coated. Wear protective clothing and respirators when applying oil-based paints or using spray equipment with any paints.
- d. Only apply paints, except water-thinned types, to surfaces that are completely free of moisture as determined by sight or touch.
- e. Thoroughly work coating materials into joints, crevices, and open spaces. Pay special attention to ensure that all edges, corners, crevices, welds, and rivets receive a film thickness equal to that of adjacent painted surfaces.
- f. Apply each coat of paint so that dry film is of uniform thickness and free from runs, drops, ridges, waves, pinholes or other voids, laps, brush marks, and variations in color, texture, and finish. Completely hide all blemishes.
- g. Touch up damaged coatings before applying subsequent coats. Broom clean and clear dust from interior areas before and during the application of coating material.
- h. Apply paint to new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metal work, and accessories. Shield sprinkler heads with protective coverings while painting is in progress. Remove sprinkler heads which have been painted and replace with new sprinkler heads. Unfinished spaces include attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and space where walls or ceiling are not painted or not constructed of a prefinished material. Upon completion of painting, remove protective covering from sprinkler heads.
- i. Piping in Unfinished Areas: Provide primed surfaces with one coat of red alkyd gloss enamel (MPI 9) applied to a minimum dry film thickness of 0.025 mm in attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and spaces where walls or ceiling are not painted or not constructed of a prefinished material.
- j. Piping in Finished Areas: Provide primed surfaces with two coats of paint to match adjacent surfaces, except provide valves and operating accessories with one coat of red alkyd gloss enamel (MPI 9) applied to a minimum dry film thickness of 0.025 mm or two component gloss

polyurethane (MPI 72) in exterior applications.

- k. Provide labeling on the surfaces of all feed and cross mains to show the pipe function such as "Sprinkler System", "Fire Department Connection", "Standpipe". For pipe sizes 100 mm and larger provide white painted stenciled letters and arrows, a minimum of 50 mm in height and visible from at least two sides when viewed from the floor. For pipe sizes less than 100 mm, provide white painted stenciled letters and arrows, a minimum of 18 mm in height and visible from the floor.
- l. All fire suppression system valves must be marked with permanent tags indicating normally open or normally closed.
- m. Drying Time: Allow time between coats, as recommended by the coating manufacturer, to permit thorough drying, but not to present topcoat adhesion problems. Provide each coat in specified condition to receive next coat.
- n. Primers, and Intermediate Coats: Do not allow primers or intermediate coats to dry more than 30 days, or longer than recommended by manufacturer, before applying subsequent coats. Follow manufacturer's recommendations for surface preparation if primers or intermediate coats are allowed to dry longer than recommended by manufacturers of subsequent coatings. Cover each preceding coat or surface completely by ensuring visually perceptible difference in shades of successive coats.
- o. Finished Surfaces: Provide finished surfaces free from runs, drops, ridges, waves, laps, brush marks, and variations in colors.
- p. Thermosetting Paints: Apply topcoats over thermosetting paints (epoxies and urethanes) within the overcoat window recommended by the manufacturer.

### 3.5.2 Mixing and Thinning of Paints

Reduce paints to proper consistency by adding fresh paint, except when thinning is mandatory to suit surface, temperature, weather conditions, application methods, or for the type of paint being used. Obtain written permission from the Contracting Officer to use thinners. Verify that the written permission includes quantities and types of thinners to use.

When thinning is allowed, thin paints immediately prior to application with not more than 0.125 L of suitable thinner per liter. The use of thinner does not relieve the Contractor from obtaining complete hiding, full film thickness, or required gloss. Thinning cannot cause the paint to exceed limits on volatile organic compounds. Do not mix paints of different manufacturers.

### 3.5.3 Two-Component Systems

Mix two-component systems in accordance with manufacturer's instructions. Follow recommendation by the manufacturer for any thinning of the first coat to ensure proper penetration and sealing for each type of substrate.

### 3.5.4 Coating Systems

- a. Systems by Substrates: Apply coatings that conform to the respective specifications listed in the following Tables:

| Table for Interior Applications |                                                               |
|---------------------------------|---------------------------------------------------------------|
| MPI Division                    | Substrate Application                                         |
| MPI Division 3                  | Interior Concrete Paint Table                                 |
| MPI Division 4                  | Interior Concrete Masonry Units Paint Table                   |
| MPI Division 5                  | Interior Metal, Ferrous and Non-Ferrous Paint Table           |
| MPI Division 6                  | Interior Wood Paint Table                                     |
| MPI Division 9                  | Interior Plaster, Gypsum Board, Textured Surfaces Paint Table |

- b. Minimum Dry Film Thickness (DFT): Apply paints, primers, varnishes, enamels, undercoats, and other coatings to a minimum dry film thickness of 0.038 mm each coat unless specified otherwise in the Tables. Coating thickness, where specified, refers to the minimum dry film thickness.
- c. Coatings for Surfaces Not Specified Otherwise: Coat unspecified surfaces the same as surfaces having similar conditions of exposure.
- d. Existing Surfaces Damaged During Performance of the Work, Including New Patches In Existing Surfaces: Coat surfaces with the following:
- (1) One coat of primer.
  - (2) One coat of undercoat or intermediate coat.
  - (3) One topcoat to match adjacent surfaces.
- e. Existing Coated Surfaces To Be Painted: Apply coatings conforming to the respective specifications listed in the Tables herein, except that pretreatments, sealers and fillers need not be provided on surfaces where existing coatings are soundly adhered and in good condition. Do not omit undercoats or primers.

### 3.6 COATING SYSTEMS FOR METAL

Apply coatings of Tables in MPI Division 5 for Exterior and Interior.

- a. Apply specified ferrous metal primer to steel surfaces on the same day that surface is cleaned, to surfaces that meet all specified surface preparation requirements at time of application.
- b. Inaccessible Surfaces: Prior to erection, use one coat of specified primer on metal surfaces that will be inaccessible after erection.
- c. Shop-primed Surfaces: Touch up exposed substrates and damaged coatings to protect from rusting prior to applying field primer.



- d. Surface Previously Coated with Epoxy or Urethane: Apply MPI 101, 0.038 mm DFT immediately prior to application of epoxy or urethane coatings.
- e. Pipes and Tubing: The semitransparent film applied to some pipes and tubing at the mill is not to be considered a shop coat. Overcoat these items with the specified ferrous-metal primer prior to application of finish coats.
- f. Exposed Nails, Screws, Fasteners, and Miscellaneous Ferrous Surfaces. On surfaces to be coated with water thinned coatings, spot prime exposed nails and other ferrous metal with latex primer MPI 107.

### 3.7 COATING SYSTEMS FOR CONCRETE AND CEMENTITIOUS SUBSTRATES

Apply coatings of Tables in MPI Division 3, 4 and 9 for Exterior and Interior.

### 3.8 INSPECTION AND ACCEPTANCE

In addition to meeting previously specified requirements, demonstrate mobility of moving components, including swinging and sliding doors, cabinets, and windows with operable sash, for inspection by the Contracting Officer. Perform this demonstration after appropriate curing and drying times of coatings have elapsed and prior to invoicing for final payment.

### 3.9 WASTE MANAGEMENT

As specified in the Waste Management Plan and as follows. Do not use kerosene or any such organic solvents to clean up water based paints. Properly dispose of paints or solvents in designated containers. Close and seal partially used containers of paint to maintain quality as necessary for reuse. Store in protected, well-ventilated, fire-safe area at moderate temperature. Place materials defined as hazardous or toxic waste in designated containers. Where local options exist for leftover paint recycling, collect all waste paint by type and provide for delivery to recycling or collection facility for reuse by local organizations.

### 3.10 PAINT TABLES

All DFT's are minimum values. Acceptable products are listed in the MPI Green Approved Products List, available at <http://www.specifygreen.com/APL/ProductIdxByMPInum.asp>.

#### 3.10.1 Interior Paint Tables

##### 3.10.1.1 MPI Division 3: Interior Concrete Paint Table

- A. New and uncoated existing Concrete, vertical surfaces, not specified otherwise

|                                      |
|--------------------------------------|
| High Performance Architectural Latex |
|--------------------------------------|

| New, uncoated<br>Existing                    | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT  |
|----------------------------------------------|---------------------------------|--------|--------------|---------|-------------|
| MPI INT 3.1C-G2<br>(Flat)                    | MPI RIN 3.1J-G2<br>(Flat)       | MPI 3  | MPI 138      | MPI 138 | 100 microns |
| MPI INT 3.1C-G3<br>(Eggshell)                | MPI RIN 3.1J-G3<br>(Eggshell)   | MPI 3  | MPI 139      | MPI 139 | 100 microns |
| MPI INT 3.1C-G5<br>(Semigloss)               | MPI RIN 3.1J-G5<br>(Semigloss)  | MPI 3  | MPI 141      | MPI 141 | 100 microns |
| Topcoat: Coating to match adjacent surfaces. |                                 |        |              |         |             |

| Institutional Low Odor / Low VOC Latex       |                                 |         |              |         |             |
|----------------------------------------------|---------------------------------|---------|--------------|---------|-------------|
| New, uncoated<br>Existing                    | Existing,<br>previously painted | Primer  | Intermediate | Topcoat | System DFT  |
| MPI INT 3.1M-G2<br>(Flat)                    | MPI RIN 3.1L-G2<br>(Flat)       | MPI 149 | MPI 144      | MPI 144 | 100 microns |
| MPI INT 3.1M-G3<br>(Eggshell)                | MPI RIN 3.1L-G3<br>(Eggshell)   | MPI 149 | MPI 145      | MPI 145 | 100 microns |
| MPI INT 3.1M-G5<br>(Semigloss)               | MPI RIN 3.1L-G5<br>(Semigloss)  | MPI 149 | MPI 147      | MPI 147 | 100 microns |
| Topcoat: Coating to match adjacent surfaces. |                                 |         |              |         |             |

C. New and uncoated existing Concrete in toilets, shower areas not otherwise specified except floors

| Waterborne Light Industrial Coating          |                                 |        |              |         |             |
|----------------------------------------------|---------------------------------|--------|--------------|---------|-------------|
| New, uncoated<br>Existing                    | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT  |
| MPI INT 3.1L-G3(Eggshell)                    | MPI RIN 3.1C-G3(Eggshell)       | MPI 3  | MPI 151      | MPI 151 | 120 microns |
| MPI INT 3.1L-G5(Semigloss)                   | MPI RIN 3.1C-G5(Semigloss)      | MPI 3  | MPI 153      | MPI 153 | 120 microns |
| MPI INT 3.1L-G6(Gloss)                       | MPI RIN 3.1C-G6(Gloss)          | MPI 3  | MPI 154      | MPI 154 | 120 microns |
| Topcoat: Coating to match adjacent surfaces. |                                 |        |              |         |             |

| Alkyd                                        |                                 |        |              |         |             |
|----------------------------------------------|---------------------------------|--------|--------------|---------|-------------|
| New, uncoated<br>Existing                    | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT  |
| MPI INT 3.1D-G3<br>(Eggshell)                | MPI RIN 3.1D-G3<br>(Eggshell)   | MPI 3  | MPI 51       | MPI 51  | 112 microns |
| MPI INT 3.1D-G5<br>(Semigloss)               | MPI RIN 3.1D-G5<br>(Semigloss)  | MPI 3  | MPI 47       | MPI 47  | 112 microns |
| MPI INT 3.1D-G6<br>(Gloss)                   | MPI RIN 3.1D-G6<br>(Gloss)      | MPI 3  | MPI 48       | MPI 48  | 112 microns |
| Topcoat: Coating to match adjacent surfaces. |                                 |        |              |         |             |

| Epoxy                                                                        |                                 |        |              |         |             |
|------------------------------------------------------------------------------|---------------------------------|--------|--------------|---------|-------------|
| New, uncoated<br>Existing                                                    | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT  |
| MPI INT 3.1F-G6<br>(Gloss)                                                   | MPI RIN 3.1E-G6<br>(Gloss)      | MPI 77 | MPI 77       | MPI 77  | 100 microns |
| Note: Primer may be reduced for penetration per manufacturer's instructions. |                                 |        |              |         |             |

### 3.10.1.2 MPI Division 5: Interior Metal, Ferrous and Non-Ferrous Paint Table

#### A. Interior Steel / Ferrous Surfaces

(1) Metal, Mechanical, Electrical, Fire extinguishing sprinkler systems including valves, conduit, hangers, supports, Surfaces adjacent to painted surfaces (Match surrounding finish), and miscellaneous metal items not otherwise specified except floors, hot metal surfaces, and new prefinished equipment

| High Performance Architectural Latex         |        |              |         |             |
|----------------------------------------------|--------|--------------|---------|-------------|
| New, uncoated Existing                       | Primer | Intermediate | Topcoat | System DFT  |
| MPI INT 5.1R-G2 (Flat)                       | MPI 76 | MPI 138      | MPI 138 | 125 microns |
| MPI INT 5.1R-G3<br>(Eggshell)                | MPI 76 | MPI 139      | MPI 139 | 125 microns |
| MPI INT 5.1R-G5<br>(Semigloss)               | MPI 76 | MPI 141      | MPI 141 | 125 microns |
| Topcoat: Coating to match adjacent surfaces. |        |              |         |             |

| Alkyd                                        |        |              |         |             |
|----------------------------------------------|--------|--------------|---------|-------------|
| New, uncoated Existing                       | Primer | Intermediate | Topcoat | System DFT  |
| MPI INT 5.1E-G2 (Flat)                       | MPI 76 | MPI 49       | MPI 49  | 131 microns |
| MPI INT 5.1E-G3 (Eggshell)                   | MPI 76 | MPI 51       | MPI 51  | 131 microns |
| MPI INT 5.1E-G5 (Semigloss)                  | MPI 76 | MPI 47       | MPI 47  | 131 microns |
| MPI INT 5.1E-G6 (Gloss)                      | MPI 76 | MPI 48       | MPI 48  | 131 microns |
| Topcoat: Coating to match adjacent surfaces. |        |              |         |             |

### 3.10.1.3 MPI Division 9: Interior Plaster, Gypsum Board, Textured Surfaces Paint Table

#### A. Interior New Wallboard not otherwise specified

| High Performance Architectural Latex - High Traffic Areas |                              |        |              |         |             |
|-----------------------------------------------------------|------------------------------|--------|--------------|---------|-------------|
| New                                                       | Existing, previously painted | Primer | Intermediate | Topcoat | System DFT  |
| MPI INT 9.2B-G2 (Flat)                                    | MPI RIN 9.2B-G2 (Flat)       | MPI 50 | MPI 138      | MPI 138 | 100 microns |
| MPI INT 9.2B-G3 (Eggshell)                                | MPI RIN 9.2B-G3 (Eggshell)   | MPI 50 | MPI 139      | MPI 139 | 100 microns |
| MPI INT 9.2B-G5 (Semigloss)                               | MPI RIN 9.2B-G5 (Semigloss)  | MPI 50 | MPI 141      | MPI 141 | 100 microns |
| Topcoat: Coating to match adjacent surfaces.              |                              |        |              |         |             |

#### Institutional Low Odor / Low VOC Latex, New

| Institutional Low Odor / Low VOC Latex |         |              |         |             |
|----------------------------------------|---------|--------------|---------|-------------|
| New                                    | Primer  | Intermediate | Topcoat | System DFT  |
| MPI INT 9.2M-G2 (Flat)                 | MPI 149 | MPI 144      | MPI 144 | 100 microns |

|                                              |         |         |         |             |
|----------------------------------------------|---------|---------|---------|-------------|
| MPI INT 9.2M-G3<br>(Eggshell)                | MPI 149 | MPI 145 | MPI 145 | 100 microns |
| MPI INT 9.2M-G5<br>(Semigloss)               | MPI 149 | MPI 147 | MPI 147 | 100 microns |
| Topcoat: Coating to match adjacent surfaces. |         |         |         |             |

-- End of Section --

## SECTION 09 96 00

## HIGH-PERFORMANCE COATINGS

**11/14**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## MASTER PAINTERS INSTITUTE (MPI)

MPI ASM (2019) Architectural Painting  
Specification Manual

## SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC 7/NACE No.4 (2007) Brush-Off Blast Cleaning

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-01 Preconstruction Submittals

Equipment List; G

## SD-03 Product Data

Epoxy Coatings; G

Polyurethane Coatings; G

Chlorinated-Rubber Coatings; G

Static Control Concrete Sealer; G

## SD-04 Samples

Color Chips; G

## SD-07 Certificates

Epoxy Coatings; G

Polyurethane Coatings; G

Chlorinated-Rubber Coatings; G

Manufacturer's Printed Instructions; G

### 1.3 QUALITY CONTROL

Comply with Master Painters Institute (MPI) Standards indicated and listed in "MPI Approved Products List." Comply with the requirements in "MPI Architectural Painting Specification Manual" before any project is started.

Submit an equipment list consisting of a list of proposed equipment to be used in performance of construction work.

Submit three color chips 75 millimeter by 100 millimeter or manufacture's pull-down of each finish color and gloss as scheduled.

### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver special coating materials to the project in their original containers bearing manufacturer's name, descriptive label, and coating formulations. Provide new and unopened containers.

Store special coating materials in tightly closed containers in a covered, well-ventilated area where they are not exposed to excessive heat, fumes, sparks, flame, or direct sunlight. Protect water-based coatings against freezing.

Store solvents, thinners, and equipment cleaners with the same care as the coating materials with ambient temperatures continuously maintained at a minimum 7 degrees C.

## PART 2 PRODUCTS

### 2.1 PERFORMANCE REQUIREMENTS

Submit manufacturer's catalog data including manufacturer's name and identification. Include detailed data analysis of each special coating material required for the project, with all the coating constituents measured as percentages of the total weight of the coating. Also provide manufacturer's data concerning application, thinning, and average coverage per liter

### 2.2 MATERIALS

#### 2.2.1 Static Control Concrete Sealer

Base coat of hygroscopic static control sealer. Apply enhanced urethane sealer. Finish coating color is translucent.

##### 2.2.1.1 Concrete Surface Coatings

Apply a minimum three coats on concrete surfaces. Provide prime coats as recommended by the coating manufacturer for the substrate to be coated and the end use of the coated surfaces. Ensure the prime coat fills concrete surface pores. Finish coats are urethane coating as specified.

## PART 3 EXECUTION

### 3.1 PREPARATION

#### 3.1.1 Surface Preparation

Protect adjacent materials and equipment against damage from spillage,

dripping, and spatter of coating materials. Leave clean building materials and equipment with all damaged surfaces corrected. Provide "WET PAINT" signs to indicate newly painted surfaces.

Protect work of other trades against damage from coating operation. Correct damage by cleaning, repairing, replacing, and recoating, as approved by the Contracting Officer, and leave in an undamaged condition. At completion of construction activities of other trades, touch up and restore damaged or defaced coated surfaces.

Provide forced ventilation for interior spaces during application and drying of coatings to prevent the buildup of toxic or explosive concentrations of solvent vapors.

Provide fire extinguishers of the required quantity and correct type to combat flammable liquid fires.

Dispose of rags that are used to wipe up coating materials, solvents, and thinners by drenching with water and placing them in a covered metal container

### 3.1.2 Cleaning

At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.

After completing coating application, clean spattered surfaces. Remove spattered coatings by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.

Clean application equipment promptly and thoroughly with a suitable solvent after each use and stored in a clean, covered, well-ventilated container.

### 3.1.3 Concrete Surfaces

Conform to MPI ASM for substrates indicated. Remove plates, machined surfaces, and similar items already in place that are not to be coated. Provide surface-applied protection before surface preparation and coating where removal is impractical or impossible. After completing coating operations, reinstall items that were removed.

Clean dirt, oil, grease, and incompatible paints from substrates to ensure bonding. Coordination of shop-applied prime coats with high-performance coatings is critical. Remove incompatible primers. Reprime substrate with compatible primers as required to produce coating systems indicated.

#### 3.1.3.1 Concrete Substrates

Remove release agents, curing compounds, efflorescence, and chalk. Maximum allowable moisture content of concrete is 12 percent. Measure moisture content with an electronic moisture meter.

Clean surfaces with pressurized water. Use pressure range of 10 350 to 27 580 kPa at 150 mm to 300 mm.

Comply with SSPC 7/NACE No.4 (NACE No. 4), "Brush-Off Blast Cleaning" for abrasive cleaning.



### 3.1.3.2 Galvanized-Metal Substrates

Remove grease and oil residue from galvanized sheet metal fabricated from coil stock by mechanical methods to produce clean, lightly etched surfaces that promote adhesion of subsequently applied coatings.

### 3.1.3.3 Aluminum Substrates

Remove surface oxidation.

### 3.1.4 Coating Material Preparation

Mix and prepare coating materials in accordance with the coating manufacturer's printed instructions for applying the particular material and coat. Keep materials which are not in actual use in closed containers.

Coating materials that have been mixed with an automatic shaker are allowed to stand to let air bubbles escape, then given a final hand mixing before application. Stir materials so as to produce a mixture of uniform density. Stir at frequent intervals during application to prevent skinning. Do not stir film which may form on the surface of the material. Remove film and strain, if necessary.

#### 3.1.4.1 Thinning

Thinning is done in accordance with coating manufacturer's printed directions for the particular material and coat.

#### 3.1.4.2 Tinting

Ensure prime and intermediate coats of paint are slightly different tints from the finish coat to facilitate identification of each coat. Tinting is done by the coating manufacturer and clearly identified as to color and coat.

## 3.2 APPLICATION

Do not perform exterior painting in damp or rainy weather. Interior painting is not allowed until the building is enclosed and has thoroughly dried out. Painting is not allowed below 10 degrees C or above 35 degrees C. Apply paint in accordance with the coating manufacturer's recommendations, and as specified.

Ensure coating application is done by skilled applicators. Apply coatings to clean and properly prepared surfaces. Apply coatings with clean, high-quality application equipment. Allow sufficient time between coats to ensure complete drying and curing. Sand and dust surfaces between coatings, as required, to produce a surface free of visible defects. Lightly sand high gloss coatings and clear finishes between coats to ensure bond of following coats.

Apply coats to the surfaces in an even film. Cloudiness, spotting, holidays, laps, application marks, runs, sags, and other similar surface imperfections are not acceptable. Remove defective coating applications and re-coat as directed.

Ensure coating lines such as wainscots are sharp, true, and well-defined. Tape may be used to establish coating lines, providing tape is removed before ragging or sawtooth edges form.

Ensure surfaces, including edges, corners, crevices, welds, and other similar changes in surface plane, meet the dry-film thickness not less than specified.

### 3.2.1 Brush Application

Use clean, proper size brushes for high-quality application of the specified coating materials. Brush out slow-dry coatings. Brush out quick-dry coatings only enough to spread out evenly.

### 3.2.2 Roller Application

Use clean roller covers of the proper nap length, nap texture, and material for high-quality application of the specified coating materials.

Ensure roller application is equivalent in all respects to the same coats applied by high-quality brush application.

### 3.2.3 Spray Application

Spray application of coatings is limited to finish coats on metal frame works, siding, decking, wire mesh, and other surfaces where hand work would be inferior. Apply spray coatings as equivalent in all respects to the same coats applied by high quality brush application. Permit each spray coat to cure before the succeeding coat is applied. Do not double back with application equipment, for the purpose of building up film thickness of two coats in one operation.

Cover surfaces adjacent to sprayed areas to prevent damage from overspray, coating rebound, and spray drift.

## 3.3 FIELD QUALITY CONTROL

### 3.3.1 Field Test

Government may take dry-film tests from time to time on finished surfaces. Apply additional coatings to surfaces where there is less than the minimum specified dry-film thickness.

### 3.3.2 Repairing

Remove damaged and unacceptable portions of completed work and replace with new work to match adjacent surfaces at no additional cost to the Government.

-- End of Section --

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## SECTION 10 14 00.10

EXTERIOR SIGNAGE  
**08/17, CHG 1: 11/18**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## CANADIAN STANDARDS ASSOCIATION (CSA)

CSA B651:23 Accessible Design for the Built Environment

## NATIONAL RESEARCH COUNCIL CANADA (NRC)

NBC (2020) National Building Code of Canada,  
Part 8, Construction Safety Measures at  
Construction and Demolition Sites

## ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System  
for Aluminum Finishes

## AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z97.1 (2015) Safety Glazing Materials Used in  
Buildings - Safety Performance  
Specifications and Methods of Test

## AMERICAN WELDING SOCIETY (AWS)

AWS C1.1M/C1.1 (2019) Recommended Practices for  
Resistance Welding

AWS D1.1/D1.1M (2025) Structural Welding Code - Steel

AWS D1.2/D1.2M (2014; Errata 1 2014; Errata 2 2020)  
Structural Welding Code - Aluminum

## ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon  
Structural Steel

ASTM A123/A123M (2024) Standard Specification for Zinc  
(Hot-Dip Galvanized) Coatings on Iron and  
Steel Products

ASTM A653/A653M (2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

|                   |                                                                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A924/A924M   | (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process                                                                              |
| ASTM A1011/A1011M | (2023) Standard Specification for Steel Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength |
| ASTM B26/B26M     | (2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings                                                                                                                       |
| ASTM B62          | (2017) Standard Specification for Composition Bronze or Ounce Metal Castings                                                                                                                 |
| ASTM B108/B108M   | (2019) Standard Specification for Aluminum-Alloy Permanent Mold Castings                                                                                                                     |
| ASTM B209M        | (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)                                                                                                       |
| ASTM B221M        | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)                                                                        |
| ASTM C1036        | (2021) Standard Specification for Flat Glass                                                                                                                                                 |
| ASTM D3841        | (2016) Standard Specification for Glass Fiber-Reinforced Polyester Plastic Panels                                                                                                            |
| ASTM E84          | (2024) Standard Test Method for Surface Burning Characteristics of Building Materials                                                                                                        |

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

|               |                              |
|---------------|------------------------------|
| NAAMM AMP 500 | (2006) Metal Finishes Manual |
|---------------|------------------------------|

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|         |                                                        |
|---------|--------------------------------------------------------|
| NFPA 70 | (2023; ERTA 1 2024; TIA 24-1) National Electrical Code |
|---------|--------------------------------------------------------|

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| SAE AMS3611 | (2011; Rev E; Stabilized (S) 2011) Plastic Sheet, Polycarbonate General Purpose |
|-------------|---------------------------------------------------------------------------------|

## 1.2 GENERAL REQUIREMENTS

All exterior signage must be provided by a single manufacturer. Exterior signage must be of the design, detail, sizes, types, and message content shown on the drawings, must conform to the requirements specified, and must be provided at the locations indicated. Submit exterior signage schedule in electronic media with spread sheet format. Spread sheet must include sign location, sign type, and message. Signs must be complete with lettering, framing as detailed, and related components for a complete

installation. Each sample must consist of a complete sign panel with letters and symbols. Samples may be installed in the work, provided each sample is identified and location recorded. Submit three color samples for each material requiring color and 305 mm square sample of sign face color sample.

#### 1.2.1 Wind Load Requirements

Exterior signage must be designed to withstand [\_\_\_\_\_] km/h windload. Submit design analysis and supporting calculations performed in support of specified signage.

#### 1.2.2 Character Proportions and Heights

Letters and numbers on indicated signs for handicapped-accessible buildings must have a width-to-height ratio between 3:5 and 1:1 and a stroke-width-to-height ratio between 1:5 and 1:10. Characters and numbers on indicated signs must be sized according to the viewing distance from which they are to be read. The minimum height is measured using an upper case letter "X". Lower case characters are permitted.

#### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are [for Contractor Quality Control approval.][for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.] Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

Approved Detail Drawings; G[, [\_\_\_\_]]

##### SD-03 Product Data

Modular Exterior Signage System

Installation

Exterior Signage; G[, [\_\_\_\_]]

Wind Load Requirements

##### SD-04 Samples

Exterior Signage; G[, [\_\_\_\_]]

##### SD-10 Operation and Maintenance Data

Protection and Cleaning; G[, [\_\_\_\_]]

#### 1.4 QUALIFICATIONS

Signs, plaques, and dimensional letters must be the standard product of a manufacturer regularly engaged in the manufacture of the products. Items of equipment must essentially duplicate equipment that has been in satisfactory use at least 2 years prior to bid opening.

## 1.5 DELIVERY AND STORAGE

Materials must be wrapped for shipment and storage, delivered to the jobsite in manufacturer's original packaging, and stored in a clean, dry area in accordance with manufacturer's instructions.

## 1.6 WARRANTY

Manufacturer's standard performance guarantees or warranties that extend beyond a one year period must be provided.

## 1.7 EXTRA STOCK

Provide [\_\_\_\_\_] extra interchangeable message panels and extra stock of the following: [[\_\_\_\_\_] message bars of each color and size for sign types [\_\_\_\_\_.] [[\_\_\_\_\_] pressure-sensitive letters in each color and size for sign type [\_\_\_\_\_.] [[\_\_\_\_\_] changeable message strips for sign type [\_\_\_\_\_.]

# PART 2 PRODUCTS

## 2.1 MODULAR EXTERIOR SIGNAGE SYSTEM

Exterior signage must consist of a system of coordinated directional, identification, and regulatory type signs located where shown. Dimensions, details, materials, message content, and design of signage must be as shown. Submit manufacturer's descriptive data and catalog cuts.

### 2.1.1 Free-Standing Base Mount Pylon/Monolith Type Signs

#### 2.1.1.1 Framing

Interior framing must consist of [aluminum] [or] [galvanized steel] tube columns welded to companion plates. Perimeter framing must consist of [aluminum] [or] [steel] angle framing welded to the post and plate system as designed. Framing members must be designed to permit [access to electrical equipment] [and] [panel removal]. Mounting must be provided as shown. Framing members of steel must be finished with semi-gloss baked enamel or two-component acrylic polyurethane. Openings must be sealed from moisture and made tamper-proof.

#### 2.1.1.2 Exterior Sheeting Panels

Modular panels must be provided in sizes shown on drawings. Panels must be fabricated a minimum of [2.3 mm thick [aluminum] [steel]] [3.2 mm thick fiberglass reinforced plastic (FRP)]. [Panels must be heliarc welded to framing system [\_\_\_\_\_.] Top and end panels must be removable and must be secured by 5 mm socket head jack nuts. Finish for metal panels must be [semi-gloss baked enamel] [two-component acrylic polyurethane] [anodized conforming to AA DAF45].

#### 2.1.1.3 Mounting

Mount by securing to concrete foundation as indicated.

#### 2.1.1.4 Finishes

Base finish must be [semi-gloss baked enamel] [or] [two-component acrylic polyurethane] [anodized conforming to AA DAF45] [\_\_\_\_\_. Metal panel

system finish must be [baked enamel or two-component acrylic polyurethane] [anodized conforming to AA DAF45 [\_\_\_\_]], as shown].

## 2.1.2 Panel And Post/Panel Type Signs

### 2.1.2.1 Posts

One-piece [aluminum] [or] [galvanized steel] posts must be provided with minimum 3.2 mm wall thickness. Posts must be designed to accept panel framing system described. The post must be designed to permit attachment of panel framing system without exposed fasteners. Caps must be provided for each post.

### 2.1.2.2 Panel Framing System

Panel framing consisting of aluminum sections and interlocking track components must be designed to interlock with posts with concealed fasteners.

### 2.1.2.3 Panels

Modular message panels must be provided in sizes shown on drawings. Panels must be fabricated a minimum of [[2.0] [2.3] [3.2] mm aluminum] [3.2 mm acrylic] [3.2 mm fiberglass reinforced plastic (FRP)]. [Panels must be designed to be interchangeable.] [Panels with metal return sheeting must have welded corners, ground smooth.] [Panels must be heliarc welded to framing system.] [Face panels must be removable to provide access to electrical components.]

### 2.1.2.4 Finishes

Post finish must be [semi-gloss baked enamel] [or] [two-component acrylic polyurethane] [anodized conforming to AA DAF45] [\_\_\_\_]. Metal panel system finish must be [baked enamel or two-component acrylic polyurethane] [anodized conforming to AA DAF45 [\_\_\_\_]], as shown].

### 2.1.2.5 Mounting

[Provide permanent mounting by embedding posts in concrete foundation as indicated.] [Provide removable mounting by [[a steel] [an aluminum]] [[sleeve] [flange]] embedded in concrete as indicated.]

## 2.1.3 Changeable Letter Directories

### 2.1.3.1 Frame and Trim

Aluminum alloy finish must be [\_\_\_\_].

### 2.1.3.2 Header Plates

[Header plate must consist of background metal matching frame and having raised letters attached through the back.] [Header plate must consist of acrylic with raised acrylic letters.] [Header plate must consist of MP plastic with raised letters.]

### 2.1.3.3 Door Glazing

Door glazing must be [clear safety or tempered glass minimum 6 mm thick.] [clear acrylic sheet 4.8 mm thick.] [clear polycarbonate sheet [4.8] [6.4]



mm thick.]

#### 2.1.3.4 Door Construction

Door frame must be of same material and finish as surrounding frame. Corners must be mitered [, reinforced] [, welded], and assembled with concealed fasteners. Hinges must be standard with manufacturer, in finish to match frames and trim. Glazing must be set in frame with resilient glazing channels.

#### 2.1.3.5 Door Locks

Door locks must be manufacturer's standard and must be keyed alike.

#### 2.1.3.6 Fabrication

Frames and trim must be assembled with corners [reinforced] [welded] and mitered to hairline fit, with no exposed fasteners. Removable changeable directory panel must consist of [6 mm thick white acrylic with clear acrylic letter tracks] [exterior grade plywood] [aluminum] [rubber] back with [vinyl] [polycarbonate] [corkboard] covering backgrooved 6 mm on centers to receive letters.

#### 2.1.3.7 Finishes

Post finish must be [semi-gloss baked enamel] [or] [two-component acrylic polyurethane] [anodized conforming to AA DAF45] [\_\_\_\_\_]. Metal panel system finish must be [baked enamel or two-component acrylic polyurethane] [anodized conforming to AA DAF45 [\_\_\_\_\_], as shown].

#### 2.1.3.8 Mounting

Directories must be mounted to supporting structures with concealed fasteners in accordance with manufacturer's instructions.

#### 2.1.3.9 Changeable Letters

Changeable letters must be upper-case or upper and lower-case [helvetica medium] [\_\_\_\_\_]. Tabbed vinyl letters and numbers must be furnished in accordance with the [drawings] [and] [schedule].

### 2.2 ILLUMINATION

Concealed lighting must be provided within panel framing members. Lighting must be controlled by a photocell device. [Top] [Back] lighting must be provided by [T-12 slimline lamps, [120] [277] [\_\_\_\_\_] volt, 60-hertz, single-phase, Type 1, or Type 2 ballast] [\_\_\_\_\_]. Ballast must be integrally mounted, high power factor and rated for use down to minus 29 degrees C ambient starting temperature. Ballast and wiring within the sign must be in metal raceways. Electrical equipment must be UL or FM listed and comply with NFPA 70. Illumination must be evenly distributed. A switch on the interior of the sign must be provided to turn off power in the sign. Switch must be readily accessible when sign is open.

### 2.3 GRAPHICS FOR EXTERIOR SIGNAGE SYSTEMS

#### 2.3.1 Graphics

Signage graphics must conform to the following:

- [ a. [Cast] [Custom fabricated] [Plate] aluminum letters, [6] [13] [\_\_\_\_\_] mm thick must be provided and fastened to the message panel with concealed fasteners. Letters must project [\_\_\_\_\_] mm from face of panel.]
- [ b. Pressure sensitive precision cut vinyl letters [with reflecting surface] [\_\_\_\_\_] must be provided.]
- [ c. Message must be applied to panel using the silkscreen process. Silkscreened images must be executed with photo screens prepared from original art. Handcut screens will not be accepted. Original art must be defined as artwork that is a first generation pattern of the original specified art. Edges and corners must be clean. Rounded corners, cut or ragged edges, edge buildup, bleeding or surfaces pinholes will not be accepted.]
- [ d. Message letters must be cut out from panel. Panel cutouts must be backed with [2.0 mm FRP] [3.2 mm acrylic] where cutouts occur.]
- [ e. Message must be cut out from panel. Acrylic letters [3] [6] [13] mm thick must be projected through the cutout area and chemically welded to 3.2 mm thick acrylic backup sheet.]
- [ f. Message must be embedded in FRP sheet and completely covered with thermosetting polyester resin. Message must be embedded minimum 0.8 mm. Sheets must be processed in one piece, in one process, to prevent delamination.]
- [ g. Message must be applied using the frisket method. Photomechanically reproduced graphic masks must be applied to the sign face which has been coated with the graphics color. A background must then be applied to the exposed surfaces. Handcut masks will not be accepted. Edges that are nicked, cut, or ragged will not be acceptable. A protective overcoat containing UV-resistant additives must be applied.]
- [ h. Message must be engraved in non-corrosive, three-ply fiberglass laminate. Message must be core color or paint filled multiple colors.]

#### 2.3.2 Messages

See [drawings] [and] [schedule] for message content. Typeface: [Helvetica medium] [\_\_\_\_\_] . Type size [\_\_\_\_\_] [as indicated].

#### 2.4 METAL PLAQUES

Design and location of plaques must be as indicated.

##### 2.4.1 Cast Metal Plaques

###### 2.4.1.1 Fabrication

Cast metal plaques must have the logo, emblem and artwork cast in the [bas relief] [flat relief] [\_\_\_\_\_] technique. Plaques must be fabricated from [prime aluminum] [bronze] [yellow brass].

###### 2.4.1.2 Size

Plaque size must be [\_\_\_\_\_] [as indicated].

#### 2.4.1.3 Border

Border must be [flat band] [plain edge] [bevel] [custom ornamental as indicated] [\_\_\_\_\_].

#### 2.4.1.4 Background

Background texture must be [leather] [fine pebble] [\_\_\_\_\_].

#### 2.4.1.5 Mounting

Mounting must be [concealed] [rosettes and anchors] [rosettes and toggle bolts] [invisible] [\_\_\_\_\_].

#### 2.4.1.6 Finish

Finishes must consist of [aluminum light colored sandblasted background. Letters must be satin polished and entire plaque must be sprayed with two coats of clear lacquer.] [aluminum with background sprayed dark gunmetal colored lacquer. Letters must be satin polished and entire plaque sprayed with two coats clear lacquer.] [bronze with dark finish oxidized background. Letters must be satin polished and entire plaque sprayed with two coats of clear lacquer.] [[aluminum] [bronze] with sprayed background. Letters must be satin polished.]

### 2.4.2 Chemically Etched Metal Plaques

#### 2.4.2.1 Fabrication

Plaque must be chemically [single-] [double-] etched one-piece [brass] [bronze] [\_\_\_\_\_] [0.8128] [1.6256] [3.175] [6.35] mm thick.

#### 2.4.2.2 Size

Plaque size must be [\_\_\_\_\_] [as shown].

#### 2.4.2.3 Finish

[Single-etched raised areas must be in [gold-tone] [silver-tone] [bronze-tone] finish and recessed areas must be colorfilled.] [Double-etched raised areas must be [gold-tone] [silver-tone] and recessed textured areas must be [gold-tone] [silver-tone] colorfilled.]

### 2.4.3 Frost and Surface Oxidized Plaques

#### 2.4.3.1 Fabrication

Plaque must be frosted and surface oxidized one - piece [anodized aluminum] [brass] [bronze] [stainless steel] [1.02] [3.175] mm thick.

#### 2.4.3.2 Size

Plaque size must be [\_\_\_\_\_] [as shown].

#### 2.4.3.3 Finish

[Material finish must be [satin] [polished].] [Frosted areas must be oxidized [black for aluminum or stainless steel] [or] [black or brown, for

brass or bronze].]

## 2.5 DIMENSIONAL BUILDING LETTERS

### 2.5.1 Fabrication

Letters must be fabricated from [cast aluminum] [cast bronze] [2.29 mm aluminum sheet] [3.17 mm aluminum sheet] [extruded aluminum] [\_\_\_\_\_]. Letters must be cleaned by chemical etching or cleaned ultrasonically in a special degreasing bath. Letters must be packaged for protection until installation.

### 2.5.2 Typeface

Typeface must be [helvetica medium] [\_\_\_\_\_] [as indicated].

### 2.5.3 Size

Letter size must be [\_\_\_\_\_] [as indicated].

### 2.5.4 Finish

[Anodized aluminum] [Baked enamel or two-component acrylic polyurethane] [[Polished] [Oxidized] bronze with clear coat] finish must be provided.

### 2.5.5 Mounting

[Threaded studs] [Steel U-bracket, cap screws, and expansion bolts] of number and size as recommended by manufacturer, must be used for concealed anchorage. Letters which project from the building line must have stud spacer sleeves. Letters, studs, and sleeves must be of the same material. Supply templates for mounting.

## 2.6 ALUMINUM ALLOY PRODUCTS

Aluminum alloy products must conform to ASTM B209M for sheet or plate, ASTM B221M for extrusions and ASTM B26/B26M or ASTM B108/B108M for castings. Aluminum extrusions must be provided at least 3 mm thick and aluminum plate or sheet at least 16 gauge thick. Welding for aluminum products must conform to AWS C1.1M/C1.1.

## 2.7 ANODIC COATING

Anodized finish must conform to AA DAF45 as follows:

- [ Clear (natural) designation AA-M10-C22-A31, Architectural Class II 0.010 mm or thicker.]
- [ Integrated color anodized designation AA-M10-C22-A32, Architectural Class 0.010 to 0.018 mm.]
- [ Electrolytically deposited color - anodized designation AA-M10-C22-A34, Architectural Class II 0.010 to 0.018 mm.]

## 2.8 ORGANIC COATING

Clean, prime and give surfaces a [semi-gloss baked enamel] [or] [two-component acrylic polyurethane] finish in accordance with NAAMM AMP 500, AMP 505, with total dry film thickness not less than 0.030

mm.

## 2.9 STEEL PRODUCTS

Structural steel products must conform to ASTM A36/A36M. Sheet and strip steel products must conform to ASTM A1011/A1011M. Welding for steel products must conform to AWS D1.2/D1.2M.

### 2.10 CAST BRONZE

Fabricate components with sharp corners, flat faces, and accurate profiles. Remove and polish burrs and rough spots. Finish faces to a uniform high luster. Cast bronze must be in accordance with ASTM B62.

### 2.11 VINYL SHEETING FOR GRAPHICS

Vinyl sheeting must be 5 to 7 year premium type and must be in accordance with the flammability requirements of ASTM E84 and must be a minimum 0.08 mm film thickness. Film must include a precoated pressure sensitive adhesive backing, Class 1, or positionable pressure sensitive adhesive backing, Class 3.

### 2.12 GLASS

Glass must be in accordance with ASTM C1036, Type I, Class 1, Quality q3 and ANSI Z97.1.

### 2.13 FIBER-REINFORCED POLYESTER (FRP) PANELS

Fiber-reinforced polyester (FRP) must be in accordance with ASTM D3841, Type II, Grade 1, Class 124, [\_\_\_\_\_] [as indicated].

### 2.14 ACRYLIC SHEET

Acrylic sheet must be in accordance with the flammability requirements of ASTM E84 and must conform to ANSI Z97.1.

### 2.15 POLYCARBONATE SHEET

Polycarbonate sheet must conform to SAE AMS3611.

### 2.16 ANCHORS AND FASTENERS

Exposed anchor and fastener materials must be compatible with metal to which applied and must match in color and finish and must be non-rusting, non-corroding, and non-staining. Exposed fasteners must be tamper-proof.

### 2.17 SHOP FABRICATION AND MANUFACTURE

#### 2.17.1 Factory Workmanship

Work must be assembled in the shop, as far as practical, ready for installation at the site. Work that cannot be shop assembled must be given a trial fit in the shop to ensure proper field assembly. Holes for bolts and screws must be drilled or punched. Drilling and punching must produce clean, true lines and surfaces. Welding to or on structural steel must be in accordance with AWS D1.1/D1.1M. Welding must be continuous along the entire area of contact. Exposed welds must be ground smooth. Exposed surfaces of work must have a smooth finish and exposed riveting

must be flush. Fastenings must be concealed where practical. Items specified to be galvanized must be by hot-dip process after fabrication if practical. Galvanization must be in accordance with ASTM A123/A123M and ASTM A653/A653M, as applicable. Other metallic coatings of steel sheet must be in accordance with ASTM A924/A924M. Joints exposed to the weather must be formed to exclude water. Drainage and weep holes must be included as required to prevent condensation buildup.

#### 2.17.2 Dissimilar Materials

Where dissimilar metals are in contact, or where aluminum is in contact with concrete, mortar, masonry, wet or pressure-treated wood, or absorptive materials subject to wetting, the surfaces must be protected with a coat of asphalt varnish or a coat of zinc-molybdate primer to prevent galvanic or corrosive action.

#### 2.17.3 Shop Painting

Surfaces of miscellaneous metal work, except nonferrous metal, corrosion resisting steel, and zinc-coated work, must be given one coat of zinc-molybdate primer or an approved rust-resisting treatment and metallic primer in accordance with manufacturer's standard practice. Surfaces of items to be embedded in concrete must not be painted. Upon completion of work, damaged surfaces must be recoated.

### 2.18 COLOR, FINISH, AND CONTRAST

Color must be [in accordance with Section 09 06 00 SCHEDULES FOR FINISHES.] [as indicated on the drawings.] [selected from manufacturers standard colors.] [[\_\_\_\_\_.] Color listed is not intended to limit the selection of equal colors from other manufacturers.] For buildings required to be handicapped-accessible, the characters and background of signs must be eggshell, matte, or other non-glare finish. Characters and symbols must contrast with their background - either light characters on a dark background or dark characters on a light background.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Signs, plaques, or dimensional letters must be installed in accordance with approved manufacturer's instructions at locations shown on the approved detail drawings; submit drawings showing elevations of each type of sign; dimensions, details, and methods of mounting or anchoring; shape and thickness of materials; and details of construction. A schedule showing the location, each sign type, and message must be included. Circuits installed underground must conform to the requirements of Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Steel conduits installed underground and illuminated signage mounted directly on buildings must be in conformance with the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Signs must be installed plumb and true at mounting heights indicated, and by method shown or specified. Signs mounted on other surfaces must not be installed until finishes on such surfaces have been completed. Submit manufacturer's installation instructions and cleaning instructions. Conform to requirements of NBC 2020, guidelines for accessibility, and to CSA B651:23, Accessible Design for the Built Environment.

### 3.1.1 Anchorage

Anchorage and fastener materials must be in accordance with approved manufacturer's instructions for the indicated substrate. Anchorage not otherwise specified or indicated must include slotted inserts, expansion shields, and powder-driven fasteners when approved for concrete; toggle bolts and through bolts for masonry; machine carriage bolts for steel; lag bolts and screws for wood.

### 3.1.2 Protection and Cleaning

The work must be protected against damage during construction. Hardware and electrical equipment must be adjusted for proper operation. Glass, frames, and other sign surfaces must be cleaned in accordance with manufacturer's instructions. After signs are completed and inspected, cover all project identification, directional, and other signs which may mislead the public. Covering must be maintained until instructed to be removed by the Contracting Officer or until the facility is to be opened for business. Submit [six] [\_\_\_\_\_] copies of maintenance instructions listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guides. The instructions must include simplified diagrams for the equipment as installed. Signs must be cleaned, as required, at time of cover removal.

## 3.2 FIELD PAINTED FINISH

Miscellaneous metals and frames must be field painted in accordance with Section 09 90 00 PAINTS AND COATINGS. Anodized metals, masonry, and glass must be protected from paint. Finish must be free of scratches or other blemishes.

-- End of Section --

## SECTION 10 14 00.20

## INTERIOR SIGNAGE

**08/20**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

## Canadian Standards Association (CSA)

CSA B651 (2023) Accessible Design for the Built Environment

## NATIONAL RESEARCH COUNCIL CANADA (NRC)

NBC (2020) National Building Code of Canada, Part 8, Construction Safety Measures at Construction and Demolition Sites

## ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System for Aluminum Finishes

AA PK-1 (2015) Pink Sheets: Designations and Chemical Composition Limits for Aluminum Alloys in the Form of Castings & Ingot

## AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 2604 (2017a) Voluntary Specification, Performance Requirements and Test Procedures for High Performance Organic Coatings on Aluminum Extrusions and Panels

## AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z97.1 (2015) Safety Glazing Materials Used in Buildings - Safety Performance Specifications and Methods of Test

## AMERICAN WELDING SOCIETY (AWS)

AWS D1.2/D1.2M (2014; Errata 1 2014; Errata 2 2020) Structural Welding Code - Aluminum

## ASTM INTERNATIONAL (ASTM)

ASTM B209M (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)

ASTM B221M (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods,



## Wire, Profiles, and Tubes (Metric)

ASTM D635 (2018) Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position

## INTERNATIONAL CODE COUNCIL (ICC)

ICC/ANSI A117.1 (2009) Accessible and Usable Buildings and Facilities

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA LD 3 (2005) Standard for High-Pressure Decorative Laminates

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 101 (2024) Life Safety Code

## NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 800-82 (2015; Rev 2) Guide to Industrial Control Systems (ICS) Security

## U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 8510.01 (2022) Risk Management Framework (RMF) for DoD Systems

DODI 8500.01 (2014) Cybersecurity

UFC 4-010-06 (2023) Cybersecurity of Facility-Related Control Systems

## U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Detail Drawings; G

## SD-03 Product Data

Room Identification And Directional Signage System; G  
Exit Door Tactile Sign; G

Door Tags; G

#### SD-04 Samples

Interior Signage; G

Software; G

Room Identification And Directional Signage System; G

Exit Door Tactile Sign; G

Door Tags; G

#### SD-10 Operation and Maintenance Data

Approved Manufacturer's Instructions; G

Protection and Cleaning; G

### 1.3 QUALITY ASSURANCE

#### 1.3.1 Samples

Submit interior signage samples of each of the following sign types showing typical quality, workmanship and color: all sign types included in project. Approved samples may be installed in the work, provided each sample is identified and location recorded.

#### 1.3.2 Detail Drawings

Submit detail drawings showing elevations of each type of sign, dimensions, details and methods of mounting or anchoring, mounting height, shape and thickness of materials, and details of construction. Include a schedule showing the location, each sign type, and message.

#### 1.3.3 Sign Fabricator

Sign Fabricator to follow room number strategies created by designer. The room numbering system to be reviewed and approved by the Contracting Officer and command end users during the shop drawing phase, and prior to fabrication. Signs to be fabricated in accordance with CSA B651 and NBC.

#### 1.3.4 Cybersecurity

- a. The Risk Management Framework (RMF) is the process by which information systems are accredited for operation by a designated official from the Using Military Department. It is the standard process under which all DoD information systems achieve and maintain their Authority To Operate. The cybersecurity process is documented in DOD 8510.01 and NIST SP 800-82. Refer to UFC 4-010-06 and DODI 8500.01 for additional requirements.
- b. All systems that are IP addressable or interface with the Assured Network required certification to operate. Coordinate with the Government to initiate and complete the accreditation process.
- c. Cybersecurity requires input from the system vendor or provider and support from the local IMD. The local IMD-IA office is the point of contact for all Cyber Security requirements. The local CMIO is the point of contact for all clinical and functional system requirements.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Package materials to prevent damage and deterioration during shipment, handling, storage and installation. Deliver products to the jobsite in manufacturer's original packaging and store in a clean, dry area in accordance with manufacturer's instructions.

#### 1.5 WARRANTY

Provide manufacturer's warranty to repair or replace defective interior signage materials and workmanship for a period of 2 years from date of final acceptance of the work.

### PART 2 PRODUCTS

#### 2.1 ROOM IDENTIFICATION AND DIRECTIONAL SIGNAGE SYSTEM

Provide signs that are standard products of manufacturers regularly engaged in the manufacture of such products that essentially duplicate signs that have been in satisfactory use at least 2 years prior to bid opening. Obtain signage from a single manufacturer with edges and corners of finished letter forms and graphics true and clean.

##### 2.1.1 Panel Sign Systems

Provide direct print acrylic with applied tactile graphics, sign is fabricated of 10 mm acrylic in two layers with smooth edge conforming to ANSI Z97.1. Provide signs that can accept raised copy and Braille with printable message inserts.

##### 2.1.2 Modular Sign Systems

Provide manufactured pre-engineered component-based sign system, consisting of a combination of aluminum extrusions and injection molded parts, pre-engineered and designed to create an updatable sign system that allows for easy and inexpensive updates and changes. Provide system with incremental widths and heights that permit the assembly of multiple inserts of variable size to create a single sign. Provide a tamper-resistant sign which requires a special tool to change inserts composed of extruded aluminum for applied graphics. Provide continuous extruded aluminum removable endcaps in square profile. Sign inserts are required to be side loading.

##### 2.1.3 Standard Room Signs

Provide signs that include tactile letters, symbols and Braille for interior rooms or spaces where the sign is not likely to change over time. Tactile text descriptions are required for pictograms that are provided to identify a permanent room. Examples include interior signs that label restrooms, stairs, room numbers or letters, and room names. These permanent room signs can include paper inserts for updatable information.

##### 2.1.3.1 Tactile Letters, Symbols and Braille

Provide ADA compliant material per 36 CFR 1191 which is raised 0.79 mm from the first surface, has a minimum 16 mm in height and is an ADA acceptable font. The color of the tactile letters is required to contrast

with the sign face color per ADA standards. The ADA required Braille has a minimum durometer reading of 90. All raised letters, numbers and symbols are to comply. Also comply with CSA B651 and NBC Canadian standards.

#### 2.1.4 Type of Mounting for Signs

Provide surface mounted signs mounted with concealed mechanical fastening through the holders. Secure inserts in holders when captured by side profiles of extruded aluminum holders.

#### 2.1.5 Character Proportions and Heights

Letters and numbers on signs conform to 36 CFR 1191.

### 2.2 EXIT DOOR TACTILE SIGN

Provide tactile sign with the message EXIT at each exit door that requires an exit sign to conform with NFPA 101. Sign tactile message is to comply with ICC/ANSI A117.1.

### 2.3 DOOR TAGS

Provide one door tag plate for each room entry door. In size as indicated on drawings. Provide room number to match architectural floor plan room number.

#### 2.3.1 Engraved Copy

Machine engrave letters, numbers, symbols, and other graphics into panel sign on face to produce precisely formed copy and sharp images, incised to uniform depth. Melamine plastic engraving stock used for ADA compliant graphic is three-ply lamination contrasting color core meeting ASTM D635.

### 2.4 MATERIALS

#### 2.4.1 Aluminum Alloy Products

Aluminum extrusions are at least 3 mm thick, and aluminum plate or sheet are at least 1.3 mm thick. Extrusions conform to ASTM B221M; plate and sheet conforms to ASTM B209M. Where anodic coatings are specified, alloy conforms to AA PK-1 alloy designation 514.0. Exposed anodized aluminum finishes are as shown. Welding for aluminum products conforms to AWS D1.2/D1.2M.

#### 2.4.2 Anodic Coating

Anodized finish conforms to AA DAF45 as follows:

- a. Clear (natural) designation AA-M10-C22-A31, Architectural Class II 0.010 mm or thicker.
- b. Integral color anodized designation AA-M10-C22-A32, Architectural Class 0.010 to 0.018 mm.

#### 2.4.3 Organic Coating

Organic coating conforms to AAMA 2604, with total dry film thickness not less than 0.030 mm.

#### 2.4.4 Plastic Laminate Sheet

ANSI/NEMA LD 3, general purpose HGS grade, 1.22 mm nominal thickness.

#### 2.4.5 Fabrication and Manufacture

##### 2.4.5.1 Factory Workmanship

Holes for bolts and screws are drilled or punched. Drilling and punching produces clean, true lines and surfaces. Exposed surfaces of work have a smooth finish; exposed riveting is flush. Conceal fastenings where practicable.

##### 2.4.5.2 Dissimilar Materials

Where dissimilar metals are in contact, protect surfaces prevent galvanic or corrosive action.

#### 2.4.6 Typeface

ADA-ABA compliant font for Room Signs.

### 2.5 GRAPHICS

Provide signage graphics for modular signs to the following:

#### 2.5.1 First Surface Copy Direct Print (Non-Tactile)

Message may be applied to panel using a direct print process. Original art is defined as artwork that is a first generation reproduction of the specified art. Provide clean edges and corners.

#### 2.5.2 Engraved Copy

Machine engrave letters, numbers, symbols, and other graphics into panel sign on face to produce precisely formed copy and sharp images, incised to uniform depth. Melamine plastic engraving stock used for ADA compliant graphic is three-ply lamination contrasting color core meeting ASTM D635.

### 2.6 COLOR, FINISH, AND CONTRAST

Provide color as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers. Finish of eggshell, matte, or other non-glare finish for all signs as required in handicapped-accessible buildings.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install signs plumb and true and in accordance with approved manufacturer's instructions at locations shown on the detail drawings. Submit operating instructions outlining the step-by-step procedures required for system operation. The instructions include simplified diagrams for the system as installed, the manufacturer's name, model number, service manual, parts list, and brief description of all equipment and their basic operating features. Provide each set permanently bound with a hard cover. The following identification must be inscribed on the covers: "OPERATING AND MAINTENANCE INSTRUCTIONS", name and location of

the facility, name of the Contractor, and contract number. Submit in accordance with Section 01 78 23 OPERATING AND MAINTENANCE DATA. Mounting height and mounting location complies with 36 CFR 1191. Install required blocking. Do not install signs on doors or other surfaces until finishes on such surfaces have been installed. Signs installed on glass surfaces are installed with matching blank back-up plates in accordance with manufacturer's instructions.

Do not install items that show visual evidence of biological growth.

#### 3.1.1 Anchorage

Provide anchorage in accordance with approved manufacturer's instructions. Anchorage not otherwise specified or shown includes slotted inserts, expansion shields, and powder-driven fasteners when approved for concrete; toggle bolts and through bolts for masonry; machine carriage bolts for steel; lag bolts and screws for wood. Provide exposed anchor and fastener materials compatible with metal to which applied with matching color and finish.

- a. Signs mounted to painted gypsum board surfaces must be removable for painting maintenance.
- c. Install signs mounted on metal surfaces with magnetic tape.

#### 3.1.2 Protection and Cleaning

Protect the work against damage during construction. Adjust hardware and electrical equipment for proper operation. Clean glass, frames, and other sign surfaces at completion of signage installation in accordance with the manufacturer's written instructions.

-- End of Section --

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## SECTION 10 26 00

## WALL AND DOOR PROTECTION

08/20

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System  
for Aluminum Finishes

## ASTM INTERNATIONAL (ASTM)

ASTM A167 (2011) Standard Specification for  
Stainless and Heat-Resisting  
Chromium-Nickel Steel Plate, Sheet, and  
Strip

## CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for  
the Testing and Evaluation of Volatile  
Organic Chemical Emissions from Indoor  
Sources using Environmental Chambers

## GREEN SEAL (GS)

GS-36 (2013) Adhesives for Commercial Use

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80 (2025) Standard for Fire Doors and Other  
Opening Protectives

## SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

## SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

## UNDERWRITERS LABORATORIES (UL)

UL 2818 (2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S"



classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Corner Guards; G

Door Protectors; G

Wall Covering and Panels; G

SD-03 Product Data

Corner Guards; G

Door Protectors; G

Wall Covering and Panels; G

Recycled content for aluminum component of corner guards; S

Recycled content for steel component of corner guards; S

SD-04 Samples

Corner Guards; G

Door Protectors; G

Wall Covering and Panels; G

SD-06 Test Reports

Fire Resistance Rating

SD-07 Certificates

Indoor air quality for adhesives; S

SD-10 Operation and Maintenance Data

Corner Guards, Data Package 1; G

Door Protectors, Data Package 1; G

Wall Covering and Panels, Data Package 1; G

1.3 CERTIFICATIONS

1.3.1 Indoor Air Quality

1.3.1.1 Wall Covering and Panels

Provide sheet and high impact resistant resilient materials certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this section. Provide current product certification

documentation from certification body.

#### 1.3.1.2 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the project site in manufacturer's original unopened containers with seals unbroken and labels and trademarks intact. Keep materials dry, protected from weather and damage, and stored under cover. Store materials at approximately 21 degrees C for at least 48 hours prior to installation.

#### 1.5 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials and workmanship for a 1 year period of one year from date of final acceptance of the work.

### PART 2 PRODUCTS

#### 2.1 STANDARD PRODUCTS

To the maximum extent possible, provide wall and door protection items that are standard products of a single manufacturer and furnished as detailed. Drawings show general configuration of products required, and items differing in minor details from those shown are acceptable.

Submit detailed shop drawings of each wall and door protection item indicated. Include elevations, dimensions, clearances, details of construction and anchorage, and details of joints and connections.

Submit manufacturers' descriptive product data for each wall and door protection item indicated. Include manufacturers' literature, finishes, profiles and thicknesses of materials.

Submit manufacturers' operations and maintenance data for each wall and door protection item indicated in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

#### 2.2 CORNER GUARDS2.2.1 Stainless Steel Corner Guards

Provide stainless steel base material that contains a minimum of 60 percent recycled content. Provide data identifying percentage of recycled content for steel component of corner guards. Fabricate stainless steel base material of 1.59 mm thick material conforming to ASTM A167, type 430 or 304. Provide corner guards that extend from floor to ceiling. Form corner guard to dimensions shown.

#### 2.3 DOOR PROTECTORS

Providedoor protection items with 1.59 mm, type 304 stainless steel for

door. Coordinate door protection material requirements with door and frame suppliers to insure fit for all components and color matching with other resilient materials. Provide adhesive as recommended by resilient material manufacturer.

## 2.4 FINISH

Submit samples indicating color and texture of materials requiring color and finish.

### 2.4.1 Aluminum Finish

Provide aluminum finish accordance with AA DAF45; exposed aluminum with designation AA-C22A31 chemically etched medium matte, with clear anodic coating. Provide Class II architectural coating that is 0.010 mm thick. Provide concealed aluminum with mill finish as fabricated, uniform in natural color and free from surface blemishes.

### 2.4.2 Stainless Steel Finish

Provide stainless steel finish in accordance with ASTM A167, Type 302 or 304, finish number 4.

## 2.5 ADHESIVES

Provide adhesive for resilient material in accordance with manufacturers recommendations. Provide sealants and non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) the VOC content requirements of SCAQMD Rule 1168, or VOC content requirements of GS-36. Provide certification of indoor air quality for adhesives.

## 2.6 COLOR

Provide color as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Do not install items that show visual evidence of biological growth. Install items on surfaces that are clean, smooth, and free of obstructions.

#### 3.1.1 Corner Guards and Wall Guards

- a. Mount guards on external corners of interior walls, partitions and columns and in accordance with manufacturer's written installation instructions.

##### 3.1.1.1 Stainless Steel Guards

- a. Mount guards on external corners of interior walls, partitions and columns and in accordance with manufacturer's recommendations.

- d. Where corner guards are installed on gypsum board, clean surfaces and anchor guards with a neoprene solvent-type contact adhesive specifically manufactured for use on gypsum board construction. Remove excess adhesive from the guard edges and allow to cure undisturbed for 24 hours.

### 3.1.2 Door Protectors

Install protectors after frames are in place, but prior to hanging of doors, in accordance with manufacturer's written instructions. Apply adhesives in controlled environment in accordance with manufacturer's written instructions. Install protection for fire doors and frames in accordance with NFPA 80.

-- End of Section --

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## SECTION 10 28 13

## TOILET ACCESSORIES

08/20

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

Canadian Standards Association (CSA)

CSA B651:23 Accessible Design for the Built Environment

NATIONAL RESEARCH COUNCIL CANADA (NRC)

NBC (2020) National Building Code of Canada,  
Part 8, Construction Safety Measures at  
Construction and Demolition Sites

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-1691 (1994; Rev F) Construction and Material  
Schedule for Military Medical and Dental  
Facilities

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Product Schedule; G

Submit product Schedule indicating types, quantities, sizes, and installation locations by room for each toilet accessory item required. Identify locations using room designations indicated on the drawings.

## SD-03 Product Data

Recycled content for stainless steel toilet accessories; S

Item A5030 Bench, Stall, Shower, Built In; G

Item A5047 Cabinet, Medicine; G

Item A5083 Dispenser, Paper Towel, Recessed; G

Item A5090 Disposal, Sanitary Napkin, SS, Surface Mounted; G

Item A5109 Grab Bar, 32 mm Dia., SS, 2 Wall, W/C Accessible; G

Item A5110 Grab Bar, 32 mm Dia., SS, 2 Wall, Shower Use; G

Item A5135 Shelf, Utility W/ Mop/Broom Holders, SS, Surf Mntd; G

Item A5145 Hook, Garment, Double, SS, Surface Mounted; G

Item A5165 Shelf, 127 mm Depth, SS, Surface Mounted; G

Item A5170 Rod, Shower Curtain, 25 mm Diameter, W/Curtain & Hooks;  
G

Item A5200 Dispenser, Toilet Tissue, SS, 2-Roll, Surface Mntd; G

Item A5205 Bar, Towel, 25 mm Diameter, SS, Surface Mounted; G

Submit catalog numbers, literature, data sheets, construction details, profiles, anchoring and mounting requirements and other pertinent data for each toilet accessory item to evaluate function, materials, dimensions and appearance.

Mirrors; G

#### SD-07 Certificates

Baby Changing Stations

#### SD-10 Operation and Maintenance Data

Item A5030 Bench, Stall, Shower, Built In; G

Item A5047 Cabinet, Medicine; G

Item A5083 Dispenser, Paper Towel, Recessed; G

Item A5090 Disposal, Sanitary Napkin, SS, Surface Mounted; G

Item A5109 Grab Bar, 32 mm Dia., SS, 2 Wall, W/C Accessible; G

Item A5110 Grab Bar, 32 mm Dia., SS, 2 Wall, Shower Use; G

Item A5135 Shelf, Utility W/ Mop/Broom Holders, SS, Surf Mntd; G

Item A5145 Hook, Garment, Double, SS, Surface Mounted; G

Item A5165 Shelf, 127 mm Depth, SS, Surface Mounted; G

Item A5170 Rod, Shower Curtain, 25 mm Diameter, W/Curtain & Hooks;  
G

Item A5200 Dispenser, Toilet Tissue, SS, 2-Roll, Surface Mntd; G

Item A5205 Bar, Towel, 25 mm Diameter, SS, Surface Mounted; G

Submit Data Package 1 for each toilet accessory item in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

### 1.3 DELIVERY, STORAGE, AND HANDLING

Wrap toilet accessories for shipment and storage, then deliver to the jobsite in manufacturer's original packaging, and store in a clean, dry area protected from construction damage and vandalism.

### 1.4 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials and workmanship for a period of one year from date of final acceptance of the work..

## PART 2 PRODUCTS

### 2.1 ACCESSORY ITEMS

Provide toilet accessories where indicated in accordance with Contractor-provided product schedule. Conform to the requirements for accessory items specified herein which are based on MIL-STD-1691 Joint Schedule Numbers (JSN). Provide each accessory item complete with the necessary mounting plates of sturdy construction with corrosion resistant surface.

Provide stainless steel products listed herein manufactured from materials containing a minimum of 50 percent recycled content. Provide data identifying percentage of recycled content for stainless steel toilet accessories.

#### 2.1.1 Anchors and Fasteners

Provide corrosion-resistant anchors and fasteners capable of developing a restraining force commensurate with the strength of the accessory to be mounted and suited for use with the supporting construction. Provide tamperproof design exposed fasteners with finish to match the accessory. Provide fasteners proposed for use for each type of wall construction and mounting.

#### 2.1.2 Finishes

Except where noted otherwise, provide the following finishes on metal:

| Metal           | Finish             |
|-----------------|--------------------|
| Stainless steel | No. 4 satin finish |
|                 |                    |

#### 2.1.3 Item A5030 Bench, Stall, Shower, Built-In

Wall mounted shower seat. Frame made of 1.27 mm stainless steel with satin finish. Seat made of one piece of 13 mm thick nonporous solid phenolic with slots to permit water to drain, secured to frame with stainless steel carriage bolts and acorn nuts. Mounting hardware included. Seat to support a minimum static load of 113 kg. Hinge seat to fold up when not in use. Seat complies with ADA guidelines.

Approximate size: 864 mm wide by 559 mm deep by 13 mm thick.



#### 2.1.4 Item A5047 Cabinet, Medicine

Medicine cabinet constructed of heavy gauge stainless steel and have a mirror mounted in swinging door, a minimum of three shelves, magnetic catch and full length piano hinge. Mirror is 6 mm thick first quality float glass electrolytically copper-plated and guaranteed against silver spoilage for 15 years. Cabinet door can be inverted to change from left to right hand swing. Unit has concealed mounting holes. Mounting hardware included.

Approximate size: 432 mm wide by 635 mm high by 102 mm deep.

#### 2.1.5 Item A5083 Dispenser, Paper Towel, Recessed

Recess mounted paper towel dispenser. Unit constructed of heavy gauge stainless steel with satin finish, all welded construction, have full length piano hinge and tumbler lock. Unit dispenses 300 C-fold or 400 multifold paper towels and be self-feeding until supply is depleted. Towel dispensing slot is snag-free. Unit is ADA compliant. Provide integral waste receptacle in the paper towel dispenser unit.

Approximate size: 385 mm wide by 1420 mm high by 180 mm deep.

#### 2.1.6 Item A5090 Disposal, Sanitary Napkin, SS, Surface Mounted

Surface mounted sanitary napkin receptacle. Unit made of stainless steel with satin finish and all welded construction. Unit has piano hinge attached at the top and an integral finger depression for opening. For use with disposable paper liners, available separately. Unit may be attached to wall or toilet partition.

Approximate size: 178 mm wide by 102 mm deep by 254 mm high.

#### 2.1.7 Item A5109 Grab Bar, 32 mm Diameter, SS, 2 Wall, W/C Accessible

Grab bar of 32 mm diameter satin finish stainless steel with peened gripping surface for use in toilet stall/room. Snap-on flange covers for concealed mounting are stainless steel and equipped with two screw holes for attachment to wall. Grab bars designed to meet and exceed ADA requirements for structural strength. Grab bars designed to withstand loads of 408 kg when properly installed. Clearance from wall to grab bar is 38 mm to meet ADA and ANSI codes.

#### 2.1.8 Item A5110 Grab Bar, 32 mm Diameter, SS, 2 Wall, Shower Use

Grab bar of 32 mm diameter satin finish stainless steel with peened gripping surface. Snap-on flange covers for concealed mounting stainless steel. Bent ends of tubing pass through the flanges and are Heliarc welded for maximum strength. Grab bars designed to meet and exceed ADA requirements for structural strength. Grab bars designed to withstand loads of 408 kg when properly installed. Clearance from wall to grab bar is 38 mm to meet ADA and ANSI codes.

#### 2.1.9 Item A5135 Shelf, Utility W/ Mop/Broom Holders, SS, Surf Mounted

Surface mounted mop/broom holder with shelf made of 1.27 mm stainless steel with all exposed surfaces in satin finish. Unit has shelf 203 mm deep with shelf support brackets of satin finish stainless steel welded to

mounting base, and a minimum of 3 hooks/3 holders. Mop holders have spring-loaded rubber cams and hold mop or broom handle with a diameter between 16 mm and 25 mm.

Approximate size: 865 mm wide by 205 mm deep.

2.1.10 Item A5145 Hook, Garment, Double, SS, Surface Mounted

Surface mounted double garment hook made of stainless steel with satin finish. For use on door back or wall. Hook comes with concealed mounting bracket secured to concealed wall plate. Mounting hardware included. Flange size is approximately 51 mm by 51 mm.

2.1.11 Item A5165 Shelf, 127 mm Depth, SS, Surface Mounted

Surface mounted shelf of 1.27 mm stainless steel with all exposed surfaces in satin finish. Shelf has minimum depth of 127 mm. Center bracket and end brackets of stainless steel, welded to shelf. Shelf length is 355 mm.

2.1.12 Item A5170 Rod, Shower Curtain, 25 mm Diameter, W/Curtain & Hooks

Shower Curtain Rod with concealed mounting. Shower curtain rod made of satin finish stainless steel, 25 mm diameter, with flanges included, and have white vinyl shower curtain, 1829 mm high, and stainless steel curtain hooks. Shower curtain has corrosion resistant grommets, reinforced heading, and treated with antibacterial and flame retardant agents. Shower hooks are stainless steel. Length as indicated on drawings.

2.1.13 Item A5200 Dispenser, Toilet Tissue, SS, 2-Roll, Surface Mounted

Concealed surface mounted, double roll, toilet tissue dispenser of stainless steel. Unit holds and dispenses two standard 133 mm diameter rolls of toilet tissue. Spindles are free-spinning for non-controlled delivery, chrome-plated plastic equipped with heavy-duty internal springs.

Approximate size: 345 mm wide by 190 mm tall by 115 mm deep.

2.1.14 Item A5205 Bar, Towel, 25 mm (1 inch) Diameter, SS, Surface Mounted

Surface mounted satin finish stainless steel towel bar of 25 mm diameter. Support posts fabricated of heavy solid cast brass with satin finish. Stainless steel set screw keeps bar from rotating in posts. Clearance between towel bar and wall is 38 mm.

2.1.15 Item Mirrors

Mirrors, Glass (MG)

Provide Type I transparent flat type, Class 1-clear glass for mirrors. Glazing Quality q1 6 mm thick conforming to ASTM C1036 and UNE-EN 1036-1. Coat glass on one surface with silver coating, copper protective coating, and mirror backing paint. Provide highly adhesive pure silver coating of a thickness which provides reflectivity of 83 percent or more of incident light when viewed through 6 mm thick glass, free of pinholes or other defects. Provide copper protective coating with pure bright reflective copper, homogeneous without sludge, pinholes or other defects, of proper thickness to prevent "adhesion pull" by mirror backing paint. Provide mirror backing paint with two coats of special scratch and abrasion-resistant paint and baked in uniform thickness to provide a

protection for silver and copper coatings which will permit normal cutting and edge fabrication

#### Mirror, Metal (MM)

Provide a brightly polished stainless steel metal mirror of 0.94 mm minimum thickness, edges turned back 6 mm and recess fitted with tempered hardboard backing, and theft-proof fasteners. Provide size in accordance with paragraph SCHEDULE.

#### Mirror, Tilt (MT)

Provide surface mounted tilt mirror with full visibility for persons in a wheelchair. Furnish fixed tilt mirror, extending at least 100 mm from the wall at the top and tapering to 25 mm at the bottom. Provide size in accordance with the drawings. Conform to ASTM C1036 and UNE-EN 12150-2 and paragraph Glass Mirrors.

### 2.1.16 Sharps Disposal

Sharps Disposal cabinet constructed of heavy gauge stainless steel and have a swinging door to allow removal of 5 quart sharps container. Unit has concealed mounting holes. Mounting hardware included. Provide two keys. Provide 5 quart sharps container.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Do not install items that show visual evidence of biological growth. Provide the same finish for the surfaces of fastening devices exposed after installation as the attached accessory. Provide oval exposed screw heads. Install accessories at the location and height indicated. Protect exposed surfaces of accessories with strippable plastic or by other means until the installation is accepted. After acceptance of accessories, remove and dispose of strippable plastic protection. Coordinate accessory manufacturer's mounting details with other trades as their work progresses. Use sealants for brackets, plates, anchoring devices and similar items in showers (a silicone sealant specified in Section 07 92 00 JOINT SEALANTS) as they are set to provide a watertight installation. After installation, thoroughly clean exposed surfaces and restore damaged work to its original condition or replace with new work. Install accessories mounted at heights that conform to the National Building Code (NBC) and CSA B651:23.

#### 3.1.1 Recessed Accessories

Fasten accessories with wood screws to studs, blocking or rough frame in wood construction. Set anchors in mortar in masonry construction. Fasten to metal studs or framing with sheet metal screws in metal construction.

#### 3.1.2 Surface Mounted Accessories

Mount on concealed backplates, unless specified otherwise. Conceal fasteners on accessories without backplates. Install accessories with corrosion-resistant fasteners as required by the construction. Install backplates in the same manner, or provide with lugs or anchors set in mortar, as required by the construction. Fasten accessories mounted on gypsum board and plaster walls without solid backing into the metal or

wood studs, or to backplates secured to metal studs.

### 3.2 CLEANING

Clean material in accordance with manufacturer's recommendations. Do not use alkaline or abrasive agents. Take precautions to avoid scratching or marring exposed surfaces.

-- End of Section --

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## SECTION 10 44 16

## FIRE EXTINGUISHERS

**11/19**

## PART 1 GENERAL

## 1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Cabinets; G

Wall Brackets; G

## SD-03 Product Data

Cabinets; G

Wall Brackets; G

## 1.2 DELIVERY, STORAGE, AND HANDLING

Protect materials from weather, soil, and damage during delivery, storage, and construction.

Deliver materials in their original packages, containers, or bundles bearing the brand name and the name and type of the material.

## PART 2 PRODUCTS

Submit fabrication drawings consisting of fabrication and assembly details performed in the factory and product data for the following items: cabinets, Wall Brackets.

## 2.1 SYSTEM DESCRIPTION

### 2.1.1 Types

## 2.2 EQUIPMENT

### 2.2.1 Cabinets

#### 2.2.1.1 Material

Provide enameled steel cabinets.

#### 2.2.1.2 Type

Provide recessed type cabinets.

Provide semi-recessed cabinet for a 100 millimeter wall.

#### 2.2.1.3 Size

Dimension cabinets to accommodate the specified fire extinguishers.

### 2.2.2 Wall Brackets

Provide wall-hook fire extinguisher wall brackets.

Provide wall bracket and accessories as approved.

## PART 3 EXECUTION

### 3.1 PROTECTION

#### 3.1.1 Repairing

Remove and replace damaged and unacceptable portions of completed work with new work at no additional cost to the Government.

#### 3.1.2 Cleaning

Clean all surfaces of the work, and adjacent surfaces which are soiled as a result of the work. Remove from the site all construction equipment, tools, surplus materials and rubbish resulting from the work.

-- End of Section --

## SECTION 10 51 13

## LOCKERS

08/24

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## Canadian Standards Association (CSA)

CSA B651 (2023) Accessible Design for the Built Environment

## ASTM INTERNATIONAL (ASTM)

ASTM D709 (2017) Standard Specification for Laminated Thermosetting Materials

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA LD 3 (2005) Standard for High-Pressure Decorative Laminates

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 286 (2024) Standard Methods of Fire Tests for Evaluating Contribution of Wall and Ceiling Interior Finish to Room Fire Growth

## U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS AA-L-00486 (Rev J) Lockers, Clothing, Steel

## U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Types; G

Location; G



## Installation

## SD-03 Product Data

Material

Handles

Finish

Locker Components

Assembly Instructions

## SD-04 Samples

Color Chips; G

## 1.3 DELIVERY, HANDLING, AND STORAGE

Deliver lockers and associated materials in their original packages, containers, or bundles bearing the manufacturer's name and the name of the material. Protect from weather, soil, and damage during delivery, storage, and construction.

## 1.4 FIELD MEASUREMENTS

To ensure proper fits, make field measurements prior to the preparation of drawings and fabrication. Verify correct location.

## 1.5 QUALITY ASSURANCE

## 1.5.1 Color Chips

Provide a minimum of three color chips, not less than 75 mm square, of each color indicated.

Government may request performance-characteristic tests on assembled lockers. Tests and results must conform to FS AA-L-00486. Phenolic core lockers must conform to applicable requirements of FS AA-L-00486 and be manufactured as high pressure phenolic panels with integral plastic laminate outer faces both sides of panels. Laminating faces to the core is not acceptable. Lockers not conforming will be rejected.

## PART 2 PRODUCTS

## 2.1 ACCESSIBILITY

Comply with 36 CFR 1191 and with CSA B651 with regard to accessibility of lockers, locker features, latching and locking, and signage.

## 2.2 TYPES

Locker must have the following type and size in the location and quantities indicated. Locker finish colors will be as scheduled.

Lockers must be manufactured as prefabricated plastic laminate faced solid phenolic core units.

### 2.2.1 Single-tier Lockers

Single-tier lockers must be as follows:

Type STC-3: Single-tier locker 457 millimeter wide, 305 millimeter deep, and 1830 millimeter high, attached to 150 millimeter high closed base

## 2.3 MATERIAL

### 2.3.1 Solid Plastic Laminate Faced Phenolic Core

ASTM D709 solid 13 mm ANSI/NEMA LD 3 plastic laminate faced phenolic core panels forming locker bodies, doors, and shelves. Include bases, sloped tops, end panels, and infill panels. Fire performance per NFPA 286.

### 2.3.2 Finish

#### 2.3.2.1 Color

As selected.

## 2.4 COMPONENTS

### 2.4.1 Built-In Locks

Built-in locks are not required.

### 2.4.2 Coat Hooks

FS AA-L-00486, nylon.

### 2.4.3 Door Handles

FS AA-L-00486. Provide stainless steel handles.

### 2.4.4 Doors

FS AA-L-00486, not less than 1.5 mm thick steel sheet. Provide closed face

#### 2.4.4.1 Hinges

In addition to the requirements of FS AA-L-00486, provide 5-knuckle hinges, minimum 50 mm high. Fabricate knuckle hinges from not less than 2 mm thick steel stainless steel. Weld or bolt hinges to the door frame to suit materials. Weld, bolt, or rivet hinges to the door to suit materials.

#### 2.4.4.2 Latching Mechanisms

FS AA-L-00486.

### 2.4.5 Latch Strikes

FS AA-L-00486. Fabricate from not less than 2 mm thick stainless steel sheet, except latch strike must be continuous from top to bottom and fabricated as part of the door framing. Provide no less than two

silencers per door strike.

#### 2.4.6 Latches

Provide recessed door latch assembly of solid phenolic panel with latching devices compatible with the locker body construction. Include necessary metal components to provide latching and locking function with latches engaging the locker body at three locations for single tier full height doors.

#### 2.4.7 Silencers

FS AA-L-00486.

#### 2.4.8 Back and Side Panels, Tops, and Bottoms

FS AA-L-00486, not less than 13 mm plastic laminate faced solid phenolic core panel.]

#### 2.4.9 Sloping Locker Tops

Provide sloping locker tops in addition to the locker-section flat tops. Sloping tops must be continuous in length. Provide fillers or closures at the exposed end of sloping tops. Fabricate sloping tops from not less than 10 mm plastic laminate faced solid phenolic core panels.

#### 2.4.10 Shelves

FS AA-L-00486. Fabricate from not less than 13 mm plastic laminate faced solid phenolic core panels.

#### 2.4.11 Base Panels

FS AA-L-00486.

#### 2.4.12 Legs

Provide lockers without legs, as indicated.

#### 2.4.13 Number Plates

Number plates are not required.

#### 2.4.14 Fastening Devices

Provide bolts, nuts, and rivets as specified in FS AA-L-00486.

### PART 3 EXECUTION

#### 3.1 ASSEMBLY AND INSTALLATION

Assemble lockers according to the locker manufacturer's instructions. Align lockers horizontally and vertically. Secure lockers to wall with screws as indicated. Bolt adjacent lockers together. Adjust doors to operate freely without sticking or binding and to ensure they close tightly.

### 3.2 FIELD QUALITY CONTROL

#### 3.2.1 Testing

Government may request performance-characteristic tests on assembled lockers in accordance with FS AA-L-00486. Lockers not conforming will be rejected.

#### 3.2.2 Repairing

Remove and replace damaged and unacceptable portions of completed work with new.

#### 3.2.3 Cleaning

Clean surfaces of the work, and adjacent surfaces soiled as a result of the work, in an approved manner. Remove equipment, surplus materials, and rubbish from the site.

-- End of Section --

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## SECTION 12 00 01.00 20

FURNITURE, FIXTURES AND EQUIPMENT (FF&E) PACKAGE  
02/23

## PART 1 GENERAL

## 1.1 CONTRACTOR'S FURNITURE, FIXTURES AND EQUIPMENT (FF&amp;E) HANDLING AND ADMINISTRATION RATE (HAR)

The HAR accounts for all administrative costs, overhead, FF&E subcontractor bonding fees deemed necessary by the Contractor, administration of subcontracts, profit, and any other costs associated with, and related to, the coordination and processing of the procurement and installation of FF&E. Offerors must propose a HAR for the FF&E not to exceed 5 percent. Do not include costs associated with INTERIOR DESIGN SERVICES in the HAR. The proposed HAR percentage is a fixed rate and will be incorporated into the contract award. The HAR will not be adjusted regardless of fluctuations from the estimate amount for the FF&E.

The Government will indicate the FF&E estimate in the FF&E Option Item in the Price Schedule based on the Final FF&E Design. The Government estimated price is to provide Offerors with the projected magnitude of effort and includes the actual estimated cost of the FF&E including freight, installation, and other applicable vendor charges. The Government estimated price is only an estimated amount and must not be altered by the Contractor during the solicitation process.

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N (2024) Navy and Marine Corps Design

UFC 3-120-10 (2018; with Change 2, 2021) Interior Design

## 1.3 DEFINITIONS - FF&amp;E TURNKEY EFFORT

The Furnishings, Fixtures, and Equipment Package may include, but is not limited to; systems and modular furniture, workstations, desks, seating, storage, filing, visual display items, accessories, artwork, command graphics, training and conference furniture, soft window treatments (draperies, valances, and cornices), shop equipment, fitness equipment, Child Development Center furniture, appliances, portable weapons cabinets, dorm and quarters furnishings and other miscellaneous items to support facility functions.

The project may also include Specialty CEQ Package for which the Activity may assist in specifying items, which may include, but not limited to: Audio Visual (AV) equipment, commercial kitchen equipment, medical equipment and wharf equipment. For the purposes of this specification, if a Specialty CEQ Package is required, the same requirements identified for the FF&E Package apply to the Specialty CEQ Package or as outlined by project-specific UFC or requiring Agency.

### 1.3.1 FF&E Option Item

As an option, provide procurement and installation coordination of the approved complete and usable Final FF&E Package.

- b. The Government is not obligated to award the FF&E Option Item. Should the Government choose to award the FF&E Option, the Item(s) will be awarded as a negotiated option to the contract or task order. The Contractor's proposed HAR will be applied to all vendor and supplier costs for the FF&E.
- c. The Government will provide separate funding for procurement and installation coordination of the FF&E Package. Construction funds will not be used. Upon receipt of required funding, the Contracting Officer will authorize the Contractor, as an option to the construction contract, to provide the Government-approved Final FF&E Package exactly as specified and complying with priorities found in the Federal Acquisition Regulation (FAR) 8.404 Use of Federal Supply Schedules. The amount of the option will be the actual cost of these items from the Federal Government price schedules and, when necessary, Open Market sources. The option amount will also include shipping, freight, handling, installation, vendor design fees, project management fees, applicable sales tax, Contractor's FF&E Handling and Administration Rate percentage as applied to the Final FF&E or Specialty CEQ total cost(s) and the NAVFAC required bond fee.
- d. If awarded, the FF&E Option Item will be awarded at least 6 months prior to the contract completion date. A minimum of 6 months is required for the Contractor to purchase, deliver and install the product without impacting the overall completion date of the project. The Contractor's schedule must assume the award of the FF&E as an option. Government Fiscal Year funding availability must be considered and incorporated into the FF&E Schedule. No schedule extensions will be granted if the option item is awarded at least 6 months prior to the contract completion date. If the Government decides to negotiate and award the option item with less than 6 months prior to the contract completion date, the Contractor may be entitled to a contract extension and extended field overhead. A contract extension and extended field overhead will only be granted in those cases where the Contractor demonstrates that the Government's award of the FF&E Option Item is within the last 6 months of the contract.
- e. The FF&E Option Item (and any Specialty Equipment and Audio Visual (A/V) Option Item(s)) are exempt from NAVFAC Procedures, Guidance and Instruction (NPGI) 17.202(a)(ii) option restriction to award within 365 days of contract award. Per NPGI 17.202(a)(ii), Note 2, "Options for Collateral Equipment (CEQ) to include Furniture, Fixtures and Equipment (FF&E) and Audio-Visual (A/V) are not limited to the 365 day time period from date of award. Written approval for a longer period to exercise options for FF&E and A/V is not required."

### 1.3.2 Authorization for Use of Government Supply Sources

The Contracting Officer will authorize the Contractor to purchase the required products through the specified sources and will provide a letter of authorization to the Contractor in accordance with FAR 51.102 Authorization to use Government supply sources (e)(1) through (5) citing the name of the furniture dealer(s) and other information to use when

accessing the Federal Government supply sources. The Contractor must procure and coordinate the installation of the approved Final FF&E Package exactly as specified.

#### 1.3.3 Buy American Act (BAA) and Trade Agreement Act (TAA)

All products under the FF&E Option Item are subject to the rules and regulations governing the acquisition of foreign supplies, e.g., Buy American Act (BAA) (FAR 25.1 Buy American-Supplies). This also includes all supplies procured through Government Supply Sources. Solicitations are required to meet the Trade Agreement Act (TAA) Designated Countries (FAR 25.4 Trade Agreements), as part of the BAA. In addition to the applicable BAA clauses and provisions for construction materials, the Contracting Officer must also insert the appropriate BAA or TAA clauses and provisions for supplies. The Contractor is responsible for ensuring the FF&E is compliant with the BAA or TAA.

#### 1.3.4 Prohibition on Procurement of Certain Items Containing Perfluorooctane Sulfonate (PFOS) or Perfluorooctanoic Acid (PFOA)

Effective April 1, 2023, in accordance with section 333 of the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, the Department of Defense must not procure any covered items that contain PFOS or PFOA. The Contractor must not provide any covered items containing PFOS or PFOA in performance of this contract, task order, delivery order, or blanket purchase agreement order. A covered item means the following:

- a. Nonstick cookware or cooking utensils for use in galleys or dining facilities.
- b. Upholstered furniture, carpets, and rugs that have been treated with stain-resistant coatings.

#### 1.3.5 Procurement and Installation

The Contractor must coordinate the building completion date with the installation dealer(s) specified in the FF&E Package. It is recommended that the Contractor order the products once the option is awarded to avoid incurring additional costs. Production and delivery dates should be requested of manufacturers or dealers at the time of order placement to coincide with the Beneficial Occupancy Date (BOD). The Contractor must anticipate possible manufacturer price increases if order placement is delayed. Any costs incurred due to manufacturer price increases as a result of order placement delays will be the burden of the Contractor.

#### 1.3.6 Deposits

The Contractor must anticipate providing a deposit of between 30 percent and 50 percent of the product costs when placing the orders with the manufacturer's dealerships. The Government will not fund deposits.

#### 1.3.7 Construction Wage Rate Requirements Statute

Prior to the Interior Designer sending out the Best Value Determination (BVD) Request for Quote (RFQ) Package to vendors, the NAVFAC Contracting Officer, Contract Specialist, Project Manager, Design Manager, NAVFAC Interior Designer and Collateral Equipment Manager made a determination as to whether the FF&E installation is covered by the Construction Wage Rate



Requirements statute. The BVD RFQ Package identified any items with installation that is subject to the Construction Wage Rate Requirements statute.

#### 1.3.7.1 Construction Wage Rate Requirements Statute is Not Applicable

The Construction Wage Rate Requirements Statute is Not Applicable if the FF&E assembly and installation work is:

- a. Segregable from the construction of the facility (i.e., procured in a separate FF&E Option (Item)); and
- b. The assembly and installation work is performed by the FF&E vendor or dealer or subcontractor(s); and
- c. The onsite FF&E work does not include a substantial amount of construction work; and
- d. The workforce for the FF&E installation and delivery is separate and distinct from the labor workforce performing the construction effort by the Contractor.

If the above a. through d. are true, the Construction Wage Rate Requirements Statute is not applicable to FF&E installed by General Services Administration (GSA), other Government Supply Sources or Open Market.

#### 1.3.7.2 Construction Wage Rate Requirements Statute is Applicable

The Construction Wage Rate Requirements Statute is Applicable if the FF&E assembly and installation work is:

- a. Performed by the Contractor; and
- b. Is not physically or functionally separate from the other construction work required by the contract; and
- c. Is not capable of being performed on a segregated basis from the other construction work required by the contract.

-OR-

- d. The assembly and installation work is performed by the FF&E vendor or dealer or subcontractor(s); and
- e. The onsite FF&E work does include a substantial amount of construction work.

If the above a. through c. are true or d. through e. are true, the Construction Wage Rate Requirements statute would be applicable to the installation of all FF&E that is installed and attached to the facility.

#### 1.3.8 Taxes

##### 1.3.8.1 Exemptions

Exemptions for certain state or local taxes may be available to the Contractor and its subcontractors. The Contractor must take maximum advantage of all exemptions, including obtaining a resale permit from

state and local taxation authorities whether available directly to the Contractor or based on an exemption afforded the Government. The responsibility for paying applicable taxes rests with the Contractor. Applicable state and local taxes to the FF&E Package must be included with the subcontractor's quote, if required by the state or locality.

#### 1.3.8.2 Construction Contract Requirements

As prescribed in FAR 29.401-3 Federal, State, and local taxes, the Contracting Officer must insert FAR clause 52.229-3, Federal, State and Local Taxes, in the construction contract. In accordance with this clause:

- a. The contract price includes all applicable Federal, State, and local taxes and duties; and
- b. The Government must, without liability, furnish evidence appropriate to establish exemption from any Federal, State, or local tax when the Contractor requests such evidence and a reasonable basis exists to sustain the exemption.

#### 1.3.9 Performance and Payment Bonds

##### 1.3.9.1 No Bond Requirement for FF&E at Time of Construction Contract Award

FAR 28.102-1(a) requires that performance and payment bonds be included in all construction contracts exceeding the Simplified Acquisition Threshold (SAT), except under the conditions listed. Since the FF&E Option Item is not part of the original contract award, the Contractor is not required to secure bond for the FF&E at the time the basic construction contract is awarded.

##### 1.3.9.2 Additional Bond at FF&E Option Item Award

FAR 28.102-2(d) requires the Government to secure additional bond from the Contractor for any contract price increases. An increase to the original contract bond is required at the time of the FF&E Option Item award to protect the Government against the Contractor not providing FF&E ordered and to ensure that the FF&E vendor(s) receive payment if the Contractor goes out of business. The increased cost of the Government required bond must come from the FF&E funding source and is not a part of the Contractor's HAR.

##### 1.3.9.3 Bonds Required by the Contractor

If the Contractor requires bonding from the FF&E subcontractor(s), the cost must be covered in the Contractor's HAR.

##### 1.3.10 Item Unique Identification (IUID) and Valuation

Item Unique Identification (IUID) and valuation is a system of marking and valuing items delivered to Department of Defense (DoD) that enhances logistics, contracting, and financial business transactions. Utilize the IUID policy for all DoD contracts that require the delivery of items. An item is a single article or a single unit formed by a grouping of subassemblies, components or constituent parts. In accordance with DFARS 211.274-2 Policy for item unique identification, IUID is required for all delivered items for which the Government's unit acquisition cost is \$5,000 or more. Items identified by the requiring activity to be serially managed, mission essential, controlled inventory or for other reasons

requiring permanent identification, must be marked with a unique item identifier.

#### 1.3.10.1 IUID

The two main steps involved in IUID requirements are:

- a. Item marking, and
- b. Delivery of data about items as a part of the acceptance and delivery process.

#### 1.3.10.2 Data Matrix Requirements

Item marking requires that qualifying items contain a data matrix either directly inscribed on the individual item or on a permanent label or data plate attached to the item. Delivery of data is executed by submitting to the IUID Registry information such as: description of the item, its value, and the date of acceptance by the government. The IUID Registry is the repository established to capture and store all IUID data.

#### 1.3.10.3 Government's Unit Acquisition Cost

In accordance with DFARS 211.274-3 Policy for valuation, Contractors must be required to identify the Government's unit acquisition cost for all items delivered even if none of the criteria for placing a unique item identification mark applies.

### 1.4 ADMINISTRATIVE REQUIREMENTS

#### 1.4.1 FF&E Pre-Installation Walk-Through

The Contractor must coordinate and participate in a walk-through to ensure that the building interior is substantially complete in all areas scheduled to receive FF&E to include floor, wall and ceiling finishes and building systems located on or in the ceilings. Attendees must include the FF&E vendor(s) in-person (if geographically feasible) or virtually and the NAVFAC Construction Manager. Approval by the NAVFAC Construction Manager to proceed with the FF&E installation is required. The walk-through date must occur prior to the commencement of the FF&E installation and accommodate sufficient time in the construction schedule for transportation of FF&E to the job site upon approval by the NAVFAC Construction Manager.

#### 1.4.2 Sequencing

The Contractor must coordinate the building completion, occupancy, and furniture installation dates with the installation dealer(s) specified in the FF&E Package. Any costs associated with or delaying furniture shipments to accommodate the construction schedule is the responsibility of the Contractor.

### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

The NAVFAC Interior Designer is required to receive a copy of all submittals with a "G" Classification.

SD-01 Preconstruction Submittals

FF&E Schedule and Schedule Updates; G

SD-04 Samples

Post Option Award FF&E Finish Submittals; G

SD-05 Design Data

SD-10 Operation and Maintenance Data

Warranty and Maintenance Information; G

SD-11 Closeout Submittals

Punch List(s) Submittal; G

1.6 QUALITY CONTROL

1.6.1 Installation Qualifications

All FF&E must be installed by the manufacturer's dealer of record and not by the Contractor.

1.7 DELIVERY, STORAGE, AND HANDLING

FF&E must be shipped to and stored at the vendor or dealer's conditioned warehouse, until all items are received unless dealer has coordinated just-in-time delivery with the Contractor. Vendor or dealer is responsible for any orders arriving before or after the agreed upon delivery and installation date with the Contractor, including storage of items.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 INTERIOR DESIGN SERVICES

Provide Interior Design FF&E services and deliverables in accordance with FC 1-300-09N, UFC 3-120-10, Interior Design Procedures: Furniture Fixtures and Equipment (FF&E) located at <https://www.wbdg.org/ffc/navy-navfac/collateral-equipment/navfac-id-procedures-ffe> and as further defined in the paragraphs below. Utilize NAVFAC templates, criteria and procedures documents located at <https://www.wbdg.org/ffc/navy-navfac/collateral-equipment>.

3.1.1 SYSTEM DESCRIPTION: CONTRACTOR'S INTERIOR DESIGN SERVICES

The Final FF&E Package submittal, as prepared by the NAVFAC Interior Designer, was developed to fully integrate with the building systems (e.g., electrical, mechanical) to include product sizes and communications, data, and electrical requirements. **The Contractor is not**

**required to hire an Interior Designer to develop or validate the Final FF&E Package.**

Provide and comply with Interior Design Procedures: Furniture Fixtures and Equipment (FF&E) CHAPTER(s) 1 through 4 and specific Design, Meeting and Submittal deliverables identified below in CHAPTER 7 ID SERVICES - DESIGN BID BUILD (DBB) - PROCESS 3.

**3.1.1.1 Final FF&E Package**

The FF&E or CEQ Package with all required FF&E items and Computer Aided Design and Drafting (CADD)/BIM drawings, will be provided by the NAVFAC Contracting Officer upon request. The Contractor must request to receive the package prior to the PRECON.

**3.1.1.2 FF&E Schedule and Schedule Updates Submittal**

For all projects, including fast track projects and renovations, the Contractor is responsible for sufficiently scheduling all FF&E meetings and deliverables early enough to:

- a. Obtain the required government approvals,
- b. Meet all FF&E funding, award, ordering and installation lead times and deadlines, and
- c. Complete the FF&E installation by the Beneficial Occupancy Date (BOD).

All submittal due dates for the FF&E and any Specialty CEQ must be noted in the FF&E Schedule and Schedule Updates Submittal and reflected in the Contractor's construction schedule. Provide a FF&E Schedule and Schedule Updates Submittal for all FF&E and Specialty CEQ services and deliverables to the NAVFAC Construction Manager, Interior Designer and Contracting Officer 7 business days after the PRECON. Submit FF&E Schedule and Schedule Updates Submittal to the NAVFAC Interior Designer within 7 days of any construction schedule revisions.

All submittal due dates for the FF&E must be reflected in the Contractor's construction schedule. Submit proposed changes to the NAVFAC Interior Designer for review. The Contracting Officer must approve all revisions or deviations.

**3.1.1.3 Services**

**3.1.1.3.1 Building Infrastructure and Systems**

The FF&E and CEQ Package(s) must be fully integrated with the construction and schedule of all building finishes and all building infrastructure and systems (such as heating, ventilation and air conditioning (HVAC), plumbing, fire protection, communications, electrical, data, architecture). All electrical, data and communications outlets, switches, fire extinguishers, thermostats, sprinkler heads and Architectural Barriers Act (ABA) clearances must be accommodated for and be fully accessible once equipment is installed. The NAVFAC Interior Designer and Equipment Specialist(s) must work with the Contractor and the electrical, telecommunication, and mechanical subcontractors to coordinate all infrastructure with the recommended vendors to include connection of power, data, communications, waterlines and gas lines, and calibration, testing and training prior to facility turnover and operation.

Provide coordination with the building infrastructure and provide revisions to the building infrastructure if required to support the FF&E Package and Specialty CEQ Package.

#### 3.1.1.3.2 Government-Furnished, Government Installed Equipment (GFGI) and Government-Furnished, Contractor-Installed Equipment (GFCI)

The FF&E and CEQ plans included in the Final FF&E Package identify all FF&E or CEQ to include specialty CEQ, existing and future Government-Furnished, Government Installed (GFGI) Equipment and Government-Furnished, Contractor-Installed Equipment (GFCI). These items may include, but are not limited to, multi-function devices, printers, shredders, safes and vending machines.

#### 3.1.2 FF&E MEETINGS, PRESENTATIONS AND SUBMITTALS

The Contractor must attend site visits, walk-throughs, coordinate with the NAVFAC Interior Designer, obtain Activity and NAVFAC approvals and provide a complete and functional FF&E Package. All meetings must be coordinated with advance notice to the FF&E vendor(s), and NAVFAC Interior Designer, so that schedule and travel can be coordinated.

These are minimum requirements and the Contractor must be prepared to provide all additional meetings and submittals that may be necessary to support the Interior Design effort and FF&E coordination. Submit any revisions or deviations caused by discontinued items or NAVFAC required changes to the NAVFAC Interior Designer for review. The Contracting Officer must approve all revisions and deviations.

##### 3.1.2.1 Interior Finish Construction Submittals

Some building interior finishes (Structural Interior Design) may need to be revised to coordinate with the Final FF&E Package. Submit any revisions or deviations to the building interior finishes to the NAVFAC Interior Designer for review. The Contracting Officer must approve any changes to the building interior finishes.

#### 3.2 POST-AWARD FF&E OPTION ITEM CHANGES

##### 3.2.1 Requests for Price Changes

The FF&E Option has been priced, negotiated, and awarded based on specific line items as detailed in the approved Final FF&E Submittal. These items have been approved considering color, specific type and quality of material, price, and usable life. The Government will expect and require the Contractor to provide exactly those items. Any requests to change FF&E items that will cause a price change must be submitted in writing for approval by the NAVFAC Contracting Officer and Interior Designer before purchase and installation.

##### 3.2.2 Post Option Award Price Increases

Reimbursement for price increases after FF&E Option award will not be considered.

##### 3.2.3 Progress Payments

Per FAR 52.232-5 Payments under Fixed-Price Construction Contracts, the

Contracting Officer may authorize a progress payment on the FF&E ordered as material delivered to the Contractor, at locations other than the site if:

- a. Consideration is specifically authorized by this contract; and
- b. The Contractor furnishes satisfactory evidence, including a copy of the order acknowledgment and proof of payment, that it has acquired title to such material, the material is stored in a secure, climate controlled location, and that the material will be used to perform this contract.

### 3.3 INSTALLATION

The FF&E Package includes the installation of all furniture and furnishings as specified in the Final FF&E Package. The FF&E installation dealer(s) specified in the FF&E Package must receive, store as required, transport to the project site, off load, inside deliver, unpack, assemble, install, clean, and dispose of all the trash for all FF&E. It is the Contractor's responsibility to coordinate the building completion, occupancy, and furniture installation dates with the installation dealer(s) specified in the FF&E Package. Any costs associated with or delaying furniture shipments to accommodate the construction schedule is the responsibility of the Contractor.

#### 3.3.1 FF&E Pre-Installation Walk-Through

The Contractor must coordinate and participate in a walk-through to ensure that the building interior is substantially complete in all areas scheduled to receive FF&E to include floor, wall and ceiling finishes and building systems located on or in the ceilings. Attendees must include the NAVFAC Construction Manager and the FF&E vendor(s) in-person (if geographically feasible) or virtually. Approval by the NAVFAC Construction Manager to proceed with the FF&E installation is required. The walk-through date must occur prior to the commencement of the FF&E installation and accommodate sufficient time in the construction schedule for transportation of FF&E to the job site upon approval by the NAVFAC Construction Manager.

#### 3.3.2 Walk Through(s) and Punch List(s) Submittal

The Contractor, NAVFAC Interior Designer, FF&E vendor(s) or dealer(s) representative and installation superintendent, other specialty consultants (if applicable), NAVFAC Construction Manager and the Activity will attend one punch list site visits. The site visit must be held at the installation dealer's 98 percent completion and must identify and record all punch list items in the Punch List(s) Submittal.

The Contractor must repair, to the Government's satisfaction, all damage to any facility finish that is a result of the furniture installation and correct all punch list items for the FF&E. Any damage or loss of FF&E after installation will be the responsibility of the Contractor and not the FF&E vendor or installer. Daily installation walk-throughs by the Contractor and the FF&E installation dealer are encouraged to document any damage and identify the responsible party(s).

### 3.4 INSTALLATION WARRANTY

Install all movable furnishings in accordance with the manufacturer's

instructions and warranty requirements. All movable furnishings must be level and aligned. All doors, drawers and accessories must be level and aligned to open, close and otherwise operate smoothly and securely.

All FF&E and Specialty CEQ must be installed by the manufacturer's dealer of record and not by the Contractor.

### 3.5 CLOSEOUT SUBMITTALS

#### 3.5.1 Warranty And Maintenance Information

Provide to the Contracting Officer at the final FF&E walk-thru one electronic copy and one binder copy of all ordering documentation, including but not limited to, quotes, purchase orders, factory order numbers (FO), IUID, and warranty and maintenance information for all products.

-- End of Section --



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## SECTION 21 13 13

WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION  
08/20

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## CSA GROUP (CSA)

|              |                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------|
| CSA B16.21   | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                                                      |
| CSA B51      | (2024) Boiler, pressure vessel, and pressure piping code                                              |
| CSA B64      | (2021) Backflow preventers and vacuum breakers                                                        |
| CSA B125.1   | (2018) Plumbing supply fittings                                                                       |
| CSA B125.3   | (2022) Plumbing fittings                                                                              |
| CSA B137.3   | (2020) Rigid polyvinylchloride (PVC) pipe and fittings for pressure applications                      |
| CSA B137.4   | (2020) Polyethylene (PE) piping systems for gas services                                              |
| CSA B131     | (1999) External Zinc-Coating of Ductile Iron Pipe (Adopted ISO 8179:1985)                             |
| CSA C22.2    | Woodworking machines – Safety – Part 8: Belt sanding and calibrating machines for straight workpieces |
| CSA G40.2    | General requirements and specific grades for structural quality steel                                 |
| CSA G40.21   | Grade 350W Class H                                                                                    |
| CSA G550     | Ductile Iron Castings                                                                                 |
| CSA G164     | Hot dip galvanizing of irregularly shaped articles                                                    |
| CAN/ULC-S502 | Gate valves used in fire-protection service                                                           |
| CAN/ULC-S524 | (2019) Standard For Installation Of Fire Alarm Systems                                                |
| CAN/ULC-S530 | (2021) Standard For Heat Actuated Fire Detectors For Fire Alarm Systems                               |

|              |                                                                            |
|--------------|----------------------------------------------------------------------------|
| CAN/ULC-S536 | (2024) Standard for Inspection and Testing of Fire Alarm Systems           |
| CAN/ULC-S560 | (2021) Standard For High-Performance And Aviation Foam Liquid Concentrates |
| CSA Z245.1   | Steel Pipe                                                                 |
| CSA Z245.30  | (2022) Field-applied external coatings for steel pipeline systems          |

## AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| ASME B16.1  | (2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250 |
| ASME B16.3  | (2021) Malleable Iron Threaded Fittings, Classes 150 and 300                |
| ASME B16.4  | (2021) Gray Iron Threaded Fittings; Classes 125 and 250                     |
| ASME B16.21 | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                            |

## AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

|           |                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------|
| ASSE 1013 | (2021) Performance Requirements for Reduced Pressure Principle Backflow Prevention Assemblies |
| ASSE 1015 | (2021) Performance Requirements for Double Check Backflow Prevention Assemblies               |

## AMERICAN WATER WORKS ASSOCIATION (AWWA)

|                  |                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------|
| AWWA C104/A21.4  | (2022) Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water                                  |
| AWWA C110/A21.10 | (2021) Ductile-Iron and Gray-Iron Fittings                                                                |
| AWWA C111/A21.11 | (2023) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings                                   |
| AWWA C203        | (2020) Coal-Tar Protective Coatings and Linings for Steel Water Pipelines - Enamel and Tape - Hot-Applied |
| AWWA M14         | (2024) Manual: Recommended Practice for Backflow Prevention and Cross-Connection Control                  |

## ASTM INTERNATIONAL (ASTM)

|               |                                                                                    |
|---------------|------------------------------------------------------------------------------------|
| ASTM A47/A47M | (1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings |
|---------------|------------------------------------------------------------------------------------|

|                 |                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------|
| ASTM A53/A53M   | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless |
| ASTM A135/A135M | (2021) Standard Specification for Electric-Resistance-Welded Steel Pipe                               |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                   |
| ASTM A183       | (2014; R 2020) Standard Specification for Carbon Steel Track Bolts and Nuts                           |
| ASTM A536       | (2024) Standard Specification for Ductile Iron Castings                                               |

## FM GLOBAL (FM)

|              |                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------|
| FM APP GUIDE | (updated on-line) Approval Guide<br><a href="https://www.approvalguide.com/">https://www.approvalguide.com/</a> |
|--------------|-----------------------------------------------------------------------------------------------------------------|

## INTELLIGENCE COMMUNITY STANDARD (ICS)

|           |                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------|
| ICS 705-1 | (2010) Physical and Technical Security Standard for Sensitive Compartmented Information Facilities |
|-----------|----------------------------------------------------------------------------------------------------|

## MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

|           |                                                                |
|-----------|----------------------------------------------------------------|
| MSS SP-71 | (2018) Gray Iron Swing Check Valves, Flanged and Threaded Ends |
|-----------|----------------------------------------------------------------|

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                            |
|----------|--------------------------------------------------------------------------------------------|
| NFPA 13  | (2022; TIA 23-4) Standard for the Installation of Sprinkler Systems                        |
| NFPA 24  | (2025) Standard for the Installation of Private Fire Service Mains and Their Appurtenances |
| NFPA 101 | (2024) Life Safety Code                                                                    |
| NFPA 291 | (2025) Recommended Practice for Fire Flow Testing and Marking of Hydrants                  |

## NATIONAL INSTITUTE FOR CERTIFICATION IN ENGINEERING TECHNOLOGIES (NICET)

|              |                                                                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NICET 1014-7 | (2012) Program Detail Manual for Certification in the Field of Fire Protection Engineering Technology (Field Code 003) Subfield of Automatic Sprinkler System Layout |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## UL SOLUTIONS (UL)

UL 668 (2004; Reprint Oct 2021) UL Standard for Safety Hose Valves for Fire-Protection Service

## UNDERWRITERS LABORATORIES (UL)

UL 199 (2020) UL Standard for Safety Automatic Sprinklers for Fire-Protection Service

UL 262 (2004; Reprint Jul 2023) Gate Valves for Fire-Protection Service

UL 312 (2022) UL Standard for Safety Check Valves for Fire-Protection Service

UL 405 (2013; Bul. 2020) UL Standard for Safety Fire Department Connection Devices

UL Fire Prot Dir UL Product IQ (updated online) at <https://productiq.ulpropsector.com/en>

## 1.2 SYSTEM DESCRIPTION

Provide wet pipe sprinkler system(s) in all areas of the building. Except as modified herein, the system must meet the requirements of NFPA 13 and CAN/ULC-S524 . Pipe sizes which are not indicated on the Contract drawings must be determined by hydraulic calculations.

## 1.2.1 Hydraulic Design

## 1.2.1.1 Basis for Calculations

A waterflow test was performed on August 2023 at Cooper Drive, Freshwater, NL Canada and resulted in a static pressure of 676 kPa with a residual pressure of 255 kPa while flowing 3,112 L/min. Perform a fire hydrant flow test prior to shop drawing submittal in accordance with NFPA 291. Results must include hydrant elevations relative to the building and hydrant number/identifiers for the tested hydrants, including which were flowed, which had a gauge. This information must be presented in a tabular form if multiple hydrants were flowed. The results must be included with the hydraulic calculations. Hydraulic calculations must be based on flow test noted in this paragraph, unless verified by the NAVFAC Atlantic Fire Protection Engineer an approved by Contracting Officer. Hydraulic calculations must be based upon the Hazen-Williams formula with a "C" value noted in NFPA 13 for piping, and for existing underground piping. Hydraulic calculations must be based on operation of the fire pump(s) provided in Section 21 30 00 FIRE PUMPS. The minimum residual pressure in a service lateral (lead-in) at the design flow rate must be 138 kPa at the inlet to the backflow preventer.

## 1.2.1.2 Hydraulic Calculations

- a. Water supply curves and system requirements must be plotted on semi-logarithmic graph ( $N^{1.85}$ ) paper so as to present a summary of the complete hydraulic calculation.
- b. Provide a summary sheet listing sprinklers in the design area and

their respective hydraulic reference points, elevations, minimum discharge pressures and minimum flows. Elevations of hydraulic reference points (nodes) must be indicated.

- c. Documentation must identify each pipe individually and the nodes connected thereto. Indicate the diameter, length, flow, velocity, friction loss, number and type fittings, total friction loss in the pipe, equivalent pipe length and Hazen-Williams coefficient for each pipe.
- d. Where the sprinkler system is supplied by interconnected risers, the sprinkler system must be hydraulically calculated using the hydraulically most demanding single riser. The calculations must not assume the simultaneous use of more than one riser.
- e. All calculations must include the backflow preventer manufacturer's stated friction loss at the design flow or 55 kPa for double check backflow preventer, whichever is greater.
- f. All calculations must be performed back to the actual location of the flow test, taking into account the direction of flow in the service main at the test location.
- g. For gridded systems, calculations must show peaking of demand area friction loss to verify that the hydraulically most demanding area is being used. A flow diagram indicating the quantity and direction of flows must be included.

#### 1.2.1.3 Design Criteria

Hydraulically design the system to discharge a minimum density of 8 L/min per square meter over the hydraulically most demanding 230 square meter of floor area as indicated on the drawings. Hydraulic calculations must be in accordance with the Area/Density Method of NFPA 13 and CAN/ULC-S524. Add an allowance for exterior hose streams of 950 L/min to the sprinkler system demand at the fire hydrant shown on the drawings closest to the point where the water service enters the building .

#### 1.2.2 Sprinkler Coverage

Sprinklers must be uniformly spaced on branch lines. Provide coverage throughout 100 percent of the building. This includes, but is not limited to, telephone rooms, electrical equipment rooms (regardless of the fire resistance rating of the enclosure), boiler rooms, switchgear rooms, transformer rooms, attached electrical vaults and other electrical and mechanical spaces. Coverage per sprinkler must be in accordance with NFPA 13 and CAN/ULC-S524. Provide sprinklers below all obstructions in accordance with NFPA 13 and CAN/ULC-S524. Exceptions are as follows:

- a. Sprinklers may be omitted from small rooms which are exempted for specific occupancies in accordance with NFPA 101 and NBC/NFC.

#### 1.2.3 Qualified Fire Protection Engineer (QFPE) (PEGNL)

An individual who is a licensed professional engineer (P.E.) who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES)

and has relevant fire protection engineering experience. Services of the QFPE must include:

- a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Working (shop) drawings and calculations must be prepared by, or prepared under the immediate supervision of, the QFPE. The QFPE must affix their professional engineering stamp with signature to the shop drawings, calculations, and material data sheets, indicating approval prior to submitting the shop drawings to the DFPE.
- b. Provide a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting all outstanding comments.
- c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).
- d. Witnessing pre-Government and final Government functional performance testing and performing a final installation review.
- e. Signing applicable certificates under SD-07.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Partial submittals and submittals not fully complying with NFPA 13 , CAN/ULC-S524, CAN/ULC-S530 and this specification section must be returned disapproved without review. SD-02, SD-03 and SD-05 must be submitted simultaneously.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the designer and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE must be returned disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Qualified Fire Protection Engineer (QFPE); G

Sprinkler System Designer; G

Sprinkler System Installer; G

#### SD-02 Shop Drawings

Shop Drawing; G

#### SD-03 Product Data

Pipe; G

Fittings; G

Valves, including gate, check, butterfly, and globe; G

Alarm Valves; G

Relief Valves; G

Sprinklers ; G

Pipe Hangers and Supports ; G

Sprinkler Alarm Switch; G

Valve Supervisory (Tamper) Switch; G

Fire Department Connection; G

Backflow Prevention Assembly; G

Air Vent; G

Hose Valve; G

Nameplates; G

#### SD-05 Design Data

Hydraulic Calculations; G

#### SD-06 Test Reports

Test Procedures; G

#### SD-07 Certificates

Verification of Compliant Installation; G

Request for Government Final Test; G

#### SD-10 Operation and Maintenance Data

Operating and Maintenance (O&M) Instructions; G

Spare Parts Data; G

#### SD-11 Closeout Submittals

As-built drawings

### 1.4 QUALITY ASSURANCE

#### 1.4.1 Preconstruction Submittals

Within 36 days of contract award but no less than 14 days prior to commencing work on site, the prime Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior



to the review and approval of the qualifications will be returned Disapproved Without Review.

#### 1.4.1.1 Shop Drawing

Two (2) copies of the shop drawings, no later than 28 days prior to the start of system installation. Working drawings conforming to the requirements prescribed in NFPA 13, CAN/ULC-S524, CAN/ULC-S530 and must be no smaller than the Contract Drawings. Each set of drawings must include the following:

- a. A descriptive index with drawings listed in sequence by number. A legend sheet identifying device symbols, nomenclature, and conventions used in the package.
- b. Floor plans drawn to a scale not less than 1:100 clearly showing locations of devices, equipment, risers, and other details required to clearly describe the proposed arrangement.
- c. Actual center-to-center dimensions between sprinklers on branch lines and between branch lines; from end sprinklers to adjacent walls; from walls to branch lines; from sprinkler feed mains, cross mains and branch lines to finished floor and roof or ceiling. A detail must show the dimension from the sprinkler and sprinkler deflector to the ceiling in finished areas.
- d. Longitudinal and transverse building sections showing typical branch line and cross main pipe routing, elevation of each typical sprinkler above finished floor and elevation of "cloud" or false ceilings in relation to the building ceilings.
- e. Plan and elevation views which establish that the equipment will fit the allotted spaces with clearance for installation and maintenance.
- f. Riser layout drawings drawn to a scale of not less than 1:25 to show details of each system component, clearances between each other and from other equipment and construction in the room.
- g. Details of each type of riser assembly, pipe hanger, and restraint of underground water main at point-of-entry into the building, and electrical devices and interconnecting wiring. The dimension from the edge of vertical piping to the nearest adjacent wall(s) must be indicated on the drawings when vertical piping is located in stairs or other portions of the means of egress.
- h. Details of each type of pipe hanger and related components.
- i. Include fire pump curve with shop drawings and hydraulic calculations.

#### 1.4.1.2 Product Data

Two (2) copies of annotated catalog data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, options, and other pertinent information, that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

#### 1.4.1.3 Hydraulic Calculations

Calculations must be as outlined in NFPA 13, CAN/ULC-S524, CAN/ULC-S530 except that calculations must be performed by computer using software intended specifically for fire protection system design using the design data shown on the drawings. Include fire pump curve with submittal.

#### 1.4.1.4 Operating and Maintenance (O&M) Instructions

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA as supplemented and modified by this specification section.

Provide six manuals and one pdf version on electronic media. The manuals must include the manufacturer's name, model number, parts list, list of parts and tools that should be kept in stock by the owner for routine maintenance, troubleshooting guide, and recommended service organization (including address and telephone number) for each item of equipment. Each service organization submitted must be capable of providing 4-hour on-site response to a service call on an emergency basis.

Submit spare parts data for each different item of material and equipment specified. The data must include a complete list of parts and supplies, and a list of parts recommended by the manufacturer to be replaced after 1-year and 3 years of service. Include a list of special tools and test equipment required for maintenance and testing of the products supplied.

#### 1.4.2 Qualifications

##### 1.4.2.1 Sprinkler System Designer

The sprinkler system designer must be certified as a Level IV Technician by National Institute for Certification in Engineering Technologies (NICET) in the Water-Based Systems Layout subfield of Fire Protection Engineering Technology in accordance with NICET 1014-7, CET, AETTNL.

##### 1.4.2.2 Sprinkler System Installer

The sprinkler system installer must be regularly engaged in the installation of the type and complexity of system specified in the contract documents, and must have served in a similar capacity for at least three systems that have performed in the manner intended for a period of not less than 6 months.

#### 1.4.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this Section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of an item or equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation are mandatory requirements.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Protect all equipment delivered and placed in storage from the weather, excessive humidity and temperature variations, dirt and dust, or other

contaminants. All pipes must be either capped or plugged until installed.

## 1.6 EXTRA MATERIALS

Spare sprinklers and wrench(es) must be provided as spare parts in accordance with NFPA 13.

## PART 2 PRODUCTS

### 2.1 MATERIALS AND EQUIPMENT

#### 2.1.1 Standard Products

Provide materials, equipment, and devices listed for fire protection service when so required by NFPA 13 or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for a classification of material. Material and equipment must be standard products of a manufacturer regularly engaged in the manufacture of the products for at least 2 years prior to bid.

#### 2.1.2 Nameplates

Major components of equipment must have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the contract number provided on a new name plate permanently affixed to the item or equipment. Nameplates must be etched metal or plastic, permanently attached by screws to control units, panels or adjacent walls.

#### 2.1.3 Identification and Marking

Pipe and fitting markings must include name or identifying symbol of manufacturer and nominal size. Pipe must be marked with ASTM designation. Valves and equipment markings must have name or identifying symbol of manufacturer, specific model number, nominal size, name of device, arrow indicating direction of flow, and position of installation (horizontal or vertical), except if valve can be installed in either position. Markings must be included on the body casting or on an etched or stamped metal nameplate permanently on the valve or cover plate.

#### 2.1.4 Pressure Ratings

Valves, fittings, couplings, alarm switches, and similar devices must be rated for the maximum working pressures that can be experienced in the system, but in no case less than 1207 kPa.

### 2.2 UNDERGROUND PIPING COMPONENTS

#### 2.2.1 Pipe

Pipe must comply with NFPA 24, CAN/ULC-S524 and CAN/ULC-S536. Minimum pipe size is 150 mm. Piping more than 1.50 meters outside the building walls must comply with Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING. A continuous section of welded stainless steel fire water service piping from a point outside the building perimeter to a flanged fitting at least 304 mm above the finished floor within the building is acceptable.

### 2.2.2 Fittings and Gaskets

Fittings must be ductile-iron conforming to AWWA C110/A21.10, CSA B137.3; CSA B137.4; CSA B131 with cement mortar lining conforming to AWWA C104/A21.4 and CSA A257.2. Gaskets must be suitable in design and size for the pipe with which such gaskets are to be used. Gaskets for ductile-iron pipe joints must conform to AWWA C111/A21.11 and CSA B131.

### 2.2.3 Gate Valve

Installation must comply with NFPA 24, CAN/ULC-S524 and CAN/ULC-S536. Gate valves for use with indicator post must conform to UL 262, CAN/ULC-S560, CAN/ULC-S502.

### 2.2.4 Valve Boxes

Except where indicator posts are provided, for each buried valve, provide a cast-iron, ductile-iron, or plastic valve box of a suitable size. Plastic boxes must be constructed of acrylonitrile-butadiene-styrene (ABS) or inorganic fiber-reinforced black polyolefin. Provide cast-iron, ductile-iron, or plastic cover for valve box with the word "WATER" cast on the cover. The minimum box shaft diameter must be 133 mm. Coat cast-iron and ductile-iron boxes with bituminous paint applied to a minimum dry-film thickness of 0.254 mm.

### 2.2.5 Buried Utility Warning and Identification Tape

Provide detectable aluminum foil plastic backed tape or detectable magnetic plastic tape manufactured specifically for warning and identification of buried piping. Tape must be detectable by an electronic detection instrument. Provide tape, 80 mm minimum width, color coded for the utility involved with warning and identification imprinted in bold block letters continuously and repeatedly over the entire tape length. Warning and identification must read "CAUTION BURIED WATER PIPING BELOW" or similar wording. Use permanent code and letter coloring unaffected by moisture and other substances contained in trench backfill material.

## 2.3 ABOVEGROUND PIPING COMPONENTS

### 2.3.1 Steel Piping Components

#### 2.3.1.1 Steel Pipe

Except as modified herein, steel pipe must be black as permitted by NFPA 13, CAN/ULC-S524, CAN/ULC-S530 and conform to the applicable provisions of ASTM A53/A53M, CSA Z245.1, ASTM A135/A135M, CSA G40.21 or ASTM A153/A153M, CSA G164.

Steel pipe must be minimum Schedule 40 for sizes 50 mm and less; and minimum Schedule 10 for sizes larger than 50 mm. Steel piping with wall thickness less than Schedule 40 must not be threaded.

#### 2.3.1.2 Fittings

Fittings must be welded, threaded, or grooved-end type. Threaded fittings must be cast-iron conforming to ASME B16.4, CSA B51, CSA B125.1, CSA B125.3, malleable-iron conforming to ASME B16.3 or ductile-iron conforming to ASTM A536 and CSA G40.2. Plain-end fittings with mechanical

couplings, fittings that use steel gripping devices to bite into the pipe, steel press fittings and field welded fittings are not permitted. Fittings, mechanical couplings, and rubber gaskets must be supplied by the same manufacturer. Threaded fittings must use Teflon tape or manufacturer's approved joint compound. Reducing couplings are not permitted except as allowed by NFPA 13, CAN/ULC-S524, CAN/ULC-S530.

#### 2.3.1.3 Grooved Mechanical Joints and Fittings

Joints and fittings must be designed for not less than 1200 kPa service and the product of the same manufacturer. Field welded fittings must not be used. Fitting and coupling housing must be malleable-iron conforming to ASTM A47/A47M, Grade 32510; ductile-iron conforming to ASTM A536, Grade 65-45-12. Rubber gasketed grooved-end pipe and fittings with mechanical couplings are permitted in pipe sizes 50 mm and larger. Gasket must be the flush type that fills the entire cavity between the fitting and the pipe. Nuts and bolts must be heat-treated steel conforming to ASTM A183 and must be cadmium-plated or zinc-electroplated.

#### 2.3.1.4 Flanges

Flanges must conform to NFPA 13, CAN/ULC-S524, CAN/ULC-S530 and ASME B16.1, CSA G550: Malleable Iron Castings. Gaskets must be non-asbestos compressed material in accordance with ASME B16.21, CSA B16.21 1.6 mm thick, and full face or self-centering flat ring type.

#### 2.3.2 Flexible Sprinkler Hose

The use of flexible hose is not permitted.

#### 2.3.3 Pipe Hangers and Supports

Provide galvanized pipe hangers and supports in accordance with NFPA 13 CAN/ULC-S524, CAN/ULC-S530..

#### 2.3.4 Valves

Provide valves of types approved for fire service. Valves must open by counterclockwise rotation.

##### 2.3.4.1 Control Valve

Manually operated sprinkler control/gate valve must be outside stem and yoke (OS&Y) type or butterfly type as indicated on the drawings and must be listed.

##### 2.3.4.2 Check Valves

Check valves must comply with UL 312. Check valves 100 mm and larger must be of the swing type, have a clear waterway and meet the requirements of MSS SP-71 CSA Z245.1/CSA B51, for Type 3 or 4. Inspection plate must be provided on valves larger than 150 mm.

##### 2.3.4.3 Hose Valve

Valve must comply with UL 668.

### 2.3.5 Riser Check Valves

Provide riser check valve, pressure gauges and main drain.

## 2.4 ALARM INITIATING AND SUPERVISORY DEVICES

### 2.4.1 Sprinkler Alarm Switch

Vane or pressure-type flow switch(es). Connection of switch must be by the fire alarm installer. Vane type alarm actuating devices must have mechanical diaphragm controlled retard device adjustable from 10 to 60 seconds and must instantly recycle.

### 2.4.2 Valve Supervisory (Tamper) Switch

Switch must be integral to the control valve or suitable for mounting to the type of control valve to be supervised open. The switch must be tamper resistant and contain SPDT (Form C) contacts arranged to transfer upon removal of the housing cover or closure of the valve of more than two rotations of the valve stem.

## 2.5 BACKFLOW PREVENTION ASSEMBLY

Double-check valve assembly backflow preventer complying with ASSE 1013, CSA B64, ASSE 1015 and AWWA M14. Each check valve must have a drain. Backflow prevention assemblies must have current "Certificate of Approval from the Foundation for Cross-Connection Control and Hydraulic Research, FCCCHR List" and be listed for fire protection use. Listing of the specific make, model, design, and size in the FCCCHR List is acceptable as the required documentation.

### 2.5.1 Backflow Preventer Test Connection

Test connection must consist of a series of listed hose valves with 65-mm National Standard male hose threads with cap and chain.

## 2.6 FIRE DEPARTMENT CONNECTION

Fire department connection must be projecting type with cast-brass body, matching wall escutcheon lettered "Auto Spkr" with a polished-brass finish. The connection must have individual self-closing clappers, caps with drip drains and chains. Female inlets must have 65-mm diameter St. John Threads. Comply with UL 405 and CAN/ULC-S560.

## 2.7 SPRINKLERS

Sprinklers must comply with UL 199, CSA C22.2 No. 199 and NFPA 13, CAN/ULC-S524, CAN/ULC-S530. Sprinklers with internal O-rings are not acceptable. Sprinklers in high heat areas including attic spaces or in close proximity to unit heaters must have temperature classification in accordance with NFPA 13, CAN/ULC-S524, CAN/ULC-S530.

### 2.7.1 Pendent Sprinkler

Pendent sprinkler must be recessed type with nominal K-factor of 115. Pendent sprinklers must have a polished chrome finish. Assembly must include an integral escutcheon.

### 2.7.2 Upright Sprinkler

Upright sprinkler must be brass and have a nominal K-factor of 80 or 115.

### 2.7.3 Concealed Sprinkler

Concealed sprinkler must be chrome-plated and have a nominal K-factor of 115. Coverplate must be white.

## 2.8 ACCESSORIES

### 2.8.1 Sprinkler Cabinet

Provide spare sprinklers in accordance with NFPA 13, CAN/ULC-S524 CAN/ULC-S530 and must be placed in a suitable metal or plastic cabinet of sufficient size to accommodate all the spare sprinklers and wrenches in designated locations. Spare sprinklers must be representative of, and in proportion to, the number of each type and temperature rating of the sprinklers installed as required by NFPA 13, CAN/ULC-S524 CAN/ULC-S530. At least one wrench of each type required must be provided.

### 2.8.2 Pendent Sprinkler Escutcheon

Escutcheon must be one-piece metallic type with a depth of less than 19 mm and suitable for installation on pendent sprinklers. The escutcheon must have a factory finish that matches the pendent sprinkler.

### 2.8.3 Pipe Escutcheon

Provide split hinge metal plates for piping entering walls, floors, and ceilings in exposed spaces. Provide polished stainless steel plates or chromium-plated finish on copper alloy plates in finished spaces. Provide paint finish on metal plates in unfinished spaces.

### 2.8.4 Sprinkler Guard

Listed guard must be a steel wire cage designed to encase the sprinkler and protect it from mechanical damage. Guards must be provided on sprinklers located within 2.1 meters of the floor and as indicated.

### 2.8.5 Relief Valve

Relief valves must be listed and installed at the riser in accordance with NFPA 13, CAN/ULC-S524, CAN/ULC-S530.

### 2.8.6 Air Vent

Air vents must be of the automatic type and piped to drain to the building exterior.

### 2.8.7 Identification Sign

Valve identification sign must be minimum 150 mm wide by 50 mm high with enamel baked finish on minimum 1.214-mm steel or 0.6-mm aluminum with red letters on a white background or white letters on red background. Wording of sign must include, but not be limited to "main drain", "auxiliary drain", "inspector's test", "alarm test", "alarm line", and similar wording as required to identify operational components. Where there is more than one sprinkler system, signage must include specific details as

to the respective system.

### PART 3 EXECUTION

#### 3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work that is dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative a condition that prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

#### 3.2 INSTALLATION

The installation must be in accordance with the applicable provisions of NFPA 13, CAN/ULC-S524, CAN/ULC-S530, NFPA 24, CAN/ULC-S524 and CAN/ULC-S536 and publications referenced therein. Locate sprinklers in a consistent pattern with ceiling grid, lights, and air supply diffusers. Install sprinkler system over and under ducts, piping and platforms when such equipment can negatively affect or disrupt the sprinkler discharge pattern and coverage.

- a. Piping offsets, fittings, and other accessories required must be furnished to provide a complete installation and to eliminate interference with other construction.
- b. Wherever the contractor's work interconnects with work of other trades the Contractor must coordinate with other Contractors to insure all Contractors have the information necessary so that they may properly install all necessary connections and equipment. Identify all work items needing access (dampers and similar equipment) that are concealed above hung ceilings by permanent color coded pins/tabs in the ceiling directly below the item.
- c. Provide required supports and hangers for piping, conduit, and equipment so that loading will not exceed allowable loadings of structure. Submittal of a bid must be a deemed representation that the contractor submitting such bid has ascertained allowable loadings and has included in his estimates the costs associated in furnishing required supports.

##### 3.2.1 Waste Removal

At the conclusion of each day's work, clean up and stockpile on site all waste, debris, and trash which may have accumulated during the day as a result of work by the contractor and of his presence on the job. Sidewalks and streets adjoining the property must be kept broom clean and free of waste, debris, trash and obstructions caused by work of the contractor, which will affect the condition and safety of streets, walks, utilities, and property.

#### 3.3 UNDERGROUND PIPING INSTALLATION

The fire protection water main must be laid, and joints anchored, in accordance with NFPA 24, CAN/ULC-S524 and CAN/ULC-S536. Minimum depth of cover must be 900 mm or the frost line, whichever is deeper. The supply line must terminate inside the building with a flanged piece, the bottom



of which must be set not less than 304 mm above the finished floor. A blind flange must be installed temporarily on top of the flanged piece to prevent the entrance of foreign matter into the supply line. A concrete thrust block must be provided at the elbow where the pipe turns up toward the floor. In addition, joints must be anchored in accordance with NFPA 24. Buried steel components must be provided with a corrosion protective coating in accordance with AWWA C203 and CSA Z245.30. Piping more than 1500 mm outside the building walls must meet the requirements of Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING.

### 3.4 ABOVEGROUND PIPING INSTALLATION

The methods of fabrication and installation of the aboveground piping must fully comply with the requirements and recommended practices of NFPA 13, CAN/ULC-S524, CAN/ULC-S530 and this specification section.

#### 3.4.1 Protection of Piping Against Earthquake Damage

Seismic restraint is not required.

#### 3.4.2 Piping in Exposed Areas

Install exposed piping without diminishing exit access widths, corridors or equipment access. Exposed horizontal piping, including drain piping, must be installed to provide maximum headroom.

#### 3.4.3 Piping in Finished Areas

In areas with suspended or dropped ceilings and in areas with concealed spaces above the ceiling, piping must be concealed above ceilings. Piping must be inspected, hydrostatically tested and approved before being concealed. Risers and similar vertical runs of piping in finished areas must be concealed.

#### 3.4.4 Pendent Sprinklers

- a. Drop nipples to pendent sprinklers must consist of minimum 25-mm pipe with a reducing coupling into which the sprinkler must be threaded.
- b. Where sprinklers are installed below suspended or dropped ceilings, drop nipples must be cut such that sprinkler ceiling plates or escutcheons are of a uniform depth throughout the finished space. The outlet of the reducing coupling must not extend below the underside of the ceiling.
- c. Recessed pendent sprinklers must be installed such that the distance from the sprinkler deflector to the underside of the ceiling must not exceed the manufacturer's listed range and must be of uniform depth throughout the finished area.
- d. Pendent sprinklers in suspended ceilings must be located in the center of the tile (plus or minus 2 inches).
- g. Where the maximum static or flowing pressure, whichever is greater at the sprinkler, applied other than through the fire department connection, exceeds 6.9 bar and a branch line above the ceiling supplies sprinklers in a pendent position below the ceiling, the cumulative horizontal length of an unsupported armover to a sprinkler or sprinkler drop must not exceed 300 mm for steel pipe and 1500 mm

for copper tube.

#### 3.4.5 Upright Sprinklers

Riser nipples or "sprigs" to upright sprinklers must contain no fittings between the branch line tee and the reducing coupling at the sprinkler.

#### 3.4.6 Pipe Joints

Pipe joints must conform to NFPA 13, CAN/ULC-S524, CAN/ULC-S530, except as modified herein. Not more than four threads must show after joint is made up. Welded joints will be permitted, only if welding operations are performed as required by NFPA 13, CAN/ULC-S524, CAN/ULC-S530 at the Contractor's fabrication shop, not at the project construction site. Flanged joints must be provided where indicated or required by NFPA 13, CAN/ULC-S524, CAN/ULC-S530. Grooved pipe and fittings must be prepared in accordance with the manufacturer's latest published specification according to pipe material, wall thickness and size. Grooved couplings, fittings and grooving tools must be products of the same manufacturer. For copper tubing, pipe and groove dimensions must comply with the tolerances specified by the coupling manufacturer. The diameter of grooves made in the field must be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe must be measured and recorded for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances.

#### 3.4.7 Reducers

Reductions in pipe sizes must be made with one-piece tapered reducing fittings. When standard fittings of the required size are not manufactured, single bushings of the face or hex type will be permitted. Where used, face bushings must be installed with the outer face flush with the face of the fitting opening being reduced. Bushings cannot be used in elbow fittings, in more than one outlet of a tee, in more than two outlets of a cross, or where the reduction in size is less than 13 mm.

#### 3.4.8 Pipe Penetrations

- a. Cutting structural members for passage of pipes or for pipe-hanger fastenings will not be permitted. Pipes that must penetrate concrete or masonry walls or concrete floors must be core-drilled and provided with pipe sleeves. Each sleeve must be Schedule 40 galvanized steel, ductile-iron or cast-iron pipe and extend through its respective wall or floor and be cut flush with each wall surface. Sleeves must provide required clearance between the pipe and the sleeve per NFPA 13. The space between the sleeve and the pipe must be firmly packed with mineral wool insulation.
- b. Where pipes and sleeves penetrate fire walls, fire partitions, or floors, pipes/sleeves must be firestopped in accordance with Section 07 84 00 FIRESTOPPING.
- c. In penetrations that are not fire-rated or not a floor penetration, the space between the sleeve and the pipe must be sealed at both ends with plastic waterproof cement that will dry to a firm but pliable mass or with a mechanically adjustable segmented elastomer seal.

- d. All penetrations through the boundary of rooms/areas identified as secure space area must meet ICS 705-1.

#### 3.4.9 Escutcheons

Escutcheons must be provided for pipe penetration in finished areas of ceilings, floors and walls. Escutcheons must be securely fastened to the pipe at surfaces through which piping passes.

#### 3.4.10 Inspector's Test Connection

Unless otherwise indicated, the test connection must consist of 25-mm pipe connected at the riser as a combination test and drain valve; a test valve located approximately 2 meters above the floor; a smooth bore brass outlet equivalent to the smallest orifice sprinkler used in the system; and a painted metal identification sign affixed to the valve with the words "Inspector's Test". All test connection piping must be inside of the building and penetrate the exterior wall at the location of the discharge orifice only. The discharge orifice must be located outside the building wall no more than 0.6 meters above finished grade, directed so as not to cause damage to adjacent construction or landscaping during full flow discharge, or to the sanitary sewer. Discharge to the exterior must not interfere with exiting from the facility. Water discharge or runoff must not cross the path of egress from the building. Do not discharge to the roof. Discharge to floor drains, janitor sinks or similar fixtures is not permitted.

Provide concrete splash blocks at all drain and inspector's test connection discharge locations if not discharging to a concrete surface. Splash blocks must be large enough to mitigate erosion and not become dislodged during a full flow of the drain. Ensure all discharged water drains away from the facility and does not cause property damage.

#### 3.4.11 Backflow Preventer

Locate within the building or in a heated enclosure in locations subject to freezing. For heated enclosures, provide a low temperature supervisory alarm connected to the facility fire alarm system. Heat trace is not permitted to be used.

Install backflow preventers so that the bottom of the assembly is a minimum of 150 mm above the finished floor/grade. Install horizontal backflow preventers so that the bottom of the assembly is no greater than 610 mm above the finished floor/grade. Install vertical backflow preventers so that the upper operating handwheel is no more than 1.8 meters above the finished floor/grade. Clearance around control valve handles must be minimum 150 mm above grade/finished floor and away from walls.

##### 3.4.11.1 Test Connection

Provide downstream of the backflow prevention assembly UL 668 hose valves with 65-mm National Standard male hose threads with cap and chain. Provide one valve for each 946 L/min of system demand or fraction thereof. Provide a permanent sign in accordance with paragraph entitled "Identification Signs" which reads, "Test Valve". Indicate location of test header. If an exterior connection, provide a control valve inside a heated mechanical room to prevent freezing.

### 3.4.12 Drains

- a. Main drain piping must be provided to discharge at a safe point outside the building, no more than 0.6 meters above finished grade. Provide a concrete splash block at drain outlet. Discharge to the exterior must not interfere with exiting from the facility. Water discharge or runoff must not cross the path of egress from the building.
- b. Auxiliary drains must be provided as required by NFPA 13, CAN/ULC-S524, CAN/ULC-S530. Auxiliary drains are permitted to discharge to a floor drain if the drain is sized to accommodate full flow (min 151 L/min). Discharge to service sinks or similar plumbing fixtures is not permitted.

### 3.4.13 Installation of Fire Department Connection

Connection must be mounted on the exterior wall approximately 900 mm above finished grade. The piping between the connection and the check valve must be provided with an automatic drip in accordance with NFPA 13, CAN/ULC-S524, CAN/ULC-S530 and piped to drain to the outside or a floor drain within the same room.

### 3.4.14 Identification Signs

Signs must be affixed to each control valve, inspector test valve, main drain, auxiliary drain, test valve, and similar valves as appropriate or as required by NFPA 13. Main drain test results must be etched into main drain identification sign. Hydraulic design data must be etched into the nameplates and permanently affixed to each sprinkler riser as specified in NFPA 13. Provide labeling on the surfaces of all feed and cross mains to show the pipe function (e.g., "Sprinkler System", "Fire Department Connection", "Standpipe") and normal valve position (e.g. "Normally Open", "Normally Closed"). For pipe sizes 100 mm and larger provide white painted stenciled letters and arrows, a minimum of 50 mm in height and visible from at least two sides when viewed from the floor. For pipe sizes less than 100 mm, provide white painted stenciled letters and arrows, a minimum of 18 mm in height and visible from the floor.

## 3.5 ELECTRICAL

Except as modified herein, electric equipment and wiring must be in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Alarm signal wiring connected to the building fire alarm control system must be by the fire alarm installer.

## 3.6 PAINTING

Color code mark piping as specified in Section 09 90 00 PAINTS AND COATINGS.

## 3.7 FIELD QUALITY CONTROL

### 3.7.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level III or IV Fire Sprinkler Technician, and the representative of the installing company, and reviewed by the QFPE 60 days prior to performing system tests. Detailed test procedures must list all components of the installed system. Test procedures must include sequence of testing, time estimate

for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forms in NFPA 13, CAN/ULC-S524, CAN/ULC-S530. The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government final testing.

- a. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

### 3.7.2 Pre-Government Testing

#### 3.7.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that equipment is functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" and "System Acceptance" as noted in NFPA 13, CAN/ULC-S524, CAN/ULC-S530. The Contractor and QFPE must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 13, CAN/ULC-S524, CAN/ULC-S530 and the test reports noted below.

- a. NFPA 13, CAN/ULC-S524,  
CAN/ULC-S530 Aboveground Material and Test Certificate
- b. NFPA 13, CAN/ULC-S524,  
CAN/ULC-S530 Underground Material and Test Certificate

#### 3.7.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the Designated Fire Protection Engineer (DFPE). Government final testing will not be scheduled until the DFPE has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the installation fire alarm reporting system have been completed and tested to confirm communications are fully functional. Submit request for test at least 30 calendar days prior to the requested test date.

### 3.7.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

### 3.7.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

- a. The manufacturer's technical representative.
- b. The contractor's Qualified Fire Protection Engineer (QFPE).
- c. Marked-up red line drawings of the system as actually installed.

Government Final Tests will be witnessed by the Designated Fire Protection Engineer. At this time, all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

### 3.8 MINIMUM SYSTEM TESTS

The system, including the underground water mains, and the aboveground piping and system components, must be tested to ensure that equipment and components function as intended. The underground and aboveground interior piping systems and attached appurtenances subjected to system working pressure must be tested in accordance with NFPA 13, CAN/ULC-S524, CAN/ULC-S530 and NFPA 24.

#### 3.8.1 Underground Piping

##### 3.8.1.1 Flushing

Underground piping must be flushed at a minimum of 3 mps in accordance with NFPA 24.

##### 3.8.1.2 Hydrostatic Test

New underground piping must be hydrostatically tested in accordance with NFPA 24.

#### 3.8.2 Aboveground Piping

##### 3.8.2.1 Hydrostatic Test

Aboveground piping must be hydrostatically tested in accordance with NFPA 13. There must be no drop in gauge pressure or visible leakage when the system is subjected to the hydrostatic test. The test pressure must be read from a gauge located at the low elevation point of the system or portion being tested.

##### 3.8.2.2 Backflow Prevention Assembly Forward Flow Test

Each backflow prevention assembly must be tested at system flow demand, including all applicable hose streams, as specified in NFPA 13, CAN/ULC-S524, CAN/ULC-S530. The Contractor must provide all equipment and instruments necessary to conduct a complete forward flow test, including 65-mm diameter hoses, playpipe nozzles or flow diffusers, calibrated pressure gauges, and pitot tube gauge. The Contractor must provide all necessary supports to safely secure hoses and nozzles during the test. At the system demand flow, the pressure readings and pressure drop (friction loss) across the assembly must be recorded. A metal placard must be provided on the backflow prevention assembly that lists the pressure readings both upstream and downstream of the assembly, total pressure drop, and the system test flow rate determined during the preliminary testing. The pressure drop must be compared to the manufacturer's data and the readings observed during the final inspections and tests.

### 3.8.3 Main Drain Flow Test

Following flushing of the underground piping, a main drain test must be made to verify the adequacy of the water supply. Static and residual pressures must be recorded on the certificate specified in paragraph SUBMITTALS.

### 3.9 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. Submit six sets of detailed as-built drawings. The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final acceptance test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the final acceptance test.

- a. Provide one set of full size paper as-built drawings and schematics. The drawings must be prepared electronically and sized no less than the contract drawings.
- b. Provide operating and maintenance (O&M) instructions.

### 3.10 ONSITE TRAINING

Conduct a training course for the responding fire department and operating and maintenance personnel as designated by the Contracting Officer. Training must be performed on two separate days (to accommodate different shifts of Fire Department personnel) for a period of 4 hours of normal working time and must start after the system is functionally complete and after the final acceptance test. The on-site training must cover all of the items contained in the approved Operating and Maintenance Instructions.

-- End of Section --

## SECTION 21 22 00

CLEAN AGENT FIRE EXTINGUISHING SYSTEMS  
04/20

## PART 1 GENERAL

## 1.1 SUMMARY

## 1.1.1 Scope

- a. Design and provide a new total flooding type clean agent fire extinguishing system conforming to NFPA 2001 and UL 2127. Pipe sizes which are not indicated on the Contract drawings shall be determined by agent discharge calculations. The system shall be complete in all ways. It shall include all mechanical and electrical installation, all detection and control equipment, agent storage containers, agent, discharge nozzles, pipe and fittings, manual release and abort stations, audible and visual alarm devices, auxiliary devices and controls, shutdowns, alarm interface, caution/advisory signs, functional checkout and testing, training and all other operations necessary for a functional system. Design discharge nozzles to uniformly distribute the clean agent throughout the hazard area. Arrange system for fully automatic and manually operated electronic control operation, with operating controls of the enclosed release type to prevent accidental operation. Discharge nozzles shall be provided, within the manufacturer's guidelines, to distribute the clean agent unobstructed throughout the protected space. The installed nozzles shall be designed to provide the proper clean agent quantity and distribution. Provide manual release stations, dead-man style abort stations and keyed override as indicated on the drawings. Provide material and equipment that are listed. Each protected area shall have audible and visible notification appliances located where shown. Electrically supervise all circuits. Provide separate and distinct audible and visual pre-discharge and discharge signals. The discharge signals shall be distinct from those used by the building fire alarm system. Design the clean agent fire extinguishing system to a concentration (by volume) as specified by NFPA 2001 and the manufacturer. Final quantity, system layout and coordination are the responsibility of the Contractor.
- b. The control system shall meet the requirements of NFPA 72. The control unit shall be listed for "Releasing Device Service".

## 1.1.2 Qualified Fire Protection Engineer (QFPE)

An individual who is a licensed professional engineer (P.E.) who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES) and has relevant fire protection engineering experience. Services of the QFPE shall include:

- a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification.
- b. Providing a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting any outstanding



comments.

- c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).
- d. Witnessing pre-Government and final Government functional performance testing and performing a final installation review.
- e. Signing applicable certificates under SD-07.

## 1.2 REFERENCES

The publications listed below form a part of this specification section to the extent referenced. The publications are referred to within the text by the basic designation only.

### CSA GROUP (CSA)

CSA B137.1 (2020) Polyethylene (PE) pipe, tubing, and fittings for cold-water pressure services

### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC (2010) Boiler and Pressure Vessels Code

### ASTM INTERNATIONAL (ASTM)

ASTM A53M (2018) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM F402 (2018) Standard Practice for Safe Handling of Solvent Cements, Primers, and Cleaners Used for Joining Thermoplastic Pipe and Fittings

### FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide  
<https://www.approvalguide.com/>

### INTELLIGENCE COMMUNITY STANDARD (ICS)

ICS 705-1 (2010) Physical and Technical Security Standard for Sensitive Compartmented Information Facilities

### INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41.1 (2002; R 2008) Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits

IEEE C62.41.2 (2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|           |                                                                               |
|-----------|-------------------------------------------------------------------------------|
| NFPA 170  | (2024; ERTA 1 2023) Standard for Fire Safety and Emergency Symbols            |
| NFPA 2001 | (2018) Standard on Clean Agent Fire Extinguishing Systems                     |
| NFPA 4    | (2018) Standard for Integrated Fire Protection and Life Safety System Testing |
| NFPA 70   | (2023; ERTA 1 2024; TIA 24-1) National Electrical Code                        |
| NFPA 72   | (2022; ERTA 22-1) National Fire Alarm and Signaling Code                      |

## NATIONAL INSTITUTE FOR CERTIFICATION IN ENGINEERING TECHNOLOGIES (NICET)

|              |                                                                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NICET 1014-7 | (2012) Program Detail Manual for Certification in the Field of Fire Protection Engineering Technology (Field Code 003) Subfield of Automatic Sprinkler System Layout |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                                     |
|--------------|---------------------------------------------------------------------|
| UFC 3-601-02 | (2021) Fire Protection Systems Inspection, Testing, and Maintenance |
|--------------|---------------------------------------------------------------------|

## UNDERWRITERS LABORATORIES (UL)

|         |                                                                                                                     |
|---------|---------------------------------------------------------------------------------------------------------------------|
| UL 1283 | (2017; Reprint Feb 2024) UL Standard for Safety Electromagnetic Interference Filters                                |
| UL 1449 | (2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices                                            |
| UL 1638 | (2023) UL Standard for Safety Visible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories |
| UL 1971 | (2002; Reprint Feb 2024) UL Standard for Safety Signaling Devices for the Deaf and Hard of Hearing                  |
| UL 2127 | (2017) UL Standard for Safety Inert Gas Clean Agent Extinguishing System Units                                      |
| UL 464  | (2023) UL Standard for Safety Audible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories |
| UL 497A | (2001; Bul. 2019) UL Standard for Safety Secondary Protectors for Communications Circuits                           |
| UL 497B | (2004; Reprint Feb 2022) UL Standard for                                                                            |

Safety Protectors for Data Communications  
and Fire Alarm Circuits

|                  |                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------|
| UL 536           | (2014) Flexible Metallic Hose                                                                                                  |
| UL 864           | (2023) UL Standard for Safety Control<br>Units and Accessories for Fire Alarm<br>Systems                                       |
| UL Fire Prot Dir | UL Product IQ (updated online) at<br><a href="https://productiq.ulpropsector.com/en">https://productiq.ulpropsector.com/en</a> |

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval.. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Partial submittals and submittals not fully complying with NFPA 2001 and this specification section shall be returned disapproved without review. SD-02, SD-03 and SD-05 shall be submitted simultaneously.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) shall be prepared by the designer and combined and submitted as one complete package. The QFPE shall review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE shall provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their registered professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE shall be returned disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-01 Preconstruction Submittals

Within 36 days of contract award but not less than 14 days prior to commencing any work on site, the Contractor shall submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications of the subcontractor and QFPE shall be returned disapproved without review. All resultant delays shall be the sole responsibility of the Contractor.

Qualified Fire Protection Engineer (QFPE); G

Clean Agent System Designer; G

Releasing System Designer; G

Releasing System Installer; G

Releasing System Supervisor; G

Releasing System Technician; G

Clean Agent System Installer; G

Test Technician; G

Releasing System Site-Specific Software Acknowledgement; G

#### SD-02 Shop Drawings

Shop drawings shall not be smaller than the Contract Drawings. Drawings shall comply with the requirements of NFPA 72 and NFPA 170. Minimum scale for floor plans shall be 1:100

Nameplates; G

Nameplate illustrations and data to obtain approval by the Contracting Officer before installation.

Instructions; G

Wiring Diagrams; G

Three (3) copies of point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams shall show connections from field devices to the releasing control unit and remote control units, initiating circuits, switches, relays and terminals, including pathway diagrams between the control unit and shared communications equipment within the protected premises. Point-to-point wiring diagrams shall be job specific and shall not indicate connections or circuits not being utilized. Provide complete riser diagrams indicating the wiring sequence of all devices and their connections to the control equipment. Include a color-code schedule for the wiring.

System Layout; G

Three (3) copies of plan view drawing showing device locations, terminal cabinet locations, junction boxes, other related equipment, conduit routing, conduit sizes, wire counts, conduit fill calculations, wire color-coding, circuit identification in each conduit, and circuit layouts for all floors. Annotate clean agent fire extinguishing system piping layout with reference points for design. Show isometric details of agent storage containers, mounting details and proposed pipe runs and sizes. Drawings shall include plan and elevation views which establish that the equipment will fit the allotted spaces with clearance for installation and maintenance. Indicate candela rating of each visual notification appliance. Clearly identify the locations of isolation modules. Indicate the addresses of all devices, modules, relays, etc. Show/identify all acoustically similar spaces.

Provide a complete description of the system operation in matrix format similar to the "Typical Input/Output Matrix" included in the Annex of NFPA 72.

For air sampling smoke detection systems, provide floor plan layouts indicating location of air aspirating system, air sampling

pipings (lengths of pipe) and sampling ports (sizes and locations). Floor plan must also indicate geographic monitor zone boundaries, location of display control unit, bar level annunciation panels if separate, and all other associated equipment that is required to provide a complete operational system.

#### System Operation; G

A complete list of device addresses and corresponding messages.

#### Notification Appliances; G

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for notification appliances. Annotate data for each circuit on the drawings.

#### Initiating Devices; G

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for initiating devices. Annotate data for each circuit on the drawings.

#### Battery Power; G

Calculations and supporting data as required in paragraph Battery Power Calculations for alarm, alert, and supervisory power requirements. Calculations including ampere-hour requirements for each system component and each panel component, and the battery recharging period, shall be included on the drawings.

#### Voltage Drop Calculations; G

Voltage drop calculations for each notification circuit indicating that sufficient voltage is available for proper operation of the system and all components, at a minimum rated voltage of the system operating on batteries. Include the calculations on the system layout drawings.

#### SD-03 Product Data

\_\_\_3\_\_\_ copies of annotated catalog data to show the specific model, type, and size of each item. Catalog cuts shall also indicate the NRTL listing. The data shall be highlighted to show model, size, options, etc., that are intended for consideration. Data shall be adequate to demonstrate compliance with all contract requirements. Product data for all equipment shall be combined into a single submittal.

Provide an equipment list identifying the type, quantity, make, and model number of each piece of equipment to be provided under this submittal. The equipment list shall include the type, quantity, make and model of spare equipment. Types and quantities of equipment submitted shall coincide with the types and quantities of equipment used in the battery calculations and those shown on the shop drawings.

#### Clean Agent; G

#### High-Pressure Containers; G

Piping and Fittings; G  
Pipe Hangers and Supports; G  
Low-Pressure Switch; G  
Discharge Nozzles; G  
Escutcheons; G  
Releasing Control Unit; G

Smoke Detectors; G  
Air Sampling Smoke Detectors; G  
Manual Release Stations; G  
Abort Stations; G  
Notification Appliances; G  
Batteries; G  
Battery Charger; G  
Surge Protective Devices; G

Wire; G  
Back Boxes and Conduit; G  
Ceiling Bridges; G  
Document Storage Cabinet; G  
Warning Signs; G  
Supplemental Notification Appliance Circuit Panels; G  
Auxiliary Power Supply Panels; G  
Addressable Interface Devices; G  
Addressable Control Modules; G  
Isolation Modules; G  
Terminal Cabinets; G

#### SD-05 Design Data

Discharge Calculations; G

Submit clean agent discharge calculations verifying total storage requirements, flooding concentrations, discharge times, flow through the piping network, pipe sizes, and nozzle orifice sizes, in accordance with the manufacturer's listed design manual and NFPA 2001. Inert gas calculations shall include the total required venting area, in square millimeters for each protected enclosure.

Air Sampling Smoke Detection System Calculations; G

Submit air sampling detection system design analysis calculations consisting of battery capacity, loading calculations, and fan speed and air flow/transport calculations. Include schematic diagrams showing pipe segments, pipe diameters, lengths of pipe, node numbers, and sample port diameters to verify the requirements are met.

#### SD-06 Test Reports

Test Procedures; G

#### SD-07 Certificates

Verification of Compliant Installation; G

Request for Government Final Test; G

## SD-09 Manufacturer's Field Reports

System Operation; G

## SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G

Six copies of the Operation and Maintenance Instructions. The O&M Instructions shall be prepared in a single volume or in multiple volumes, with each volume indexed, and may be submitted as a Technical Data Package. Manuals shall be approved prior to training. The Releasing System Operation and Maintenance Instructions shall include the following:

- a. "Manufacturer Data Package 5" as specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA.
- b. Operating manual outlining step-by-step procedures required for system startup, operation, and shutdown. The manual shall include the manufacturer's name, model number, service manual, parts list, and preliminary equipment list complete with description of equipment and their basic operating features.
- c. Maintenance manual listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide. The manuals shall include conduit layout, equipment layout and simplified wiring, and control diagrams of the system as installed.
- d. Complete procedures for system revision and expansion, detailing both equipment and software requirements.
- e. Software delivered for this project on each type of CD/DVD media utilized.
- f. Printouts of configuration settings for all devices.
- g. Routine maintenance checklist. The routine maintenance checklist shall be arranged in a columnar format. The first column shall list all installed devices, the second column shall state the maintenance activity or state no maintenance required, the third column shall state the frequency of the maintenance activity, and the fourth column provided for additional comments or reference. All data (devices, testing frequencies, etc.) shall comply with UFC 3-601-02.
- h. A final Equipment List shall be submitted with the Operating and Maintenance (O&M) manual.

Instruction of Government Employees; G

## SD-11 Closeout Submittals

Spare parts

As-built drawings

The drawings shall show the system as installed, including deviations from both the project drawings and the approved shop

drawings. These drawings shall be submitted within two weeks after the final Government test of the system. At least one set of the as-built (marked-up) drawings shall be provided at the time of, or prior to the final Government test.

#### 1.4 SYSTEM OPERATION

##### 1.4.1 System Sequence

Automatic operation of each protected area shall be as follows:

- a. Actuation of Alarm Level 1 (Alert) within the system shall:
  - (1) Activate the "ALERT" indicator lamp on the air aspirating smoke detection control unit.
  - (2) Transmit a supervisory signal to the releasing control unit (RCU).
- b. Actuation of Alarm Level 2 (Action) within the system shall:
  - (1) Activate the "ALARM" indicator lamp on the releasing control unit (RCU).
  - (2) Energize the alarm audible and visible notification device located in the hazard area for evacuation of the protected area.
  - (3) Operate door holder/closures on access doors.
  - (4) Transmit an alarm signal to the main fire alarm control unit (FACU).
  - (5) Activate the building fire alarm system.
- c. Actuation of Alarm Level 3 (Fire 1) within the system shall:
  - (1) Activate the "PRE-DISCHARGE" indicator on the RCU.
  - (2) Energize the pre-discharge audible and visible notification device located in the hazard area.
  - (3) Shut down the HVAC system and/or close dampers.
  - (4) Start time-delay sequence (not to exceed 60 seconds).
  - (5) System abort sequence is enabled at this time.
  - (6) Shunt power to protected equipment
- d. After completion of the time-delay sequence, the clean agent fire extinguishing system shall activate and the following shall occur:
  - (1) Activate the "RELEASE" indicator on the RCU.
  - (2) Shutdown power to all equipment in hazard area.
  - (3) Energize the visible notification device(s) outside the hazard area in which the discharge occurred.
  - (5) Transmit a system release signal to the FACU.
- e. The system shall be capable of being actuated by manual release stations located at each hazard exit. Operation of a manual release station shall duplicate the sequence description above except that the time delay and abort functions SHALL be bypassed.

##### 1.4.2 Circuit Requirements

Connect alarm initiating devices to initiating device circuits (IDC),



Class B or to signal line circuits (SLC), Class B, in accordance with NFPA 72. Alarm notification or indicating appliances shall be connected to notification appliance circuit (NAC), Class B in accordance with NFPA 72. Provide a separate circuit for actuation of each clean agent system. Fully supervise the circuits that actuate the water control valves so that the occurrence of a single open or a single ground fault condition in the interconnecting conductors will be indicated at the control unit.

#### 1.5 TECHNICAL DATA AND SITE-SPECIFIC SOFTWARE

Technical data and site-specific software (meaning technical data that relates to computer software) that are specifically identified in this project, and may be required in other specifications, shall be delivered, strictly in accordance with the CONTRACT CLAUSES. The releasing system manufacturer must submit written confirmation of this contract provision as "Releasing System Site-Specific Software Acknowledgement". Identify data delivered by reference to the specification paragraph against which it is furnished. Data to be submitted shall include complete system, equipment, and software descriptions. Descriptions shall show how the equipment will operate as a system to meet the performance requirements of this contract. The site-specific software data package shall also include the following:

- a. Items identified in NFPA 72, titled "Site-Specific Software".
- b. Identification of programmable portions of the system equipment and capabilities.
- c. Description of system revision and expansion capabilities and methods of implementation detailing both equipment and software requirements.
- d. Provision of operational software data on all modes of programmable portions for the releasing control system.
- e. Description of releasing control system equipment operation.
- f. Description of auxiliary and remote equipment operations.
- g. Library of application software.
- h. Operation and maintenance manuals.

#### 1.6 QUALITY ASSURANCE

##### 1.6.1 Qualifications

###### 1.6.1.1 Releasing System Designer

The system designer shall be certified as a Level III (minimum) Technician by National Institute for Certification in Engineering Technologies (NICET) in the Fire Alarm Systems subfield of Fire Protection Engineering Technology in accordance with NICET 1014-7, or meet the qualifications for a QFPE.

###### 1.6.1.2 Releasing System Supervisor

A NICET Level III or IV fire alarm technician shall supervise the installation of the releasing system, including the air sampling smoke detection system. A fire alarm technician with a minimum of eight years

of experience shall supervise the installation of the releasing system. The fire alarm technicians supervising the installation of equipment shall be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

#### 1.6.1.3 Releasing System Technician

Releasing system technicians with a minimum of four years of experience shall be utilized to install and terminate devices, cabinets and panels. The fire alarm technicians installing the equipment shall be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings, and must be thoroughly experienced in the design and installation of air sampling detection systems.

#### 1.6.1.4 Releasing System Installer

Releasing system installer with a minimum of two years of experience utilized to assist in the installation of devices, cabinets and panels. NICET Level II technician to assist in the installation of devices, cabinets and panels. A licensed electrician shall be allowed to install wire, cable, conduit and backboxes for the system. The fire alarm installer shall be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

#### 1.6.1.5 Clean Agent System Designer

The designer shall be certified as a Level IV (minimum) Technician by National Institute for Certification in Engineering Technologies (NICET) in the Special Hazards subfield of Fire Protection Engineering Technology in accordance with NICET 1014-7, or meet the qualifications for a QFPE.

#### 1.6.1.6 Clean Agent System Installer

The installing contractor shall be trained by the supplier to install, test and maintain a clean agent system. The installing contractor shall be an experienced firm regularly engaged in the installation of automatic clean agent, or similar, fire extinguishing systems in strict accordance with all applicable standards. The installing contractor must have a minimum of 5 years experience in the design, installation and testing of clean agent, or similar, fire extinguishing systems.

#### 1.6.1.7 Clean Agent Recharging Service

The installing contractor shall maintain, or have access to, a clean agent recharging service. The installing contractor shall provide proof of ability to recharge the system with clean agent within 24 hours of notification during the warranty period of the system.

#### 1.6.1.8 Test Technician

Releasing system technicians with a minimum of eight years of experience and NICET Level IV utilized in testing and certification of the installation of the devices, cabinets and panels. The fire alarm technicians testing the equipment shall be factory trained in the installation, adjustment, testing, and operation of the equipment installed as part of this project.

#### 1.6.1.9 Manufacturer

Components shall be of current design and shall be in regular and recurrent production at the time of installation. Provide design, materials, and devices for a protected premises fire alarm system, complete, conforming to NFPA 72 and NFPA 2001, except as specified herein.

#### 1.6.2 Regulatory Requirements

Equipment and material shall be listed or approved. Listed or approved, as used in this Section, shall mean listed, labeled, or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of any item of equipment described shall not be construed as waiving this requirement. All listings or approval by testing laboratories shall be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation shall be considered as mandatory requirements.

#### 1.7 DELIVERY, STORAGE, AND HANDLING

Protect equipment delivered and placed in storage from the weather, humidity, and temperature variation, dirt and dust, and other contaminants.

#### 1.8 EXTRA MATERIALS

##### 1.8.1 Spare Parts

Furnish the following spare parts in the manufacturers original unopened containers:

- a. Five complete sets of system keys.
- b. Two of each type of fuse required by the system.
- c. One manual stations.
- d. Two of each type of detector installed.
- e. Two of each type of detector base and head installed.
- g. One of each type of abort station.
- h. Two air sampling smoke detection system filter assemblies.
- i. Two of each type of audible and visual alarm device installed.
- j. One textual visual notification appliance.
- k. Two low voltage, one ethernet, and one 120 VAC surge protective devices.
- n. 1 spare zone modules for modular type control units in addition to those installed in the panel.

### 1.8.2 Special Tools

Software, connecting cables and proprietary equipment, necessary for the maintenance, testing, and reprogramming of the equipment shall be furnished to the Contracting Officer, prior to the instruction of Government employees.

## 1.9 EXCLUSIONS

The work listed below shall be provided by others:

- a. 120 VAC or 240 VAC power supply to the RCU.
- b. Interlock wiring and conduit for shutdown of the HVAC, dampers, and/or electric power supplies.
- c. Connection to local/remote fire alarm system(s) and listed central station(s).
- d. Mechanical ventilation, including overpressure relief.
- e. Structural review of floor assembly to verify adequate support for cylinders.

## PART 2 PRODUCTS

### 2.1 MATERIALS AND EQUIPMENT

#### 2.1.1 Standard Products

Provide materials, equipment, and devices that have been tested by a nationally recognized testing laboratory and listed for fire protection service when so required by NFPA 2001 or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for any particular classification of materials. Material and equipment shall be the standard products of a manufacturer regularly engaged in the manufacture of the products for at least 2 years prior to bid opening.

#### 2.1.2 Nameplates

Major components of equipment shall have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the Contract number provided on a new name plate permanently affixed to the item or equipment. Major components include, but are not limited to, the following:

- a. Releasing control unit.
- b. Clean agent tank.

Nameplates shall be etched metal or plastic, permanently attached by screws to panels or adjacent walls.

#### 2.1.3 Identification and Marking

Pipe and fitting markings shall include name or identifying symbol of manufacturer and nominal size. Pipe shall be marked with ASTM

designation. Valves and equipment markings shall have name or identifying symbol of manufacturer, specific model number, nominal size, name of device, arrow indicating direction of flow, and position of installation (horizontal or vertical), except if valve can be installed in either position. Markings shall be included on the body casting or on an etched or stamped metal nameplate permanently on the valve or cover plate.

#### 2.1.4 Keys

Keys and locks for equipment, panels and devices shall be identical.

#### 2.1.5 Instructions

Provide a typeset printed or typewritten instruction card mounted behind a Lexan plastic or glass cover in a stainless steel or aluminum frame. Install the instructions on the interior of the RCU. The card shall show those steps to be taken by an operator when a signal is received as well as the functional operation of the system under all conditions, normal, alarm, supervisory, and trouble. The instructions and their mounting location shall be approved by the Contracting Officer before being posted.

### 2.2 PIPING COMPONENTS

#### 2.2.1 Piping and Fittings

Pipe and fittings shall be steel conforming to ASTM A53M, nonferrous drawn seamless copper tubing conforming to CSA B137.1 or flexible metallic hose conforming to UL 536.

Fittings shall be used for changes in direction of piping and for all connections. Reduce pipe sizes in the fitting. Do not use flush bushings. Fuse brazed joints, when used, with an alloy with a melting point above 537 degrees C.

For 13-mm and 19-mm iron pipe size (ips), provide Schedule 40. For 19.25-mm or greater, use only Schedule 90 pipe. Standard malleable iron banded fittings or ductile-iron fittings are to be used up through 19-mm ips. Use extra heavy malleable-iron or ductile-iron fittings through 50-mm ips. Use forged steel fittings in all sizes over 50 mm.

#### 2.2.2 Valves

For flanged valves, the class and style of flanges required to match the valve's flanged connection shall be used. All gaskets, o-rings, sealants and other valve components shall be constructed of materials that are compatible with the agent. Valves shall be protected against mechanical, chemical, or other damage.

#### 2.2.3 Pipe Hangers and Supports

Pipe hangers and supports shall conform to NFPA 2001 and manufacturer's recommendations.

#### 2.2.4 Escutcheons

Escutcheons shall be listed for piping passing through floors, walls and ceilings, consisting of one-piece or split-type. Provide chrome plated escutcheons where pipe passes through walls and ceilings in finished areas. Other escutcheons may be steel or cast iron, with aluminum paint

finish. Securely fasten escutcheons in place with setscrews or other positive means.

#### 2.2.5 Discharge Nozzle

Permanently mark discharge nozzles to identify the nozzle and to show the equivalent single orifice diameter regardless of shape and number of orifices. Nozzles shall be available in NPT sizes 8 mm to 50 mm and shall be available in 180-degree and 360-degree distribution patterns.

#### 2.2.6 High-Pressure Containers

Each container shall meet the following:

- a. A pressure gauge to provide visual and electrical supervision of the container pressure. The pressure gauge shall be color-coded to provide an easy, visual indication of container pressure.
- b. A pressure relief provision that automatically operates when the internal temperature exceeds 54 degrees C. Each container shall have a safety device to relieve excess pressure safely, in advance of the rated cylinder test pressure.
- c. The requirements of U.S. Department of Transportation specifications if used as shipping containers. If not used as a shipping container, each container shall be designed, fabricated, inspected, certified, and stamped in accordance with Section VIII of ASME BPVC.
- d. Provided with support racks that anchor to walls and floors.
- e. A permanent nameplate or other permanent marking that indicates the following:
  - (1) For halocarbon agent containers, the agent, tare and gross weights and super-pressurization level (where applicable) of the container.
  - (2) For inert gas agent containers, the agent, pressurization level of the container, and nominal agent volume.

#### 2.2.7 Low-Pressure Switch

The clean agent tanks shall have a low-pressure switch to warn of clean agent tank depressurization. The low-pressure switch shall be wired to the RCP to provide an audible and visual "Supervisory" signal in the event the container pressure drops 5 percent below normal operating pressure.

#### 2.3 RELEASING CONTROL UNIT (RCU)

The RCU shall be listed, analog-addressable and compatible with the system, devices and functions specified. Route all supervision of the RCU through the main FACU. RCU and the FACU shall be connected with a fiber optic network cable. The RCU shall perform all functions necessary to operate the system detection, actuation, and auxiliary functions independent of the FACU. The control system shall be microprocessor-based utilizing distributed processing concept. A single microprocessor failure shall not impact operation of additional modules on the system. The control system shall be capable of supporting an air aspirating smoke detection system. The control system's initiating circuits shall be capable of Class A or Class B operation. The solenoid must be listed for

use with the releasing unit and extinguishing equipment. Each circuit shall be capable of monitoring contact devices configured for manual release, system abort, trouble input or auxiliary (non-fire) input. The RCU shall contain release circuits for activation of an extinguishing system. Each release circuit shall be capable of Class B operation. The RCU shall contain at least two notification appliance circuits for annunciation. Each notification circuit shall be capable of Class A or Class B operation. Provide alarm/trouble reset switches to reset a cleared device in alarm or trouble. Alarm or trouble signals are not to be self-restoring without activating the switch. The system shall be capable of providing a "PRE-ALARM" feature that can give advanced warning of a possible alarm condition.

An operator at the RCU, having the proper access level shall have the capability to manually access the following information for each initiating device:

- a. Primary status.
- b. Device type.
- c. Present average value.
- d. Present sensitivity selected.
- e. Sensor range (normal, dirty, etc).

## 2.4 SMOKE DETECTORS

### 2.4.1 Air Sampling Smoke Detectors

The addressable air sampling smoke system shall consist of a detector assembly housing an integral aspiration fan, filter, laser-based detection chamber and control, output and supervision circuitry. The system shall consist of a piping or tubing distribution network that runs from the detector assembly(s) to the protected area(s) and is supported by air sampling smoke detection system calculations from a computer-based design modeling tool. The system shall include configurable alarm and trouble relay outputs for interface to other systems where required.

- a. System shall be complete in all ways. It shall include all engineering, and electrical installation, all detection and control equipment, auxiliary devices and controls, alarm interface, functional checkout and testing, training and all other operations necessary for a functional system.
- b. System base detectors and modules shall each accommodate up to microbore sampling tubes where each tube has a sampling point at the end. Additional modules may be used to provide up to 20 sampling holes per system.
- c. Program alarm thresholds to the following values unless the results of the system acceptance tests indicate a clear need to change them. In the event that such a need is indicated, notify the Contracting Officer and provide complete documentation concerning the need to deviate from these values. Include within the deviation documentation request, information that complies with the paragraph entitled

"Sensitivity Verification Test". Ensure initial threshold levels are approved prior to the final acceptance test.

- (1) Alarm Level 1: set ALERT at 0.200 percent obscuration/foot
  - (2) Alarm Level 2: set FIRE 1 at 1.500 percent obscuration/foot
  - (3) Alarm Level 3: set FIRE 2 at 3.000 percent obscuration/foot
- d. All air sampling smoke detection devices and associated components shall be new, standard products or the manufacturer's latest design and suitable to perform the functions intended.
- e. The laser detection chamber shall be of the mass light scattering type and capable of detecting a wide range of smoke particle types of varying size. A particle counting method shall be employed for the purposes of:
- (1) Preventing large particles from affecting the true smoke reading.
  - (2) Monitoring contamination of the filter (dust and dirt, etc.) to automatically notify when maintenance is required. The particle counting method shall not be used for the purpose of smoke density measurement.
- f. Detector(s) shall be self-monitoring for filter contamination and provide indication through system fault when replacement is necessary. Detectors which allow automatic reset of filter status upon removal and re-insertion are not permitted.
- g. Detector(s) shall contain relays for alarm and fault conditions. The relays shall be software programmable to the required functions.
- h. Detector(s) shall permit configuration by programmers that are either integral to the system, portable or PC based.
- i. Detector(s) shall allow programming of:
- (1) Smoke threshold alarm levels; ALERT, PRE-ALARM, FIRE 1 and FIRE 2.
  - (2) Time delays. Ensure the display control panel contains individual adjustable alarm time delay features for each of the alarm threshold levels. Provide an adjustment range between 0 and 60 seconds. Program the alarm threshold time delays to 30 seconds for alarm levels 1 and 2, and 15 seconds for alarm levels 3 and 4.
  - (3) Faults, including airflow, detector, power, filter and network, as well as an indication of the urgency of the fault.
  - (4) Configuration of relay outputs for remote indication of alarm and fault conditions.
  - (5) General purpose input functionality.

## 2.5 MANUAL RELEASE STATIONS

Provide metal or plastic, semi-flush mounted, double-action, addressable



manual stations, that are not subject to operation by jarring or vibration. Stations shall be equipped with screw terminals for each conductor. Stations that require the replacement of any portion of the device after activation are not permitted. Stations shall be finished in yellow with molded raised lettering operating instructions of contrasting color. The use of a key shall be required to reset the station. Stations shall have a separate screw terminal for each conductor. Manual release stations shall be a different color from building fire alarm pull stations.

## 2.6 ABORT STATIONS

Guarded, spring-loaded, "dead-man" type which operates only when pressure is manually applied to the switch. Upon release of manual pressure, switch de-activates allowing delayed functions to resume. After start of agent, abort switch has no effect. Activation of the abort switch during normal (non-alarm) conditions causes activation of system supervisory signal.

## 2.7 ADDRESSABLE INTERFACE DEVICES

The initiating device being monitored shall be configured as a Class "B" initiating device circuit. The module shall be listed as compatible with the control unit. The module shall provide address setting means compatible with the control unit's SLC supervision and store an internal identifying code. Monitor module shall contain an integral LED that flashes each time the monitor module is polled and is visible through the device cover plate. Pull stations with a monitor module in a common backbox are not required to have an LED. Modules shall be listed for the environmental conditions in which they will be installed.

## 2.8 ADDRESSABLE CONTROL MODULES

The control module shall be capable of operating as a relay (dry contact form C) for interfacing the control unit with other systems, and to control door holders. The module shall be listed as compatible with the control unit. The indicating device or the external load being controlled shall be configured as a Class "B" notification appliance circuits. The system shall be capable of supervising, audible, visual and dry contact circuits. The control module shall have both an input and output address. The supervision shall detect a short on the supervised circuit and shall prevent power from being applied to the circuit. The control model shall provide address setting means compatible with the control unit's SLC supervision and store an internal identifying code. The control module shall contain an integral LED that flashes each time the control module is polled and is visible through the device cover plate. Control Modules shall be listed for the environmental conditions in which they will be installed.

## 2.9 ISOLATION MODULES

- a. Provide isolation modules to subdivide each signaling line circuit in accordance with NFPA 72 between adjacent isolation modules.
- b. Isolation modules shall provide short circuit isolation for signaling line circuit wiring.
- c. Power and communications shall be supplied by the SLC and shall report faults to the RCU.

- d. After the wiring fault is repaired, the fault isolation modules shall test the lines and automatically restore the connection.

## 2.10 NOTIFICATION APPLIANCES

Notification appliances shall be suitable for connection to supervised notification appliance circuits. Appliance shall have a separate screw terminal for each conductor. The surface of the appliance shall be red in color. Each signal device shall have a rigid plastic or metal identification sign with lettering a minimum of 40 millimeters high. Label the pre-discharge alarm "FIRE" and the discharge alarm "Clean Agent Fire Extinguishing System DISCHARGE". Locate post-discharge visual alarms outside all entrances to the protected areas, and provide signs reading "Clean Agent DISCHARGED WHEN FLASHING - DO NOT ENTER".

### 2.10.1 Horns

Horns shall conform to the applicable requirements of UL 464. Horns shall be semi-flush mounted, with the matching mounting backbox surface mounted vibrating type suitable for use in an electrically supervised circuit. Horns shall produce a sound rating of at least 85 dBA at 3 meters. Horns used in exterior locations shall be specifically listed or approved for outdoor use and be provided with metal housing and protective grilles.

### 2.10.2 Visual Notification Appliances

Visual notification appliances shall conform to the applicable requirements of UL 1638, UL 1971 and conform to the Architectural Barriers Act (ABA). Visual Notification Appliances shall have clear high intensity optic lens, xenon flash tubes, light emitting diode (LED) or rotating beacon. The light pattern shall be disbursed so that it is visible above and below the strobe and from a 90 degree angle on both sides of the strobe. Strobe flash rate shall be 1 flash per second and a minimum of 15 candela (based on the UL 1971 test. Strobe shall be semi-flush mounted. Strobes shall be synchronized with the building fire alarm strobes per NFPA 72.

## 2.11 ELECTRIC POWER

### 2.11.1 Primary Power

Power shall be 120 VAC 60 Hz service for the RCU from the AC service to the building in accordance with NFPA 72.

### 2.12 SECONDARY POWER SUPPLY

Provide for system operation in the event of primary power source failure. Transfer from normal to auxiliary (secondary) power or restoration from auxiliary to normal power shall be automatic and shall not cause transmission of a false alarm.

#### 2.12.1 Batteries

Provide sealed, maintenance-free, valve regulated lead acid batteries as the source for emergency power to the RCU. Batteries shall contain suspended electrolyte. The battery system shall be maintained in a fully charged condition by means of a solid state battery charger. Provide an automatic transfer switch to transfer the load to the batteries in the event of the failure of primary power.

### 2.12.1.1 Capacity

Battery size shall apply to every panel associated with this system, including supplemental notification appliance circuit panels, auxiliary power supply panels, and fire alarm transmitters.

- a. Sufficient capacity to operate the releasing system under supervisory and trouble conditions, including audible trouble signal devices for 48 hours and audible and visual signal devices under alarm conditions for an additional 15 minutes (or minimum required time for clean agent discharge), whichever is greater.

### 2.12.1.2 Battery Power Calculations

- a. Verify that battery capacity exceeds supervisory and alarm power requirements for the criteria noted in the paragraph "Capacity" above.
  - (1) Substantiate the battery calculations for alarm and supervisory power requirements. Include ampere-hour requirements for each system component and each panel component, and compliance with UL 864.
  - (2) Provide complete battery calculations for both the alarm and supervisory power requirements. Submit ampere-hour requirements for each system component with the calculations.
  - (3) Provide voltage drop calculations to indicate that sufficient voltage is available for proper operation of the system and all components, at the minimum rated voltage of the system operating on batteries. Calculations shall be performed using the minimum rated voltage of each component.
- b. For battery calculations assume a starting voltage of 24 VDC for starting the calculations to size the batteries. Calculate the required Amp-Hours for the specified standby time, and then calculate the required Amp-Hours for the specified alarm time. Using 20.4 VDC as starting voltage, perform a voltage drop calculation for circuits containing device and/or appliances remote from the power sources.

### 2.12.2 Battery Chargers

Provide a solid state, fully automatic, variable charging rate battery charger. The charger shall be capable of providing 120 percent of the connected system load and shall maintain the batteries at full charge. In the event the batteries are fully discharged (20.4 Volts dc), the charger shall recharge the batteries back to 95 percent of full charge within 48 hours after a single discharge cycle as described in paragraph CAPACITY above. Provide pilot light to indicate when batteries are manually placed on a high rate of charge as part of the unit assembly if a high rate switch is provided.

## 2.13 SURGE PROTECTIVE DEVICES

Surge protective devices shall be provided to suppress all voltage transients which might damage RCU components. Systems having circuits located outdoors, communications equipment shall be protected against surges induced on any signaling line circuit. Cables and conductors, that serve as communications links, shall have surge protection circuits

installed at each end. The surge protective device shall wire in series to the power supply of the protected equipment with screw terminations. Line voltage surge arrestor shall be installed directly adjacent to the power panel where the releasing control unit breaker is located.

- a. Surge protective devices for nominal 120 VAC shall be UL 1449 listed with a maximum 500 volt suppression level and have a maximum response time of 5 nanoseconds. The surge protective device shall also meet IEEE C62.41.1 and IEEE C62.41.2 category B tests for surge capacity. The surge protective device shall feature multi-stage construction and be provided with a long-life indicator lamp (either light emitting diode or neon) which extinguishes upon failure of protected components. Any unit fusing shall be externally accessible.
- b. Surge protective devices for nominal 24 VAC, fire alarm telephone dialer, or ethernet connection shall be UL 497B listed, meet IEEE C62.41.1, and have a maximum response time of 1-nanosecond. The surge protective device shall feature multi-stage construction and be self-resetting. The surge protective device shall be a base and plug style. The base assembly shall have screw terminals for releasing system wiring. The base assembly shall accept "plug-in" surge protective module.
- c. All surge protective devices (SPD) shall be the standard product of a single manufacturer and be equal or better than the following:
  - (1) For 120 VAC nominal line voltage: UL 1449 and UL 1283 listed, series connected 120 VAC, 20A rated, surge protective device in a NEMA 4x enclosure. Minimum 50,000 amp surge current rating with EMI/RFI filtering and a dry contact circuit for remote monitoring of surge protection status.
  - (2) For 24-volt nominal line voltage: UL 497B listed, series connected low voltage, 24-volt, 5A rated, loop circuit protector, base and replaceable module.
  - (3) For alarm telephone dialers: UL 497A listed, series connected, 130-volt, 150 mA rated with self-resetting fuse, dialer circuit protector with modular plug and play.
  - (4) For IP-DACTS: UL 497B listed, series connected, 6.4-volt, 1.5A rated with 20 kA/pair surge current, data network protector with modular plug and play.

## 2.14 ELECTRICAL

### 2.14.1 Wire

Provide wiring materials under this section as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM with the additions and modifications specified herein.

### 2.14.2 Alarm Wiring

IDC wiring shall be solid copper cable in accordance with the manufacturers requirements. Copper initiating device circuit field wiring shall be No. 16 AWG size conductors at a minimum. Visual notification appliance circuit conductors, that contain audible alarm appliances, shall be copper No. 14 AWG size conductors at a minimum. Wire size shall be

sufficient to prevent voltage drop problems. Circuits operating at 24 VDC shall not operate at less than the listed voltages for the detectors and/or appliances. Power wiring, operating at 120 VAC minimum, shall be a minimum No. 12 AWG solid copper having similar insulation. Acceptable power-limited cables are FPL, FPLR or FPLP as appropriate with red colored covering. Nonpower-limited cables shall comply with NFPA 70.

## 2.15 WARNING SIGNS

Signs shall be three-layer red-white-red micarta engraved to show white letters on a red background. Letters are to be uppercase. Warning signs are to be 0.3175 mm thick with beveled edges. Make letters for "WARNING" 3.81 cm tall, and all other lettering 2.54 cm tall.

## PART 3 EXECUTION

### 3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work is in any way dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative any condition which prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

### 3.2 INSTALLATION

The installation shall be in accordance with the applicable provisions of NFPA 72 and NFPA 2001. Provide each system complete and ready for operation. Conceal piping to the maximum extent possible. Inspect and test pipe and receive Contracting Officer approval before pipe is concealed. Provide each system with a listed pressure-relief device designed to operate between 13.79 and 22.75 megapascals and located between the storage cylinder manifolds and any normally closed valve.

Recess mount the releasing control panel.

### 3.3 ELECTRICAL

Except as modified herein, electric equipment and wiring shall be in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Alarm signal wiring connected to the building fire alarm control system shall be by the fire alarm installer.

#### 3.3.1 Overcurrent and Surge Protection

All equipment connected to alternating current circuits shall be protected from surges. Cables and conductors that serve as communications links, except fiber optics, shall have surge protection circuits installed at each end. Fuses shall not be used for surge protection.

#### 3.3.2 Grounding

Grounding shall be provided to building ground.

### 3.3.3 System Field Wiring

#### 3.3.3.1 Wiring within Cabinets, Enclosures, and Boxes

Provide wiring installed in a neat and workmanlike manner and installed parallel with or at right angles to the sides and back of any box, enclosure, or cabinet. Conductors that are terminated, spliced, or otherwise interrupted in any enclosure, cabinet, mounting, or junction box shall be connected to screw-type terminal blocks. Mark each terminal in accordance with the wiring diagrams of the system. The use of wire nuts or similar devices is prohibited. Wiring to conform with NFPA 70.

Indicate the following in the wiring diagrams:

- a. Point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams shall show connections from field devices to the RCU and remote RCU, initiating circuits, switches, relays and terminals.
- b. Complete riser diagrams indicating the wiring sequence of devices and their connections to the control equipment. Include a color code schedule for the wiring. Include floor plans showing the locations of devices and equipment.

#### 3.3.3.2 Terminal Cabinets

Provide a terminal cabinet at the base of any circuit riser, on each floor at each riser, and where indicated on the drawings. Terminal size shall be appropriate for the size of the wiring to be connected. Conductor terminations shall be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection shall be permanently mounted in the terminal cabinet. Minimum size is 200 mm by 200 mm. Only screw-type terminals are permitted. Provide an identification label, that displays "RELEASING SYSTEM TERMINAL CABINET" with 50-mm lettering, on the front of the terminal cabinet.

#### 3.3.3.3 Alarm Wiring

- a. Voltages shall not be mixed in any junction box, housing or device, except those containing power supplies and control relays.
- b. Utilize shielded wiring where recommended by the manufacturer. For shielded wiring, ground the shield at only one point, in or adjacent to the RCU.
- c. Pigtail or T-tap connections to initiating device circuits, supervisory alarm circuits, and notification appliance circuits are prohibited.
- d. Color coding is required for circuits and shall be maintained throughout the circuit. Conductors used for the same functions shall be similarly color coded. Conform wiring to NFPA 70.
- e. Pull all conductors splice free. The use of wire nuts, crimped connectors, or twisting of conductors is prohibited. Where splices are unavoidable, the location of the junction box or pull box where they occur shall be identified on the as-built drawings. The number

and location of splices shall be subject to approval by the NAVFAC Atlantic Designated Fire Protection Engineer (DFPE).

#### 3.3.3.4 Back Boxes and Conduit

In addition to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, provide all wiring in rigid metal conduit or intermediate metal conduit unless specifically indicated otherwise. Minimum conduit size shall be 19 mm in diameter. Do not use electrical non-metallic tubing (ENT) or flexible non-metallic tubing and associated fittings.

- a. Galvanized rigid steel (GRS) conduit shall be utilized where exposed to weather, where subject to physical damage, and where exposed on exterior of buildings. Intermediate metal conduit (IMC) may be used in lieu of GRS as allowed by NFPA 70.
- b. Electrical metallic tubing (EMT) is permitted above suspended ceilings or exposed where not subject to physical damage. Do not use EMT underground, encased in concrete, mortar, or grout, in hazardous locations, where exposed to physical damage, outdoors or in fire pump rooms. Use die-cast compression connectors.
- c. For rigid metallic conduit (RMC), only threaded type fitting are permitted for wet or damp locations.
- d. Flexible metal conduit is permitted for initiating device circuits 2 meters in length or less. Flexible metal conduit is prohibited for notification appliance circuits and signaling line circuits. Use liquid tight flexible metal conduit in damp and wet locations.
- e. Schedule 40 (minimum) polyvinyl chloride (PVC) is permitted where conduit is routed underground or underground below floor slabs. Convert non-metallic conduit, other than PVC Schedule 40 or 80, to plastic-coated rigid, or IMC, steel conduit before turning up through floor slab.
- f. Exterior wall penetrations shall be weathertight. Conduit shall be sealed to prevent the infiltration of moisture.

#### 3.3.3.5 Conductor Terminations

Labeling of conductors at terminal blocks in terminal cabinets, and RCU shall be provided at each conductor connection. Each conductor or cable shall have a shrink-wrap label to provide a unique and specific designation. Each terminal cabinet and RCU shall contain a laminated drawing that indicates each conductor, its label, circuit, and terminal. The laminated drawing shall be neat, using 12-point lettering minimum size, and mounted within each cabinet, panel, or unit so that it does not interfere with the wiring or terminals. Maintain existing color code scheme where connecting to existing equipment.

### 3.4 RELEASING CONTROL SYSTEM

#### 3.4.1 Releasing Control Unit (RCU)

The RCU shall be located in a year round environmentally conditioned space and not in the hazard area served but adjacent to it. Locate the RCU where indicated on the drawings. Recess the enclosure with the top of the cabinet meters above the finished floor or center the cabinet at 1.5

meters, whichever is lower. Conductor terminations shall be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection shall be permanently mounted in the RCU. Locate the document storage cabinet adjacent to the RCU unless the Contracting Officer directs otherwise.

### 3.4.2 Air Sampling Smoke Detector

Locate air sampling smoke detectors in accordance with the manufacturer's instructions. Air sampling smoke detectors shall be installed as follows:

#### a. Air Sampling Smoke Detector Assembly:

- (1) Detector assembly shall be mounted to a wall at a height between 1200 to 1800 mm to top of detector measured above the finished floor.
- (2) Mounting shall be in a fully accessible and visible location.
- (3) Mounting or attachment to site equipment, cable trays, movable walls, other equipment or equipment supports is not permitted.
- (4) Piping network insertion into the detector inlet shall not be glued.
- (5) Air sampling smoke detector assembly shall be installed in accordance with this specification section and the manufacturer's installation and instruction manuals.
- (6) Flexible tubing for termination of the sampling pipe network into detector inlet is not permitted unless allowed by its listing.
- (7) Provide red background with white lettering labels that are plastic or phenolic type with a minimum of 6.4 mm block lettering to indicate detector and zone. For example: "AIR SAMPLING SMOKE DETECTOR No. 1-1 No. 5".
- (8) Provide a typeset printed or typewritten instruction card mounted behind a Lexan plastic or glass cover in a stainless steel or aluminum frame. Install the frame in a conspicuous location observable from the ASD panel. The card shall show those steps to be taken by an operator when a signal is received as well as the functional operation of the system under all conditions, normal, alarm, supervisory, and trouble. The instructions shall be approved by the Contracting Officer before being posted.

#### b. Pipe and Sampling Tube Mounting:

- (1) The pipe and sampling tubing detection network shall be mounted as per the design and manufacturer's specification. The hardware used for mounting will depend upon the design and site requirements.
- (2) To minimize flexing, pipes shall be secured every 1.5 meters.
- (3) Pipes shall be suspended between 25 and 100 mm below the ceiling. In areas with a suspended ceiling, the pipe network shall be installed above the ceiling utilizing the manufacturer's capillary sample port supported by the ceiling.



- (4) The sampling tubes shall be of the same length or use the manufacturer's guidelines to run tubes of the required lengths.
- (5) When installing a pipe network in areas subject to high temperature fluctuations allow for the contraction and expansion of pipes.
- (6) Where expansion or contraction of pipes is likely either after installation or on a continuous basis, do not place pipe clips adjacent to couplings and socket unions as these may interfere with the movement of the pipe.
- (7) No bends are permitted within the first 450 mm from the detector inlet.
- (8) The routing of the piping and sample tube network shall be coordinated with potential obstructions, including cable trays, grounding bars, and HVAC ductwork.
- (9) All changes in direction shall be made with standard elbows or tees.
- (10) All joints shall be air-tight and made by using solvent cement, except at the entry to the detector assembly. Refer to ASTM F402.
- (11) All pipes shall be supported by mechanical hangers attached to the structure of the building. Not more than 300 mm of pipe shall extend beyond the last hanger of each sampling pipe. The final installation shall result in no noticeable deflection in the piping network.
- (12) Attachment of air sampling pipes to cable trays, "gray iron", and telecommunications equipment is prohibited.
- (13) Clearly label pipe network to distinguish the pipe from other facility pipe work or protective cabling enclosures. For example: "SMOKE DETECTION SAMPLING TUBE - DO NOT DISTURB". In open rooms and exposed areas, provide labels at no greater than 6.1-meter intervals. Provide labels every 3 meters where piping is installed above suspended ceilings and every 609 mm, centered in the floor panels, where piping is installed within the raised floor cavity.
- (14) Placement of the sampling tube shall take into consideration appropriate sampling point locations and spacing.

c. Air Sampling Points:

- (1) Open area ceiling sampling points shall be oriented downward and shall be within 25 to 100 mm below the underside of the ceiling above where the ceiling is smooth.
- (2) Label all air sampling points with a round red label, each with a center hole to match the diameter of the drilled sampling point. For example: "AIR SAMPLING POINT DIA 3.2 MM". Indicate fractional dimensions in decimal format with a minimum of three decimal places.

### 3.4.3 Manual Stations

Locate manual stations as required by NFPA 72 and NFPA 2001. Mount stations so they are located no farther than 1.5 meters from the exit door they serve, measured horizontally. Manual stations shall be mounted at 1067 mm measured to the operating handle.

### 3.4.4 Notification Appliances

- a. Locate notification appliance devices as required by NFPA 72 and NFPA 2001. Where more than two strobe appliances (releasing and/or building) are located in the same room or corridor or field of view, provide synchronized operation. Devices shall use screw terminals for all field wiring. Audible and visual notification appliances mounted in acoustical ceiling tiles shall be centered in the tiles +/- 50 mm.
- b. Audible and visual notification appliances mounted on the exterior of the building, within unconditioned spaces, or in the vicinity of showers shall be listed weatherproof appliances installed on weatherproof backboxes.

### 3.4.5 Smoke or Fire/Smoke Dampers

Activation of smoke or fire/smoke dampers shall occur upon second zone detection, or upon activation of clean agent discharge by manual release station. Smoke or fire/smoke dampers are specified in Section 23 00 00 AIR SUPPLY, DISTRIBUTION, VENTILATION, AND EXHAUST SYSTEMS.

### 3.4.6 Ceiling Bridges

Provide ceiling bridges for ceiling-mounted appliances. Ceiling bridges shall be as recommended/required by the manufacturer of the ceiling-mounted notification appliance.

## 3.5 CLEAN AGENT EXTINGUISHING SYSTEM

### 3.5.1 Distribution Piping and Fittings

Distribution piping and fittings shall be installed in accordance with the manufacturer's requirements, NFPA 2001 and approved piping standards and guidelines. All distribution piping shall be installed by qualified individuals using good, accepted practices and quality procedures. All piping shall be adequately supported and anchored at all directional changes and nozzle locations. All piping shall be reamed, blown clear and swabbed with suitable solvents to remove burrs, mill varnish and cutting oils before assembly. All pipe threads shall be sealed with Teflon tape pipe sealant applied to the male thread ONLY.

#### 3.5.1.1 Pipe Penetrations

- a. Where pipes or conduits penetrate fire walls, fire partitions, or floors, pipes/conduits shall be firestopped in accordance with Section 07 84 00 FIRESTOPPING.
- b. In penetrations that are not fire-rated or not a floor penetration, the space between the sleeve and the pipe shall be sealed at both ends with plastic waterproof cement that will dry to a firm but pliable mass or with a mechanically adjustable segmented elastomer seal.

- c. All penetrations through the boundary of rooms/areas identified as secure space area must meet ICS 705-1.

#### 3.5.1.2 Escutcheons

Escutcheons shall be listed for piping passing through floors, walls and ceilings, consisting of one-piece or split-type. Provide chrome plated escutcheons where pipe passes through walls and ceilings in finished areas. Other escutcheons may be steel or cast-iron, with aluminum paint finish. Securely fasten escutcheons in place with setscrews or other positive means.

#### 3.5.2 Discharge Nozzle

Where clogging by external foreign materials is likely, discharge nozzles shall be provided with frangible discs, blow-off caps, or other suitable devices.

### 3.6 WARNING SIGNS

Provide three-layer red-white-red micarta engraved to show white letters on a red background, warning signs. Letters are to be uppercase. Warning signs are to be 0.3175 mm thick with beveled edges. Make letters for "WARNING" 3.81 cm tall, and all other lettering 2.54 cm tall.

#### 3.6.1 Protected Space

Permanently affix a sign adjacent to every audible/visual alarm activated due to agent discharge reading:

**WARNING**

**THIS SPACE IS PROTECTED BY A CLEAN AGENT FIRE EXTINGUISHING SYSTEM. DO NOT ENTER WITHOUT AUTHORIZATION DURING OR AFTER DISCHARGE. THIS STROBE INDICATES DISCHARGE.**

#### 3.6.2 Manual Release Station Activation

Place warning signs, "Clean Agent Fire Extinguishing System" manual release, at each station indicating that operation of the station initiates immediate fire extinguishing clean agent discharge. Where building fire alarm pull stations are also mounted at the exits from the protected areas, separate them from clean agent manual release stations by at least 1-meter horizontally. If 1-meter separation is not possible, provide additional precautions to avoid accidental activation. Provide labels to clearly distinguish building fire alarm stations from clean agent manual release stations. Ensure clean agent manual release stations are a different color (yellow or orange) from building fire alarm pull stations and are consistent throughout the facility. Place a sign at every location where manual operation of the system may occur reading:

**WARNING**

**ACTUATION OF THIS DEVICE WILL CAUSE CLEAN AGENT GAS TO DISCHARGE. BEFORE ACTUATING, ENSURE THAT PERSONNEL ARE CLEAR OF THE AREA.**

#### 3.6.3 System Controls

Provide an electrical and mechanical actuating control system contained in

a RCU as modified below.

#### 3.6.3.1 Disconnect Switch

Connect the positive and negative conductors of the Class B solenoid/actuator/electric release head circuitry in series to a lock switch. Provide and install the switch in an enclosure inside the facility. A clearly visible sign on the enclosure, or immediately adjacent, is to explicitly indicate its purpose as "CLEAN AGENT FIRE EXTINGUISHING SYSTEM DISCONNECT SWITCH". The disconnect switch shall be supervised by the RCU.

#### 3.6.4 Operating Instructions

Provide operating instructions for clean agent fire protection systems consisting of raised or embossed white letter on red rigid plastic or enameled steel background and of adequate size to permit them to be easily read. Provide operating instructions at each RCU. Instruction are to clearly indicate necessary steps for the operating of the system.

### 3.7 FIRESTOPPING

Provide firestopping for holes at conduit and pipe penetrations through floor slabs, fire-rated walls, partitions with fire-rated doors, corridor walls, and vertical service shafts in accordance with Section 07 84 00 FIRESTOPPING.

### 3.8 PAINTING

- a. In unfinished areas (including areas above drop ceilings), paint all exposed electrical conduit (serving releasing system equipment), releasing system conduit, surface metal raceway, junction boxes and covers red. In lieu of painting conduit, the contractor may utilize red conduit with a factory applied finish.
- b. In finished areas, paint exposed electrical conduit (serving releasing system equipment), releasing system conduit, surface metal raceways, junction boxes, and electrical boxes to match adjacent finishes. The inside cover of the junction box must be identified as "Releasing System" and the conduit must have painted red bands 19 mm wide at 3-meter centers and at each side of a floor, wall, or ceiling penetration.
- c. Painting shall comply with Section 09 90 00 PAINTS AND COATINGS.

### 3.9 FIELD QUALITY CONTROL

#### 3.9.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level IV Fire Alarm Technician, and the representative of the installing company, and reviewed by the QFPE 60 days prior to performing system tests. Detailed test procedures shall list all components of the installed system such as initiating devices and circuits, notification appliances and circuits, control devices/equipment, batteries, power sources/supply, annunciators, special hazard equipment, and interface equipment. Test procedures shall include sequence of testing, time estimate for each test, and sample test data forms. The test data forms shall be in a check-off format (pass/fail with space to add applicable test data; similar to the

forma in NFPA 72 and NFPA 4.) The test procedures and accompanying test data forms shall be used for the pre-Government testing and the final Government testing. The test data forms shall record the test results and shall:

- a. Identify the NFPA Class of all initiating device circuits (IDC), notification appliance circuits (NAC, and signaling line circuits (SLC)).
- b. Identify each test required by NFPA 72 Test Methods and NFPA 2001 and required test herein to be performed on each component, and describe how these tests shall be performed.
- c. Identify each component and circuit as to type, location within the facility, and unique identity within the installed system. Provide necessary floor plan sheets showing each component location, test location, and alphanumeric identity.
- d. Identify all test equipment and personnel required to perform each test (including equipment necessary for smoke detector testing. The use of magnets is not permitted.
- e. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

### 3.9.2 Pre-Government Testing

#### 3.9.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that devices and circuits are functioning properly. Releasing system tests shall meet the requirements of paragraph entitled "Minimum System Tests" as required by NFPA 72. The Contractor and an authorized representative from each supplier of equipment shall be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter shall include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 72 and NFPA 2001 and the test reports noted below.

- a. NFPA 72 Record of Completion
- b. NFPA 72 Record of Inspection and Testing
- c. NFPA 2001 Clean Agent System Acceptance Test Report.
- d. Audibility test results with marked-up test floor plans.
- e. Documentation that all tests identified in the paragraph "Minimum System Tests" are complete.

#### 3.9.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the Designated Fire Protection Engineer (DFPE). Government final testing will not be scheduled until the DFPE has received copies of the request for Government

final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the installation-wide fire reporting system has been completed and tested to confirm communications are fully functional. Submit request for test at least 30 calendar days prior to the requested test date.

### 3.9.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests shall be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

### 3.9.4 Government Final Tests

The tests shall be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

- a. The manufacturer's technical representative.
- b. The contractor's Qualified Fire Protection Engineer (QFPE).
- c. Marked-up red line drawings of the system as actually installed.
- d. Loop resistance test results.
- e. Complete program printout including input/output addresses.
- f. Copy of pre-Government Test Certificate, test procedures and completed test data forms.
- g. Audibility test results with marked-up floor plans.

Government Final Tests will be witnessed by the NAVFAC Fire Protection Engineer, and Qualified Fire Protection Engineer (QFPE). At this time, any and all required tests noted in the paragraph "Minimum System Tests" shall be repeated at their discretion.

## 3.10 MINIMUM SYSTEM TESTS

### 3.10.1 System Tests

Test the system in accordance with the procedures outlined in NFPA 72. The required tests are as follows:

- a. Loop Resistance Tests: Measure and record the resistance of each circuit with each pair of conductors in the circuit short-circuited at the farthest point from the circuit origin. The tests shall be witnessed by the Contracting Officer and test results recorded for use at the final acceptance test.
- b. Verify the absence of unwanted voltages between circuit conductors and ground. The tests shall be accomplished at the preliminary test with results available at the final system test.
- c. Verify that the control unit is in the normal condition as detailed in

the manufacturer's O&M manual.

- d. Test each initiating device and notification appliance and circuit for proper operation and response at the control unit. Smoke detectors shall be tested in accordance with manufacturer's recommended calibrated test method. Use of magnets is prohibited. Testing of duct smoke detectors shall comply with the requirements of NFPA 72 except disconnect at least 20 percent of devices. If there is a failure at these devices, then supervision shall be tested at each device.
- e. Test the system for specified functions in accordance with the contract drawings and specifications and the manufacturer's O&M manual.
- f. Test both primary power and secondary power. Verify, by test, the secondary power system is capable of operating the system for the time period and in the manner specified.
- g. Determine that the system is operable under trouble conditions as specified.
- h. Visually inspect wiring.
- i. Test the battery charger and batteries.
- j. Verify that software control and data files have been entered or programmed into the releasing control unit. Hard copy records of the software shall be provided to the Contracting Officer.
- k. Verify that red-line drawings are accurate.
- l. Measure the current in circuits to ensure there is the calculated spare capacity for the circuits.
- m. Measure voltage readings for circuits to ensure that voltage drop is not excessive.
- n. Disconnect the verification feature for smoke detectors during tests to minimize the amount of smoke needed to activate the sensor. Testing of smoke detectors shall be conducted using real smoke or the use of canned smoke which is permitted.
- o. Measure the voltage drop at the most remote appliance (based on wire length) on each notification appliance circuit.
- p. Verify the documentation cabinet is installed and contains all as-built shop drawings, product data sheets, design calculations, site-specific software data package, and all documentation required by paragraph titled "Test Reports".

### 3.10.2 Audibility Tests

Sound pressure levels from audible notification appliances shall not exceed 110 dBA in any occupiable area. The provisions for audible notification (audibility) must be met with doors, fire shutters, movable partitions, and similar devices closed.

### 3.10.3 Room Pressurization Test

After all construction work is complete, conduct a room pressurization

test in accordance with NFPA 2001 in each clean agent fire extinguishing system hazard area. Test shall confirm enclosure's ability to retain the agent concentration level for the required hold time. If the test fails, the clean agent fire extinguishing system contractor shall coordinate room sealing with the general contractor. Additional tests shall be conducted until successful test results are achieved. Include final test results with the final record drawings.

- a. A room pressurization test shall be conducted, in each protected space, to determine the presence of openings, which would affect the agent system concentration levels. The test(s) shall be conducted using the Retro-Tec Corp. Door Fan system, or equivalent, with integrated computer program. All testing shall be in accordance with NFPA 2001.
- b. If room pressurization testing indicates openings exist that would result in leakage and/or loss of the clean agent fire extinguishing agent, the installing contractor shall be responsible for coordinating the proper sealing of the protected space(s) by the general contractor or his subcontractor or agent. The general contractor shall be responsible for adequately sealing all protected space(s) against agent loss or leakage. The installing contractor shall inspect all work to ascertain that the protected space(s) have been adequately and properly sealed. THE CLEAN AGENT FIRE EXTINGUISHING SYSTEM INSTALLING CONTRACTOR SHALL BE RESPONSIBLE FOR THE SUCCESS OF THE ROOM PRESSURIZATION TESTS. If the first room pressurization test is not successful, in accordance with this specification, the installing contractor shall inform the general contractor to determine and correct the cause of the test failure. If, after inspection the additional sealing performed, the installing contractor is satisfied that the protected space is ready for retesting, the installing contractor shall conduct additional room pressurization tests, at no additional cost, until a successful test is obtained. Copies of successful test results shall be submitted to the Contracting Officer for record.

#### 3.10.4 Clean Agent Fire Extinguishing Systems Test

Using nitrogen or other inert gas, perform a flow test on the piping network to verify that flow is continuous and unobstructed through piping and nozzles and that the piping has been cleaned and swabbed.

#### 3.10.5 Pressure/Leak Test

Pneumatically test the piping in a closed circuit for a period of 10 minutes at 276 kPa. At the end of 10 minutes, the pressure drop shall not fall below 80 percent of the test pressure. Repair leaks and retest until no leaks exist.

#### 3.11 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals shall be delivered to the Contracting Officer for review and acceptance. The drawings shall show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings shall be submitted within two weeks after the final Government test of the system. At least one set of as-built (marked-up) drawings shall be provided at the time of, or prior to the final Government test.



- a. Furnish one set of PDFs containing software back-up and CAD based drawings in latest version of AutoCAD, DXF and portable document formats of as-built drawings and schematics.
- b. Include complete wiring diagrams showing connections between devices and equipment, both factory and field wired.
- c. Include a riser diagram and drawings showing the as-built location of devices and equipment.
- d. Provide operation and maintenance (O&M) instructions.

### 3.12 INSTRUCTION OF GOVERNMENT EMPLOYEES

#### 3.12.1 Instructor

Provide the services of an instructor, who has received specific training from the manufacturer for the training of other persons regarding the operation, inspection, testing, and maintenance of the system provided. The instructor shall train the Government employees designated by the Contracting Officer, in the care, adjustment, maintenance, and operation of the releasing system. The instructor shall be thoroughly familiar with all parts of this installation. The instructor shall be trained in operating theory as well as in practical O&M work. Submit the instructor's information and qualifications including the training history.

#### 3.12.2 Required Instruction Time

Provide 8 hours of instruction after final acceptance of the system. The instruction shall be given during regular working hours on such dates and times selected by the Contracting Officer. The instruction may be divided into two or more periods at the discretion of the Contracting Officer. The on-site training shall allow for rescheduling for unforeseen maintenance and/or fire department responses.

##### 3.12.2.1 Technical Training

Equipment manufacturer or a factory representative shall provide 1 days of on-site and 5 days of technical training to the Government at the manufacturing facility. Training shall allow for classroom instruction as well as individual hands on troubleshooting and diagnostics exercises. Training shall occur within 6 months of system acceptance.

#### 3.12.3 Technical Training Manual

Provide, in manual format, lesson plans, operating instructions, maintenance procedures, and training data for the training courses. The operations training shall familiarize designated government personnel with proper operation of the installed system. The maintenance training course shall provide the designated government personnel adequate knowledge required to diagnose, repair, maintain, and expand functions inherent to the system.

### 3.13 EXTRA MATERIALS

#### 3.13.1 Repair Service/Replacement Parts

Repair services and replacement parts for the system shall be available for a period of 10 years after the date of final acceptance of this work

by the Contracting Officer. During the warranty period, the service technician shall be on-site within 24 hours after notification. All repairs shall be completed within 24 hours of arrival on-site.

During the warranty period, the installing releasing system contractor is responsible for conducting all required testing and maintenance in accordance with the requirements and recommended practices of NFPA 72, NFPA 2001 and the system manufacturer. Installing releasing system contractor is NOT responsible for any damage resulting from abuse, misuse, or neglect of equipment by the end user.

### 3.13.2 Spare Parts

Spare parts furnished shall be directly interchangeable with the corresponding components of the installed system. Spare parts shall be suitably packaged and identified by nameplate, tagging, or stamping. Spare parts shall be delivered to the Contracting Officer at the time of the final acceptance testing and shall be accompanied by an inventory list.

### 3.13.3 Document Storage Cabinet

Upon completion of the project, but prior to project close-out, place in the document storage cabinet copies of the following record documentation:

- a. As-built shop drawings
- b. Product data sheets
- c. Design calculations
- e. All documentation required by SD-06.

-- End of Section --

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## SECTION 21 30 00

## FIRE PUMPS

04/08, CHG 1: 08/13

## PART 1 GENERAL

## 1.1 SUMMARY

Except as modified in this Section or on the drawings, install fire pumps in conformance with NFPA 20, NFPA 70, and NFPA 72. In the event of a conflict between specific provisions of this specification and applicable NFPA standards, this specification governs. Devices and equipment for fire protection service must be UL Fire Prot Dir listed or FM APP GUIDE approved. Interpret all reference to the authority having jurisdiction to mean the Contracting Officer or the NAVFAC Atlantic, Naval Facilities Engineering Systems Command, Fire Protection Engineer for Navy projects.

## 1.2 SEQUENCING

## 1.2.1 Primary Fire Pump

Provide primary fire pump that automatically operates when the pressure drops to 758 kPa sprinkler system, or manually when the starter is operated.

Provide pump that automatically shut down after a running time of 10 minutes unless manually shutdown. The fire pump must automatically stop operating when the system pressure reaches 862 kPa and after the fire pump has operated for the minimum pump run time specified herein.

## 1.2.2 Pressure Maintenance Pump

Operate pressure maintenance pump when the system pressure drops to 793 kPa. Stop pump automatically when the system pressure reaches 862\_\_ kPa and after the pump has operated for the minimum pump run time specified herein.

## 1.3 FIRE PUMP INSTALLATION RELATED SUBMITTALS

The Fire Protection Specialist must prepare a list of the submittals, from the Contract Submittal Register, that relate to the successful installation of the fire pump(s), no later than 7 days after the approval of the Fire Protection Specialist and the Manufacturer's Representative. Include a letter of approval signed and dated by the Fire Protection Specialist with the submittals identified on this list when submitted to the Government.

## 1.4 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 (2020) Pipe Flanges and Flanged Fittings  
NPS 1/2 Through NPS 24 Metric/Inch Standard

ASME B16.39 (2020) Standard for Malleable Iron  
Threaded Pipe Unions; Classes 150, 250,

and 300

ASME B31.1 (2024) Power Piping

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA 10084 (2017) Standard Methods for the Examination of Water and Wastewater

AWWA B300 (2024) Hypochlorites

AWWA B301 (2024) Liquid Chlorine

AWWA C110/A21.10 (2021) Ductile-Iron and Gray-Iron Fittings

AWWA C111/A21.11 (2023) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings

AWWA C500 (2019) Metal-Seated Gate Valves for Water Supply Service

ASTM INTERNATIONAL (ASTM)

ASTM A47/A47M (1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings

ASTM A183 (2014; R 2020) Standard Specification for Carbon Steel Track Bolts and Nuts

ASTM A193/A193M (2024a) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications

ASTM A194/A194M (2024) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both

ASTM A449 (2014; R 2020) Standard Specification for Hex Cap Screws, Bolts, and Studs, Steel, Heat Treated, 120/105/90 ksi Minimum Tensile Strength, General Use

ASTM A536 (2024) Standard Specification for Ductile Iron Castings

ASTM A795/A795M (2021) Standard Specification for Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Fire Protection Use

ASTM B135/B135M (2017) Standard Specification for Seamless Brass Tube

ASTM F436M (2011) Hardened Steel Washers (Metric)

## FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide  
<https://www.approvalguide.com/>

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 20 (2025) Standard for the Installation of  
Stationary Pumps for Fire Protection

NFPA 24 (2025) Standard for the Installation of  
Private Fire Service Mains and Their  
Appurtenances

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National  
Electrical Code

NFPA 72 (2022; ERTA 22-1) National Fire Alarm and  
Signaling Code

NFPA 1963 (2019) Standard for Fire Hose Connections

NATIONAL INSTITUTE FOR CERTIFICATION IN ENGINEERING TECHNOLOGIES  
(NICET)

NICET 1014-7 (2012) Program Detail Manual for  
Certification in the Field of Fire  
Protection Engineering Technology (Field  
Code 003) Subfield of Automatic Sprinkler  
System Layout

## UL SOLUTIONS (UL)

UL 262 (2004; Reprint Jul 2023) Gate Valves for  
Fire-Protection Service

UL 448 (2020) UL Standard for Safety Centrifugal  
Stationary Pumps for Fire-Protection  
Service

UL 1247 (2023) Diesel Engines for Driving  
Stationary Fire Pumps

UL Fire Prot Dir UL Product IQ (updated online) at  
<https://productiq.ulpropsector.com/en>

## 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

The NAVFAC Atlantic, Naval Facilities Engineering Systems Command, Fire Protection Engineer, will review and approve all submittals in this

section requiring Government approval.

SD-01 Preconstruction Submittals

Fire Pump Installation Related Submittals

Fire Protection Specialist; G

No later than 14 days after the Notice to Proceed and prior to the submittal of the fire pump installation drawings

SD-02 Shop Drawings

Installation Drawings; G

3 copies

As-Built Drawings; G

Piping Layout; G

Pump Room; G

SD-03 Product Data

Catalog Data; G

Spare Parts

Preliminary Tests

At least 14 days prior to the proposed date and time to begin Preliminary Tests

Field Tests; G

At least 2 weeks before starting field tests

Manufacturer's Representative; G

Field Training; G

Army Final Acceptance Test

Navy Formal Inspection and Tests

SD-06 Test Reports

Preliminary Tests

3 copies of the completed Preliminary Tests Reports, no later than 7 days after the completion of the Preliminary Tests.

Navy Formal Inspection and Tests; G

SD-07 Certificates

Fire Protection Specialist

No later than 14 days after the Notice to Proceed and prior to the submittal of the fire pump installation drawings

Qualifications of Welders

Qualifications of Installer

Preliminary Test Certification

Final Test Certification

#### SD-10 Operation and Maintenance Data

Operating and Maintenance Instructions; G

At least 14 days prior to conducting field training

Flow Meter

Submit Data Package 2 for flow meter and controllers in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

### 1.6 EXTRA MATERIALS

Submit Spare Parts data for each different item of equipment and material specified. Include a complete list of parts and supplies, with current unit prices and source of supply, and a list of parts recommended by the manufacturer to be replaced after 1 year and 3 years of service. Include a list of special tools and test equipment required for maintenance and testing of the products supplied by the Contractor.

### 1.7 QUALITY ASSURANCE

#### 1.7.1 Fire Protection Specialist

Perform work specified in this section under the supervision of and certified by the Fire Protection Specialist. Submit the name and documentation of certification of the proposed Fire Protection Specialists. The Fire Protection Specialist is an individual who is a registered professional engineer and a Full Member of the Society of Fire Protection Engineers or who is certified as a Level IV Technician by National Institute for Certification in Engineering Technologies (NICET) in the Automatic Sprinkler System Layout subfield of Fire Protection Engineering Technology in accordance with NICET 1014-7. Engage the Fire Protection Specialist regularly in the design and installation of the type and complexity of system specified in the Contract documents. The specialist is an individual who has served in a similar capacity for at least three systems that have performed in the manner intended for a period of not less than 6 months.

#### 1.7.2 Qualifications of Welders

Submit certificates of each welder's qualifications prior to site welding; certifications more than one year old are not acceptable.

#### 1.7.3 Qualifications of Installer

Prior to installation, submit data for approval showing that the Contractor has successfully installed fire pumps and associated equipment



of the same type and design as specified herein, or that he has a firm contractual agreement with a subcontractor having such required experience. Include the names and locations of at least two installations where the Contractor, or the subcontractor referred to above, has installed such systems. Indicate the type and design of each system and certify that each system has performed satisfactorily in the manner intended for a period of not less than 18 months.

#### 1.7.4 Preliminary Test Certification

When preliminary tests have been completed and corrections made, submit a signed and dated certificate with a request for a formal inspection and tests.

#### 1.7.5 Final Test Certification

Concurrent with the Final Acceptance Test Report, submit certification by the Fire Protection Specialist that the fire pump installation is in accordance with the contract requirements, including signed approval of the Preliminary and Final Acceptance Test Reports. Submit data for approval showing the name and certification of all involved individuals with such qualifications at or prior to submittal of drawings.

#### 1.7.6 Manufacturer's Representative

Perform work specified in this section under the supervision of and certified by a representative of the fire pump manufacturer. Submit the name and documentation of certification of the proposed Manufacturer's Representative, concurrent with submittal of the Fire Protection Specialist Qualifications. Engage the Manufacturer's Representative regularly in the installation of the type and complexity of fire pump(s) specified in the Contract documents. The representative is an individual who has served in a similar capacity for at least three systems that have performed in the manner intended for a period of not less than 6 months.

### 1.8 DELIVERY, STORAGE, AND HANDLING

Protect all equipment delivered and placed in storage from the weather, excessive humidity and temperature variations, dirt and dust, or other contaminants. Additionally, either cap or plug all pipes until installed.

## PART 2 PRODUCTS

### 2.1 MATERIALS AND EQUIPMENT

- a. Provide materials and equipment that are standard products of a manufacturer regularly engaged in the manufacture of such products and that essentially duplicates items that have been in satisfactory use for at least 2 years prior to bid opening.
- b. Submit manufacturer's catalog data included with the Fire Pump Installation Drawings for each separate piece of equipment proposed for use in the system. Indicate the name of the manufacturer of each item of equipment, with data annotated to indicate model to be provided. In addition, provide a complete equipment list that includes equipment description, model number and quantity. Catalog data for material and equipment includes, but is not limited to, the following:

- (1) Fire pumps, drivers and controllers including manufacturer's certified shop test characteristic curve for each pump. Shop test curve may be submitted after approval of catalog data but must be submitted prior to the final tests.
  - (2) Pressure maintenance pump and controller.
  - (3) Piping components.
  - (4) Valves, including gate, check, globe and relief valves.
  - (5) Gauges.
  - (6) Hose valve manifold test header and hose valves.
  - (7) Flow meter.
  - (8) Restrictive orifice union.
  - (9) Associated devices and equipment.
- c. Provide a nameplate that identifies the manufacturer's name, address, type or style, model or serial number, contract number and accepted date; capacity or size; system in which installed and system which it controls and catalog number for all equipment. Affix standard nameplates securely in a conspicuous place and easy to read to pumps and motors. Provide nameplates and markings in accordance with UL 448. Provide nameplate and markings in accordance with UL 1247 for diesel driver. Provide electric motor nameplates with the minimum information required by NFPA 70, Section 430-7.

## 2.2 FIRE PUMP

Provide electric motor driven fire pump. Provide pump capacity rated at \_\_\_ 31.55\_\_\_ L/second with a rated net pressure of \_\_\_476\_\_\_ kPa. Furnish no less than 150 percent of rated flow capacity at not less than 65 percent of rated net pressure. Provide centrifugal horizontal split case fire pump. Equip horizontal pump with automatic air release devices. Use maximum rated pump speed of 2100 rpm when driving the pump at rated capacity. Provide pump that is manual pushbutton start and stop and automatic start and automatic stop. Provide pump conforming to the requirements of UL 448. Ensure fire pump discharge and suction gauges are oil-filled type.

## 2.3 REQUIREMENTS FOR FIRE PROTECTION SERVICE

### 2.3.1 General Requirements

Provide Materials and Equipment that have been tested by Underwriters Laboratories, Inc. and are listed in UL Fire Prot Dir or approved by Factory Mutual and listed in FM APP GUIDE. Where the terms "listed" or "approved" appear in this specification, such means listed in UL Fire Prot Dir or FM APP GUIDE.

### 2.3.2 Alarms

Provide audible and visual alarms as required by NFPA 20 on the controller. Provide remote supervision as required by NFPA 20, in accordance with NFPA 72 under Section 26.5. Provide remote alarm devices

as indicated on the drawings. Activate alarm signal upon the following conditions: electric motor controller has operated into a pump running condition, loss of electrical power to electric motor starter, and phase reversal on line side of motor starter. Provide weatherproof exterior alarm devices. Provide alarm silencing switch and red signal lamp, with signal lamp arranged to come on when switch is placed in OFF position.

## 2.4 UNDERGROUND PIPING COMPONENTS

### 2.4.1 Pipe and Fittings

Provide outside-coated, cement mortar-lined, ductile-iron pipe (with a rated working pressure of 1207 kPa) conforming to NFPA 24 for piping under the building and less than 1.50 m outside of the building walls. Anchor the joints in accordance with NFPA 24; provide concrete thrust block at the elbow where the pipe turns up toward the floor, and restrain the pipe riser with steel rods from the elbow to the flange above the floor. Provide minimum pipe size of 150 mm. Provide minimum cover depth as required by NFPA 24, but no less than 1 m. Piping more than 1.50 m outside of the building walls must be provided under Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING.

### 2.4.2 Fittings and Gaskets

Provide ductile iron fittings conforming to AWWA C110/A21.10. Provide gaskets that are suitable in design and size for the pipe with which such gaskets are to be used. Provide gaskets for ductile iron pipe joints conforming to AWWA C111/A21.11.

### 2.4.3 Valves and Valve Boxes

Provide gate valves conforming to AWWA C500 or UL 262. Provide valves with cast-iron body and bronze trim and that open by counterclockwise rotation. Except for post indicator valves, provide all underground valves with an adjustable cast-iron or ductile iron valve box of a size suitable for the valve on which the box is to be used, but no less than 133 mm in diameter. Coat the box with bituminous coating. Provide a cast-iron or ductile-iron cover with the word "WATER" cast on the cover for each box.

### 2.4.4 Gate Valve and Indicator Posts

Provide inside screw type gate valves for underground installation with counterclockwise rotation to open. Where indicating type valves are shown or required, use indicating valves consisting of gate valves with an approved indicator post of a length to permit the top of the post to be located 900 mm above finished grade. Provide gate valves and indicator posts with one coat of primer and two coats of red enamel paint and that are listed in UL Fire Prot Dir or FM APP GUIDE.

### 2.4.5 Buried Utility Warning and Identification Tape

Provide detectable aluminum foil plastic-backed tape or detectable magnetic plastic tape manufactured specifically for warning and identification of buried piping for all buried piping. Ensure tape is detectable by an electronic detection instrument. Provide tape in rolls, 80 mm minimum width, color-coded for the utility involved and imprinted in bold black letters continuously and repeatedly over the entire tape length. Warning and identification must be "CAUTION BURIED WATER PIPING

BELOW" or similar wording. Use permanent code and lettering that is unaffected by moisture and other substances contained in the trench backfill material. Bury tape at a depth of 300 mm below the top surface of earth or the top surface of the subgrade under pavement.

## 2.5 ABOVEGROUND PIPING COMPONENTS

### 2.5.1 Pipe Sizes 65 mm and Larger

#### 2.5.1.1 Pipe

Provide piping consisting of ASTM A795/A795M, Weight Class STD (Standard), Schedule 40 (except for Schedule 30 for pipe sizes 200 mm and greater in diameter), Type E or Type S, Grade A; black steel pipe. Join steel pipe by means of flanges welded to the pipe or mechanical grooved joints only. Do not join piping by welding or weld fittings. Galvanize suction piping on the inside in accordance with NFPA 20.

#### 2.5.1.2 Grooved Mechanical Joints and Fittings

Provide joints and fittings that are designed for no less than 1200 kPa service and are the product of the same manufacturer. Provide malleable iron fitting and coupling houses conforming to ASTM A47/A47M, Grade 32510; ductile iron conforming to ASTM A536, Grade 65-45-12. Provide flush type gasket that fills the entire cavity between the fitting and the pipe. Provide cadmium plated or zinc electroplated, heat-treated steel nuts and bolts conforming to ASTM A183.

#### 2.5.1.3 Flanges

Provide ASME B16.5, Class 150 flanges. Provide flanges at valves, connections to equipment, and where indicated.

#### 2.5.1.4 Gaskets

Provide AWWA C111/A21.11, cloth inserted red rubber gaskets.

#### 2.5.1.5 Bolts

Provide ASTM A449, Type 1 or 2, or ASTM A193/A193M, Grade B7 bolts. Do not extend bolts less than three full threads beyond the nut with bolts tightened to the required torque.

#### 2.5.1.6 Nuts

Provide ASTM A194/A194M, Grade 7 nuts.

#### 2.5.1.7 Washers

Provide washers meeting the requirements of ASTM F436M. Provide flat circular washers under all bolt heads and nuts.

### 2.5.2 Piping Sizes 50 mm and Smaller

#### 2.5.2.1 Steel Pipe

Provide ASTM A795/A795M, Weight Class STD (Standard), Schedule 40, Type E or Type S, Grade A, zinc-coated steel pipe with threaded end connections. Provide ASME B16.39, Class 150, zinc-coated threaded fittings. Provide

ASME B16.39, Class 150, zinc-coated unions.

### 2.5.3 Pipe Hangers and Supports

Provide adjustable hangers and supports that are UL listed UL Fire Prot Dir or FM approved FM APP GUIDE. Finish of rods, nuts, washers, hangers, and supports must be zinc-plated after fabrication.

### 2.5.4 Valves

Provide valves that are UL listed UL Fire Prot Dir or FM approved FM APP GUIDE for fire protection service and with flange or threaded end connections.

#### 2.5.4.1 Gate Valves and Control Valves

Provide outside screw and yoke (O.S.&Y.) type gate valves and control valves which open by counterclockwise rotation. Butterfly-type control valves are not permitted.

#### 2.5.4.2 Tamper Switch

Equip the suction control valves, the discharge control valves, valves to test header and flow meter, and the by-pass control valves with valve tamper switches for monitoring by the fire alarm system.

#### 2.5.4.3 Check Valve

Check valve must be clear open, swing type check valve with flange or threaded inspection plate.

#### 2.5.4.4 Relief Valve

Provide pilot operated or spring operated type relief valve conforming to NFPA 20. Provide a means of detecting water motion in the relief lines where the discharge is not visible within the pump house.

#### 2.5.4.5 Circulating Relief Valve

Provide an adjustable circulating relief valve for each fire pump in accordance with NFPA 20.

#### 2.5.4.6 Suction Pressure Regulating Valve

Provide suction pressure regulating valve that is FM approved FM APP GUIDE. Monitor suction pressure through a pressure line to the controlling mechanism of the regulating valve. Arrange valve in accordance with the manufacturer's recommendations.

### 2.5.5 Hose Valve Manifold Test Header

Construct header of steel pipe. Provide ASME B16.5, Class 150 flanged inlet connection to hose valve manifold assembly. Provide approved bronze hose gate valve with 65 mm National Standard male hose threads with cap and chain; locate one meter above grade in the horizontal position for each test header outlet. Ensure welding is metallic arc process in accordance with ASME B31.1.

### 2.5.6 Pipe Sleeves

Provide a pipe sleeve at each location where piping passes entirely through walls, ceilings, roofs, and floors, including pipe entering buildings from the exterior. Secure sleeves in position and location during construction. Provide sleeves of sufficient length to pass through entire thickness of walls, ceilings, and floors. Provide 25 mm minimum clearance between exterior of piping or pipe insulation, and interior of sleeve or core-drilled hole. Firmly pack space with mineral wool insulation. Seal space at both ends of the sleeve or core-drilled hole with plastic waterproof cement which will dry to a firm but pliable mass, or provide a mechanically adjustable segmented elastomeric seal. In fire walls and fire floors, provide a fire seal between the pipe and the sleeve in accordance with Section 07 84 00 FIRESTOPPING.

- a. Sleeves in Masonry and Concrete Walls, Ceilings, Roofs, and Floors: Provide hot-dip galvanized steel, ductile-iron, or cast-iron pipe sleeves. Core drilling of masonry and concrete may be provided in lieu of pipe sleeves provided that cavities in the core-drilled hole be completely grouted smooth.
- b. Sleeves in Other Than Masonry and Concrete Walls, Ceilings, Roofs, and Floors: Provide galvanized steel sheet pipe not less than 4.4 kg/square m.

### 2.5.7 Escutcheon Plates

Provide one-piece or split-hinge metal plates for piping entering floors, walls, and ceilings in exposed areas. Provide polished stainless steel or chromium-plated finish on copper alloy plates in finished spaces. Provide paint finish on plates in unfinished spaces. Secure plates in place.

## 2.6 DISINFECTING MATERIALS

### 2.6.1 Liquid Chlorine

Provide liquid chlorine conforming to AWWA B301.

### 2.6.2 Hypochlorites

Provide calcium hypochlorite and sodium hypochlorite conforming to AWWA B300.

## 2.7 ELECTRIC MOTOR DRIVER

Provide motors, controllers, contactors, and disconnects with their respective pieces of equipment, as specified herein and provide electrical connections under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide controllers and contactors with a maximum of 120-volt control circuits, and auxiliary contacts for use with the controls furnished. When motors and equipment furnished are larger than sizes indicated, include the cost of providing additional electrical service and related work under this section. Provide motor conforming to NEMA MG 1 Design B type. Provide premium efficiency type integral size motors in accordance with NEMA MG 1. Provide motor wattage of sufficient size so that the nameplate wattage rating will not be exceeded throughout the entire published pump characteristic curve. The motor and fire pump controller must be fully compatible.

## 2.8 FIRE PUMP CONTROLLER

Provide automatic type controller which is UL listed UL Fire Prot Dir or FM approved FM APP GUIDE for fire pump service. Arrange pump for automatic start and stop, and manual push-button stop. Accomplish automatic stopping only after all starting causes have returned to normal and after a minimum pump run time has elapsed. Provide controllers that are completely terminally wired, ready for field connections, and mounted in a NEMA Type 2 drip-proof enclosure arranged so that controller current carrying parts will not be less than 300 mm above the floor. Provide controller with voltage surge arresters installed in accordance with NFPA 20. Equip controller with a bourdon tube pressure switch or a solid state pressure switch with independent high and low adjustments, automatic starting relay actuated from normally closed contacts, visual alarm lamps and supervisory power light. Equip controller with a thermostat switch with adjustable setting to monitor the pump room temperature and to provide an alarm when temperatures falls below 5 degrees C. Provide controller which is factory-equipped with a heater operated by thermostat to prevent moisture in the cabinet.

### 2.8.1 Controller for Electric Motor Driven Fire Pump

Controller must be electronic soft start starting type. Design controller for 35 kW at 480 volts. Provide controller and integral transfer switch with a short circuit rating of 150 kA r.m.s. symmetrical at 480 volts a.c..

Provide an automatic transfer switch (ATS) for each fire pump in compliance with NFPA 20 and specifically listed for fire pump service. The ATS must transfer source of power to the alternate source upon loss of normal power. Controller must monitor pump running, loss of a phase or line power, phase reversal and pump room temperature. Provide ATS integral to controller as one equipment enclosure. Display alarms individually in front of panel by lighting of visual lamps. Label each lamp with rigid etched plastic labels. Equip controller with terminals for remote monitoring of pump running, pump power supply trouble (loss of power or phase and phase reversal), and pump room trouble (pump room temperature ), and for remote start. Limited service fire pump controllers are not permitted, except for fire pumps driven by electric motors rated less than 11 kW. Equip controller with a 7-day electric pressure recorder with 24-hour spring wound back-up. Provide a readout of the system pressure from 0 to 207 Pa, time, and date. Controller must require the pumps to run for ten minutes for pumps with driver motors under 149 kW and for 15 minutes for pumps with motors 149 kW and greater, prior to automatic shutdown. Equip the controller with an externally operable isolating switch which manually operates the motor circuit. Provide means in the controller for measuring current for all motor circuit conductors.

## 2.9 PRESSURE SENSING LINE

Provide a completely separate pressure sensing line for each fire pump and for the jockey pump. Arrange the sensing line in accordance with Figure A-7-5.2.1. of NFPA 20. Provide sensing line of 13 mmH58 brass tubing complying with ASTM B135/B135M. Equip the sensing line with two restrictive orifice unions each. Provide restricted orifice unions consisting of ground-face unions with brass restricted diaphragms drilled for a 2.4 mm. Mount restricted orifice unions in the horizontal position, no less than 1.5 m apart on the sensing line. Provide two test connections for each sensing line. Test connections consist of two brass 13 mm globe valves and 8 mm gauge connection tee arranged in accordance with NFPA 20. Equip one of the test connections with a 0 to 2100 kPa

water oil-filled gauge. Connect sensing line to the pump discharge piping between the discharge piping control valve and the check valve.

## 2.10 PRESSURE MAINTENANCE PUMP

### 2.10.1 General

Provide pressure maintenance pump that is electric motor driven, centrifugal type with a rated discharge of 0.32 L/second at 586 kPa. Pump must draft from the suction supply side of the suction pipe gate valve of the fire pump and discharge into the system at the downstream side of the pump discharge gate valve. Provide an approved indicating gate valve of the outside screw and yoke (O.S.&Y.) type in the maintenance pump discharge and suction piping. Provide oil-filled water pressure gauge and approved check valve in the maintenance pump discharge piping. Provide swing type check valve with removable inspection plate.

### 2.10.2 Pressure Maintenance Pump Controller

Arrange pressure maintenance pump controller for automatic and manual starting and stopping and equip with a "manual-off-automatic" switch. Provide controller which is completely prewired, ready for field connections, and wall-mounted in a NEMA Type 2 drip-proof enclosure. Equip the controller with a bourdon tube pressure switch or a solid state pressure switch with independent high and low adjustments for automatic starting and stopping. Provide a sensing line connected to the pressure maintenance pump discharge piping between the control valve and the check valve. Provide sensing line conforming to paragraph, PRESSURE SENSING LINE. The sensing line must be completely separate from the fire pump sensing lines. Provide an adjustable run timer to prevent frequent starting and stopping of the pump motor. Set the run timer for 2 minutes.

## 2.11 PUMP BASE PLATE AND PAD

Provide a common base plate for each horizontal-shaft fire pump for mounting pump and driver unit. Construct the base plate of cast iron with raised lip tapped for drainage or welded steel shapes with suitable drainage. Provide each base plate for the horizontal fire pumps with a 25 mmgalvanized steel drain line piped to the nearest floor drain. For vertical shaft pumps, provide pump head with a cast-iron base plate which serves as the sole plate for mounting the discharge head assembly. Mount pump units and bases on a raised 100 mmreinforced concrete pad that is an integral part of the reinforced concrete floor.

## 2.12 HOSE VALVE MANIFOLD TEST HEADER

Connect hose valve test header by ASME B16.5, Class 150 flange inlet connection. Provide UL listed UL Fire Prot Dir or FM approved FM APP GUIDE bronze hose gate valves with 65 mm American National Fire Hose Connection Screw Standard Threads (NH) in accordance with NFPA 1963. Provide number of valves in accordance with NFPA 20. Equip each hose valve with a cap and chain, and locate no more than 900 mm and no less than 600 mm above grade.

## 2.13 FLOW METER

Provide meter that is UL listed UL Fire Prot Dir or FM approved FM APP GUIDE as flow meters for fire pump installation with direct flow readout device. Provide flow meter that is capable of metering any waterflow quantities between 50 percent and 150 percent of the rated flow



of the pumps. Arrange piping to permit flow meter to discharge to pump suction and to discharge through test header. The meter throttle valve and the meter control valves must be O.S.&Y. valves. Provide automatic air release if flow meter piping between pump discharge and pump suction forms an inverted "U". Provide venturi type meter.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

#### 3.2 INSPECTION BY FIRE PROTECTION SPECIALIST

Periodically perform a thorough inspection of the fire pump installation, including visual observation of the pump while running, to assure that the installation conforms to the contract requirements. Excessive vibration, leaks (oil or water), unusual noises, overheating, or other potential problems are not acceptable. Inspection includes piping and equipment clearance, access, supports, and guards. Bring any discrepancy to the attention of the Contracting Officer in writing, no later than three working days after the discrepancy is discovered. The Fire Protection Specialist must witness the preliminary and final acceptance tests and, after completion of the inspections and a successful final acceptance test, sign test results and certify in writing that the installation the fire pump installation is in accordance with the contract requirements.

#### 3.3 INSTALLATION

Provide equipment, materials, workmanship, fabrication, assembly, erection, installation, examination, inspection and testing in accordance NFPA 20, except as modified herein. In addition, install the fire pump and engine in accordance with the written instructions of the manufacturer.

##### 3.3.1 Installation Drawings

Submit Fire Pump Installation Drawings consisting of a detailed plan view, detailed elevations and sections of the pump room, equipment and piping, drawn to a scale of not less than 1:20. Indicate equipment, piping, and associated pump equipment to scale. Indicate all clearance, such as those between piping and equipment; between equipment and walls, ceiling and floors; and for electrical working distance clearance around all electrical equipment. Include a legend identifying all symbols, nomenclatures, and abbreviations. Indicate a complete piping and equipment layout including elevations and/or section views of the following:

- a. Fire pumps, controllers, piping, valves, and associated equipment.
- b. Sensing line for each pump including the pressure maintenance pump.
- c. Engine fuel system for diesel driven pumps.
- d. Engine cooling system for diesel driven pumps.
- e. Pipe hangers and sway bracing including support for diesel muffler and exhaust piping.

- f. Restraint of underground water main at entry-point to the building including details of pipe clamps, tie rods, mechanical retainer glands, and thrust blocks.
- g. A one-line schematic diagram indicating layout and sizes of all piping, devices, valves and fittings.
- h. A complete point-to-point connection drawing of the pump power, control and alarm systems, as well as interior wiring schematics of each controller.

### 3.3.2 Pump Room Configuration

Provide detail plan view of the pump room including elevations and sections showing the fire pumps, associated equipment, and piping. Submit working drawings on sheets not smaller than A1 594 by 841 mm; include data for the proper installation of each system. Show piping schematic of pumps, devices, valves, pipe, and fittings. Provide an isometric drawing of the fire pump and all associated piping. Show point to point electrical wiring diagrams. Show piping layout and sensing piping arrangement. Show engine fuel and cooling system. Include:

- a. Pumps, drivers, and controllers
- b. Hose valve manifold test header
- c. Circuit diagrams for pumps
- d. Wiring diagrams of each controller

### 3.3.3 Accessories

Furnish tank supports, piping offsets, fittings, and required accessories as specified to provide a complete installation and to eliminate interference with other construction.

## 3.4 PIPE AND FITTINGS

Piping must be inspected, tested and approved before burying, covering, or concealing. Provide fittings for changes in direction of piping and for all connections. Make changes in piping sizes using tapered reducing pipe fittings. Do not use bushings. Photograph all piping prior to burying, covering, or concealing.

### 3.4.1 Cleaning of Piping

Interior and ends of piping must be clean and free of any water or foreign material. Keep piping clean during installation by means of plugs or other approved methods. When work is not in progress, securely close open ends of the piping so that no water or foreign matter will enter the pipes or fittings. Inspect piping before placing in position.

### 3.4.2 Threaded Connections

Provide jointing compound for pipe threads consisting of Teflon pipe thread paste and apply to male threads only. Provide exposed ferrous pipe threads with one coat of zinc molybdate primer applied to a minimum of dry film thickness of 0.025 mm.

### 3.4.3 Pipe Hangers and Supports

Provide additional hangers and supports for concentrated loads in aboveground piping, such as for valves and risers.

#### 3.4.3.1 Vertical Piping

Support piping at each floor, at no more than 3 meters intervals.

#### 3.4.3.2 Horizontal Piping

Space horizontal piping supports as follows:

| MAXIMUM SPACING (METERS) |              |     |     |    |     |     |     |     |     |      |
|--------------------------|--------------|-----|-----|----|-----|-----|-----|-----|-----|------|
| Nominal Pipe Size (mm)   | 25 and Under | 32  | 40  | 50 | 65  | 80  | 90  | 100 | 125 | 150+ |
| Copper Tube              | 1.8          | 2   | 2.4 |    |     |     |     |     |     |      |
| Steel Pipe               | 2            | 2.4 | 2.7 | 3  | 3.3 | 3.6 | 3.9 | 4.2 | 4.8 | 5.0  |

#### 3.4.4 Underground Piping

Install underground piping and fittings in conformance to NFPA 24. Anchor joints in accordance with NFPA 24. Provide concrete thrust block at elbow where pipe turns up towards floor, and restrain the pipe riser with steel rods from the elbow to the flange above the floor. After installation in accordance with NFPA 24, clean rods and nuts thoroughly and coat with asphalt or other corrosion-retard material approved by the Contracting Officer. Provide minimum cover depth of 900 mm.

#### 3.4.5 Grooved Mechanical Joint

Prepare grooves according to the coupling manufacturer's instructions. Use grooved fittings, couplings, and grooving tools which are products of the same manufacturer. Ensure pipe and groove dimensions comply with the tolerances specified by the coupling manufacturer. Measure the diameter of grooves made in the field using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Measure groove width and dimension of groove from end of pipe for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances. Do not use grooved joints in concealed locations, such as behind solid walls or ceilings, unless an access panel is shown on the drawings for servicing or adjusting the joint.

### 3.5 ELECTRICAL WORK

Provide electric motor and controls in accordance with NFPA 20, NFPA 72

and NFPA 70, unless more stringent requirements are specified herein or are indicated on the drawings. Provide electrical wiring and associated equipment in accordance with NFPA 20 and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide wiring in rigid metal conduit or intermediate metal conduit, except electrical metallic tubing conduit may be provided in dry locations not enclosed in concrete or where not subject to mechanical damage.

### 3.6 PIPE COLOR CODE MARKING

Color code marking of piping as specified in Section 09 90 00 PAINTS AND COATINGS.

### 3.7 FLUSHING

Flush the fire pump suction and discharge piping at 150 percent of rated capacity of each pump. Where the pump installation consists of more than one pump, the flushing is the total quantity of water flowing when all pumps are discharging at 150 percent of their rated capacities. The new pumps may be used to attain the required flushing volume. Do not flush underground piping using the fire pumps. Continue flushing operations until water is clear, but no less than 10 minutes. Submit a signed and dated flushing certificate before requesting field testing.

### 3.8 FIELD TESTS

Submit system diagrams that show the layout of equipment, piping, and storage units, and typed condensed sequence of operation, wiring and control diagrams, and operation manuals explaining preventative maintenance procedures, methods of checking the system for normal, safe operation, and procedures for safely starting and stopping the system must be framed under glass or laminated plastic. After approval, post these items where directed.

#### 3.8.1 Hydrostatic Test

Hydrostatically test piping at 1379 kPa for a period of 2-hours, or at least 345 kPa in excess of the maximum pressure, when the maximum pressure in the system is in excess of 1207 kPa in accordance with NFPA 20.

#### 3.8.2 Preliminary Tests

Submit proposed procedures for Preliminary Tests prior to the proposed date and time to begin Preliminary Tests. The Fire Protection Specialist must take all readings and measurements. The Manufacturer's Representative, a representative of the fire pump controller manufacturer, and a representative of the diesel engine manufacturer (when supplied) must witness the complete operational testing of the fire pump and drivers. The fire pump controller manufacturer's representative and the diesel engine manufacturer's representative must each be an experienced technician employed by the respective manufacturers and capable of demonstrating operation of all features of respective components including trouble alarms and operating features. Thoroughly inspect and test fire pumps, drivers and equipment to insure that the system is correct, complete, and ready for operation. Ensure that pumps are operating at rated capacity, pressure and speed. Include manual starting and running to ensure proper operation and to detect leakage or other abnormal conditions, flow testing, automatic start testing, testing of automatic settings, sequence of operation check, test of required accessories; test

of pump alarms devices and supervisory signals, test of pump cooling, operational test of relief valves, and test of automatic power transfer, if provided. Pumps must run without abnormal noise, vibration or heating. If any component or system was found to be defective, inoperative, or not in compliance with the contract requirements during the tests and inspection, make corrections and repeat the entire preliminary. Submit Preliminary Tests Reports, to include both the Contractor's Material and Test Certificate for Underground Piping and the Contractor's Material and Test Certificate for Aboveground Piping. All items in the Report must be signed by the Fire Protection Specialist and the Manufacturer's Representative.

### 3.8.3 Navy Formal Inspection and Tests

The NAVFAC Atlantic, Naval Facilities Engineering Systems Command, Fire Protection Engineer will witness formal tests and approve all systems before they are accepted. Submit the request for formal inspection at least 30 days prior to the date the inspection is to take place. An experienced technician regularly employed by the pump installer must be present during the inspection. Where pumps are engine driven, an experienced technician regularly employed by the engine manufacturer capable of demonstrating that all engine trouble alarms and operating features perform as required must be present. Submit proposed date and time to begin Navy Formal Inspection and Tests, with the Acceptance Procedures. Provide notification at least 30 days prior to the proposed start of the test. Include a copy of the Contractor's Material & Test Certificates. Submit 3 copies of the completed Navy Formal Inspection and Tests Reports, no later than 7 days after the completion of the tests. All items in the reports must be signed by the Fire Protection Specialist and the Manufacturer's Representative. Test reports in booklet form (each copy furnished in a properly labeled three ring binder) showing all field tests and measurements taken during the preliminary and final testing, and documentation that proves compliance with the specified performance criteria, upon completion of the installation and final testing of the installed system. Indicate the final position of the controls and pressure switches in each test report. Include the description of the hydrostatic test conducted on the piping and flushing of the suction and discharge piping. Include a copy of the manufacturer's certified pump curve for each fire pump in the report.

#### 3.8.3.1 Full Water Flow Test

Include a full water flow test in the acceptance test. The securing of all hoses and nozzles during the tests is the responsibility of the Contractor. Conduct water flow testing in a safe manner with no destruction to the existing facility or new construction. Tests include 100 and 150 percent capacity flows and pressures, and no-flow pressures for compliance with manufacturer's characteristic curves. At this inspection repeat the required tests as directed.

#### 3.8.3.2 Correcting Defects

Correct defects in the work, and make additional tests until the Contractor has demonstrated that the system complies with the contract requirements.

#### 3.8.3.3 Documentation of Test

Manufacturer's certified shop test characteristic curves for each pump

being tested must be furnished by the Contractor at the time of the pump acceptance test.

#### 3.8.4 Test Equipment

Provide all equipment and instruments necessary to conduct a complete final test, including 65 mm diameter hoses, playpipe nozzles, pitot tube gauges, portable digital tachometer, voltage and ampere meters, and calibrated oil-filled water pressure gauges. Provide all necessary supports to safely secure hoses and nozzles during the test. The Government will furnish water for the tests.

### 3.9 DISINFECTION

After all system components are installed including pumps, piping, and other associated work, and all hydrostatic tests are successfully completed, thoroughly flush the pumps and all piping to be disinfected with potable water until there is no visible sign of dirt or other residue, and hydrostatic test are successfully completed, flush each portion of the piping specified in this Section system to be disinfected thoroughly with potable water until all entrained dirt and other foreign materials are removed before introducing chlorinating material.

#### 3.9.1 Chlorination

Provide chlorinating material consisting of hypochlorites or liquid chlorine. Feed the chlorinating material into the sprinkler piping at a constant rate of 50 parts per million (ppm). Use a properly adjusted hypochlorite solution injected into the system with a hypochlorinator, or liquid chlorine injected into the system through a solution-fed chlorinator and booster pump. Continue chlorination application until the entire system is filled. Keep the water in the system for a minimum of 24 hours. Open and close each valve in the system several times to ensure its proper disinfection. Following the 24-hour period, allow no less than 25 ppm chlorine residual to remain in the system.

#### 3.9.2 Flushing

Flush the system with clean water until the residual chlorine is reduced to less than one part per million. Samples of water in disinfected containers for bacterial examination will be taken from several system locations which are approved by the Contracting Officer.

#### 3.9.3 Sample Testing

Test samples for total coliform organisms (coliform bacteria, fecal coliform, streptococcal, and other bacteria) in accordance with AWWA 10084. Use either the multiple-tube fermentation technique or the membrane-filter technique testing method. Repeat disinfection until tests indicate the absence of coliform organisms (zero mean coliform density per 100 milliliters) in the samples for at least 2 full days. The system will not be accepted until satisfactory bacteriological results have been obtained.

### 3.10 SYSTEM STARTUP

Fully enclose or properly guard coupling, rotating parts, gears, projecting equipment, etc. so as to prevent possible injury to persons that come in close proximity of the equipment. Conduct testing of the fire pumps in a safe manner and ensure that all equipment is safely

secured. Use hoses and nozzles to conduct flow tests that are in excellent condition and safely anchor and secure to prevent any misdirection of the hose streams.

Post operating instructions for pumps, drivers, controllers, and flow meters.

### 3.11 CLOSEOUT ACTIVITIES

#### 3.11.1 Field Training

Conduct a training course for operating and maintenance personnel as designated by the Contracting Officer. Submit the proposed schedule for field training at least 14 days prior to the start of related training. Provide training for a period of 8 hours of normal working time and start after the fire pump installation is functionally complete and after the Final Acceptance Test. Cover all of the items contained in the approved Operating and Maintenance Instructions. Submit manuals listing step-by-step procedures required for system startup, operation, shutdown, and routine maintenance. Include the manufacturer's name, model number, parts list, list of parts and tools that should be kept in stock by the owner for routine maintenance including the name of a local supplier, simplified wiring and controls diagrams, troubleshooting guide, and recommended service organization (including address and telephone number) for each item of equipment. Submit Data Package 3 for fire pumps and drivers in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. Submit service organization that is capable of providing 4 hour onsite response to a service call on an emergency basis.

#### 3.11.2 As-Built Drawings

Submit As-Built Drawings, no later than 14 days after completion of the Final Tests. Update the Fire Pump Installation Drawings to reflect as-built conditions after all related work is completed and prepare drawings on reproducible full-size mylar film.

### 3.12 PROTECTION

Carefully remove materials so as not to damage material which is to remain. Replace existing work damaged by the Contractor's operations with new work of the same construction.

-- End of Section --

## SECTION 22 00 00

PLUMBING, GENERAL PURPOSE  
05/24

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## Canadian General Standards Board (CGSB)

|                 |                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| CAN/CGSB-19.13  | Sealing Compound, One Component, Elastomeric, Chemical Curing                                                         |
| CAN/CGSB-37.2-M | (1988) Emulsified Asphalt, Mineral - Colloid Type, Unfilled, for Dampproofing and Waterproofing and for Roof Coatings |

## CSA GROUP (CSA)

|                   |                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| CSA B304          | (2018) Liquid Chlorine                                                                                                    |
| CSA B45.1         | (2018) Ceramic plumbing fixtures                                                                                          |
| CSA B16           | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                                                                          |
| CSA B45.5         | (2022) Plastic Plumbing Fixtures                                                                                          |
| CSA B51           | (2024) Boiler, pressure vessel, and pressure piping code                                                                  |
| CSA B52           | (2023) Mechanical refrigeration code                                                                                      |
| CSA B64           | (2021) Backflow preventers and vacuum breakers                                                                            |
| CSA B70           | (2019) Cast iron soil pipe, fittings, and means of joining                                                                |
| CSA B79           | (2008) Commercial and residential drains and cleanouts                                                                    |
| CSA B125.1        | (2018) Plumbing supply fittings                                                                                           |
| CSA B125.3        | (2022) Plumbing fittings                                                                                                  |
| CSA B126          | (2023) Water cisterns                                                                                                     |
| CSA B128.1        | Des(2021) Design and installation of non-potable water systems/Maintenance and field testing of non-potable water systems |
| CSA B128.1/B128.2 | (2006) Design and installation of                                                                                         |



|                   |                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------|
|                   | non-potable water systems/Maintenance and field testing of non-potable water systems                                |
| CSA B137.1        | (2020) Polyethylene (PE) pipe, tubing, and fittings for cold-water pressure services                                |
| CSA B137.3        | (2020) Rigid polyvinylchloride (PVC) pipe and fittings for pressure applications                                    |
| CSA B137.4        | (2020) Polyethylene (PE) piping systems for gas services                                                            |
| CSA B137.6        | (2020) Chlorinated polyvinylchloride (CPVC) pipe, tubing, and fittings for hot- and cold-water distribution systems |
| CSA B137.11       | (2020) Polypropylene (PP-R & PP-RCT) pipe and fittings for pressure applications                                    |
| CSA B137.5        | (2020) Crosslinked polyethylene (PEX) tubing systems for pressure applications                                      |
| CSA B149.3        | Field approval of fuel-related components on appliances and equipment                                               |
| CSA B181.1        | (1996) ABS Drain, Waste, and Vent Pipe and Pipe Fittings                                                            |
| CSA B181.2        | PVC Drain, Waste, and Vent Pipe and Pipe Fittings                                                                   |
| CSA B182.1        | Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)                                           |
| CSA B182.2        | PVC Sewer Pipe and Fittings (PSM Type)                                                                              |
| CSA B182.11       | Recommended Practice for the Installation of Thermoplastic Drain, Storm, and Sewer Pipe and Fittings                |
| CSA B306          | (2023) Portable engine fuelling system for marine use                                                               |
| CSA B339          | (2023) Cylinders, spheres, and tubes for the transportation of dangerous goods                                      |
| CAN/CSA B483.1    | Drinking water treatment systems                                                                                    |
| CSA B651          | (2023) Accessible Design for the Built Environment                                                                  |
| CSA B813          | (Latest Edition) Soldering Fluxes for Copper and Copper Alloy Tube                                                  |
| CSA C22.2 No. 0.3 | (2023) Test methods for electrical wires and cables                                                                 |
| CSA C22.2 No. 61  | (2021) Household cooking ranges                                                                                     |

|                    |                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| CSA C22.2 No. 94.1 | (2020) Enclosures for electrical equipment, non-environmental considerations (Tri-national standard with NMX-J-235/1-ANCE-2015 and UL 50) |
| CSA C22.2 No. 94.2 | (2025) Enclosures for electrical equipment, environmental considerations (Trinational standard with NMX-J-235/2-ANCE-2020 and UL 50E)     |
| CSA C22.2 NO. 120  | (2023) Refrigeration equipment                                                                                                            |
| CSA C390           | Test Methods, Marking Requirements, and Energy Efficiency Levels for Three-phase Induction Motors                                         |
| CSA G40.21         | Grade 350W Class H                                                                                                                        |
| CSA S16            | Design of steel structures                                                                                                                |
| CSA W47.1          | Certification of companies for fusion welding of steel                                                                                    |
| CSA W117.2         | (2023) Safety in welding, cutting, and allied processes                                                                                   |
| CSA Z245.15        | (2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends                                                        |
| CSA Z245.30        | (2022) Field-applied external coatings for steel pipeline systems                                                                         |
| CSA Z259           | (2021) Connecting components for personal fall-arrest systems (PFAS)                                                                      |
| CSA Z317.1         | (2021) Special requirements for plumbing installations in health care facilities                                                          |
| CSA Z358.1         | (2014) Emergency Eyewash & Shower Standard                                                                                                |

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| ANSI Z21.22/CSA 4.4 | (2015; R 2020) Relief Valves for Hot Water Supply Systems |
|---------------------|-----------------------------------------------------------|

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

|                  |                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------|
| ANSI/ASHRAE 18   | (2008; R 2013) Methods Of Testing For Rating Drinking Water Coolers With Self-Contained Mechanical Refrigeration |
| ASHRAE 90.1 - SI | (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings                                       |

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|               |                                    |
|---------------|------------------------------------|
| ASME A112.1.2 | (2012; R 2017; R 2022) Air Gaps in |
|---------------|------------------------------------|

|                           |                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
|                           | Plumbing Systems (For Plumbing Fixtures and Water-Connected Receptors)                                                   |
| ASME A112.6.1M            | (1997; R 2017) Floor Affixed Supports for Off-the-Floor Plumbing Fixtures for Public Use                                 |
| ASME A112.6.3             | (2022) Floor Drains                                                                                                      |
| ASME A112.14.1            | (2003; R 2017; R 2022) Backwater Valves                                                                                  |
| ASME A112.18.1/CSA B125.1 | (2018) Plumbing Supply Fittings                                                                                          |
| ASME A112.19.2/CSA B45.1  | (2018; ERTA 2018) Standard for Vitreous China Plumbing Fixtures and Hydraulic Requirements for Water Closets and Urinals |
| ASME A112.19.3/CSA B45.4  | (2022) Stainless Steel Plumbing Fixtures                                                                                 |
| ASME A112.36.2M           | (2022) Cleanouts                                                                                                         |
| ASME B1.20.1              | (2013; R 2018) Pipe Threads, General Purpose (Inch)                                                                      |
| ASME B16.5                | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard                                     |
| ASME B16.21               | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                                                                         |
| ASME B16.34               | (2021) Valves - Flanged, Threaded and Welding End                                                                        |
| ASME B16.50               | (2021) Wrought Copper and Copper Alloy Braze-Joint Pressure Fittings                                                     |
| ASME B31.1                | (2024) Power Piping                                                                                                      |
| ASME B31.5                | (2022) Refrigeration Piping and Heat Transfer Components                                                                 |
| ASME B31.9                | (2020) Building Services Piping                                                                                          |
| ASME B40.100              | (2022) Pressure Gauges and Gauge Attachments                                                                             |
| ASME BPVC                 | (2010) Boiler and Pressure Vessels Code                                                                                  |
| ASME BPVC SEC IV          | (2017) BPVC Section IV-Rules for Construction of Heating Boilers                                                         |
| ASME BPVC SEC IX          | (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications                                           |
| ASME CSD-1                | (2021) Control and Safety Devices for Automatically Fired Boilers                                                        |

## AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

|           |                                                                                                                                   |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| ASSE 1001 | (2021) Performance Requirements for Atmospheric Type Vacuum Breakers                                                              |
| ASSE 1003 | (2020) Performance Requirements for Water Pressure Reducing Valves for Domestic Water Distribution Systems - (ANSI approved 2010) |
| ASSE 1010 | (2021) Performance Requirements for Water Hammer Arresters                                                                        |
| ASSE 1011 | (2017) Performance Requirements for Hose Connection Vacuum Breakers                                                               |
| ASSE 1012 | (2023) Performance Requirements for Backflow Preventer with an Intermediate Atmospheric Vent                                      |
| ASSE 1013 | (2021) Performance Requirements for Reduced Pressure Principle Backflow Prevention Assemblies                                     |
| ASSE 1016 | (2017) Performance Requirements for Automatic Compensating Valves for individual Showers and Tub/Shower Combinations              |
| ASSE 1017 | (2023) Performance Requirements for Temperature Actuated Mixing Valves for Hot Water Distribution Systems - (ANSI approved 2010)  |
| ASSE 1018 | (2023) Performance Requirements for Trap Seal Primer Valves - Potable Water Supplied (ANSI Approved 2002)                         |
| ASSE 1019 | (2023) Performance Requirements for Wall Hydrant with Backflow Protection and Freeze Resistance                                   |
| ASSE 1020 | (2020) Performance Requirements for Pressure Vacuum Breaker Assemblies                                                            |
| ASSE 1070 | (2020) Performance Requirements for Water Temperature Limiting Devices                                                            |
| ASSE 1071 | (2012) Performance Requirements for Temperature Actuated Mixing Valves for Plumbed Emergency Equipment                            |

## AMERICAN WATER WORKS ASSOCIATION (AWWA)

|            |                                                                     |
|------------|---------------------------------------------------------------------|
| AWWA 10084 | (2017) Standard Methods for the Examination of Water and Wastewater |
| AWWA B300  | (2024) Hypochlorites                                                |

|           |                                                                                                           |
|-----------|-----------------------------------------------------------------------------------------------------------|
| AWWA B301 | (2024) Liquid Chlorine                                                                                    |
| AWWA C203 | (2020) Coal-Tar Protective Coatings and Linings for Steel Water Pipelines - Enamel and Tape - Hot-Applied |
| AWWA C606 | (2022) Grooved and Shouldered Joints                                                                      |
| AWWA C651 | (2014) Standard for Disinfecting Water Mains                                                              |
| AWWA C652 | (2019) Disinfection of Water-Storage Facilities                                                           |
| AWWA D100 | (2021) Welded Steel Tanks for Water Storage                                                               |

AMERICAN WELDING SOCIETY (AWS)

|                |                                                                          |
|----------------|--------------------------------------------------------------------------|
| AWS B2.2/B2.2M | (2016) Specification for Brazing Procedure and Performance Qualification |
|----------------|--------------------------------------------------------------------------|

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A74        | (2021) Standard Specification for Cast Iron Soil Pipe and Fittings                                                                                       |
| ASTM A105/A105M | (2023) Standard Specification for Carbon Steel Forgings for Piping Applications                                                                          |
| ASTM A193/A193M | (2024a) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications |
| ASTM A515/A515M | (2017; R2022) Standard Specification for Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher-Temperature Service                          |
| ASTM A516/A516M | (2017) Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service                                      |
| ASTM B32        | (2020) Standard Specification for Solder Metal                                                                                                           |
| ASTM B88M       | (2020) Standard Specification for Seamless Copper Water Tube (Metric)                                                                                    |
| ASTM B117       | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus                                                                                        |
| ASTM B370       | (2022) Standard Specification for Copper Sheet and Strip for Building Construction                                                                       |
| ASTM B813       | (2024) Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube                                                  |

|               |                                                                                                                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C564     | (2020a) Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings                                                                                                                          |
| ASTM C920     | (2018; R 2024) Standard Specification for Elastomeric Joint Sealants                                                                                                                                            |
| ASTM D638     | (2014) Standard Test Method for Tensile Properties of Plastics                                                                                                                                                  |
| ASTM D1004    | (2013) Initial Tear Resistance of Plastic Film and Sheeting                                                                                                                                                     |
| ASTM D1248    | (2016) Standard Specification for Polyethylene Plastics Extrusion Materials for Wire and Cable                                                                                                                  |
| ASTM D2235    | (2004; R 2016) Standard Specification for Solvent Cement for Acrylonitrile-Butadiene-Styrene (ABS) Plastic Pipe and Fittings                                                                                    |
| ASTM D2564    | (2020) Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems                                                                                                         |
| ASTM D2855    | (2020) Standard Practice for the Two-Step (Primer and Solvent Cement) Method of Joining Poly (Vinyl Chloride) (PVC) or Chlorinated Poly (Vinyl Chloride) (CPVC) Pipe and Piping Components with Tapered Sockets |
| ASTM D3138    | (2004; R 2016) Standard Specification for Solvent Cements for Transition Joints Between Acrylonitrile-Butadiene-Styrene (ABS) and Poly(Vinyl Chloride) (PVC) Non-Pressure Piping Components                     |
| ASTM D3139    | (2019) Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals                                                                                                                                       |
| ASTM D3212    | (2020) Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals                                                                                                     |
| ASTM D3311    | (2017) Standard Specification for Drain, Waste, and Vent (DWV) Plastic Fittings Patterns                                                                                                                        |
| ASTM D4551    | (2017) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Flexible Concealed Water-Containment Membrane                                                                                              |
| ASTM E1       | (2014) Standard Specification for ASTM Liquid-in-Glass Thermometers                                                                                                                                             |
| ASTM E96/E96M | (2024a) Standard Test Methods for                                                                                                                                                                               |

Gravimetric Determination of Water Vapor  
Transmission Rate of Materials

|            |                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| ASTM F409  | (2022) Standard Specification for Thermoplastic Accessible and Replaceable Plastic Tube and Tubular Fittings             |
| ASTM F477  | (2014; R 2021) Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe                           |
| ASTM F493  | (2022) Standard Specification for Solvent Cements for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe and Fittings |
| ASTM F2389 | (2023) Standard Specification for Pressure-rated Polypropylene (PP) Piping Systems                                       |

## CAST IRON SOIL PIPE INSTITUTE (CISPI)

|           |                                                                                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| CISPI 310 | (2012) Coupling for Use in Connection with Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

## COPPER DEVELOPMENT ASSOCIATION (CDA)

|           |                                    |
|-----------|------------------------------------|
| CDA A4015 | (2016; 14/17) Copper Tube Handbook |
|-----------|------------------------------------|

## CSA GROUP (CSA)

## INTERNATIONAL CODE COUNCIL (ICC)

|            |                                                                               |
|------------|-------------------------------------------------------------------------------|
| ICC A117.1 | (2017) Standard And Commentary Accessible and Usable Buildings and Facilities |
| ICC IPC    | (2024) International Plumbing Code                                            |

## INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

|                  |                                                                                      |
|------------------|--------------------------------------------------------------------------------------|
| ANSI/ISEA Z358.1 | (2014; R 2020) American National Standard for Emergency Eyewash and Shower Equipment |
|------------------|--------------------------------------------------------------------------------------|

## U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                         |
|--------------|-------------------------|
| UFC 3-420-01 | (2021) Plumbing Systems |
|--------------|-------------------------|

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS  
INDUSTRY (MSS)

|           |                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------|
| MSS SP-25 | (2018) Standard Marking System for Valves, Fittings, Flanges and Unions                                        |
| MSS SP-58 | (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation |
| MSS SP-67 | (2022) Butterfly Valves                                                                                        |

|            |                                                                                    |
|------------|------------------------------------------------------------------------------------|
| MSS SP-70  | (2011) Gray Iron Gate Valves, Flanged and Threaded Ends                            |
| MSS SP-71  | (2018) Gray Iron Swing Check Valves, Flanged and Threaded Ends                     |
| MSS SP-72  | (2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service          |
| MSS SP-78  | (2011) Cast Iron Plug Valves, Flanged and Threaded Ends                            |
| MSS SP-80  | (2019) Bronze Gate, Globe, Angle and Check Valves                                  |
| MSS SP-85  | (2011) Gray Iron Globe & Angle Valves Flanged and Threaded Ends                    |
| MSS SP-110 | (2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends |

#### NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| NEMA 250   | (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)                     |
| NEMA MG 1  | (2021) Motors and Generators                                                        |
| NEMA MG 11 | (1977; R 2012) Energy Management Guide for Selection and Use of Single Phase Motors |

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                  |
|----------|----------------------------------------------------------------------------------|
| NFPA 90A | (2024) Standard for the Installation of Air Conditioning and Ventilating Systems |
|----------|----------------------------------------------------------------------------------|

#### NSF INTERNATIONAL (NSF)

|             |                                                                          |
|-------------|--------------------------------------------------------------------------|
| NSF 372     | (2022) Drinking Water System Components - Lead Content                   |
| NSF/ANSI 14 | (2023) Plastics Piping System Components and Related Materials           |
| NSF/ANSI 60 | (2022; Addenda 2022) Drinking Water Treatment Chemicals - Health Effects |
| NSF/ANSI 61 | (2022) Drinking Water System Components - Health Effects                 |

#### PLASTIC PIPE AND FITTINGS ASSOCIATION (PPFA)

|               |                                                                  |
|---------------|------------------------------------------------------------------|
| PPFA Fire Man | (2016) Firestopping: Plastic Pipe in Fire Resistive Construction |
|---------------|------------------------------------------------------------------|



## PLUMBING AND DRAINAGE INSTITUTE (PDI)

PDI WH 201 (2010) Water Hammer Arresters Standard

## SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE J1508 (2023) Hose Clamp Specifications

## U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

PL 93-523 (1974; A 1999) Safe Drinking Water Act

## U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 430 Energy Conservation Program for Consumer Products

40 CFR 141.80 National Primary Drinking Water Regulations; Control of Lead and Copper; General Requirements

## UNDERWRITERS LABORATORIES (UL)

UL 174 (2004; Reprint Oct 2023) UL Standard for Safety Household Electric Storage Tank Water Heaters

UL 399 (2017; Reprint May 2019) UL Standard for Safety Drinking Water Coolers

UL 430 (2015; Reprint Sep 2021) UL Standard for Safety Waste Disposers

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Plumbing System; G

## SD-03 Product Data

Backflow Prevention Assemblies; G

## Fixtures

Shower Faucets; G

Flush Valve Water Closets

WaterSense Label for Flush Valve Water Closet; S

Countertop LavatoriesWaterSense Label for Lavatory Faucet; S  
Kitchen Sinks

Drinking-Water Coolers; G

WaterSense Label for Showerhead; S

Water Heaters; G

Pumps; G

Welding

Vibration-Absorbing Features; G

Plumbing System

Thermostatic Mixing Valves

Temperature Controlled Mixing Valve

Point-Of-Use Mixing Valve

Automatic Temperature and Pressure Compensating Valve

Temperature Actuated Mixing Valve

#### SD-06 Test Reports

Tests, Flushing and Disinfection

Test of Backflow Prevention Assemblies; G.

#### SD-07 Certificates

Materials and Equipment

Bolts

#### SD-10 Operation and Maintenance Data

Plumbing System; G

### 1.3 STANDARD PRODUCTS

Specified materials and equipment must be standard products of a manufacturer regularly engaged in the manufacture of such products. Specified equipment must essentially duplicate equipment that has performed satisfactorily at least 2 years prior to bid opening. Standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

Where equipment is specified to conform to requirements of the ASME Boiler and Pressure Vessel Code, the design, fabrication, and installation must conform to the code and a certificate must be submitted for the equipment.

#### 1.3.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

#### 1.3.2 Service Support

The equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

#### 1.3.3 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

#### 1.3.4 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

##### 1.3.4.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions must be considered mandatory, the word "should" must be interpreted as "must." Reference to the "code official" must be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" must be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" must be interpreted to mean the "lessor." References to the "permit holder" must be interpreted to mean the "Contractor."

#### 1.3.4.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, must be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

#### 1.5 PERFORMANCE REQUIREMENTS

##### 1.5.1 Welding

Piping must be welded in accordance with qualified procedures using performance-qualified welders and welding operators. Submit a copy of qualified procedures and a list of names and identification symbols of qualified welders and welding operators. Procedures and welders must be qualified in accordance with ASME BPVC SEC IX. Welding procedures qualified by others, and welders and welding operators qualified by another employer, may be accepted as permitted by ASME B31.9, unless temperature and pressure limits specified then may be accepted as permitted by ASME B31.1. The Contracting Officer must be notified 24 hours in advance of tests, and the tests must be performed at the work site if practicable. Welders or welding operators must apply their assigned symbols near each weld they make as a permanent record.

#### 1.6 REGULATORY REQUIREMENTS

Unless otherwise required herein, plumbing work must be in accordance with ICC IPC.

#### 1.7 PROJECT/SITE CONDITIONS

The Contractor must become familiar with details of the work, verify dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

#### 1.8 INSTRUCTION TO GOVERNMENT PERSONNEL

Furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work. Submit Plumbing System Operation and Maintenance Data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government

for regular operation. The number of man-days (8 hours per day) of instruction furnished must be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system. When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

## 1.9 ACCESSIBILITY OF EQUIPMENT

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Materials for various services must be in accordance with TABLES I and II.

Pipe schedules must be selected based on service requirements. Pipe fittings must be compatible with the applicable pipe materials. Plastic pipe, fittings, and solvent cement must meet NSF/ANSI 14 and must be NSF listed for the service intended. Plastic pipe, fittings, and solvent cement used for potable hot and cold water service must bear the NSF seal "NSF-PW." Polypropylene pipe and fittings must conform to dimensional requirements of Schedule 40, Iron Pipe size and must comply with NSF/ANSI 14, NSF/ANSI 61 and ASTM F2389. Polypropylene piping that will be exposed to UV light must be provided with a Factory applied UV resistant coating. Pipe threads (except dry seal) must conform to ASME B1.20.1. Grooved pipe couplings and fittings must be from the same manufacturer. Material or used in any potable water system intended for human consumption be certified in accordance with NSF 372 with the exception of brazing materials. In line devices such as water meters, building valves, check valves, meter stops, valves, fittings and back flow preventers must comply with PL 93-523 and NSF/ANSI 61, Section 8. End point devices such as drinking water fountains, lavatory faucets, kitchen and bar faucets, residential ice makers, supply stops and end point control valves used to dispense water for drinking must meet the requirements of NSF/ANSI 61, Section 9. Hubless cast-iron soil pipe must not be installed underground, under concrete floor slabs, or in crawl spaces below kitchen floors (for Air Force/Army/Navy projects). Plastic pipe must not be installed in air plenums. Plastic pipe must not be installed in a pressure piping system in buildings greater than three stories including any basement levels.

#### 2.1.1 Pipe Joint Materials

Grooved pipe and hubless cast-iron soil pipe must not be used underground. Solder containing lead must not be used with copper pipe. Cast iron soil pipe and fittings must be marked with the collective trademark of the Cast Iron Soil Institute. Joints and gasket materials must conform to the following:

- a. Coupling for Cast-Iron Pipe: for hub and spigot type ASTM A74, CSA B70, AWWA C606, CSA B137.1. For hubless type: CISPI 310 CSA B70.
- d. Flange Gaskets: Gaskets must be made of non-asbestos material in

accordance with ASME B16.21 CSA B51. Gaskets must be flat, 1.6 mm thick, and contain Aramid fibers bonded with Styrene Butadiene Rubber (SBR) or Nitro Butadiene Rubber (NBR). Gaskets must be the full face or self centering flat ring type. Gaskets used for hydrocarbon service must be bonded with NBR.

- g. Solder Material: Solder metal must conform to ASTM B32 CSA B339.
- h. Solder Flux: Flux must be liquid form, non-corrosive, and conform to ASTM B813, Standard Test 1 CSA B813.
- i. PTFE Tape: PTFE Tape, for use with Threaded Metal or Plastic Pipe.
- j. Rubber Gaskets for Cast-Iron Soil-Pipe and Fittings (hub and spigot type and hubless type): ASTM C564 CSA B813.
- l. Flexible Elastomeric Seals: ASTM D3139, ASTM D3212 or ASTM F477 CSA B137.3, CSA B137.4, CSA B181.2/CSA B182.2.
- n. Solvent Cement for Transition Joints between ABS and PVC Nonpressure Piping Components: ASTM D3138 CSA B181.1/CSA B181.2.
- o. Plastic Solvent Cement for ABS Plastic Pipe: ASTM D2235 CSA B181.1/CSA B181.2.
- p. Plastic Solvent Cement for PVC Plastic Pipe: ASTM D2564 and ASTM D2855 CSA B137.3.
- q. Plastic Solvent Cement for CPVC Plastic Pipe: ASTM F493 CSA B137.6.
- r. Flanged fittings including, but not limited to, flanges, bolts, nuts and bolt patterns must be in accordance with ASME B16.5, CSA B51: CSA B16, CSA B125.3: class 150 and must have the manufacturer's trademark affixed in accordance with MSS SP-25 CSA B125.1/CSA Z245.15. Flange material must conform to ASTM A105/A105M CSA G40.21. Blind flange material must conform to ASTM A516/A516M CSA G40.21 cold service and ASTM A515/A515M CSA G40.21 for hot service. Bolts must be high strength or intermediate strength with material conforming to ASTM A193/A193M CSA B182.1. Submit written certification by the bolt manufacturer that the bolts furnished comply with the specified requirements.
- u. Copper tubing must conform to ASTM B88M, Type K, L or M CSA B182.1.
- v. Heat-fusion joints for polypropylene piping: ASTM F2389 CSA B137.11.

#### 2.1.2 Miscellaneous Materials

Miscellaneous materials must conform to the following:

- a. Water Hammer Arrestor: PDI WH 201.piston type.
- b. Copper, Sheet and Strip for Building Construction: ASTM B370.
- d. Hose Clamps: SAE J1508 CSA.
- e. Supports for Off-The-Floor Plumbing Fixtures: ASME A112.6.1M CSA B45.5.
- f. Metallic Cleanouts: ASME A112.36.2M CSA B45.5.

- g. Plumbing Fixture Setting Compound: A preformed flexible ring seal molded from hydrocarbon wax material. The seal material must be nonvolatile nonasphaltic and contain germicide and provide watertight, gastight, odorproof and verminproof properties.
- h. Coal-Tar Protective Coatings and Linings for Steel Water Pipelines: AWWA C203 CSA Z245.30.
- i. Hypochlorites: AWWA B300 NSF/ANSI 60.
- j. Liquid Chlorine: AWWA B301 CSA B304.
- k. Gauges - Pressure and Vacuum Indicating Dial Type - Elastic Element: ASME B40.100 CSA B51/CAN/CSA B40.100.
- l. Thermometers: ASTM E1. Mercury must not be used in thermometers.

### 2.1.3 Pipe Insulation Material

Insulation must be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

## 2.2 PIPE HANGERS, INSERTS, AND SUPPORTS

Pipe hangers, inserts, and supports must conform to MSS SP-58 CSA Z259/CSA W47.1.

## 2.3 VALVES

Valves must be provided on supplies to equipment and fixtures. Valves 65 mm and smaller must be bronze with threaded bodies for pipe and solder-type connections for tubing. Valves 80 mm and larger must have flanged iron bodies and bronze trim. Pressure ratings must be based upon the application. Grooved end valves may be provided if the manufacturer certifies that the valves meet the performance requirements of applicable MSS standard. Valves must conform to the following standards:

| Description                                                                 | Standard                          |
|-----------------------------------------------------------------------------|-----------------------------------|
| Butterfly Valves                                                            | CSA Z245.15 MSS SP-67             |
| Cast-Iron Gate Valves, Flanged and Threaded Ends                            | MSS SP-70 CSA B51/CSA Z245.15     |
| Cast-Iron Swing Check Valves, Flanged and Threaded Ends                     | MSS SP-71 CSA Z245.15/CSA B51     |
| Ball Valves with Flanged Butt-Welding Ends for General Service              | MSS SP-72 CSA Z245.15/CSA B51     |
| Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends | MSS SP-110 CSA Z245.15/CSA B125.1 |
| Cast-Iron Plug Valves, Flanged and Threaded Ends                            | CSA Z245.15 MSS SP-78             |

|                                                                                  |                                                                                                         |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Bronze Gate, Globe, Angle, and Check Valves                                      | MSS SP-80 CSA B125.1/CSA Z245.15                                                                        |
| Steel Valves, Socket Welding and Threaded Ends                                   | ASME B16.34 CSA B51<br>CSA B16                                                                          |
| Cast-Iron Globe and Angle Valves, Flanged and Threaded Ends                      | MSS SP-85 CSA B125.1/CSA Z245.15                                                                        |
| Backwater Valves                                                                 | CSA B181.2 ASME A112.14.1                                                                               |
| Vacuum Relief Valves                                                             | ANSI Z21.22/CSA 4.4                                                                                     |
| Water Pressure Reducing Valves                                                   | CSA B126 ASSE 1003                                                                                      |
| Water Heater Drain Valves                                                        | CSA B51, ASME BPVC SEC IV Part HLW-810:<br>Requirements for Potable-Water Heaters<br>Bottom Drain Valve |
| Trap Seal Primer Valves                                                          | CSA B126 ASSE 1018                                                                                      |
| Temperature and Pressure Relief Valves for Hot Water Supply Systems              | ANSI Z21.22/CSA 4.4                                                                                     |
| Temperature and Pressure Relief Valves for Automatically Fired Hot Water Boilers | CSA B51 ASME CSD-1 CSA B149.3<br>Safety Code No., Part CW, Article 5                                    |

### 2.3.1 Backwater Valves

Backwater valves must be either separate from the floor drain or a combination floor drain, P-trap, and backwater valve, as indicated on drawings. Valves must have cast-iron bodies with cleanouts large enough to permit removal of interior parts. Valves must be of the flap type, hinged or pivoted, with revolving disks. Hinge pivots, disks, and seats must be nonferrous metal. Disks must be slightly open in a no-flow no-backwater condition. Cleanouts must extend to finished floor and be fitted with threaded countersunk plugs.

### 2.3.2 Wall Faucets

Wall faucets with vacuum-breaker backflow preventer must be brass with 20 mm male inlet threads, hexagon shoulder, and 20 mm hose connection. Faucet handle must be securely attached to stem.



### 2.3.3 Wall Hydrants (Frostproof)

ASSE 1019 with vacuum-breaker backflow preventer must have a nickel-brass or nickel-bronze wall plate and flange with nozzle and detachable key handle. A brass or bronze operating rod must be provided within a galvanized iron casing of sufficient length to extend through the wall so that the valve is inside the building, and the portion of the hydrant between the outlet and valve is self-draining. A brass or bronze valve with coupling and union elbow having metal-to-metal seat must be provided. Valve rod and seat washer must be removable through the face of the hydrant. The hydrant must have 20 mm exposed hose thread on spout and 20 mm male pipe thread on inlet.

### 2.3.4 Relief Valves

Water heaters and hot water storage tanks must have a combination pressure and temperature (P&T) relief valve. The pressure relief element of a P&T relief valve must have adequate capacity to prevent excessive pressure buildup in the system when the system is operating at the maximum rate of heat input. The temperature element of a P&T relief valve must have a relieving capacity which is at least equal to the total input of the heaters when operating at their maximum capacity. Relief valves must be rated according to ANSI Z21.22/CSA 4.4. Relief valves for systems where the maximum rate of heat input is less than 59 kW must have 20 mm minimum inlets, and 20 mm outlets. Relief valves for systems where the maximum rate of heat input is greater than 59 kW must have 25 mm minimum inlets, and 25 mm outlets. The discharge pipe from the relief valve must be the size of the valve outlet.

### 2.3.5 Thermostatic Mixing Valves

Each valve must be constructed to control the mixing of hot and cold water and to deliver water at a desired temperature regardless of pressure or input temperature changes. The control element must be of an approved type. The body must be of heavy cast bronze, and interior parts must be brass, bronze, corrosion-resisting steel or copper. The valve must be equipped with necessary stops, check valves, unions, and sediment strainers on the inlets.

### 2.3.6 Temperature Controlled Mixing Valve

Provide an ASSE 1017 CSA B125.3/CAN/CSA Z317.1 temperature controlled (master mixing valve) to lower temperature per UFC 3-420-01 NPC immediately downstream of the water heater or storage tank as close to the fixture(s) as possible in order to reduce the amount of stored tempered water in the pipes in accordance with ASSE 1017 CSA B125.3/CAN/CSA Z317.1.

### 2.3.7 Point-of-Use Mixing Valve

Provide a point-of-use mixing valve in accordance with ASSE 1070 CSA B125.3 at lavatories, hand-wash sinks, and washfountain locations. Point-of-use mixing valve must limit the tempered water to not greater than 43.4 degrees C.

### 2.3.8 Automatic Temperature and Pressure Compensating Valve

Provide automatic temperature and/or pressure compensating valve installed at each individual shower or tub/shower combination in accordance with ASSE 1016/CSA B125.3. The compensating valve must be integrated with each

individual shower or tub/shower combination valve and must be installed at point of use. The valve must be equipped with a device to limit the maximum temperature to 48.9 degrees C.

#### 2.3.9 Temperature Actuated Mixing Valve

Provide ASSE 1071 temperature actuated mixing valve to provide tepid water for emergency eyewash, shower/eyewash, drench shower and combination units that comply with ANSI/ISEA Z358.1 CSA Z358.1-15 (R2020), CSA Z358.1-15 (R2020) Section 4, CSA Z358.1-15 (R2020) Section 5, CSA Z358.1-15 (R2020), Section 6, CSA Z358.1-15 (R2020) Section 7. Valves must have a means to limit the maximum outlet temperature and must include a means of preventing cross-flow. Valves can be located in a wall-mounted bracket.

### 2.4 FIXTURES

Submit list of fixtures with manufacturer, model, and flow rate. Water closet replacements in major renovations may have a flush valve of up to 6.1 LPF to accommodate existing plumbing capacity. Fixtures for use by the physically handicapped must be in accordance with ICC A117.1 CSA B651. Vitreous China, nonabsorbent, hard-burned, and vitrified throughout the body must be provided. Porcelain enameled ware must have specially selected, clear white, acid-resisting enamel coating evenly applied on surfaces. No fixture will be accepted that shows cracks, crazes, blisters, thin spots, or other flaws. Fixtures must be equipped with appurtenances such as traps, faucets, stop valves, and drain fittings. Each fixture and piece of equipment requiring connections to the drainage system, except grease interceptors, must be equipped with a trap. Brass expansion or toggle bolts capped with acorn nuts must be provided for supports, and polished chromium-plated pipe, valves, and fittings must be provided where exposed to view. Fixtures with the supply discharge below the rim must be equipped with backflow preventers. Internal parts of flush valves and flushometer valves, shower mixing valves, shower head face plates, pop-up stoppers of lavatory waste drains, and pop-up stoppers and overflow tees and shoes of bathtub waste drains must be copper alloy with all visible surfaces chrome plated. Plastic in contact with hot water must be suitable for 82 degrees C water temperature.

#### 2.4.1 Lavatories

Enameled cast-iron lavatories must be provided with two cast-iron or steel brackets secured to the underside of the apron and drilled for bolting to the wall in a manner similar to the hanger plate. Exposed brackets must be porcelain enameled. Vitreous china lavatories must be provided with two integral molded lugs on the back-underside of the fixture and drilled for bolting to the wall in a manner similar to the hanger plate. Provide public faucet with a maximum flow rate of 1.9 L/min at a maximum flowing pressure of 414 kPa. Provide WaterSense labeled private faucet with a maximum flow rate of 5.7 L/min and a minimum flow rate of 3 L/min at a maximum flowing pressure of 414 kPa. Provide WaterSense labeled metering faucet with a maximum flow rate of 1.0 L per metering cycle. Provide data identifying WaterSense label for lavatory faucet.

#### 2.4.2 Flush Valve Water Closets

##### 2.4.2.1 Flush Valve Water Closet Type and Mount

ASME A112.19.2/CSA B45.1, white vitreous china, siphon jet, elongated bowl, floor-mounted, floor outlet. Top of toilet seat height above floor must be 356 to 381 mm, except 432 to 483 mm for wheelchair water closets.

Provide wax bowl ring including plastic sleeve. Provide white solid plastic elongated open-front seat.

#### 2.4.2.2 Flush Valve Water Closet Flush and Flow

Water flushing volume of the water closet and flush valve combination must not exceed 4.85 liters per flush. Water closets must meet the EPA WaterSense product definition and must be EPA WaterSense labeled products. Provide data identifying WaterSense label for flush valve water closet. Fixtures that require the WaterSense label but are located in countries where the label is unavailable, see table below or see <https://uwla.eu/for> WaterSense label substitutes.

| WaterSense Equivalencies |            |
|--------------------------|------------|
| Australia                | WELS       |
| Canada                   | WaterSense |

| Non-Mandatory Labels* |                      |
|-----------------------|----------------------|
| European Union        | European Water Label |
| Taiwan                | Water Saving Label   |
| China, Hong Kong      | WELS                 |

\*Water efficiency labels recognized as market transformational, and are recommended, but not required for LEED projects.

#### 2.4.2.3 Flush Valve Water Closet Control Valve

Provide large diameter flush valve including angle control-stop valve, vacuum breaker, tail pieces, slip nuts, and wall plates; exposed to view components must be chromium-plated or polished stainless steel. Flush valves must be nonhold-open type. Mount flush valves not less than 279 mm above the fixture. Mounted height of flush valve must not interfere with the hand rail in ADA stalls.

#### 2.4.3 Countertop Lavatories

ASME A112.19.2/CSA B45.1, CSA B45.1, white vitreous china, ASME A112.19.3/CSA B45.4 CSA B79 302 stainless steel, minimum dimensions of 483 mm wide by 432 mm front to rear, with supply openings for use with top mounted centerset faucets. Furnish template and mounting kit by lavatory manufacturer. Provide lavatory faucets and accessories meeting the flow rate and product requirements of the paragraph LAVATORIES. Mount counter with the top surface 864 mm above floor and with 737 mm minimum clearance from bottom of the counter face to floor.

#### 2.4.4 Lavatory Systems, Single or Multiple Stations

ASME A112.18.1/CSA B125.1, provide wall hung, solid surface counter with integral bowls and 410 mm access panel. Provide with one bowls. Height to rim dimensions 864 mm, provide manufacturer's standard, chrome plated solid brass faucet at each bowl with manual control. Provide ASSE 1070 point-of-use mixing valve with check stops per single user station, set to

maximum 43.3 degrees C.

#### 2.4.5 Kitchen Sinks

ASME A112.19.3/CSA B45.4, single compartment, 20 gage stainless steel with integral mounting rim for flush installation, minimum dimensions of 425 mm wide by 320 mm front to rear, with undersides fully sound deadened, with supply openings for use with top mounted washerless sink faucets with hose spray, and with 89 mm drain outlet. Water flow rate must not exceed 8.3 L per minute when measured at a flowing water pressure of 414 kPa. Provide stainless steel drain outlets and stainless steel cup strainers. Provide separate 38 mm P-trap and drain piping to vertical vent piping from each compartment. Provide top mounted washerless sink faucets with hose spray. Provide UL 430 waste disposer in right compartment.

#### 2.4.6 Drinking-Water Coolers for Standing Person

CSA C22.2 NO. 120. Dual or Bi-Level water coolers to conform to the requirements of DRINKING-WATER COOLER FOR STANDING PERSON and WHEELCHAIR DRINKING WATER COOLER conforming to ICC A117.1, CSA B651, ANSI/ASHRAE 18, CSA C22.2 NO. 120, UL 399, NSF/ANSI 61 CAN/CSA B483.1. Wall mounted height water cooler with more than a single thickness of metal between the potable water and the refrigerant in the heat exchanger, wall-hung, bubbler style, air-cooled condensing unit, 8.4 ml per second minimum capacity, stainless steel splash receptor and basin, bottle filler, vandal resistant, touchless control, filter, and stainless steel cabinet. Bubblers must be controlled by push levers or push bars, front mounted or side mounted near the front edge of the cabinet. Bubbler spouts must be mounted at maximum of 965 mm to 1092 mm above floor. Spouts must direct water flow at least 102 mm above unit basin and trajectory parallel or nearly parallel to the front of unit. Provide ASME A112.6.1M concealed steel pipe chair carriers.

#### 2.4.7 Wheelchair Drinking Water Cooler

ANSI/ASHRAE 18, UL 399, NSF/ANSI 61 CSA C22.2 NO. 120, CAN/CSA B483.1, CSA B651, and CSA B651, ADA and ICC A117.1. Dual height water cooler with more than a single thickness of metal between the potable water and the refrigerant in the heat exchanger, wall-mounted bubbler style with ASME A112.6.1M CSA B45.5 concealed chair carrier, air-cooled condensing unit, 4.99 mL per second minimum capacity and 8.4 ml per second minimum capacity, stainless steel splash receptor, and all stainless steel cabinet, with 686 mm minimum knee clearance from front bottom of unit to floor and 914 mm maximum spout height above floor for ADA side and 965 mm to 1092 mm above floor for standing person side and bottle filler vandal resistant touchless control filter cane apron. Bubblers must also be controlled by push levers, by push bars, or touch pads one on each side or one on front and both sides of the cabinet.

2.4.8 Shower Faucets

Provide ball joint self-cleaning shower heads. Provide WaterSense labeled showerhead with a maximum flow rate of 6.6 L/min. Provide data identifying WaterSense label for showerhead. Fixtures that require the WaterSense label but are located in countries where the label is unavailable, see table below or see <https://uwla.eu/> for WaterSense label substitutes.

|                          |                      |
|--------------------------|----------------------|
| WaterSense Equivalencies |                      |
| Australia                | WELS                 |
| Canada                   | WaterSense           |
| Non-Mandatory Labels*    |                      |
| European Union           | European Water Label |
| Taiwan                   | Water Saving Label   |
| China, Hong Kong         | WELS                 |

\*Water efficiency labels recognized as market transformational, and are recommended, but not required for LEED projects.

#### 2.4.9 Precast Terrazzo Shower Floors

Terrazzo must be made of marble chips cast in white portland cement to produce 25 mPa minimum compressive strength 7 days after casting. Provide floor or wall outlet copper alloy body drain cast integral with terrazzo, with polished stainless steel strainers.

#### 2.4.10 Precast Terrazzo Mop Sinks

Terrazzo must be made of marble chips cast in white portland cement to produce 25 mPa minimum compressive strength 7 days after casting. Terrazzo can include corner type stainless steel curb cap wall guard additions with minimum dimensions of 660 mm wide by 660 mm length by 305 mm height. Provide floor or wall outlet copper alloy body drain cast integral with terrazzo, with polished stainless steel strainers.

#### 2.4.11 Emergency Eyewash

ANSI/ISEA Z358.1, CSA Z358.1 floor and wall-mounted, freeze protected, nonclogging eye-wash with quick opening, full-flow valves, dust covers, stainless steel eye-wash receptor. Unit must deliver a minimum of 1.51 L/min of water at 207kPa flow pressure, with eye wash nozzles 838 mm to 1143 mm above finished floor. Provide copper alloy control valves. Provide an air-gap with the lowest potable eye wash water outlet located above the overflow rim by not less than the International Plumbing Code minimum. Provide a pressure-compensated tempering valve, with leaving water temperature setpoint adjustable throughout the range 15.6 degrees C to 35 degrees C. Provide packaged, UL listed, alarm system; including an amber strobe lamp, horn with externally adjustable loudness and horn silencing switch, mounting hardware, and waterflow service within NEMA Type 3 or 4 CSA TYPE 3 OR 4 enclosures and for explosion proof service within NEMA Type 7 or 9 CSA TYPE 7 OR 9 enclosures.

### 2.5 BACKFLOW PREVENTERS

Backflow prevention devices must be approved by the State or local regulatory agencies. If there is no State or local regulatory agency requirements, the backflow prevention devices must be listed by the Foundation for Cross-Connection Control & Hydraulic Research, or any other approved testing laboratory having equivalent capabilities for both

laboratory and field evaluation of backflow prevention devices and assemblies. Reduced pressure principle assemblies, double check valve assemblies, atmospheric (nonpressure) type vacuum breakers, and pressure type vacuum breakers must meet the above requirements.

Backflow preventers with intermediate atmospheric vent must conform to ASSE 1012 CSA B64.3. Reduced pressure principle backflow preventers must conform to ASSE 1013 CSA B64.4. Hose connection vacuum breakers must conform to ASSE 1011 CSA B64.2.1. Pipe applied atmospheric type vacuum breakers must conform to ASSE 1001. Pressure vacuum breaker assembly must conform to ASSE 1020. Air gaps in plumbing systems must conform to ASME A112.1.2.

## 2.6 DRAINS

### 2.6.1 Floor and Shower Drains

Floor and shower drains must consist of a galvanized body, integral seepage pan, and adjustable perforated or slotted chromium-plated bronze, nickel-bronze, or nickel-brass strainer, consisting of grate and threaded collar. Floor drains must be cast iron except where metallic waterproofing membrane is installed. Drains must be of double drainage pattern for embedding in the floor construction. The seepage pan must have weep holes or channels for drainage to the drainpipe. The seepage pan and weep holes must not be filled or plugged and have pervious materials (such as pea gravel) or a weep hole protection device to allow drainage. The strainer must be adjustable to floor thickness. A clamping device for attaching flashing or waterproofing membrane to the seepage pan without damaging the flashing or waterproofing membrane must be provided when required. Drains must be provided with threaded connection. Between the drain outlet and waste pipe, a neoprene rubber gasket conforming to ASTM C564 CSA C22.2 No. 0.3 may be installed, provided that the drain is specifically designed for the rubber gasket compression type joint. Floor and shower drains must conform to ASME A112.6.3 CSA B79. Provide drain with trap primer connection, trap primer, and connection piping. Primer must meet ASSE 1018.

#### 2.6.1.1 Metallic Shower Pan Drains

Where metallic shower pan membrane is installed, polyethylene drain with corrosion-resistant screws securing the clamping device must be provided. Polyethylene drains must have fittings to adapt drain to waste piping. Polyethylene for floor drains must conform to ASTM D1248 CSA C22.2 No. 0.3. Drains must have separate cast-iron "P" trap, circular body, seepage pan, and strainer, unless otherwise indicated.

#### 2.6.1.2 Drains and Backwater Valves

Drains and backwater valves installed in connection with waterproofed floors or shower pans must be equipped with bolted-type device to securely clamp flashing.

### 2.6.2 Bathtub and Shower Faucets and Drain Fittings

Provide single control pressure equalizing bathtub and shower faucets with body mounted from behind the wall with threaded connections. Provide ball joint self-cleaning shower heads. Provide WaterSense labeled showerhead with a maximum flow rate of 6.6 L/min. Provide showerheads meeting the requirements in the paragraph SHOWER FAUCETS. Provide tubing mounted from

behind the wall between bathtub faucets and shower heads and bathtub diverter spouts. Provide separate globe valves or angle valves with union connections in each supply to faucet. Provide trip-lever pop-up drain fittings for above-the-floor drain installations. The top of drain pop-ups, drain outlets, tub overflow outlet, and; control handle for pop-up drain must be chromium-plated or polished stainless steel. Linkage between drain pop-up and pop-up control handle at bathtub overflow outlet must be copper alloy or stainless steel. Provide 40 mm copper alloy adjustable tubing with slip nuts and gaskets between bathtub overflow and drain outlet; chromium-plated finish is not required. Provide bathtub and shower valve with ball type control handle.

### 2.6.3 Floor Sinks

Floor sinks must be square, with 300 mm nominal overall width or diameter and 250 mm nominal overall depth. Floor sink must have an acid-resistant enamel interior finish with cast-iron body, aluminum ABS sediment bucket, and perforated grate of cast iron in industrial areas and stainless steel in finished areas. The outlet pipe size must be as indicated or of the same size as the connecting pipe.

## 2.7 SHOWER PAN

Shower pan may be copper, or nonmetallic material.

### 2.7.1 Sheet Copper

Sheet copper must be 4.9 kg per square meter weight.

### 2.7.2 Plasticized Polyvinyl Chloride Shower Pan Material

Material must be sheet form. The material must be 1.016 mm minimum thickness of plasticized polyvinyl chloride or chlorinated polyethylene and must be in accordance with ASTM D4551.

### 2.7.3 Nonplasticized Polyvinyl Chloride (PVC) Shower Pan Material

Material must consist of a plastic waterproofing membrane in sheet form. The material must be 1.016 mm minimum thickness of nonplasticized PVC and must have the following minimum properties:

a. or ASTM D638:

|                            |             |
|----------------------------|-------------|
| Ultimate Tensile Strength: | 1.79 MPa    |
| Ultimate Elongation:       | 398 percent |
| 100 Percent Modulus:       | 3.07 MPa    |

b. ASTM D1004:

|                |                          |
|----------------|--------------------------|
| Tear Strength: | 53 kilonewtons per meter |
|----------------|--------------------------|

c. ASTM E96/E96M:

|                     |                                  |
|---------------------|----------------------------------|
| Permeance:<br>meter | 0.46 ng per Pa per second per sq |
|---------------------|----------------------------------|

d. Other Properties:

|                   |      |
|-------------------|------|
| Specific Gravity: | 1.29 |
|-------------------|------|

|                       |                    |
|-----------------------|--------------------|
| PVC Solvent:          | Weldable           |
| Cold Crack:           | minus 47 degrees C |
| Dimensional stability | 100 degrees C      |
| Hardness, Shore A:    | 89                 |

## 2.8 TRAPS

Unless otherwise specified, traps must be plastic per ASTM F409 CSA B181.2. Traps must be without a cleanout. Provide traps with removable access panels for easy clean-out at sinks and lavatories.

## 2.9 INTERCEPTORS

### 2.9.1 Oil Interceptor

Cast iron or welded steel, coated inside and outside with white acid resistant epoxy, with internal air relief bypass, bronze cleanout plug, double wall trap seal, removable combination pressure equalizing and flow diffusing baffle and sediment bucket, horizontal baffle, adjustable oil draw-off and vent connections on either side, gas and watertight gasketed nonskid cover, and flow control fitting.

## 2.10 WATER HEATERS

Water heater types and capacities must be as indicated. Each water heater must have replaceable anodes. Each primary water heater must have controls with an adjustable range that includes 32 to 71 degrees C. Each gas-fired water heater and booster water heater must have controls with an adjustable range that includes 49 to 82 degrees C. Hot water systems utilizing recirculation systems must be tied into building off-hour controls. The thermal efficiencies and standby heat losses must conform to TABLE III in PART 3 of this Section for each type of water heater specified. The only exception is that storage water heaters and hot water storage tanks having more than 2000 liters storage capacity need not meet the standard loss requirement if the tank surface area is insulated to R-12.5 and if a standing light is not used. Plastic materials polyetherimide (PEI) and polyethersulfone (PES) are forbidden to be used for vent piping of combustion gases. A factory pre-charged expansion tank must be installed on the cold water supply to each water heater. Expansion tanks must be specifically designed for use on potable water systems and must be rated for 93 degrees C water temperature and 1034 kPa working pressure. The expansion tank size and acceptance volume must be 11 L as indicated.

### 2.10.1 Automatic Storage Type

Heaters must be complete with control system, temperature gauge, and pressure gauge, and must have ASME rated combination pressure and temperature relief valve.

#### 2.10.1.1 Electric Type

Electric type water heaters must conform to UL 174 CSA C22.2 No. 61 with dual heating elements. Each element must be 4.5 KW. The elements must be wired so that only one element can operate at a time.

## 2.11 HOT-WATER STORAGE TANKS

Hot-water storage tanks must be constructed by one manufacturer, ASME



stamped for the working pressure, and must have the National Board (ASME) registration. The specification AWWA D100 CSA S16 and CSA B51 provides standards for water storage tanks, not domestic hot water storage tanks under domestic water pressure. The statement prior describes the tanks being ASME, it must be ASME BPVC CSA B51 (Boiler and Pressure Vessel Code), stamped for working pressure. The two statements combined describe a hot water storage tank under domestic water pressure. The tank must be cement-lined or glass-lined steel type in accordance with AWWA D100 CSA S16 and CSA B51. The heat loss must conform to TABLE III in PART 3 of this Section as determined by the requirements of ASHRAE 90.1 - SI NECB. Each tank must be equipped with a thermometer, conforming to ASTM E1, Type I, Class 3, Range C, style and form as required for the installation, and with 175 mm scale. Thermometer must have a separable socket suitable for a 20 mm tapped opening. Tanks must be equipped with a pressure gauge 155 mm minimum diameter face. Insulation must be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Storage tank capacity must be as shown. Install hot water storage tanks with access hatches in order to facilitate annual cleaning and inspections.

## 2.12 PUMPS

### 2.12.1 Circulating Pumps

Domestic hot water circulating pumps must be electrically driven, single-stage, centrifugal, with mechanical seals, suitable for the intended service. Pump and motor must be integrally mounted on a cast-iron or steel subbase, close-coupled with an overhung impeller, or supported by the piping on which it is installed. The shaft must be one-piece, heat-treated, corrosion-resisting steel with impeller and smooth-surfaced housing of bronze.

#### 2.12.1.1 Circulating Pump Enclosure

Motor must be totally enclosed, fan-cooled and must have sufficient wattage for the service required. Each pump motor must be equipped with an across-the-line magnetic controller in a NEMA 250 CSA C22.2 No. 94.1, CSA C22.2 No. 94.2, Type 1 enclosure with "START-STOP" switch in cover.

#### 2.12.1.2 Circulating Pump Size and Efficiency

Integral size motors must be premium efficiency type in accordance with CSA C390. Pump motors smaller than 746 W must have integral thermal overload protection in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Guards must shield exposed moving parts. 2.12.2 Flexible Connectors

Flexible connectors must be provided at the suction and discharge of each pump that is 1 hp or larger. Connectors must be constructed of neoprene, rubber, or braided bronze, with Class 150 standard flanges. Flexible connectors must be line size and suitable for the pressure and temperature of the intended service. 2.13 DOMESTIC WATER SERVICE METER

The requirements for metering and submetering are specified in Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING.

## 2.14 ELECTRICAL WORK

Provide electrical motor driven equipment specified complete with motors, motor starters, and controls as specified herein and in Section 26 20 00

INTERIOR DISTRIBUTION SYSTEM. Provide internal wiring for components of packaged equipment as an integral part of the equipment. Provide single-phase, fractional-horsepower alternating-current motors, including motors that are part of a system, corresponding to the applications in accordance with NEMA MG 11. Provide motors in accordance with NEMA MG 1 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor.

#### 2.14.1 Electrical Requirements for Motors

Motors must be rated for continuous duty with the enclosure specified. Motor duty requirements must allow for maximum frequency start-stop operation and minimum encountered interval between start and stop. Motor torque must be capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Motor bearings must be fitted with grease supply fittings and grease relief to outside of the enclosure.

#### 2.14.2 Electrical Requirements for Controllers and Contactors

Controllers and contactors must have auxiliary contacts for use with the controls provided. Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices specified, but not shown, must be provided. For packaged equipment, the manufacturer must provide controllers, including the required monitors and timed restart.

#### 2.14.3 Electrical Requirements for Power Wiring

Power wiring and conduit for field installed equipment must be provided under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

### 2.15 MISCELLANEOUS PIPING ITEMS

#### 2.15.1 Escutcheon Plates

Provide one piece or split hinge metal plates for piping entering floors, walls, and ceilings in exposed spaces. Provide chromium-plated on copper alloy plates or polished stainless steel finish in finished spaces. Provide paint finish on plates in unfinished spaces.

#### 2.15.2 Pipe Sleeves

Provide where piping passes entirely through walls, ceilings, roofs, and floors. Sleeves are not required where supply drain, waste, and vent (DWV) piping passes through concrete floor slabs located on grade, except where penetrating a membrane waterproof floor.

##### 2.15.2.1 Sleeves in Masonry and Concrete

Provide steel pipe sleeves or schedule 40 PVC plastic pipe sleeves. Sleeves are not required where drain, waste, and vent (DWV) piping passes through concrete floor slabs located on grade. Core drilling of masonry and concrete may be provided in lieu of pipe sleeves when cavities in the core-drilled hole are completely grouted smooth.

#### 2.15.2.2 Sleeves Not in Masonry and Concrete

Provide 26 gage galvanized steel sheet or PVC plastic pipe sleeves.

#### 2.15.3 Pipe Hangers (Supports)

Provide MSS SP-58 Type 1 with adjustable type steel support rods, except as specified or indicated otherwise. Attach to steel joists with Type 19 or 23 clamps and retaining straps. Attach to Steel W or S beams with Type 21, 28, 29, or 30 clamps. Attach to steel angles and vertical web steel channels with Type 20 clamp with beam clamp channel adapter. Attach to horizontal web steel channel and wood with drilled hole on centerline and double nut and washer. Attach to concrete with Type 18 insert or drilled expansion anchor. Provide Type 40 insulation protection shield for insulated piping.

#### 2.15.4 Nameplates

Provide 3.2 mm thick melamine laminated plastic nameplates, black matte finish with white center core, for equipment, gages, thermometers, and valves; valves in supplies to faucets will not require nameplates. Accurately align lettering and engrave minimum of 6.4 mm high normal block lettering into the white core. Minimum size of nameplates must be 25 by 63 mm. Key nameplates to a chart and schedule for each system. Frame charts and schedules under glass and place where directed near each system. Furnish two copies of each chart and schedule.

### PART 3 EXECUTION

#### 3.1 GENERAL INSTALLATION REQUIREMENTS

Piping located in air plenums must conform to NFPA 90A NFPA 90A requirements. Piping located in shafts that constitute air ducts or that enclose air ducts must be noncombustible in accordance with NFPA 90A. Installation of plastic pipe where in compliance with NFPA may be installed in accordance with PPFA Fire Man. The plumbing system must be installed complete with necessary fixtures, fittings, traps, valves, and accessories. Submit diagrams, instructions, and other sheets proposed for posting for Plumbing System. Submit for Plumbing System manufacturer's recommendations for the installation of bell and spigot and hubless joints for cast iron soil pipe. Water and drainage piping must be extended 1.5 m outside the building, unless otherwise indicated. A full port ball valve and drain must be installed on the water service line inside the building approximately 150 mm above the floor from point of entry. Piping must be connected to the exterior service lines or capped or plugged if the exterior service is not in place. Sewer and water pipes must be laid in separate trenches, except when otherwise shown. Exterior underground utilities must be at least 300 mm below the finish grade or as indicated on the drawings. If trenches are closed or the pipes are otherwise covered before being connected to the service lines, the location of the end of each plumbing utility must be marked with a stake or other acceptable means. Valves must be installed with control no lower than the valve body. Avoid the use of combination double wye and one-eighth bend fittings in the horizontal orientation below grade or below slab.

### 3.1.1 Water Pipe, Fittings, and Connections

#### 3.1.1.1 Utilities

The piping must be extended to fixtures, outlets, and equipment. The hot-water and cold-water piping system must be arranged and installed to permit draining. The supply line to each item of equipment or fixture, except faucets, flush valves, or other control valves which are supplied with integral stops, must be equipped with a shutoff valve to enable isolation of the item for repair and maintenance without interfering with operation of other equipment or fixtures. Supply piping to fixtures, faucets, hydrants, shower heads, and flushing devices must be anchored to prevent movement.

#### 3.1.1.2 Cutting and Repairing

The work must be carefully laid out in advance, and unnecessary cutting of construction must be avoided. Damage to building, piping, wiring, or equipment as a result of cutting must be repaired by mechanics skilled in the trade involved.

#### 3.1.1.3 Protection of Fixtures, Materials, and Equipment

Pipe openings must be closed with caps or plugs during installation. Fixtures and equipment must be tightly covered and protected against dirt, water, chemicals, and mechanical injury. Upon completion of the work, the fixtures, materials, and equipment must be thoroughly cleaned, adjusted, and operated. Safety guards must be provided for exposed rotating equipment.

#### 3.1.1.4 Mains, Branches, and Runouts

Piping must be installed as indicated. Pipe must be accurately cut and worked into place without springing or forcing. Structural portions of the building must not be weakened. Aboveground piping must run parallel with the lines of the building, unless otherwise indicated. Branch pipes from service lines may be taken from top, bottom, or side of main, using crossover fittings required by structural or installation conditions. Supply pipes, valves, and fittings must be kept a sufficient distance from other work and other services to permit not less than 12 mm between finished covering on the different services. Bare and insulated water lines must not bear directly against building structural elements so as to transmit sound to the structure or to prevent flexible movement of the lines. Water pipe must not be buried in or under floors unless specifically indicated or approved. Changes in pipe sizes must be made with reducing fittings. Use of bushings will not be permitted except for use in situations in which standard factory fabricated components are furnished to accommodate specific accepted installation practice. Change in direction must be made with fittings, except that bending of pipe 100 mm and smaller will be permitted, provided a pipe bender is used and wide sweep bends are formed. The center-line radius of bends must be not less than six diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be acceptable.

#### 3.1.1.5 Pipe Drains

Pipe drains indicated must consist of 20 mm hose bibb with renewable seat and full port ball valve ahead of hose bibb. At other low points, 20 mm brass plugs or caps must be provided. Disconnection of the supply piping

at the fixture is an acceptable drain.

#### 3.1.1.6 Expansion and Contraction of Piping

Allowance must be made throughout for expansion and contraction of water pipe. Each hot-water and hot-water circulation riser must have expansion loops or other provisions such as offsets and changes in direction where indicated and required. Risers must be securely anchored as required or where indicated to force expansion to loops. Branch connections from risers must be made with ample swing or offset to avoid undue strain on fittings or short pipe lengths. Horizontal runs of pipe over 15 m in length must be anchored to the wall or the supporting construction about midway on the run to force expansion, evenly divided, toward the ends. Sufficient flexibility must be provided on branch runouts from mains and risers to provide for expansion and contraction of piping. Flexibility must be provided by installing one or more turns in the line so that piping will spring enough to allow for expansion without straining. If mechanical grooved pipe coupling systems are provided, the deviation from design requirements for expansion and contraction may be allowed pending approval of Contracting Officer.

#### 3.1.1.7 Thrust Restraint

Plugs, caps, tees, valves and bends deflecting 11.25 degrees or more, either vertically or horizontally, in waterlines 100 mm in diameter or larger must be provided with thrust blocks, where indicated, to prevent movement. Thrust blocking must be concrete of a mix not leaner than: 1 cement, 2-1/2 sand, 5 gravel; and having a compressive strength of not less than 14 MPa after 28 days. Blocking must be placed between solid ground and the fitting to be anchored. Unless otherwise indicated or directed on drawings, the base and thrust bearing sides of the thrust block must be poured against undisturbed earth. The side of the thrust block not subject to thrust must be poured against forms. The area of bearing will be as indicated on drawings. Blocking must be placed so that the joints of the fitting are accessible for repair. Steel rods and clamps, protected by galvanizing or by coating with bituminous paint, must be used to anchor vertical down bends into gravity thrust blocks.

#### 3.1.1.8 Commercial-Type Water Hammer Arresters

Commercial-type water hammer arresters must be provided on hot- and cold-water supplies and must be located as generally indicated, with precise location and sizing to be in accordance with PDI WH 201 CSA B125.3. Water hammer arresters, where concealed, must be accessible by means of access doors or removable panels. Commercial-type water hammer arresters must conform to ASSE 1010 CSA B125.3. Vertical capped pipe columns will not be permitted.

#### 3.1.2 Joints

Installation of pipe and fittings must be made in accordance with the manufacturer's recommendations. Mitering of joints for elbows and notching of straight runs of pipe for tees will not be permitted. Joints must be made up with fittings of compatible material and made for the specific purpose intended.

##### 3.1.2.1 Threaded

Threaded joints must have American Standard taper pipe threads conforming

to ASME B1.20.1 CSA B51. Only male pipe threads must be coated with graphite or with an approved graphite compound, or with an inert filler and oil, or must have a polytetrafluoroethylene tape applied.

#### 3.1.2.2 Unions and Flanges

Unions, flanges and mechanical couplings must not be concealed in walls, ceilings, or partitions. Unions must be used on pipe sizes 65 mm and smaller; flanges must be used on pipe sizes 80 mm and larger.

#### 3.1.2.3 Cast Iron Soil, Waste and Vent Pipe

Bell and spigot compression and hubless gasketed clamp joints for soil, waste and vent piping must be installed per the manufacturer's recommendations.

#### 3.1.2.4 Copper Tube and Pipe

- a. Brazed. Brazed joints must be made in conformance with CSA W117.2, AWS B2.2/B2.2M, CSA B51, ASME B16.50, CSA B16 and CDA A4015 CSA B125.3 with flux and are acceptable for all pipe sizes. Copper to copper joints must include the use of copper-phosphorus or copper-phosphorus-silver brazing metal without flux. Brazing of dissimilar metals (copper to bronze or brass) must include the use of flux with either a copper-phosphorus, copper-phosphorus-silver or a silver brazing filler metal.
- b. Soldered. Soldered joints must be made with flux and are only acceptable for piping 50 mm and smaller. Soldered joints must conform to ASME B31.5 CSA B52 and CDA A4015, CSA B339, CSA B306, ASTM B813. Soldered joints must not be used in compressed air piping between the air compressor and the receiver.
- c. Copper Tube Extracted Joint. Mechanically extracted joints must be made in accordance with ICC IPC.

#### 3.1.2.5 Plastic Pipe

Acrylonitrile-Butadiene-Styrene (ABS) pipe must have joints made with solvent cement. PVC and CPVC pipe must have joints made with solvent cement elastomeric, threading, (threading of Schedule 80 Pipe is allowed only where required for disconnection and inspection; threading of Schedule 40 Pipe is not allowed), or mated flanged.

#### 3.1.3 Dissimilar Pipe Materials

Connections between ferrous and non-ferrous copper water pipe must be made with dielectric unions or flange waterways. Dielectric waterways must have temperature and pressure rating equal to or greater than that specified for the connecting piping. Waterways must have metal connections on both ends suited to match connecting piping. Dielectric waterways must be internally lined with an insulator specifically designed to prevent current flow between dissimilar metals. Dielectric flanges must meet the performance requirements described herein for dielectric waterways. Connecting joints between plastic and metallic pipe must be made with transition fitting for the specific purpose.

### 3.1.4 Pipe Sleeves and Flashing

Pipe sleeves must be furnished and set in their proper and permanent location.

#### 3.1.4.1 Sleeve Requirements

Unless indicated otherwise, provide pipe sleeves meeting the following requirements:

- a. Secure sleeves in position and location during construction. Provide sleeves of sufficient length to pass through entire thickness of walls, ceilings, roofs, and floors.
- b. A modular mechanical type sealing assembly may be installed in lieu of a waterproofing clamping flange and caulking and sealing of annular space between pipe and sleeve. The seals must consist of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe and sleeve using galvanized steel bolts, nuts, and pressure plates. The links must be loosely assembled with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut. After the seal assembly is properly positioned in the sleeve, tightening of the bolt must cause the rubber sealing elements to expand and provide a watertight seal between the pipe and the sleeve. Each seal assembly must be sized as recommended by the manufacturer to fit the pipe and sleeve involved.
- c. Sleeves must not be installed in structural members, except where indicated or approved. Rectangular and square openings must be as detailed. Each sleeve must extend through its respective floor, or roof, and must be cut flush with each surface, except for special circumstances. Pipe sleeves passing through floors in wet areas such as mechanical equipment rooms, lavatories, kitchens, and other plumbing fixture areas must extend a minimum of 100 mm above the finished floor.
- d. Unless otherwise indicated, sleeves must be of a size to provide a minimum clearance between bare pipe or insulation and inside of sleeve or between insulation and inside of sleeve. Sleeves in bearing walls and concrete slab on grade floors must be steel pipe or cast-iron pipe. Sleeves in nonbearing walls or ceilings may be steel pipe, cast-iron pipe, galvanized sheet metal with lock-type longitudinal seam, or plastic.
- e. Except as otherwise specified, the annular space between pipe and sleeve, or between jacket over insulation and sleeve, must be sealed as indicated with sealants conforming to ASTM C920 CAN/CGSB-19.13 and with a primer, backstop material and surface preparation as specified in Section 07 92 00 JOINT SEALANTS. The annular space between pipe and sleeve, between bare insulation and sleeve or between jacket over insulation and sleeve must not be sealed for interior walls which are not designated as fire rated.
- f. Sleeves through below-grade walls in contact with earth must be recessed 12 mm from wall surfaces on both sides. Annular space between pipe and sleeve must be filled with backing material and sealants in the joint between the pipe and concrete or masonry wall as specified above. Sealant selected for the earth side of the wall must be compatible with dampproofing/waterproofing materials that are to be

applied over the joint sealant. Pipe sleeves in fire-rated walls must conform to the requirements in Section 07 84 00 FIRESTOPPING.

#### 3.1.4.2 Flashing Requirements

Pipes passing through roof must be installed through a 4.9 kg per square meter copper flashing, each within an integral skirt or flange. Flashing must be suitably formed, and the skirt or flange must extend not less than 200 mm from the pipe and must be set over the roof or floor membrane in a solid coating of bituminous cement. The flashing must extend up the pipe a minimum of 250 mm. For cleanouts, the flashing must be turned down into the hub and caulked after placing the ferrule. Pipes passing through pitched roofs must be flashed, using lead or copper flashing, with an adjustable integral flange of adequate size to extend not less than 200 mm from the pipe in all directions and lapped into the roofing to provide a watertight seal. The annular space between the flashing and the bare pipe or between the flashing and the metal-jacket-covered insulation must be sealed as indicated. Flashing for dry vents must be turned down into the pipe to form a waterproof joint. Pipes, up to and including 250 mm in diameter, passing through roof or floor waterproofing membrane may be installed through a cast-iron sleeve with caulking recess, anchor lugs, flashing-clamp device, and pressure ring with brass bolts. Flashing shield must be fitted into the sleeve clamping device. Pipes passing through wall waterproofing membrane must be sleeved as described above. A waterproofing clamping flange must be installed.

#### 3.1.4.3 Waterproofing

Waterproofing at floor-mounted water closets must be accomplished by forming a flashing guard from soft-tempered sheet copper. The center of the sheet must be perforated and turned down approximately 40 mm to fit between the outside diameter of the drainpipe and the inside diameter of the cast-iron or steel pipe sleeve. The turned-down portion of the flashing guard must be embedded in sealant to a depth of approximately 40 mm; then the sealant must be finished off flush to floor level between the flashing guard and drainpipe. The flashing guard of sheet copper must extend not less than 200 mm from the drainpipe and must be lapped between the floor membrane in a solid coating of bituminous cement. If cast-iron water closet floor flanges are used, the space between the pipe sleeve and drainpipe must be sealed with sealant and the flashing guard must be upturned approximately 40 mm to fit the outside diameter of the drainpipe and the inside diameter of the water closet floor flange. The upturned portion of the sheet fitted into the floor flange must be sealed.

#### 3.1.4.4 Optional Counterflashing

Instead of turning the flashing down into a dry vent pipe, or caulking and sealing the annular space between the pipe and flashing or metal-jacket-covered insulation and flashing, counterflashing may be accomplished by utilizing the following:

- a. A standard roof coupling for threaded pipe up to 150 mm in diameter.
- b. A tack-welded or banded-metal rain shield around the pipe.

#### 3.1.4.5 Pipe Penetrations of Slab on Grade Floors

Where pipes, fixture drains, floor drains, cleanouts or similar items penetrate slab on grade floors, except at penetrations of floors with



waterproofing membrane as specified in paragraphs FLASHING REQUIREMENTS and WATERPROOFING, a groove 6 to 13 mm wide by 6 to 10 mm deep must be formed around the pipe, fitting or drain. The groove must be filled with a sealant as specified in Section 07 92 00 JOINT SEALANTS.

#### 3.1.4.6 Pipe Penetrations

Provide sealants for all pipe penetrations. All pipe penetrations must be sealed to prevent infiltration of air, insects, and vermin.

#### 3.1.5 Fire Seal

Where pipes pass through fire walls, fire-partitions, fire-rated pipe chase walls or floors above grade, a fire seal must be provided as specified in Section 07 84 00 FIRESTOPPING.

#### 3.1.6 Supports

##### 3.1.6.1 General

Hangers used to support piping 50 mm and larger must be fabricated to permit adequate adjustment after erection while still supporting the load. Pipe guides and anchors must be installed to keep pipes in accurate alignment, to direct the expansion movement, and to prevent buckling, swaying, and undue strain. Piping subjected to vertical movement when operating temperatures exceed ambient temperatures must be supported by variable spring hangers and supports or by constant support hangers. In the support of multiple pipe runs on a common base member, a clip or clamp must be used where each pipe crosses the base support member. Spacing of the base support members must not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run. Threaded sections of rods must not be formed or bent.

##### 3.1.6.2 Pipe Supports and Structural Bracing, Seismic Requirements

Piping and attached valves must be supported and braced to resist seismic loads as specified in Section 22 05 48.00 20 MECHANICAL SOUND, VIBRATION, AND SEISMIC CONTROL. Structural steel required for reinforcement to properly support piping, headers, and equipment, but not shown, must be provided. Material used for supports must be as specified in Section 05 12 00 STRUCTURAL STEEL.

##### 3.1.6.3 Pipe Hangers, Inserts, and Supports

Installation of pipe hangers, inserts and supports must conform to CSA Z259 /CSA W47.1 except as modified herein.

- a. Types 5, 12, and 26 must not be used.
- b. Type 3 must not be used on insulated pipe.
- c. Type 18 inserts must be secured to concrete forms before concrete is placed. Continuous inserts which allow more adjustment may be used if they otherwise meet the requirements for type 18 inserts.
- d. Type 19 and 23 C-clamps must be torqued per MSS SP-58 and must have both locknuts and retaining devices furnished by the manufacturer. Field-fabricated C-clamp bodies or retaining devices are not acceptable.

- e. Type 20 attachments used on angles and channels must be furnished with an added malleable-iron heel plate or adapter.
- f. Type 24 may be used only on trapeze hanger systems or on fabricated frames.
- g. Type 39 saddles must be used on insulated pipe 100 mm and larger when the temperature of the medium is 15 degrees C or higher. Type 39 saddles must be welded to the pipe.
- h. Type 40 shields must:
  - (1) Be used on insulated pipe less than 100 mm.
  - (2) Be used on insulated pipe 100 mm and larger when the temperature of the medium is 15 degrees C or less.
  - (3) Have a high density insert for all pipe sizes. High density inserts must have a density of 128 kg per cubic meter or greater.
- i. Horizontal pipe supports must be spaced as specified in CSA Z259/ CSA W47.1 and a support must be installed not over 300 mm from the pipe fitting joint at each change in direction of the piping. Pipe supports must be spaced not over 1.5 m apart at valves. Operating temperatures in determining hanger spacing for PVC or CPVC pipe must be 49 degrees C for PVC and 82 degrees C for CPVC. Horizontal pipe runs must include allowances for expansion and contraction.
- j. Vertical pipe must be supported at each floor, except at slab-on-grade, at intervals of not more than 4.5 m nor more than 2 m from end of risers, and at vent terminations. Vertical pipe risers must include allowances for expansion and contraction.
- k. Type 35 guides using steel, reinforced polytetrafluoroethylene (PTFE) or graphite slides must be provided to allow longitudinal pipe movement. Slide materials must be suitable for the system operating temperatures, atmospheric conditions, and bearing loads encountered. Lateral restraints must be provided as needed. Where steel slides do not require provisions for lateral restraint the following may be used:
  - (1) On pipe 100 mm and larger when the temperature of the medium is 15 degrees C or higher, a Type 39 saddle, welded to the pipe, may freely rest on a steel plate.
  - (2) On pipe less than 100 mm a Type 40 shield, attached to the pipe or insulation, may freely rest on a steel plate.
  - (3) On pipe 100 mm and larger carrying medium less than 15 degrees C a Type 40 shield, attached to the pipe or insulation, may freely rest on a steel plate.
- l. Pipe hangers on horizontal insulated pipe must be the size of the outside diameter of the insulation. The insulation must be continuous through the hanger on all pipe sizes and applications.
- m. Where there are high system temperatures and welding to piping is not desirable, the type 35 guide must include a pipe cradle, welded to the guide structure and strapped securely to the pipe. The pipe must be

separated from the slide material by at least 100 mm or by an amount adequate for the insulation, whichever is greater.

- n. Hangers and supports for plastic pipe must not compress, distort, cut or abrade the piping, and must allow free movement of pipe except where otherwise required in the control of expansion/contraction.

#### 3.1.6.4 Structural Attachments

Attachment to building structure concrete and masonry must be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors must be applied with a safety factor not less than 5. Supports must not be attached to metal decking. Supports must not be attached to the underside of concrete filled floor or concrete roof decks unless approved by the Contracting Officer. Masonry anchors for overhead applications must be constructed of ferrous materials only.

#### 3.1.7 Welded Installation

Plumbing pipe weldments must be as indicated. Changes in direction of piping must be made with welding fittings only; mitering or notching pipe to form elbows and tees or other similar type construction will not be permitted. Branch connection may be made with either welding tees or forged branch outlet fittings. Branch outlet fittings must be forged, flared for improvement of flow where attached to the run, and reinforced against external strains. Beveling, alignment, heat treatment, and inspection of weld must conform to ASME B31.1 CSA B51. Weld defects must be removed and repairs made to the weld, or the weld joints must be entirely removed and rewelded. After filler metal has been removed from its original package, it must be protected or stored so that its characteristics or welding properties are not affected. Electrodes that have been wetted or that have lost any of their coating must not be used.

#### 3.1.8 Pipe Cleanouts

Pipe cleanouts must be the same size as the pipe except that cleanout plugs larger than 100 mm will not be required. A cleanout installed in connection with cast-iron soil pipe must consist of a long-sweep 1/4 bend or one or two 1/8 bends extended to the place shown. An extra-heavy cast-brass or cast-iron ferrule with countersunk cast-brass head screw plug must be caulked into the hub of the fitting and must be flush with the floor. Cleanouts in connection with other pipe, where indicated, must be T-pattern, 90-degree branch drainage fittings with cast-brass screw plugs, except plastic plugs must be installed in plastic pipe. Plugs must be the same size as the pipe up to and including 100 mm. Cleanout tee branches with screw plug must be installed at the foot of soil and waste stacks, at the foot of interior downspouts, on each connection to building storm drain where interior downspouts are indicated, and on each building drain outside the building. Cleanout tee branches may be omitted on stacks in single story buildings with slab-on-grade construction or where less than 450 mm of crawl space is provided under the floor. Cleanouts on pipe concealed in partitions must be provided with chromium plated bronze, nickel bronze, nickel brass or stainless steel flush type access cover plates. Round access covers must be provided and secured to plugs with securing screw. Square access covers may be provided with matching frames, anchoring lugs and cover screws. Cleanouts in finished walls must have access covers and frames installed flush with the finished wall. Cleanouts installed in finished floors subject to foot traffic must be provided with a chrome-plated cast brass, nickel brass, or nickel bronze

cover secured to the plug or cover frame and set flush with the finished floor. Heads of fastening screws must not project above the cover surface. Where cleanouts are provided with adjustable heads, the heads must be plastic.

### 3.2 WATER HEATERS AND HOT WATER STORAGE TANKS

#### 3.2.1 Relief Valves

No valves must be installed between a relief valve and its water heater or storage tank. The P&T relief valve must be installed where the valve actuator comes in contact with the hottest water in the heater. Whenever possible, the relief valve must be installed directly in a tapping in the tank or heater; otherwise, the P&T valve must be installed in the hot-water outlet piping. A vacuum relief valve must be provided on the cold water supply line to the hot-water storage tank or water heater and mounted above and within 150 mm above the top of the tank or water heater.

#### 3.2.2 Heat Traps

Piping to and from each water heater and hot water storage tank must be routed horizontally and downward a minimum of 610 mm before turning in an upward direction.

#### 3.2.3 Connections to Water Heaters

Connections of metallic pipe to water heaters must be made with dielectric unions or flanges. Use of dielectric nipples is prohibited unless internally lined and specifically designed to limit current flow between dissimilar metals.

#### 3.2.4 Expansion Tank

A pre-charged expansion tank must be installed on the cold water supply between the water heater inlet and the cold water supply shut-off valve. The Contractor must adjust the expansion tank air pressure, as recommended by the tank manufacturer, to match incoming water pressure.

#### 3.2.5 Direct Fired and Domestic Water Heaters

Notify the Contracting Officer when any direct fired domestic water heater over 117,124.2 Watts is operational and ready to be inspected and certified.

### 3.3 FIXTURES AND FIXTURE TRIMMINGS

Polished chromium-plated pipe, valves, and fittings must be provided where exposed to view. Angle stops, straight stops, stops integral with the faucets, or concealed type of lock-shield, and loose-key pattern stops for supplies with threaded, sweat or solvent weld inlets must be furnished and installed with fixtures. Where connections between copper tubing and faucets are made by rubber compression fittings, a beading tool must be used to mechanically deform the tubing above the compression fitting. Exposed traps and supply pipes for fixtures and equipment must be connected to the rough piping systems at the wall, unless otherwise specified under the item. Floor and wall escutcheons must be as specified. Drain lines and hot water lines of fixtures for handicapped personnel must be insulated and do not require polished chrome finish. Plumbing fixtures and accessories must be installed within the space shown.

### 3.3.1 Fixture Connections

Where space limitations prohibit standard fittings in conjunction with the cast-iron floor flange, special short-radius fittings must be provided. Connections between earthenware fixtures and flanges on soil pipe must be made gastight and watertight with a closet-setting compound or neoprene gasket and seal. Use of natural rubber gaskets or putty will not be permitted. Fixtures with outlet flanges must be set the proper distance from floor or wall to make a first-class joint with the closet-setting compound or gasket and fixture used.

### 3.3.2 Flushometer Valves

Flushometer valves must be secured to prevent movement by anchoring the long finished top spud connecting tube to wall adjacent to valve with approved metal bracket. Flushometer valves for water closets must be installed 1 m above the floor, except at water closets intended for use by the physically handicapped where flushometer valves must be mounted at approximately 760 mm above the floor and arranged to avoid interference with grab bars. In addition, for water closets intended for handicap use, the flush valve handle must be installed on the wide side of the enclosure.

### 3.3.3 Height of Fixture Rims Above Floor

Lavatories must be mounted with rim 775 mm above finished floor. Wall-hung drinking fountains and water coolers must be installed with rim 1020 mm above floor. Wall-hung service sinks must be mounted with rim 700 mm above the floor. Installation of fixtures for use by the physically handicapped must be in accordance with ICC A117.1 CSA B651.

### 3.3.4 Shower Bath Outfits

The area around the water supply piping to the mixing valves and behind the escutcheon plate must be made watertight by caulking or gasketing.

### 3.3.5 Fixture Supports

Fixture supports for off-the-floor lavatories, urinals, water closets, and other fixtures of similar size, design, and use, must be of the chair-carrier type. The carrier must provide the necessary means of mounting the fixture, with a foot or feet to anchor the assembly to the floor slab. Adjustability must be provided to locate the fixture at the desired height and in proper relation to the wall. Support plates, in lieu of chair carrier, must be fastened to the wall structure only where it is not possible to anchor a floor-mounted chair carrier to the floor slab.

#### 3.3.5.1 Support for Solid Masonry Construction

Chair carrier must be anchored to the floor slab. Where a floor-anchored chair carrier cannot be used, a suitable wall plate must be imbedded in the masonry wall.

#### 3.3.5.2 Support for Concrete-Masonry Wall Construction

Chair carrier must be anchored to floor slab. Where a floor-anchored chair carrier cannot be used, a suitable wall plate must be fastened to the concrete wall using through bolts and a back-up plate.

### 3.3.5.3 Support for Steel Stud Frame Partitions

Chair carrier must be used. The anchor feet and tubular uprights must be of the heavy duty design; and feet (bases) must be steel and welded to a square or rectangular steel tube upright. Wall plates, in lieu of floor-anchored chair carriers, must be used only if adjoining steel partition studs are suitably reinforced to support a wall plate bolted to these studs.

### 3.3.5.4 Support for Wood Stud Construction

Where floor is a concrete slab, a floor-anchored chair carrier must be used. Where entire construction is wood, wood crosspieces must be installed. Fixture hanger plates, supports, brackets, or mounting lugs must be fastened with not less than No. 10 wood screws, 6 mm thick minimum steel hanger, or toggle bolts with nut. The wood crosspieces must extend the full width of the fixture and must be securely supported.

### 3.3.6 Backflow Prevention Devices

Plumbing fixtures, equipment, and pipe connections must not cross connect or interconnect between a potable water supply and any source of nonpotable water. Backflow preventers must be installed where indicated and in accordance with ICC IPC at all other locations necessary to preclude a cross-connect or interconnect between a potable water supply and any nonpotable substance. In addition backflow preventers must be installed at all locations where the potable water outlet is below the flood level of the equipment, or where the potable water outlet will be located below the level of the nonpotable substance. Backflow preventers must be located so that no part of the device will be submerged. Backflow preventers must be of sufficient size to allow unrestricted flow of water to the equipment, and preclude the backflow of any nonpotable substance into the potable water system. Bypass piping must not be provided around backflow preventers. Access must be provided for maintenance and testing. Each device must be a standard commercial unit.

### 3.3.7 Access Panels

Access panels must be provided for concealed valves and controls, or any item requiring inspection or maintenance. Access panels must be of sufficient size and located so that the concealed items may be serviced, maintained, or replaced. Access panels must be as specified in Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

### 3.3.8 Traps

Each trap must be placed as near the fixture as possible, and no fixture must be double-trapped. Traps installed on cast-iron soil pipe must be cast iron. Traps installed on steel pipe or copper tubing must be recess-drainage pattern, or brass-tube type. Traps installed on plastic pipe may be plastic conforming to ASTM D3311. Traps for acid-resisting waste must be of the same material as the pipe.

### 3.3.9 Shower Pans

Before installing shower pan, subfloor must be free of projections such as nail heads or rough edges of aggregate. Drain must be a bolt-down, clamping-ring type with weepholes, installed so the lip of the subdrain is

flush with subfloor.

#### 3.3.9.1 General

The floor of each individual shower, the shower-area portion of combination shower and drying room, and the entire shower and drying room where the two are not separated by curb or partition, must be made watertight with a shower pan fabricated in place. The shower pan material must be cut to size and shape of the area indicated, in one piece to the maximum extent practicable, allowing a minimum of 150 mm for turnup on walls or partitions, and must be folded over the curb with an approximate return of 1/4 of curb height. The upstands must be placed behind any wall or partition finish. Subflooring must be smooth and clean, with nailheads driven flush with surface, and must be sloped to drain. Shower pans must be clamped to drains with the drain clamping ring.

#### 3.3.9.2 Metal Shower Pans

When a shower pan of required size cannot be furnished in one piece, metal pieces must be joined with a flintlock seam and soldered or burned. The corners must be folded, not cut, and the corner seam must be soldered or burned. Pans, including upstands, must be coated on all surfaces with one brush coat of asphalt. Asphalt must be applied evenly at not less than 1 liter per square meter. A layer of felt covered with building paper must be placed between shower pans and wood floors. The joining surfaces of metal pan and drain must be given a brush coat of asphalt after the pan is connected to the drain.

#### 3.3.9.3 Plasticized Chlorinated Polyethylene Shower Pans

Corners of plasticized chlorinated polyethylene shower pans must be folded against the upstand by making a pig-ear fold. Hot-air gun or heat lamp must be used in making corner folds. Each pig-ear corner fold must be nailed or stapled 12 mm from the upper edge to hold it in place. Nails must be galvanized large-head roofing nails. On metal framing or studs, approved duct tape must be used to secure pig-ear fold and membrane. Where no backing is provided between the studs, the membrane slack must be taken up by pleating and stapling or nailing to studding 12 mm from upper edge. To adhere the membrane to vertical surfaces, the back of the membrane and the surface to which it will be applied must be coated with adhesive that becomes dry to the touch in 5 to 10 minutes, after which the membrane must be pressed into place. Surfaces to be solvent-welded must be clean. Surfaces to be joined with xylene must be initially sprayed and vigorously cleaned with a cotton cloth, followed by final coating of xylene and the joining of the surfaces by roller or equivalent means. If ambient or membrane temperatures are below 4 degrees C the membrane and the joint must be heated prior to application of xylene. Heat may be applied with hot-air gun or heat lamp, taking precautions not to scorch the membrane. Adequate ventilation and wearing of gloves are required when working with xylene. Membrane must be pressed into position on the drain body, and must be cut and fit to match so that membrane can be properly clamped and an effective gasket-type seal provided. On wood subflooring, two layers of 0.73 kg per square meter dry felt must be installed prior to installation of shower pan to ensure a smooth surface for installation.

#### 3.3.9.4 Nonplasticized Polyvinyl Chloride (PVC) Shower Pans

Nonplasticized PVC must be turned up behind walls or wall surfaces a

distance of not less than 150 mm in room areas and 75 mm above curb level in curbed spaces with sufficient material to fold over and fasten to outside face of curb. Corners must be pig-ear type and folded between pan and studs. Only top 25 mm of upstand must be nailed to hold in place. Nails must be galvanized large-head roofing type. Approved duct tape must be used on metal framing or studs to secure pig-ear fold and membrane. Where no backing is provided between studs, the membrane slack must be taken up by pleating and stapling or nailing to studding at top inch of upstand. To adhere the membrane to vertical surfaces, the back of the membrane and the surface to which it is to be applied must be coated with adhesive that becomes dry to the touch in 5 to 10 minutes, after which the membrane must be pressed into place. Trim for drain must be exactly the size of drain opening. Bolt holes must be pierced to accommodate bolts with a tight fit. Adhesive must be used between pan and subdrain. Clamping ring must be bolted firmly. A small amount of gravel or porous materials must be placed at weepholes so that holes remain clear when setting bed is poured. Membrane must be solvent welded with PVC solvent cement. Surfaces to be solvent welded must be clean (free of grease and grime). Sheets must be laid on a flat surface with an overlap of about 50 mm. Top edge must be folded back and surface primed with a PVC primer. PVC cement must be applied and surfaces immediately placed together, while still wet. Joint must be lightly rolled with a paint roller, then as the joint sets must be rolled firmly but not so hard as to distort the material. In long lengths, about 600 or 900 mm at a time must be welded. On wood subflooring, two layers of 0.73 kg per square meter felt must be installed prior to installation of shower pan to ensure a smooth surface installation.

### 3.4 VIBRATION-ABSORBING FEATURES

Mechanical equipment, including compressors and pumps, must be isolated from the building structure by approved vibration-absorbing features, unless otherwise shown. Submit details of vibration-absorbing features, including arrangement, foundation plan, dimensions and specifications. Each foundation must include an adequate number of standard isolation units. Each unit must consist of machine and floor or foundation fastening, together with intermediate isolation material, and must be a standard product with printed load rating. Piping connected to mechanical equipment must be provided with flexible connectors. Isolation unit installation must limit vibration to 10 percent of the lowest equipment rpm.

### 3.5 WATER METER REMOTE READOUT REGISTER

The remote readout register must be mounted at the location indicated or as directed by the Contracting Officer.

### 3.6 IDENTIFICATION SYSTEMS

#### 3.6.1 Identification Tags

Identification tags made of brass, engraved laminated plastic, or engraved anodized aluminum, indicating service and valve number must be installed on valves, except those valves installed on supplies at plumbing fixtures. Tags must be 35 mm minimum diameter, and marking must be stamped or engraved. Indentations must be black, for reading clarity. Tags must be attached to valves with No. 12 AWG, copper wire, chrome-plated beaded chain, or plastic straps designed for that purpose.



### 3.6.2 Pipe Color Code Marking

Color code marking of piping must be as specified in Section 09 90 00 PAINTS AND COATINGS.

### 3.6.3 Color Coding Scheme for Locating Hidden Utility Components

Scheme must be provided in buildings having suspended grid ceilings. The color coding scheme must identify points of access for maintenance and operation of operable components which are not visible from the finished space and installed in the space directly above the suspended grid ceiling. The operable components must include valves, dampers, switches, linkages and thermostats. The color coding scheme must consist of a color code board and colored metal disks. Each colored metal disk must be approximately 12 mm in diameter and secured to removable ceiling panels with fasteners. The fasteners must be inserted into the ceiling panels so that the fasteners will be concealed from view. The fasteners must be manually removable without tools and must not separate from the ceiling panels when panels are dropped from ceiling height. Installation of colored metal disks must follow completion of the finished surface on which the disks are to be fastened. The color code board must have the approximate dimensions of 1 m width, 750 mm height, and 12 mm thickness. The board must be made of wood fiberboard and framed under glass or 1.6 mm transparent plastic cover. Unless otherwise directed, the color code symbols must be approximately 20 mm in diameter and the related lettering in 12 mm high capital letters. The color code board must be mounted and located in the mechanical or equipment room. The color code system must be as indicated below:

| Color | System | Item | Location |
|-------|--------|------|----------|
|       |        |      |          |

## 3.7 ESCUTCHEONS

Escutcheons must be provided at finished surfaces where bare or insulated piping, exposed to view, passes through floors, walls, or ceilings, except in boiler, utility, or equipment rooms. Escutcheons must be fastened securely to pipe or pipe covering and must be satin-finish, corrosion-resisting steel, polished chromium-plated zinc alloy, or polished chromium-plated copper alloy. Escutcheons must be either one-piece or split-pattern, held in place by internal spring tension or setscrew.

## 3.8 PAINTING

Painting of pipes, hangers, supports, and other iron work, either in concealed spaces or exposed spaces, is specified in Section 09 90 00 PAINTS AND COATINGS.

### 3.8.1 Painting of New Equipment

New equipment painting must be factory applied or shop applied, and must be as specified herein, and provided under each individual section.

#### 3.8.1.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject

to certification that the factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors must withstand 500 hours in a salt-spray fog test. Salt-spray fog test must be in accordance with ASTM B117, and for that test the acceptance criteria must be as follows: immediately after completion of the test, the paint must show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen must show no signs of rust creepage beyond 3 mm on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment must not be less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 50 degrees C, the factory painting system must be designed for the temperature service.

#### 3.8.1.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except metal surfaces subject to temperatures in excess of 50 degrees C must be cleaned to bare metal.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Color of finish coat must be aluminum or light gray.

- a. Temperatures Less Than 50 Degrees C: Immediately after cleaning, the metal surfaces subject to temperatures less than 50 degrees C must receive one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm, one coat of primer applied to a minimum dry film thickness of 0.0255 mm; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm per coat.
- b. Temperatures Between 50 and 205 Degrees C: Metal surfaces subject to temperatures between 50 and 205 degrees C must receive two coats of 205 degrees C heat-resisting enamel applied to a total minimum thickness of 0.05 mm.
- c. Temperatures Greater Than 205 Degrees C: Metal surfaces subject to temperatures greater than 205 degrees C must receive two coats of 315 degrees C heat-resisting paint applied to a total minimum dry film thickness of 0.05 mm.

### 3.9 TESTS, FLUSHING AND DISINFECTION

Submit test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, completion and testing of the installed system. Each test report must indicate the final position of controls.

#### 3.9.1 Plumbing System

The following tests must be performed on the plumbing system in accordance with ICC IPC, except that the drainage and vent system final test must include the smoke test. The Contractor has the option to perform a

peppermint test in lieu of the smoke test. If a peppermint test is chosen, the Contractor must submit a testing procedure and reasons for choosing this option in lieu of the smoke test to the Contracting Officer for approval.

- a. Drainage and Vent Systems Test. The final test must include a smoke test.
- b. Building Sewers Tests.
- c. Water Supply Systems Tests.

#### 3.9.1.1 Test of Backflow Prevention Assemblies

Certification of proper operation must be submitted and must be as accomplished in accordance with state regulations by an individual certified by the state to perform such tests. If no state requirement exists, the Contractor must have the manufacturer's representative test the device to ensure the unit is properly installed and performing as intended. The Contractor must provide written documentation of the tests performed and signed by the individual performing the tests. Backflow prevention assembly must be tested using gauges specifically designed for the testing of backflow prevention assemblies.

Backflow prevention assembly test gauges must be tested annually for accuracy in accordance with the requirements of State or local regulatory agencies. If there is no State or local regulatory agency requirements, gauges must be tested annually for accuracy in accordance with the requirements of University of Southern California's Foundation of Cross Connection Control and Hydraulic Research or the American Water Works Association Manual of Cross Connection (Manual M-14), or any other approved testing laboratory having equivalent capabilities for both laboratory and field evaluation of backflow prevention assembly test gauges. Report form for each assembly must include, as a minimum, the following:

|                        |                                       |
|------------------------|---------------------------------------|
| Data on Device         | Data on Testing Firm                  |
| Type of Assembly       | Name                                  |
| Manufacturer           | Address                               |
| Model Number           | Certified Tester                      |
| Serial Number          | Certified Tester No.                  |
| Size                   | Date of Test                          |
| Location               |                                       |
| Test Pressure Readings | Serial Number and Test Data of Gauges |

If the unit fails to meet specified requirements, the unit must be repaired and retested.

### 3.9.1.2 Shower Pans

After installation of the pan and finished floor, the drain must be temporarily plugged below the weep holes. The floor area must be flooded with water to a minimum depth of 25 mm for a period of 24 hours. Any drop in the water level during test, except for evaporation, will be reason for rejection, repair, and retest.

### 3.9.2 Defective Work

If inspection or test shows defects, such defective work or material must be replaced or repaired as necessary and inspection and tests must be repeated. Repairs to piping must be made with new materials. Caulking of screwed joints or holes will not be acceptable.

### 3.9.3 System Flushing

#### 3.9.3.1 During Flushing

Before operational tests or disinfection, potable water piping system must be flushed with potable water. Sufficient water must be used to produce a water velocity that is capable of entraining and removing debris in all portions of the piping system. This requires simultaneous operation of all fixtures on a common branch or main in order to produce a flushing velocity of approximately 1.2 meters per second through all portions of the piping system. In the event that this is impossible due to size of system, the Contracting Officer (or the designated representative) must specify the number of fixtures to be operated during flushing. Contractor must provide adequate personnel to monitor the flushing operation and to ensure that drain lines are unobstructed in order to prevent flooding of the facility. Contractor must be responsible for any flood damage resulting from flushing of the system. Flushing must be continued until entrained dirt and other foreign materials have been removed and until discharge water shows no discoloration. All faucets and drinking water fountains, to include any device considered as an end point device by NSF/ANSI 61, Section 9, must be flushed a minimum of 1 L per 24 hour period, ten times over a 14 day period.

#### 3.9.3.2 After Flushing

System must be drained at low points. Strainer screens must be removed, cleaned, and replaced. After flushing and cleaning, systems must be prepared for testing by immediately filling water piping with clean, fresh potable water. Any stoppage, discoloration, or other damage to the finish, furnishings, or parts of the building due to the Contractor's failure to properly clean the piping system must be repaired by the Contractor. When the system flushing is complete, the hot-water system must be adjusted for uniform circulation. Flushing devices and automatic control systems must be adjusted for proper operation according to manufacturer's instructions. Flow rates on fixtures must not exceed those stated in PART 2 of this Section. Unless more stringent local requirements exist, lead levels must not exceed limits established by 40 CFR 141.80 (c)(1). The water supply to the building must be tested separately to ensure that any lead contamination found during potable water system testing is due to work being performed inside the building.

### 3.9.4 Operational Test

Upon completion of flushing and prior to disinfection procedures, the

Contractor must subject the plumbing system to operating tests to demonstrate satisfactory installation, connections, adjustments, and functional and operational efficiency. Such operating tests must cover a period of not less than 8 hours for each system and must include the following information in a report with conclusion as to the adequacy of the system:

- a. Time, date, and duration of test.
- b. Water pressures at the most remote and the highest fixtures.
- c. Operation of each fixture and fixture trim.
- d. Operation of each valve, hydrant, and faucet.
- e. Pump suction and discharge pressures.
- f. Temperature of each domestic hot-water supply.
- g. Operation of each floor and roof drain by flooding with water.
- h. Operation of each vacuum breaker and backflow preventer.

### 3.9.5 Disinfection

After all system components are provided and operational tests are complete, the entire domestic hot- and cold-water distribution system must be disinfected. Before introducing disinfecting chlorination material, entire system must be flushed with potable water until any entrained dirt and other foreign materials have been removed.

Water chlorination procedure must be in accordance with AWWA C651 CSA B128.1/B128.2 and AWWA C652 CSA B128.1 as modified and supplemented by this specification. The chlorinating material must be hypochlorites or liquid chlorine. The chlorinating material must be fed into the water piping system at a constant rate at a concentration of at least 50 parts per million (ppm). Feed a properly adjusted hypochlorite solution injected into the system with a hypochlorinator, or inject liquid chlorine into the system through a solution-feed chlorinator and booster pump until the entire system is completely filled.

#### 3.9.5.1 Water Chlorination Monitoring

Test the chlorine residual level in the water at 6 hour intervals for a continuous period of 24 hours. If at the end of a 6 hour interval, the chlorine residual has dropped to less than 25 ppm, flush the piping including tanks with potable water, and repeat the above chlorination procedures. During the chlorination period, each valve and faucet must be opened and closed several times. After the second 24 hour period, verify that no less than 25 ppm chlorine residual remains in the treated system. The 24 hour chlorination procedure must be repeated until no less than 25 ppm chlorine residual remains in the treated system.

#### 3.9.5.2 Water Chlorination Flushing

Upon the specified verification, the system including tanks must then be flushed with potable water until the residual chlorine level is reduced to

less than one part per million. During the flushing period, each valve and faucet must be opened and closed several times.

#### 3.9.5.3 Sample Testing

Take additional samples of water in disinfected containers, for bacterial examination, at locations specified by the Contracting Officer. Test these samples for total coliform organisms (coliform bacteria, fecal coliform, streptococcal, and other bacteria) in accordance with AWWA 10084 CSA B128.1. The testing method used must be EPA approved for drinking water systems and must comply with applicable local and state requirements.

#### 3.9.5.4 Disinfection Acceptance

Disinfection must be repeated until bacterial tests indicate the absence of coliform organisms (zero mean coliform density per 100 milliliters) in the samples for at least 2 full days. The system will not be accepted until satisfactory bacteriological results have been obtained.

### 3.10 POSTED INSTRUCTIONS

Framed instructions under glass or in laminated plastic, including wiring and control diagrams showing the complete layout of the entire system, must be posted where directed. Condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation, and procedures for safely starting and stopping the system must be prepared in typed form, framed as specified above for the wiring and control diagrams and posted beside the diagrams. The framed instructions must be posted before acceptance testing of the systems.

### 3.11 PERFORMANCE OF WATER HEATING EQUIPMENT

Standard rating condition terms are as follows:

DP = Draw pattern

EF = Energy factor, minimum overall efficiency.

ET = Minimum thermal efficiency with 21 degrees C delta T.

SL = Standby loss is maximum (Btu/h) based on a 21 degrees C temperature difference between stored water and ambient requirements.

UEF = Uniform energy factor

V = Rated volume in gallons

Q = Nameplate input rate in kW (Btu/h)

#### 3.11.1 Storage Water Heaters

##### 3.11.1.1 Electric

- b. Storage capacity more than 208 liters and less than or equal to 454 liters with a draw pattern (DP) of: very small must have a uniform energy factor (UEF) of 1.9236 -  $(0.0011 \times V)$ , low must have a uniform

energy factor (UEF) of 2.0440 -  $(0.0011 \times V)$ , medium must have a uniform energy factor (UEF) of 2.1171 -  $(0.0011 \times V)$ , high must have a uniform energy factor (UEF) of 2.418 -  $(0.0011 \times V)$  per 10 CFR 430.

### 3.11.2 Unfired Hot Water Storage

All volumes and inputs: must meet or exceed R-12.5.

### 3.12 TABLES

| TABLE I                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                 |           |           |           |           |           |           |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| PIPE AND FITTING MATERIALS FOR DRAINAGE, WASTE, VENT AND CONDENSATE DRAIN PIPING SYSTEMS                                                                                                                                                                                                                                                  |                                                                                                                                                 |           |           |           |           |           |           |           |
| It #                                                                                                                                                                                                                                                                                                                                      | Pipe and Fitting Materials                                                                                                                      | SERVICE A | SERVICE B | SERVICE C | SERVICE D | SERVICE E | SERVICE F | SERVICE G |
| 1                                                                                                                                                                                                                                                                                                                                         | Cast iron soil pipe and fittings, hub and spigot, CSA B70 with compression gaskets. Pipe and fittings shall be marked with the CISPI trademark. | X         | X         | X         | X         | X         |           |           |
| 2                                                                                                                                                                                                                                                                                                                                         | Cast iron soil pipe and fittings hubless, CSA B70. Pipe and fittings shall be marked with the CISPI trademark.                                  |           | X         | X         | X         | X         |           |           |
| 3                                                                                                                                                                                                                                                                                                                                         | Cast Iron pipe and fittings, CSA B70 piping                                                                                                     | X         | X         | X         | X         | X         | X         |           |
| SERVICE:<br><br>A - Underground Building Soil, Waste and Storm Drain<br>B - Aboveground Soil, Waste, Drain In Buildings<br>C - Underground Vent<br>D - Aboveground Vent<br>E - Interior Rainwater Conductors Aboveground<br>F - Corrosive Waste And Vent Above And Belowground<br>G - Condensate Drain Aboveground<br><br>* - Hard Temper |                                                                                                                                                 |           |           |           |           |           |           |           |

| TABLE II                                               |                                                                                                             |           |           |           |           |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|
| PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS |                                                                                                             |           |           |           |           |
| Item #                                                 | Pipe and Fitting Materials                                                                                  | SERVICE A | SERVICE B | SERVICE C | SERVICE D |
| 1                                                      | Seamless copper pipe, ASTM B42                                                                              | X         | X         |           | X         |
| 2                                                      | Seamless copper water tube, CSA B137.1                                                                      | X**       | X**       | X**       | X***      |
| 3                                                      | Wrought copper and bronze solder-joint pressure fittings, CSA B51, CSA B125.1 for use with Items 5, 7 and 8 | X         | X         | X         | X         |
| 4                                                      | Cast copper alloy solder-joint pressure fittings, CSA B51: CSA B125.1 CSA B125.3 for use with Item 8        | X         | X         | X         | X         |
| 5                                                      | Polyethylene (PE) plastic pipe, Schedules 40 and 80, based on outside diameter                              | X         |           |           | X         |
| 6                                                      | Polyethylene (PE) plastic pipe (SDR-PR), based on controlled outside diameter, CSA B137.1/ CSA B137.4       | X         |           |           | X         |
| 7                                                      | Polyethylene (PE) plastic pipe (SIDR-PR), based on controlled inside diameter, CSA B137.1/ CSA B137.4       | X         |           |           | X         |
| 8                                                      | Butt fusion polyethylene (PE) plastic pipe fittings, CSA B137.4 for use with Items 14, 15, and 16           | X         |           |           | X         |
| 9                                                      | Polyethylene (PE) plastic tubing, CSA B137.1/CSA B137.4                                                     | X         |           |           | X         |
| 10                                                     | Chlorinated polyvinyl chloride (CPVC) plastic hot and cold water distribution system, CSA B137.6            | X         | X         |           | X         |
| 11                                                     | Chlorinated polyvinyl chloride (CPVC) plastic pipe, Schedule 40 and 80, CSA B137.6                          | X         | X         |           | X         |
| 12                                                     | Chlorinated polyvinyl chloride (CPVC) plastic pipe (SDR-PR) CSA B137.6                                      | X         | X         |           | X         |



| TABLE II                                               |                                                                                                                     |           |           |           |           |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|
| PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS |                                                                                                                     |           |           |           |           |
| Item #                                                 | Pipe and Fitting Materials                                                                                          | SERVICE A | SERVICE B | SERVICE C | SERVICE D |
| 13                                                     | Polyvinyl chloride (PVC) plastic pipe, Schedules 40, 80, and 120, CSA B137.3                                        | X         |           |           | X         |
| 14                                                     | Polyvinyl chloride (PVC) pressure-rated pipe (SDR Series), CSA B137.3                                               | X         |           |           | X         |
| 15                                                     | Polyvinyl chloride (PVC) plastic pipe fittings, Schedule 40, CSA B137.3                                             | X         |           |           | X         |
| 16                                                     | Socket-type polyvinyl chloride (PVC) plastic pipe fittings, schedule 80, CSA B137.3 for use with Items 26 and 27    | X         |           |           | X         |
| 17                                                     | Joints for IPS PVC pipe using solvent cement, CSA B137.3                                                            | X         |           |           | X         |
| 18                                                     | Polypropylene (PP) plastic pipe and fittings; CSA B182.11                                                           | X         | X         |           | X         |
| 19                                                     | Fittings: brass or bronze; CSA B51: CSA B125.1, CSA B125.3, and CSA B51: CSA B125.1, CSA B125.3 and CAN/CGSB-37.2-M | X         | X         |           |           |
| 20                                                     | Crosslinked Polyethylene (PEX) Plastic Pipe CSA B137.5                                                              | X         | X         |           | X         |
| 21                                                     | Press Fittings                                                                                                      | X         | X         |           |           |

| TABLE II                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |           |           |           |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|
| PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS |                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |           |           |           |
| Item #                                                 | Pipe and Fitting Materials                                                                                                                                                                                                                                                                                                                                                                                                                    | SERVICE A | SERVICE B | SERVICE C | SERVICE D |
|                                                        | SERVICE:<br>A - Cold Water Service Aboveground<br>B - Hot and Cold Water Distribution 82 degrees C<br>Maximum Aboveground<br>C - Compressed Air Lubricated<br>D - Cold Water Service Belowground<br>Indicated types are minimum wall thicknesses.<br>** - Type L - Hard<br>*** - Type K - Hard temper with brazed joints only or type K-soft<br>temper without joints in or under floors<br>**** - In or under slab floors only brazed joints |           |           |           |           |

| TABLE III                                                                              |                                |                     |                       |                             |  |
|----------------------------------------------------------------------------------------|--------------------------------|---------------------|-----------------------|-----------------------------|--|
| STANDARD RATING CONDITIONS AND MINIMUM PERFORMANCE RATINGS FOR WATER HEATING EQUIPMENT |                                |                     |                       |                             |  |
| <u>FUEL</u>                                                                            | <u>STORAGE CAPACITY LITERS</u> | <u>INPUT RATING</u> | <u>TEST PROCEDURE</u> | <u>REQUIRED PERFORMANCE</u> |  |
| A. STORAGE WATER HEATERS                                                               |                                |                     |                       |                             |  |
|                                                                                        |                                |                     |                       |                             |  |
|                                                                                        |                                |                     |                       |                             |  |
|                                                                                        |                                |                     |                       |                             |  |
|                                                                                        |                                |                     |                       |                             |  |
|                                                                                        |                                |                     |                       |                             |  |

| TABLE III                                                                              |                                        |                         |                           |                             |                          |
|----------------------------------------------------------------------------------------|----------------------------------------|-------------------------|---------------------------|-----------------------------|--------------------------|
| STANDARD RATING CONDITIONS AND MINIMUM PERFORMANCE RATINGS FOR WATER HEATING EQUIPMENT |                                        |                         |                           |                             |                          |
| <u>FUEL</u>                                                                            | <u>STORAGE<br/>CAPACITY<br/>LITERS</u> | <u>INPUT<br/>RATING</u> | <u>TEST<br/>PROCEDURE</u> | <u>REQUIRED PERFORMANCE</u> |                          |
| Elect.                                                                                 | >208 L and ≤<br>454 L                  |                         | 10 CFR 430                | DP                          | UEF                      |
|                                                                                        |                                        |                         |                           | Very Small                  | 1.9236 -<br>(0.0011 x V) |
|                                                                                        |                                        |                         |                           | Low                         | 2.0440 -<br>(0.0011 x V) |
|                                                                                        |                                        |                         |                           | Medium                      | 2.1171 -<br>(0.0011 x V) |
|                                                                                        |                                        |                         |                           | High                        | 2.2418 -<br>(0.0011 x V) |
|                                                                                        |                                        |                         |                           |                             |                          |
|                                                                                        |                                        |                         |                           |                             |                          |
|                                                                                        |                                        |                         |                           |                             |                          |
|                                                                                        |                                        |                         |                           |                             |                          |
| B. Unfired Hot Water Storage, R = 2.2 minimum                                          |                                        |                         |                           |                             |                          |

-- End of Section --

## SECTION 22 05 48.00 20

## MECHANICAL SOUND, VIBRATION, AND SEISMIC CONTROL

04/06, CHG 1: 05/15

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## Canadian General Standards Board (CGSB)

CGSB 41-GP-25M (2012) Pipe, Polyethylene, for the  
Transport of Liquids

## CSA GROUP (CSA)

CSA G40.21 Grade 350W Class H  
CSA G164 Hot dip galvanizing of irregularly shaped  
articles  
CSA S16 Design of steel structures  
CAN/ULC-S102 Surface burning characteristics  
CSA W59 Welded steel construction

## AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 360 (2016) Specification for Structural Steel  
Buildings

## AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2025) Structural Welding Code - Steel

## ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon  
Structural Steel  
ASTM A123/A123M (2024) Standard Specification for Zinc  
(Hot-Dip Galvanized) Coatings on Iron and  
Steel Products  
ASTM A653/A653M (2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process  
ASTM C94/C94M (2025) Standard Specification for  
Ready-Mixed Concrete  
ASTM D471 (2016a) Standard Test Method for Rubber

## Property - Effect of Liquids

ASTM D2240 (2015; R 2021) Standard Test Method for Rubber Property - Durometer Hardness

ASTM E84 (2024) Standard Test Method for Surface Burning Characteristics of Building Materials

## SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1793 (2012) Architectural Sheet Metal Manual, 7th Edition

SMACNA 1966 (2020) HVAC Duct Construction Standards Metal and Flexible, 4th Edition

SMACNA 1981 (2020) Seismic Restraint Manual Guidelines for Mechanical Systems, 4th Edition

## 1.2 RELATED REQUIREMENTS

The provisions of Section 23 03 00.00 20 BASIC MECHANICAL MATERIALS AND METHODS apply to this section.

## 1.3 DEFINITIONS

## 1.3.1 Decibels dB

Measure of sound level. Decibels are referenced to either 20 uPa for sound pressure levels or one pW for sound power levels. dBA is the overall "A" weighted sound level.

## 1.3.2 Machinery

The vibration or noise producing equipment that must be isolated.

## 1.3.3 Manufacturer

The fabricator or supplier of vibration-isolation or seismic-protection materials and equipment. For mechanical equipment and machinery the term machinery manufacturer will be used.

## 1.3.4 Micropascal uPa

10 to the minus 6 power newtons per square meter.

## 1.3.5 Picowatt pW

10 to the minus 12 power watts.

## 1.4 SYSTEM DESCRIPTION

## 1.4.1 Spring Isolator Data

For each type and size of spring isolator, submit the spring outside diameter, deflection, operating spring height, unloaded spring height, solid spring height, the ratio of the outside diameter to the operating

spring height, the load to deflection ratio of the springs, and weight and sizes of structural steel members.

#### 1.4.2 Machinery Manufacturer's Sound Data

For each piece of indicated machinery to be vibration isolated, the calculated sound power test data or sound pressure test data as levels in dB in the eight octave bands between 63 and 8,000 Hz. Refer sound power levels to one pW and sound pressure levels to 20 uPa. Submit the overall "A" weighted scale sound pressure level in dB. Submit the standard test procedure used to obtain the sound power or pressure data for the applicable vibration isolation equipment size.

#### 1.4.3 Machinery

For each item of machinery, compare spring static deflections with the specified minimum static deflection, to show that the calculated spring static deflections are not less than the minimum static deflections specified. Rated spring static deflections are not acceptable in lieu of calculated spring static deflections.

#### 1.4.4 Machinery Over 136 Kilograms

For machinery items over 136 kg, provide calculations for shear, pull-up, primary overturning, and secondary overturning.

#### 1.4.5 Machinery Vibration Criteria

| TABLE 1A                                                     |                             |                |                 |                |                 |                |
|--------------------------------------------------------------|-----------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection       |                             |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported. |                             |                |                 |                |                 |                |
| Column Spacing                                               | Slab on earth and 0-9 meter |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                    | Type (Note (1))             | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Absorption Refrigerative Machines                            | SV-R                        | 25.40          | SV-R            | 44.45          | SV-R            | 69.85          |
| Centrifugal Chillers or Heat Pumps                           |                             |                |                 |                |                 |                |
| Hermetic Type                                                | SV-B                        | 44.45          | SV-B            | 63.50          | SV-B            | 88.90          |
| Open Type                                                    | SV-1                        | 44.45          | SV-I            | 63.50          | SV-I            | 88.90          |

| TABLE 1A                                                     |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection       |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported. |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| Column Spacing                                               | Slab on earth and 0-9 meter                                                                                                                                                                                                           |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                    | Type (Note (1))                                                                                                                                                                                                                       | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Reciprocating Air or Refrigeration Compressors               |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| 500 to 750 rpm                                               | S-R                                                                                                                                                                                                                                   | 44.45          | S-R             | 63.50          | S-R             | 88.90          |
| 751 rpm and up                                               | S-R                                                                                                                                                                                                                                   | 38.10          | S-R             | 63.50          | S-R             | 88.90          |
| Reciprocating Chillers or Heat Pumps                         |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| 500 to 750 rpm                                               | SV-R                                                                                                                                                                                                                                  | 44.45          | SV-R            | 63.50          | SV-R            | 88.90          |
| 751 rpm and up                                               | SV-R                                                                                                                                                                                                                                  | 38.10          | SV-R            | 63.50          | SV-R            | 88.90          |
| Packaged Boilers                                             | SV                                                                                                                                                                                                                                    | 25.40          | SV              | 63.50          | SV-R            | 88.90          |
| Closed Coupled Pumps                                         |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| Up to 5 1/2 kw                                               | S-I                                                                                                                                                                                                                                   | 25.40          | S-I             | 25.40          | S-I             | 25.40          |
| Over 5 1/2 kw                                                | S-I                                                                                                                                                                                                                                   | 38.10          | S-I             | 63.50          | S-I             | 63.50          |
| Base Mounted Pumps                                           |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| Up to 15 kw                                                  | S-I                                                                                                                                                                                                                                   | 38.10          | S-I             | 63.50          | S-I             | 63.50          |
| 15 to 56 kw                                                  | S-I                                                                                                                                                                                                                                   | 38.10          | S-I             | 63.50          | S-I             | 88.90          |
| Over 56 kw                                                   | S-I                                                                                                                                                                                                                                   | 63.50          | S-I             | 88.90          | S-I             | 88.90          |
| Cooling Towers and Evaporative Condensers                    | SV with deflections specified for centrifugal blowers when springs are supported on beams. Use deflection listed for column supported floors with up to 9 meters column spacing when springs are located on columns or bearing walls. |                |                 |                |                 |                |

| TABLE 1A                                                                |                             |                                                     |                 |                |                 |                |
|-------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                  |                             |                                                     |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.            |                             |                                                     |                 |                |                 |                |
| Column Spacing                                                          | Slab on earth and 0-9 meter |                                                     | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                               | Type (Note (1))             | MSD (Note (1))                                      | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Factory Assembled Air Handling Equipment AH, AC and HV Units (Note (2)) |                             |                                                     |                 |                |                 |                |
| Suspended Units                                                         |                             |                                                     |                 |                |                 |                |
| Up to 3 3/4 kw                                                          | H                           | 25.40                                               | H               | 25.40          | H               | 25.40          |
| Over 3 3/4 kw                                                           |                             |                                                     |                 |                |                 |                |
| Up to 400 rpm                                                           | H                           | 44.45                                               | H               | 44.45          | H               | 44.45          |
| Over 401 rpm                                                            | H                           | 25.40                                               | H               | 38.10          | H               | 63.50          |
| Floor Mounted Units                                                     |                             |                                                     |                 |                |                 |                |
| Up to 3 3/4 kw                                                          | S                           | 25.40                                               | S               | 25.40          | S               | 25.40          |
| Over 3 3/4 kw                                                           |                             |                                                     |                 |                |                 |                |
| Up to 400 rpm                                                           | S-R                         | 44.45                                               | S-R             | 44.45          | S-R             | 63.50          |
| Over 401 rpm                                                            | S-R                         | 25.40                                               | S-R             | 38.10          | S-R             | 63.50          |
| Centrifugal Blowers                                                     |                             |                                                     |                 |                |                 |                |
| 175 - 224 rpm                                                           | S-B                         | 120.65                                              | S-B             | 120.65         | S-B             | 120.65         |
| 225 - 299 rpm                                                           | S-B                         | 95.25                                               | S-B             | 120.65         | S-B             | 120.65         |
| 300 - 374 rpm                                                           | S-B                         | 69.85                                               | S-B             | 114.30         | S-B             | 120.65         |
| 375 - 499 rpm                                                           | S-B                         | 63.50                                               | S-B             | 88.90          | S-B             | 114.30         |
| Over 500 rpm                                                            | S-B                         | 44.45                                               | S-B             | 63.50          | S-B             | 88.90          |
| Tubular Centrifugal and Axial Fans (Note (2))                           |                             |                                                     |                 |                |                 |                |
| Suspended                                                               |                             | H with deflection specified for centrifugal blowers |                 |                |                 |                |



| TABLE 1A                                                                                                             |                             |                                                                                       |                 |                |                 |                |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                                               |                             |                                                                                       |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.                                                         |                             |                                                                                       |                 |                |                 |                |
| Column Spacing                                                                                                       | Slab on earth and 0-9 meter |                                                                                       | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                                                                            | Type (Note (1))             | MSD (Note (1))                                                                        | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Floor Mounted Arrangements 1 & 9                                                                                     |                             | S-B with deflections specified for centrifugal blowers                                |                 |                |                 |                |
| Utility Fans (Note (2))                                                                                              |                             |                                                                                       |                 |                |                 |                |
| Suspended                                                                                                            |                             | H with deflections specified for centrifugal blowers but not to exceed 69.85 mm       |                 |                |                 |                |
| Floor-Mounted                                                                                                        |                             | S-R with deflections not specified for centrifugal blowers but not to exceed 69.85 mm |                 |                |                 |                |
| High Pressure Fans (Over 1494 Pa Static Pressure) and Other Machineries Producing Thrust (Note (2))                  |                             | HR recommended for minimizing undesirable thrust effects                              |                 |                |                 |                |
| Internal Combustion Engines and Engine Driven Equip                                                                  |                             |                                                                                       |                 |                |                 |                |
| 750 rpm and over                                                                                                     | S                           | 38.10                                                                                 | S               | 63.50          | S               | 88.90          |
| Dimmer Banks and Transformers                                                                                        |                             |                                                                                       |                 |                |                 |                |
| Up to 454 kg                                                                                                         | NM                          | 8.89                                                                                  | NM              | 8.89           | NM              | 88.90          |
| Over 454 kg                                                                                                          | SV                          | 25.40                                                                                 | SV              | 25.40          | SV              | 25.40          |
| NOTES:                                                                                                               |                             |                                                                                       |                 |                |                 |                |
| (1) Equipment Vibration Isolation Schedule Designations (Hyphenated designations are combinations of the following:) |                             |                                                                                       |                 |                |                 |                |
| B - Welded structural steel bases.                                                                                   |                             |                                                                                       |                 |                |                 |                |
| H - Spring isolators (suspended equipment and piping). Where required, provide with adjustable preloading devices.   |                             |                                                                                       |                 |                |                 |                |
| HR - Thrust restraints                                                                                               |                             |                                                                                       |                 |                |                 |                |
| I - Concrete inertia bases with steel forms.                                                                         |                             |                                                                                       |                 |                |                 |                |

| TABLE 1A                                                                                                                                                                                                                                                   |                             |                |                 |                |                 |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                                                                                                                                                                                     |                             |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.                                                                                                                                                                                               |                             |                |                 |                |                 |                |
| Column Spacing                                                                                                                                                                                                                                             | Slab on earth and 0-9 meter |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                                                                                                                                                                                                                  | Type (Note (1))             | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| NM - Neoprene mounts.                                                                                                                                                                                                                                      |                             |                |                 |                |                 |                |
| NP - Neoprene pads.                                                                                                                                                                                                                                        |                             |                |                 |                |                 |                |
| R - Structural steel rail for equipment mounts.                                                                                                                                                                                                            |                             |                |                 |                |                 |                |
| S - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                               |                             |                |                 |                |                 |                |
| SV - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                              |                             |                |                 |                |                 |                |
| SX - Freestanding spring isolators with adjustable cushioned vertical stops and cushioned horizontal stops (floor-mounted equipment. Protected spring isolators SX may be substituted wherever S or SV is specified and must meet all requirements.        |                             |                |                 |                |                 |                |
| (2) Fans                                                                                                                                                                                                                                                   |                             |                |                 |                |                 |                |
| a. When fan motors are 56 kW or larger, use the deflection requirements for the next wider column spacing. Except for building slab on grade a minimum of 63.50 mm should be used unless larger deflections are specified in the centrifugal blower table. |                             |                |                 |                |                 |                |
| b. Provide sway brace isolators for tubular centrifugal and axial fans when the fan pressure exceeds 996 Pa.                                                                                                                                               |                             |                |                 |                |                 |                |
| c. Provide inertia bases for all fans in lieu of structural steel bases or rails specified above when the fan pressure exceeds 996 Pa.                                                                                                                     |                             |                |                 |                |                 |                |

| TABLE 1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                             |                |                 |                |                 |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |                |                 |                |                 |                |
| Column Spacing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Slab on earth and 0-9 meter |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Type (Note (1))             | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                             |                |                 |                |                 |                |
| <p>d. With attaching brackets, suspension spring isolators bridge between the structure and the thrust-producing machinery such as high-pressure fan. Both types H and HR normally provide reaction in tension, while types S, SV, and SX normally provide reaction in compression. Thrust restraints are low-cost and effective components available from manufacturers. Use thrust restraints to eliminate the need for or reduce the magnitude of inertia mass when the mass is only used to reduce the displacement effects of the thrust.</p> |                             |                |                 |                |                 |                |

| TABLE 1B                                                        |                 |      |
|-----------------------------------------------------------------|-----------------|------|
| Class II Vibration Isolator Types and Minimum Static Deflection |                 |      |
| (MSD, mm) for basements below grade and floor slabs on earth    |                 |      |
| Equipment                                                       | Type (Note (1)) | MSD  |
| Absorption Refrigeration Machines                               | NP              | 6.35 |
|                                                                 | NM              | 8.89 |
| Centrifugal Chillers or Heat Pumps                              |                 |      |
| Hermetic Type                                                   | NP              | 6.35 |
|                                                                 | NM              | 8.89 |
| Open Type                                                       | NM-I            | 8.89 |

| TABLE 1B                                                                |                 |       |
|-------------------------------------------------------------------------|-----------------|-------|
| Class II Vibration Isolator Types and Minimum Static Deflection         |                 |       |
| (MSD, mm) for basements below grade and floor slabs on earth            |                 |       |
| Equipment                                                               | Type (Note (1)) | MSD   |
| Reciprocating Air or Refrigeration Compressors                          |                 |       |
| 500 to 750 rpm                                                          | S               | 25.40 |
| 751 rpm and up                                                          | S               | 25.40 |
| Reciprocating Chillers or Heat Pumps                                    |                 |       |
| 500 to 750 rpm                                                          | SV              | 25.40 |
| 751 rpm and up                                                          | SV              | 25.40 |
| Packaged Boilers                                                        | NP              | 6.35  |
|                                                                         | NM              | 8.89  |
| Pumps                                                                   |                 |       |
| Closed Coupled                                                          | NP              | 6.35  |
| Up to 5 1/2 kW                                                          | NM              | 8.89  |
| Over 5 1/2 kW                                                           | S-I             | 25.40 |
| Base Mounted                                                            |                 |       |
| Up to 15 kW                                                             | S-I             | 25.40 |
| 15 to 56 kW                                                             | S-I             | 25.40 |
| Over 56 kW                                                              | S-I             | 25.40 |
| Cooling Towers and<br>Evaporative Condensers                            | NP              | 6.35  |
|                                                                         | NM              | 8.89  |
| Factory Assembled Air Handling Equipment AH, AC and HV Units (Note (2)) |                 |       |
| Suspended Units                                                         |                 |       |
| Up to 3 3/4 kW                                                          | H               | 25.40 |
| Over 3 3/4 kW                                                           |                 |       |
| Up to 400 rpm                                                           | H               | 44.45 |
| Over 401 rpm                                                            | H               | 25.40 |

| TABLE 1B                                                                                               |                                                                |       |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-------|
| Class II Vibration Isolator Types and Minimum Static Deflection                                        |                                                                |       |
| (MSD, mm) for basements below grade and floor slabs on earth                                           |                                                                |       |
| Equipment                                                                                              | Type (Note (1))                                                | MSD   |
| Floor Mounted Units                                                                                    |                                                                |       |
| Up to 3 3/4 kw                                                                                         | NP                                                             | 6.35  |
|                                                                                                        | NM                                                             | 8.89  |
| Over 3 3/4 kw                                                                                          |                                                                |       |
| Up to 400 rpm                                                                                          | NM                                                             | 8.89  |
| Over 401 rpm                                                                                           | NM                                                             | 8.89  |
| Centrifugal Blowers                                                                                    |                                                                |       |
| 175 - 224 rpm                                                                                          | NM-B                                                           | 8.89  |
| 225 - 299 rpm                                                                                          | NM-B                                                           | 8.89  |
| 300 - 374 rpm                                                                                          | NM-B                                                           | 8.89  |
| 375 - 499 rpm                                                                                          | NM-B                                                           | 8.89  |
| Over 500 rpm                                                                                           | NM-B                                                           | 8.89  |
| Tubular Centrifugal and Axial Fans (Note (2))                                                          |                                                                |       |
| Suspended                                                                                              | H with deflections specified for centrifugal blowers           |       |
| Floor Mounted Arrangements<br>1 & 9                                                                    | NM                                                             | 8.89  |
| Utility Fans (Note (2))                                                                                |                                                                |       |
| Suspended and centrifugal                                                                              | H with deflections specified for                               |       |
| Floor-Mounted                                                                                          | NM                                                             | 8.89  |
| High Pressure Fans (Over 1494 Pa Static Pressure) and<br>Other Machineries Producing Thrust (Note (2)) | HR recommended for<br>minimizing undesirable<br>thrust effects |       |
| Internal Combustion Engines and Engine Driven Equip                                                    |                                                                |       |
| 750 rpm and over                                                                                       | S                                                              | 25.40 |
| Dimmer Banks and Transformers                                                                          |                                                                |       |

| TABLE 1B                                                        |                 |       |
|-----------------------------------------------------------------|-----------------|-------|
| Class II Vibration Isolator Types and Minimum Static Deflection |                 |       |
| (MSD, mm) for basements below grade and floor slabs on earth    |                 |       |
| Equipment                                                       | Type (Note (1)) | MSD   |
| Up to 454 kg                                                    | NP              | 6.35  |
|                                                                 | NM              | 8.89  |
| Over 454 kg                                                     | SV              | 25.40 |
| NOTES: Note (1) and Note (2) are same as for TABLE 1A.          |                 |       |

Provide vibration isolators for mechanical and electrical machinery and associated piping and ductwork, to minimize transmission of vibrations and structure borne noise to the building structure or spaces or from the building structure to the machinery. Comply with the following vibration schedule.

#### 1.4.6 Machinery Airborne Sound Level Criteria

| TABLE 2A              |                                         |     |     |     |      |      |      |      |
|-----------------------|-----------------------------------------|-----|-----|-----|------|------|------|------|
| Sound Data Schedule   |                                         |     |     |     |      |      |      |      |
| Equipment             | Maximum Sound Power Level (dB)          |     |     |     |      |      |      |      |
|                       | Octave Band Level Center Frequency (Hz) |     |     |     |      |      |      |      |
|                       | 63                                      | 125 | 250 | 500 | 1000 | 2000 | 4000 | 8000 |
| Air Handling Unit     | 94                                      | 90  | 89  | 89  | 89   | 84   | 82   | 79   |
| Make-Up Air Fan       | 91                                      | 91  | 80  | 84  | 82   | 76   | 71   | 65   |
| Air Conditioning Unit | 100                                     | 96  | 90  | 89  | 86   | 80   | 75   | 72   |
| Boiler                | 75                                      | 72  | 72  | 75  | 76   | 63   | 55   | 50   |

| TABLE 2A            |                                         |     |     |     |      |      |      |      |
|---------------------|-----------------------------------------|-----|-----|-----|------|------|------|------|
| Sound Data Schedule |                                         |     |     |     |      |      |      |      |
| Equipment           | Maximum Sound Power Level (dB)          |     |     |     |      |      |      |      |
|                     | Octave Band Level Center Frequency (Hz) |     |     |     |      |      |      |      |
|                     | 63                                      | 125 | 250 | 500 | 1000 | 2000 | 4000 | 8000 |
| Chiller             | 98                                      | 98  | 96  | 95  | 93   | 94   | 88   | 81   |
| Cooling Tower       | 110                                     | 110 | 105 | 102 | 98   | 95   | 92   | 87   |
| Air Compressor      | 90                                      | 89  | 92  | 93  | 92   | 92   | 90   | 81   |
| Pump                | 85                                      | 80  | 82  | 82  | 80   | 77   | 74   | 72   |
| Fan                 | 55                                      | 50  | 48  | 47  | 48   | 46   | 42   | 37   |

#### 1.4.6.1 Basic Criteria

For each piece of machinery in the human work environment, do not exceed the maximum airborne sound levels 84 dB A-weighted scale, continuous or intermittent, or 140 dB peak sound pressure-level, impact or impulse, noise.

#### 1.4.6.2 Sound Data Schedule

| TABLE 2A              |                                         |     |     |     |      |      |      |      |
|-----------------------|-----------------------------------------|-----|-----|-----|------|------|------|------|
| Sound Data Schedule   |                                         |     |     |     |      |      |      |      |
| Equipment             | Maximum Sound Power Level (dB)          |     |     |     |      |      |      |      |
|                       | Octave Band Level Center Frequency (Hz) |     |     |     |      |      |      |      |
|                       | 63                                      | 125 | 250 | 500 | 1000 | 2000 | 4000 | 8000 |
| Air Handling Unit     | 94                                      | 90  | 89  | 89  | 89   | 84   | 82   | 79   |
| Make-Up Air Fan       | 91                                      | 91  | 80  | 84  | 82   | 76   | 71   | 65   |
| Air Conditioning Unit | 100                                     | 96  | 90  | 89  | 86   | 80   | 75   | 72   |
| Boiler                | 75                                      | 72  | 72  | 75  | 76   | 63   | 55   | 50   |
| Chiller               | 98                                      | 98  | 96  | 95  | 93   | 94   | 88   | 81   |
| Cooling Tower         | 110                                     | 110 | 105 | 102 | 98   | 95   | 92   | 87   |

| TABLE 2A            |                                         |     |     |     |      |      |      |      |
|---------------------|-----------------------------------------|-----|-----|-----|------|------|------|------|
| Sound Data Schedule |                                         |     |     |     |      |      |      |      |
| Equipment           | Maximum Sound Power Level (dB)          |     |     |     |      |      |      |      |
|                     | Octave Band Level Center Frequency (Hz) |     |     |     |      |      |      |      |
|                     | 63                                      | 125 | 250 | 500 | 1000 | 2000 | 4000 | 8000 |
| Air Compressor      | 90                                      | 89  | 92  | 93  | 92   | 92   | 90   | 81   |
| Pump                | 85                                      | 80  | 82  | 82  | 80   | 77   | 74   | 72   |
| Fan                 | 55                                      | 50  | 48  | 47  | 48   | 46   | 42   | 37   |

#### 1.4.7 Seismic Protection Criteria

#### 1.4.8 Welding

AWS D1.1/D1.1M CSA W59.

### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Inertia Bases

Machinery Bases

Saddles

#### SD-03 Product Data

Isolators

Flexible Connectors

Flexible Duct Connectors

Pipe Guides

Vertical Stops

Thrust Restraints

Inertia Bases

Machinery Bases

Machinery Foundations and Subbases

Saddles



## Machinery Manufacturer's Sound Data

### SD-05 Design Data

Inertia Bases

Machinery Bases

Saddles

Each Item of Machinery

Each Item of Machinery over 136 Kilograms

Submit design calculations for inertia bases, machinery base, and saddles, either by the machinery manufacturer for the recommended machinery mounting or by the vibration-isolation equipment manufacturer.

### SD-06 Test Reports

Equipment Vibration Tests

Protected Spring Isolators

### SD-08 Manufacturer's Instructions

Vibration and Noise Isolation Components

Seismic Protection Components

## 1.6 QUALITY ASSURANCE

### 1.6.1 Vibration Isolator Procurement

For each piece of machinery to be isolated from vibration, supply the inertia base, machinery base, saddles, vibration isolators and other associated materials and equipment as a coordinated package by a single manufacturer or by the machinery manufacturer. Select isolators that provide uniform deflection even when machinery weight is not evenly distributed. This requirement does not include the flexible connectors or the hangers for the associated piping and ductwork.

### 1.6.2 Unitized Machinery Assemblies

Mounting of unitized assemblies directly on vibration isolation springs is acceptable if machinery manufacturer certifies that the end supports of the assemblies have been designed for such installation.

## PART 2 PRODUCTS

### 2.1 CORROSION PROTECTION FOR STEEL PARTS

ASTM A123/A123M CSA G164 or ASTM A653/A653M CSA G164 hot-dipped galvanized, or equivalent manufacturer standard coatings. Where steel parts are exposed to the weather, provide galvanized coating of at least 0.61 kg of zinc per square meter of surface. Coat springs with neoprene.

## 2.2 NEOPRENE

ASTM D471 CSA G40.21 and ASTM D2240, CGSB 41-GP-25M Grade Durometer 40, 50, or 60, and oil resistant.

## 2.3 FLOOR-MOUNTED ISOLATORS

### 2.3.1 Neoprene Isolation Pads

Provide pads at least 6 mm thick with cross-ribbed or waffle design. For concentrated loads, provide steel bearing plates bonded or cold cemented to the pads.

### 2.3.2 Neoprene Isolators

Provide molded neoprene isolators having steel base plates with mounting holes and, at the top, steel mounting plates with mounting holes or threaded inserts. Provide elements of type and size coded with molded letters or color-coded for capacity identification. Embed metal parts completely in neoprene.

## 2.4 SPRING ISOLATORS AND PROTECTED SPRING ISOLATORS

Provide spring isolators or protected spring isolators that are adjustable and laterally stable with free-standing springs of horizontal stiffness at minimum 80 percent of the vertical (axial) stiffness. For machine-attached and floor-attached restraining elements, separate from metal-to-metal contact by neoprene cushions 3 mm thick minimum. Provide neoprene acoustic friction pads at least 6 mm thick.

### 2.4.1 Springs

Provide springs with base and compression plates, to keep spring ends parallel during and after deflection to operating height. Provide outside coil diameters at least 0.8 of the operating height. At operating height, springs must have additional travel to complete (solid) compression equal to at least 50 percent of the operating deflection.

### 2.4.2 Mounting and Adjustment

Provide base and compression plates with mounting holes or threaded fittings. Bolt leveling adjustment bolts to machinery or base.

## 2.5 SUSPENSION ISOLATORS

Provide hangers with suspension isolators encased in open steel brackets. Isolate hanger rods from isolator steel brackets with neoprene-lined opening.

### 2.5.1 Suspension Neoprene Isolators

Provide double-deflection elements with minimum 10 mm deflection.

### 2.5.2 Suspension Spring Isolators

Provide hangers with springs and molded neoprene elements in series. Provide isolators with adjustable spring-preloading devices where required to maintain constant pipe elevations during installation and when pipe operational loads are transferred to the springs.

## 2.6 MACHINERY BASES SADDLES

ASTM A36/A36M CSA G40.21 300W and AISC 360 CSA S16.

## 2.7 INERTIA BASES

ASTM A36/A36M CSA G40.21 300W steel, ASTM C94/C94M ( 20 MPa) concrete.

## 2.8 FLEXIBLE CONNECTORS FOR PIPING

Straight or elbow flexible connectors rated for temperatures, pressures, and fluids to be conveyed. Provide flexible connectors with the strength 4 times operating pressure at highest system operating temperature. Provide elbow flexible connectors with a permanently set angle.

### 2.8.1 Elastomeric Flexible Connectors

Fabricated of multiple plies of tire cord fabric and elastomeric materials with integral reinforced elastomeric flanges with galvanized malleable iron back up rings.

### 2.8.2 Metal Flexible Connectors

Fabricated of Grade E phosphor bronze, monel or corrugated stainless steel tube covered with comparable bronze or stainless steel braid restraining and pressure cover.

## 2.9 FLEXIBLE DUCT CONNECTORS

Provide flexible duct connectors fabricated in accordance with SMACNA 1966.

## 2.10 PIPE GUIDES

Factory-fabricated. Weld steel bar guides to the pipe at a maximum radial spacing of 60 degrees. The outside diameter around the guide bars must be smaller than the inside diameter of the guide sleeve in accordance with standard field construction practice. For pipe temperatures below 16 degrees C, provide metal sleeve, minimum 16 kg per cubic meter density insulation.

## 2.11 THRUST RESTRAINTS

Adjustable spring thrust restraints, able to resist the thrust force with at least 25 percent unused capacity. The operating spring deflection must be not less than 50 percent of the static deflection of the isolation supporting the machinery.

## 2.12 SEISMIC PROTECTION COMPONENTS FOR PIPING AND DUCTWORK

Section 23 03 00.00 20 BASIC MECHANICAL MATERIALS AND METHODS. SMACNA 1981.

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 Vibration and Noise Isolation Components

Install vibration-and-noise isolation materials and equipment in

accordance with machinery manufacturer's instructions.

### 3.1.2 Suspension Vibration Isolators

Provide suspension isolation hangers for piping, suspended equipment, and suspended equipment platforms in mechanical equipment rooms, as indicated and as specified. For operating load static deflections of 6 mm or less, provide neoprene pads or single deflection neoprene isolators. For operating load static deflections over 8 to 10 mm, provide double-deflection neoprene element isolators. For operating load static deflections over 10 mm, provide isolators with spring and neoprene elements in series.

### 3.1.3 Vertical Stops

For machinery affected by wind pressure or having an operational weight different from installed weight, provide resilient vertical limit stops which prevent spring extension when weight is removed. Provide vertical stops for machinery containing liquid, such as water chillers, evaporative coolers, boilers, and cooling towers. Spring isolated or protected spring isolated machinery must rock and move freely within limits of stops or seismic restraint devices.

### 3.1.4 Thrust Restraints

Where required, provide pairs of thrust restraints, symmetrically installed on both sides of the steady state line of thrust.

### 3.1.5 Flexible Pipe and Duct Connectors

Install flexible connectors in accordance with the manufacturer's instructions. When liquid pulsation dampening is required, flexible connectors with spherical configuration may be used.

### 3.1.6 Machinery

Provide vibration isolators, flexible connectors in accordance with manufacturer's recommendations. Machinery with spring isolators or protected spring isolators must rock or move freely within limits of stops.

#### 3.1.6.1 Stability

Isolators must be stable during starting and stopping of machinery without traverse and eccentric movement of machinery that would damage or adversely affect the machinery or attachments.

#### 3.1.6.2 Lateral Motion

The installed vibration isolation system for each piece of floor or ceiling mounted machinery must have a maximum lateral motion under machinery start up and shut down conditions of not more than 6 mm. Restrain motions in excess by approved spring mountings.

#### 3.1.6.3 Unbalanced Machinery

Provide foundation suspension systems specifically designed to resist horizontal forces for machinery with large unbalanced horizontal forces. Vibration isolator systems must conform to the machinery manufacturer's recommendations.

## 3.1.6.4 Nonrotating Machinery

Mount nonrotating machinery in systems which includes rotating or vibrating machinery on isolators having the same deflection as the hangers and supports for the pipe connected to.

## 3.1.6.5 Unitized Machinery Assemblies

| TABLE 3A                                                     |                             |                |                 |                |                 |                |
|--------------------------------------------------------------|-----------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection       |                             |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported. |                             |                |                 |                |                 |                |
| Column Spacing                                               | Slab on earth and 0-9 meter |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                    | Type (Note (1))             | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Absorption Refrigerative Machines                            | SV-R                        | 25.40          | SV-R            | 44.45          | SV-R            | 69.85          |
| Centrifugal Chillers or Heat Pumps                           |                             |                |                 |                |                 |                |
| Hermetic Type                                                | SV-B                        | 44.45          | SV-B            | 63.50          | SV-B            | 88.90          |
| Open Type                                                    | SV-1                        | 44.45          | SV-I            | 63.50          | SV-I            | 88.90          |
| Reciprocating Air or Refrigeration Compressors               |                             |                |                 |                |                 |                |
| 500 to 750 rpm                                               | S-R                         | 44.45          | S-R             | 63.50          | S-R             | 88.90          |
| 751 rpm and up                                               | S-R                         | 38.10          | S-R             | 63.50          | S-R             | 88.90          |
| Reciprocating Chillers or Heat Pumps                         |                             |                |                 |                |                 |                |
| 500 to 750 rpm                                               | SV-R                        | 44.45          | SV-R            | 63.50          | SV-R            | 88.90          |
| 751 rpm and up                                               | SV-R                        | 38.10          | SV-R            | 63.50          | SV-R            | 88.90          |
| Packaged Boilers                                             | SV                          | 25.40          | SV              | 63.50          | SV-R            | 88.90          |
| Closed Coupled Pumps                                         |                             |                |                 |                |                 |                |

| TABLE 3A                                                                |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                  |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.            |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| Column Spacing                                                          | Slab on earth and 0-9 meter                                                                                                                                                                                                           |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                               | Type (Note (1))                                                                                                                                                                                                                       | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Up to 5 1/2 kw                                                          | S-I                                                                                                                                                                                                                                   | 25.40          | S-I             | 25.40          | S-I             | 25.40          |
| Over 5 1/2 kw                                                           | S-I                                                                                                                                                                                                                                   | 38.10          | S-I             | 63.50          | S-I             | 63.50          |
| Base Mounted Pumps                                                      |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| Up to 15 kw                                                             | S-I                                                                                                                                                                                                                                   | 38.10          | S-I             | 63.50          | S-I             | 63.50          |
| 15 to 56 kw                                                             | S-I                                                                                                                                                                                                                                   | 38.10          | S-I             | 63.50          | S-I             | 88.90          |
| Over 56 kw                                                              | S-I                                                                                                                                                                                                                                   | 63.50          | S-I             | 88.90          | S-I             | 88.90          |
| Cooling Towers and Evaporative Condensers                               | SV with deflections specified for centrifugal blowers when springs are supported on beams. Use deflection listed for column supported floors with up to 9 meters column spacing when springs are located on columns or bearing walls. |                |                 |                |                 |                |
| Factory Assembled Air Handling Equipment AH, AC and HV Units (Note (2)) |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| Suspended Units                                                         |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| Up to 3 3/4 kw                                                          | H                                                                                                                                                                                                                                     | 25.40          | H               | 25.40          | H               | 25.40          |
| Over 3 3/4 kw                                                           |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| Up to 400 rpm                                                           | H                                                                                                                                                                                                                                     | 44.45          | H               | 44.45          | H               | 44.45          |
| Over 401 rpm                                                            | H                                                                                                                                                                                                                                     | 25.40          | H               | 38.10          | H               | 63.50          |
| Floor Mounted Units                                                     |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| Up to 3 3/4 kw                                                          | S                                                                                                                                                                                                                                     | 25.40          | S               | 25.40          | S               | 25.40          |
| Over 3 3/4 kw                                                           |                                                                                                                                                                                                                                       |                |                 |                |                 |                |
| Up to 400 rpm                                                           | S-R                                                                                                                                                                                                                                   | 44.45          | S-R             | 44.45          | S-R             | 63.50          |

| TABLE 3A                                                                                            |                             |                                                                                       |                 |                |                 |                |
|-----------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                              |                             |                                                                                       |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.                                        |                             |                                                                                       |                 |                |                 |                |
| Column Spacing                                                                                      | Slab on earth and 0-9 meter |                                                                                       | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                                                           | Type (Note (1))             | MSD (Note (1))                                                                        | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Over 401 rpm                                                                                        | S-R                         | 25.40                                                                                 | S-R             | 38.10          | S-R             | 63.50          |
| Centrifugal Blowers                                                                                 |                             |                                                                                       |                 |                |                 |                |
| 175 - 224 rpm                                                                                       | S-B                         | 120.65                                                                                | S-B             | 120.65         | S-B             | 120.65         |
| 225 - 299 rpm                                                                                       | S-B                         | 95.25                                                                                 | S-B             | 120.65         | S-B             | 120.65         |
| 300 - 374 rpm                                                                                       | S-B                         | 69.85                                                                                 | S-B             | 114.30         | S-B             | 120.65         |
| 375 - 499 rpm                                                                                       | S-B                         | 63.50                                                                                 | S-B             | 88.90          | S-B             | 114.30         |
| Over 500 rpm                                                                                        | S-B                         | 44.45                                                                                 | S-B             | 63.50          | S-B             | 88.90          |
| Tubular Centrifugal and Axial Fans (Note (2))                                                       |                             |                                                                                       |                 |                |                 |                |
| Suspended                                                                                           |                             | H with deflection specified for centrifugal blowers                                   |                 |                |                 |                |
| Floor Mounted Arrangements 1 & 9                                                                    |                             | S-B with deflections specified for centrifugal blowers                                |                 |                |                 |                |
| Utility Fans (Note (2))                                                                             |                             |                                                                                       |                 |                |                 |                |
| Suspended                                                                                           |                             | H with deflections specified for centrifugal blowers but not to exceed 69.85 mm       |                 |                |                 |                |
| Floor-Mounted                                                                                       |                             | S-R with deflections not specified for centrifugal blowers but not to exceed 69.85 mm |                 |                |                 |                |
| High Pressure Fans (Over 1494 Pa Static Pressure) and Other Machineries Producing Thrust (Note (2)) |                             | HR recommended for minimizing undesirable thrust effects                              |                 |                |                 |                |
| Internal Combustion Engines and Engine Driven Equip                                                 |                             |                                                                                       |                 |                |                 |                |
| 750 rpm and over                                                                                    | S                           | 38.10                                                                                 | S               | 63.50          | S               | 88.90          |

| TABLE 3A                                                                                                                                                                                                                                                   |                             |                |                 |                |                 |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                                                                                                                                                                                     |                             |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.                                                                                                                                                                                               |                             |                |                 |                |                 |                |
| Column Spacing                                                                                                                                                                                                                                             | Slab on earth and 0-9 meter |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                                                                                                                                                                                                                  | Type (Note (1))             | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Dimmer Banks and Transformers                                                                                                                                                                                                                              |                             |                |                 |                |                 |                |
| Up to 454 kg                                                                                                                                                                                                                                               | NM                          | 8.89           | NM              | 8.89           | NM              | 88.90          |
| Over 454 kg                                                                                                                                                                                                                                                | SV                          | 25.40          | SV              | 25.40          | SV              | 25.40          |
| NOTES:                                                                                                                                                                                                                                                     |                             |                |                 |                |                 |                |
| (1) Equipment Vibration Isolation Schedule Designations (Hyphenated designations are combinations of the following:)                                                                                                                                       |                             |                |                 |                |                 |                |
| B - Welded structural steel bases.                                                                                                                                                                                                                         |                             |                |                 |                |                 |                |
| H - Spring isolators (suspended equipment and piping). Where required, provide with adjustable preloading devices.                                                                                                                                         |                             |                |                 |                |                 |                |
| HR - Thrust restraints                                                                                                                                                                                                                                     |                             |                |                 |                |                 |                |
| I - Concrete inertia bases with steel forms.                                                                                                                                                                                                               |                             |                |                 |                |                 |                |
| NM - Neoprene mounts.                                                                                                                                                                                                                                      |                             |                |                 |                |                 |                |
| NP - Neoprene pads.                                                                                                                                                                                                                                        |                             |                |                 |                |                 |                |
| R - Structural steel rail for equipment mounts.                                                                                                                                                                                                            |                             |                |                 |                |                 |                |
| S - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                               |                             |                |                 |                |                 |                |
| SV - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                              |                             |                |                 |                |                 |                |
| SX - Freestanding spring isolators with adjustable cushioned vertical stops and cushioned horizontal stops (floor-mounted equipment). Protected spring isolators SX may be substituted wherever S or SV is specified and must meet all requirements.       |                             |                |                 |                |                 |                |
| (2) Fans                                                                                                                                                                                                                                                   |                             |                |                 |                |                 |                |
| a. When fan motors are 56 kW or larger, use the deflection requirements for the next wider column spacing. Except for building slab on grade a minimum of 63.50 mm should be used unless larger deflections are specified in the centrifugal blower table. |                             |                |                 |                |                 |                |
| b. Provide sway brace isolators for tubular centrifugal and axial fans when the fan pressure exceeds 996 Pa.                                                                                                                                               |                             |                |                 |                |                 |                |



| TABLE 3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                             |                |                 |                |                 |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                             |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                             |                |                 |                |                 |                |
| Column Spacing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Slab on earth and 0-9 meter |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Type (Note (1))             | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| c. Provide inertia bases for all fans in lieu of structural steel bases or rails specified above when the fan pressure exceeds 996 Pa.                                                                                                                                                                                                                                                                                                                                                                                                      |                             |                |                 |                |                 |                |
| d. With attaching brackets, suspension spring isolators bridge between the structure and the thrust-producing machinery such as high-pressure fan. Both types H and HR normally provide reaction in tension, while types S, SV, and SX normally provide reaction in compression. Thrust restraints are low-cost and effective components available from manufacturers. Use thrust restraints to eliminate the need for or reduce the magnitude of inertia mass when the mass is only used to reduce the displacement effects of the thrust. |                             |                |                 |                |                 |                |

Unitized assemblies such as chillers with evaporator and condenser, and top mounted centrifugal compressor or unitized absorption refrigeration machines, structurally designed with end supports, may be mounted on steel rails and springs in lieu of steel bases and springs. Where the slab or deck is less than 100 mm thick, provide spring isolation units with the deflection double that of the vibration isolation schedule, up to a maximum static deflection of 127 mm.

## 3.1.6.6 Roof and Upper Floor Mounted Machinery

| TABLE 3A                                                     |                             |                |                 |                |                 |                |
|--------------------------------------------------------------|-----------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection       |                             |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported. |                             |                |                 |                |                 |                |
| Column Spacing                                               | Slab on earth and 0-9 meter |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                    | Type (Note (1))             | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Absorption Refrigeration Machines                            | SV-R                        | 25.40          | SV-R            | 44.45          | SV-R            | 69.85          |
| Centrifugal Chillers or Heat Pumps                           |                             |                |                 |                |                 |                |
| Hermetic Type                                                | SV-B                        | 44.45          | SV-B            | 63.50          | SV-B            | 88.90          |
| Open Type                                                    | SV-1                        | 44.45          | SV-I            | 63.50          | SV-I            | 88.90          |
| Reciprocating Air or Refrigeration Compressors               |                             |                |                 |                |                 |                |
| 500 to 750 rpm                                               | S-R                         | 44.45          | S-R             | 63.50          | S-R             | 88.90          |
| 751 rpm and up                                               | S-R                         | 38.10          | S-R             | 63.50          | S-R             | 88.90          |
| Reciprocating Chillers or Heat Pumps                         |                             |                |                 |                |                 |                |
| 500 to 750 rpm                                               | SV-R                        | 44.45          | SV-R            | 63.50          | SV-R            | 88.90          |
| 751 rpm and up                                               | SV-R                        | 38.10          | SV-R            | 63.50          | SV-R            | 88.90          |
| Packaged Boilers                                             | SV                          | 25.40          | SV              | 63.50          | SV-R            | 88.90          |
| Closed Coupled Pumps                                         |                             |                |                 |                |                 |                |
| Up to 5 1/2 kW                                               | S-I                         | 25.40          | S-I             | 25.40          | S-I             | 25.40          |
| Over 5 1/2 kW                                                | S-I                         | 38.10          | S-I             | 63.50          | S-I             | 63.50          |

| TABLE 3A                                                                |                             |                                                                                                                                                                                                                                       |                 |                |                 |                |
|-------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                  |                             |                                                                                                                                                                                                                                       |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.            |                             |                                                                                                                                                                                                                                       |                 |                |                 |                |
| Column Spacing                                                          | Slab on earth and 0-9 meter |                                                                                                                                                                                                                                       | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                               | Type (Note (1))             | MSD (Note (1))                                                                                                                                                                                                                        | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Base Mounted Pumps                                                      |                             |                                                                                                                                                                                                                                       |                 |                |                 |                |
| Up to 15 kW                                                             | S-I                         | 38.10                                                                                                                                                                                                                                 | S-I             | 63.50          | S-I             | 63.50          |
| 15 to 56 kW                                                             | S-I                         | 38.10                                                                                                                                                                                                                                 | S-I             | 63.50          | S-I             | 88.90          |
| Over 56 kW                                                              | S-I                         | 63.50                                                                                                                                                                                                                                 | S-I             | 88.90          | S-I             | 88.90          |
| Cooling Towers and Evaporative Condensers                               |                             | SV with deflections specified for centrifugal blowers when springs are supported on beams. Use deflection listed for column supported floors with up to 9 meters column spacing when springs are located on columns or bearing walls. |                 |                |                 |                |
| Factory Assembled Air Handling Equipment AH, AC and HV Units (Note (2)) |                             |                                                                                                                                                                                                                                       |                 |                |                 |                |
| Suspended Units                                                         |                             |                                                                                                                                                                                                                                       |                 |                |                 |                |
| Up to 3 3/4 kW                                                          | H                           | 25.40                                                                                                                                                                                                                                 | H               | 25.40          | H               | 25.40          |
| Over 3 3/4 kW                                                           |                             |                                                                                                                                                                                                                                       |                 |                |                 |                |
| Up to 400 rpm                                                           | H                           | 44.45                                                                                                                                                                                                                                 | H               | 44.45          | H               | 44.45          |
| Over 401 rpm                                                            | H                           | 25.40                                                                                                                                                                                                                                 | H               | 38.10          | H               | 63.50          |
| Floor Mounted Units                                                     |                             |                                                                                                                                                                                                                                       |                 |                |                 |                |
| Up to 3 3/4 kW                                                          | S                           | 25.40                                                                                                                                                                                                                                 | S               | 25.40          | S               | 25.40          |
| Over 3 3/4 kW                                                           |                             |                                                                                                                                                                                                                                       |                 |                |                 |                |
| Up to 400 rpm                                                           | S-R                         | 44.45                                                                                                                                                                                                                                 | S-R             | 44.45          | S-R             | 63.50          |
| Over 401 rpm                                                            | S-R                         | 25.40                                                                                                                                                                                                                                 | S-R             | 38.10          | S-R             | 63.50          |
| Centrifugal Blowers                                                     |                             |                                                                                                                                                                                                                                       |                 |                |                 |                |

| TABLE 3A                                                                                            |                             |                                                                                       |                 |                |                 |                |
|-----------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                              |                             |                                                                                       |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.                                        |                             |                                                                                       |                 |                |                 |                |
| Column Spacing                                                                                      | Slab on earth and 0-9 meter |                                                                                       | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                                                           | Type (Note (1))             | MSD (Note (1))                                                                        | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| 175 - 224 rpm                                                                                       | S-B                         | 120.65                                                                                | S-B             | 120.65         | S-B             | 120.65         |
| 225 - 299 rpm                                                                                       | S-B                         | 95.25                                                                                 | S-B             | 120.65         | S-B             | 120.65         |
| 300 - 374 rpm                                                                                       | S-B                         | 69.85                                                                                 | S-B             | 114.30         | S-B             | 120.65         |
| 375 - 499 rpm                                                                                       | S-B                         | 63.50                                                                                 | S-B             | 88.90          | S-B             | 114.30         |
| Over 500 rpm                                                                                        | S-B                         | 44.45                                                                                 | S-B             | 63.50          | S-B             | 88.90          |
| Tubular Centrifugal and Axial Fans (Note (2))                                                       |                             |                                                                                       |                 |                |                 |                |
| Suspended                                                                                           |                             | H with deflection specified for centrifugal blowers                                   |                 |                |                 |                |
| Floor Mounted Arrangements 1 & 9                                                                    |                             | S-B with deflections specified for centrifugal blowers                                |                 |                |                 |                |
| Utility Fans (Note (2))                                                                             |                             |                                                                                       |                 |                |                 |                |
| Suspended                                                                                           |                             | H with deflections specified for centrifugal blowers but not to exceed 69.85 mm       |                 |                |                 |                |
| Floor-Mounted                                                                                       |                             | S-R with deflections not specified for centrifugal blowers but not to exceed 69.85 mm |                 |                |                 |                |
| High Pressure Fans (Over 1494 Pa Static Pressure) and Other Machineries Producing Thrust (Note (2)) |                             | HR recommended for minimizing undesirable thrust effects                              |                 |                |                 |                |
| Internal Combustion Engines and Engine Driven Equip                                                 |                             |                                                                                       |                 |                |                 |                |
| 750 rpm and over                                                                                    | S                           | 38.10                                                                                 | S               | 63.50          | S               | 88.90          |
| Dimmer Banks and Transformers                                                                       |                             |                                                                                       |                 |                |                 |                |
| Up to 454 kg                                                                                        | NM                          | 8.89                                                                                  | NM              | 8.89           | NM              | 88.90          |

| TABLE 3A                                                                                                                                                                                                                                                   |                             |                |                 |                |                 |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                                                                                                                                                                                     |                             |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.                                                                                                                                                                                               |                             |                |                 |                |                 |                |
| Column Spacing                                                                                                                                                                                                                                             | Slab on earth and 0-9 meter |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                                                                                                                                                                                                                  | Type (Note (1))             | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
| Over 454 kg                                                                                                                                                                                                                                                | SV                          | 25.40          | SV              | 25.40          | SV              | 25.40          |
| NOTES:                                                                                                                                                                                                                                                     |                             |                |                 |                |                 |                |
| (1) Equipment Vibration Isolation Schedule Designations (Hyphenated designations are combinations of the following:)                                                                                                                                       |                             |                |                 |                |                 |                |
| B - Welded structural steel bases.                                                                                                                                                                                                                         |                             |                |                 |                |                 |                |
| H - Spring isolators (suspended equipment and piping). Where required, provide with adjustable preloading devices.                                                                                                                                         |                             |                |                 |                |                 |                |
| HR - Thrust restraints                                                                                                                                                                                                                                     |                             |                |                 |                |                 |                |
| I - Concrete inertia bases with steel forms.                                                                                                                                                                                                               |                             |                |                 |                |                 |                |
| NM - Neoprene mounts.                                                                                                                                                                                                                                      |                             |                |                 |                |                 |                |
| NP - Neoprene pads.                                                                                                                                                                                                                                        |                             |                |                 |                |                 |                |
| R - Structural steel rail for equipment mounts.                                                                                                                                                                                                            |                             |                |                 |                |                 |                |
| S - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                               |                             |                |                 |                |                 |                |
| SV - Freestanding spring isolators (floor-mounted equipment).                                                                                                                                                                                              |                             |                |                 |                |                 |                |
| SX - Freestanding spring isolators with adjustable cushioned vertical stops and cushioned horizontal stops (floor-mounted equipment). Protected spring isolators SX may be substituted wherever S or SV is specified and must meet all requirements.       |                             |                |                 |                |                 |                |
| (2) Fans                                                                                                                                                                                                                                                   |                             |                |                 |                |                 |                |
| a. When fan motors are 56 kW or larger, use the deflection requirements for the next wider column spacing. Except for building slab on grade a minimum of 63.50 mm should be used unless larger deflections are specified in the centrifugal blower table. |                             |                |                 |                |                 |                |
| b. Provide sway brace isolators for tubular centrifugal and axial fans when the fan pressure exceeds 996 Pa.                                                                                                                                               |                             |                |                 |                |                 |                |
| c. Provide inertia bases for all fans in lieu of structural steel bases or rails specified above when the fan pressure exceeds 996 Pa.                                                                                                                     |                             |                |                 |                |                 |                |

| TABLE 3A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                             |                |                 |                |                 |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|-----------------|----------------|-----------------|----------------|
| Vibration Isolator Types and Minimum Static Deflection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |                |                 |                |                 |                |
| (MSD, mm) for 100-200 mm slab on grade and column supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |                |                 |                |                 |                |
| Column Spacing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Slab on earth and 0-9 meter |                | 9.1-12 meters   |                | 12.1-15 meters  |                |
| Equipment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Type (Note (1))             | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) | Type (Note (1)) | MSD (Note (1)) |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                             |                |                 |                |                 |                |
| <p>d. With attaching brackets, suspension spring isolators bridge between the structure and the thrust-producing machinery such as high-pressure fan. Both types H and HR normally provide reaction in tension, while types S, SV, and SX normally provide reaction in compression. Thrust restraints are low-cost and effective components available from manufacturers. Use thrust restraints to eliminate the need for or reduce the magnitude of inertia mass when the mass is only used to reduce the displacement effects of the thrust.</p> |                             |                |                 |                |                 |                |

| TABLE 1B                                                        |                 |      |
|-----------------------------------------------------------------|-----------------|------|
| Class II Vibration Isolator Types and Minimum Static Deflection |                 |      |
| (MSD, mm) for basements below grade and floor slabs on earth    |                 |      |
| Equipment                                                       | Type (Note (1)) | MSD  |
| Absorption Refrigeration Machines                               | NP              | 6.35 |
|                                                                 | NM              | 8.89 |
| Centrifugal Chillers or Heat Pumps                              |                 |      |
| Hermetic Type                                                   | NP              | 6.35 |
|                                                                 | NM              | 8.89 |
| Open Type                                                       | NM-I            | 8.89 |
| Reciprocating Air or Refrigeration Compressors                  |                 |      |

| TABLE 1B                                                                |                        |            |
|-------------------------------------------------------------------------|------------------------|------------|
| Class II Vibration Isolator Types and Minimum Static Deflection         |                        |            |
| (MSD, mm) for basements below grade and floor slabs on earth            |                        |            |
| <u>Equipment</u>                                                        | <u>Type (Note (1))</u> | <u>MSD</u> |
| 500 to 750 rpm                                                          | S                      | 25.40      |
| 751 rpm and up                                                          | S                      | 25.40      |
| Reciprocating Chillers or Heat Pumps                                    |                        |            |
| 500 to 750 rpm                                                          | SV                     | 25.40      |
| 751 rpm and up                                                          | SV                     | 25.40      |
| Packaged Boilers                                                        | NP                     | 6.35       |
|                                                                         | NM                     | 8.89       |
| Pumps                                                                   |                        |            |
| Closed Coupled                                                          | NP                     | 6.35       |
| Up to 5 1/2 kW                                                          | NM                     | 8.89       |
| Over 5 1/2 kW                                                           | S-I                    | 25.40      |
| Base Mounted                                                            |                        |            |
| Up to 15 kW                                                             | S-I                    | 25.40      |
| 15 to 56 kW                                                             | S-I                    | 25.40      |
| Over 56 kW                                                              | S-I                    | 25.40      |
| Cooling Towers and<br>Evaporative Condensers                            | NP                     | 6.35       |
|                                                                         | NM                     | 8.89       |
| Factory Assembled Air Handling Equipment AH, AC and HV Units (Note (2)) |                        |            |
| Suspended Units                                                         |                        |            |
| Up to 3 3/4 kW                                                          | H                      | 25.40      |
| Over 3 3/4 kW                                                           |                        |            |
| Up to 400 rpm                                                           | H                      | 44.45      |
| Over 401 rpm                                                            | H                      | 25.40      |
| Floor Mounted Units                                                     |                        |            |

| TABLE 1B                                                                                               |                                                      |                                                                |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------|
| Class II Vibration Isolator Types and Minimum Static Deflection                                        |                                                      |                                                                |
| (MSD, mm) for basements below grade and floor slabs on earth                                           |                                                      |                                                                |
| Equipment                                                                                              | Type (Note (1))                                      | MSD                                                            |
| Up to 3 3/4 kw                                                                                         | NP                                                   | 6.35                                                           |
|                                                                                                        | NM                                                   | 8.89                                                           |
| Over 3 3/4 kw                                                                                          |                                                      |                                                                |
| Up to 400 rpm                                                                                          | NM                                                   | 8.89                                                           |
| Over 401 rpm                                                                                           | NM                                                   | 8.89                                                           |
| Centrifugal Blowers                                                                                    |                                                      |                                                                |
| 175 - 224 rpm                                                                                          | NM-B                                                 | 8.89                                                           |
| 225 - 299 rpm                                                                                          | NM-B                                                 | 8.89                                                           |
| 300 - 374 rpm                                                                                          | NM-B                                                 | 8.89                                                           |
| 375 - 499 rpm                                                                                          | NM-B                                                 | 8.89                                                           |
| Over 500 rpm                                                                                           | NM-B                                                 | 8.89                                                           |
| Tubular Centrifugal and Axial Fans (Note (2))                                                          |                                                      |                                                                |
| Suspended                                                                                              | H with deflections specified for centrifugal blowers |                                                                |
| Floor Mounted Arrangements<br>1 & 9                                                                    | NM                                                   | 8.89                                                           |
| Utility Fans (Note (2))                                                                                |                                                      |                                                                |
| Suspended and centrifugal                                                                              | H with deflections specified for                     |                                                                |
| Floor-Mounted                                                                                          | NM                                                   | 8.89                                                           |
| High Pressure Fans (Over 1494 Pa Static Pressure) and<br>Other Machineries Producing Thrust (Note (2)) |                                                      | HR recommended for<br>minimizing undesirable<br>thrust effects |
| Internal Combustion Engines and Engine Driven Equip                                                    |                                                      |                                                                |
| 750 rpm and over                                                                                       | S                                                    | 25.40                                                          |
| Dimmer Banks and Transformers                                                                          |                                                      |                                                                |



| TABLE 1B                                                        |                 |       |
|-----------------------------------------------------------------|-----------------|-------|
| Class II Vibration Isolator Types and Minimum Static Deflection |                 |       |
| (MSD, mm) for basements below grade and floor slabs on earth    |                 |       |
| Equipment                                                       | Type (Note (1)) | MSD   |
| Up to 454 kg                                                    | NP              | 6.35  |
|                                                                 | NM              | 8.89  |
| Over 454 kg                                                     | SV              | 25.40 |
| NOTES: Note (1) and Note (2) are same as for TABLE 3A.          |                 |       |

On the roof or upper floors, mount machinery on isolators with vertical stops. Rest isolators on beams or structures designed and installed in accordance with the SMACNA 1793, Plate 61.

### 3.1.7 Piping

Provide vibration isolation for piping. The isolator deflections must be equal to or greater than the static deflection of the vibration isolators provided for the connected machinery as follows:

#### 3.1.7.1 Piping Connected to Vibration Isolated Machinery

For a distance of 15 meters or 50 pipe diameters, whichever is greater.

#### 3.1.7.2 Chilled, Hot, and Dual Temperature Piping

For risers from pumps and for the first 6 meters of the branch connection of the main supply and return piping at each floor.

### 3.1.8 Water and Steam Distribution Piping Application

Resiliently support piping with combination spring and neoprene isolation hangers. Provide spring elements with 16 mm static deflection; install the hanger with spacing so that the first harmonic natural frequency is not less than 360 Hz. Provide double-deflection neoprene elements. For the first two isolation hangers from the rotating equipment of 90 mm and smaller piping systems, ensure a deflection equal to the equipment-isolation static deflection. For the first four piping isolation hanger supports from rotating equipment of 100 mm and larger piping systems, use resilient hanger-rod isolators at a fixed elevation regardless of load changes. Incorporate an adjustable preloading device to transfer the load to the spring element within the hanger mounting

after the piping system has been filled with water.

### 3.1.9 Pipe Hanger and Support Installation

#### 3.1.9.1 Pipe Hangers

Provide eye-bolts or swivel joints for pipe hangers to permit pipe thermal or mechanical movement without angular misalignment of hanger vibration isolator.

#### 3.1.9.2 High Temperatures

Where neoprene elements of vibration isolator may be subjected to high pipe temperatures, above 71 degrees C, provide metal heat shields or thermal isolators.

#### 3.1.9.3 Valves

Provide vibration isolation hangers and supports at modulating, pressure reducing, or control valves which will induce fluid pulsations. When required or indicated, isolate valves with flexible connectors.

#### 3.1.9.4 Machinery Without Flexible Connections

When piping is not connected to vibrating machinery with flexible connectors, provide the first four hangers with isolation elements designed for deflections equal to equipment vibration isolator deflections (including static, operating, and start-up).

#### 3.1.9.5 Pipe Risers

Provide pipe riser supports with bearing plates and two layers of 6 mm thick ribbed or waffled neoprene pad loaded to not more than 345 kPa. Separate isolation pads with 6 mm steel plate. Weld pipe riser clamps at anchor points to the pipe and to pairs of vertical acoustical pipe anchor mountings which must be rigidly fastened to the steel framing.

#### 3.1.9.6 Supports at Base of Pipe Risers

Piping isolation supports at the base of risers must be two layers of 13 mm thick heavy-duty neoprene pad separated by 6 mm thick steel plate. Use bearing plates sized to provide a pad loading of not more than 3447 kPa. Weld the stanchion between the pipe and isolation support to the pipe and weld or bolt to the isolation support. Bolt isolation support to the floor slab with resilient sleeves and washers. Where supplementary steel is required to support piping, provide a maximum deflection of 2 mm at the mid-span of this steel under the load. Rigidly support piping from the supplementary steel with the supplementary steel isolated from the building structure with isolators.

#### 3.1.9.7 Pipe Anchors

Attach each end of the pipe anchor to an omni-directional pipe isolator which in turn must be rigidly fastened to the steel framing or structural concrete. Provide a telescoping pipe isolator of two sizes of steel tubing separated by a minimum 13 mm thick pad of heavy-duty neoprene or heavy-duty neoprene and canvas. Provide vertical restraints by similar material to prevent vertical travel in either direction. The load on the isolation material must not exceed 3447 kPa.

### 3.1.10 Equipment Room Sound Isolation

Do not allow direct contact between pipe or ducts and walls, floor slabs, roofs, ceilings or partitions of equipment rooms.

#### 3.1.10.1 Pipe Penetrations

Provide galvanized Schedule 40 pipe sleeves and tightly pack annular space between sleeves and pipe with insulation having a flame spread rating not more than 25 and a smoke developed rating not more than 50 when tested in accordance with ASTM E84, CAN/ULC-S102 maximum effective temperature 538 degrees C, bulk density 96 kg/cu. meter minimum. Provide uninsulated pipe with a 25 mm thick mineral fiber sleeve the full length of the penetration and seal each end with an interior or exterior and weather resistant non-hardening compound. Provide sealant and mineral-fiber sleeve of a flame spread rating not more than 25 and a smoke developed rating not more than 50 when tested in accordance with ASTM E84.

#### 3.1.10.2 Duct Penetrations

Pack openings around ducts with mineral fiber insulation the full length of the penetration having a flame spread rating not more than 25 and a smoke developed rating not more than 50 when tested in accordance with ASTM E84. At each end of duct opening provide sealing collars and seal with an interior or exterior and weather resistant non-hardening compound.

#### 3.1.10.3 Ducts Passing Through Equipment Rooms

Provide with sound insulation equal to the sound attenuation value of the wall, floor, or ceiling penetrated.

### 3.1.11 Machinery Foundations and Subbases

Provide cast in place anchor bolts as recommended by the machinery manufacturer.

#### 3.1.11.1 Machinery Subbases

Provide concrete subbases at least 102 mm high for floor mounted equipment. Rest subbases on structural floor and reinforce with steel rods interconnected with floor reinforcing bars by tie bars hooked at both ends. Provide at least 50 mm clearance between subbases and inertia bases, steel bases, and steel saddles with machinery in operation.

#### 3.1.11.2 Common Machinery Foundations

Mount electrical motors on the same foundations as driven machinery. Support piping connections, strainers, valves, and risers on the same foundation as the pumps.

#### 3.1.11.3 Foundation and Subbase Concrete

Cast concrete foundations and subbases of ASTM C94/C94M 20 MPa concrete reinforced with steel bars as indicated or recommended by machinery manufacturer.

#### 3.1.11.4 Anchor Bolts and Grout

Secure machinery to foundations and inertia bases with anchor bolts. Grout equipment with baseplates, the full area under baseplates with premixed non-shrinking grout. After grout has set, remove wedges, shims, and jack bolts and fill spaces with grout.

#### 3.1.12 Inertia Bases

Install inertia bases in accordance with the recommendations of the machinery manufacturer or inertia base manufacturer, as applicable.

#### 3.1.13 Suspended Machinery Platforms

Provide with vibration-isolation hangers.

#### 3.1.14 Electrical Connections

Provide flexible conduit or multiple conductor cable connections for machinery with sufficient extra length to permit 50 mm minimum displacement in any direction without damage.

#### 3.1.15 Systems Not To Be Vibration Isolated

Do not provide vibration isolation for electrical raceways and conduits or for fire protection, storm, sanitary, and domestic water piping systems which do not include pumps or other vibrating, rotating, or pulsating equipment including control and pressure reducing valves.

### 3.2 FIELD QUALITY CONTROL

Provide equipment and apparatus required for performing inspections and tests. Notify Contracting Officer 14 days prior to machinery sound vibration seismic testing. Rebalance, adjust, or replace machinery with noise or vibration levels in excess of those given in the machinery specifications, or machinery manufacturer's data.

#### 3.2.1 Field Inspections

Prior to initial operation, inspect the vibration isolators for conformance to drawings, specifications, and manufacturer's data and instructions. Check for vibration and noise transmission through connections, piping, ductwork, foundations, and walls. Check connector alignment before and after filling of system and during operation. Correct misalignment without damage to connector and in accordance with manufacturer's recommendations.

#### 3.2.2 Spring Isolator Inspection

After installation of spring isolators or protected spring isolators, and seismic restraint devices, the machinery must rock freely on its spring isolators within limits of stops or seismic restraint devices. Eliminate or correct interferences.

#### 3.2.3 Tests

Adjust, repair, or replace isolators as required to reduce vibration and noise transmissions to specified levels.

### 3.2.3.1 Equipment Vibration Tests

Perform vibration tests to determine conformance with vibration isolation schedule specified specified.

-- End of Section --

## SECTION 23 03 00.00 20

## BASIC MECHANICAL MATERIALS AND METHODS

**08/10, CHG 3: 08/18**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## ASTM INTERNATIONAL (ASTM)

ASTM B117 (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

## INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2 (2023) National Electrical Safety Code

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

NEMA MG 10 (2017) Energy Management Guide for Selection and Use of Fixed Frequency Medium AC Squirrel-Cage Polyphase Induction Motors

NEMA MG 11 (1977; R 2012) Energy Management Guide for Selection and Use of Single Phase Motors

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

## 1.2 RELATED REQUIREMENTS

This section applies to all sections of Divisions: 21, FIRE SUPPRESSION; 22, PLUMBING; and 23, HEATING, VENTILATING, AND AIR CONDITIONING of this project specification, unless specified otherwise in the individual section.

## 1.3 QUALITY ASSURANCE

## 1.3.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

### 1.3.2 Alternative Qualifications

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

### 1.3.3 Service Support

The equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

### 1.3.4 Manufacturer's Nameplate

For each item of equipment, provide a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

### 1.3.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

#### 1.3.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions must be considered mandatory, the word "should" is interpreted as "must." Reference to the "code official" must be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" must be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" must be interpreted to mean the "lessor." References to the "permit holder" must be interpreted to mean the "Contractor."

#### 1.3.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, must be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

## 1.4 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace

damaged or defective items.

## 1.5 ELECTRICAL INSTALLATION REQUIREMENTS

Electrical installations must conform to IEEE C2, NFPA 70, and requirements specified herein.

### 1.5.1 New Work

Provide electrical components of mechanical equipment, such as motors, motor starters, control or push-button stations, float or pressure switches, solenoid valves, integral disconnects, and other devices functioning to control mechanical equipment, as well as control wiring and conduit for circuits rated 100 volts or less, to conform with the requirements of the section covering the mechanical equipment. Extended voltage range motors are not to be permitted. The interconnecting power wiring and conduit, control wiring rated 120 volts (nominal) and conduit, and the electrical power circuits must be provided under Division 26, except internal wiring for components of package equipment must be provided as an integral part of the equipment. When motors and equipment furnished are larger than sizes indicated, provide any required changes to the electrical service as may be necessary and related work as a part of the work for the section specifying that motor or equipment.

### 1.5.2 Modifications to Existing Systems

Where existing mechanical systems and motor-operated equipment require modifications, provide electrical components under Division 26.

### 1.5.3 High Efficiency Motors

#### 1.5.3.1 High Efficiency Single-Phase Motors

Unless otherwise specified, single-phase fractional-horsepower alternating-current motors must be high efficiency types corresponding to the applications listed in NEMA MG 11.

#### 1.5.3.2 High Efficiency Polyphase Motors

Unless otherwise specified, polyphase motors must be selected based on high efficiency characteristics relative to the applications as listed in NEMA MG 10. Additionally, polyphase squirrel-cage medium induction motors with continuous ratings must meet or exceed energy efficient ratings in accordance with Table 12-6C of NEMA MG 1.

### 1.5.4 Three-Phase Motor Protection

Provide controllers for motors rated one 1.34 kilowatts and larger with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage. Provide protection for motors from immediate restart by a time adjustable restart relay.

## 1.6 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system.



Instructors must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work.

Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished must be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

### 1.7 ACCESSIBILITY

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

## PART 2 PRODUCTS

Not Used

## PART 3 EXECUTION

### 3.1 PAINTING OF NEW EQUIPMENT

New equipment painting must be factory applied or shop applied, and must be as specified herein, and provided under each individual section.

#### 3.1.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors must withstand 500 hours in a salt-spray fog test. Salt-spray fog test must be in accordance with ASTM B117, and for that test the acceptance criteria must be as follows: immediately after completion of the test, the paint must show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen must show no signs of rust creepage beyond 3 mm on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment must not be less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 50 degrees C, the factory painting system must be designed for the temperature service.

#### 3.1.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except metal surfaces subject to temperatures in excess of 50 degrees C must be cleaned to bare

metal.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Color of finish coat must be aluminum or light gray.

- a. Temperatures Less Than 50 Degrees C: Immediately after cleaning, the metal surfaces subject to temperatures less than 50 degrees C must receive one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm, one coat of primer applied to a minimum dry film thickness of 0.0255 mm; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm per coat.
- b. Temperatures Between 50 and 205 Degrees C: Metal surfaces subject to temperatures between 50 and 205 degrees C must receive two coats of 205 degrees C heat-resisting enamel applied to a total minimum thickness of 0.05 mm.
- c. Temperatures Greater Than 205 Degrees C: Metal surfaces subject to temperatures greater than 205 degrees C must receive two coats of 315 degrees C heat-resisting paint applied to a total minimum dry film thickness of 0.05 mm.

-- End of Section --

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## SECTION 23 05 93

TESTING, ADJUSTING, AND BALANCING FOR HVAC  
05/25

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 203 (1990; R 2011) Field Performance  
Measurements of Fan Systems

## ASSOCIATED AIR BALANCE COUNCIL (AABC)

AABC MN-1 (2016, 7th ed) National Standards for  
Total System Balance

## NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)

NEBB PROCEDURAL STANDARDS (2015) Procedural Standards for TAB  
(Testing, Adjusting and Balancing)  
Environmental Systems

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70E (2024) Standard for Electrical Safety in  
the Workplace

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION  
(SMACNA)

SMACNA 016 (2012) HVAC Air Duct Leakage Test Manual -  
2nd Edition

SMACNA 1780 (2023) HVAC Systems - Testing, Adjusting  
and Balancing, 4th Edition

## 1.2 DEFINITIONS

In some instances, terminology differs between this Contract and the TAB certifying body or the TAB Standard primarily because this section intends to use the industry standards specified, along with additional requirements listed within this section, to produce optimal results.

- a. AABC: Associated Air Balance Council
- b. Certifying Body: An independent body responsible for issuing a formal recognition, such as a certificate, proving that the certification meets specific national or international requirements.
- c. COTR: Contracting Officer's Technical Representative

- d. DALT: Duct Air Leakage Testing (of HVAC duct systems such as supply air (SA), return air (RA), exhaust air (EA), relief air (LA), and outdoor air (OA)).
- e. DALT Failure: This phrase means "a leakage rate measured during DALT Field Work and Acceptance Testing which exceeds the allowable leakage rate for the duct test pressure and leakage class indicated."
- f. Duct System: When applied to DALT, this phrase means "complete duct system, inclusive of all ductwork, plenums, mains, branches, fittings and duct-mounted components and appurtenances, such as manual balancing dampers, control dampers, access doors, fire dampers, and duct-mounted coils, up to, but excluding flexible ducts and air-handling equipment, such as air-handling units (AHUs), dedicated outdoor air system (DOAS) units, fan coil units (FCUs), water source heat pumps (WSHPs), and variable-air volume (VAV) terminal units."
- g. HVAC: Heating, ventilating, and air-conditioning
- h. TAB: Testing, Adjusting, and Balancing (of HVAC systems).
- i. TAB Failure: This phrase means "a measurement taken during TAB Field Acceptance Testing which does not fall within the specified Acceptance Tolerance for a specific parameter."
- j. TAB Firm: AABC Certified Member Agency, NEBB Certified Firm, TABB Certified Contractor
- k. TAB Standard: Standard utilized by TAB Team based on TAB Firm's certifying body:
  - (1) AABC - National Standards for Total System Balance (AABC MN-1)
  - (2) NEBB - Procedural Standard for Testing, Adjusting and Balancing of Environmental Systems (NEBB PROCEDURAL STANDARDS)
  - (3) TABB - HVAC Systems Testing, Adjusting & Balancing Manual (SMACNA 1780)
- l. TAB Team Supervisor: AABC Test and Balance Engineer (TBE), NEBB TAB Certified Professional (TAB CP), TABB TAB Certified Supervisor
- m. TAB Team: TAB Firm personnel
- n. TABB: Testing, Adjusting and Balancing Bureau

### 1.3 GENERAL REQUIREMENTS

The work includes DALT and TAB of new HVAC systems, which are located within, on, under, between, and adjacent to buildings.

Perform TAB in accordance with the requirements of the TAB Standard identified by the TAB Firm's certifying body, as supplemented and modified by this specification section. Perform DALT in compliance with the procedures specified in SMACNA 016, as supplemented and modified by this specification section.

### 1.3.1 Support Personnel

Ensure the TAB Team Supervisor or TAB Team Field Leader is present full-time on the Contract site for overall management and direct observation of the DALT and TAB Field Work. Certified TAB Team Field Technicians may work independently of direct observation by the TAB Team Supervisor or Field Leader.

Ensure the technical personnel, such as factory representatives and HVAC controls contractor technicians, to operate HVAC equipment are present to enable the TAB Team to accomplish the DALT and the TAB measurement work. Provide the personnel to install field designated test ports and test apparatus connections.

### 1.3.2 Coordination

Coordinate testing schedules with work in other Sections performed on the components or systems to be tested. Ensure support personnel are present at the times required by the TAB Team and cause no delay in the DALT and the TAB Field Work.

## 1.4 RELATED REQUIREMENTS

Requirements for climate conditions during building pressure testing are specified in Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING.

Requirements for thermal insulation of ductwork, piping, fittings, and valves are specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

Requirements for coordination with commissioning are specified in Section 23 08 00 COMMISSIONING OF MECHANICAL AND PLUMBING SYSTEMS.

Requirements for calibration and testing of HVAC controls system devices are specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

Requirements for Detail Drawings, construction and sealing of HVAC air distribution systems are specified in Section 23 30 00 HVAC AIR DISTRIBUTION.

## 1.5 SUBMITTALS

Unless otherwise specified or directed by the Contracting Officer, provide submittals in electronic format.

Government approval is required for submittals with a "G" or "S" classification.

### SD-01 Preconstruction Submittals

TAB Firm And Personnel Qualifications; G

Design Review Report; G

TAB Pre-Field Engineering Plan; G

### SD-06 Test Reports

Certified Final DALT Report; G

## 1.6 TAB FIRM AND PERSONNEL QUALIFICATIONS

Engage the services of a qualified TAB Firm to perform DALT and TAB work for this project. Only approved firm and personnel are allowed to perform DALT and TAB work on this Contract. Comply with the following:

- a. Certification: Maintain TAB Firm and personnel certifications for the entire duration of duties specified. Renew Certifications which expire prior to completion of the TAB work in a timely manner so there is no lapse in certification. Provide written notice to the Contracting Officer, with a copy of renewed certificates of the TAB Firm and personnel, prior to expiration of the previous certificate(s). If, for any reason, the TAB Firm or its employee loses subject certification during this Contract, immediately notify the Contracting Officer in writing and comply with the requirements for the replacement of TAB Firm and TAB Team Members. An approved successor must validate all work performed for this project by the TAB Firm or TAB Team Member who lost certification.
- b. Replacement of TAB Firm and TAB Team Members: Replacement or substitution of firm or individual members may occur if each complies with the applicable qualifications and is approved by the Contracting Officer prior to the performance of its duties on this Contract. Submit replacement TAB Firm and Personnel Qualifications in accordance with this section.

Submit the TAB Firm and Personnel Qualifications within 60 calendar days after the date of Design Acceptance.

### 1.6.1 TAB Firm

Provide documentation that the TAB Firm is a first tier subcontractor to the prime Contractor and an independent company, not participating in any other work on this Contract or affiliated to any other company participating in work on this Contract, including, but not limited to, design, furnishing equipment, or construction. No work may be performed by a second tier subcontractor.

Provide a copy of current certificate, with expiration date, to certify that the TAB Firm is certified by AABC, NEBB, or TABB in categories and functions of the DALT and TAB field work specified. Designate the single certifying body to be utilized when the firm is certified by more than one.

### 1.6.2 TAB Personnel

For each TAB Team role below, provide at least one qualified, full-time employee of the TAB Firm for approval. Include the proposed role(s) of each employee with their qualification documentation. It is acceptable for a TAB Team member to perform duties of another TAB Team role if they meet the qualifications of both.

#### a. TAB Team Supervisor

Provide documentation, including full name and a copy of current certificate, that the TAB Team Supervisor is a AABC TBE, a NEBB TAB CP, or a TABB TAB Certified Supervisor.

#### b. TAB Team Field Leader

Include full name, a copy of current certificate, and a list of projects, roles performed, and associated dates.

c. TAB Team Field Technician

Include full name, a copy of current certificate when applicable, and a list of projects, roles performed, and associated dates.

1.7 TAB STANDARD

All recommendations and suggested practices contained in the TAB Standard are mandatory. Maintain a hard copy and an electronic PDF file copy of the standard on site.

All quality assurance provisions of the TAB Standard, such as performance guarantees, are part of this Contract. For systems or system components not covered in the TAB Standard, TAB procedures and report forms must be developed by the TAB Team Supervisor and included in the TAB Pre-Field Engineering Plan. Where new procedures and requirements, applicable to the Contract requirements have been published or adopted by the certifying body responsible for the TAB Standard, the more stringent of the requirements and recommendations contained in these procedures and requirements are considered mandatory.

1.8 INSTRUMENTATION

It is the responsibility of the TAB Firm to use instrumentation that meets the requirements of the TAB Standard. Select appropriate instruments to achieve the applicable TAB tolerances. Instrumentation used for taking air wet-bulb temperature readings must provide accuracy of a minimum of plus or minus 5 percent at the measured face velocities. Where the instrument manufacturer calibration recommendations are more stringent than those listed in the TAB Standard, adhere to the manufacturer's recommendations. Instrumentation must be in proper operating condition and must be applied in accordance with the instrumentation's manufacturer recommendations.

All instrumentation must bear a valid National Institute of Standards and Technology (NIST) traceable calibration certificate, dated within one year, during field work and Field Acceptance Testing.

1.9 DESIGN REVIEW

The TAB Team Supervisor must review the Contract Plans and Specifications to advise the Contracting Officer of any omissions and deficiencies that would prevent the effective and accurate TAB of the systems in compliance with the requirements of this section. Accomplish the following:

- a. Verify in the documents the presence and location of permanently installed test ports and other devices needed, including gauge cocks, thermometer wells, flow control devices, flow measuring devices, balancing valves, and manual balancing dampers.
- b. Confirm the inclusion of the design values necessary to complete the DALT and TAB report forms required for the TAB Pre-Field Engineering Plan.
- c. Prepare a report containing a detailed listing of the items identified



during the review. Provide the omission and deficiency listing in a tabular format, with each item sequentially numbered and with a reference to the applicable design document. For each omission and deficiency, provide a complete explanation including supporting documentation detailing the design deficiency. If no deficiencies are evident, state so in the report. Include the TAB Related Submittals List in the report.

Submit the Design Review Report to the Contracting Officer for approval within 30 calendar days after approval of the TAB Firm and Personnel Qualifications.

#### 1.10 TAB PRE-FIELD ENGINEERING PLAN

The TAB Team Supervisor must utilize the TAB Standard, Contract Plans, Contract Specifications, approved contract revisions, and contract change orders to prepare a plan for the DALT and TAB field work, including procedures, report forms, diagrams, instrumentation and other supporting documents in compliance with the following:

- a. Project Summary: Provide a detailed narrative describing the DALT and TAB Field Work to be performed, including systems and equipment. Clearly distinguish between DALT information and TAB information.
- b. Step-by-Step Procedures: Provide a list of the intended procedural steps for DALT and TAB Field Work from start to finish. Include the following:
  - (1) The intended procedural steps for TAB work for systems, subsystems, and system components. TAB Standard procedure excerpts may be provided to supplement this requirement.
  - (2) A description of how each type of data measurement will be obtained.
  - (3) A list of Contractor support personnel required for each step, and tasks they need to perform.
- c. Report Forms: Provide the report forms to be used in the DALT and TAB reports to document the DALT and TAB Field Work, with the preliminary information and design values filled in. Utilize contract document design HVAC systems values. For TAB report forms, use forms inclusive of all data on TAB Standard sample report forms, or generate where no TAB Standard form exists, for all data collection requirements. Ensure report forms contain data fields for each HVAC controls system flow measuring device's calibration, including design flow rate, actual flow rate, controls system reading and factor. For DALT report forms, use forms comparable to the "Air Duct Leakage Test Summary Report Forms" located in SMACNA 016, with an example of design and test data in the plan. Coordinate the key numbering used in the report forms with that used in the TAB Schematic Drawings.
- d. TAB Schematic Drawings: Provide the schematic drawings to be used in the required reports, which may include building floor plans, mechanical room plans, equipment schedules, duct system plans, and equipment elevations. Use the contract drawings and, if available, ductwork shop drawings to provide the following:
  - (1) Air moving equipment and pump schedules.

- (2) Thermal energy transfer equipment schedules.
- (3) Duct Construction and Leakage Testing Schedule.
- (4) Air system plans showing the location and airflow quantities of all supply, return, exhaust, relief, outdoor, and transfer air terminal devices such as registers, grilles, diffusers, louvers, and rooftop hoods and their respective airflow quantities.
- (5) A key numbering system on the drawings which identifies each piece of equipment and each air terminal device contained in the air terminal airflow report sheets. It is acceptable to utilize the contract drawing numbering system when provided for equipment.
- (6) A unique number or mark for each water flow measurement fitting and balancing valve. It is acceptable to utilize the contract drawing numbering system when provided for valves and fittings.
- (7) Intended TAB measurement locations, including where test ports need to be provided by the Contractor.

For Static Pressure Profile and Duct Traverse Equipment Diagrams, provide diagrams, representative of the actual equipment configuration, developed from TAB Standard forms, contract drawings or approved equipment submittals.

For DALT schematic drawings, provide a representative diagram matching the example DALT report form in this plan.

- e. Instrumentation: Provide a list of DALT and TAB instruments for this project. Include the instrument name and description, manufacturer, model number, scale range, published accuracy, most recent calibration date, and what the instrument will be used for on this project. Submit instrument manufacturer's literature to document instrument accuracy performance is in compliance with that specified. Provide a copy of the calibration certificate for each calibrated instrument to be used.
- f. Pre-DALT Field Work Checklist: Provide a thorough checklist of the work items and inspections that need to be accomplished before DALT Field Work can be performed. Include signature and date lines for the Contractor's quality control representative and mechanical contractor's representative acknowledging completion of the marked checklist items.
- g. Pre-TAB Field Work Checklist: Include items such as completion of applicable HVAC controls systems installation, cleaning of hydronic system including but not limited to strainers, and replacement of HVAC system air filters. Each section must include signature and date lines for the Contractor's quality control representative and mechanical contractor's representative acknowledging completion of the marked checklist items.

Submit the TAB Pre-Field Engineering Plan to the Contracting Officer for approval a minimum of 60 calendar days in advance of commencement of DALT TAB field work.

### 1.11 PRE-DALT/TAB MEETING

Meet with the Contracting Officer's Technical Representative (COTR) to develop a mutual understanding relative to the details of the DALT and TAB work requirements. Requirements to be discussed include required submittals, work schedules, and field quality control. Ensure that the TAB Team Supervisor and a representative for the sheet metal contractor, mechanical contractor, electrical contractor, and HVAC controls system contractor are present at this meeting. Conduct the Pre-DALT/TAB Meeting a minimum of 30 calendar days in advance of the commencement of ductwork installation.

Provide written notice to the Contracting Officer of the proposed date of the Pre-DALT/TAB Meeting at a minimum of 14 calendar days prior to the proposed date.

### 1.12 REPORTS

#### 1.12.1 TAB Reports

- a. **Project Summary:** Provide a project narrative, including systems and equipment tested, describing the outcome of the testing and balancing and include a list of items documenting system variations deviating from the design tolerances, any abnormalities discovered, and any outstanding items with proposed corrective action. When applicable, provide notations describing how actual field procedures differed from those approved in the TAB Pre-Field Engineering Plan. If no procedures differed, state so in the report.
- b. **Certification:** Include the TAB Team Supervisor's physical or electronic stamp, issued by the certifying body, and dated signature.
- c. **Data from TAB Field Work:** Prepare the TAB field data using the report forms approved in the TAB Pre-Field Engineering Plan and any other data sheets necessary to document the required final system balance and demonstrate accuracy and repeatability during TAB Field Acceptance Testing. For each unique calibration factor utilized, report the traverse airflow, the flow measuring instrument reading (e.g., flow hood airflow), the resulting correction factor, and the location of the traverse taken. Include notation on each form with the date the form's data was collected and with any Supporting Document reference number. Provide a detailed explanation wherever a final measurement did not achieve the required value or fall within the required range of values. Include the following:
  - (1) **System Configuration:** Clearly identify system configurations and conditions affecting data for all reported data. Include system operational parameters such as device positioning, system diversity, modes of operation, and setpoints necessary to set up and duplicate the system configuration during the TAB Field Acceptance Testing. Include final hydronic differential pressure setpoint, hydronic system fill pressure, glycol percentage, pump and fan motor frequency in maximum, fan motor frequency in minimum, fan brake horsepower, calibration coefficients and factors, control damper positions, and primary air static pressure setpoint.
  - (2) **Static Pressure Profiles:** Provide static pressure profiles in accordance with this section. Illustrate pressures in an

Equipment Diagram representative of the actual installation. Include design static pressure and differential pressure values for comparison where provided.

- (3) Duct Traverses: Provide duct traverse measurement data in accordance with this section. Include individual velocities from the duct traverses taken. Illustrate, in an Equipment Diagram, the actual entering and leaving duct traverse airflows for each piece of air-handling equipment exceeding 944 L/S.
- (4) Air and Water Systems' Open Paths: Identify air and hydronic system open paths in the report form.
- (5) Motor Data: Provide motor current (amperage) and voltage for each phase of motors, including those powered by VFDs. Include current setting for thermal overload protection for motors, including starter or VFD brand, model, enclosure type, installed overload devices, rating(s) or rating range, and setpoints, and revised device ratings and setpoints when applicable.
- (6) Thermal Performance Data: Include capacity field data and design and calculated capacity, along with ambient outdoor temperature data, on thermal performance report forms. Include notation to identify system configuration settings that deviate from the design sequences of operation and setpoints when necessary to obtain the design capacity.
- d. Flow Measuring Device Calibration: Provide calibration data at each scheduled design volumetric flow rate.
- e. Instruments: Provide an updated list of the instruments used to measure the TAB data. Include in the listing each instrument's unique identification number, calibration date, and calibration expiration date. Include a copy of the instrument calibration certificate when updated from the copy in the approved TAB Pre-Field Engineering Plan.
- f. Performance Curves: Provide copies of the factory pump curves and fan curves for pumps and fans included on the job, and manufacturer equipment data curves or tables correlating pressure drop and water flow rate. Include marks or notation indicating the reported final balance values on each curve.
- g. Calibration Charts: Provide a factory calibration chart, table, graph, or diagram, as applicable, for each applicable balancing valve and flow measuring device such as a venturi flow meter and an orifice flow meter.
- h. TAB Schematic Drawings: Provide updated drawings and diagrams with the final installed locations of all terminals and devices, any numbering changes, and actual test locations including duct traverse and static pressure measurements. Indicate the air terminal device and water coil of the open paths for those identified.
- i. Mechanical Equipment Schedules: Include a copy of the mechanical equipment schedules.
- j. Supporting Documents: Provide copies of any request for information (RFI) with the RFI response; summaries of implemented change order(s); meeting minutes with participants; electronic mail with addresses; and, other documentation substantiating any deviations of the reported

design values from the initial contract design documents. Include this documentation in an appendix to the TAB report with sequential numbering of each separate document for reference in the data presentation forms.

- k. Guarantee: Provide documentation substantiating the inclusion of the quality assurance program such as a certificate of conformance or performance guarantee issued by the TAB Firm's certifying body.

#### 1.13 SEQUENCING AND SCHEDULING

##### 1.13.1 Submittal and Work Schedule

Ensure sufficient time is scheduled to complete each item, including Government review period for submittals. The order of items listed below does not imply a specified sequence:

- a. Submission of TAB Firm and Personnel Qualifications
- b. Submission of Design Review Report
- c. Submission of TAB Pre-Field Engineering Plan
- d. Pre-DALT/TAB Meeting
- e. DALT Field Work
- f. Submission of Pre-Final DALT Reports
- g. DALT Field Acceptance Testing
- h. Submission of Certified Final DALT Report
- i. Proportional Balancing TAB Field Work

#### PART 2 PRODUCTS

Not used.

#### PART 3 EXECUTION

##### 3.1 DALT PROCEDURES

###### 3.1.1 Systems Requiring DALT

All duct systems, regardless of duct pressure classification, are subject to DALT including supply, return, exhaust, relief, and outdoor air. From each duct system indicated as subject to DALT, the Contracting Officer will randomly select sections of each completed duct system for testing such that the duct selections will not exceed 25 percent of the total measured linear footage of each duct system. Include notation within the associated Advance Notice indicating the specific unit(s) when electing the following for any single unit or quantity of similar units:

- a. It is acceptable for an entire duct system associated with a single unit (e.g., 100 percent of DOAS-1 supply air ductwork) to be leak tested instead of disassembling that system in order to DALT only the 25 percent duct selection specified. Testing an entire duct system for a unit does not reduce the DALT requirements for any other duct

system or selection.

- b. It is acceptable for the duct systems (e.g., FCU supply air and return air) for 25 percent of the total quantity of similar terminal units and light-duty air-handling equipment (e.g., WSHPs less than 5 tons cooling capacity, VAVs, and FCUs) to be leak tested instead of disassembling 25 percent of each duct system for each of these units.

### 3.1.2 Prerequisites for DALT Field Work

Complete the following prior to starting DALT Field Work:

- a. Submission and approval of the SD-01 Preconstruction Submittals.
- b. Installation and sealing in conformance with the contract documents of those duct systems indicated completed and ready for testing.
- c. Confirmation ductwork to be air leakage tested is accessible and not concealed or insulated.
- d. Work items and inspections indicated in the Pre-DALT Field Work Checklist.
- e. Furnish the TAB Team a copy of the ductwork layout Detail Drawings indicating the completed duct systems available for DALT.

### 3.1.3 Advance Notice for Pre-Final DALT Field Work

Provide the Contracting Officer with written notification including the proposed date for DALT and a copy of the ductwork layout Detail Drawings with indication and dimensions of the duct systems, or portion thereof, completed and ready for DALT. Include with the notice the signed and dated Pre-DALT Field Work Checklist, with all work items certified by the Contractor and mechanical contractor as completed.

Provide the notice after completion of the installation of duct systems indicated as ready for testing.

### 3.1.4 Ductwork Selections

From each duct system indicated within the Advance Notice of Pre-Final DALT Field Work, the Contracting Officer will select sections of each completed duct system for testing in accordance with this section. Maintain a hard copy of the drawings indicating ductwork selections on site.

Sealing of the ductwork selections is prohibited, with exception of temporary end caps and connection for test apparatus, from the time the Contractor is notified of the selections until DALT measurements are recorded.

### 3.1.5 DALT Field Work

Conduct DALT Field work on the ductwork selections in compliance with the procedures specified in SMACNA 016, as supplemented and modified by this section. Use a duct test pressure equivalent to each ductwork selection's pressure classification. Calculate the ductwork selection allowable leakage using the more stringent, lower numerical value, of the duct system leakage class for the duct pressure classification indicated on the

drawings or the maximum permitted leakage class defined as follows:

- a. Round/Oval Leakage Class = 3 and Rectangular Leakage Class = 4 at duct pressure classification of 1000 Pa and above.
- b. Round/Oval Leakage Class = 4 and Rectangular Leakage Class = 8 at duct pressure classification of 500 Pa and 750 Pa.
- c. Round/Oval Leakage Class = 8 and Rectangular Leakage Class = 16 at duct pressure classification of 250 Pa.

Complete DALT Field Work on the ductwork selections within 48 hours from receipt of selections, including disconnection, capping, measuring, and testing of ductwork. If any of the duct sections tested is a DALT failure, terminate testing for that system and report the duct selection as failed and the duct system as DALT terminated in the Pre-Final DALT Report.

### 3.1.6 Pre-Final DALT Report

Upon completion of the DALT Field Work, prepare a Pre-Final DALT Report. Prepare separate Pre-Final DALT Reports when DALT is not completed in a single event and differentiate these with a sequential numerical suffix such as Pre-Final DALT Report 1.

### 3.1.7 Quality Assurance

#### 3.1.7.1 DALT Field Acceptance Testing

In the presence of the Contracting Officer and TAB Team Field Leader, conduct field acceptance testing of a maximum of 50 percent of the duct sections in the approved Pre-Final DALT Report as selected by the Contracting Officer. If any of the duct sections tested is a DALT failure, terminate data checking for that section; the associated Pre-Final DALT Report data for the given duct selection will be disapproved.

Provide the Contracting Officer with written notification of the proposed date for DALT Field Acceptance Testing within 14 calendar days of approval of the Pre-Final DALT Report and a minimum of 7 calendar days in advance of proposed field acceptance testing date.

#### 3.1.7.2 Retesting and Additional Testing

For any duct section indicated as a DALT failure, make the necessary repairs. Repairs must be applied to similar conditions in all untested duct sections. For each DALT failure, and after repairs are completed, conduct DALT on the untested duct selection(s) of the terminated duct system, on the failed section, and one additional duct section as selected by the Contracting Officer. Prepare and resubmit the Pre-Final DALT Report with the original, retesting, and additional selection test results for approval. Reschedule field acceptance testing of the approved revised report with the Contracting Officer.

### 3.1.8 Certified Final DALT Report

Prepare the Certified Final DALT Report using data furnished by the TAB Team in the Pre-Final DALT Report(s).

Submit the Certified Final DALT Report to the Contracting Officer for approval within 14 calendar days of successful completion of all DALT Field Acceptance Testing.

### 3.2 TAB PROCEDURES

#### 3.2.1 Systems Requiring TAB

#### 3.2.2 TAB Tolerances

##### 3.2.2.1 System Tolerances

System tolerances apply to the design flow rates as specified or indicated in the contract documents. HVAC systems' tolerance is plus or minus 5 percent, except as indicated below:

##### a. Air Systems:

- (1) OA 23.5 L/s or less: plus 2.35 L/s to minus zero L/s
- (2) EA 23.5 L/s or less: plus zero L/s to minus 2.35 L/s
- (3) SA and RA of 100 percent recirculating single-zone air-handling equipment: plus or minus 10 percent
- (4) SA and RA 23.5 L/s or less: plus or minus 1.42 L/s

##### 3.2.2.2 Acceptance Tolerances

Field Acceptance Testing measurements must remain within the associated System Tolerance of the design flow rate. When the associated System Tolerance is plus or minus 10 percent, Field Acceptance Testing measurements must remain within the associated System Tolerance of the design flow rate and repeat within plus or minus 5 percent of the approved Certified TAB Report data.

#### 3.2.3 Advance Notice for TAB Field Work

Provide the Contracting Officer with written notification including the proposed dates for TAB Field Work and a list of the systems ready for TAB on the proposed start date. Include with the notice the signed and dated Pre-TAB Field Work Checklist, with the applicable work items certified by the Contractor and mechanical contractor as having been completed and working as designed.

#### 3.2.4 Proportional Balancing TAB Field Work

Conduct TAB Field Work in conformance with the TAB Standard, as supplemented and modified by this section. Test, adjust, and balance the HVAC systems until measured flow rates (air and water flow) are within the specified System Tolerances. Do not exceed maximum rated revolutions per minute (RPM) and rated full load amperage for equipment during any mode of operation, including with loaded filters and wet cooling coils. Variable speed drives must be capable of controlling to maximum and minimum flows with a drive operational range not less than 25 Hz. Operation exceeding 60 Hz is permitted only for direct-drive plenum fans rated for operation at higher frequencies. Follow the procedures specified in NFPA 70E during electrical measurements on exposed energized conductors or circuits. Additionally, this work includes the following:



- a. Adjustment of flow control and balancing devices such as balancing valves, balancing dampers, and sheaves.
- b. Adjustment of variable frequency drives and speed controllers.
- c. Calibration of each controls system flow measuring device at each scheduled design volumetric flow rate.
- d. Changing out fan sheaves if required to obtain flow rates specified or indicated.

#### 3.2.4.1 Prerequisites for TAB Field Work

Complete the following prior to starting Field Work:

- a. DALT Field Work and obtain approval of the Certified Final DALT Report.
- b. Installation of pressure and temperature test ports adjacent to inlet and outlet of every pump, heat exchanger, chiller, boiler, and coil, and as required by the TAB Team Supervisor.
- c. Work items and inspections indicated in the Pre-TAB Field Work Checklist, including providing new air filters and cleaning screens of hydronic system strainers.
- d. Enclosure of the building envelope according to the contract documents with the Air Barrier Pressure Test completed and the Air Leakage Test Reports and Diagnostic Test Reports approved. Hydronic system TAB field work may commence prior to air barrier test completion.
- e. Manufacturer's equipment start-up forms for each piece of equipment to be tested.
- f. HVAC controls system installation and start-up for each piece of equipment to be tested.

#### 3.2.4.2 TAB Air Distribution Systems

Balance systems to minimize throttling losses with at least one open path, with fully open balancing dampers from the fan discharge to an outlet and from fan suction to an inlet, for fans greater than 1 hp.

##### a. Static Pressure Profile Work

Measure and record static pressure profiles under maximum airflow conditions for air systems of 944 L/S and greater.

Measure static pressure data for supply, return, relief, exhaust, and outdoor air ducts for the systems. The static pressure measurement data must include, in addition to TAB Standard requirements, the following:

- (1) Inlet and discharge static pressures for fans.
- (2) Static pressure drop across coils and heat reclaim devices installed in unit cabinetry or the system ductwork.
- (3) Static pressure drop across automatic control dampers.

- (4) Static pressure drop across air filters, acoustic silencers, moisture eliminators, airflow straighteners, airflow measuring stations, and other pressure drop producing specialty items installed in unit cabinetry or in the system ductwork. Examples of these specialty items are smoke detectors, white sound generators, RF shielding, wave guides, security bars, blast valves, and duct mounted humidifiers.

Do not report static pressure drop across duct fittings provided for the sole purpose of conveying air, such as elbows with or without turning vanes, transitions, offsets, plenums, manual dampers, and branch take-offs.

- (5) Report static pressure drop across louvers.
- (6) Report static pressure readings in the duct at the point where the ducts connect to air-handling equipment.

b. Duct Traverse Work

Perform duct traverses, in addition to TAB Standard requirements, for main supply, return, exhaust, relief and outdoor air ducts for each piece of air-handling equipment exceeding 472 L/s. Include those ducts which lack 7 1/2 duct diameters upstream and 3 duct diameters downstream of straight duct unobstructed by duct fittings, offsets, and elbows. Perform additional duct traverses as shown on the contract drawings. Evaluate the suitability of the duct traverse measurement based on satisfying the qualifications for a satisfactory pitot traverse plane as defined by AMCA 203.

c. Building Pressurization Work

Perform measurements upon completion of air system balancing and during climate conditions suitable for a pressure test in accordance with Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING. Record building differential pressure for all sides of the building on each floor where openings allow measurements by instrumentation. Report all system setup parameters affecting building pressure measurement (e.g., exhaust, relief and outdoor airflow, damper positions, and fan speeds) and indicate wind speed during time of building pressure measurements. Measure in maximum and minimum building systems configuration.

Measurements must occur during building system setup with building exterior doors and windows closed and as follows:

- (1) Occupied Mode; air-handling equipment and fans in maximum cooling, maximum design airflow.
- (2) Occupied Mode; air-handling equipment in Demand Controlled Ventilation Occupied-Standby Mode.

3.2.4.3 TAB Hydronic Distribution Systems

Balance systems to minimize throttling losses with at least one path, with fully open balancing valves, from the pump discharge to a terminal device and back to the pump suction, for pumps greater than 1 hp. Measure and record flow rates at all balancing valves and flow measuring devices.

#### 3.2.4.4 Marking of Settings and Test Ports

Upon completing each item of work for TAB Field Work specified and prior to TAB Field Acceptance Testing, the TAB Team is to permanently mark the settings of HVAC adjustment devices including valves, gauges, and dampers so that adjustment can be restored if disturbed at any time. Secure balancing damper handles and set valve memory stops. Label variable frequency drives with final frequency (Hz) and control setpoint. Provide permanent markings clearly indicating the settings on the adjustment devices which result in the data reported on the Certified TAB Report. Permanently and legibly mark and identify the duct test ports. If the ducts have external insulation, make these markings on the exterior side of the insulation. Show the location of test ports on the as-built mechanical drawings with dimensions given where the test port is covered by exterior insulation.

#### 3.2.5 Thermal Performance TAB Field Work

##### 3.2.5.1 Prerequisites for Thermal Performance Field Work

Complete the following prior to starting Field Work:

- a. Proportional Balancing TAB Field Work.
- b. Sound Assessment Field Work.
- c. Work items and inspections indicated in the Pre-TAB Field Work Checklist, including confirming clean filters and strainers.

##### 3.2.5.2 Thermal Performance Testing

Conduct thermal performance (capacity) tests to verify thermal energy transfer components, devices, and equipment meet the indicated design capacity. This work includes:

- a. Verifying, using the same procedures and instruments for proportional balancing TAB field work, flows (air and water) remain within the System Tolerances. If the measurements exceed the specified System Tolerances, report these as deficiencies in accordance with this section.
- b. Measuring water temperatures through immersion into fluid stream.
- c. For ducted units, measuring airflows via duct traverse.
- d. Measuring the range, from minimum to maximum, of the outdoor ambient dry-bulb temperature and corresponding dew-point temperature within which the TAB data was recorded. Record these temperature ranges on each day of data measurements.

Report all design data, actual field measurements, and capacity calculations for the components, devices, and equipment below.

##### 3.2.5.3 Units with Coils

Report heating and cooling performance tests for heating and cooling coils such as chilled-water, direct-expansion:

- a. For units with capacities greater than 26,370 Watts cooling, such as

factory manufactured units, central built-up units and rooftop units, determine the apparent air-side coil capacity by calculations utilizing direct measurement of airflow, and measurements of entering and leaving wet-bulb and dry-bulb temperatures for cooling capacity and dry-bulb temperatures only for heating capacity. Calculate water-side coil capacity utilizing direct measurements of water flow rate, and entering and leaving water temperatures. Measure and record coil water pressure drop.

Measure and report dry-bulb temperature and relative humidity within the supply air duct downstream of each cooling coil. Report the corresponding dewpoint temperature.

- b. For units with capacities of 26,370 Watts or less, such as fan coil units, duct mounted reheat coils associated with VAV terminal units, and unitary units, determine the apparent air-side coil capacity by calculations using direct measurement of airflow, single point measurement centered within the airstream of entering and leaving wet-bulb and dry-bulb temperatures for cooling capacity and dry-bulb temperatures only for heating capacity.

#### 3.2.5.4 Units with Heat Recovery Devices

Report heating and cooling energy recovery performance tests for energy recovery devices such as wheels, coils, and fixed plate exchangers.

Report total cooling capacity, heating capacity, and energy recovery effectiveness. For outdoor air and exhaust airstreams measure and report airflows and entering and leaving wet-bulb and dry-bulb temperatures across heat recovery device. For water-side calculations, utilize direct measurements of water flow rate and entering and leaving water temperatures. Measure and record water pressure drops.

#### 3.2.5.5 Thermal Energy Transfer Equipment

Report heating and cooling performance tests for thermal energy transfer equipment (e.g., boilers, chillers, and cooling towers). Measure and report water flow rates, water-side pressure drops, and entering and leaving water temperatures. Report capacities.

#### 3.2.6 Deficiencies

Strive to meet the intent of this section to maximize the performance of the equipment as designed and installed. However, if deficiencies in equipment or system design or installation prevent TAB work from being accomplished in accordance with this section, the TAB Team Supervisor must submit written notification within five working days from discovery of all such deficiencies for resolution. The TAB Firm must issue notice concurrently to the Contractor for corrective action and to the Contracting Officer for information. Provide a complete explanation, including supporting documentation, detailing the deficiencies and proposed corrective actions in the notification. Where deficiencies are encountered that adversely impact successful completion of:

- a. TAB Field Work: Resolve deficiencies prior to submission of the Certified TAB Report.
- b. TAB Field Acceptance Testing: The Contractor must, within five working days of the TAB Firm notice, submit written notification

directly to the Contracting Officer, with a separate copy to the TAB Firm, of all such deficiencies, the intended or implemented corrective action, and the planned or actual date(s) for completion of each corrective action.

### 3.2.7 Certified TAB Reports

Upon completing each item of work for TAB Field Work specified, prepare a TAB Report in accordance with this section using data furnished by the TAB Team. If there are no comments generated during the Certified TAB Report review process, state so in the report.

Provide a copy of the approved Certified TAB Report data in an electronic spreadsheet file, along with all supporting graphs, charts, and curves, within 21 calendar days of receipt of TAB Report approval.

### 3.2.8 Quality Assurance

#### 3.2.8.1 TAB Field Acceptance Testing

In the presence of the Contracting Officer and TAB Team Field Leader, conduct field acceptance testing of points and areas in the approved Certified TAB Report as selected by the Contracting Officer. The selections include data types such as water and air volumetric flow rates, velocity, pressure, rotational speed (RPM), temperature, and electrical current recorded in the TAB Report. Measurement and test procedures are the same as required for TAB Field Work. If any of the measurements is a TAB failure, the Contracting Officer may terminate data verification. The sample group will be disapproved and require retesting and additional testing.

#### 3.2.8.2 Sample Strategy

Field acceptance testing includes verification of selections using the sample strategy indicated for each Category. For each of the following Categories, the Contracting Officer will select a sample group for testing:

Category 1: 100 percent of the air system equipment such as DOAS units, AHUs (rooftop and central stations), return fans, computer room air-conditioning (CRAC) units, and energy recovery units (ERUs) and 25 percent of the associated air terminal devices.

Category 2: 100 percent of the water system equipment such as chillers, boilers, cooling towers, pumps, and Category 1 equipment's coils.

Category 3: 25 percent of the air terminal units (e.g., VAV terminal units, FCUs), and of the exhaust fans, supply fans, and relief fans and 100 percent of the selected equipment's air terminal devices.

Category 4: 25 percent of the hydronic systems' flow measuring devices (e.g., manual balancing valves and flow meters) and Category 3 equipment's hydronic coils.

Quantity is rounded up to a whole number for a sample size and is a minimum of three units or devices when three or more exist.

#### 3.2.8.3 Retesting and Additional Testing

For any data indicated as a TAB Failure and when the data verification is

terminated, make the necessary corrections, conduct TAB Field Work, set and secure balancing devices, and mark settings. Prepare and resubmit the applicable TAB Report with the corrected test results for approval. Reschedule TAB Field Acceptance Testing of the approved revised report with the Contracting Officer. For each TAB Failure, and after the revised report is approved, conduct field acceptance testing of the failed data point, one additional selection by the Contracting Officer from the same Category for each failure, and new selections by the Contracting Officer in the same Category using the Category's sample strategy. If there is insufficient quantity of data in the same Category, the Contracting Officer may select data from a different Category.

### 3.3 PREREQUISITE FOR SYSTEMS ACCEPTANCE

Compliance with field acceptance testing requirements of this section is a prerequisite for the final Contracting Officer acceptance of the HVAC systems.

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## SECTION 23 07 00

THERMAL INSULATION FOR MECHANICAL SYSTEMS  
08/24

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. At the discretion of the Government, the manufacturer of any material supplied will be required to furnish test reports pertaining to any of the tests necessary to assure compliance with the standard or standards referenced in this specification.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING  
ENGINEERS (ASHRAE)

|                  |                                                                                  |
|------------------|----------------------------------------------------------------------------------|
| ASHRAE 90.1 - SI | (2019) Energy Standard for Buildings<br>Except Low-Rise Residential Buildings    |
| ASHRAE 90.2      | (2024; Errata 2024) Energy-Efficient<br>Design of Low-Rise Residential Buildings |

## ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A240/A240M | (2025) Standard Specification for Chromium<br>and Chromium-Nickel Stainless Steel Plate,<br>Sheet, and Strip for Pressure Vessels and<br>for General Applications |
| ASTM A580/A580M | (2023) Standard Specification for<br>Stainless Steel Wire                                                                                                         |
| ASTM B209M      | (2014) Standard Specification for Aluminum<br>and Aluminum-Alloy Sheet and Plate (Metric)                                                                         |
| ASTM C195       | (2007; R 2024) Standard Specification for<br>Mineral Fiber Thermal Insulating Cement                                                                              |
| ASTM C450       | (2008) Standard Practice for Fabrication<br>of Thermal Insulating Fitting Covers for<br>NPS Piping, and Vessel Lagging                                            |
| ASTM C533       | (2017; R 2023) Standard Specification for<br>Calcium Silicate Block and Pipe Thermal<br>Insulation                                                                |
| ASTM C534/C534M | (2024) Standard Specification for<br>Preformed Flexible Elastomeric Cellular<br>Thermal Insulation in Sheet and Tubular<br>Form                                   |
| ASTM C547       | (2022a) Standard Specification for Mineral<br>Fiber Pipe Insulation                                                                                               |
| ASTM C552       | (2022) Standard Specification for Cellular                                                                                                                        |



## Glass Thermal Insulation

|            |                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------|
| ASTM C553  | (2024) Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications      |
| ASTM C591  | (2022) Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation                     |
| ASTM C610  | (2017; R 2023) Standard Specification for Molded Expanded Perlite Block and Pipe Thermal Insulation                        |
| ASTM C612  | (2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation                                 |
| ASTM C647  | (2008; R 2013) Properties and Tests of Mastics and Coating Finishes for Thermal Insulation                                 |
| ASTM C755  | (2019b) Standard Practice for Selection of Water Vapor Retarders for Thermal Insulation                                    |
| ASTM C795  | (2008; R 2023) Standard Specification for Thermal Insulation for Use in Contact with Austenitic Stainless Steel            |
| ASTM C916  | (2020) Standard Specification for Adhesives for Duct Thermal Insulation                                                    |
| ASTM C920  | (2018; R 2024) Standard Specification for Elastomeric Joint Sealants                                                       |
| ASTM C921  | (2010; R 2015) Standard Practice for Determining the Properties of Jacketing Materials for Thermal Insulation              |
| ASTM C1126 | (2024) Standard Specification for Faced or Unfaced Rigid Cellular Phenolic Thermal Insulation                              |
| ASTM C1136 | (2023) Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation                           |
| ASTM C1290 | (2016; R 2021) Standard Specification for Flexible Fibrous Glass Blanket Insulation Used to Externally Insulate HVAC Ducts |
| ASTM C1710 | (2011) Standard Guide for Installation of Flexible Closed Cell Preformed Insulation in Tube and Sheet Form                 |
| ASTM D882  | (2012) Tensile Properties of Thin Plastic Sheeting                                                                         |

|               |                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D2863    | (2019) Standard Test Method for Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)                               |
| ASTM D5590    | (2017; R 2021) Standard Test Method for Determining the Resistance of Paint Films and Related Coatings to Fungal Defacement by Accelerated Four-Week Agar Plate Assay |
| ASTM E84      | (2024) Standard Test Method for Surface Burning Characteristics of Building Materials                                                                                 |
| ASTM E96/E96M | (2024a) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials                                                             |
| ASTM E2231    | (2021) Standard Practice for Specimen Preparation and Mounting of Pipe and Duct Insulation Materials to Assess Surface Burning Characteristics                        |

## CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

## FM GLOBAL (FM)

|              |                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------|
| FM APP GUIDE | (updated on-line) Approval Guide<br><a href="https://www.approvalguide.com/">https://www.approvalguide.com/</a> |
|--------------|-----------------------------------------------------------------------------------------------------------------|

## GREEN SEAL (GS)

|       |                                     |
|-------|-------------------------------------|
| GS-36 | (2013) Adhesives for Commercial Use |
|-------|-------------------------------------|

## MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

|           |                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------|
| MSS SP-58 | (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation |
|-----------|----------------------------------------------------------------------------------------------------------------|

## MIDWEST INSULATION CONTRACTORS ASSOCIATION (MICA)

|                      |                                                                |
|----------------------|----------------------------------------------------------------|
| MICA Insulation Stds | (8th Ed) National Commercial & Industrial Insulation Standards |
|----------------------|----------------------------------------------------------------|

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                       |
|----------|---------------------------------------------------------------------------------------|
| NFPA 90A | (2024) Standard for the Installation of Air Conditioning and Ventilating Systems      |
| NFPA 90B | (2024) Standard for the Installation of Warm Air Heating and Air Conditioning Systems |

## SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

## SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

## TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

TAPPI T403 OM (2022) Bursting Strength of Paper

## U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-A-3316 (1987; Rev C; Am 2 1990) Adhesives,  
Fire-Resistant, Thermal Insulation

MIL-A-24179 (1969; Rev A; Am 2 1980; Notice 1 1987;  
Notice 2 2020) Adhesive, Flexible  
Unicellular-Plastic Thermal Insulation

MIL-PRF-19565 (1988; Rev C) Coating Compounds, Thermal  
Insulation, Fire- and Water-Resistant,  
Vapor-Barrier

## UNDERWRITERS LABORATORIES (UL)

UL 94 (2023; Reprint Jan 2024) UL Standard for  
Safety Tests for Flammability of Plastic  
Materials for Parts in Devices and  
Appliances

UL 723 (2020) UL Standard for Safety Test for  
Surface Burning Characteristics of  
Building Materials

UL 2818 (2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

## 1.2 SYSTEM DESCRIPTION

## 1.2.1 General

Provide field-applied insulation and accessories on mechanical systems as specified herein; factory-applied insulation is specified under the piping, duct or equipment to be insulated. . Provide field applied insulation materials required for use on Government-furnished items as listed in the SPECIAL CONTRACT REQUIREMENTS.

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Submit the three SD types, SD-02 Shop Drawings, SD-03 Product Data, and

SD-08 Manufacturer's Instructions at the same time for each system.

SD-02 Shop Drawings

MICA Plates; G

Pipe Insulation Systems and Associated Accessories

Duct Insulation Systems and Associated Accessories

Equipment Insulation Systems and Associated Accessories

Recycled content for insulation materials; S

SD-03 Product Data

Pipe Insulation Systems; G

Duct Insulation Systems; G

Equipment Insulation Systems; G

SD-04 Samples

Thermal Insulation; G

Display Samples; G

SD-07 Certificates

Indoor air quality for adhesives; S

SD-08 Manufacturer's Instructions

Pipe Insulation Systems; G

Duct Insulation Systems; G

Equipment Insulation Systems; G

## 1.4 CERTIFICATIONS

### 1.4.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

## 1.5 QUALITY ASSURANCE

### 1.5.1 Installer Qualification

Qualified installers must have successfully completed three or more similar type jobs within the last 5 years.

## 1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in the manufacturer's unopened containers. Protect materials delivered and placed in storage from weather, humidity, dirt, dust and other contaminants. The Contracting Officer may reject insulation material and supplies that become dirty, dusty, wet, or contaminated by some other means. Attach manufacturer's stamp or label giving the name of the manufacturer and brand, and a description of the material, date codes, and approximate shelf life (if applicable) to packages or standard containers of insulation, jacket material, cements, adhesives, and coatings delivered for use, and samples required for approval. Insulation packages and containers must be asbestos free.

## PART 2 PRODUCTS

### 2.1 STANDARD PRODUCTS

Provide materials which are the standard products of manufacturers regularly engaged in the manufacture of such products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening and is an acceptable product per this specification. Submit a complete list of materials, including manufacturer's descriptive technical literature, performance data, catalog cuts, and installation instructions. Include the product number, k-value, thickness and furnished accessories including adhesives, sealants and jackets for each mechanical system requiring insulation. The product data must be copyrighted, have an identifying or publication number, and have been published prior to the issuance date of this solicitation. Submit materials furnished under this section together in a booklet.

#### 2.1.1 Insulation System

Provide insulation systems in accordance with the approved MICA National Insulation Standards plates as supplemented by this specification. Provide field-applied insulation for heating, ventilating, and cooling (HVAC) air distribution systems and piping systems that are located within, on, under, and adjacent to buildings; and for plumbing systems. Provide CFC and HCFC free insulation.

#### 2.1.2 Surface Burning Characteristics

Unless otherwise specified, insulation must have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Determine flame spread, and smoke developed indexes, by ASTM E84 or UL 723. Test insulation in the same density and installed thickness as the material to be used in the actual construction. Prepare and mount test specimens according to ASTM E2231.

### 2.2 MATERIALS

Provide insulation that meets or exceeds the requirements of ASHRAE 90.1 - SI. Ensure insulation exterior is cleanable, grease resistant, non-flaking and non-peeling. Provide compatible materials that do not contribute to corrosion, soften, or otherwise attack surfaces to which applied in either wet or dry state. Use materials on stainless steel surfaces meeting ASTM C795 requirements. Do not use calcium silicate on chilled or cold water systems. Use asbestos free materials. Provide product recognized under UL 94 (if containing plastic) and listed

in FM APP GUIDE.

### 2.2.1 Adhesives

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168 (HVAC duct sealants must meet limit requirements of "Other" category within SCAQMD Rule 1168 sealants table). Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for adhesives.

#### 2.2.1.1 Acoustical Lining Insulation Adhesive

Provide a nonflammable, fire-resistant adhesive conforming to ASTM C916, Type I.

#### 2.2.1.2 Mineral Fiber Insulation Cement

Provide cement in accordance with ASTM C195.

#### 2.2.1.3 Lagging Adhesive

Lagging is the material used for thermal insulation, especially around a cylindrical object. This may include the insulation as well as the cloth/material covering the insulation. To resist mold/mildew, use lagging adhesive meeting ASTM D5590 with 0 growth rating. Provide nonflammable and fire-resistant lagging adhesives that have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Ensure adhesive is MIL-A-3316, Class 1, pigmented white or red and suitable, as listed by the manufacturer, for bonding fibrous glass cloth to faced and unfaced fibrous glass insulation board; for bonding cotton brattice cloth to faced and unfaced fibrous glass insulation board; for sealing edges of and bonding glass tape to joints of fibrous glass board; for bonding lagging cloth to thermal insulation; or Class 2 for attaching fibrous glass insulation to metal surfaces. Apply lagging adhesives in strict accordance with the manufacturer's recommendations for pipe and duct insulation.

#### 2.2.1.4 Contact Adhesive

Adhesives may be any of, but not limited to, the neoprene based, rubber based, or elastomeric type that have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Ensure adhesive does not adversely affect, initially or in service, the insulation to which it is applied, nor cause any corrosive effect on metal to which it is applied. Ensure that any solvent dispersing medium or volatile component of the adhesive has no objectionable odor and does not contain any benzene or carbon tetrachloride. Ensure dried adhesive does not emit nauseous, irritating, or toxic volatile matters or aerosols when the adhesive is heated to any temperature up to 100 degrees C. The dried adhesive must be nonflammable and fire resistant. Flexible Elastomeric Adhesive: Comply with MIL-A-24179, Type II, Class I. Provide product listed in FM APP GUIDE.

### 2.2.2 Caulking

ASTM C920, Type S, Grade NS, Class 25, Use A.

### 2.2.3 Corner Angles

Nominal 0.406 mm aluminum 25 by 25 mm with factory applied kraft backing. Aluminum must be ASTM B209M, Alloy 3003, 3105, or 5005.

### 2.2.4 Fittings

Fabricated Fittings are the prefabricated fittings for flexible elastomeric pipe insulation systems in accordance with ASTM C1710. Together with the flexible elastomeric tubes, they provide complete system integrity for retarding heat gain and controlling condensation drip from chilled-water and refrigeration systems. Flexible elastomeric, fabricated fittings provide thermal protection (0.25 k) and condensation resistance (0.05 Water Vapor Transmission factor). For satisfactory performance, use properly installed protective vapor retarder/barriers and vapor stops on high relative humidity and below ambient temperature applications to reduce movement of moisture through or around the insulation to the colder interior surface.

### 2.2.5 Finishing Cement

ASTM C450: Mineral fiber hydraulic-setting thermal insulating and finishing cement. All cements that may come in contact with Austenitic stainless steel must comply with ASTM C795.

### 2.2.6 Fibrous Glass Cloth and Glass Tape

Provide fibrous glass cloth, with 20X20 maximum mesh size, and glass tape with maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Provide tape consisting of 100 mm wide rolls. Provide Class 3 tape that is 0.15 kg/square m. Elastomeric Foam Tape: Black vapor-retarder foam tape with acrylic adhesive containing an anti-microbial additive.

### 2.2.7 Staples

Outward clinching type monel.

### 2.2.8 Jackets

#### 2.2.8.1 Aluminum Jackets

Provide aluminum jackets consisting of corrugated, embossed or smooth sheet, 0.406 mm nominal thickness; ASTM B209M, Temper H14, Temper H16, Alloy 3003, 5005, or 3105. Do not use corrugated aluminum jacket outdoors. Aluminum jacket securing bands must be Type 304 stainless steel, 0.396 mm thick, 13 mm wide for pipe under 300 mm diameter and 19 mm wide for pipe over 300 mm and larger diameter. Aluminum jacket circumferential seam bands must be 50.8 by 0.406 mm aluminum matching jacket material. Ensure bands for insulation below ground are 19 by 0.508 mm thick stainless steel, or fiberglass reinforced tape. The jacket may, at the option of the Contractor, be provided with a factory fabricated Pittsburgh or "Z" type longitudinal joint. When the "Z" joint is used, use bands at the circumferential joints that are designed by the manufacturer to seal the joints and hold the jacket in place.

#### 2.2.8.2 Polyvinyl Chloride (PVC) Jackets

Polyvinyl chloride (PVC) jacket and fitting covers must have high impact strength, ultraviolet (UV) resistant rating or treatment and moderate chemical resistance. The minimum thickness for a Standard jacket is 0.508mm and where high abuse resistance is required use Heavy Duty PVC jacket 0.762 mm. Standard PVC fittings will have an initial core thickness (before thermoforming) of 0.635 mm and heavy duty fittings will have an initial core thickness of 0.762 mm.

#### 2.2.8.3 Vapor Barrier/Vapor Retarder

Apply the following criteria to determine which system is required.

- a. On ducts, equip piping and equipment operating below 40 degrees C or located outside with a vapor barrier.
- b. Install ducts, pipes and equipment that are located inside and that always operate above 40 degrees C with a vapor retarder where required as stated in paragraph VAPOR RETARDER REQUIRED.

#### 2.2.9 Vapor Retarder Required

ASTM C921, Type I, minimum puncture resistance 50 Beach units on all surfaces except concealed ductwork, where a minimum puncture resistance of 25 Beach units is acceptable. Minimum tensile strength, 6.1 N/mm width. ASTM C921, Type II, minimum puncture resistance 25 Beach units, tensile strength minimum 3.5 N/mm width. Use jackets on insulation exposed in finished areas that have white finish suitable for painting without sizing. Based on the application, insulation materials that require manufacturer or fabricator applied pipe insulation jackets are cellular glass, when all joints are sealed with a vapor barrier mastic, and mineral fiber. Ensure all non-metallic jackets have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Flexible elastomerics require (in addition to vapor barrier skin) vapor retarder jacketing for high relative humidity and below ambient temperature applications.

##### 2.2.9.1 White Vapor Retarder All Service Jacket (ASJ)

ASJ is for use on hot/cold pipes, ducts, or equipment indoors or outdoors if covered by a suitable protective jacket. Provide product which meets all physical property and performance requirements of ASTM C1136, Type I, except a minimum burst strength of 585 kPa. ASTM D2863 Limited Oxygen Index (LOI) is a minimum of 31.

In addition, do not use paper or other moisture-sensitive material for the outer exposed surface or the inner-most surface contacting the insulation. Ensure the outer exposed surface is white and has an emittance no less than 0.80.

##### 2.2.9.2 Vapor Retarder/Vapor Barrier Mastic Coatings

###### 2.2.9.2.1 Vapor Barrier

The vapor barrier must be self adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) greater than 3 plies standard grade, silver, white, black and embossed white jacket for use on hot/cold pipes. Ensure permeability is



less than 0.02 when tested in accordance with ASTM E96/E96M. Provide products meeting UL 723 or ASTM E84 flame and smoke requirements and that are UV resistant.

#### 2.2.9.2.2 Vapor Retarder

Provide fire and water resistant vapor retarder coating appropriately selected for either outdoor or indoor service. Color must be white. Ensure the water vapor permeance of the compound is in accordance with ASTM C755. Provide nonflammable, fire resistant coating. To resist mold/mildew, provide coating meeting ASTM D5590 with 0 growth rating. Ensure coating meets MIL-PRF-19565 Type II (if selected for indoor service) and is Qualified Products Database listed. Determine all other application and service properties pursuant to ASTM C647.

#### 2.2.9.3 Laminated Film Vapor Retarder

ASTM C1136, Type I, maximum moisture vapor transmission 0.02 perms, minimum puncture resistance 50 Beach units on all surfaces except concealed ductwork; where Type II, maximum moisture vapor transmission 0.02 perms, a minimum puncture resistance of 25 Beach units is acceptable. Provide vapor retarder with a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Flexible Elastomeric exterior foam with factory applied UV Jacket. Construction of laminate designed to provide UV resistance, high puncture, tear resistance and an excellent WVT rate.

#### 2.2.9.4 Polyvinylidene Chloride (PVDC) Film Vapor Retarder

Provide PVDC film vapor retarder with a maximum moisture vapor transmission of 0.02 perms, minimum puncture resistance of 150 Beach units, a minimum tensile strength in any direction of 5.3 kN/m when tested in accordance with ASTM D882, and a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

#### 2.2.9.5 Polyvinylidene Chloride Vapor Retarder Adhesive Tape

Requirements must meet the same as specified for Laminated Film Vapor Retarder above.

#### 2.2.9.6 Vapor Barrier/Weather Barrier

Ensure the vapor barrier is greater than 3 ply self adhesive laminate -white vapor barrier jacket- superior performance (permeability less than 0.02 when tested in accordance with ASTM E96/E96M). Provide vapor barrier meeting UL 723 or ASTM E84 25 flame and 50 smoke requirements; and UV resistant. Minimum burst strength 1.3 MPa in accordance with TAPPI T403 OM. Tensile strength 0.12 kg/m width (PSTC-1000). Provide tape as specified for laminated film vapor barrier above.

#### 2.2.10 Vapor Retarder Not Required

ASTM C921, Type II, Class D, minimum puncture resistance 50 Beach units on all surfaces except ductwork, where Type IV, maximum moisture vapor transmission 0.10, a minimum puncture resistance of 25 Beach units is acceptable. Provide jacket with a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

### 2.2.11 Wire

Soft annealed ASTM A580/A580M Type 302, 304 or 316 stainless steel, 16 or 18 gauge.

### 2.2.12 Insulation Bands

Provide 13 mm wide; 26 gauge stainless steel insulation bands.

### 2.2.13 Sealants

Choose sealants from the butyl polymer type, the styrene-butadiene rubber type, or the butyl type of sealants. Provide sealants with a maximum permeance of 0.02 perms based on Procedure B for ASTM E96/E96M, and a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

## 2.3 PIPE INSULATION SYSTEMS

Conform insulation materials to Table 1 and minimum insulation thickness as listed in Table 2 and meet or exceed the requirements of ASHRAE 90.1 - SIASHRAE 90.1 - IP. Limit pipe insulation materials to those listed herein and meeting the following requirements:

### 2.3.1 Recycled Materials

Provide insulation materials containing the following minimum percentage of recycled material content by weight:

- Rock Wool: 75 percent slag of weight
- Fiberglass: 20 percent glass cullet
- Rigid Foam: 9 percent recovered material
- Phenolic Rigid Foam: 9 percent recovered material

Provide data identifying percentage of recycled content for insulation materials.

### 2.3.2 Aboveground Cold Pipeline (-34 to 16 deg. C)

Provide insulation for outdoor, indoor, exposed or concealed applications, as follows:

#### 2.3.2.1 Cellular Glass

ASTM C552, Type II, and Type III. Supply the insulation from the fabricator with (paragraph WHITE VAPOR RETARDER ALL SERVICE JACKET (ASJ)) ASJ vapor retarder and installed with all longitudinal overlaps sealed and all circumferential joints ASJ taped or supply the insulation unfaced from the fabricator and install with all longitudinal and circumferential joints sealed with vapor barrier mastic.

#### 2.3.2.2 Flexible Elastomeric Cellular Insulation

Closed-cell, foam- or expanded-rubber materials containing anti-microbial additive, complying with ASTM C534/C534M, Grade 1, Type I or II. Type I, Grade 1 for tubular materials. Type II, Grade 1, for sheet materials. Ensure Type I and II have vapor retarder/vapor barrier skin on one or both sides of the insulation, and require an additional exterior vapor retarder covering for high relative humidity and below ambient temperature

applications.

#### 2.3.2.3 Polyisocyanurate Insulation

ASTM C591, Type I. Polyisocyanurate insulation may only be used on systems operating at temperatures not to exceed 60 degrees C . Supply the insulation with a factory applied vapor retarder/barrier that complies with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. The insulation and all covering must pass the flame spread index of 25 and the smoke developed index of 50 when tested in accordance with ASTM E84.

#### 2.3.3 Aboveground Hot Pipeline (Above 16 deg. C)

Provide insulation for outdoor, indoor, exposed or concealed applications meeting the following requirements. Supply the insulation with manufacturer's recommended factory-applied jacket/vapor barrier.

##### 2.3.3.1 Mineral Fiber

ASTM C547, Types I, II or III, supply the insulation with manufacturer's recommended factory-applied jacket.

##### 2.3.3.2 Calcium Silicate

ASTM C533, Type I indoor only, or outdoors above 121 degrees C pipe temperature. Supply insulation with the manufacturer's recommended factory-applied jacket/vapor barrier.

##### 2.3.3.3 Cellular Glass

ASTM C552, Type II and Type III. Supply the insulation with manufacturer's recommended factory-applied jacket.

##### 2.3.3.4 Flexible Elastomeric Cellular Insulation

Closed-cell, foam- or expanded-rubber materials containing anti-microbial additive, complying with ASTM C534/C534M, Grade 1, Type I or II to 105 degrees C service. Type I for tubular materials. Type II for sheet materials.

##### 2.3.3.5 Phenolic Insulation

ASTM C1126 Type III to 121 degrees C service must comply with ASTM C795. Supply the insulation with manufacturer's recommended factory-applied jacket/vapor barrier.

##### 2.3.3.6 Perlite Insulation

ASTM C610

##### 2.3.3.7 Polyisocyanurate Insulation

ASTM C591, Type I. Supply the insulation with a factory applied vapor retarder/barrier that complies with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. The insulation and all covering must pass the flame spread index of 25 and the smoke developed index of 50 when tested in accordance with ASTM E84.

#### 2.3.4 Below-ground Pipeline Insulation

For below-ground pipeline insulation, use cellular glass, ASTM C552, type II.

### 2.4 DUCT INSULATION SYSTEMS

#### 2.4.1 Factory Applied Insulation

Provide factory-applied ASTM C534/C534M Grade 1, Type II, flexible elastomeric closed cell insulation according to manufacturer's recommendations for insulation with insulation manufacturer's standard reinforced fire-retardant vapor barrier, with identification of installed thermal resistance (R) value and out-of-package R value.

#### 2.4.2 Field Applied Insulation

##### 2.4.2.1 General

Provide the minimum of 50 mm thickness in compliance with ASHRAE 90.2 ASHRAE 90.1 - SI.

##### 2.4.3 Rigid Insulation

Mineral-Fiber Board Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C612, Type IA or Type IB; 3.0 pcf. Provide insulation with factory applied ASJandfactory-applied FSK jacket. Minimum thickness in accordance with ASHRAE 90.1 - SIASHRAE 90.2Tables 3 and 4

##### 2.4.4 Blanket Insulation

Mineral-Fiber Blanket Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C553, Type II and ASTM C1290, Type III with factory-applied FSK jacket; 3/4 pcf. Minimum thickness in accordance with ASHRAE 90.1 - SIASHRAE 90.2Tables 3 and 4.

##### 2.4.5 Acoustical Duct Lining

##### 2.4.5.1 General

For ductwork indicated or specified in Section 23 30 00 HVAC AIR DISTRIBUTION to be acoustically lined, provide external insulation in accordance with this specification section and in addition to the acoustical duct lining if R-value would not be met as required in Table 3 and 4. The installation of internal acoustical duct insulation is prohibited.

##### 2.4.5.2 Duct Liner

Flexible Elastomeric Acoustical and Conformable Duct Liner Materials: Flexible Elastomeric Thermal, Acoustical and Conformable Insulation Compliance with ASTM C534/C534M Grade 1, Type II; and NFPA 90A or NFPA 90B as applicable.

## 2.4.6 Duct Insulation Jackets

### 2.4.6.1 All-Purpose Jacket

Provide insulation with insulation manufacturer's standard reinforced fire-retardant jacket with or without integral vapor barrier as required by the service. In exposed locations, provide jacket with a white surface.

### 2.4.6.2 Metal Jackets

#### 2.4.6.2.1 Aluminum Jackets

ASTM B209M, Temper H14, minimum thickness of 27 gauge (0.41 mm), with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide smooth surface jackets for jacket outside dimension 200 mm and larger. Provide corrugated surface jackets for jacket outside dimension 200 mm and larger. Provide stainless steel bands, minimum width of 13 mm.

#### 2.4.6.2.2 Stainless Steel Jackets

ASTM A240/A240M; Type 304, minimum thickness of 33 gauge (0.25 mm), smooth surface with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide stainless steel bands, minimum width of 13 mm.

### 2.4.6.3 Vapor Barrier/Weatherproofing Jacket

Provide vapor barrier/weatherproofing jacket that is laminated self-adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) less than 0.02 permeability, (greater than 3 ply, standard grade, silver, white, black and embossed or greater than 8 ply (minimum 0.072 mm adhesive), heavy duty white or natural).

## 2.4.7 Weatherproof Duct Insulation

Provide ASTM C534/C534M Grade 1, Type II, flexible elastomeric cellular insulation, and weatherproofing as specified in manufacturer's instruction. Multi-ply, Polymeric Blend Laminate Jacketing: Construction of laminate designed to provide UV resistance, high puncture, tear resistance and an excellent WVT rate.

## 2.5 EQUIPMENT INSULATION SYSTEMS

Insulate equipment and accessories as specified in Tables 5 and 6. In outside locations, provide insulation 13 mm thicker than specified. Increase the specified insulation thickness for equipment where necessary to equal the thickness of angles or other structural members to make a smooth, exterior surface. Submit a booklet containing manufacturer's published installation instructions for the insulation systems. The instructions must be copyrighted, have an identifying or publication number, and have been published prior to the issuance date of this solicitation. A booklet is also required by paragraphs titled: Pipe Insulation Systems and Duct Insulation Systems.

## PART 3 EXECUTION

### 3.1 APPLICATION - GENERAL

Apply insulation to unheated and uncooled piping and equipment. Do not

compress flexible elastomeric cellular insulation at joists, studs, columns, ducts, and hangers. The insulation must not pull apart after a one hour period; replace any insulation found to pull apart after one hour.

### 3.1.1 Display Samples

Submit and display, after approval of materials, actual sections of installed systems, properly insulated in accordance with the specification requirements. Such actual sections must remain accessible to inspection throughout the job and will be reviewed from time to time for controlling the quality of the work throughout the construction site. Identify each material used by indicating on an attached sheet the specification requirement for the material and the material by each manufacturer intended to meet the requirement. The Contracting Officer will inspect display sample sections at the jobsite. Keep approved display sample sections on display at the jobsite during the construction period. Upon completion of construction, the display sample sections will be closed and sealed.

#### 3.1.1.1 Pipe Insulation Display Sections

Include as a minimum an elbow or tee, a valve, dielectric waterways and flanges, a hanger with protection shield and insulation insert, or dowel as required, at support point, method of fastening and sealing insulation at longitudinal lap, circumferential lap, butt joints at fittings and on pipe runs, and terminating points for each type of pipe insulation used on the job, and for hot pipelines and cold pipelines, both interior and exterior, even when the same type of insulation is used for these services.

#### 3.1.1.2 Duct Insulation Display Sections

Display sample sections for rigid and flexible duct insulation used on the job. Use a temporary covering to enclose and protect display sections for duct insulation exposed to weather

### 3.1.2 Installation

Except as otherwise specified, install material in accordance with the manufacturer's written instructions. Do not apply insulation materials until tests and heat tracing specified in other paragraphs of this specification are completed. Remove material such as rust, scale, dirt and moisture from surfaces to receive insulation. Keep insulation clean and dry. Do not remove insulation from its shipping containers until the day it is ready to use and return to like containers or equally protect from dirt and moisture at the end of each workday. Thoroughly clean insulation that becomes dirty prior to use. If insulation becomes wet or if cleaning does not restore the surfaces to like new condition, reject the insulation, and immediately remove from the jobsite. Stagger joints on multi layer insulation. Mix mineral fiber thermal insulating cement with demineralized water when used on stainless steel surfaces. Install insulation, jacketing and accessories in accordance with MICA Insulation Stds plates except where modified herein or on the drawings.

### 3.1.3 Firestopping

Where pipes and ducts pass through fire walls, fire partitions, above grade floors, and fire rated chase walls, seal the penetration with fire stopping materials as specified in Section 07 84 00 FIRESTOPPING. The

protection of ducts at point of passage through firewalls must be in accordance with NFPA 90A and NFPA 90B. All other penetrations, such as piping, conduit, and wiring, through firewalls must be protected with a material or system of the same hourly rating that is listed by UL, FM, or a NRTL.

#### 3.1.4 Installation of Flexible Elastomeric Cellular Insulation

Install flexible elastomeric cellular insulation with seams and joints sealed with rubberized contact adhesive. Do not use flexible elastomeric cellular insulation on surfaces greater than 105 degrees C. Stagger seams when applying multiple layers of insulation. Protect insulation exposed to weather with metal jacketing as recommended by the manufacturer after the adhesive is dry and cured.

##### 3.1.4.1 Adhesive Application

Apply a brush coating of adhesive to both butt ends to be joined and to both slit surfaces to be sealed. Allow the adhesive to set until dry to touch but tacky under slight pressure before joining the surfaces. Ensure insulation seals at seams and joints are not capable of being pulled apart one hour after application. Replace insulation that can be pulled apart one hour after installation.

##### 3.1.4.2 Adhesive Safety Precautions

Use natural cross-ventilation, local (mechanical) pickup, or general area (mechanical) ventilation to prevent an accumulation of solvent vapors, keeping in mind the ventilation pattern must remove any heavier-than-air solvent vapors from lower levels of the workspaces. Gloves and spectacle-type safety glasses are recommended in accordance with safe installation practices.

#### 3.1.5 Welding

Welding is not permitted on piping, duct or equipment without written approval of the Contracting Officer. The capacitor discharge welding process may be used for securing metal fasteners to duct.

#### 3.1.6 Pipes/Ducts/Equipment That Require Insulation

Insulation is required on all pipes, ducts, or equipment, except for omitted items as specified.

### 3.2 PIPE INSULATION SYSTEMS INSTALLATION

#### 3.2.1 Pipe Insulation

##### 3.2.1.1 General

Install pipe insulation on aboveground hot and cold pipeline systems as specified below to form a continuous thermal retarder/barrier, including straight runs, fittings and appurtenances unless specified otherwise. Install full length units of insulation using a single cut piece to complete a run. Do not use cut pieces or scraps abutting each other. Omit pipe insulation on the following:

- a. Air chambers.

- b. Adjacent insulation.
- c. ASME stamps.
- d. Access plates of fan housings.

#### 3.2.1.2 Pipes Passing Through Walls, Roofs, and Floors

Provide continuous pipe insulation through the sleeve.

Provide an aluminum jacket or vapor barrier/weatherproofing self adhesive jacket (minimum 0.05 mm adhesive, 0.075 mm embossed) less than 0.0000 permeability, greater than 3 ply standard grade, silver, white, black and embossed with factory applied moisture retarder over the insulation wherever penetrations require sealing.

##### 3.2.1.2.1 Penetrate Interior Walls

Provide aluminum jacket or vapor barrier/weatherproofing - self adhesive jacket (minimum 0.05 mm adhesive, 0.075 mm embossed) less than 0.02 permeability, greater than 3 plies standard grade, silver, white, black and embossed which extends 50 mm beyond either side of the wall and secure on each end with a band.

##### 3.2.1.2.2 Penetrating Floors

Extend the aluminum jacket from a point below the backup material to a point 250 mm above the floor with one band at the floor and one not more than 25 mm from the end of the aluminum jacket.

##### 3.2.1.2.3 Penetrating Waterproofed Floors

Extend the aluminum jacket from below the backup material to a point 50 mm above the flashing with a band 25 mm from the end of the aluminum jacket.

##### 3.2.1.2.4 Penetrating Exterior Walls

Continue the aluminum jacket required for pipe exposed to weather through the sleeve to a point 50 mm beyond the interior surface of the wall.

##### 3.2.1.2.5 Penetrating Roofs

Insulate pipe as required for interior service to a point flush with the top of the flashing and sealed with flashing sealant. Tightly butt the insulation for exterior application to the top of flashing and interior insulation. Extend the exterior aluminum jacket 50 mm down beyond the end of the insulation to form a counter flashing. Seal the flashing and counter flashing underneath with metal jacketing/flashing sealant.

##### 3.2.1.2.6 Domestic Cold Water Pipes Supplying Lavatories or Other Similar Cooling Service

Terminate the insulation on the finished side of the wall (i.e., insulation must cover the pipe throughout the wall penetration). Protect the insulation with two coats of weather barrier mastic (breather emulsion type weatherproof mastic impermeable to water and permeable to air) with a minimum total thickness of 2 mm. Extend the mastic out onto the insulation 50 mm and seal the end of the insulation. Caulk the annular space



between the outer surface of the pipe insulation and the wall penetration with an approved fire stop material having vapor retarder properties. Cover the pipe and wall penetration with a properly sized (well fitting) escutcheon plate. Ensure the escutcheon plate overlaps the wall penetration by at least 10 mm.

### 3.2.1.3 Pipes Passing Through Hangers

Ensure insulation, whether hot or cold application, is continuous through hangers. Support all horizontal pipes 50 mm and smaller on hangers with the addition of a Type 40 protection shield to protect the insulation in accordance with MSS SP-58. Whenever insulation shows signs of being compressed, or when the insulation or jacket shows visible signs of distortion at or near the support shield, install insulation inserts as specified below for piping larger than 50 mm, or factory insulated hangers (designed with a load bearing core) can be used.

#### 3.2.1.3.1 Horizontal Pipes Larger Than 50 mm at 16 Degrees C and Above

Supported on hangers in accordance with MSS SP-58, and Section 22 00 00 PLUMBING, GENERAL PURPOSE.

#### 3.2.1.3.2 Horizontal Pipes Larger Than 50 mm and Below 16 Degrees C

Supported on hangers with the addition of a Type 40 protection shield in accordance with MSS SP-58. Install an insulation insert of cellular glass, prefabricated insulation pipe hangers, or perlite above 27 degrees C above each shield. Ensure insert covers no less than the bottom 180-degree arc of the pipe. Provide inserts that are the same thickness as the insulation, and extend 50 mm on each end beyond the protection shield. When insulation inserts are required in accordance with the above, and the insulation thickness is less than 25 mm, wooden or cork dowels or blocks may be installed between the pipe and the shield to prevent the weight of the pipe from crushing the insulation, as an option to installing insulation inserts. Ensure the insulation jacket is continuous over the wooden dowel, wooden block, or insulation insert.

#### 3.2.1.3.3 Vertical Pipes

Supported with either Type 8 or Type 42 riser clamps with the addition of two Type 40 protection shields in accordance with MSS SP-58 covering the 360-degree arc of the insulation. Install an insulation insert of cellular glass or calcium silicate between each shield and the pipe. Ensure the insert covers the 360-degree arc of the pipe. Provide inserts that are the same thickness as the insulation, and extend 50 mm on each end beyond the protection shield. When insulation inserts are required in accordance with the above, and the insulation thickness is less than 25 mm, wooden or cork dowels or blocks may be installed between the pipe and the shield to prevent the hanger from crushing the insulation, as an option instead of installing insulation inserts. Ensure the insulation jacket is continuous over the wooden dowel, wooden block, or insulation insert. Support the vertical weight of the pipe with hangers located in a horizontal section of the pipe. When the pipe riser is longer than 9 m, support the weight of the pipe additionally with hangers in the vertical run of the pipe that are directly clamped to the pipe, penetrating the pipe insulation. Use insulated hangers and seal the insulation jacket as indicated herein for anchors in a similar service.

#### 3.2.1.3.4 Inserts

Covered with a jacket material of the same appearance and quality as the adjoining pipe insulation jacket, overlap the adjoining pipe jacket 38 mm, and seal as required for the pipe jacket. Use jacket material to cover inserts in flexible elastomeric cellular insulation conforming to ASTM C1136, Type 1, and is allowed to be of a different material than the adjoining insulation material.

#### 3.2.1.4 Flexible Elastomeric Cellular Pipe Insulation

Use tubular form flexible elastomeric cellular pipe insulation for pipe sizes 150 mm and less. Grade 1, Do not stretch Type II sheet insulation used on pipes larger than 150 mm around the pipe. On pipes larger than 300 mm, adhere the insulation directly to the pipe on the lower 1/3 of the pipe. Stagger seams when applying multiple layers of insulation. Insulate sweat fittings with miter-cut pieces the same size as on adjacent piping. Insulate screwed fittings with sleeved fitting covers fabricated from miter-cut pieces and overlap and seal to the adjacent pipe insulation. Type II requires an additional exterior vapor retarder/barrier covering for high relative humidity and below ambient temperature applications.

#### 3.2.1.5 Pipes in High Abuse Areas

In high abuse areas such as janitor closets and traffic areas in equipment rooms, kitchens, and mechanical rooms, utilize welded PVC, stainless steel, aluminum or flexible laminate cladding (comprised of elastomeric, plastic or metal foil laminate) laminated self-adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) vapor barrier/weatherproofing jacket, - less than 0.02 permeability; (greater than 3 ply, standard grade, silver, white, black and embossed) aluminum jackets. Protect pipe insulation to the 2 m level.

#### 3.2.1.6 Pipe Insulation Material and Thickness

Pipe insulation materials must be as listed in Table 1 and must meet or exceed the requirements of ASHRAE 90.1 - SI. Insulate these refrigerant lines according to the temperature of the pipe contents while the Variable Refrigerant Flow (VRF) system is fully operating following the temperature parameters as stated in Table 1 and Table 2 of this specification.

| TABLE 1                                                                  |                               |                 |      |       |             |
|--------------------------------------------------------------------------|-------------------------------|-----------------|------|-------|-------------|
| Insulation Material for Piping                                           |                               |                 |      |       |             |
| Service                                                                  |                               |                 |      |       |             |
|                                                                          | Material                      | Specification   | Type | Class | VR/VB Req'd |
| Chilled Water (Supply & Return, Dual Temperature Piping, 4.44 C nominal) |                               |                 |      |       |             |
|                                                                          |                               |                 |      |       |             |
|                                                                          | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | Yes         |
|                                                                          |                               |                 |      |       |             |

| TABLE 1                                                                                            |                               |                 |      |       |             |
|----------------------------------------------------------------------------------------------------|-------------------------------|-----------------|------|-------|-------------|
| Insulation Material for Piping                                                                     |                               |                 |      |       |             |
| Service                                                                                            |                               |                 |      |       |             |
|                                                                                                    | Material                      | Specification   | Type | Class | VR/VB Req'd |
|                                                                                                    |                               |                 |      |       |             |
|                                                                                                    |                               |                 |      |       |             |
|                                                                                                    |                               |                 |      |       |             |
|                                                                                                    |                               |                 |      |       |             |
|                                                                                                    |                               |                 |      |       |             |
|                                                                                                    |                               |                 |      |       |             |
| Cold Domestic Water Piping, Makeup Water, and Condenser Water Piping                               |                               |                 |      |       |             |
|                                                                                                    | Cellular Glass                | ASTM C552       | II   | 2     | No          |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | No          |
| Hot Domestic Water Supply & Recirculating Piping (Max 93 C)                                        |                               |                 |      |       |             |
|                                                                                                    | Mineral Fiber                 | ASTM C547       | I    | 1     | No          |
|                                                                                                    |                               |                 |      |       |             |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | No          |
|                                                                                                    | Faced Phenolic Foam           | ASTM C1126      | III  |       | Yes         |
| Refrigerant Suction Piping (1.67 degrees C nominal)                                                |                               |                 |      |       |             |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | No          |
|                                                                                                    |                               |                 |      |       |             |
| Compressed Air Discharge, Steam and Condensate Return (94 to 121 Degrees C)                        |                               |                 |      |       |             |
|                                                                                                    | Cellular Glass                | ASTM C552       | II   |       | No          |
|                                                                                                    | Mineral Fiber                 | ASTM C547       | I    | 1     | No          |
|                                                                                                    | Calcium Silicate              | ASTM C533       | I    |       | No          |
|                                                                                                    | Faced Phenolic Foam           | ASTM C1126      | III  |       | Yes         |
|                                                                                                    | Perlite                       | ASTM C610       |      |       | No          |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    | 2     | No          |
| Exposed Lavatory Drains, Exposed Domestic Water Piping & Drains to Areas for Handicapped Personnel |                               |                 |      |       |             |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | No          |
| Condensate Drain Located Inside Building                                                           |                               |                 |      |       |             |

| TABLE 1                                                                   |                               |                 |      |       |             |
|---------------------------------------------------------------------------|-------------------------------|-----------------|------|-------|-------------|
| Insulation Material for Piping                                            |                               |                 |      |       |             |
| Service                                                                   |                               |                 |      |       |             |
|                                                                           | Material                      | Specification   | Type | Class | VR/VB Req'd |
|                                                                           | Cellular Glass                | ASTM C552       | II   | 2     | No          |
|                                                                           | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | No          |
| Medium Temperature Hot Water, Steam and Condensate (122 to 176 Degrees C) |                               |                 |      |       |             |
|                                                                           | Mineral Fiber                 | ASTM C547       | I    | 1     | No          |
|                                                                           | Calcium Silicate              | ASTM C533       | I    |       | No          |
|                                                                           |                               |                 |      |       |             |
|                                                                           |                               |                 |      |       |             |
|                                                                           |                               |                 |      |       | No          |
|                                                                           |                               |                 |      |       |             |
|                                                                           |                               |                 |      |       |             |
|                                                                           |                               |                 |      |       |             |
|                                                                           |                               |                 |      |       |             |
|                                                                           |                               |                 |      |       |             |
|                                                                           |                               |                 |      |       |             |
|                                                                           |                               |                 |      |       |             |
|                                                                           |                               |                 |      |       |             |
| Note: VR/VB = Vapor Retarder/Vapor Barrier                                |                               |                 |      |       |             |

| TABLE 2                                                                                                                                                                                                                                                                                                   |                               |                                                  |                         |        |         |          |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------|-------------------------|--------|---------|----------|------------|
| Piping Insulation Thickness (mm)<br>For flexible cellular foam the thickness should be 13mm instead of 15mm. Economic thickness or prevention of condensation is the basis of these tables. If prevention of condensation is the criterion, the ambient temperature and relative humidity must be stated. |                               |                                                  |                         |        |         |          |            |
| Service                                                                                                                                                                                                                                                                                                   |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           | Material                      | Conductivity<br>K-value<br>(W/m <sup>2</sup> -K) | Tube And Pipe Size (mm) |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  | <25                     | 25-<40 | 40-<100 | 100-<200 | > or = 200 |
| Chilled Water (Supply & Return, 4.44 Degrees C nominal)                                                                                                                                                                                                                                                   |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           | Flexible Elastomeric Cellular | 0.034                                            | 25                      | 25     | 25      | N/A      | N/A        |
|                                                                                                                                                                                                                                                                                                           | Polyisocyanurate              | 0.026                                            | 13                      | 25     | 25      | 25       | 40         |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
| Cold Domestic Water Piping, Makeup Water, and Condenser Water                                                                                                                                                                                                                                             |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           | Cellular Glass                | 0.043                                            | 40                      | 40     | 40      | 40       | 40         |
|                                                                                                                                                                                                                                                                                                           | Flexible Elastomeric Cellular | 0.034                                            | 25                      | 25     | 25      | N/A      | N/A        |
| Hot Domestic Water Supply & Recirculating Piping (Max 93 C)                                                                                                                                                                                                                                               |                               |                                                  |                         |        |         |          |            |

| TABLE 2                                                                                                                                                                                                                                                                                                   |                               |                                                  |                         |        |         |          |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------|-------------------------|--------|---------|----------|------------|
| Piping Insulation Thickness (mm)<br>For flexible cellular foam the thickness should be 13mm instead of 15mm. Economic thickness or prevention of condensation is the basis of these tables. If prevention of condensation is the criterion, the ambient temperature and relative humidity must be stated. |                               |                                                  |                         |        |         |          |            |
| Service                                                                                                                                                                                                                                                                                                   |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           | Material                      | Conductivity<br>K-value<br>(W/m <sup>2</sup> -K) | Tube And Pipe Size (mm) |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  | <25                     | 25-<40 | 40-<100 | 100-<200 | > or = 200 |
|                                                                                                                                                                                                                                                                                                           | Mineral Fiber                 | 0.037                                            | 40                      | 40     | 50      | 50       | 50         |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           | Flexible Elastomeric Cellular | 0.043                                            | 40                      | 40     | 50      | N/A      | N/A        |
| Refrigerant Suction Piping (1.67 degrees C nominal)                                                                                                                                                                                                                                                       |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           | Flexible Elastomeric Cellular | 0.034                                            | 25                      | 25     | 25      | N/A      | N/A        |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |                               |                                                  |                         |        |         |          |            |
| Exposed Lavatory Drains, Exposed Domestic Water Piping & Drains to Areas for Handicapped Personnel                                                                                                                                                                                                        |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           | Flexible Elastomeric Cellular |                                                  | 13                      | 13     | 13      | 13       | 13         |
| Condensate Drain Located Inside Building                                                                                                                                                                                                                                                                  |                               |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           | Cellular Glass                | 0.043                                            | 40                      | 40     | 40      | 40       | 40         |

[illegible]

| TABLE 2                                                                                                                                                                                                                                                                                                   |          |                                                  |                         |        |         |          |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------|-------------------------|--------|---------|----------|------------|
| Piping Insulation Thickness (mm)<br>For flexible cellular foam the thickness should be 13mm instead of 15mm. Economic thickness or prevention of condensation is the basis of these tables. If prevention of condensation is the criterion, the ambient temperature and relative humidity must be stated. |          |                                                  |                         |        |         |          |            |
| Service                                                                                                                                                                                                                                                                                                   |          |                                                  |                         |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           | Material | Conductivity<br>K-value<br>(W/m <sup>2</sup> -K) | Tube And Pipe Size (mm) |        |         |          |            |
|                                                                                                                                                                                                                                                                                                           |          |                                                  | <25                     | 25-<40 | 40-<100 | 100-<200 | > or = 200 |
|                                                                                                                                                                                                                                                                                                           |          |                                                  |                         |        |         |          |            |

### 3.2.2 Aboveground Cold Pipelines

Insulate the following cold pipelines for minus 34 to plus 16 degrees C in accordance with Table 2 except those piping listed in subparagraph Pipe Insulation in PART 3 as to be omitted. This includes but is not limited to the following:

- a. Make-up water.
- b. Horizontal and vertical portions of interior roof drains.
- c. Refrigerant suction lines.
- d. Chilled water.
- e. Air conditioner condensate drains.
- f. Condenser water piping where pipe temperature may fall below ambient dew point.
- g. Exposed lavatory drains and domestic water lines serving plumbing fixtures for handicap persons.
- h. Domestic cold and chilled drinking water.

#### 3.2.2.1 Insulation Material and Thickness

Determine insulation thickness for cold pipelines using Table 2.



### 3.2.2.2 Factory or Field applied Jacket

Cover insulation with a factory applied vapor retarder jacket/vapor barrier or field applied seal welded PVC jacket or greater than 3 ply laminated self-adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) vapor barrier/weatherproofing jacket - less than 0.02 permeability, standard grade, silver, white, black and embossed for use with Mineral Fiber, Cellular Glass, and Phenolic Foam Insulated Pipe. For insulation inside the building, to be protected with an aluminum jacket, PVC jacket or fittings (installed over a vapor retarder jacket), or greater than 3 ply vapor barrier/weatherproofing self-adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) product, less than 0.02 permeability, standard grade, Embossed Silver, White & Black, install the insulation and vapor retarder jacket as specified herein. Install the aluminum jacket, PVC jacket or fittings (installed over a vapor retarder jacket), or greater than 3 ply vapor barrier/weatherproofing self-adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) product, less than 0.02 permeability, standard grade, embossed silver, White & Black, as specified for piping exposed to weather, except sealing of the laps of the aluminum jacket is not required. In high abuse areas such as janitor closets and traffic areas in equipment rooms, kitchens, and mechanical rooms, provide aluminum jackets, PVC jacket or fittings (installed over a vapor retarder jacket), or greater than 3 ply vapor barrier/weatherproofing self-adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) product, less than 0.02 permeability, standard grade, embossed silver, white & black, for pipe insulation to the 1.8 m level.

### 3.2.2.3 Installing Insulation for Straight Runs Hot and Cold Pipe

Apply insulation to the pipe with tight butt joints. Seal all butted joints and ends with joint sealant and seal with a vapor retarder coating, greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape or PVDC adhesive tape.

#### 3.2.2.3.1 Longitudinal Laps of the Jacket Material

Overlap not less than 38 mm. Provide butt strips 75 mm wide for circumferential joints.

#### 3.2.2.3.2 Laps and Butt Strips

Secure with adhesive and staple on 100 mm centers if not factory self-sealing. If staples are used, seal in accordance with paragraph STAPLES below. Note that staples are not required with cellular glass systems.

#### 3.2.2.3.3 Factory Self-Sealing Lap Systems

May be used when the ambient temperature is between 4 and 50 degrees C during installation. Install the lap system in accordance with manufacturer's recommendations. Use a stapler only if specifically recommended by the manufacturer. Where gaps occur, replace the section or repair the gap by applying adhesive under the lap and then stapling.

#### 3.2.2.3.4 Staples

Coat all staples, including those used to repair factory self-seal lap systems, with a vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - 0.0000 perm adhesive tape. Coat all seams,

except those on factory self-seal systems, with vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape.

#### 3.2.2.3.5 Breaks and Punctures in the Jacket Material

Patch by wrapping a strip of jacket material around the pipe and secure it with adhesive, staple, and coat with vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape. Extend the patch not less than 38 mm past the break.

#### 3.2.2.3.6 Penetrations Such as Thermometers

Fill the voids in the insulation and seal with vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape.

#### 3.2.2.3.7 Flexible Elastomeric Cellular Pipe Insulation

Install by slitting the tubular sections and applying them onto the piping or tubing. Alternately, whenever possible slide un-slit sections over the open ends of piping or tubing. Secure all seams and butt joints and seal with adhesive. When using self seal products, secure only the butt joints with adhesive. Push insulation on the pipe, never pulled. Stretching of insulation may result in open seams and joints. Clean cut all edges. Rough or jagged edges of the insulation are not be permitted. Use proper tools such as sharp knives. Do not stretch Grade 1, Type II sheet insulation around the pipe when used on pipe larger than 150 mm. On pipes larger than 300 mm, adhere sheet insulation directly to the pipe on the lower 1/3 of the pipe.

#### 3.2.2.4 Insulation for Fittings and Accessories

- a. Butt pipe insulation tightly to the insulation of the fittings and accessories. Seal the butted joints and ends with joint sealant and seal with a vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape.
- b. Place precut or preformed insulation around all fittings and accessories and conform to MICA plates except as modified herein: 5 for anchors; 10, 11, and 13 for fittings; 14 for valves; and 17 for flanges and unions. Insulation must be the same insulation as the pipe insulation, including same density, thickness, and thermal conductivity. Where precut/preformed is unavailable, rigid preformed pipe insulation sections may be segmented into the shape required. Use insulation of the same thickness and conductivity as the adjoining pipe insulation. If nesting size insulation is used, overlap the insulation 50 mm or one pipe diameter. Elbows insulated using segments must conform to MICA Tables 12.20 "Mitered Insulation Elbow". Submit a booklet containing completed MICA Insulation Stds plates detailing each insulating system for each pipe, duct, or equipment insulating system, after approval of materials and prior to applying insulation.
  - (1) Ensure MICA plates detail the materials to be installed and the specific insulation application. Submit all MICA plates required showing the entire insulating system, including plates required to show insulation penetrations, vessel bottom and top heads, legs, and skirt insulation as applicable. Present all variations

of insulation systems including locations, materials, vaporproofing, jackets and insulation accessories.

- (2) If the Contractor elects to submit detailed drawings instead of edited MICA Plates, ensure the detail drawings are technically equivalent to the edited MICA Plate submittal.
- c. Upon completion of insulation installation on flanges, unions, valves, anchors, fittings and accessories, terminations, seams, joints and insulation not protected by factory vapor retarder jackets or PVC fitting covers must be protected with PVDC or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape or two coats of vapor retarder coating with a minimum total thickness of 2 mm, applied with glass tape embedded between coats. Overlap tap seams 25 mm. Extend the coating out onto the adjoining pipe insulation 50 mm. Protect fabricated insulation with a factory vapor retarder jacket with either greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape, standard grade, silver, white, black and embossed or PVDC adhesive tape or two coats of vapor retarder coating with a minimum thickness of 2 mm and with a 50 mm wide glass tape embedded between coats. Where fitting insulation butts to pipe insulation, seal the joints with a vapor retarder coating and a 100 mm wide ASJ tape which matches the jacket of the pipe insulation.
- d. Insulate anchors attached directly to the pipe for a sufficient distance to prevent condensation but no less than 150 mm from the insulation surface.
- e. Mark insulation to show the location of unions, strainers, and check valves.

### 3.2.2.5 Optional PVC Fitting Covers

At the option of the Contractor, premolded, one or two piece PVC fitting covers may be used in lieu of the vapor retarder and embedded glass tape. Factory precut insulation inserts or premolded insulation segments must be used under the fitting covers for elbows, tees and other miscellaneous configurations. Insulation inserts must be the same installed insulation thickness. One insert must be installed for each inch of pipe insulation. The installed and compressed insert must have the same thermal performance as the pipe insulation. Premolded insulation segments must be the same insulation as the pipe insulation including same density, thickness, and thermal conductivity. PVC fitting covers must be secured by PVC vapor retarder tape, adhesive, seal welding or with tacks made for securing PVC covers. For above ambient pipe temperatures, secure PVC covers using tacks. As an option circumferential joints can be covered with PVC vapor retarder tape. For below ambient conditions seal system using PVC vapor retarder tape, and vapor retarder mastic with vapor dams. Longitudinal seams may be sealed with vapor retarder mastic, PVC vapor retarder tape or solvent weld adhesive to obtain a continuous vapor seal. For severe service conditions install PVC jackets over the existing ASJ jacket and seal using a solvent weld adhesive.

### 3.2.3 Aboveground Hot Pipelines

#### 3.2.3.1 General Requirements

Insulate all hot pipe lines above 16 degrees C, except those piping listed in subparagraph Pipe Insulation in PART 3 as to be omitted, in accordance

with Table 2. This includes but is not limited to the following:

- a. Domestic hot water supply & re-circulating system.
- b. Condensate & compressed air discharge.

Cover insulation, in accordance with manufacturer's recommendations, with a factory applied Type I jacket or field applied aluminum where required or seal welded PVC.

### 3.2.3.2 Insulation for Fittings and Accessories

Butt pipe insulation tightly to the insulation of the fittings and accessories. Seal butted joints and ends with joint sealant. Mark insulation to show the location of unions, strainers, check valves and other components that would otherwise be hidden from view by the insulation.

#### 3.2.3.2.1 Precut or Preformed

Place precut insulation inserts or preformed insulation inserts around all fittings and accessories. Insulation inserts must be the same installed insulation thickness as the pipe insulation. One insert must be installed for each inch of pipe insulation thickness. The installed and compressed insert must have the same thermal performance as the pipe insulation. Preformed insulation inserts must be the same insulation as the pipe insulation including same density, thickness, and thermal conductivity.

#### 3.2.3.2.2 Rigid Preformed

Where precut/preformed is unavailable, rigid preformed pipe insulation sections may be segmented into the shape required. Use insulation that is the same thickness and conductivity as the adjoining pipe insulation. If nesting size insulation is used, do not overlap insulation 50 mm or one pipe diameter. Elbows insulated using segments must conform to MICA Tables 12.20 "Mitered Insulation Elbow".

### 3.2.4 Piping Exposed to Weather

Insulate and jacket piping exposed to weather as specified for the applicable service inside the building. After this procedure, apply a laminated self-adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) vapor barrier/weatherproofing jacket - less than 0.02 permeability (greater than 3 ply, standard grade, silver, white, black and embossed aluminum jacket, stainless steel or PVC jacket).

PVC jacketing requires no factory-applied jacket beneath it, however apply an all service jacket if factory applied jacketing is not furnished.

#### 3.2.4.1 Aluminum Jacket

The jacket for hot piping may be factory applied. Overlap the jacket no less than 50 mm at longitudinal and circumferential joints and secure with bands at no more than 300 mm centers. Overlap longitudinal joints down to shed water and locate at 4 or 8 o'clock positions. Seal joints on piping 16 degrees C and below with metal jacketing/flashing sealant while overlapping to prevent moisture penetration. Where jacketing on piping 16 degrees C and below abuts an un-insulated surface, caulk joints to prevent moisture penetration. Seal joints on piping above 16 degrees C with a

moisture retarder.

#### 3.2.4.2 Insulation for Fittings

Insulate and finish flanges, unions, valves, fittings, and accessories as specified for the applicable service. Apply two coats of breather emulsion type weatherproof mastic (impermeable to water, permeable to air) recommended by the insulation manufacturer with glass tape embedded between coats. Overlap tape no less than 25 mm and the adjoining aluminum jacket no less than 50 mm. Factory preformed aluminum jackets may be used in lieu of the above. Provide molded PVC fitting covers when PVC jackets are used for straight runs of pipe. PVC fitting covers must be sealed based upon the pipe temperature in combination with ambient conditions and if the pipe insulation will be exposed to severe service conditions. For below ambient conditions, the pipe insulation and insulation fittings must be covered and sealed with a vapor barrier or weatherproofing jacket with a minimum 0.02 permeability as defined per ASTM C1136 before installing the PVC jacket and fitting cover. When in doubt as to the pipe temperature or service conditions, secure the PVC jacket using a weld adhesive using the manufacturer's recommended adhesive.

#### 3.2.4.3 PVC Jacket

Provide ultraviolet resistant PVC jacket that is adhesive welded weather tight with manufacturer's recommended adhesive. Include provision for thermal expansion.

#### 3.2.4.4 Stainless Steel Jackets

ASTM A240/A240M; Type 304, minimum thickness of 33 gauge (0.25 mm), smooth surface with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide stainless steel bands, minimum width of 13 mm.

#### 3.2.5 Below Ground Pipe Insulation

Insulate below ground pipes in accordance with Table 2, except as precluded in paragraph PIPE INSULATION in PART 3. This includes, but is not limited to the following:

- a. Chiller Water.

##### 3.2.5.1 Type of Insulation

Insulate below ground pipe with Cellular Glass insulation, in accordance with manufacturer's instructions for application with thickness as determined from Table 2 (whichever is the most restrictive).

##### 3.2.5.2 Installation of Below ground Pipe Insulation

- a. Coat bore surfaces of the insulation with a thin coat of gypsum cement of a type recommended by the insulation manufacturer. Ensure coating thickness is sufficient to fill surface cells of insulation. Do not use mastic type materials for this coating. Note that unless this is for a cyclic application (i.e., one that fluctuates between high and low temperature on a daily process basis) there is no need to bore coat the material.
- b. Use stainless steel bands, 19 mm wide by 0.508 mm thick to secure

insulation in place. Apply a minimum of two bands per section of insulation. As an alternate, fiberglass reinforced tape may be used to secure insulation on piping up to 300 mm in diameter. Apply a minimum of two bands per section of insulation.

- c. Terminate insulation at anchor blocks but continue through sleeves and manholes.
- d. At point of entry to buildings, terminate underground insulation 50 mm inside the wall or floor, butt tightly against the aboveground insulation and seal the butt joint with high temperature silicone sealant and cover with fibrous glass tape.
- e. Make provision for expansion and contraction of the insulation system in accordance with the insulation manufacturer's recommendations.
- f. Insulate flanges, couplings, valves, and fittings with factory pre-molded, prefabricated, or field-fabricated sections of insulation of the same material and thickness as the adjoining pipe insulation. Secure insulation sections as recommended by the manufacturer.
- g. Finish insulation, including fittings, with three coats of asphaltic mastic, with 6 by 5.5 mesh synthetic reinforcing fabric embedded between coats. Overlap fabric a minimum of 50 mm at joints. Ensure total film thickness is a minimum of 4.7 mm. As an alternate, apply a prefabricated bituminous laminated jacket, reinforced with internal reinforcement mesh, to the insulation. Use jacketing material and application procedures that match manufacturer's written instructions. Vapor barrier - less than 0.02 permeability self adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) jacket greater than 3 ply, standard grade, silver, white, black and embossed or greater than 8 ply (minimum 0.072 adhesive), heavy duty, white or natural). Use application procedures that match the manufacturer's written instructions.
- h. At termination points, other than building entrances, use mastic and cloth or tape to cover the ends of insulation and extend 50 mm along the bare pipe.

### 3.3 DUCT INSULATION SYSTEMS INSTALLATION

Except for oven hood exhaust duct insulation, install corner angles on external corners of insulation on ductwork in exposed finished spaces before covering with jacket. Air conditioned spaces are defined as those spaces directly supplied with cooled conditioned air (or provided with a cooling device such as a fan-coil unit) and heated conditioned air (or provided with a heating device such as a unit heater, radiator or convector).

#### 3.3.1 Duct Insulation Minimum Thickness

Duct insulation minimum thickness in accordance with Table 4.

| Table 4 - Minimum Duct Insulation (mm) |    |
|----------------------------------------|----|
| Cold Air Ducts                         | 50 |
| Relief Ducts                           | 40 |
| Fresh Air Intake Ducts                 | 40 |
|                                        |    |
| Warm Air Ducts                         | 50 |
| Relief Ducts                           | 40 |
| Fresh Air Intake Ducts                 | 40 |

### 3.3.2 Insulation and Vapor Retarder/Vapor Barrier for Cold Air Duct

Provide insulation and vapor retarder/vapor barrier for the following cold air ducts and associated equipment.

- a. Supply ducts.
- b. Return air ducts.
- c. Relief ducts.
- d. Flexible run-outs (field-insulated).
- e. Plenums.
- f. Duct-mounted coil casings.
- g. Coil headers and return bends.
- h. Coil casings.
- i. Fresh air intake ducts.
- j. Filter boxes.
- k. Mixing boxes (field-insulated).
- l. Supply fans (field-insulated).
- m. Site-erected air conditioner casings.
- n. Ducts exposed to weather.
- o. Combustion air intake ducts.
- p. Exhaust ducts passing through concealed spaces, unconditioned spaces, or semi-heated spaces exhausting conditioned air.

Use insulation for rectangular ducts that is flexible type where concealed, minimum density 12 kg/cubic m, and rigid type where exposed, minimum density 48 kg/cubic m. Provide insulation for both concealed or exposed round/oval ducts that is flexible type, minimum density 12 kg/cubic m or a semi rigid board, minimum density 48 kg/cubic m, formed or fabricated to a tight fit, edges beveled and joints tightly butted and

staggered. Provide insulation for all exposed ducts with either a white, paint-able, factory-applied Type I jacket or a field applied vapor retarder/vapor barrier jacket coating finish as specified. Ensure the total field applied dry film thickness is approximately 2 mm. Provide insulation on all concealed duct with a factory-applied Type I or II vapor retarder/vapor barrier jacket. Continue duct insulation through sleeves and prepare openings except firewall penetrations. Duct insulation terminating at fire dampers, must be continuous over the damper collar and retaining angle of fire dampers, which are exposed to unconditioned air and which may be prone to condensate formation. Provide duct insulation and vapor retarder/vapor barrier to cover the collar, neck, and un-insulated surfaces of diffusers, registers and grills. Apply vapor retarder/vapor barrier materials to form a complete unbroken vapor seal over the insulation. Seal sheet metal duct in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION.

#### 3.3.2.1 Installation on Concealed Duct

- a. For rectangular, oval or round ducts, attach flexible insulation by applying adhesive around the entire perimeter of the duct in 150 mm wide strips on 300 mm centers.
- b. For rectangular and oval ducts, 600 mm and larger, additionally secure insulation to bottom of ducts using mechanical fasteners. Space fasteners on 400 mm centers and no more than 400 mm from duct corners.
- c. For rectangular, oval and round ducts, provide mechanical fasteners on sides of duct risers for all duct sizes. Space fasteners on 400 mm centers and no more than 400 mm from duct corners.
- d. Impale insulation on the mechanical fasteners (self stick pins) where used and press thoroughly into the adhesive. Take care to ensure vapor retarder/vapor barrier jacket joints overlap 50 mm. Do not compress insulation to a thickness less than that specified. Carry insulation over standing seams and trapeze-type duct hangers.
- e. Where mechanical fasteners are used, install self-locking washers and trim and bend the pin over.
- f. Secure jacket overlaps with staples and tape as necessary to ensure a secure seal. Coat staples, tape and seams with a brush coat of vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate (minimum 0.05 mm adhesive, 0.075 mm embossed) - less than 0.0000 perm adhesive tape.
- g. Cover breaks in the jacket material with patches of the same material as the vapor retarder jacket. Do not extend patches less than 50 mm beyond the break or penetration in all directions and secure with tape and staples. Seal staples and tape joints with a brush coat of vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate (minimum 0.05 mm adhesive, 0.075 mm embossed) - less than 0.0000 perm adhesive tape.
- h. At jacket penetrations such as hangers, thermometers, and damper operating rods, fill voids in the insulation and seal the penetration with a brush coat of vapor retarder coating or PVDC adhesive tape greater than 3 ply laminate (minimum 0.05 mm adhesive, 0.075 mm embossed) - less than 0.0000 perm adhesive tape.



- i. Seal insulation terminations and pin punctures and flash with a reinforced vapor retarder coating finish or tape with a brush coat of vapor retarder coating.. Ensure the coating overlaps the adjoining insulation and un-insulated surface 50 mm. Extend pin puncture coatings 50 mm from the puncture in all directions.
- j. Where insulation standoff brackets occur, extend insulation under the bracket and terminate the jacket at the bracket.

### 3.3.2.2 Installation on Exposed Duct Work

- a. For rectangular ducts, secure rigid insulation to the duct by mechanical fasteners on all four sides of the duct, space no more than 300 mm apart and no more than 75 mm from the edges of the insulation joints. Provide a minimum of two rows of fasteners for each side of duct 300 mm and larger. Provide one row for each side of duct less than 300 mm. Provide mechanical fasteners that are corrosion resistant as G60 coated galvanized steel, and indefinitely sustain a 22.7 kg tensile dead load test perpendicular to the duct wall.
- b. Form duct insulation with minimum jacket seams. Fasten each piece of rigid insulation to the duct using mechanical fasteners. When the height of projections is less than the insulation thickness, bring insulation up to standing seams, reinforcing, and other vertical projections and do not carry over. Continue vapor retarder/barrier jacket across seams, reinforcing, and projections. When height of projections is greater than the insulation thickness, carry over insulation and jacket . Apply insulation with joints tightly butted. Neatly bevel insulation around name plates and access plates and doors.
- c. Impale insulation on the fasteners; install self-locking washers and trim and bend the pin over.
- d. Seal joints in the insulation jacket with a 100 mm wide strip of tape. Seal taped seams with a brush coat of vapor retarder coating.
- e. Cover breaks and ribs or standing seam penetrations in the jacket material with a patch of the same material as the jacket. Do not extend patches less than 50 mm beyond the break or penetration and secure with tape and staple. Seal staples and joints with a brush coat of vapor retarder coating.
- f. At jacket penetrations such as hangers, thermometers, and damper operating rods, fill the voids in the insulation and seal the penetrations with a flashing sealant.
- g. Seal and flash insulation terminations and pin punctures with a reinforced vapor retarder coating finish. Ensure coating overlaps the adjoining insulation and un-insulated surface 50 mm. Extend pin puncture coatings 50 mm from the puncture in all directions.
- h. Insulation for both concealed or exposed round or oval ducts must be flexible type, minimum density 12 kg/cubic m or a semi rigid board, minimum density 48 kg/cubic m, formed or fabricated to a tight fit, edges beveled and joints tightly butted and staggered. Insulation for all exposed ducts must be provided with either a white, factory-applied Type I jacket or a field applied vapor retarder or vapor barrier jacket coating finish as specified, the total field applied dry film thickness must be approximately 2 mm.

- i. Insulate oval and round ducts, flexible type, with factory Type I jacket insulation with minimum density of 12 kg per cubic meter, attach in accordance with MICA standards.

### 3.3.3 Insulation for Warm Air Duct

Provide insulation and vapor barrier for the following warm air ducts and associated equipment:.

- a. Supply ducts.
- b. Return air ducts.
- c. Relief air ducts
- d. Flexible run-outs (field insulated).
- e. Plenums.
- f. Duct-mounted coil casings.
- g. Coil-headers and return bends.
- h. Coil casings.
- i. Fresh air intake ducts.
- j. Filter boxes.
- k. Mixing boxes.
- l. Supply fans.
- m. Site-erected air conditioner casings.
- n. Ducts exposed to weather.
- o. Exhaust ducts passing through concealed spaces, unconditioned spaces, or semi-heated spaces exhausting conditioned air.

Provide insulation for rectangular ducts that is flexible type where concealed, and rigid type where exposed. Provide insulation on exposed ducts with a white, paint-able, factory-applied Type II jacket, or finish with adhesive finish. Use flexible type insulation for round ducts, with a factory-applied Type II jacket. Provide insulation on concealed duct with a factory-applied Type II jacket. Accomplish adhesive finish where indicated to be used by applying two coats of adhesive with a layer of glass cloth embedded between the coats. Ensure total dry film thickness is approximately 2.0 mm. Continue duct insulation through sleeves and prepare openings. Terminate duct insulation at fire dampers and flexible connections.

#### 3.3.3.1 Installation on Concealed Duct

- a. For rectangular, oval and round ducts, attach insulation by applying adhesive around the entire perimeter of the duct in 150 mm wide strips on 300 mm centers.

- b. For rectangular and oval ducts 600 mm and larger, secure insulation to the bottom of ducts using mechanical fasteners. Space fasteners on 450 mm centers and no more than 450 mm from duct corner.
- c. For rectangular, oval and round ducts, provide mechanical fasteners on sides of duct risers for all duct sizes. Space fasteners on 450 mm centers and no more than 450 mm from duct corners.
- d. Impale insulation on the mechanical fasteners where used. Do not compress insulation to a thickness less than that specified. Carry insulation over standing seams and trapeze-type hangers.
- e. Install self-locking washers where mechanical fasteners are used and trim and bend the pin over.
- f. Do not overlap insulation jacket less than 50 mm at joints and secure the lap and staple on 100 mm centers.

#### 3.3.3.2 Installation on Exposed Duct

- a. For rectangular ducts, secure the rigid insulation to the duct using mechanical fasteners on all four sides of the duct, space no more than 400 mm apart and no more than 150 mm from the edges of the insulation joints. Provide a minimum of two rows of fasteners for each side of duct 300 mm and larger and a minimum of one row for each side of duct less than 300 mm.
- b. Form duct insulation with factory-applied jacket with minimum jacket seams, and fasten each piece of rigid insulation to the duct using mechanical fasteners. When the height of projection is less than the insulation thickness, bring insulation up to standing seams, reinforcing, and other vertical projections and do not carry over the projection. Continue jacket across seams, reinforcing, and projections. Where the height of projections is greater than the insulation thickness, carry insulation and jacket over the projection.
- c. Impale insulation on the fasteners; install self-locking washers and trim and bend the pin over.
- d. Seal joints on jacketed insulation with a 100 mm wide strip of tape and brush with vapor retarder coating.
- e. Cover breaks and penetrations in the jacket material with a patch of the same material as the jacket. Extend patches no less than 50 mm beyond the break or penetration and secure with adhesive and staple.
- f. Seal insulation terminations and pin punctures with tape and brush with vapor retarder coating.
- g. Insulate oval and round ducts, flexible type, with factory Type I jacket insulation, minimum density of 12 kg per cubic meter attach by staples spaced no more than 400 mm and no more than 150 mm from the degrees of joints. Seal joints in accordance with item "d." above.

#### 3.3.4 Ducts Handling Air for Dual Purpose

For air handling ducts for dual purpose below and above 16 degrees C, insulate ducts as specified for cold air duct.

### 3.3.5 Insulation for Evaporative Cooling Duct

Insulate evaporative cooling supply duct located in spaces not evaporatively cooled. Use material and installation requirements as specified for duct insulation for warm air duct.

### 3.3.6 Duct Test Holes

After duct systems have been tested, adjusted, and balanced, repair breaks in the insulation and jacket in accordance with the applicable section of this specification for the type of duct insulation to be repaired. After the insulation is repaired, identify on the outside of the insulation with marker, sticker, or other means "TAB test ports."

### 3.3.7 Duct Exposed to Weather

#### 3.3.7.1 Installation

Insulate and finish ducts exposed to weather as specified for the applicable service for exposed duct inside the building. After the above is accomplished, further finish the insulation as detailed in the following subparagraphs.

#### 3.3.7.2 Round Duct

Laminated self-adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) vapor barrier/weatherproofing jacket - Less than 0.02 permeability, (greater than 3 ply, standard grade, silver, white, black and embossed or greater than 8 ply, heavy duty, white and natural) membrane must be applied overlapping material by 75 mm no bands or caulking needed - see manufacturer's recommended installation instructions. Aluminum jacket with factory applied moisture retarder must be applied with the joints lapped no less than 75 mm and secured with bands located at circumferential laps and at no more than 300 mm intervals throughout. Lap horizontal joints down to shed water and located at 4 or 8 o'clock position. Seal joints with metal jacketing sealant to prevent moisture penetration. Where jacketing abuts an un-insulated surface, seal joints with metal jacketing sealant.

#### 3.3.7.3 Fittings

Finish fittings and other irregular shapes as specified for rectangular ducts.

#### 3.3.7.4 Rectangular Ducts

Apply two coats of weather barrier mastic reinforced with fabric or mesh for outdoor application to the entire surface. Ensure each coat of weatherproof mastic has a minimum thickness of 2 mm. Ensure exterior is a metal jacketing applied for mechanical abuse and weather protection, and secure with screws or vapor barrier/weatherproofing jacket less than 0.02 permeability greater than 3 ply, standard grade, silver, white, black, and embossed or greater than 8 ply, heavy duty white and natural. Apply membrane overlapping material by 75 mm. No bands or caulking needed-see manufacturing recommend installation instructions.

### 3.4 EQUIPMENT INSULATION SYSTEMS INSTALLATION

#### 3.4.1 General

Provide removable insulation sections to cover parts of equipment that must be opened periodically for maintenance including vessel covers, fasteners, flanges and accessories. Omit equipment insulation on the following:

- a. Hand-holes.
- b. Boiler manholes.
- c. Cleanouts.
- d. ASME stamps.
- e. Manufacturer's nameplates.
- f. Duct Test/Balance Test Holes.

#### 3.4.2 Insulation for Cold Equipment

Cold equipment below 16 degrees C: Furnish insulation on equipment handling media below 16 degrees C including the following:

- a. Pumps.
- b. Refrigeration equipment parts that are not factory insulated.
- c. Drip pans under chilled equipment.
- d. Cold water storage tanks.
- e. Water softeners.
- f. Duct mounted coils.
- g. Cold and chilled water pumps.
- h. Pneumatic water tanks.
- i. Roof drain bodies.
- j. Air handling equipment parts that are not factory insulated.
- k. Expansion and air separation tanks.

##### 3.4.2.1 Insulation Type

Provide insulation suitable for the temperature encountered. Provide material and thicknesses as shown in Table 5:

| TABLE 5                                           |                               |                |
|---------------------------------------------------|-------------------------------|----------------|
| Insulation Thickness for Cold Equipment (mm)      |                               |                |
| Equipment handling media at indicated temperature |                               |                |
|                                                   | Material                      | Thickness (mm) |
| 2 to 16 degrees C                                 |                               |                |
|                                                   |                               |                |
|                                                   | Flexible Elastomeric Cellular | 25             |
| Minus 18 to 1 degree C                            |                               |                |
|                                                   |                               |                |
|                                                   | Flexible Elastomeric Cellular | 40             |
| Minus 34 to minus 17 degrees C                    |                               |                |
|                                                   |                               |                |
|                                                   | Flexible Elastomeric Cellular | 45             |

#### 3.4.2.2 Pump Insulation

- a. Insulate pumps by forming a box around the pump housing. Construct the box by forming the bottom and sides using joints that do not leave raw ends of insulation exposed. Join joints between sides and between sides and bottom by adhesive with lap strips for rigid mineral fiber and contact adhesive for flexible elastomeric cellular insulation. Ensure box conforms to the requirements of MICA Insulation Stds plate No. 49 when using flexible elastomeric cellular insulation. Ensure joints between top cover and sides fit tightly forming a female shiplap joint on the side pieces and a male joint on the top cover, thus making the top cover removable.
- b. Protect exposed insulation corners with corner angles.
- c. Upon completion of installation of the insulation, including removable sections, apply two coats of vapor retarder coating with a layer of glass cloth embedded between the coats. The total dry thickness of the finish must be 2 mm. Provide a parting line between the box and the removable sections allowing the removable sections to be removed without disturbing the insulation coating. Apply flashing sealant to parting line, between equipment and removable section insulation, and at all penetrations.

#### 3.4.2.3 Other Equipment

- a. Form or fabricate insulation to fit the equipment. To ensure a tight fit on round equipment, bevel edges and tightly butt and stagger joints.

- b. Secure insulation in place with bands or wires at intervals as recommended by the manufacturer but no more than 300 mm centers except adhere flexible elastomeric cellular with contact adhesive. Protect insulation corners under wires and bands with suitable corner angles.
- c. Install cellular glass in accordance with manufacturer's instructions. Seal joints and ends with joint sealant, and seal with a vapor retarder coating.
- d. Use removable insulation on heads of heat exchangers. Fabricate removable section joints using a male-female shiplap type joint. Finish the entire surface of the removable section by applying two coats of vapor retarder coating with a layer of glass cloth embedded between the coats. The total dry thickness of the finish must be 2 mm.
- e. Protect exposed insulation corners with corner angles.
- f. Apply insulation on equipment with ribs over 150 by 150 mm by 12 gauge welded wire fabric which has been cinched in place, or if approved by the Contracting Officer, spot weld to the equipment over the ribs. Secure insulation to the fabric with J-hooks and 50 by 50 mm washers or securely band or wire in place on 300 mm centers.

#### 3.4.2.4 Vapor Retarder/Vapor Barrier

Upon completion of installation of insulation, caulk penetrations. Apply two coats of vapor retarder coating or vapor barrier jacket over insulation, including removable sections, with a layer of open mesh synthetic fabric embedded between the coats. Ensure the total dry thickness of the finish is 2 mm. Apply flasing sealant or vapor barrier tape to parting line between equipment and removable section insulation.

#### 3.4.3 Insulation for Hot Equipment

Furnish insulation on equipment handling media above 16 degrees C including the following:

- a. Converters.
- b. Water heaters.
- c. Pumps handling media above 54 degrees C.
- d. Air separation tanks.

##### 3.4.3.1 Insulation

Provide insulation suitable for the temperature encountered. Insulate shell and tube-type heat exchangers for the temperature of the shell medium.

Determine insulation thickness for hot equipment using Table 6:

| TABLE 6                                                                      |                                       |                |
|------------------------------------------------------------------------------|---------------------------------------|----------------|
| Insulation Thickness for Hot Equipment (mm)                                  |                                       |                |
| Equipment handling steam or media at indicated pressure or temperature limit |                                       |                |
|                                                                              | Material                              | Thickness (mm) |
| 103 kPa or 121 degrees C                                                     |                                       |                |
|                                                                              | Rigid Mineral Fiber                   | 50             |
|                                                                              | Flexible Mineral Fiber                | 50             |
|                                                                              | Calcium Silicate/Perlite              | 100            |
|                                                                              |                                       |                |
|                                                                              | Faced Phenolic Foam                   | 40             |
|                                                                              | Flexible Elastomeric Cellular (<93 C) | 25             |
|                                                                              |                                       |                |
|                                                                              |                                       |                |
|                                                                              |                                       |                |
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|                                                                              |                                       |                |
|                                                                              |                                       |                |
|                                                                              |                                       |                |
|                                                                              |                                       |                |

### 3.4.3.2 Insulation of Pumps

Insulate pumps by forming a box around the pump housing. Construct the box by forming the bottom and sides using joints that do not leave raw ends of insulation exposed. Band bottom and sides to form a rigid housing that does not rest on the pump. Ensure joints between top cover and sides fit tightly. The top cover must have a joint forming a female shiplap joint on the side pieces and a male joint on the top cover, making the top cover removable. Apply two coats of Class I adhesive over insulation, including removable sections, with a layer of glass cloth embedded between the coats. Provide a parting line between the box and the removable sections allowing the removable sections to be removed without disturbing



the insulation coating. The total dry thickness of the finish must be 2 mm. Apply caulking to parting line of the removable sections and penetrations.

#### 3.4.3.3 Other Equipment

- a. Form or fabricate insulation to fit the equipment. To ensure a tight fit on round equipment, bevel edges and tightly butt and stagger joints.
- b. Secure insulation in place with bands or wires at intervals as recommended by the manufacturer but no greater than 300 mm centers except adhere flexible elastomeric cellular. Protect insulation corners under wires and bands with suitable corner angles.
- c. On high vibration equipment, set cellular glass insulation in a coating of bedding compound as recommended by the manufacturer, and seal joints with bedding compound. Fill mineral fiber joints with finishing cement.
- d. Provide removable insulation on heads of heat exchangers. Fabricate the removable section joint using a male-female shiplap type joint. Finish the entire surface of the removable section as specified.
- e. Protect exposed insulation corners with corner angles.
- f. On equipment with ribs, such as boiler flue gas connection, draft fans, and fly ash or soot collectors, apply insulation over 150 by 150 mm by 12 gauge welded wire fabric which has been cinched in place, or if approved by the Contracting Officer, spot weld to the equipment over the ribs. Secure insulation to the fabric with J-hooks and 50 by 50 mm washers or securely band or wire in place on 300 mm (maximum) centers.
- g. On equipment handling media above 316 degrees C, apply insulation in two or more layers with staggered joints.
- h. Upon completion of installation of insulation, caulk penetrations. Apply two coats of adhesive over insulation, including removable sections, with a layer of glass cloth embedded between the coats. The total dry thickness of the finish must be 2 mm. Apply caulking to parting line between equipment and removable section insulation.

#### 3.4.4 Equipment Exposed to Weather

##### 3.4.4.1 Installation

Insulate equipment exposed to weather and finish in accordance with the requirements for ducts exposed to weather in paragraph DUCT INSULATION INSTALLATION.

##### 3.4.4.2 Optional Panels

At the option of the Contractor, prefabricated metal insulation panels may be used in lieu of the insulation and finish previously specified. Thermal performance must be equal to or better than that specified for field applied insulation. Provide panels that are the standard catalog product of a manufacturer of metal insulation panels. Provide fastenings, flashing, and support system conforming to published recommendations of the manufacturer for weatherproof installation and that prevent moisture

from entering the insulation. Design panels to accommodate thermal expansion and to support a 1112 N walking load without permanent deformation or permanent damage to the insulation. Exterior metal cover sheet must be aluminum and exposed fastenings must be stainless steel or aluminum.

-- End of Section --

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## SECTION 23 08 00

## COMMISSIONING OF MECHANICAL AND PLUMBING SYSTEMS

05/23, CHG 1: 08/24

## PART 1 GENERAL

Building Commissioning is a systematic, quality-focused process for enhancing the delivery of a project that focuses on verifying and documenting that all of the commissioned systems and assemblies are planned, designed, installed, tested, operated, and maintained to meet the project requirements. The purpose is to reduce the cost and performance risks associated with delivering facilities projects, and to increase value to owners, occupants, and users. Comply with specification Section 01 91 00.15 BUILDING COMMISSIONING.

## 1.1 SEQUENCING AND SCHEDULING

Complete functional performance testing prior to performance verification testing required by Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. Complete the following prior to starting Functional Performance Tests of mechanical systems:

- a. All equipment and systems completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with contract documents and construction plans and specifications
- b. Final DALT Report submitted and approved in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- c. Pre-final Testing, Adjusting, and Balancing Report submitted in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- d. The Certificate of Readiness submitted and approved in accordance with Section 01 91 00.15 BUILDING COMMISSIONING
- e. Air Leakage Test Reports and Diagnostic Test Reports submitted and approved in accordance with Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING
- f. Tests, Flushing, and Disinfection in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Test Equipment; G

SD-06 Test Reports

Pipe Flushing, Testing, And Water Treatment Reports; G

Completed Pre-Functional Checklists; G

Seasonal Test Report; G

Post-Construction Trend Log Report; G

#### SD-07 Certificates

Certificate Of Readiness; G

### 1.3 ACCESSIBILITY REQUIREMENTS

Equipment, systems, and devices for commissioned systems must be accessible. Make necessary modifications if systems and devices are not accessible for inspections and testing.

Assist commissioning team in testing by removing equipment covers, opening access panels, and other required activities that assist with visual oversight. Furnish ladders, flashlights, meters, gauges, or other inspection equipment as necessary.

### 1.4 COORDINATION

Refer to Section 01 91 00.15 BUILDING COMMISSIONING for requirements pertaining to coordination during the commissioning process. Coordinate with the Commissioning Provider in accordance with Section 01 91 00.15 and in accordance with the Commissioning Plan to schedule inspections as required to support the commissioning process. Furnish additional information requested by the Commissioning Provider. Coordinate scheduling of Functional Performance Testing with the commissioning team. Provide plans, reports, notes, and other documentation to the Commissioning Provider as specified in the commissioning plan, as it is completed.

### 1.5 PIPE FLUSHING, TESTING, AND WATER TREATMENT REPORTS

Test requirements are specified in Division 22 and 23 piping Sections. Prepare a pipe system cleaning, flushing, and hydrostatic testing log. Provide cleaning, flushing, testing, and water treatment log and final reports.

Include the following in the pipe system cleaning, flushing, and hydrostatic testing log:

- a. Minimum flushing water velocity.
- b. Water treatment reports.
- c. Tracking checklist for managing and ensuring that all pipe sections have been cleaned, flushed, hydrostatically tested, and chemically treated.

### 1.6 CERTIFICATE OF READINESS

Submit Certificate of Readiness documentation in accordance with Section 01 91 00.15 BUILDING COMMISSIONING for all equipment and systems including

start-up reports; completed Pre-Functional Checklists; Testing, Adjusting, and Balancing (TAB) Report; Issues Log; HVAC Controls Start-Up Reports. Do not schedule Functional Performance Tests for the system until the Certificate of Readiness for that system receives approval by the Government. The CxC, Quality Control Manager, and the Mechanical, Electrical, Controls, and TAB subcontractor representatives must sign and date the Certificate of Readiness.

## PART 2 PRODUCTS

### 2.1 TEST EQUIPMENT

Provide all testing equipment required to perform testing for the systems to be commissioned, except for equipment specific to and used by TAB as required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Provide a sufficient quantity of two-way radios for each subcontractor. Submit list of Test Equipment and instrumentation to be used for testing including equipment/instrument identification number, equipment application or planned use, manufacturer, make, model, and serial number, and calibration history with certificates. Also list special equipment and proprietary tools specific to a piece of equipment required for testing.

#### 2.1.1 Proprietary Equipment

Provide manufacturer's proprietary test equipment and software required by any equipment manufacturer for programming and start-up, whether specified or not. Provide manufacturer test equipment, demonstrate its use, and assist in the commissioning process as needed. Provide data logging equipment and software required to test equipment.

#### 2.1.2 Calibration and Accuracy

Comply with equipment manufacturer's test equipment calibration procedures and intervals. Recalibrate test instruments immediately after instruments have been repaired resulting from being dropped or damaged. Affix calibration tags to test instruments. Furnish calibration records to Contracting Officer upon request.

Provide all testing equipment of sufficient quality and accuracy to test and measure system performance with the tolerances specified. Unless otherwise noted, the following minimum requirements apply: Provide temperature sensors and digital thermometers with a certified calibration within the past year to an accuracy of 0.3 degrees C and a resolution of plus or minus 0.1 degrees C. Provide pressure sensors with an accuracy of plus or minus 2.0 percent of the value range being measured (not full range of meter) and calibrated within the last year.

## PART 3 EXECUTION

### 3.1 MEETINGS

Attend all meetings in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

Provide timely updates on construction schedule changes to allow the Commissioning Provider to execute commissioning process efficiently. Notify Contracting Officer of anticipated construction delays to commissioning activities not yet performed or not yet scheduled.

### 3.2 COMMISSIONING CONSTRUCTION OBSERVATION CHECKLISTS

Commissioning construction observation checklists include Pre-Functional Checklists, Integrated Systems Test Checklists, and Functional Performance Test Checklists. Provide commissioning construction observation checklists for the Interim and Final Construction Phase Commissioning Plan in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

#### 3.2.1 Pre-Functional Checklists

The Pre-Functional Checklists must include items for physical inspection or testing that demonstrate that installation and start-up of equipment and systems is complete. Refer to paragraph PRE-FUNCTIONAL CHECKS. Pre-functional checklists must be tailored to verify the specific installation requirements and details of the construction documents and manufacturer's instructions.

Use the Pre-Functional Checklists prepared by the CxC for physical inspection or testing to demonstrate that installation and start-up of equipment and systems is complete. Refer to paragraph PRE-FUNCTIONAL CHECKS.

#### 3.2.2 Functional Performance Test Checklists

Functional Performance Test Checklists must include procedures that explain, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING. Include the following sections and details appropriate to the systems being tested in the Functional Performance Test Checklists:

- a. Notable system features including information about controls to facilitate understanding of system operation and maintenance
- b. Conclusions and recommendations. Conclusions must clearly indicate if system does or does not perform in accordance with contract requirements and specifications. Recommendation must clearly indicate that the system should or should not be approved by the Government.
- c. Test conditions including date, beginning and ending time, and beginning and ending outdoor air conditions
- d. Attendees
- e. Identification of the equipment involved in the test
- f. **System feature identification**
- g. Point-to-point observations including demonstrating system flow meters and sensors have been calibrated and are correctly displayed on the Operator work station
- h. Actuator operation observations demonstrating actuator responses to commands from the control system
- i. As-found condition of the system operation

- j. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.
- k. Space for comments for each test item.

Use the Functional Performance Test Checklists prepared by the CxC that list, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING.

### 3.2.3 Integrated Systems Test Checklists

Integrated Systems Test Checklists must include test procedures that explain, step-by-step, the actions and expected results that will demonstrate that the interactive operations between systems performs in accordance with the contract. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING. Include the following sections in the Integrated Systems Test Checklists:

- a. Notable features of the interconnected systems organized by discipline including information to facilitate understanding of system operation
- b. Conclusions and recommendations. Conclusions must clearly indicate if the systems do or do not perform in accordance with contract requirements and specifications. Recommendation must clearly indicate that the systems should or should not be approved by the Government
- c. Test conditions including date and beginning and ending time
- d. Identification of the equipment and systems involved in the test
- e. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.
- f. Space for comments for each test item.

Use the Integrated Systems Test Checklists prepared by the CxC that list, step-by-step, the actions and expected results that will demonstrate that the interactive operations between systems performs in accordance with the contract. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING.

### 3.3 PRE-FUNCTIONAL CHECKS

Pre-Functional Checks are a type of Commissioning Inspection in accordance with Section 01 91 00.15 BUILDING COMMISSIONING. Complete one Pre-Functional Checklist for each individual item of equipment or system for each system required to be commissioned including, but not limited to, ductwork, piping, equipment, fixtures, and controls. Include manufacturer start-up checklists associated with equipment with the submission of the Pre-Functional Checklists. Provide manufacturer's installation manual for each type of unit. Indicate commissioning team member inspection and validation of each Pre-Functional Checklist item by initials. Validation of each Pre-Functional Checklist item by each team member indicates that item conforms to the contract documents and validated design in their area of responsibility. Commissioning Specialist validation of each



Pre-Functional Checklist item indicates that each item has been installed correctly and in accordance with contract documents. Required commissioning team members for Pre-Functional Checks includes the CxC, Mechanical Commissioning Specialist, Quality Control Manager, sub-contractor representative for each trade responsible for construction/installation of the checklist item, and the TAB representative (for items impacting TAB). Submit the initialed and Completed Pre-Functional Checklists no later than 7 calendar days after completion of inspection of all checklist items for each system.

### 3.4 STARTUP AND INITIAL CHECKOUT

Document start-up and initial testing procedures, and include in the Completed Pre-Functional Checklists submittal, including:

- a. Startup tests and factory testing reports.
- b. Manufacturer's representative start-up, operating, troubleshooting and maintenance procedures.
- c. Additional documentation necessary for third party certification programs.
- d. Perform and clearly document system operational checks and quality control checks as they are completed, and providing a copy to the commissioning team.
- e. Correct deficiencies and sign the Certificate of Readiness for each system before functional performance testing

### 3.5 DUCT AIR LEAKAGE TEST (DALT) REPORT REVIEW AND VERIFICATION

#### 3.5.1 Duct Air Leakage Test (DALT) Report Review

The Mechanical System Technical Commissioning Specialist must review the pre-final DALT Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The Commissioning Specialist is responsible for reviewing the pre-final DALT Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to DALT Report approval.

#### 3.5.2 Duct Air Leakage Test (DALT) Report Verification

The Mechanical System Technical Commissioning Specialist must witness the DALT Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The Commissioning Specialist is responsible for witnessing the DALT Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to DALT Report approval.

### 3.6 TESTING, ADJUSTING, AND BALANCING REPORT REVIEW AND VERIFICATION

#### 3.6.1 Testing, Adjusting, and Balancing (TAB) Report Review

The Mechanical System Technical Commissioning Specialist must review the pre-final TAB Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The Commissioning Specialist is responsible for reviewing the pre-final TAB Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to TAB Report approval.

#### 3.6.2 Testing, Adjusting, and Balancing (TAB) Report Verification

The Mechanical System Technical Commissioning Specialist must witness the TAB Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The Commissioning Specialist is responsible for witnessing the TAB Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to TAB Report approval.

### 3.7 HVAC CONTROLS START-UP AND PERFORMANCE VERIFICATION TEST REVIEW

The Mechanical System Technical Commissioning Specialist must review the Start-Up Testing Report, PVT Procedures and PVT Reports, including endurance testing trend submittals, required by Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. The Mechanical System Technical Commissioning Specialist must review each submittal and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The Commissioning Specialist is responsible for reviewing the Start-Up Testing Report, PVT Procedures and PVT Reports including endurance testing trend data required by Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to final acceptance.

### 3.8 FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING

Functional Performance and Integrated Systems Testing are a type of Commissioning Testing in accordance with Section 01 91 00.15 BUILDING COMMISSIONING. Conduct Functional Performance Testing in accordance with Section 01 91 00.15 BUILDING COMMISSIONING and requirements in this Section. Prior to Functional Performance Testing, complete all prerequisites in accordance with paragraph SEQUENCING AND SCHEDULING. Demonstrate that all system components have been installed, that each control device and item of equipment operates, and that the systems operate and perform, including interactive operation between systems, in accordance with contract documents.

Perform Integrated Systems Tests only after the Functional Performance Tests for each associated system are completed with all deficiencies

resolved and after the related Functional Performance Test Checklists have been signed by each commissioning team member.

### 3.8.1 Test Scheduling and Coordination

Schedule and conduct Initial Functional Performance Tests as soon as all contract work is completed, regardless of the season. Develop and implement means of artificial loading to demonstrate, to a reasonable level of confidence, the ability of the HVAC systems to handle peak seasonal loads. Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. Correct all deficiencies identified through any prior review, inspection, or test activity before the start of Functional Performance Tests.

Functional Performance Tests and Integrated Systems Tests must be performed with the CxC present. Government reserves the right to witness all tests. Coordinate test schedule with Government representatives.

### 3.8.2 Preparation

Put equipment and systems into operation and continue operation during each working day of functional performance and integrated systems testing, as required. Verify temperature and pressure taps in accordance with Contract Documents. Provide a pressure/temperature plug at each water sensor which is an input point to control system.

Perform minor adjustments to equipment and systems during Functional Performance Tests as deemed necessary by the commissioning team. Where calibrated DDC sensors cannot be used to record test data, provide measuring instruments, logging devices, and data acquisition equipment to record data for the complete range of test data for the required test period.

### 3.8.3 Testing Procedures

Provide all necessary materials and system modifications to produce the necessary flows, pressures, temperatures, and other conditions necessary to execute the test according to the specified conditions. At completion of the test, return the affected building equipment and systems to their pre-test condition.

Follow the Functional Performance Test from the approved Final Construction Phase Commissioning Plan. Perform Functional Performance Tests for each item of equipment and each system required to be commissioned. Verify all sensor calibrations, control responses, safeties, interlocks, operating modes, sequences of operation, capacities, and all other performance requirements comply with contract, regardless of the specific items listed within the checklists provided. In general, testing must progress from equipment or components to subsystems to systems to interlocks and connections between systems. Commissioning Specialists are responsible for determining the order of components and systems to be tested.

Indicate validation of each item of equipment and systems tested by signature of each commissioning team member for each test. The Quality Control Representative, Commissioning Specialists, and Contracting Officer's Representative, if present, must indicate validation after the equipment and systems are free of deficiencies.

### 3.8.4 Simulating Conditions

Functional performance testing is conducted by simulating conditions at control devices to initiate a control system response. Before testing, calibrate all sensors, transducers and devices. Over-writing control input values through the control system is not acceptable unless approved by the Contracting Officer. Perform each test under conditions that simulate actual conditions as close as is practically possible. Specific examples of simulating conditions are provided below. Do not simulate conditions when damage to the system or building may result.

- a. When varying static pressures inside ductwork cannot be simulated within the duct, and where a sensor signals the controls system to initiate sequences at various duct static pressures, it is acceptable to simulate the various pressures with a Pneumatic Squeeze-Bulb Type Signaling Device with gauge temporarily attached to the sensing tube leading to the transmitter. It is not acceptable to reset the various set-points, nor to simulate an electric analog signal (unless approved as noted above).
- b. Dirty filter pressure drops can be simulated by partially blocking filter face.
- c. Freeze-stat safeties can be simulated by packing portion of sensor with ice.
- d. High outside air temperatures can be simulated with a hair blower.
- e. Raising entering cooling coil temperatures by activating a heating/preheat coil can be used to simulate entering cooling coil conditions.
- f. Do not use signal generators to simulate sensor signals unless approved by the Contracting Officer, as noted above, for special cases.
- g. Control set points can be altered. For example, to see the air conditioning compressor lockout work at an outside air temperature below 12 degrees C, when the outside air temperature is above 12 degrees C, temporarily change the lockout set point to be minus 17 degrees C above the current outside air temperature. Caution: Set points are not to be raised or lowered to a point to cause damage to the components, systems, or the building structure and contents.
- h. Test duct mounted smoke detectors in accordance with the manufacturer's recommendations. Perform the tests with air system at minimum airflow condition.
- i. Test current sensing relays used for fan and pump status signals to control system to indicate unit failure and run status by resetting the set point on the relay to simulate a lost belt or unit failure while the unit is running. Confirm that the failure alarm was generated and received at the control system. After the test is conducted, return the set point to its original set-point or a set-point as indicated by the Contracting Officer.

### 3.8.5 Manufacturer's Representative

Provide a factory trained representative authorized by the equipment

manufacturer to perform Functional Performance Testing for the following equipment:

Chillers

Packaged Direct-Expansion Refrigeration Equipment, including variable refrigerant flow (VRF) systems

Packaged Computer Room Air Handlers (CRAH) Air Conditioners (CRAC)

Booster Pumps

Ensure the test representative reviews, approves, and signs the completed field test report. Include person's name with signatures.

### 3.8.6 Integrated Systems Tests

Follow the Integrated Systems Test Checklists from the approved Final Construction Phase Commissioning Plan. Integrated Systems Tests must be performed for the interactive operation between HVAC systems and other systems such as fire protection systems, smoke control systems, or back-up electrical supply. Verify correct interactive operation, acceptable speed of response, and other contract requirements for both normal and failure modes. Examples of Integrated Systems Tests include the correct operation of HVAC systems during emergency system activation or correct operation of uninterruptible power supplies or energy generators and connected systems.

### 3.8.7 Sample Strategy

Refer to Section 01 91 00.15 BUILDING COMMISSIONING for sample strategy. Complete a Functional Performance Test Checklist for each item of equipment or system to be tested.

#### 3.8.7.1 100 Percent Sample Procedures

Systems or equipment for which 100 percent sample size are tested fail if one or more of the test procedures results in discovery of a deficiency and the deficiency cannot be resolved within 5 minutes during the test.

Re-test to the extent necessary to confirm that the deficiencies have been corrected without negatively impacting the performance of the rest of the system.

#### 3.8.7.2 Less than 100 Percent Sample Procedures

Randomly test each sample group of identical equipment. If 10 percent of the units in the first sample fail the functional performance tests, test a second sample group, the same size as the first sample group. The second sample must not include any units from the first sample group.

If 10 percent of the units in the second sample fail, test all remaining units. If at any point frequent failures occur the CxC may stop the testing and require the contractor to perform and document a checkout of the remaining units prior to continuing functional testing.

### 3.9 INTEGRATED SYSTEMS TEST

Perform Integrated Systems Tests only after the Functional Performance Tests for each associated system are completed with all deficiencies

resolved and after the related Functional Performance Test Checklists have been signed by each commissioning team member.

### 3.10 RETESTING REQUIREMENTS

Abort tests if any deficiency prevents successful completion of the test or if any required commissioning team member is not present for the test. Re-test only after all deficiencies identified during the original tests have been corrected.

Contracting Officer may withhold payment equivalent to lost time, re-testing, and aborted tests. These costs may include salary, travel costs, and per diem for Government team members. Correct deficiencies as identified by the commissioning team and retest the systems to be commissioned.

### 3.11 SYSTEM ACCEPTANCE

Systems may be partially accepted prior to seasonal testing if they comply with all construction contract and accepted design requirements that can be tested during initial Functional Performance Tests. All test procedures must be successful completed prior to full systems acceptance.

### 3.12 SEASONAL TESTS

Perform Initial Functional Performance Tests as soon as all contract work is completed, but prior to facility turnover, regardless of the season.

In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems during peak heating and cooling seasons during outdoor air condition design extremes. Schedule Seasonal Functional Performance Tests in coordination with the Contracting Officer. Submit Seasonal Test Report within 14 days of test completion. Include seasonal test report in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

Execute seasonal functional performance testing, witnessed by the Contracting Officer. Correct deficiencies and make adjustments to O&M manuals and as-built drawings for applicable issues identified in any seasonal testing.

### 3.13 TRAINING

The Mechanical Systems Technical Commissioning Specialist must review the training plan required by Section 01 78 00 OPERATION AND MAINTENANCE DATA and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel.

The Commissioning Provider is responsible for overseeing and approving the training plan required by Section 01 78 00 OPERATION AND MAINTENANCE DATA and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel.

Coordinate, schedule, and document all required training. At a minimum, include the following items in the training report for commissioned systems:

- a. Complete commissioning documentation

- b. Complete O&M data
- c. Complete Training
- d. Purpose of equipment.
- e. Principle of how the equipment works.
- f. Important parts and assemblies.
- g. How the equipment achieves its purpose and necessary operating conditions.
- h. Most likely failure modes, causes and corrections.
- i. On site demonstration.
- j. Provide updates to O&M manuals based on field modifications.
- k. Provide training of the post-occupancy operations and maintenance staff.

### 3.14 COMMISSIONING REPORT

Include all completed Pre-Functional Checklists and Functional Performance and Integrated Systems Test Checklists in the Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING. Include the approved TAB Report.

### 3.15 POST-CONSTRUCTION ENDURANCE TEST

Perform an Endurance Test in accordance with the paragraph ENDURANCE TESTING in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC once during the peak heating season and once during the peak cooling season during outdoor air condition extremes with the exception that network bandwidth usage measurement and recording is not required.

The Mechanical System Commissioning Specialists must review the trend logs from the Endurance Tests to ensure that the systems have stable operation and operate as required by the construction contract, the accepted design. The Commissioning Specialists must provide a Post-Construction Trend Log Report that identifies any deficiencies noted in operation, recommendations for correction, and includes a graphical representation of the trends. Provide one Trend Log Report for the peak cooling season and one Trend Log Report for the peak heating season. Submit the Post-Construction Trend Log Reports no later than 14 calendar days following receipt of the trend log data by the Commissioning Specialist. Provide a Post-Construction Trend Log Report meeting the same requirements as the Endurance Testing Results Format in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. Provide one Trend Log Report for the peak cooling season and one Trend Log Report for the peak heating season. Submit the Post-Construction Trend Log Reports no later than 14 calendar days following download of the trended data. Include the Post-Construction Trend Log Reports in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

-- End of Section --

## SECTION 23 09 00

INSTRUMENTATION AND CONTROL FOR HVAC  
08/24

## PART 1 GENERAL

## 1.1 SUMMARY

Provide a complete Direct Digital Control (DDC) system suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as indicated and shown and in accordance with Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet or Niagara BACnet systems, Section 25 05 11.01 CYBERSECURITY FOR FACILITY-RELATED CONTROL HVAC SYSTEMS, and other referenced Sections.

## 1.1.1 Proprietary Systems

## 1.1.1.1 Proprietary Systems Exempted From Open Protocol Requirements

The following systems are specifically exempted from the open protocol requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS:

- a. A simple split (DX) system consisting of a single indoor unit and a single outdoor unit from the same manufacturer.

## 1.1.1.2 Implementation of Proprietary Systems

For proprietary systems exempted from open protocol requirements, a proprietary network and DDC hardware communicating via a proprietary protocol are permitted. For these systems a building control network meeting the requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS must also be provided, along with a gateway or interface to connect the proprietary system to the open building control network.

The proprietary system gateway or interface must provide the required functionality as shown on the points schedule. Scheduling, alarming, trending, overrides, network inputs, network outputs and other protocol related requirements must be met on the open protocol control system as specified in Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

## 1.1.2 System Requirements

Provide systems meeting the requirements this Section and other Sections referenced by this Section, and which have the following characteristics:

- a. The system implements the control sequences of operation shown in the Contract Drawings using DDC hardware to control mechanical and electrical equipment
- b. The system meet the requirements of this specification as a



stand-alone system and does not require connection to any other system.

- c. Control sequences reside in DDC hardware in the building. The building control network is not dependent upon connection to a Utility Monitoring and Control System (UMCS) Front End or to any other system for performance of control sequences. To the greatest extent practical, the hardware performs control sequences without reliance on the building network.
- d. The hardware is installed such that individual control equipment can be replaced by similar control equipment from other equipment manufacturers with no loss of system functionality.
- e. All necessary documentation, configuration information, programming tools, programs, drivers, and other software are licensed to and otherwise remain with the Government such that the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer.
- f. Sufficient documentation and data, including rights to documentation and data, are provided such that the Government or their agents can execute work to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer.
- g. Hardware is installed and configured such that the Government or their agents are able to perform repair, replacement, and upgrades of individual hardware without further interaction with the Contractor, Vendor or Manufacturer.

#### 1.1.3 End to End Accuracy

Select products, install and configure the system such that the maximum error of a measured value as read from the DDC Hardware over the network is less than the maximum allowable error specified for the sensor or instrumentation.

#### 1.1.4 Verification of Dimensions

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

#### 1.1.5 Drawings

The Government will not indicate all offsets, fittings, and accessories that may be required on the drawings. Carefully investigate the mechanical, electrical, and finish conditions that could affect the work to be performed, arrange such work accordingly, and provide all work necessary to meet such conditions.

### 1.2 RELATED SECTIONS

Related work specified elsewhere:

- a. Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet systems with or without Niagara Framework.

- b. Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC
- c. Section 25 05 11.01 CYBERSECURITY FOR FACILITY-RELATED CONTROLHVAC SYSTEMS
- d. Section 23 81 00 DECENTRALIZED UNITARY HVAC EQUIPMENT
- e. Section 01 91 00.15 BUILDING COMMISSIONING

### 1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135 (2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-24 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control Networks

ASHRAE FUN SI (2021) Fundamentals Handbook, SI Edition

#### INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41 (1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

#### NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NFPA 90A (2024) Standard for the Installation of Air Conditioning and Ventilating Systems

#### UNDERWRITERS LABORATORIES (UL)

UL 5085-3 (2006; Reprint Jan 2022) UL Standard for Safety Low Voltage Transformers - Part 3: Class 2 and Class 3 Transformers

### 1.4 DEFINITIONS

The following list of definitions includes terms used in Sections referenced by this Section and are included here for completeness. The definitions contained in this Section may disagree with how terms are defined or used in other documents, including documents referenced by this

Section. The definitions included here are the authoritative definitions for this Section and all Sections referenced by this Section.

After each term the protocol related to that term is included in parenthesis.

#### 1.4.1 Alarm Generation (All protocols)

Alarm Generation is the monitoring of a value, comparison of the value to alarm conditions and the creation of an alarm when the conditions set for the alarm are met.

#### 1.4.2 Building Automation and Control Network (BACnet) (BACnet)

The term BACnet is used in two ways. First meaning the BACnet Protocol Standard - the communication requirements as defined by ASHRAE 135 including all annexes and addenda. The second to refer to the overall technology related to the ASHRAE 135 protocol.

#### 1.4.3 BACnet Advanced Application Controller (B-AAC) (BACnet)

A hardware device BTL Listed as a B-AAC, which is required to support BACnet Interoperability Building Blocks (BIBBs) for scheduling and alarming, but is not required to support as many BIBBs as a B-BC.

#### 1.4.4 BACnet Application Specific Controller (B-ASC) (BACnet)

A hardware device BTL Listed as a B-ASC, with fewer BIBB requirements than a B-AAC. It is intended for use in a specific application.

#### 1.4.5 BACnet Building Controller (B-BC) (BACnet)

A hardware device BTL Listed as a B-BC. A general-purpose, field-programmable device capable of carrying out a variety of building automation and control tasks including control and monitoring via direct digital control (DDC) of specific systems and data storage for trend information, time schedules, and alarm data. Like the other BTL Listed controller types (B-AAC, B-ASC etc.) a B-BC device is required to support the server ("B") side of the ReadProperty and WriteProperty services, but unlike the other controller types it is also required to support the client ("A") side of these services. Communication between controllers requires that one of them support the client side and the other support the server side, so a B-BC is often used when communication between controllers is needed.

#### 1.4.6 BACnet Broadcast Management Device (BBMD) (BACnet)

A communications device, typically combined with a BACnet router. A BBMD forwards BACnet broadcast messages to BACnet/IP devices and other BBMDs connected to the same BACnet/IP network. Each IP subnet that is part of a BACnet/IP network must have at least one BBMD. Note there are additional restrictions when multiple BBMDs share an IP subnet.

#### 1.4.7 BACnet/IP (BACnet)

An extension of BACnet, Annex J, defines the use of a reserved UDP socket to transmit BACnet messages over IP networks. A BACnet/IP network is a collection of one or more IP subnets that share the same BACnet network number. See also paragraph BACNET BROADCAST MANAGEMENT DEVICE.

#### 1.4.8 BACnet Internetwork (BACnet)

Two or more BACnet networks, connected with BACnet routers. In a BACnet Internetwork, there exists only one message path between devices.

#### 1.4.9 BACnet Interoperability Building Blocks (BIBBs) (BACnet)

A BIBB is a collection of one or more ASHRAE 135 Services intended to define a higher level of interoperability. BIBBs are combined to build the BACnet functional requirements for a device in a specification. Some BIBBs define additional requirements (beyond requiring support for specific services) in order to achieve a level of interoperability. For example, the BIBB DS-V-A (Data Sharing-View-A), which would typically be used by a front-end, not only requires the client to support the ReadProperty Service, but also provides a list of data types (Object / Properties) which the client must be able to interpret and display for the user.

In the BIBB shorthand notation, -A is the client side and -B is the server side.

| The following is a list of some BIBBs used by this or referenced Sections: |                                                         |
|----------------------------------------------------------------------------|---------------------------------------------------------|
| DS-COV-A                                                                   | Data Sharing-Change of Value (A side)                   |
| DS-COV-B                                                                   | Data Sharing-Change of Value (B side)                   |
| NM-RC-B                                                                    | Network Management-Router Configuration (B side)        |
| DS-RP-A                                                                    | Data Sharing-Read Property (A side)                     |
| DS-RP-B                                                                    | Data Sharing-Read Property (B side)                     |
| DS-RPM-A                                                                   | Data Sharing-Read Property Multiple (A Side)            |
| DS-RPM-B                                                                   | Data Sharing-Read Property Multiple (B Side)            |
| DS-WP-A                                                                    | Data Sharing-Write Property (A Side)                    |
| DM-TS-B                                                                    | Device Management-Time Synchronization (B Side)         |
| DM-UTC-B                                                                   | Device Management-UTC Time Synchronization (B Side)     |
| DS-WP-B                                                                    | Data Sharing-Write Property (B side)                    |
| SCHED-E-B                                                                  | Scheduling-External (B side)                            |
| DM-OCD-B                                                                   | Device Management-Object Creation and Deletion (B side) |
| AE-N-I-B                                                                   | Alarm and Event-Notification Internal (B Side)          |
| AE-N-E-B                                                                   | Alarm and Event-Notification External (B Side)          |
| T-VMT-I-B                                                                  | Trending-Viewing and Modifying Trends Internal (B Side) |

|                                                                            |                                                         |
|----------------------------------------------------------------------------|---------------------------------------------------------|
| The following is a list of some BIBBs used by this or referenced Sections: |                                                         |
| T-VMT-E-B                                                                  | Trending-Viewing and Modifying Trends External (B Side) |

#### 1.4.10 BACnet Network (BACnet)

In BACnet, a portion of the control Internetwork consisting of one or more segments connected by repeaters. Networks are separated by routers.

#### 1.4.11 BACnet Operator Display (B-OD) (BACnet)

A basic operator interface with limited capabilities relative to a B-OWS. It is not intended to perform direct digital control. A B-OD profile could be used for LCD devices, displays affixed to BACnet devices, handheld terminals or other very simple user interfaces.

#### 1.4.12 BACnet Segment (BACnet)

One or more physical segments interconnected by repeaters (ASHRAE 135).

#### 1.4.13 BACnet Smart Actuator (B-SA) (BACnet)

A simple actuator device with limited resources intended for specific applications.

#### 1.4.14 BACnet Smart Sensor (B-SS) (BACnet)

A simple sensing device with limited resources.

#### 1.4.15 BACnet Testing Laboratories (BTL) (BACnet)

Established by BACnet International to support compliance testing and interoperability testing activities and consists of BTL Manager and the BTL Working Group (BTL-WG). BTL also publishes Implementation Guidelines.

#### 1.4.16 BACnet Testing Laboratories (BTL) Listed (BACnet)

A device that has been listed by BACnet Testing Laboratory. Devices may be certified to a specific device profile, in which case the listing indicates that the device supports the required capabilities for that profile, or may be listed as "other".

#### 1.4.17 Binary (All protocols)

A two-state system where an "ON" condition is represented by a high signal level and an "OFF" condition is represented by a low signal level. 'Digital' is sometimes used interchangeably with 'binary'.

#### 1.4.18 Broadcast (BACnet)

Unlike most messages, which are intended for a specific recipient device, a broadcast message is intended for all devices on the network.

#### 1.4.19 Building Control Network (BCN) (All protocols)

The network connecting all DDC Hardware within a building (or specific group of buildings).

#### 1.4.20 Building Point of Connection (BPOC) (All protocols)

A FPOC for a Building Control System. (This term is being phased out of use in preference for FPOC but is still used in some specifications and criteria. When it was used, it typically referred to a piece of control hardware. The current FPOC definition typically refers instead to IT hardware.)

#### 1.4.21 Commandable (All protocols)

See Overridable.

#### 1.4.22 Commandable Objects (BACnet)

Commandable Objects have a Commandable Property, Priority\_Array, and Relinquish\_Default Property as defined in ASHRAE 135, Clause 19.2, Command Prioritization.

#### 1.4.23 Configurable (All protocols)

A property, setting, or value is configurable if it can be changed via hardware settings on the device, via the use of engineering software or over the control network from the front end, and is retained through (after) loss of power.

- a. In a BACnet system, a property, setting, or value is configurable if it can be changed via one or more of:
  - (1) via BACnet services (including proprietary BACnet services)
  - (2) via hardware settings on the device

Note this is more stringent than the ASHRAE 135 definition.

#### 1.4.24 Control Logic Diagram (All protocols)

A graphical representation of control logic for multiple processes that make up a system.

#### 1.4.25 Device (BACnet)

A Digital Controller that contains a BACnet Device Object and uses BACnet to communicate with other devices.

#### 1.4.26 Device Object (BACnet)

Every BACnet device requires one Device Object, whose properties represent the network visible properties of that device. Every Device Object requires a unique Object Identifier number on the BACnet Internetwork. This number is often referred to as the device instance or device ID.

#### 1.4.27 Device Profile (BACnet)

A collection of BIBBs determining minimum BACnet capabilities of a device, defined in ASHRAE 135. Standard device profiles include BACnet Advanced Workstations (B-AWS), BACnet Building Controllers (B-BC), BACnet Advanced

Application Controllers (B-AAC), BACnet Application Specific Controllers (B-ASC), BACnet Smart Actuator (B-SA), and BACnet Smart Sensor (B-SS).

#### 1.4.28 Digital Controller (All protocols)

An electronic controller, usually with internal programming logic and digital and analog input/output capability, which performs control functions.

#### 1.4.29 Direct Digital Control (DDC) (All protocols)

Digital controllers performing control logic. Usually the controller directly senses physical values, makes control decisions with internal programs, and outputs control signals to directly operate switches, valves, dampers, and motor controllers.

#### 1.4.30 Field Point of Connection (FPOC) (All protocols)

The FPOC is the point of connection between the UMCS IP Network and the field control network (either an IP network, a non-IP network, or a combination of both). The hardware at this location which provides the connection is generally an IT device such as a switch, IP router, or firewall.

In general, the term "FPOC Location" means the place where this connection occurs, and "FPOC Hardware" means the device that provides the connection. Sometimes the term "FPOC" is used to mean either and its actual meaning (i.e. location or hardware) is determined by the context in which it is used.

#### 1.4.31 Gateway (All protocols)

A device that translates from one protocol application data format to another. Devices that change only the transport mechanism of the protocol - "translating" from TP/FT-10 to Ethernet/IP or from BACnet MS/TP to BACnet over IP for example - are not gateways as the underlying data format does not change. Gateways are also called Communications Bridges or Protocol Translators.

#### 1.4.32 IEEE 802.3 Ethernet (All protocols)

A family of local-area-network technologies providing high-speed networking features over various media, typically Cat 5, 5e or Cat 6 twisted pair copper or fiber optic cable.

#### 1.4.33 Internet Protocol (IP, TCP/IP, UDP/IP) (All protocols)

A communication method, the most common use is the World Wide Web. At the lowest level, it is based on Internet Protocol (IP), a method for conveying and routing packets of information over various LAN media. Two common protocols using IP are User Datagram Protocol (UDP) and Transmission Control Protocol (TCP). UDP conveys information to well-known "sockets" without confirmation of receipt. TCP establishes connections, also known as "sessions", which have end-to-end confirmation and guaranteed sequence of delivery.

#### 1.4.34 Input/Output (I/O) (All protocols)

Physical inputs and outputs to and from a device, although the term

sometimes describes network or "virtual" inputs or outputs. See also "Points".

#### 1.4.35 I/O Expansion Unit (All protocols)

An I/O expansion unit provides additional point capacity to a digital controller

#### 1.4.36 IP subnet (All protocols)

A group of devices which share a defined range IP addresses. Devices on a common IP subnet can share data (including broadcasts) directly without the need for the traffic to traverse an IP router.

#### 1.4.37 Local-Area Network (LAN) (All protocols)

A communication network that spans a limited geographic area and uses the same basic communication technology throughout.

#### 1.4.38 Local Display Panels (LDPs) (All protocols)

A DDC Hardware with a display and navigation buttons, and must provide display and adjustment of points as shown on the Points Schedule and as indicated.

#### 1.4.39 MAC Address (All protocols)

Media Access Control address. The physical device address that identifies a device on a Local Area Network.

#### 1.4.40 Master-Slave/Token-Passing (MS/TP) (BACnet)

Data link protocol as defined by the BACnet standard. Multiple speeds (data rates) are permitted by the BACnet MS/TP standard.

#### 1.4.41 Monitoring and Control (M&C) Software (All protocols)

The UMCS 'front end' software which performs supervisory functions such as alarm handling, scheduling and data logging and provides a user interface for monitoring the system and configuring these functions.

#### 1.4.42 Network Number (BACnet)

A site-specific number assigned to each network. This network number must be unique throughout the BACnet Internetwork.

#### 1.4.43 Object (BACnet)

An ASHRAE 135 Object. The concept of organizing BACnet information into standard components with various associated Properties. Examples include Analog Input objects and Binary Output objects.

#### 1.4.44 Object Identifier (BACnet)

A grouping of two Object properties: Object Type (e.g. Analog Value, Schedule, etc.) and Object Instance (in this case, a number). Object Identifiers must be unique within a device.



#### 1.4.45 Object Instance (BACnet)

See paragraph OBJECT IDENTIFIER

#### 1.4.46 Object Properties (BACnet)

Attributes of an object. Examples include present value and high limit properties of an analog input object. Properties are defined in ASHRAE 135; some are optional and some are required. Objects are controlled by reading from and writing to object properties.

#### 1.4.47 Operator Configurable (All protocols)

Operator configurable values are values that can be changed from a single common front end user interface across multiple vendor systems.

For non Niagara-based BACnet systems, a property, setting, or value in a device is Operator Configurable when it is Configurable and is either:

- a. a Writable Property of a Standard BACnet Object; or
- b. a Property of a Standard BACnet Object that is Writable when Out\_Of\_Service is TRUE and Out\_Of\_Service is Writable.

#### 1.4.48 Override (All protocols)

Changing the value of a point outside of the normal sequence of operation where the change has priority over the sequence and where there is a mechanism for releasing the change such that the point returns to the normal value. Overrides persist until released or overridden at the same or higher priority but are not required to persist through a loss of power. Overrides are often used by operators to change values, and generally originate at a user interface (workstation or local display panel).

#### 1.4.49 Packaged Equipment (All protocols)

Packaged equipment is a single piece of equipment provided by a manufacturer in a substantially complete and operable condition, where the controls (DDC Hardware) are factory installed, and the equipment is sold and shipped from the manufacturer as a single entity. Disassembly and reassembly of a large piece of equipment for shipping does not prevent it from being packaged equipment. Package units may require field installation of remote sensors. Packaged equipment is also called a "packaged unit".

Note industry may use the term "Packaged System" to mean a collection of equipment that is designed to work together where each piece of equipment is packaged equipment and there is a network that connects the equipment together. A "packaged system" of this type is NOT packaged equipment; it is a collection of packaged equipment, and each piece of equipment must individually meet specification requirements.

#### 1.4.50 Packaged Unit (All protocols)

See packaged equipment.

#### 1.4.51 Performance Verification Test (PVT) (All protocols)

The procedure for determining if the installed BAS meets design criteria prior to final acceptance. The PVT is performed after installation, testing, and balancing of mechanical systems. Typically the PVT is performed by the Contractor in the presence of the Government.

#### 1.4.52 Physical Segment (BACnet)

A single contiguous medium to which BACnet devices are attached (ASHRAE 135).

#### 1.4.53 Polling (All protocols)

A device periodically requesting data from another device.

#### 1.4.54 Points (All protocols)

Physical and virtual inputs and outputs. See also paragraph INPUT/OUTPUT (I/O).

#### 1.4.55 Proportional, Integral, and Derivative (PID) Control Loop (All protocols)

Three parameters used to control modulating equipment to maintain a setpoint. Derivative control is often not required for HVAC systems (leaving "PI" control).

#### 1.4.56 Proprietary (BACnet)

Within the context of BACnet, any extension of or addition to object types, properties, PrivateTransfer services, or enumerations specified in ASHRAE 135. Objects with Object\_Type values of 128 and above are Proprietary Objects. Properties with Property\_Identifier of 512 and above are proprietary Properties.

#### 1.4.57 Protocol Implementation Conformance Statement (PICS) (BACnet)

A document, created by the manufacturer of a device, which describes which portions of the BACnet standard may be implemented by a given device. ASHRAE 135 requires that all ASHRAE 135 devices have a PICS, and also defines a minimum set of information that must be in it. A device as installed for a specific project may not implement everything in its PICS.

#### 1.4.58 Repeater (All protocols)

A device that connects two control network segments and retransmits all information received on one side onto the other.

#### 1.4.59 Router (All protocols)

A device that connects two ASHRAE 135 networks and controls traffic between the two by retransmitting signals received from one side onto the other based on the signal destination. Routers are used to subdivide a BACnet internetwork and to limit network traffic.

#### 1.4.60 Segment (All protocols)

A 'single' section of a control network that contains no repeaters or

routers. There is generally a limit on the number of devices on a segment, and this limit is dependent on the topology/media and device type.

#### 1.4.61 Standard BACnet Objects (BACnet)

Objects with Object\_Type values below 128 and specifically enumerated in Clause 21 of ASHRAE 135. Objects which are not proprietary. See paragraph PROPRIETARY.

#### 1.4.62 Standard BACnet Properties (BACnet)

Properties with Property\_Identifier values below 512 and specifically enumerated in Clause 21 of ASHRAE 135. Properties which are not proprietary. See Proprietary.

#### 1.4.63 Standard BACnet Services (BACnet)

ASHRAE 135 services other than ConfirmedPrivateTransfer or UnconfirmedPrivateTransfer. See paragraph PROPRIETARY.

#### 1.4.64 UMCS (All protocols)

UMCS stands for Utility Monitoring and Control System. The term refers to all components by which a project site monitors, manages, and controls real-time operation of HVAC and other building systems. These components include the UMCS "front-end" and all field building control systems connected to the front-end. The front-end consists of Monitoring and Control Software (user interface software), browser-based user interfaces and network infrastructure.

The network infrastructure (the "UMCS Network"), is an IP network connecting multiple building or facility control networks to the Monitoring and Control Software.

#### 1.4.65 UMCS Network (All protocols)

The UMCS Network connects multiple building or facility control networks to the Monitoring and Control Software.

#### 1.4.66 Writable Property (BACnet)

A Property is Writable when it can be changed through the use of one or more of the WriteProperty services defined in ASHRAE 135, Clause 15 regardless of the value of any other Property. Note that in the ASHRAE 135 standard, some Properties may be writable when the Out of Service Property is TRUE; for purposes of this Section, Properties that are only writable when the Out of Service Property is TRUE are not considered to be Writable.

### 1.5 PROJECT SEQUENCING

TABLE II: PROJECT SEQUENCING lists the sequencing of submittals as specified in paragraph SUBMITTALS (denoted by an 'S' in the 'TYPE' column) and activities as specified in PART 3 EXECUTION (denoted by an 'E' in the 'TYPE' column). TABLE II does not specify overall project milestone and completion dates; these dates are specified in the contract documents.

a. Sequencing for Submittals: The sequencing specified for submittals is

the deadline by which the submittal must be initially submitted to the Government. Following submission there will be a Government review period as specified in Section 01 33 00 SUBMITTAL PROCEDURES. If the submittal is not accepted by the Government, revise the submittal and resubmit it to the Government within 14 days of notification that the submittal has been rejected. Upon resubmittal there will be an additional Government review period. If the submittal is not accepted the process repeats until the submittal is accepted by the Government.

- b. Sequencing for Activities: The sequencing specified for activities indicates the earliest the activity may begin.
- c. Abbreviations: In TABLE II the abbreviation AAO is used for 'after approval of' and 'ACO' is used for 'after completion of'.

| TABLE II. PROJECT SEQUENCING |      |                                                |                                               |
|------------------------------|------|------------------------------------------------|-----------------------------------------------|
| ITEM #                       | TYPE | DESCRIPTION                                    | SEQUENCING (START OF ACTIVITY OR DEADLINE FOR |
| 1                            | S    | Existing Conditions Report                     |                                               |
| 2                            | S    | DDC Contractor Design Drawings                 |                                               |
| 3                            | S    | Manufacturer's Product Data                    |                                               |
| 4                            | S    | Pre-construction QC Checklist                  |                                               |
| 5                            | E    | Install Building Control System                | AAO #1 thru #4                                |
| 6                            | E    | Start-Up and Start-Up Testing                  | ACO #5                                        |
| 7                            | S    | Post-Construction QC Checklist                 | ACO #6                                        |
| 8                            | S    | Programming Software<br>Configuration Software | ACO #6                                        |
| 9                            | S    | Draft As-Built Drawings                        | ACO #6                                        |
| 10                           | S    | Start-Up Testing Report                        | ACO #6                                        |
| 11                           | S    | PVT Procedures                                 | before schedule start of #12 and AAO #10      |

| TABLE II. PROJECT SEQUENCING |      |                                                                      |                                                      |
|------------------------------|------|----------------------------------------------------------------------|------------------------------------------------------|
| ITEM #                       | TYPE | DESCRIPTION                                                          | SEQUENCING (START OF ACTIVITY OR DEADLINE FOR        |
| 12                           | S,E  | Execute PVT PVT Testing Activities                                   | AAO #9 and #11As indicated in PART 3 of this Section |
| 13                           | S    | PVT Report                                                           | ACO #12<br>As indicated in PART 3 of this Section    |
| 14                           | S    | Controller Application Programs<br>Controller Configuration Settings | AAO #13                                              |
| 15                           | S    | Final As-Built Drawings                                              | AAO #13                                              |
| 16                           | S    | O&M Instructions                                                     | AAO #15                                              |
| 17                           | S    | Training Documentation                                               | AAO #10 and before scheduled start of #18            |
| 18                           | E    | Training                                                             | AAO #16 and #17                                      |
| 19                           | S    | Closeout QC Checklist                                                | ACO #18                                              |

## 1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

DDC Contractor Design Drawings; G

Draft As-Built Drawings; G

Final As-Built Drawings; G

SD-03 Product Data

Programming Software; G

Controller Application Programs; G

Configuration Software; G

Controller Configuration Settings; G

Manufacturer's Product Data; G

SD-05 Design Data

SD-06 Test Reports

Pre-Construction Quality Control (QC) Checklist; G

Post-Construction Quality Control (QC) Checklist; G

Start-Up Testing Report; G

PVT Procedures; G

PVT Report; G

Control Contractor's Performance Verification Testing Plan; G

Equipment Supplier's Performance Verification Testing Plan; G

Endurance Testing Results; G

Performance Verification Test Report; G

SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G

Training Documentation; G

SD-11 Closeout Submittals

Enclosure Keys; G

Password Summary Report; G

Closeout Quality Control (QC) Checklist; G

1.7 DATA PACKAGE AND SUBMITTAL REQUIREMENTS

Technical data packages consisting of technical data and computer software (meaning technical data which relates to computer software) which are specifically identified in this project and which may be defined/required in other specifications must be delivered strictly in accordance with the CONTRACT CLAUSES and in accordance with the Contract Data Requirements List, DD Form 1423. Data delivered must be identified by reference to the particular specification paragraph against which it is furnished. All

submittals not specified as technical data packages are considered 'shop drawings' under the Federal Acquisition Regulation Supplement (FARS) and must contain no proprietary information and be delivered with unrestricted rights.

## 1.8 SOFTWARE FOR DDC HARDWARE AND GATEWAYS

Provide all software related to the programming and configuration of DDC Hardware and Gateways as indicated. License all Software to the project site. The term "controller" as used in these requirements means both DDC Hardware and Gateways.

### 1.8.1 Configuration Software

For each type of controller, provide the configuration tool software in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit hard copies of the software user manuals for each software with the software submittal.

Submit Configuration Software on CD-ROM as a Technical Data Package. Submit 2 hard copies of the software user manual for each piece of software.

### 1.8.2 Controller Configuration Settings

For each controller, provide copies of the installed configuration settings as source code compatible with the configuration tool software for that controller in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Controller Configuration Settings on CD-ROM as a Technical Data Package. Include on the CD-ROM a list or table of contents clearly indicating which files are associated with each device. Submit 2 copies of the Controller Configuration Settings CD-ROM.

### 1.8.3 Programming Software

For each type of programmable controller, provide the programming software in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit hard copies of software user manuals for each software with the software submittal.

Submit Programming Software on CD-ROM as a Technical Data Package. Submit 2 hard copies of the software user manual for each piece of software.

### 1.8.4 Controller Application Programs

For each programmable controller, provide copies of the application program as source code compatible with the programming software for that controller in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Controller Application Programs on CD-ROM as a Technical Data Package. Include on the CD-ROM a list or table of contents clearly indicating which application program is associated with each device. Submit 2 copies of the Controller Application Programs CD-ROM.

## 1.9 QUALITY CONTROL CHECKLISTS

The QC Checklist for BACnet Systems in APPENDIX A of this Section must be completed by the Contractor's Chief Quality Control (QC) Representative and submitted as indicated.

The QC Representative must verify each item indicated and initial in the space provided to indicate that the requirement has been met. The QC Representative must sign and date the Checklist prior to submission to the Government.

### 1.9.1 Pre-Construction Quality Control (QC) Checklist

Complete items indicated as Pre-Construction QC Checklist items in the QC Checklist. Submit four copies of the Pre-Construction QC Checklist.

### 1.9.2 Post-Construction Quality Control (QC) Checklist

Complete items indicated as Post-Construction QC Checklist items in the QC Checklist. Submit four copies of the Post-Construction QC Checklist.

### 1.9.3 Closeout Quality Control (QC) Checklist

Complete items indicated as Closeout QC Checklist items in the QC Checklist. Submit four copies of the Closeout QC Checklist.

## PART 2 PRODUCTS

Provide products meeting the requirements of Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet or Niagara BACnet systems, other referenced Sections, and this Section.

### 2.1 GENERAL PRODUCT REQUIREMENTS

Units of the same type of equipment must be products of a single manufacturer. Each major component of equipment must have the manufacturer's name and address, and the model and serial number in a conspicuous place. Materials and equipment must be standard products of a manufacturer regularly engaged in the manufacturing of these and similar products. The standard products must have been in a satisfactory commercial or industrial use for two years prior to use on this project. The two year use must include applications of equipment and materials under similar circumstances and of similar size. DDC Hardware not meeting the two-year field service requirement is acceptable provided it has been successfully used by the Contractor in a minimum of two previous projects. The equipment items must be supported by a service organization. Items of the same type and purpose must be identical, including equipment, assemblies, parts and components.

### 2.2 PRODUCT DATA

Provide manufacturer's product data sheets documenting compliance with product specifications for each product provided under Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS, or this Section. Provide product data for all products in a single indexed compendium, organized by product type.



For all BACnet hardware: for each manufacturer, model and version (revision) of DDC Hardware provide the Protocol Implementation Conformance Statement (PICS) in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Manufacturer's Product Data on CD-ROM.

## 2.3 OPERATION ENVIRONMENT

Unless otherwise specified, provide products rated for continuous operation under the following conditions:

- a. Pressure: Pressure conditions normally encountered in the installed location.
- b. Vibration: Vibration conditions normally encountered in the installed location.
- c. Temperature:
  - (1) Products installed indoors: Ambient temperatures in the range of 0 to 50 degrees C and temperature conditions outside this range normally encountered at the installed location.
  - (2) Products installed outdoors or in unconditioned indoor spaces: Ambient temperatures in the range of -37 to +66 degrees C and temperature conditions outside this range normally encountered at the installed location.
- d. Humidity: 10 to 95 percent relative humidity, noncondensing and humidity conditions outside this range normally encountered at the installed location.

## 2.4 WIRELESS CAPABILITY

Wireless capabilities must comply with Section 25 05 11.01 CYBERSECURITY FOR FACILITY-RELATED CONTROL HVAC SYSTEMS

## 2.5 ENCLOSURES

Enclosures supplied as an integral (pre-packaged) part of another product are acceptable. Provide two Enclosure Keys for each lockable enclosure on a single ring per enclosure with a tag identifying the enclosure the keys operate. Provide enclosures meeting the following minimum requirements:

### 2.5.1 Outdoors

For enclosures located outdoors, provide enclosures meeting NEMA 250 Type 4 requirements.

### 2.5.2 Mechanical and Electrical Rooms

For enclosures located in mechanical or electrical rooms, provide enclosures meeting NEMA 250 Type 2 requirements.

### 2.5.3 Other Locations

For enclosures in other locations including but not limited to occupied

spaces, above ceilings, and in plenum returns, provide enclosures meeting NEMA 250 Type 1 requirements.

## 2.6 WIRE AND CABLE

Provide wire and cable meeting the requirements of NFPA 70 and NFPA 90A in addition to the requirements of this specification and referenced specifications.

### 2.6.1 Terminal Blocks

For terminal blocks which are not integral to other equipment, provide terminal blocks which are insulated, modular, feed-through, clamp style with recessed captive screw-type clamping mechanism, suitable for DIN rail mounting, and which have enclosed sides or end plates and partition plates for separation.

### 2.6.2 Control Wiring for Binary Signals

For Control Wiring for Binary Signals, provide 18 AWG (1.02 mm diameter) copper or thicker wire rated for 300-volt service.

### 2.6.3 Control Wiring for Analog Signals

For Control Wiring for Analog Signals, provide 18 AWG (1.02 mm diameter) or thicker, copper, single- or multiple-twisted wire meeting the following requirements:

- a. minimum 50 mm (2 inch) lay of twist
- b. 100 percent shielded pairs
- c. at least 300-volt insulation
- d. each pair has a 20 AWG tinned-copper drain wire and individual overall pair insulation
- e. cables have an overall aluminum-polyester or tinned-copper cable-shield tape, overall 20 AWG tinned-copper cable drain wire, and overall cable insulation.

### 2.6.4 Power Wiring for Control Devices

For 24-volt circuits, provide insulated copper 18 AWG or thicker wire rated for 300 VAC service. For 120-volt circuits, provide 14 AWG or thicker stranded copper wire rated for 600-volt service.

### 2.6.5 Transformers

Provide UL 5085-3 approved transformers. Select transformers sized so that the connected load is no greater than 80 percent of the transformer rated capacity.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Fully install and test the control system in accordance Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET

DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet or Niagara BACnet systems, and this Section.

#### 3.1.1 Dielectric Isolation

Provide dielectric isolation where dissimilar metals are used for connection and support. Install control system in a manner that provides clearance for control system maintenance by maintaining access space required to calibrate, remove, repair, or replace control system devices. Install control system such that it does not interfere with the clearance requirements for mechanical and electrical system maintenance.

#### 3.1.2 Penetrations in Building Exterior

Make all penetrations through and mounting holes in the building exterior watertight.

#### 3.1.3 Device Mounting Criteria

Install devices in accordance with the manufacturer's recommendations and as indicated and shown. Provide a weathershield for all devices installed outdoors. Provide clearance for control system maintenance by maintaining access space required to calibrate, remove, repair, or replace control system devices. Provide clearance for mechanical and electrical system maintenance; do not not interfere with the clearance requirements for mechanical and electrical system maintenance.

#### 3.1.4 Labels and Tags

Key all labels and tags to the unique identifiers shown on the As-Built drawings. For labels exterior to protective enclosures provide engraved plastic labels mechanically attached to the enclosure or DDC Hardware. Labels inside protective enclosures may be attached using adhesive, but must not be hand written. For tags, provide plastic or metal tags mechanically attached directly to each device or attached by a metal chain or wire.

- a. Label all Enclosures and DDC Hardware.
- b. Tag Airflow measurement arrays (AFMA) with flow rate range for signal output range, duct size, and pitot tube AFMA flow coefficient.
- c. Tag duct static pressure taps at the location of the pressure tap

#### 3.1.5 Surge Protection

##### 3.1.5.1 Power-Line Surge Protection

Protect equipment connected to AC circuits to withstand power-line surges in accordance with IEEE C62.41. Do not use fuses for surge protection.

##### 3.1.5.2 Surge Protection for Transmitter and Control Wiring

Protect DDC hardware against or provided DDC hardware capable of withstanding surges induced on control and transmitter wiring installed outdoors and as shown. Protect equipment against the following two waveforms:

- a. A waveform with a 10-microsecond rise time, a 1000-microsecond decay

time and a peak current of 60 amps.

- b. A waveform with an 8-microsecond rise time, a 20-microsecond decay time and a peak current of 500 amperes.

### 3.1.6 Basic Cybersecurity Requirements

#### 3.1.6.1 Passwords

Password development and the Password Summary Report submittal must comply with Section 25 05 11.01 CYBERSECURITY FOR FACILITY-RELATED CONTROL HVAC SYSTEMS.

#### 3.1.6.2 Wireless Capability

Wireless capabilities must comply with Section 25 05 11.01 CYBERSECURITY FOR FACILITY-RELATED CONTROL HVAC SYSTEMS..

#### 3.1.6.3 IP Network Physical Security

IP network physical security must comply with Section 25 05 11.01 CYBERSECURITY FOR FACILITY-RELATED CONTROL HVAC SYSTEMS.

### 3.2 DRAWINGS AND CALCULATIONS

Provide drawings in the form and arrangement indicated and shown. Use the same abbreviations, symbols, nomenclature and identifiers shown. Assign a unique identifier as shown to each control system element on a drawing. When packaging drawings, group schedules by system. When space allows, it is permissible to include multiple schedules for the same system on a single sheet. Except for drawings covering all systems, do not put information for different systems on the same sheet.

Submit hardcopy drawings on ISO A1 841 by 594 mm sheets, and electronic drawings in PDF and in AutoCAD format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule.

- a. Submit DDC Contractor Design Drawings consisting of each drawing indicated with pre-construction information depicting the intended control system design and plans. Submit DDC Contractor Design Drawings as a single complete package: 2 hard copies and 3 copies on CD-ROM.
- b. Submit Draft As-Built Drawings consisting of each drawing indicated updated with as-built data for the system prior to PVT. Submit Draft As-Built Drawings as a single complete package: 2 hard copies and 3 copies on CD-ROM.
- c. Submit Final As-Built Drawings consisting of each drawing indicated updated with all final as-built data. Final As-Built Drawings as a single complete package: 2 hard copies and 3 copies on CD-ROM.

#### 3.2.1 Sample Drawings

Sample drawings in electronic format are available at the Whole Building Design Guide page for this section:

<https://www.wbdg.org/dod/ufgs/ufgs-23-09-00>

These drawings may prove useful in demonstrating expected drawing formatting and example content and are provided for illustrative purposes only. Note that these drawings do not meet the content requirements of this Section and must be completed to meet project requirements.

### 3.2.2 Drawing Index and Legend

Provide an HVAC Control System Drawing Index showing the name and number of the building, military site, State or other similar designation, and Country. In the Drawing Index, list all Contractor Design Drawings, including the drawing number, sheet number, drawing title, and computer filename when used. In the Design Drawing Legend, show and describe all symbols, abbreviations and acronyms used on the Design Drawings. Provide a single Index and Legend for the entire drawing package.

### 3.2.3 Thermostat and Occupancy Sensor Schedule

Provide a thermostat and occupancy sensor schedule containing each thermostat's unique identifier, room identifier and control features and functions as shown. Provide a single thermostat and occupancy sensor schedule for the entire project.

### 3.2.4 Valve Schedule

Provide a valve schedule containing each valve's unique identifier, size, flow coefficient Kv (Cv), pressure drop at specified flow rate, spring range, positive positioner range, actuator size, close-off pressure to torque data, dimensions, and access and clearance requirements data. In the valve schedule include actuator selection data supported by calculations of the force required to move and seal the valve, access and clearance requirements. Provide a single valve schedule for the entire project.

### 3.2.5 Damper Schedule

Provide a damper schedule containing each damper's unique identifier, type (opposed or parallel blade), nominal and actual sizes, orientation of axis and frame, direction of blade rotation, actuator size and spring ranges, operation rate, positive positioner range, location of actuators and damper end switches, arrangement of sections in multi-section dampers, and methods of connecting dampers, actuators, and linkages. Include the AMCA 511 maximum leakage rate at the operating static-pressure differential for each damper in the Damper Schedule. Provide a single damper schedule for the entire project.

### 3.2.6 Project Summary Equipment Schedule

Provide a project summary equipment schedule containing the manufacturer, model number, part number and descriptive name for each control device, hardware and component provided under this specification. Provide a single project equipment schedule for the entire project.

### 3.2.7 Equipment Schedule

Provide system equipment schedules containing the unique identifier, manufacturer, model number, part number and descriptive name for each control device, hardware and component provided under this specification. Provide a separate equipment schedule for each HVAC system.

### 3.2.8 Occupancy Schedule

Provide an occupancy schedule drawing containing the same fields as the occupancy schedule Contract Drawing with Contractor updated information. Provide a single occupancy schedule for the entire project.

### 3.2.9 DDC Hardware Schedule

Provide a single DDC Hardware Schedule for the entire project and including following information for each device.

#### 3.2.9.1 DDC Hardware Identifier

The Unique DDC Hardware Identifier for the device.

#### 3.2.9.2 HVAC System

The system "name" used to identify a specific system (the name used on the system schematic drawing for that system).

#### 3.2.9.3 BACnet Device Information

##### 3.2.9.3.1 Device Object Identifier

The Device Object Identifier: The Object\_Identifier of the Device Object

##### 3.2.9.3.2 Network Number

The Network Number for the device.

##### 3.2.9.3.3 MAC Address

The MAC Address for the device

##### 3.2.9.3.4 BTL Listing

The BTL Listing of the device. If the device is listed under multiple BTL Profiles, indicate the profile that matches the use and configuration of the device as installed.

##### 3.2.9.3.5 Proprietary Services Information

If the device uses non-standard ASHRAE 135 services as defined and permitted in Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS, indicate that the device uses non-standard services and include a description of all non-standard services used. Describe usage and content such that a device from another vendor can interoperate with the device using the non-standard service. Provide descriptions with sufficient detail to allow a device from a different manufacturer to be programmed to both read and write the non-standard service request:

- a. read: interpret the data contained in the non-standard service and;
- b. write: given similar data, generate the appropriate non-standard service request.

#### 3.2.9.3.6 Alarming Information

Indicate whether the device is used for alarm generation, and which types of alarm generation the device implements: intrinsic, local algorithmic, remote algorithmic.

#### 3.2.9.3.7 Scheduling Information

Indicate whether the device is used for scheduling.

#### 3.2.9.3.8 Trending Information

Indicate whether the device is used for trending, and indicate if the device is used to trend local values, remote values, or both.

#### 3.2.10 Points Schedule

Provide a Points Schedule in tabular form for each HVAC system, with the indicated columns and with each row representing a hardware point, network point or configuration point in the system.

- a. When a Points Schedule was included in the Contract Drawing package, use the same fields as the Contract Drawing with updated information in addition to the indicated fields.
- b. When Point Schedules are included in the contract package, items requiring contractor verification or input have been shown in angle brackets ("<" and ">"), such as <\_\_\_> for a required entry or <value> for a value requiring confirmation. Complete all items in brackets as well as any blank cells. Do not modify values which are not in brackets without approval.

Points Schedule Columns must include:

##### 3.2.10.1 Point Name

The abbreviated name for the point using the indicated naming convention.

##### 3.2.10.2 Description

A brief functional description of the point such as "Supply Air Temperature".

##### 3.2.10.3 DDC Hardware Identifier

The Unique DDC Hardware Identifier shown on the DDC Hardware Schedule and used across all drawings for the DDC Hardware containing the point.

##### 3.2.10.4 Settings

The value and units of any setpoints, configured setpoints, configuration parameters, and settings related to each point.

##### 3.2.10.5 Range

The range of values, including units, associated with the point, including but not limited to a zone temperature setpoint adjustment range, a sensor measurement range, occupancy values for an occupancy input, or the status of a safety.

### 3.2.10.6 Input or Output (I/O) Type

The type of input or output signal associated with the point. Use the following abbreviations for entries in this column:

- a. AI: The value comes from a hardware (physical) Analog Input
- b. AO: The value is output as a hardware (physical) Analog Output
- c. BI: The value comes from a hardware (physical) Binary Input
- d. BO: The value is output as a hardware (physical) Binary Output
- e. PULSE: The value comes from a hardware (physical) Pulse Accumulator Input
- f. NET-IN: The value is provided from the network (generally from another device). Use this entry only when the value is received from another device as part of scheduling or as part of a sequence of operation, not when the value is received on the network for supervisory functions such as trending, alarming, override or display at a user interface.
- g. NET-OUT: The value is provided to another controller over the network. Use this entry only when the value is transmitted to another device as part of scheduling or as part of a sequence of operation, not when the value is transmitted on the network for supervisory functions such as trending, alarming, override or display at a user interface.

### 3.2.10.7 Object and Property Information

The Object Type and Instance Number for the Object associated with the point. If the value of the point is not in the Present\_Value Property, then also provide the Property ID for the Property containing the value of the point. Any point that is displayed at the front end or on an LDP, is trended, is used by another device on the network, or has an alarm condition must be documented here.

### 3.2.10.8 Network Data Exchange Information (Gets Data From, Sends Data To)

Provide the DDC Hardware Identifier of other DDC Hardware the point is shared with.

### 3.2.10.9 Override Information (Object Type and Instance Number)

For each point requiring an Override, indicate if the Object for the point is Commandable or, if the use of a separate Object was specifically approved by the Contracting Officer, provide the Object Type and Instance Number of the Object to be used in overriding the point.

### 3.2.10.10 Trend Object Information

For each point requiring a trend, indicate if the trend is Local or Remote, the trend Object type and the trend Object instance number. For remote trends provide the DDC Hardware Identifier for the device containing the trend Object in the Points Schedule notes.



### 3.2.10.11 Alarm Information

Indicate the Alarm Generation Type, Event Enrollment Object Instance Number, and Notification Class Object Instance Number for each point requiring an alarm. (Note that not all alarms will have Event Enrollment Objects.)

### 3.2.10.12 Configuration Information

Indicate the means of configuration associated with each point.

- a. For Operator Configurable Points indicate BACnet Object and Property information (Name, Type, Identifiers) containing the configurable value. Indicate whether the property is writable always, or only when Out\_Of\_Service is TRUE.
- b. For Configurable Points indicate the BACnet Object and Property information as for Operator Configurable points, or identification of the configurable settings from within the engineering software for the device or identification of the hardware settings on the device.

### 3.2.11 Riser Diagram

The Riser Diagram of the Building Control Network may be in tabular form, and must show all DDC Hardware and all Network Hardware, including network terminators. For each item, provide the unique identifier, common descriptive name, physical sequential order (previous and next device on the network), room identifier and location within room. A single riser diagram must be submitted for the entire system.

### 3.2.12 Control System Schematics

Provide control system schematics in the same form as the control system schematic Contract Drawing with Contractor updated information. Provide a control system schematic for each HVAC system.

### 3.2.13 Sequences of Operation

Provide HVAC control system sequence of operation and in the same format as the Contract Drawings. Within these drawings, refer to devices by their unique identifiers. Submit sequences of operation for each HVAC system

### 3.2.14 Controller, Motor Starter and Relay Wiring Diagram

Provide controller wiring diagrams as functional wiring diagrams which show the interconnection of conductors and cables to each controller and to the identified terminals of input and output devices, starters and package equipment. Show necessary jumpers and ground connections and the labels of all conductors. Identify sources of power required for control systems and for packaged equipment control systems back to the panel board circuit breaker number, controller enclosures, magnetic starter, or packaged equipment control circuit. Show each power supply and transformer not integral to a controller, starter, or packaged equipment. Show the connected volt-ampere load and the power supply volt-ampere rating. Provide wiring diagrams for each HVAC system.

### 3.3 CONTROLLER TUNING

Tune each controller in a manner consistent with that described in the ASHRAE FUN SI and in the manufacturer's instruction manual. Tuning must consist of adjustment of the proportional, integral, and where applicable, the derivative (PID) settings to provide stable closed-loop control. Each loop must be tuned while the system or plant is operating at a high gain (worst case) condition, where high gain can generally be defined as a low-flow or low-load condition. Upon final adjustment of the PID settings, in response to a change in controller setpoint, the controlled variable must settle out at the new setpoint with no more than two (2) oscillations above and below setpoint. Upon settling out at the new setpoint the controller output must be steady. With the exception of naturally slow processes such as zone temperature control, the controller must settle out at the new setpoint within five (5) minutes. Set the controller to its correct setpoint and record and submit the final PID configuration settings with the O&M Instructions and on the associated Points Schedule.

### 3.4 START-UP

#### 3.4.1 Start-Up Test

Perform the following startup tests for each control system to ensure that the described control system components are installed and functioning per this specification.

Adjust, calibrate, measure, program, configure, set the time schedules, and otherwise perform all necessary actions to ensure that the systems function as indicated and shown in the sequence of operation and other contract documents.

##### 3.4.1.1 Systems Check

An item-by-item check must be performed for each HVAC system

##### 3.4.1.1.1 Step 1 - System Inspection

With the system in unoccupied mode and with fan hand-off-auto switches in the OFF position, verify that power and main air are available where required and that all output devices are in their failsafe and normal positions. Inspect each local display panel and each M&C Client to verify that all displays indicate shutdown conditions.

##### 3.4.1.1.2 Step 2 - Calibration Accuracy Check

Perform a two-point accuracy check of the calibration of each HVAC control system sensing element and transmitter by comparing the value from the test instrument to the network value provided by the DDC Hardware. Use digital indicating test instruments, such as digital thermometers, motor-driven psychrometers, and tachometers. Use test instruments with accuracy at least twice as accurate as the specified sensor accuracy and with calibration traceable to National Institute of Standards and Technology standards. Check one the first check point in the bottom one-third of the sensor range, and the second in the top one-third of the sensor range. Verify that the sensing element-to-DDC readout accuracies at two points are within the specified product accuracy tolerances, and if not recalibrate or replace the device and repeat the calibration check.

#### 3.4.1.1.3 Step 3 - Actuator Range Check

With the system running, apply a signal to each actuator through the DDC Hardware controller. Verify proper operation of the actuators and positioners for all actuated devices and record the signal levels for the extreme positions of each device. Vary the signal over its full range, and verify that the actuators travel from zero stroke to full stroke within the signal range. Where applicable, verify that all sequenced actuators move from zero stroke to full stroke in the proper direction, and move the connected device in the proper direction from one extreme position to the other. For valve actuators and damper actuators, perform the actuator range check under normal system pressures.

#### 3.4.1.2 Weather Dependent Test

Perform weather dependent test procedures in the appropriate climatic season.

#### 3.4.2 Start-Up Testing Report

Submit 4 copies of the Start-Up Testing Report. The report may be submitted as a Technical Data Package documenting the results of the tests performed and certifying that the system is installed and functioning per this specification, and is ready for the Performance Verification Test (PVT).

### 3.5 PERFORMANCE VERIFICATION TEST (PVT)

#### 3.5.1 PVT Procedures

Prepare PVT Procedures explaining step-by-step, the actions and expected results that will demonstrate that the control system performs in accordance with the sequences of operation, and other contract documents. Submit 4 copies of the PVT Procedures. The PVT Procedures may be submitted as a Technical Data Package.

##### 3.5.1.1 Sensor Accuracy Checks

Include a one-point accuracy check of each sensor in the PVT procedures.

##### 3.5.1.2 Endurance Test

Include a one-week endurance test as part of the PVT during which the system is operated continuously.

Use the building control system BACnet Trend Log or Trend Log Multiple Objects to trend all points shown as requiring a trend on the Point Schedule for the entire endurance test. If insufficient buffer capacity exists to trend the entire endurance test, upload trend logs during the course of the endurance test to ensure that no trend data is lost.

##### 3.5.1.3 PVT Equipment List

Include in the PVT procedures a control system performance verification test equipment list that lists the equipment to be used during performance verification testing. For each piece of equipment, include manufacturer name, model number, equipment function, the date of the latest calibration, and the results of the latest calibration

### 3.5.2 PVT Execution

Demonstrate compliance of the control system with the contract documents. Using test plans and procedures approved by the Government, software capable of reading and writing COV Notification Subscriptions, Notification Class Recipient List Properties, event enrollments, demonstrate all physical and functional requirements of the project. Show, step-by-step, the actions and results demonstrating that the control systems perform in accordance with the sequences of operation. Do not start the performance verification test until after receipt of written permission by the Government, based on Government approval of the PVT Plan and Draft As-Built's and completion of balancing. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. Do not conduct tests during scheduled seasonal off periods of base heating and cooling systems. If the system experiences any failures during the endurance test portion of the PVT, repair the system repeat the endurance test portion of the PVT until the system operates continuously and without failure for the specified endurance test period.

### 3.5.3 PVT Report

Prepare and submit a PVT report documenting all tests performed during the PVT and their results. Include all tests in the PVT procedures and any additional tests performed during PVT. Document test failures and repairs conducted with the test results.

Submit four copies of the PVT Report. The PVT Report may be submitted as a Technical Data Package.

## 3.6 PERFORMANCE VERIFICATION TESTING

### 3.6.1 General

PVT testing must demonstrate compliance of controls work with contract document requirements and must be performed by the Controls Contractor and Equipment Suppliers. No less than 14 calendar days prior to start of controls system installation, meet with the Contracting Office's technical representative (COTR), the Contractor's QA representative, the Contractor's Controls Contractor representative, and the control system Owner to develop a mutual understanding relate to the details of the PVT work requirements, including required submittals, work schedule, and field quality control.

### 3.6.2 Performance Verification Testing and Commissioning

PVT testing is a Government quality assurance function that includes systems trending and field tests. Commissioning is a quality control function that is the Commissioning Team's responsibility to the extent required by this contract.

### 3.6.3 Performance Verification Testing of Equipment with Packaged Controls

Controls Contractor and Equipment Supplier(s) must share and coordinate PVT testing responsibilities for equipment provided with on-board factory packaged controls such as boiler controllers, dedicated outside air systems (DOAS's), and packaged pumping systems.

### 3.6.3.1 Controls Contractor Responsibilities

The Controls Contractor must provide a PVT Plan separate from Equipment Supplier's performance verification testing plan, perform endurance testing, and perform PVT testing concurrent with Equipment Suppliers' testing for equipment provided with on-board factory packaged controls to demonstrate the following:

- a. Equipment enabling and disabling.
- b. Equipment standard and optional control points necessary to accomplish functionality regardless if specified in contract documents or not.
- c. Equipment standard and optional alarms critical to safe operation regardless if specified in contract documents or not.
- d. All control points added by Controls Contractor in addition to onboard factory packaged controls regardless if specified in contract documents or not.

Refer to paragraphs titled "Performance Verification Test Plan" and "Endurance Testing" for additional information.

### 3.6.3.2 Equipment Supplier Responsibilities

Each Equipment Supplier must provide PVT Plans separate from Controls Contractor's plans and perform PVT testing concurrent with Controls Contractor's testing for their equipment provided with on-board factory packaged controls to demonstrate the following:

- a. Equipment standard and optional control features necessary to accomplish functionality regardless if specified in contract documents or not.
- b. Equipment standard and optional operation modes necessary to accomplish functionality regardless if specified in contract documents or not.
- c. Equipment standard and optional alarm conditions for safe operation regardless if specified in contract documents or not.

Refer to all paragraphs under paragraph titled "Performance Verification Testing" except for section titled "Endurance Testing" for additional information.

### 3.6.4 Sequencing of Performance Verification Testing Activities

PVT activities must be sequenced with major activities listed below for Test and Balance (TAB) Contractor, Equipment Suppliers, Commissioning Specialists, and others to demonstrate fully functioning systems. Refer to Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS). Complete the items in TABLE III: SEQUENCING OF PVT TESTING ACTIVITIES as schedule activities or milestones.

| TABLE III: SEQUENCING OF PVT TESTING ACTIVITIES |                                                                                                                  |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| SEQUENCE                                        | ITEM                                                                                                             |
| 1                                               | Submission, review, and approval of Control Contractors PVT Plans.                                               |
| 2                                               | Submission, review, and approval of Equipment Suppliers PVT Plans.                                               |
| 3                                               | Submission, review, and approval of certified final Test and Balance Report.                                     |
| 4                                               | Conduct commissioning functional performance tests.                                                              |
| 5                                               | Submission, review, and approval of all of the Commissioning Specialists completed functional performance tests. |
| 6                                               | Request Contracting Officer to allow beginning of Government-witnessed PVT testing.                              |
| 7                                               | Contracting Officers approval to begin PVT testing.                                                              |
| 8                                               | Conduct PVT field work.                                                                                          |
| 9                                               | Governments verbal approval of PVT field work for all systems.                                                   |
| 10                                              | Conduct Test and Balance verification field work.                                                                |
| 11                                              | Governments written approval of Test and Balance verification field work.                                        |
| 12                                              | Submission, review, and approval of endurance testing.                                                           |
| 13                                              | Governments written approval of PVT field work for all systems.                                                  |
| 14                                              | Facility acceptance recommendation.                                                                              |
| 15                                              | Submission, review, and approval of Control Contractors PVT Report.                                              |
| 16                                              | Submission, review, and approval of Equipment Suppliers PVT Report.                                              |
| 17                                              | Conduct applicable re-testing and seasonal testing within 10 months of beneficial occupancy.                     |

#### 3.6.4.1 PVT Testing for Multi-Phase Construction

For air moving systems except outside air systems serving multiple phases, all major activities listed in TABLE III through Government's verbal approval of Test and Balance verification field work can be completed by phase if all ductwork construction is completed for that phase.

For primary systems such as chilled water systems, HVAC heating hot water systems, and outside air systems serving multiple phases, all major activities listed in TABLE III through Government's verbal approval of Test and Balance verification field work for all air moving systems served by that primary system for that phase must be completed prior to conducting PVT field work for that primary system.

### 3.6.5 Control Contractor's Performance Verification Testing Plan

Submit a detailed PVT Plan of the proposed control systems testing in this contract for approval prior to its use. Develop and use a single PVT Plan for each system with a unique control sequence. Systems sharing an identical control sequence can be tested using copies of the PVT Plan intended for these systems.

PVT Plans must include system-based, step-by-step test methods demonstrating system performs in accordance with contract document requirements. The Government may provide sample PVT Plans upon request. PVT Plans must include the following:

- a. Control sequences from contract documents segmented such that each control algorithm, operation mode, and alarm condition is immediately followed by numbered test methods required to initiate a response, expected response, space for comments, and "pass" or "fail" indication for each expected response.
- b. PVT Plans with control sequences from contract documents that are not segmented into parts will not be accepted.
- c. Indication where assisting personnel are required such as Mechanical Contractor.
- d. Signature and date lines for the Contractor's PVT administrator, Contractor's quality assurance representative, and Contracting Officer's representative acknowledging completion of testing.

### 3.6.6 Performance Verification Testing Sample Size

PVT testing sample sizes will be as follows:

- a. 100-Percent of the following systems:
  - (1) primary systems including, but not limited to, chilled water and HVAC heating hot water systems
  - (2) air handling unit systems including all associated fans except for remote exhaust air fans
  - (3) DOAS's including all associated fans except for remote exhaust air fans
- b. 20-Percent of each set of systems with a shared identical control sequence for systems such as:
  - (1) air terminal units
  - (2) exhaust air fans
  - (3) terminal equipment such as fan coil units and unit heaters

#### 3.6.6.1 Selection of Systems to Test

For sample sets less than 100-percent, the Government will choose which systems will be tested. The Government may require additional testing if previous testing results are inconsistent or demonstrate improper system control as follows:

- a. An additional 25-percent after five-percent failure rate of first sample set.
- b. 100-percent after any failures occurring in additional sample set.

### 3.6.7 Conducting Performance Verification Testing

At least 15 days prior to preferred test date, request the Contracting Officer to allow the beginning of Government-witnessed PVT testing. Provide an estimated time table required to perform testing of each system. Furnish personnel, equipment, instrumentation, and supplies necessary to perform all aspects of testing. Testing personnel must be regularly employed in the testing and calibration of control systems. After receipt of Contracting Officer's approval to begin testing, perform PVT testing using project's as-built (shop) control system drawings, project's design drawings, and approved PVT Plans.

During testing, identify deficiencies that do not meet contract document requirements. Deficiencies must be investigated, corrected with corrections documented, and re-tested at a later date following procedures for the initial PVT testing. The Government may require re-testing of any control system components affected by the original failed test.

### 3.6.8 Endurance Testing

#### 3.6.8.1 General

Conduct endurance testing in conjunction with the PVT to demonstrate control loop stability and accuracy. For all control loops tested, record trend data of the control variables over time, demonstrating that the control loop responds to a sudden change of the control variable set point without excessive overshoot or undershoot. Conduct endurance testing for each system subject to PVT testing. Systems must be operating as normally anticipated during occupancy throughout endurance testing.

Endurance testing results must clearly demonstrate control loop stability and accuracy. Controlled loop outputs must be stable and accurately maintain each setpoint.

#### 3.6.8.2 Hardware

Use Government furnished hardware for testing if available when endurance testing begins. If unavailable, the Contractor must provide suitable hardware for required testing.

If insufficient buffer capacity exists to trend the entire endurance test, upload trend data during the course of endurance testing to ensure all trend data is retained. Lost trend data will require retesting of all control points for affected system(s).

#### 3.6.8.3 Endurance Testing Results Format

Submit endurance testing results for each tested system in a graphical format complete with clear indication of value(s) for y-axis, value for x-axis, and legend identifying each trended control point. The number of control points contained on a single graph must be such that all control points can be clearly visible. Control points must be logically grouped such that related points appear on a single graph. In addition, submit a



separate comma separated value (CSV) file of raw trend data for each trended system. Each trended control point in CSV file must be clearly identified.

For control points recorded based on change of value, change of value for recording data must be clearly identified for each control point.

#### 3.6.8.4 Endurance Testing Start, Duration, and Frequency

Trending of all control points for a given system must start at an identical date and time regardless of the basis of data collection. Duration of all endurance tests must be at least one-week.

Unless specified otherwise for control points recorded based on time, frequency of data collection must be 15-minutes. Frequency of data collection for specific types of control points is as follows:

##### 3.6.8.4.1 Points Trended at One Minute Intervals

- a. Temperature for supply air, return air, mixed air, supply water, and return water
- b. Temperature for outside air, supply air, return air and exhaust air entering and leaving energy recovery device
- c. Flow for supply air, return air, outside air, chilled water, and HVAC heating hot water
- d. Flow for exhaust air associated with energy recovery
- e. Relative humidity for outside air and return air
- f. Relative humidity for outside air, supply air, return air and exhaust air entering and leaving energy recovery device
- g. Command and status for control dampers and control valves
- h. Speed for fans and pumps
- i. Pressure for fans and pumps

##### 3.6.8.4.2 Points Trended at 15 Minute Intervals

- a. Temperature and relative humidity for zones
- b. Temperature and relative humidity for outside air not associated with energy recovery
- c. Command and status for equipment
- d. Pressure relative to the outside for facility

#### 3.6.8.5 Trended Control Points

Trended control points for each system must demonstrate each system performs in accordance with contract document requirements. Trended control points must include, but not be limited to, control points listed in contract document points list.

Minimum control points that are required to be trended for selected systems are listed below. These control points must be trended as applicable to this contract in addition to control points necessary to demonstrate systems perform in accordance with contract document requirements and those listed in contract document's points list.

3.6.8.5.1 Air-Cooled Chiller Chilled Water System.

- a. Chiller(s) command and status
- b. Chiller isolation valve(s) command and status
- c. Chilled water pump(s) actual speed
- d. Chilled water pump(s) setpoint and actual differential pressure
- e. Minimum flow bypass control valve command
- f. Minimum system flow setpoint and actual flow
- g. Chilled water supply setpoint and actual temperature
- h. Chilled water return actual temperature
- i. Chilled water actual flow
- j. Outside air actual dry-bulb temperature

3.6.8.5.2 Dedicated Outside Air System (DOAS)

- a. Outside air actual dry-bulb temperature
- b. Outside air actual relative humidity
- c. Outside air isolation damper command and status
- d. Outside air setpoint and actual airflow
- e. Energy recovery core command, status, and actual speed
- f. Energy recovery core OA bypass control damper command and status
- g. Energy recovery core defrost cycle command and status
- h. Energy recovery core OA discharge air actual dry-bulb temperature
- i. Energy recovery core OA discharge air actual relative humidity
- j. Preheat coil leaving air setpoint and actual temperature
- k. Preheat coil control actuator command
- l. Cooling coil leaving air setpoint and actual temperature
- m. Cooling coil control valve command
- n. Supply air fan actual speed
- o. Reheat coil control valve command

- p. Discharge air setpoint and actual temperature
- q. Supply air fan setpoint and actual static pressure
- r. Facility setpoint and actual relative pressure
- s. Return air actual dry-bulb temperature
- t. Return air actual relative humidity
- u. Energy recovery core EA bypass control damper command and status
- v. Energy recovery core EA discharge air actual dry-bulb temperature
- w. Energy recovery core EA discharge air actual relative humidity
- x. Exhaust air fan actual speed
- y. Exhaust air isolation damper command and status

#### 3.6.8.6 Endurance Testing Sample Size

Endurance Testing sample sizes were as follows:

- a. 100-Percent of the following systems:
  - (1) primary systems including, but not limited to, chilled water and HVAC heating hot water systems
  - (2) air handling unit systems including all associated fans except for remote exhaust air fans
  - (3) DOAS's including all associated fans except for remote exhaust air fans
- b. 20-Percent of each set of systems with a shared identical control sequence for systems such as:
  - (1) air terminal units
  - (2) exhaust air fans
  - (3) terminal equipment such as fan coil units and unit heaters

##### 3.6.8.6.1 Selection of Systems to Test

For sample sets less than 100-percent, the Government will choose which systems will be tested. The Government may require additional testing if previous testing results are inconsistent or demonstrate improper system control as follows:

- a. An additional 25-percent after five-percent failure rate of first sample set.
- b. 100-percent after any failures occurring in additional sample set.

### 3.6.9 Performance Verification Test Report

Submit a PVT Report after receiving Government's written approval of PVT field work that is intended to document test results and final control system sequences and settings prior to turnover. The PVT Report must contain the following:

- a. Executive summary that briefly discusses results of each system's endurance testing and PVT testing and conclusions for each system.
- b. Endurance testing for each system.
- c. Completed PVT Plan for each system used during testing that includes hand written field notes and participant signatures.
- d. Blank PVT Plan for each system approved prior to testing that is edited to reflect changes occurring during testing. Edits must be typed and must reflect changes to control sequences from contract documents, must reflect changes to numbered test methods required to initiate a response, and must reflect changes to expected response. Only one blank PVT Plan is required for each set of systems sharing an identical control sequence, such as air terminal units, exhaust air fans, fan coil units and unit heaters.
- e. Written certification that the installation and testing of all systems are complete and meet all contract document requirements.

### 3.7 OPERATION AND MAINTENANCE (O&M) INSTRUCTIONS

Provide HVAC control System Operation and Maintenance Instructions which include:

- a. "Data Package 3" as indicated in Section 01 78 23 OPERATION AND MAINTENANCE DATA for each piece of control equipment.
- b. "Data Package 4" as described in Section 01 78 23 OPERATION AND MAINTENANCE DATA for all air compressors.
- c. HVAC control system sequences of operation formatted as indicated.
- d. Procedures for the HVAC system start-up, operation and shut-down including the manufacturer's supplied procedures for each piece of equipment, and procedures for the overall HVAC system.
- e. As-built HVAC control system detail drawings formatted as indicated.
- f. Routine maintenance checklist. Provide the routine maintenance checklist arranged in a columnar format, where the first column lists all installed devices, the second column states the maintenance activity or that no maintenance required, the third column states the frequency of the maintenance activity, and the fourth column is used for additional comments or reference.
- g. Qualified service organization list, including at a minimum company name, contact name and phone number.
- h. Start-Up Testing Report.
- i. Performance Verification Test (PVT) Procedures and Report.

Submit 2 copies of the Operation and Maintenance Instructions, indexed and in booklet form. The Operation and Maintenance Instructions may be submitted as a Technical Data Package.

### 3.8 MAINTENANCE AND SERVICE

Provide services, materials and equipment as necessary to maintain the entire system in an operational state as indicated for a period of three years from the date of final acceptance of the project. Minimize impacts on facility operations.

- a. The integration of the system specified in this section into a Utility Monitoring and Control System must not, of itself, void the warranty or otherwise alter the requirement for the three year maintenance and service period. Integration into a UMCS includes but is not limited to establishing communication between devices in the control system and the front end or devices in another system.
- b. The changing of configuration properties must not, of itself, void the warranty or otherwise alter the requirement for the three year maintenance and service period.

#### 3.8.1 Description of Work

Provide adjustment and repair of the system including the manufacturer's required sensor and actuator (including transducer) calibration, span and range adjustment.

#### 3.8.2 Personnel

Use only service personnel qualified to accomplish work promptly and satisfactorily. Advise the Government in writing of the name of the designated service representative, and of any changes in personnel.

#### 3.8.3 Scheduled Inspections

Perform two inspections at six-month intervals and provide work required. Perform inspections in June and December. During each inspection perform the indicated tasks:

- a. Perform visual checks and operational tests of equipment.
- b. Clean control system equipment including interior and exterior surfaces.
- c. Check and calibrate each field device. Check and calibrate 50 percent of the total analog inputs and outputs during the first inspection. Check and calibrate the remaining 50 percent of the analog inputs and outputs during the second major inspection. Certify analog test instrumentation accuracy to be twice the specified accuracy of the device being calibrated. Randomly check at least 25 percent of all binary inputs and outputs for proper operation during the first inspection. Randomly check at least 25 percent of the remaining binary inputs and outputs during the second inspection. If more than 20 percent of checked inputs or outputs failed the calibration check during any inspection, check and recalibrate all inputs and outputs during that inspection.

- d. Run system software diagnostics and correct diagnosed problems.
- e. Resolve any previous outstanding problems.

#### 3.8.4 Scheduled Work

This work must be performed during regular working hours, Monday through Friday, excluding Federal holidays.

#### 3.8.5 Emergency Service

The Government will initiate service calls when the system is not functioning properly. Qualified personnel must be available to provide service to the system. A telephone number where the service supervisor can be reached at all times must be provided. Service personnel must be at the site within 24 hours after receiving a request for service. The control system must be restored to proper operating condition as required per Section 01 78 00 CLOSEOUT SUBMITTALS.

#### 3.8.6 Operation

After performing scheduled adjustments and repairs, verify control system operation as demonstrated by the applicable tests of the performance verification test.

#### 3.8.7 Records and Logs

Keep dated records and logs of each task, with cumulative records for each major component, and for the complete system chronologically. Maintain a continuous log for all devices, including initial analog span and zero calibration values and digital points. Keep complete logs and provide logs for inspection onsite, demonstrating that planned and systematic adjustments and repairs have been accomplished for the control system.

#### 3.8.8 Work Requests

Record each service call request as received and include its location, date and time the call was received, nature of trouble, names of the service personnel assigned to the task, instructions describing what has to be done, the amount and nature of the materials to be used, the time and date work started, and the time and date of completion. Submit a record of the work performed within 5 days after work is accomplished.

#### 3.8.9 System Modifications

Submit recommendations for system modification in writing. Do not make system modifications, including operating parameters and control settings, without prior approval of the Government.

### 3.9 TRAINING

Conduct a training course for operating staff members designated by the Government in the maintenance and operation of the system, including specified hardware and software. Conduct 32 hours of training at the project site within 30 days after successful completion of the performance verification test. The Government reserves the right to make audio and visual recordings (using Government supplied equipment) of the training sessions for later use. Provide audiovisual equipment and other training materials and supplies required to conduct training. A training day is

defined as 8 hours of classroom instruction, including two 15 minute breaks and excluding lunchtime, Monday through Friday, during the daytime shift in effect at the training facility.

### 3.9.1 Training Documentation

Prepare training documentation consisting of:

- a. Course Attendee List: Develop the list of course attendees in coordination with and signed by the ControlsHVAC shop supervisor.
- b. Training Manuals: Provide training manuals which include an agenda, defined objectives for each lesson, and a detailed description of the subject matter for each lesson. When presenting portions of the course material by audiovisuals, deliver copies of those audiovisuals as a part of the printed training manuals.

### 3.9.2 Training Course Content

For guidance in planning the required instruction, assume that attendees will have a high school education, and are familiar with HVAC systems. During the training course, cover all of the material contained in the Operating and Maintenance Instructions, the layout and location of each controller enclosure, the layout of one of each type of equipment and the locations of each, the location of each control device external to the panels, the location of the compressed air station, preventive maintenance, troubleshooting, diagnostics, calibration, adjustment, commissioning, tuning, and repair procedures. Typical systems and similar systems may be treated as a group, with instruction on the physical layout of one such system. Present the results of the performance verification test and the Start-Up Testing Report as benchmarks of HVAC control system performance by which to measure operation and maintenance effectiveness.

### 3.9.3 Training Documentation Submittal Requirements

Submit hardcopy training manuals and all training materials on CD-ROM. Provide one hardcopy manual for each trainee on the Course Attendee List and 2 additional copies for archive at the project site. Provide 2 copies of the Course Attendee List with the archival copies. Training Documentation may be submitted as a Technical Data Package.

**APPENDIX A**

| <b><u>QC CHECKLIST FOR BACNET SYSTEMS</u></b>                                                                                                                                                                                                  |                                                                                                                   |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------|
| <p>This checklist is not all-inclusive of the requirements of this specification and should not be interpreted as such.</p> <p>Instructions: Initial each item in the space provided ( ____ ) verifying that the requirement has been met.</p> |                                                                                                                   |      |
| <p>This checklist is for (circle one:)</p> <p>Pre-Construction QC Checklist Submittal</p> <p>Post-Construction QC Checklist Submittal</p> <p>Close-out QC Checklist Submittal</p>                                                              |                                                                                                                   |      |
| Items verified for Pre-Construction, Post-Construction and Closeout QC Checklist Submittals:                                                                                                                                                   |                                                                                                                   |      |
| 1                                                                                                                                                                                                                                              | All DDC Hardware is numbered on Control System Schematic Drawings.                                                | ____ |
| 2                                                                                                                                                                                                                                              | Signal lines on Control System Schematic are labeled with the signal type.                                        | ____ |
| 3                                                                                                                                                                                                                                              | Local Display Panel (LDP) Locations are shown on Control System Schematic drawings.                               | ____ |
| Items verified for Post-Construction and Closeout QC Checklist Submittals:                                                                                                                                                                     |                                                                                                                   |      |
| 4                                                                                                                                                                                                                                              | All sequences are performed as specified using DDC Hardware.                                                      | ____ |
| 5                                                                                                                                                                                                                                              | Training schedule and course attendee list has been developed and coordinated with shops and submitted.           | ____ |
| Items verified for Closeout QC Checklist Submittal:                                                                                                                                                                                            |                                                                                                                   |      |
| 6                                                                                                                                                                                                                                              | Final As-built Drawings, including all Points Schedule drawings, accurately represent the final installed system. | ____ |
| 7                                                                                                                                                                                                                                              | Programming software has been submitted for all programmable controllers.                                         | ____ |
| 8                                                                                                                                                                                                                                              | All software has been licensed to the Government.                                                                 | ____ |
| 9                                                                                                                                                                                                                                              | O&M Instructions have been completed and submitted.                                                               | ____ |
| 10                                                                                                                                                                                                                                             | Training course has been completed.                                                                               | ____ |





## SECTION 23 09 13

INSTRUMENTATION AND CONTROL DEVICES FOR HVAC  
11/15, CHG 2: 05/21

## PART 1 GENERAL

## 1.1 SUMMARY

This section provides for the instrumentation control system components excluding direct digital controllers, network controllers, gateways etc. that are necessary for a completely functional automatic control system. When combined with a Direct Digital Control (DDC) system, the Instrumentation and Control Devices covered under this section must be a complete system suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as specified and indicated.

- a. Install hardware to perform the control sequences as specified and indicated and to provide control of the equipment as specified and indicated.
- b. Install hardware such that individual control equipment can be replaced by similar control equipment from other equipment manufacturers with no loss of system functionality.
- c. Install and configure hardware such that the Government or their agents are able to perform repair, replacement, and upgrades of individual hardware without further interaction with the installing Contractor.

## 1.1.1 Verification of Dimensions

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

## 1.1.2 Drawings

The Government will not indicate all offsets, fittings, and accessories that may be required on the drawings. Carefully investigate the mechanical, electrical, and finish conditions that could affect the work to be performed, arrange such work accordingly, and provide all work necessary to meet such conditions.

## 1.2 RELATED SECTIONS

Related work specified elsewhere.

Section 01 30 00 ADMINISTRATIVE REQUIREMENTS

Section 23 30 00 HVAC AIR DISTRIBUTION

Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM

### 1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

- |            |                                                                     |
|------------|---------------------------------------------------------------------|
| AMCA 500-D | (2018) Laboratory Methods of Testing<br>Dampers for Rating          |
| AMCA 511   | (2010; R 2016) Certified Ratings Program<br>for Air Control Devices |

#### AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

- |            |                                                                        |
|------------|------------------------------------------------------------------------|
| ANSI C12.1 | (2014; Errata 2016) Electric Meters - Code<br>for Electricity Metering |
|------------|------------------------------------------------------------------------|

#### AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

- |              |                                                                   |
|--------------|-------------------------------------------------------------------|
| ASME B16.15  | (2024) Cast Copper Alloy Threaded Fittings<br>Classes 125 and 250 |
| ASME B16.34  | (2021) Valves - Flanged, Threaded and<br>Welding End              |
| ASME B40.100 | (2022) Pressure Gauges and Gauge<br>Attachments                   |

#### ASTM INTERNATIONAL (ASTM)

- |           |                                                            |
|-----------|------------------------------------------------------------|
| ASTM A536 | (2024) Standard Specification for Ductile<br>Iron Castings |
|-----------|------------------------------------------------------------|

#### FLUID CONTROLS INSTITUTE (FCI)

- |          |                                   |
|----------|-----------------------------------|
| FCI 70-2 | (2021) Control Valve Seat Leakage |
|----------|-----------------------------------|

#### INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- |          |                                                                                                                           |
|----------|---------------------------------------------------------------------------------------------------------------------------|
| IEEE 142 | (2007; Errata 2014) Recommended Practice<br>for Grounding of Industrial and Commercial<br>Power Systems - IEEE Green Book |
|----------|---------------------------------------------------------------------------------------------------------------------------|

#### NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

- |             |                                                                           |
|-------------|---------------------------------------------------------------------------|
| ANSI C12.20 | (2015; E 2018) Electricity Meters - 0.1,<br>0.2, and 0.5 Accuracy Classes |
| NEMA 250    | (2020) Enclosures for Electrical Equipment<br>(1000 Volts Maximum)        |

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- |          |                                                           |
|----------|-----------------------------------------------------------|
| NFPA 70  | (2023; ERTA 1 2024; TIA 24-1) National<br>Electrical Code |
| NFPA 90A | (2024) Standard for the Installation of                   |

## Air Conditioning and Ventilating Systems

## UNDERWRITERS LABORATORIES (UL)

|           |                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------|
| UL 555    | (2006; Reprint Aug 2016) UL Standard for Safety Fire Dampers                                                        |
| UL 555S   | (2014; Reprint Oct 2020) UL Standard for Safety Smoke Dampers                                                       |
| UL 5085-3 | (2006; Reprint Jan 2022) UL Standard for Safety Low Voltage Transformers - Part 3: Class 2 and Class 3 Transformers |

## 1.4 SUBMITTALS

Submittal requirements are specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

## 1.5 DELIVERY AND STORAGE

Store and protect products from the weather, humidity, and temperature variations, dirt and dust, and other contaminants, within the storage condition limits published by the equipment manufacturer.

## 1.6 INPUT MEASUREMENT ACCURACY

Select, install and configure sensors, transmitters and DDC Hardware such that the maximum error of the measured value at the input of the DDC hardware is less than the maximum allowable error specified for the sensor or instrumentation.

## 1.7 SUBCONTRACTOR SPECIAL REQUIREMENTS

Perform all work in this section in accordance with the paragraph entitled CONTRACTOR SPECIAL REQUIREMENTS in Section 01 30 00 ADMINISTRATIVE REQUIREMENTS.

## PART 2 PRODUCTS

## 2.1 EQUIPMENT

## 2.1.1 General Requirements

All products used to meet this specification must meet the indicated requirements, but not all products specified here will be required by every project. All products must meet the requirements both Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and this Section.

## 2.1.2 Operation Environment Requirements

Unless otherwise specified, provide products rated for continuous operation under the following conditions:

## 2.1.2.1 Pressure

Pressure conditions normally encountered in the installed location.

#### 2.1.2.2 Vibration

Vibration conditions normally encountered in the installed location.

#### 2.1.2.3 Temperature

- a. Products installed indoors: Ambient temperatures in the range of 0 to 50 degrees C and temperature conditions outside this range normally encountered at the installed location.
- b. Products installed outdoors or in unconditioned indoor spaces: Ambient temperatures in the range of -37 to +66 degrees C and temperature conditions outside this range normally encountered at the installed location.

#### 2.1.2.4 Humidity

10 to 95 percent relative humidity, non-condensing and also humidity conditions outside this range normally encountered at the installed location.

### 2.2 WEATHERSHIELDS

Provide weathershields constructed of galvanized steel painted white, unpainted aluminum, aluminum painted white, or white PVC.

### 2.3 WIRE AND CABLE

Provide wire and cable meeting the requirements of NFPA 70 and NFPA 90A in addition to the requirements of this specification and referenced specifications.

#### 2.3.1 Terminal Blocks

For terminal blocks which are not integral to other equipment, provide terminal blocks which are insulated, modular, feed-through, clamp style with recessed captive screw-type clamping mechanism, suitable for DIN rail mounting, and which have enclosed sides or end plates and partition plates for separation.

#### 2.3.2 Control Wiring for Binary Signals

For Control Wiring for Binary Signals, provide 18 AWG (1.02 mm diameter) copper or thicker wire rated for 300-volt service.

#### 2.3.3 Control Wiring for Analog Signals

For Control Wiring for Analog Signals, provide 18 AWG (1.02 mm diameter) or thicker, copper, single- or multiple-twisted wire meeting the following requirements:

- a. minimum 50 mm (2 inch) lay of twist
- b. 100 percent shielded pairs
- c. at least 300-volt insulation
- d. each pair has a 20 AWG tinned-copper drain wire and individual overall

pair insulation

- e. cables have an overall aluminum-polyester or tinned-copper cable-shield tape, overall 20 AWG tinned-copper cable drain wire, and overall cable insulation.

#### 2.3.4 Power Wiring for Control Devices

For 24-volt circuits, provide insulated copper 18 AWG or thicker wire rated for 300 VAC service. For 120-volt circuits, provide 14 AWG or thicker stranded copper wire rated for 600-volt service.

#### 2.3.5 Transformers

Provide UL 5085-3 approved transformers. Select transformers sized so that the connected load is no greater than 80 percent of the transformer rated capacity.

### 2.4 AUTOMATIC CONTROL VALVES

Provide valves with stainless-steel stems and stuffing boxes with extended necks to clear the piping insulation. Provide valves with bodies meeting ASME B16.34 or ASME B16.15 pressure and temperature class ratings based on the design operating temperature and 150 percent of the system design operating pressure. Unless otherwise specified or indicated, provide valves meeting FCI 70-2 Class III leakage rating. Provide valves rated for modulating or two-position service as indicated, which close against a differential pressure indicated as the Close-Off pressure and which are Normally-Open, Normally-Closed, or Fail-In-Last-Position as indicated.

#### 2.4.1 Valve Type

##### 2.4.1.1 Liquid Service 150 Degrees F or Less

Use either globe valves or ball valves except that butterfly valves may be used for sizes 100 mm and larger.

#### 2.4.2 Valve Flow Coefficient and Flow Characteristic

##### 2.4.2.1 Two-Way Modulating Valves

Provide the valve coefficient (Kv) indicated. Provide equal-percentage flow characteristic for liquid service except for butterfly valves.

##### 2.4.2.2 Three-Way Modulating Valves

Provide the valve coefficient (Kv) indicated. Provide linear flow characteristic with constant total flow throughout full plug travel.

#### 2.4.3 Two-Position Valves

Use full line size full port valves with maximum available (Kv).

#### 2.4.4 Globe Valves

##### 2.4.4.1 Liquid Service Not Exceeding 66 Degrees C

- a. Valve body and body connections:

- (1) valves 38 mm and smaller: brass or bronze body, with threaded or union ends
- (2) valves from 51 mm to 76 mm inclusive: brass, bronze, or iron bodies. 51 millimeters valves with threaded connections; 63 to 76 mm valves with flanged connections

- b. Internal valve trim: Brass or bronze.
- c. Stems: Stainless steel.
- d. Provide valves compatible with a solution of 50 percent ethylene or propylene glycol.

#### 2.4.5 Ball Valves

##### 2.4.5.1 Liquid Service Not Exceeding 66 Degrees C

- a. Valve body and connections:
  - (a) 1) valves 38 mm and smaller: bodies of brass or bronze, with threaded or union ends
  - (b) 2) valves from 51 mm to 76 mm inclusive: bodies of brass, bronze, or iron. 50 mm valves with threaded connections; valves from 63 to 76 mm with flanged connections.
- b. Ball: Stainless steel or nickel-plated brass or chrome-plated brass.
- c. Seals: Reinforced Teflon seals and EPDM O-rings.
- d. Stem: Stainless steel, blow-out proof.
- e. Provide valves compatible with a solution of 50 percent ethylene or propylene glycol.

#### 2.4.6 Butterfly Valves

Provide butterfly valves which are threaded lug type suitable for dead-end service and modulation to the fully-closed position, with carbon-steel bodies or with ductile iron bodies in accordance with ASTM A536. Provide butterfly valves with non-corrosive discs, stainless steel shafts supported by bearings, and EPDM seats suitable for temperatures from -28.9 to +121.1 degrees C. Provide valves with rated Kv of the Kv at 70 percent (60 degrees) open position. Provide valves meeting FCI 70-2 Class VI leakage rating.

#### 2.4.7 Pressure Independent Control Valves (PICV)

Provide pressure independent control valves which include a regulator valve which maintains the differential pressure across a flow control valve. Pressure independent control valves must accurately control the flow from 0-100 percent full rated flow regardless of changes in the piping pressure and not vary the flow more than plus or minus 5 percent at any given flow control valve position when the PICV differential pressure lies between the manufacturer's stated minimum and maximum. The rated minimum differential pressure for steady flow must not exceed 34.5 kPa across the PICV. Provide either globe or ball type valves meeting the

indicated requirements for globe and ball valves. Provide valves with a flow tag listing full rated flow and minimum required pressure drop. Provide valves with factory installed Pressure/Temperature ports ("Pete's Plugs") to measure the pressure drop to determine the valve flow rate.

#### 2.4.8 Duct-Coil and Terminal-Unit-Coil Valves

For duct or terminal-unit coils provide control valves with either flare-type, screw type, or solder-type ends. Provide flare nuts for each flare-type end valve.

### 2.5 DAMPERS

#### 2.5.1 Damper Assembly

Provide single damper sections with blades no longer than 1.2 m and which are no higher than 1.8 m and damper blade width of 203 mm or less. When larger sizes are required, combine damper sections. Provide dampers made of steel, or other materials where indicated and with assembly frames constructed of 2.8 mm minimum thickness stainless steel channels with mitered and welded corners. Steel channel frames constructed of 1.5 mm minimum thickness are acceptable provided the corners are reinforced.

- a. Flat blades must be made rigid by folding the edges. Blade-operating linkages must be within the frame so that blade-connecting devices within the same damper section must not be located directly in the air stream.
- b. Damper axles must be 13 mm minimum, plated steel rods supported in the damper frame by stainless steel or bronze bearings. Blades mounted vertically must be supported by thrust bearings.
- c. Provide dampers which do not exceed a pressure drop through the damper of 10 Pa at 5 m/s in the wide-open position. Provide dampers with frames not less than 50 mm in width. Provide dampers which have been tested in accordance with AMCA 500-D.

#### 2.5.2 Operating Linkages

For operating links external to dampers, such as crank arms, connecting rods, and line shafting for transmitting motion from damper actuators to dampers, provide links able to withstand a load equal to at least 300 percent of the maximum required damper-operating force without deforming. Rod lengths must be adjustable. Links must be brass, bronze, zinc-coated steel, or stainless steel. Working parts of joints and clevises must be brass, bronze, or stainless steel. Adjustments of crank arms must control the open and closed positions of dampers.

#### 2.5.3 Damper Types

##### 2.5.3.1 Flow Control Dampers

Provide parallel-blade or opposed blade type dampers for outside air, return air, relief air, exhaust, face and bypass dampers as indicated on the Damper Schedule. Blades must have interlocking edges. The channel frames of the dampers must be provided with jamb seals to minimize air leakage. Unless otherwise indicated, dampers must meet AMCA 511 Class 1 requirements. Outside air damper seals must be suitable for an operating temperature range of -40 to +75 degrees C. Dampers must be rated at not



less than 10 m/s air velocity.

#### 2.5.3.2 Mechanical Rooms and Other Utility Space Ventilation Dampers

Provide utility space ventilation dampers as indicated. Unless otherwise indicated provide AMCA 511 class 3 dampers. Provide dampers rated at not less than 7.6 m/s air velocity.

#### 2.5.3.3 Smoke Dampers

Provide smoke-damper and actuator assemblies which meet the current requirements of NFPA 90A, UL 555, and UL 555S. For combination fire and smoke dampers provide dampers rated for 121 degrees C Class II leakage per UL 555S.

### 2.6 SENSORS AND INSTRUMENTATION

Unless otherwise specified, provide sensors and instrumentation which incorporate an integral transmitter. Sensors and instrumentation, including their transmitters, must meet the specified accuracy and drift requirements at the input of the connected DDC Hardware's analog-to-digital conversion.

#### 2.6.1 Analog and Binary Transmitters

Provide transmitters which match the characteristics of the sensor. Transmitters providing analog values must produce a linear 4-20 mAdc, 0-10 Vdc signal corresponding to the required operating range and must have zero and span adjustment. Transmitters providing binary values must have dry contacts rated at 1A at 24 Volts AC.

#### 2.6.2 Network Transmitters

Sensors and Instrumentation incorporating an integral network connection are considered DDC Hardware and must meet the DDC Hardware requirements of Section BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS when used in a BACnet network.

#### 2.6.3 Temperature Sensors

Provide the same sensor type throughout the project. Temperature sensors may be provided without transmitters. Where transmitters are used, the range must be the smallest available from the manufacturer and suitable for the application such that the range encompasses the expected range of temperatures to be measured. The end to end accuracy includes the combined effect of sensitivity, hysteresis, linearity and repeatability between the measured variable and the end user interface (graphic presentation) including transmitters if used.

##### 2.6.3.1 Sensor Accuracy and Stability of Control

###### 2.6.3.1.1 Conditioned Space Temperature

Plus or minus 0.3 degrees C over the operating range.

###### 2.6.3.1.2 Unconditioned Space Temperature

a. Plus or minus 0.6 degrees C over the range of -1 to +55 degrees C AND

- b. Plus or minus 2 degrees C over the rest of the operating range.

#### 2.6.3.1.3 Duct Temperature

Plus or minus 0.3 degrees C

#### 2.6.3.1.4 Outside Air Temperature

- a. Plus or minus 1 degree C over the range of -35 to +55 degrees C AND

- b. Plus or minus 0.6 degrees C over the range of -1 to +40 degrees C.

#### 2.6.3.1.5 Chilled Water

Plus or minus 0.4 degrees C over the range of 2 to 18 degrees C.

#### 2.6.3.2 Transmitter Drift

The maximum allowable transmitter drift: 0.1 degrees C per year.

#### 2.6.3.3 Point Temperature Sensors

Point Sensors must be encapsulated in epoxy, series 300 stainless steel, anodized aluminum, or copper.

#### 2.6.3.4 Temperature Sensor Details

##### 2.6.3.4.1 Room Type

Provide the sensing element components within a decorative protective cover suitable for surrounding decor.

##### 2.6.3.4.2 Duct Probe Type

Ensure the probe is long enough to properly sense the air stream temperature.

##### 2.6.3.4.3 Duct Averaging Type

Continuous averaging sensors must be one foot in length for each 0.1 square m of duct cross-sectional area, and a minimum length of 1.5 m.

##### 2.6.3.4.4 Pipe Immersion Type

For pipes with larger than 7.6 cm diameter, provide minimum 7.6 cm immersion. For pipes with less than 7.6 cm diameter, provide immersion at least half the diameter of the pipe. Provide each sensor with a corresponding pipe-mounted sensor well, unless indicated otherwise. Sensor wells must be stainless steel when used in steel piping, and brass when used in copper piping.

##### 2.6.3.4.5 Outside Air Type

Provide the sensing element rated for outdoor use

#### 2.6.4 Relative Humidity Sensor

Relative humidity sensors must use bulk polymer resistive or thin film capacitive type non-saturating sensing elements capable of withstanding a saturated condition without permanently affecting calibration or

sustaining damage. The sensors must include removable protective membrane filters. Where required for exterior installation, sensors must be capable of surviving below freezing temperatures and direct contact with moisture without affecting sensor calibration. When used indoors, the sensor must be capable of being exposed to a condensing air stream (100 percent relative humidity) with no adverse effect to the sensor's calibration or other harm to the instrument. The sensor must be of the wall-mounted or duct-mounted type, as required by the application, and must be provided with any required accessories. Sensors used in duct high-limit applications must have a bulk polymer resistive sensing element. Duct-mounted sensors must be provided with a duct probe designed to protect the sensing element from dust accumulation and mechanical damage. Relative humidity (RH) sensors must measure relative humidity over a range of 0 percent to 100 percent with an accuracy of plus or minus 3 percent. RH sensors must function over a temperature range of 4.4 to 57.2 degrees C and must not drift more than 1 percent per year.

#### 2.6.5 Carbon Dioxide (CO2) Sensors

Provide photometric type CO2 sensors with integral transducers and linear output. Carbon dioxide (CO2) sensors must measure CO2 concentrations between 0 to 2000 parts per million (ppm) using non-dispersible infrared (NDIR) technology with an accuracy of plus or minus 50 ppm and a maximum response time of 1 minute. The sensor must be rated for operation at ambient air temperatures within the range of 0 to 50 degrees C and relative humidity within the range of 20 to 95 percent (non-condensing). The sensor must have a maximum drift of 2 percent per year. The sensor chamber must be manufactured with a non-corrosive material that does not affect carbon dioxide sample concentration. Duct mounted sensors must be provided with a duct probe designed to protect the sensing element from dust accumulation and mechanical damage. The sensor must have a calibration interval no less than 5 years.

#### 2.6.6 Differential Pressure Instrumentation

##### 2.6.6.1 Differential Pressure Sensors

Provide Differential Pressure Sensors with ranges as indicated or as required for the application. Pressure sensor ranges must not exceed the high end range indicated on the Points Schedule by more than 50 percent. The over pressure rating must be a minimum of 150 percent of the highest design pressure of either input to the sensor. The accuracy must be plus or minus 1 percent of full scale. The sensor must have a maximum drift of 2 percent per year.

##### 2.6.6.2 Differential Pressure Switch

Provide differential pressure switches with a user-adjustable setpoint which are sized for the application such that the setpoint is between 25 percent and 75 percent of the full range. The over pressure rating must be a minimum of 150 percent of the highest design pressure of either input to the sensor. The switch must have two sets of contacts and each contact must have a rating greater than it's connected load. Contacts must open or close upon rise of pressure above the setpoint or drop of pressure below the setpoint as indicated.

## 2.6.7 Flow Sensors

### 2.6.7.1 Airflow Measurement Array (AFMA)

#### 2.6.7.1.1 Airflow Straightener

Provide AFMAs which contain an airflow straightener if required by the AFMA manufacturer's published installation instructions. The straightener must be contained inside a flanged sheet metal casing, with the AFMA located as specified according to the published recommendation of the AFMA manufacturer. In the absence of published documentation, provide airflow straighteners if there is any duct obstruction within 5 duct diameters upstream of the AFMA. Air-flow straighteners, where required, must be constructed of 3 mm aluminum honeycomb and the depth of the straightener must not be less than 40 mm.

#### 2.6.7.1.2 Resistance to Airflow

The resistance to air flow through the AFMA, including the airflow straightener must not exceed 20 Pa at an airflow of 10 m/s. AFMA construction must be suitable for operation at airflows of up to 25 m/s over a temperature range of 4 to 49 degrees C.

#### 2.6.7.1.3 Outside Air Temperature

In outside air measurement or in low-temperature air delivery applications, provide an AFMA certified by the manufacturer to be accurate as specified over a temperature range of -29 to +49 degrees C.

#### 2.6.7.1.4 Pitot Tube AFMA

Each Pitot Tube AFMA must contain an array of velocity sensing elements. The velocity sensing elements must be of the multiple pitot tube type with averaging manifolds. The sensing elements must be distributed across the duct cross section in the quantity and pattern specified or recommended by the published installation instructions of the AFMA manufacturer.

- a. Pitot Tube AFMAs for use in airflows over 3.0 m/s must have an accuracy of plus or minus 5 percent over a range of 2.5 to 12.5 m/s.
- b. Pitot Tube AFMAs for use in airflows under 3.0 m/s must have an accuracy of plus or minus 5 percent over a range of 0.6 to 12.5 m/s.

#### 2.6.7.1.5 Electronic AFMA

Each electronic AFMA must consist of an array of velocity sensing elements of the resistance temperature detector (RTD) or thermistor type. The sensing elements must be distributed across the duct cross section in the quantity and pattern specified or recommended by the published application data of the AFMA manufacturer. Electronic AFMAs must have an accuracy of plus or minus 5 percent over a range of 0.6 to 12.5 m/s and the output must be temperature compensated over a range of 0 to 100 degrees C.

#### 2.6.7.1.6 Fan Inlet Measurement Devices

Fan inlet measurement devices cannot be used unless indicated on the drawings or schedules.

#### 2.6.7.2 Orifice Plate

Orifice plate must be made of an austenitic stainless steel sheet of 3. mm nominal thickness with an accuracy of plus or minus 1 percent of full flow. The orifice plate must be flat within 0.1 mm. The orifice surface roughness must not exceed 0.5  $\mu\text{m}$ . The thickness of the cylindrical face of the orifice must not exceed 2 percent of the pipe inside diameter or 12.5 percent of the orifice diameter, whichever is smaller. The upstream edge of the orifice must be square and sharp. Where orifice plates are used, concentric orifice plates must be used in all applications except steam flow measurement in horizontal pipelines.

#### 2.6.7.3 Flow Nozzle

Flow nozzle must be made of austenitic stainless steel with an accuracy of plus or minus 1 percent of full flow. The inlet nozzle form must be elliptical and the nozzle throat must be the quadrant of an ellipse. The thickness of the nozzle wall and flange must be such that distortion of the nozzle throat from strains caused by the pipeline temperature and pressure, flange bolting, or other methods of installing the nozzle in the pipeline must not cause the accuracy to degrade beyond the specified limit. The outside diameter of the nozzle flange or the design of the flange facing must be such that the nozzle throat must be centered accurately in the pipe.

#### 2.6.7.4 Venturi Tube

Venturi tube must be made of cast iron or cast steel and must have an accuracy of plus or minus 1 percent of full flow. The throat section must be lined with austenitic stainless steel. Thermal expansion characteristics of the lining must be the same as that of the throat casting material. The surface of the throat lining must be machined to a plus or minus 1.2  $\mu\text{m}$  finish, including the short curvature leading from the converging entrance section into the throat.

#### 2.6.7.5 Annular Pitot Tube

Annular pitot tube must be made of austenitic stainless steel with an accuracy of plus or minus 2 percent of full flow and a repeatability of plus or minus 0.5 percent of measured value. The unit must have at least one static port and no less than four total head pressure ports with an averaging manifold.

#### 2.6.7.6 Insertion Turbine Flowmeter

Provide dual axial turbine flowmeter with all installation hardware necessary to enable insertion and removal of the meter without system shutdown. All parts must meet or exceed the pressure classification of the pipe system it is installed in. Insertion Turbine Flowmeter accuracy must be plus or minus 0.5 percent of rate at calibrated velocity., within plus or minus of rate over a 10:1 turndown and within plus or minus 2 percent of rate over a 50:1 turndown. Repeatability must be plus or minus 0.25 percent of reading. The meter flow sensing element must operate over a range suitable for the installed location with a pressure loss limited to 1 percent of operating pressure at maximum flow rate. The flowmeter ,must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc outputs. The turbine rotor assembly must be constructed of Series 300 stainless steel and use Teflon seals.

#### 2.6.7.7 Vortex Shedding Flowmeter

Vortex Shedding Flowmeter accuracy must be within plus or minus 0.8 percent of the actual reading over the range of the meter. Steam meters must contain density compensation by direct measurement of temperature. Mass flow inferred from specified steam pressure are not acceptable. The flow meter body must be made of austenitic stainless steel and include a weather tight NEMA 4X electronics enclosure. The vortex shedding flowmeter body must not require removal from the piping in order to replace the shedding sensor.

#### 2.6.7.8 Ultrasonic Flow Meter

Provide Ultrasonic Flow Meters complete with matched transducers, self aligning installation hardware and transducer cables. Ultrasonic transducers must be optimized for the specific pipe and process conditions for the application. The flow meter accuracy must plus or minus 1 percent of rate from 0.3 to 12 meters/sec. The flowmeter must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc output.

#### 2.6.7.9 Insertion Magnetic Flow Meter

Provide insertion type magnetic flowmeters with all installation hardware necessary to enable insertion and removal of the meter without system shutdown. All parts must meet or exceed the pressure classification of the pipe system it is installed in. Flowmeter accuracy must be no greater than plus or minus 1 percent of rate from 0.6 to 6 meters/sec. Wetted material parts must be 300 series stainless steel. The flowmeter must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc outputs.

#### 2.6.7.10 Positive Displacement Flow Meter

The flow meter must be a direct reading, gerotor, nutating disc or vane type displacement device rated for liquid service as indicated. A counter must be mounted on top of the meter, and must consist of a non-resettable mechanical totalizer for local reading, and a pulse transmitter for remote reading. The totalizer must have a six digit register to indicate the volume passed through the meter in liters, and a sweep-hand dial to indicate down to 1 L. The pulse transmitter must have a hermetically sealed reed switch which is activated by magnets fixed on gears of the counter. The meter must have a bronze body with threaded or flanged connections as required for the application. Output accuracy must be plus or minus 2 percent of the flow range. The maximum pressure drop at full flow must be 34 kPa.

#### 2.6.7.11 Flow Meters, Paddle Type

Sensor must be non-magnetic, with forward curved impeller blades designed for water containing debris. Sensor accuracy must be plus or minus 1 percent of rate of flow, minimum operating flow velocity must be 0.3 meters/second. Sensor repeatability and linearity must be plus or minus 1 percent. Materials which will be wetted must be made from non-corrosive materials and must not contaminate water. The sensor must be rated for installation in pipes of 76 mm to 1 m diameters. The transmitter housing must be a NEMA 250 Type 4 enclosure.

#### 2.6.7.12 Flow Switch

Flow switch must have a repetitive accuracy of plus or minus 10 percent of actual flow setting. Switch actuation must be adjustable over the operating flow range, and must be sized for the application such that the setpoint is between 25 percent and 75 percent of the full range. The switch must have Form C snap-action contacts, rated for the application. The flow switch must have non flexible paddle with magnetically actuated contacts and be rated for service at a pressure greater than the installed conditions. Flow switch for use in sewage system must be rated for use in corrosive environments encountered.

#### 2.6.8 Electrical Instruments

Provide Electrical Instruments with an input range as indicated or sized for the application. Unless otherwise specified, AC instrumentation must be suitable for 60 Hz operation.

##### 2.6.8.1 Current Transducers

Current transducers must accept an AC current input and must have an accuracy of plus or minus 2 percent of full scale. The device must have a means for calibration. Current transducers for variable frequency applications must be rated for variable frequency operation.

##### 2.6.8.2 Current Sensing Relays (CSRs)

Current sensing relays (CSRs) must provide a normally-open contact with a voltage and amperage rating greater than its connected load. Current sensing relays must be of split-core design. The CSR must be rated for operation at 200 percent of the connected load. Voltage isolation must be a minimum of 600 volts. The CSR must auto-calibrate to the connected load or be adjustable and field calibrated. Current sensors for variable frequency applications must be rated for variable frequency operation.

##### 2.6.8.3 Voltage Transducers

Voltage transducers must accept an AC voltage input and have an accuracy of plus or minus 0.25 percent of full scale. The device must have a means for calibration. Line side fuses for transducer protection must be provided.

##### 2.6.8.4 Energy Metering

###### 2.6.8.4.1 Watt or Watthour Transducers

Watt transducers must measure voltage and current and must output kW or kWh or both kW and kWh as indicated. kW outputs must have an accuracy of plus or minus 0.5 percent over a power factor range of 0.1 to 1. kWh outputs must have an accuracy of plus or minus 0.5 percent over a power factor range of 0.1 to 1.

###### 2.6.8.4.2 Watthour Revenue Meter (with and without Demand Register)

All Watthour revenue meters must measure voltage and current and must be in accordance with ANSI C12.1 with an ANSI C12.20 Accuracy class of 0.2 and must have pulse initiators for remote monitoring of Watthour consumption. Pulse initiators must consist of form C contacts with a current rating not to exceed two amperes and voltage not to exceed 500 V,

with combinations of VA not to exceed 100 VA, and a life rating of one billion operations. Watthour revenue meters with demand registers must output instantaneous demand in addition to the pulse initiators.

#### 2.6.8.4.3 Hydronic BTU Meters

The BTU meter is to be supplied with wall mount hardware and be capable of being installed remote from the flow meter. The BTU meter must include an LCD display for local indication of energy rate and for display of parameters and settings during configuration. Each BTU meter must be factory configured for its specific application and be completely field configurable by the user via a front panel keypad (no special interface device or computer required). The unit must output Energy Rate, Energy Total, Flow Rate, Supply Temperature, and Return Temperature. An integral transmitter is to provide a linear analog or configurable pulse output signal representing the energy rate; and the signal must be compatible with building automation system DDC Hardware to which the output is connected.

#### 2.6.9 pH Sensor

The sensor must be suitable for applications and chemicals encountered in water treatment systems of boilers, chillers and condenser water systems. Construction, wiring, fittings and accessories must be corrosion and chemical resistant with fittings for tank or suspension installation. Housing must be polyvinylidene fluoride with O-rings made of chemical resistant materials which do not corrode or deteriorate with extended exposure to chemicals. The sensor must be encapsulated. Periodic replacement must not be required for continued sensor operation. Sensors must use a ceramic junction and pH sensitive glass membrane capable of withstanding a pressure of 689 kPa at 66 degrees C. The reference cell must be double junction configuration. Sensor range must be 0 to 12 pH, stability 0.05, sensitivity 0.02, and repeatability of plus or minus 0.05 pH value, response of 90 percent of full scale in one second and a linearity of 99 percent of theoretical electrode output measured at 24 degrees C.

#### 2.6.10 Hydrogen Gas Detector

Provide a complete hydrogen monitoring system with visual and audible alarms. The system shall be complete with the main display, hydrogen gas sensors each with 8M of cable, push button diagnostic test, and junction box. The system shall have a 1% mechanical relay for warning and a 2% for alarm signaling. The system be capable energizing an associated ventilation system fan at 1% hydrogen concentration. This unit can be mounted directly to a wall or to an electrical box. System shall have the capability to monitor up to 3 sensor inputs. Provide a test kit including regulator, 1% and 2% hydrogen, tubing and storage case.

#### 2.6.11 Temperature Switch

##### 2.6.11.1 Duct Mount Temperature Low Limit Safety Switch (Freezestat)

Duct mount temperature low limit switches (Freezestats) must be manual reset, low temperature safety switches at least 3 meters long per square meter of coverage which must respond to the coldest 450 mm segment with an accuracy of plus or minus 2 degrees C. The switch must have a field-adjustable setpoint with a range of at least -1 +10 degrees C. The switch must have two sets of contacts, and each contact must have a rating



greater than its connected load. Contacts must open or close upon drop of temperature below setpoint as indicated and must remain in this state until reset.

#### 2.6.11.2 Pipe Mount Temperature Limit Switch (Aquastat)

Pipe mount temperature limit switches (aquastats) must have a field adjustable setpoint between 15 and 32 degrees C, an accuracy of plus or minus 2 degrees C and a 5 degrees C fixed deadband. The switch must have two sets of contacts, and each contact must have a rating greater than its connected load. Contacts must open or close upon change of temperature above or below setpoint as indicated.

#### 2.6.12 Damper End Switches

Each end switch must be a hermetically sealed switch with a trip lever and over-travel mechanism. The switch enclosure must be suitable for mounting on the duct exterior and must permit setting the position of the trip lever that actuates the switch. The trip lever must be aligned with the damper blade.

End switches integral to an electric damper actuator are allowed as long as at least one is adjustable over the travel of the actuator.

### 2.7 INDICATING DEVICES

All indicating devices must display readings in metric (SI) units.

#### 2.7.1 Thermometers

Provide bi-metal type thermometers at locations indicated. Thermometers must have either 230 mm long scales or 90 mm diameter dials, with insertion, immersion, or averaging elements. Provide matching thermowells for pipe-mounted installations. Select scale ranges suitable for the intended service, with the normal operating temperature near the scale's midpoint. The thermometer's accuracy must be plus or minus 2 percent of the scale range.

##### 2.7.1.1 Piping System Thermometers

Piping system thermometers must have brass, malleable iron or aluminum alloy case and frame, clear protective face, permanently stabilized glass tube with indicating-fluid column, white face, black numbers, and a 230 mm scale. Piping system thermometers must have an accuracy of plus or minus 1 percent of scale range. Thermometers for piping systems must have rigid stems with straight, angular, or inclined pattern. Thermometer stems must have expansion heads as required to prevent breakage at extreme temperatures. On rigid-stem thermometers, the space between bulb and stem must be filled with a heat-transfer medium.

##### 2.7.1.2 Air-Duct Thermometers

Air-duct thermometers must have perforated stem guards and 45-degree adjustable duct flanges with locking mechanism.

#### 2.7.2 Pressure Gauges

Provide pipe-mounted pressure gauges at the locations indicated. Gauges must conform to ASME B40.100 and have a 100 mm diameter dial and shutoff cock. Select scale ranges suitable for the intended service, with the

normal operating pressure near the scale's midpoint. The gauge's accuracy must be plus or minus 2 percent of the scale range.

Gauges must be suitable for field or panel mounting as required, must have black legend on white background, and must have a pointer traveling through a 270-degree arc. Gauge range must be suitable for the application with an upper end of the range not to exceed 150 percent of the design upper limit. Accuracy must be plus or minus 3 percent of scale range. Gauges must meet requirements of ASME B40.100.

### 2.7.3 Low Differential Pressure Gauges

Gauges for low differential pressure measurements must be a minimum of 90 mm (nominal) size with two sets of pressure taps, and must have a diaphragm-actuated pointer, white dial with black figures, and pointer zero adjustment. Gauge range must be suitable for the application with an upper end of the range not to exceed 150 percent of the design upper limit. Accuracy must be plus or minus two percent of scale range.

## 2.8 OUTPUT DEVICES

### 2.8.1 Actuators

Actuators must be electric (electronic) . All actuators must be normally open (NO), normally closed (NC) or fail-in-last-position (FILP) as indicated. Normally open and normally closed actuators must be of mechanical spring return type. Electric actuators must have an electronic cut off or other means to provide burnout protection if stalled. Actuators must have a visible position indicator. Electric actuators must provide position feedback to the controller as indicated. Actuators must operate smoothly and fully open or close the devices to which they are applied. Electric actuators must have a full stroke response time in both directions of 90 seconds or less at rated load. Electric actuators must be of the foot-mounted type with an oil-immersed gear train or the direct-coupled type. Where multiple electric actuators operate from a common signal, the actuators must provide an output signal identical to its input signal to the additional devices. All actuators must be rated for their operating environment. Actuators used outdoors must be designed and rated for outdoor use. Actuators under continuous exposure to water, such as those used in sumps, must be submersible.

Actuators incorporating an integral network connection are considered DDC Hardware and must meet the DDC Hardware requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

#### 2.8.1.1 Valve Actuators

Valve actuators must provide shutoff pressures and torques as indicated on the Valve Schedule.

#### 2.8.1.2 Damper Actuators

Damper actuators must provide the torque necessary per damper manufacturer's instructions to modulate the dampers smoothly over its full range of operation and torque must be at least 7.3 Nm/square m of damper area for opposed blade dampers and 10.9 Nm/square m of damper area for parallel blade dampers.

### 2.8.1.3 Electric Actuators

Each actuator must have distinct markings indicating the full-open and full-closed position. Each actuator must deliver the torque required for continuous uniform motion and must have internal end switches to limit the travel, or be capable of withstanding continuous stalling without damage. Actuators must function properly within 85 to 110 percent of rated line voltage. Provide actuators with hardened steel running shafts and gears of steel or copper alloy. Fiber or reinforced nylon gears may be used for torques less than 16 inch-pounds..

- a. Two-position actuators must be single direction, spring return, or reversing type. Two position actuator signals may either be the control power voltage or line voltage as needed for torque or appropriate interlock circuits.
- b. Modulating actuators must be capable of stopping at any point in the cycle, and starting in either direction from any point. Actuators must be equipped with a switch for reversing direction, and a button to disengage the clutch to allow manual adjustments. Provide the actuator with a hand crank for manual adjustments, as applicable. Modulating actuator input signals can either be a 4 to 20 mAdc or a 0-10 VDC signal.
- c. Floating or pulse width modulation actuators are acceptable for non-fail safe applications unless indicated otherwise provided that the floating point control (timed actuation) must have a scheduled re-calibration of span and position no more than once a day and no less than once a week. The schedule for the re-calibration should not affect occupied conditions and be staggered between equipment to prevent falsely loading or unloading central plant equipment.

### 2.8.2 Relays

Relays must have contacts rated for the intended application, indicator light, and dust proof enclosure. The indicator light must be lit when the coil is energized and off when coil is not energized.

Control relay contacts must have utilization category and ratings selected for the application. Each set of contacts must incorporate a normally open (NO), normally closed (NC) and common contact. Relays must be rated for a minimum life of one million operations.

## 2.9 USER INPUT DEVICES

User Input Devices, including potentiometers, switches and momentary contact push-buttons. Potentiometers must be of the thumb wheel or sliding bar type. Momentary Contact Push-Buttons may include an adjustable timer for their output. User input devices must be labeled for their function.

## 2.10 MULTIFUNCTION DEVICES

Multifunction devices are products which combine the functions of multiple sensor, user input or output devices into a single product. Unless otherwise specified, the multifunction device must meet all requirements of each component device. Where the requirements for the component devices conflict, the multifunction device must meet the most stringent of the requirements.

### 2.10.1 Current Sensing Relay Command Switch

The Current Sensing Relay portion must meet all requirements of the Current Sensing Relay input device. The Command Switch portion must meet all requirements of the Relay output device except that it must have at least one normally-open (NO) contact.

Current Sensing Relays used for Variable Frequency Drives must be rated for Variable Frequency applications unless installed on the source side of the drive. If used in this situation, the threshold for showing status must be set to allow for the VFD's control power when the drive is not enabled and provide indication of operation when the drive is enabled at minimum speed.

### 2.10.2 Space Sensor Module

Space Sensor Modules must be multifunction devices incorporating a temperature sensor and one or more of the following as specified and indicated on the Space Sensor Module Schedule:

- a. A temperature indicating device.
- b. A User Input Device which must adjust a temperature setpoint output.
- c. A User Input Momentary Contact Button and an output to the control system indicating zone occupancy.
- d. A three position User Input Switch labeled to indicate heating, cooling and off positions ('HEAT-COOL-OFF' switch) and providing corresponding outputs to the control system.
- e. A two position User Input Switch labeled with 'AUTO' and 'ON' positions and providing corresponding output to the control system..
- f. A multi-position User Input Switch with 'OFF' and at least two fan speed positions and providing corresponding outputs to the control system.

Space Sensor Modules cannot contain mercury (Hg).

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 General Installation Requirements

Perform the installation under the supervision of competent technicians regularly employed in the installation of DDC systems.

##### 3.1.1.1 Device Mounting Criteria

All devices must be installed in accordance with manufacturer's recommendations and as specified and indicated. Control devices to be installed in piping and ductwork must be provided with required gaskets, flanges, thermal compounds, insulation, piping, fittings, and manual valves for shutoff, equalization, purging, and calibration. Strap-on temperature sensing elements must not be used except as specified. Spare thermowells must be installed adjacent to each thermowell containing a

sensor and as indicated. Devices located outdoors must have a weathershield.

#### 3.1.1.2 Labels and Tags

Match labels and tags to the unique identifiers indicated on the As-Built drawings. Label all enclosures and instrumentation. Tag all sensors and actuators in mechanical rooms. Tag airflow measurement arrays to show flow rate range for signal output range, duct size, and pitot tube AFMA flow coefficient. Tag duct static pressure taps at the location of the pressure tap. Provide plastic or metal tags, mechanically attached directly to each device or attached by a metal chain or wire. Labels exterior to protective enclosures must be engraved plastic and mechanically attached to the enclosure or instrumentation. Labels inside protective enclosures may be attached using adhesive, but must not be hand written.

#### 3.1.2 Weathershield

Provide weathershields for sensors located outdoors. Install weathershields such that they prevent the sun from directly striking the sensor and prevent rain from directly striking or dripping onto the sensor. Install weather shields with adequate ventilation so that the sensing element responds to the ambient conditions of the surroundings. When installing weathershields near outside air intake ducts, install them such that normal outside air flow does not cause rainwater to strike the sensor.

#### 3.1.3 Room Instrument Mounting

Mount room instruments, including but not limited to wall mounted non-adjustable space sensor modules and sensors located in occupied spaces, above the floor unless otherwise indicated. Install adjustable devices to be ADA compliant unless otherwise indicated on the Room Sensor Schedule:

- a. Space Sensor Modules for Fan Coil Units may be either unit or wall mounted but not mounted on an exterior wall.
- b. Wall mount all other Space Sensor Modules.

#### 3.1.4 Indication Devices Installed in Piping and Liquid Systems

Install thermometers and temperature sensing elements in liquid systems in thermowells. Provide spare Pressure/Temperature Ports (Pete's Plug) for all temperature and pressure sensing elements installed in liquid systems for calibration/testing.

#### 3.1.5 Occupancy Sensors

Provide a sufficient quantity of occupancy sensors to provide complete coverage of the area (room or space). Occupancy sensors are to be ceiling mounted. Install occupancy sensors in accordance with NFPA 70 requirements and the manufacturer's instructions. Do not locate occupancy sensors within 2 m of HVAC outlets or heating ducts, or where they can "see" beyond any doorway. Installation above doorway(s) is preferred. Do not use ultrasonic sensors in spaces containing ceiling fans. Install sensors to detect motion to within 600 mm of all room entrances and to not trigger due to motion outside the room. Set the off-delay timer to 15 minutes unless otherwise indicated. Adjust sensors prior to beneficial

occupancy, but after installation of furniture systems, shelving, partitions, etc. For each controlled area, provide one hundred percent coverage capable of detecting small hand-motion movements, accommodating all occupancy habits of single or multiple occupants at any location within the controlled room.

### 3.1.6 Switches

#### 3.1.6.1 Temperature Limit Switch

Provide a temperature limit switch (freezestat) to sense the temperature at the location indicated. Provide a sufficient number of temperature limit switches (freezestats) to provide complete coverage of the duct section but no less than 3 m in length per square meter of cross sectional area. Install manual reset limit switches in approved, accessible locations where they can be reset easily. Install temperature limit switch (freezestat) sensing elements in a side-to-side (not top-to-bottom) serpentine pattern with the relay section at the highest point and in accordance with the manufacturer's installation instructions.

#### 3.1.6.2 Hand-Off Auto Switches

Wire safety controls such as smoke detectors and freeze protection thermostats to protect the equipment during both hand and auto operation.

### 3.1.7 Temperature Sensors

Install temperature sensors in locations that are accessible and provide a good representation of sensed media. Installations in dead spaces are not acceptable. Calibrate and install sensors according to manufacturer's instructions. Select sensors only for intended application as designated or recommended by manufacturer.

#### 3.1.7.1 Room Temperature Sensors

Mount the sensors on interior walls to sense the average room temperature at the locations indicated. Avoid locations near heat sources such as copy machines or locations by supply air outlet drafts. Mount the center of all user-adjustable sensors 1.5 m above the finished floor, 1220 mm above the floor to meet ADA requirements. Non user-adjustable sensors can be mounted as indicated in paragraph ROOM INSTRUMENT MOUNTING.

#### 3.1.7.2 Duct Temperature Sensors

##### 3.1.7.2.1 Probe Type

Place tip of the sensor in the middle of the airstream or in accordance with manufacturer's recommendations or instructions. Provide a gasket between the sensor housing and the duct wall. Seal the duct penetration air tight. When installed in insulated duct, provide enclosure or stand off fitting to accommodate the thickness of duct insulation to allow for maintenance or replacement of the sensor and wiring terminations. Seal the duct insulation penetration vapor tight.

##### 3.1.7.2.2 Averaging Type

Weave the sensing element in a serpentine fashion from side to side perpendicular to the flow, across the duct or air handler cross-section, using durable non-metal supports in accordance with manufacturer's

installation instructions. Avoid tight radius bends or kinking of the sensing element. Prevent contact between the sensing element and the duct or air handler internals. Provide a duct access door at the sensor location. The access door must be hinged on the side, factory insulated, have cam type locks, and be as large as the duct will permit, maximum 18 by 18 inches. For sensors inside air handlers, the sensors must be fully accessible through the air handler's access doors without removing any of the air handler's internals.

#### 3.1.7.3 Immersion Temperature Sensors

Provide thermowells for sensors measuring piping, tank, or pressure vessel temperatures. Locate wells to sense continuous flow conditions. Do not install wells using extension couplings. When installed on insulated piping, provide stand enclosure or stand off fitting to accommodate the thickness of the pipe insulation and allow for maintenance or replacement of the sensor or wiring terminations. Where piping diameters are smaller than the length of the wells, provide wells in piping at elbows to sense flow across entire area of well. Wells must not restrict flow area to less than 70 percent of pipe area. Increase piping size as required to avoid restriction. Provide the sensor well with a heat-sensitive transfer agent between the sensor and the well interior ensuring contact between the sensor and the well.

#### 3.1.7.4 Outside Air Temperature Sensors

Provide outside air temperature sensors on the building's north side with a protective weather shade that does not inhibit free air flow across the sensing element, and protects the sensor from snow, ice, and rain. Location must not be near exhaust hoods and other areas such that it is not influenced by radiation or convection sources which may affect the reading. Provide a shield to shade the sensor from direct sunlight.

#### 3.1.8 Air Flow Measurement Arrays (AFMA)

Locate Outside Air AFMAs downstream from the Outside Air filters.

Install AFMAs with the manufacturer's recommended minimum distances between upstream and downstream disturbances. Airflow straighteners may be used to reduce minimum distances as recommended by the AFMA manufacturer.

#### 3.1.9 Duct Static Pressure Sensors

Locate the duct static pressure sensing tap at 75 percent of the distance between the first and last air terminal units as indicated on the design documents. If the transmitter output is a 0-10Vdc signal, locate the transmitter in the same enclosure as the air handling unit (AHU) controller for the AHU serving the terminal units. If a remote duct static pressure sensor is to be used, run the signal wire back to the controller for the air handling unit.

#### 3.1.10 Relative Humidity Sensors

Install relative humidity sensors in supply air ducts at least 3 m downstream of humidity injection elements.

### 3.1.11 Meters

#### 3.1.11.1 Flowmeters

Install flowmeters to ensure minimum straight unobstructed piping for at least 10 pipe diameters upstream and at least 5 pipe diameters downstream of the flowmeter, and in accordance with the manufacturer's installation instructions.

#### 3.1.11.2 Energy Meters

Locate energy meters as indicated. Connect each meter output to the DDC system, to measure both instantaneous demand/energy and other variables as indicated.

### 3.1.12 Dampers

#### 3.1.12.1 Damper Actuators

Provide spring return actuators which fail to a position that protects the served equipment and space on all control dampers related to freeze protection or force protection. For all outside, makeup and relief dampers provide dampers which fail closed. Terminal fan coil units, terminal VAV units, convectors, and unit heaters may be non-spring return unless indicated otherwise. Do not mount actuators in the air stream. Do not connect multiple actuators to a common drive shaft. Install actuators so that their action seal the damper to the extent required to maintain leakage at or below the specified rate and so that they move the blades smoothly throughout the full range of motion.

#### 3.1.12.2 Damper Installation

Install dampers straight and true, level in all planes, and square in all dimensions. Dampers must move freely without undue stress due to twisting, racking (parallelogramming), bowing, or other installation error. External linkages must operate smoothly over the entire range of motion, without deformation or slipping of any connecting rods, joints or brackets that will prevent a return to its normal position. Blades must close completely and leakage must not exceed that specified at the rated static pressure. Provide structural support for multi-section dampers. Acceptable methods of structural support include but are not limited to U-channel, angle iron, corner angles and bolts, bent galvanized steel stiffeners, sleeve attachments, braces, and building structure. Where multi-section dampers are installed in ducts or sleeves, they must not sag due to lack of support. Do not use jackshafts to link more than three damper sections. Do not use blade to blade linkages. Install outside and return air dampers such that their blades direct their respective air streams towards each other to provide for maximum mixing of air streams.

### 3.1.13 Valves

Install the valves in accordance with the manufacturer's instructions.

#### 3.1.13.1 Valve Actuators

Provide spring return actuators on all control valves where freeze protection is required. Spring return actuators for terminal fan coil units, terminal VAV units, convectors, and unit heaters are not required unless indicated otherwise.



### 3.1.14 Thermometers and Gauges

#### 3.1.14.1 Thermometers

Mount devices to allow reading while standing on the floor or ground, as applicable.

### 3.1.15 Wire and Cable

Provide complete electrical wiring for the Control System, including wiring to transformer primaries. Wire and Cable must be installed without splices between control devices and in accordance with NFPA 70 and NFPA 90A. Instrumentation grounding must be installed per the device manufacturer's instructions and as necessary to prevent ground loops, noise, and surges from adversely affecting operation of the system. Test installed ground rods as specified in IEEE 142. Cables and conductor wires must be tagged at both ends, with the identifier indicated on the shop drawings. Electrical work must be as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and as indicated. Wiring external to enclosures must be run in raceways, except low-voltage control and low-voltage network wiring may be installed as follows:

- a. plenum rated cable in suspended ceilings over occupied spaces may be run without raceways
- b. nonmetallic-sheathed cables or metallic-armored cables may be installed as permitted by NFPA 70.

Install control circuit wiring not in raceways in a neat and safe manner. Wiring must not use the suspended ceiling system (including tiles, frames or hangers) for support. Where conduit or raceways are required, control circuit wiring must not run in the same conduit/raceway as power wiring over 50 volts. Run all circuits over 50 volts in conduit, metallic tubing, covered metal raceways, or armored cable.

### 3.1.16 Copper Tubing

Provide hard-drawn copper tubing in exposed areas and either hard-drawn or annealed copper tubing in concealed areas. Use only tool-made bends. Use only brass or copper solder joint type fittings, except for connections to apparatus. For connections to apparatus use brass compression type fittings.

### 3.1.17 Plastic Tubing

Install plastic tubing within covered raceways or conduit except when otherwise specified. Do not use plastic tubing for applications where the tubing could be subjected to a temperature exceeding 55 degrees C. For fittings, use brass or acetal resin of the compression or barbed push-on type for instrument service. Except in walls and exposed locations, plastic multitube instrument tubing bundle without conduit or raceway protection may be used where a number of air lines run to the same points, provided the multitube bundle is enclosed in a protective sheath, is run parallel to the building lines and is adequately supported as specified.

-- End of Section --

## SECTION 23 09 23.02

BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS  
08/24

## PART 1 GENERAL

## 1.1 SUMMARY

Provide a complete Direct Digital Control (DDC) system, except for the front end which is specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

## 1.1.1 System Requirements

Provide a system meeting the requirements of both Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and this Section and with the following characteristics:

- a. Except for Gateways, the control system must be an open implementation of BACnet technology using ASHRAE 135 as the communications protocol. The system must use standard ASHRAE 135 Objects and Properties. The system must use standard ASHRAE 135 Services exclusively for communication over the network. Gateways to packaged units must communicate with other DDC hardware using ASHRAE 135 exclusively and may communicate with packaged equipment using other protocols. The control system must be installed such that any two devices on the Internetwork can communicate using standard ASHRAE 135 Services.
- b. Install and configure control hardware to provide ASHRAE 135 Objects and Properties as indicated and as needed to meet the requirements of this specification.

## 1.1.2 Verification of Specification Requirements

Review all specifications related to the control system installation and advise the Contracting Officer of any discrepancies before performing any work. If Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC or any other Section referenced in this specification is not included in the project specifications advise the Contracting Officer and either obtain the missing Section or obtain Contracting Officer approval before performing any work.

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135

(2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-24 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control

## Networks

### BACNET INTERNATIONAL (BTL)

BTL Guide (v.50; 2022) BACnet Testing Laboratory  
Implementation Guidelines

### INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.3 (2022; CORR 2024) Ethernet

### TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-485 (1998a; R 2012) Electrical Characteristics  
of Generators and Receivers for Use in  
Balanced Digital Multipoint Systems

### U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC Part 15 Radio Frequency Devices (47 CFR 15)

### UNDERWRITERS LABORATORIES (UL)

UL 916 (2015; Reprint Oct 2021) UL Standard for  
Safety Energy Management Equipment

## 1.3 DEFINITIONS

For definitions related to this section, see Section 23 09 00  
INSTRUMENTATION AND CONTROL FOR HVAC.

## 1.4 SUBMITTALS

Submittal requirements related to this Section are specified in Section  
23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

## PART 2 PRODUCTS

All products used to meet this specification must meet the indicated  
requirements, but not all products specified here will be required by  
every project. All products must meet the requirements of Section 23 09 00  
INSTRUMENTATION AND CONTROL FOR HVAC, Section 25 05 11.01 CYBERSECURITY  
FOR FACILITY-RELATED CONTROL HVAC SYSTEMS and this Section.

### 2.1 NETWORK HARDWARE

#### 2.1.1 BACnet Router

All BACnet Routers must be BACnet/IP Routers and must perform layer 3  
routing of ASHRAE 135 packets over an IP network in accordance with  
ASHRAE 135 Annex J and Clause 6. The router must provide the appropriate  
connection to the IP network and connections to one or more ASHRAE 135  
MS/TP networks. Devices used as BACnet Routers must meet the  
requirements for DDC Hardware, and must support the NM-RC-B BIBB.

#### 2.1.2 BACnet Gateways

In addition to the requirements for DDC Hardware, the BACnet Gateway must  
meet the following requirements:

- a. It must perform bi-directional protocol translation from one non-ASHRAE 135 protocol to ASHRAE 135. BACnet Gateways must incorporate a network connection to an ASHRAE 135 network (either BACnet over IP in accordance with Annex J or MS/TP) and a separate connection appropriate for the non-ASHRAE 135 protocol and media.
- b. It must retain its configuration after a power loss of an indefinite time, and must automatically return to their pre-power loss state once power is restored.
- c. It must allow bi-directional mapping of data between the non-ASHRAE 135 protocol and Standard Objects as defined in ASHRAE 135. It must support the DS-RP-B BIBB for Objects requiring read access and the DS-WP-B BIBB for Objects requiring write access.
- d. It must support the DS-COV-B BIBB.

Although Gateways must meet DDC Hardware requirements they are not DDC Hardware and must not be used when DDC Hardware is required.

### 2.1.3 Ethernet Switch

Ethernet Switches must be as specified in Section 25 05 11.01  
CYBERSECURITY FOR FACILITY-RELATED CONTROLHVAC SYSTEMS

## 2.2 CONTROL NETWORK WIRING

- a. BACnet MS/TP communications wiring must be in accordance with ASHRAE 135. The wiring must use shielded, three wire (twisted-pair with reference) cable with characteristic impedance between 100 and 120 ohms. Distributed capacitance between conductors must be less than 100 pF per meter.
- b. Building Control Network Backbone IP Network must use Ethernet media. Ethernet cables must be CAT-5e at a minimum and meet all requirements of IEEE 802.3 and ASHRAE 135.

## 2.3 DIRECT DIGITAL CONTROL (DDC) HARDWARE

### 2.3.1 General Requirements

All DDC Hardware must meet the following requirements:

- a. It must be locally powered and must incorporate a light to indicate the device is receiving power.
- b. It must conform to the BTL Guide
- c. It must be BACnet Testing Laboratory (BTL) Listed.
- d. The Manufacturer's Product Data submittal for each piece of DDC Hardware must include the Protocol Implementation Conformance Statement (PICS) for that hardware as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.
- e. It must communicate and be interoperable in accordance with ASHRAE 135 and have connections for BACnet IP or MS/TP control network wiring.

- f. Other than devices controlling terminal units or functioning solely as a BACnet Router, it must support DS-COV-B, DS-RPM-A and DS-RPM-B BIBBs.
- g. Devices supporting the DS-RP-A BIBB must also support the DS-COV-A BIBB.
- h. Application programs, configuration settings and communication information must be stored in a manner such that they persist through loss of power:
  - (1) Application programs must persist regardless of the length of time power is lost.
  - (2) Configured settings must persist for any loss of power less than 2,500 hours.
  - (3) Communication information, including but not limited to COV subscriptions, event reporting destinations, Notification Class Object settings, and internal communication settings, must persist for any loss of power less than 2,500 hours.
- i. Internal Clocks:
  - (1) Clocks in DDC Hardware incorporating a Clock must continue to function for 120 hours upon loss of power to the DDC Hardware.
  - (2) DDC Hardware incorporating a Clock must support the DM-TS-B or DM-UTC-B BIBB.
- j. It must have all functionality indicated and required to support the application (Sequence of Operation or portion thereof) in which it is used, including but not limited to providing Objects as specified and as indicated on the Points Schedule.
- k. In addition to these general requirements and the DDC Hardware Input-Output (I/O) Function requirements, all DDC Hardware must also meet any additional requirements for the application in which it is used (e.g. scheduling, alarming, trending, etc.).
- l. It must meet FCC Part 15 requirements and have UL 916 or equivalent safety listing.
- m. Device must support Commandable Objects to support Override requirements as detailed in PART 3 EXECUTION
- n. User interfaces which allow for modification of Properties or settings must be password-protected.
- o. Devices communicating BACnet MS/TP must meet the following requirements:
  - (1) Must have a configurable Max\_Master Property.
  - (2) DDC Hardware other than hardware controlling a single terminal unit must have a configurable Max\_Info\_Frames Property.
  - (3) Must respond to any valid request within 50 msec with either the appropriate response or with a response of "Reply Postponed".

- (4) Must use twisted pair with reference and shield (3-wire media) wiring, or twisted pair with shield (2-wire media) wiring and use half-wave rectification.
- p. Devices communicating BACnet/IP must use UDP Port 0xBAC0. Devices with configurable UDP Ports must default to 0xBAC0.
- q. All Device IDs, Network Numbers, and BACnet MAC addresses of devices must be fully configurable without limitation, except MS/TP MAC addresses may be limited by ASHRAE 135 requirements.
- r. DDC Hardware controlling a single terminal unit must have:
  - (1) Objects (including the Device Object) with an Object Name Property of at least 8 characters in length.
  - (2) A configurable Device Object Name.
  - (3) A configurable Device Object Description Property at least 16 characters in length.
- s. Except for Objects in DDC Hardware controlling a single terminal unit, all Objects (including Device Objects) must:
  - (1) Have a configurable Object Name Property of at least 12 characters in length.
  - (2) Have a configurable Object Description Property of at least 24 characters in length.
- t. For programmable DDC Hardware, provide and license to the project site all programming software required to program the Hardware in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.
- u. For programmable DDC Hardware, provide copies of the installed application programs (all software that is not common to every controller of the same manufacturer and model) as source code compatible with the supplied programming software in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. The submitted application program must be the complete application necessary for controller to function as installed and be sufficient to allow replacement of the installed controller with another controller of the same type.

### 2.3.2 Hardware Input-Output (I/O) Functions

DDC Hardware incorporating hardware input-output (I/O) functions must meet the following requirements:

#### 2.3.2.1 Analog Inputs

DC Hardware analog inputs (AIs) must be implemented using ASHRAE 135 Analog Input Objects and perform analog to digital (A-to-D) conversion with a minimum resolution of 8 bits plus sign or better as needed to meet the accuracy requirements specified in Section 23 09 00. Signal conditioning including transient rejection must be provided for each analog input. Analog inputs must be capable of being individually calibrated for zero and span. Calibration via software scaling performed as part of point configuration is acceptable. The AI must incorporate

common mode noise rejection of at least 50 dB from 0 to 100 Hz for differential inputs, and normal mode noise rejection of at least 20 dB at 60 Hz from a source impedance of 10,000 ohms.

#### 2.3.2.2 Analog Outputs

DDC Hardware analog outputs (AOs) must be implemented using ASHRAE 135 Analog Output Objects and perform digital to analog (D-to-A) conversion with a minimum resolution of 8 bits plus sign, and output a signal with a range of 4-20 mAdc or 0-10 Vdc. Analog outputs must be capable of being individually calibrated for zero and span. Calibration via software scaling performed as part of point configuration is acceptable. DDC Hardware with Hand-Off-Auto (H-O-A) switches for analog outputs must provide for overriding the output to 0 percent and to 100 percent.

#### 2.3.2.3 Binary Inputs

DDC Hardware binary inputs (BIs) must be implemented using ASHRAE 135 Binary Input Objects and accept contact closures and must ignore transients of less than 5 milli-second duration. Protection against a transient 50VAC must be provided.

#### 2.3.2.4 Binary Outputs

DDC Hardware binary outputs (BOs) must be implemented using ASHRAE 135 Binary Output Objects and provide relay contact closures or triac outputs for momentary and maintained operation of output devices. DDC Hardware with H-O-A switches for binary outputs must provide for overriding the output open or closed.

##### 2.3.2.4.1 Relay Contact Closures

Closures must have a minimum duration of 0.1 second. Relays must provide at least 180V of isolation. Electromagnetic interference suppression must be provided on all output lines to limit transients to 50 Vac. Minimum contact rating must be 0.5 amperes at 24 Vac.

##### 2.3.2.4.2 Triac Outputs

Triac outputs must provide at least 180 V of isolation. Minimum contact rating must be 0.5 amperes at 24 Vac.

#### 2.3.2.5 Pulse Accumulator

DDC Hardware pulse accumulators must be implemented using either an ASHRAE 135 Accumulator Object or an ASHRAE 135 Analog Value Object where the Present\_Value is the totalized pulse count. Pulse accumulators must accept contact closures, ignore transients less than 5 msec duration, protect against transients of 50 VAC, and accept rates of at least 20 pulses per second.

#### 2.3.2.6 ASHRAE 135 Objects for Hardware Inputs and Outputs

The requirements for use of ASHRAE 135 objects for hardware input and outputs includes devices where the hardware sensor or actuator is integral to the controller (e.g. a VAV box with integral damper actuator, a smart sensor, a VFD, etc.)

### 2.3.2.7 Integrated H-O-A Switches

Where integrated H-O-A switches are provided on hardware outputs, controller must provide means of monitoring position or status of H-O-A switch. This feedback may be provided via any valid BACnet method, including the use of proprietary Objects, Properties, or Services.

### 2.3.3 Local Display Panel (LDP)

The Local Display Panels (LDPs) must be DDC Hardware with a display and navigation buttons or a touch screen display, and must provide display and adjustment of ASHRAE 135 Properties as indicated on the Points Schedule and as specified. LDPs must be either BTL Listed as a B-OD, B-OWS, B-AWS, or be an integral part of another piece of DDC Hardware listed as a B-BC. For LDPs listed as B-OWS or B-AWS, the hardware must be BTL listed and the product must come factory installed with all applications necessary for the device to function as an LDP.

The adjustment of values using display and navigation buttons must be password protected.

### 2.3.4 Expansion Modules and Tethered Hardware

A single piece of DDC Hardware may consist of a base unit and also:

- a. An unlimited number of hardware expansion modules, where the individual hardware expansion modules are designed to directly connect, both mechanically and electrically, to the base unit hardware. The expansion modules must be commercially available as an optional add-on to the base unit.
- b. A single piece of hardware connected (tethered) to a base unit by a single cable where the cable carries a proprietary protocol between the base unit and tethered hardware. The tethered hardware must not contain control logic and be commercially available as an optional add-on to the base unit as a single package.

Note that this restriction on tethered hardware does not apply to sensors or actuators using standard binary or analog signals (not a communications protocol); sensors or actuators using standard binary or analog signals are not considered part of the DDC Hardware.

Hardware capable of being installed stand-alone, or without a separate base unit, is DDC Hardware and must not be used as expansion modules or tethered hardware.

### 2.3.5 Supervisory Control Requirements

#### 2.3.5.1 Scheduling Hardware

DDC Hardware used for scheduling must meet the following requirements:

- a. It must be BTL Listed as a B-BC and support the SCHED-E-B BIBB.
- b. It is preferred, but not required, that devices support the DM-OCD-B BIBB on all Calendar and Schedule Objects, such that a front end BTL listed as a B-AWS may create or delete Calendar and Schedule Objects. It is also preferred but not required that devices supporting the DM-OCD-B BIBB accept any valid value for properties of Calendar and



Schedule Objects. Note that there are additional requirements in the EXECUTION Part of this Section for Devices which do not support the DM-OCD-B BIBB as specified.

- c. The Date\_List property of all Calendar Objects must be writable.
- d. The Present\_Value Property of Schedule must support the following values: 1, 2, 3, 4.

#### 2.3.5.2 Alarm Generation Hardware

DDC Hardware used for alarm generation must meet the following requirements:

- a. Device must support the AE-N-I-B BIBB
- b. The Recipient\_List Property must be Writable for all Notification Class Objects used for alarm generation.
- c. For all Objects implementing Intrinsic Alarming, the following Properties must be Writable:
  - (1) Time\_Delay
  - (2) High\_Limit
  - (3) Low\_Limit
  - (4) Deadband
  - (5) Event\_Enable
  - (6) If the issue date of this project specification is after 1 January 2016, Time\_Delay\_Normal must be writable.
- d. For Event Enrollment Objects used for alarm generation, the following Properties must be Writable:
  - (1) Event\_Parameters
  - (2) Event\_Enable
  - (3) If the issue date of this project specification is after 1 January 2016, Time\_Delay\_Normal must be writable.
- e. It is preferred, but not required, that devices support the DM-OCD-B BIBB on all Notification Class Objects and Event Enrollment Objects, such that a front end BTL listed as a B-AWS may create or delete Notification Class Objects and Event Enrollment Objects. It is also preferred, but not required that devices supporting the DM-OCD-B BIBB accept any valid value as an initial value for properties of Notification Class Objects and Event Enrollment Objects. Note that there are additional requirements in the EXECUTION Part of this Section for devices which do not support the DM-OCD-B BIBB as specified.
- f. Devices provided to meet the the requirements indicated under "Support for Future Alarm Generation" in the EXECUTION part of this specification must support the AE-N-E-B BIBB.

#### 2.3.5.3 Trending Hardware

DDC Hardware used for collecting trend data must meet the following requirements:

- a. Device must support Trend Log or Trend Log Multiple Objects.

- b. Device must support the T-VMT-I-B BIBB.
- c. Devices provided to meet the EXECUTION requirement for support of Future Trending must support the T-VMT-E-B BIBB.
- d. The following properties of all Trend Log or Trend Log Multiple Objects must be present and Writable:
  - Start\_Time
  - Stop\_Time
  - Log\_DeviceObjectProperty
  - Log Interval Log interval must support an interval of at least 60 minutes duration.
- e. Trend Log Objects must support using Intrinsic Reporting to send a BUFFER\_FULL event.
- f. The device must have a Notification Class Object for the BUFFER\_FULL event. The Recipient\_List Property must be Writable.
- g. Devices must support values of at least 1,000 for Buffer\_Size Properties.
- h. It is preferred, but not required, that devices support the DM-OCD-B BIBB on all Trend Log Objects, such that a front end BTL listed as a A-AWS may create or delete Trend Log Objects. It is also preferred, but not required that devices supporting the DM-OCD-B BIBB accept any valid value as an initial value for properties of Trend Log Objects. Note that there are additional EXECUTION requirements for devices which do not support the DM-OCD-B BIBB as specified.

## PART 3 EXECUTION

### 3.1 CONTROL SYSTEM INSTALLATION3.1.1 Building Control Network (BCN)

Install the Building Control Network (BCN) as a single BACnet Internetwork consisting of a single IP network as the BCN Backbone and zero or more BACnet MS/TP networks. Note that in some cases there may only be a single device on the BCN Backbone.

Except as permitted for the non-BACnet side of Gateways, use exclusively ASHRAE 135 networks.

#### 3.1.1.1 Building Control Network IP Backbone

Install IP Network Cabling in conduit. Install Ethernet Switches in lockable enclosures. Install the Building Control Network (BCN) IP Backbone such that it is available at the Facility Point of Connection (FPOC) location as indicated. When the FPOC location is a room number, provide sufficient additional media to ensure that the Building Control Network (BCN) IP Backbone can be extended to any location in the room.

Use UDP port 0xBAC0 for all BACnet traffic on the IP network.

#### 3.1.1.2 BACnet MS/TP Networks

When using MS/TP, provide MS/TP networks in accordance with ASHRAE 135 and in accordance with the ASHRAE 135 figure "Mixed Devices on 3-Conductor Cable with Shield" (Figure 9-1.4 in the 2020 version of ASHRAE 135).

Ground the shield at the BACnet Router and at no other point. Ground the reference wire at the BACnet Router through a 100 ohm resistor and do not ground it at any other point. In addition:

- a. Provide each segment in a doubly terminated bus topology in accordance with TIA-485.
- b. Provide each segment with 2 sets of network bias resistors in accordance with ASHRAE 135, with one set of resistors at each end of the MS/TP network.
- c. Use 3 wire (twisted pair and reference) with shield media for all MS/TP media installed inside. Use fiber optic isolation in accordance with ASHRAE 135 for all MS/TP media installed outside buildings, or between multiple buildings.
- d. For 18 AWG cable, use segments with a maximum length of 1200 m. When using greater distances or different wire gauges comply with the electrical specifications of TIA-485.
- e. For each controller that does not use the reference wire provide transient suppression at the network connection of the controller if the controller itself does not incorporate transient suppression.
- f. Install no more than 32 devices on each MS/TP segment. Do not use MS/TP to MS/TP routers.
- g. Connect each MS/TP network to the BCN backbone via a BACnet Router.
- h. For BACnet Routers, configure the MS/TP MAC address to 0. Assign MAC Addresses to other devices consecutively beginning at 1, with no gaps.
- i. Configure the Max\_Master Property of all devices to be 31.

#### 3.1.1.3 Building Control Network (BCN) Installation

Provide a building control network meeting the following requirements:

- a. Install all DDC Hardware connected to the Building Control Network.
- b. Where multiple pieces of DDC Hardware are used to execute one sequence, install all DDC Hardware executing that sequence on a single MS/TP network dedicated to that sequence.
- c. Traffic between BACnet networks must be exclusively via BACnet routers.

#### 3.1.2 DDC Hardware

Install all DDC Hardware that connects to an IP network in lockable enclosure. Install other DDC Hardware that is not in suspended ceilings in lockable enclosures. For all DDC hardware with a user interface, coordinate with site to determine proper passwords and configure passwords into device.

- a. Except for zone sensors (thermostats), install all Tethered Hardware within 2 m of its base unit.
- b. Install and configure all BTL-Listed devices in a manner consistent with their BTL Listing such that the device as provided still meets

all requirements necessary for its BTL Listing.

- c. Install and configure all BTL-Listed devices in a manner consistent with the BTL Device Implementation Guidelines such that the device as provided meets all those Guidelines.

#### 3.1.2.1 Device Identifiers, Network Addresses, and IP addresses

- a. Do not use any Device Identifier or Network Number already used by another BACnet system at the project site. Coordinate Device IDs and Network Numbers with the installation.
- b. Coordinate device IP addresses with installation. The installation POC is with the IT Department.

#### 3.1.2.2 Object Name Property and Object Description Property

Configure the Object\_Names and Object\_Descriptions properties of all Objects (including Device Objects) as indicated on the Points Schedule (Point Name and Point Description) and as specified. At a minimum:

- a. Except for DDC Hardware controlling a single terminal unit, configure the Object\_Name and Object\_Description properties of all Objects (including Device Objects) as indicated on the Points Schedule and as specified.
- b. In DDC Hardware controlling a single terminal unit, configure the Device Object\_Name and Device Object\_Description as indicated on the Points Schedule and as specified.

When Points Schedule entries exceed the length limitations in the device, notify the Contracting Officer Representative and provide recommended alternatives for approval.

#### 3.1.2.3 Hand-Off-Auto (H-O-A) Switches

Provide Hand-Off-Auto (H-O-A) switches for all DDC Hardware analog outputs and binary outputs used for control of systems other than terminal units, as specified and as indicated on the Points Schedule. Provide H-O-A switches that are integral to the controller hardware, an external device co-located with (in the same enclosure as) the controller, integral to the controlled equipment, or an external device co-located with (in the same enclosure as) the controlled equipment.

- a. For H-O-A switches integral to DDC Hardware, meet the requirements specified in paragraph DIRECT DIGITAL CONTROL (DDC) HARDWARE.
- b. For external H-O-A switches used for binary outputs, provide for overriding the output open or closed.
- c. For external H-O-A switches used for analog outputs, provide for overriding through the range of 0 percent to 100 percent.

#### 3.1.2.4 Local Display Panels

Provide LDPs to display and override values of ASHRAE 135 Object Properties as indicated on the Points Schedule. Install LDPs displaying points for anything other than a terminal unit in the same room as the equipment. Install LDPs displaying points for only terminal units in a

mechanical room central to the group of terminal units it serves. For LDPs using WriteProperty to commandable objects to implement an override, write values with priority 9.

#### 3.1.2.5 MS/TP Slave Devices

Configure all MS/TP devices as Master devices. Do not configure any devices to act as slave devices.

#### 3.1.2.6 Change of Value (COV) and Read Property

- a. To the greatest extent possible, configure all devices to support the SubscribeCOV service (the DS-COV-B BIBB). At a minimum, all devices supporting the DS-RP-B BIBB, other than devices controlling only a single terminal unit, must be configured to support the DS-COV-B BIBB.
- b. Whenever supported by the server side, configure client devices to use the DS-COV-A BIBB.

#### 3.1.2.7 Engineering Units

#### 3.1.2.8 Occupancy Modes

Use the following correspondence between value and occupancy mode whenever an occupancy state or value is required:

- a. OCCUPIED mode: a value of one
- b. UNOCCUPIED mode: a value of two
- c. WARM-UP/COOL-DOWN (PRE-OCCUPANCY) mode: a value of three

Note that elsewhere in this Section the Schedule Object is required to also support a value of four, which is reserved for future use. Also note that the behavior of a system in each of these occupancy modes is indicated in the sequence of operation for the system.

#### 3.1.2.9 Use of BACnet Objects

Use only standard non-proprietary ASHRAE 135 Objects and services to accomplish the project scope of work as follows:

- a. Use Analog Input or Analog Output Objects for all analog hardware I/O. Do not use Analog Value Object for analog hardware I/O).
- b. Use Binary Input or Binary Output Objects for all binary hardware I/O. Do not use Binary Value Objects for binary hardware I/O.
- c. Use Analog Value Objects for analog setpoints.
- d. Use Accumulator Objects or Analog Value Objects for pulse inputs.
- e. For occupancy modes, use Multistate Value Objects and the correspondence between value and occupancy mode specified in paragraph OCCUPANCY MODES.
- f. Use Schedule Objects and Calendar Objects for all scheduling. Use Trend Log Objects or Trend Log Multiple Objects for all trending and Notification Class Objects for trend log upload. Use a combination of Event Enrollment Objects, Intrinsic Alarming, and Notification Class Objects for alarm generation.

- g. For all other points shown on the Points Schedule as requiring an ASHRAE 135 Object, use the Object type shown on the Points Schedule or, if no Object Type is shown, use a standard Object appropriate to the point.

#### 3.1.2.10 Use of Standard BACnet Services

Except as noted in this paragraph, for all DDC Hardware use Standard BACnet Services as defined in this specification (which excludes some ASHRAE 135 services) exclusively for application control functionality and communication.

DDC Hardware that cannot meet this requirement may use non-standard services provided they can provide identical functionality using Standard BACnet Services when communicating with BACnet devices from a different vendor. When implementing non-standard services, document all non-standard services in the DDC Hardware Schedule as specified and as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

#### 3.1.2.11 Device Application Configuration

- a. For every property, setting or value shown on the Points Schedule or otherwise indicated as Configurable, provide a value that is retained through loss of power and can be changed via one or more of:
  - (1) BACnet services (including proprietary services)
  - (2) Hardware settings on the device
- b. For every property, setting or value shown on the Points Schedule or otherwise indicated as Operator Configurable, provide a value that is retained through loss of power and can be changed via one or more of:
  - (1) A Writable Property of a standard BACnet Object
  - (2) A Property of a standard BACnet Object that is Writable when Out\_Of\_Service is TRUE and Out\_Of\_Service is Writable.

### 3.1.3 Scheduling, Alarming, Trending, and Overrides

#### 3.1.3.1 Scheduling

Configure schedules in BACnet Scheduling Objects to schedule systems as indicated on the Points Schedule and as specified using the indicated correspondence between value and occupancy mode. If no devices supports both the SCHED-E-B and DM-OCD-B BIBBS for Schedule Objects, provide 5 blank Schedule Objects in DDC Hardware BTL listed as B-BCs and supporting the SCHED-E-B BIBB for later use by the site.

Provide a separate schedule for each AHU including it's associated Terminal Units and for each stand-alone Terminal Unit (those not dependent upon AHU service) or group of stand-alone Terminal Units acting according to a common schedule as indicated.

#### 3.1.3.2 Configuration of Alarm Generation

- a. Send alarm events as Alarms (not Events).

- b. Use the ConfirmedNotification Service for alarm events.
- c. For alarm generation, support two priority levels for alarms: critical and non-critical. Configure the Priority of Notification Class Objects to use Priority 112 for critical and 224 for non-critical alarms.
- d. Number of Notification Class Objects for Alarm Generation:
  - (1) If the device implements non-critical alarms, or if any Object in the device supports Intrinsic Alarms, then provide a single Notification Class Object specifically for (shared by) all non-critical alarms.
  - (2) If the device implements critical alarms, provide a single Notification Class Object specifically for (shared by) all critical alarms.
  - (3) If the device implements both critical and non-critical alarms, provide both Notification Class Objects (one for critical, one for non-critical).
  - (4) If the device controls equipment other than a single terminal unit, provide both Notification Class Objects (one for critical, one for non-critical) even if no alarm generation is required at time of installation.
- e. For all intrinsic alarms configure the Limit\_Enable Property to set both HighLimitEnable and LowLimitEnable to TRUE. If the specified alarm conditions are for a single-sided alarm (only High\_Limit used or only Low\_Limit used) assign a value to the unused limit such that the unused alarm condition will not occur.
- f. For all objects supporting intrinsic alarming, even if no alarm generation is required during installation, configure the following Properties as follows:
  - (1) Notification\_Class to point to the non-Critical Notification Class Object in that device.
  - (2) Limit\_Enable to enable both the HighLimitEnable and LowLimitEnable
  - (3) Notify\_Type to Alarm
- g. Use of alarm generation types:
  - (1) Only use algorithmic alarm generation when intrinsic alarm generation is not supported by the device or object, or when the specific alarm conditions cannot be implemented using intrinsic alarm generation.
  - (2) Only use remote alarm generation when the alarm cannot be generated using intrinsic or local algorithmic alarm generation on the device containing the referenced property. If remote alarm generation is used, use the same DDC Hardware for all remote alarm generation within a single sequence.

### 3.1.3.3 Support for Future Alarm Generation

For every piece of DDC Hardware, support future alarm generation capabilities by supporting either intrinsic or additional algorithmic alarming. Provide one of the following:

- a. Support intrinsic alarming for every Object used by the application in that device.
- b. Support additional Event\_Enrollment Objects. For DDC hardware controlling a single terminal unit, support at least one additional object. Otherwise, support at least 4 additional Objects. Support additional Event\_Enrollment Objects via one of the following:
  - (1) Provide unused Event\_Enrollment Objects on that device.
  - (2) Support the DM-OCD-B BIBB and the creation of sufficient Event\_Enrollment Objects on that device.
  - (3) Provide one or more devices in the IP network that support the AE-N-E-B BIBB and have unused Event\_Enrollment Objects.
  - (4) Provide one or more devices on the IP network that support the AE-N-E-B BIBB, the DM-OCD-B BIBB, and the creation of sufficient Event\_Enrollment Objects.

The total number of Event\_Enrollment Objects required by the project is the sum of the individual device requirements, and the distribution of Event\_Enrollment Objects among devices is not further restricted. (Note this allows a single device to contain many Event\_Enrollment Objects satisfying the requirements for multiple devices.)

### 3.1.3.4 Trend Log Configuration

- a. Configure trends in Trend Log or Trend Log Multiple Objects as indicated on the Points Schedule and as specified.
- b. Configure all trend logs (including any provided to support future trends) to save data on regular intervals using the BUFFER\_FULL event to request trend upload from the front end.
- c. Configure Trend Log Objects with a minimum Buffer\_Size property value of 1,000 and Trend Log Multiple Objects with a minimum Buffer\_Size property value of 1,000 per point trended (for example, a Trend Log Multiple Object used to trend 3 points must have a Buffer\_Size Property value of at least 3,000).
- d. Configure a Notification Class Object in devices doing trending (including devices supporting future trends) to handle the BUFFER\_FULL event.
- e. When possible, trend each point using an Object in the device containing the point. When it is necessary to trend using a an Object in another device, all trends not on the same Device as the Object being trended must be on a single device (i.e. all Trend Log and Trend Log Multiple Objects used for remote trending within a sequence must be on the same device).
- f. For each trend log, including any trend logs provided to support



future trending, configure the following properties as specified:

- (1) Logging\_Type: Set to Polling
- (2) Stop\_When\_Full: Set to Wrap Around
- (3) Buffer\_Size: Set to 400 or greater.
- (4) Notification\_Threshold: Set to 90 percent of full
- (5) Notification\_Class: Set to the Notification Class Object in that device
- (6) Event\_Enable: Set to TRUE
- (7) Log\_Interval: Set to 15 minutes.

g. Future Trending support. Provide support for future trending:

- (1) Provide one or more devices on the Building Control Network Backbone IP network which support both the T-VMT-E-B and DM-OCD-B BIBBs for Trend Log Objects. Provide sufficient devices to support the creation of at least one additional Trend Log Object for every terminal unit plus 4 additional Trend Log Objects for every non-terminal unit.
- (2) Provide one additional Trend Log Object for every terminal unit plus 4 additional Trend Log Objects for every non-terminal unit in one or more devices on the Building Control Network Backbone IP network that support the T-VMT-E-B BIBB for later use by the site.
- (3) A combination of these two methods is permitted provided the total required number of Trend Log Objects is met.

#### 3.1.3.5 Overrides

Provide an override for each point shown on the Points Schedule as requiring an override.

Unless otherwise approved, provide Commandable Objects to support all Overrides . With specific approval from the Contracting Officer, Overrides for points which are not hardware outputs and which are in DDC hardware controlling a single terminal unit may support overrides via an additional Object provided for the override. No other means of implementing Overrides may be used.

- a. Where Commandable Objects are used, ensure that WriteProperty service requests with a Priority of 10 or less take precedence over the SEQUENCE VALUE and that WriteProperty service request with a priority of 11 or more have a lower precedence than the SEQUENCE VALUE.
- b. For devices implementing overrides via additional Objects, provide Objects which are NOT Written to as part of the normal Sequence of Operations and are Writable when Out\_Of\_Service is TRUE and Out\_Of\_Service is Writable. Use this point as an Override of the normal value when Out\_Of\_Service is TRUE and the normal value otherwise. Note these Objects may be modified as part of the sequence via local processes, but must not be modified by local processes when

Out\_Of\_Service is TRUE.

#### 3.1.4 BACnet Gateways

The requirements in this paragraph do not themselves permit the installation of hardware not meeting the other requirements of this section. Except for proprietary systems specifically indicated in Section 23 09 00, all control hardware installed under this project must meet the requirements of this specification, including the control hardware providing the network interface for a package unit or split system specified under another section. Only use gateways to connect to pre-existing control devices, and to proprietary systems specifically permitted by Section 23 09 00.

##### 3.1.4.1 General Gateway Requirements

Provide BACnet Gateways to connect non-BACnet control hardware in accordance with the following:

- a. Configure gateways to map writable data points in the controlled equipment to Writable Properties of Standard Objects as indicated in the Points Schedule and as specified.
- b. Configure gateway to map readable data points in the controlled equipment to Readable Properties of Standard Objects as indicated in the Points Schedule and as specified.
- c. Configure gateway to support the DS-COV-B BIBB for all points mapped to BACnet Objects.
- d. Do not use non-BACnet control hardware for controlling built-up units or any other equipment that was not furnished with factory-installed controls.
- e. Do not use non-BACnet control hardware for system scheduling functions.
- f. Each gateway must communicate with and perform protocol translation for non-BACnet control hardware controlling one and only one package unit or a single non-BACnet system specifically permitted by Section 23 09 00.
- g. Connect one network port on the gateway to the Building Control Backbone IP Network or to a BACnet MS/TP network and the other port to the single piece of controlled equipment or the non-BACnet system specifically permitted by Section 23 09 00..
- h. For gateways to existing package units or simple split systems, non-BACnet network wiring connecting the gateway to the package unit must not exceed 3 meters in length and must connect to exactly two devices: the controlled equipment (packaged unit) or split system interface and the gateway.

-- End of Section --

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## SECTION 23 23 00

## REFRIGERANT PIPING

**08/21**

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## Canadian General Standards Board (CGSB)

CAN/CGSB-1.181                      Zinc dust pigment used in protective coatings

## CSA GROUP (CSA)

|                 |                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------|
| CSA B16.22      | (2021) Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings                                |
| CSA B32         | (2020) Standard Specification for Solder Metal                                                       |
| CAN/CSA B40.100 | (2022) Pressure Gauges and Gauge Attachments                                                         |
| CSA B51         | (2024) Boiler, pressure vessel, and pressure piping code                                             |
| CSA B52         | (2023) Mechanical refrigeration code                                                                 |
| CSA B125.3      | (2022) Plumbing fittings                                                                             |
| CSA B137.1      | (2020) Polyethylene (PE) pipe, tubing, and fittings for cold-water pressure services                 |
| CSA B242        | Pipe Hangers and Supports                                                                            |
| CSA B280        | Standard Specification for Seamless Copper Tube for Air Conditioning and Refrigeration Field Service |
| CSA C743        | (2025) Performance standard for rating water-chilling and heat pump water-heating packages           |
| CSA FB3         | (2012) Specification for Fluxes for Brazing and Braze Welding                                        |
| CSA G164        | Hot dip galvanizing of irregularly shaped articles                                                   |
| CSA G345.2      | Seamless And Welded Steel Tubes For Low-Temperature Service, Equivalent In Scope to ASTM A334/A334M  |

|             |                                                                      |
|-------------|----------------------------------------------------------------------|
| CAN/CSA-HP3 | Copper Alloy Sand Castings                                           |
| CSA W47.1   | Certification of companies for fusion welding of steel               |
| CSA W48     | (2023) Filler metals and allied materials for metal arc welding      |
| CSA W117.2  | (2023) Safety in welding, cutting, and allied processes              |
| CSA Z245.1  | Steel Pipe                                                           |
| CSA Z259    | (2021) Connecting components for personal fall-arrest systems (PFAS) |

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

|              |                                                                                 |
|--------------|---------------------------------------------------------------------------------|
| AHRI 711     | (2009) Performance Rating of Liquid-Line Driers                                 |
| AHRI 720     | (2002) Refrigerant Access Valves and Hose Connectors                            |
| AHRI 751 SI  | (2016) Performance Rating of Thermostatic Refrigerant Expansion Valves          |
| AHRI 760 I-P | (2014) Performance Rating of Solenoid Valves for Use with Volatile Refrigerants |
| AHRI 1371 SI | (2017) Performance Rating of Electronic Expansion Valves                        |

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

|           |                                                                                |
|-----------|--------------------------------------------------------------------------------|
| ASHRAE 17 | (2022) Method of Testing Capacity of Thermostatic Refrigerant Expansion Valves |
|-----------|--------------------------------------------------------------------------------|

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|              |                                                                        |
|--------------|------------------------------------------------------------------------|
| ASME B16.22  | (2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings  |
| ASME B16.26  | (2018) Standard for Cast Copper Alloy Fittings for Flared Copper Tubes |
| ASME B31.1   | (2024) Power Piping                                                    |
| ASME B31.5   | (2022) Refrigeration Piping and Heat Transfer Components               |
| ASME B40.100 | (2022) Pressure Gauges and Gauge Attachments                           |

AMERICAN WELDING SOCIETY (AWS)

|                |                                            |
|----------------|--------------------------------------------|
| AWS A5.8/A5.8M | (2019) Specification for Filler Metals for |
|----------------|--------------------------------------------|

## Brazing and Braze Welding

AWS A5.31/A5.31M

(2012) Specification for Fluxes for  
Brazing and Braze Welding

AWS BRH

(2007; 5th Ed) Brazing Handbook

## ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M

(2024) Standard Specification for Pipe,  
Steel, Black and Hot-Dipped, Zinc-Coated,  
Welded and Seamless

ASTM A653/A653M

(2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

ASTM B32

(2020) Standard Specification for Solder  
Metal

ASTM B62

(2017) Standard Specification for  
Composition Bronze or Ounce Metal Castings

ASTM B75/B75M

(2020) Standard Specification for Seamless  
Copper Tube

ASTM B117

(2019) Standard Practice for Operating  
Salt Spray (Fog) Apparatus

ASTM B280

(2023) Standard Specification for Seamless  
Copper Tube for Air Conditioning and  
Refrigeration Field Service

ASTM B813

(2024) Standard Specification for Liquid  
and Paste Fluxes for Soldering of Copper  
and Copper Alloy Tube

ASTM D520

(2000; R 2019) Zinc Dust Pigment

ASTM D3308

(2012; R 2017) Standard Specification for  
PTFE Resin Skived TapeMANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS  
INDUSTRY (MSS)

MSS SP-58

(2018) Pipe Hangers and Supports -  
Materials, Design and Manufacture,  
Selection, Application, and Installation

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

## Refrigerant Piping System

## Verification of Dimensions

## SD-06 Test Reports

## Refrigerant Piping Tests

## Maintenance; G

## 1.3 QUALITY ASSURANCE

## 1.3.1 Contract Drawings

Because of the small scale of the drawings, it is not possible to indicate all offsets, fittings, and accessories that may be required. Carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

## 1.4 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation is the Contractor's responsibility. Replace any materials found to be damaged at the Contractor's expense. During installation, cap piping and similar openings to keep out dirt and other foreign matter.

## 1.5 MAINTENANCE

## 1.5.1 General

Submit Data Package 2 plus operation and maintenance data complying with the requirements of Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein.

## PART 2 PRODUCTS

## 2.1 STANDARD COMMERCIAL PRODUCTS

- a. Provide materials and equipment which are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship and that have been in satisfactory commercial or industrial use for 2 years prior to bid opening.
- b. The 2 year use must include applications of equipment and materials under similar circumstances and of similar size. The 2 years' experience must be satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures. Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown.
- c. Exposed equipment moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that

may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices so that proper operation of equipment is not impaired. Welding and cutting safety requirements must be in accordance with CSA W117.2.

## 2.2 ELECTRICAL WORK

Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices specified, but not shown, must be provided.

## 2.3 REFRIGERANT PIPING SYSTEM

Provide refrigerant piping, valves, fittings, and accessories in accordance with CSA B52, except as specified herein. Refrigerant piping, valves, fittings, and accessories must be compatible with the fluids used and capable of withstanding the pressures and temperatures of the service. Refrigerant piping, valves, and accessories used for refrigerant service must be cleaned, dehydrated, and sealed (capped or plugged) prior to shipment from the manufacturer's plant.

## 2.4 PIPE, FITTINGS AND END CONNECTIONS (JOINTS)

### 2.4.1 Steel Tubing

Tubing must be cold-rolled, electric-forged, welded-steel in accordance with CSA G345.2, Grade 1. Joints and fittings must be socket type provided by the steel tubing manufacturer.

### 2.4.2 Copper Tubing

Provide copper tubing conforming to ASTM B280 CSA B280 annealed or hard drawn as required. Copper tubing must bear the product identification markings in accordance with ASTM B280, CSA B280 "ACR" must be present on copper tubing. Copper tubing must be soft annealed where bending is required and hard drawn where no bending is required. Soft annealed copper tubing must not be used in sizes larger than 35 mm. Joints must be brazed except that joints on lines 22 mm and smaller may be flared. Cast copper alloy fittings for flared copper tube must conform to ASME B16.26 CSA B125.3 and ASTM B62 CAN/CSA-HP3. Wrought copper and bronze solder-joint pressure fittings must conform to ASME B16.22 CSA B16.22 and ASTM B75/B75M CSA B137.1. Joints and fittings for brazed joint must be wrought-copper or forged-brass sweat fittings. Cast sweat-type joints and fittings are not allowed for brazed joints. Brass or bronze adapters for brazed tubing may be used for connecting tubing to flanges and to threaded ends of valves and equipment.

### 2.4.3 Solder

Solder must conform to ASTM B32 CSA B32 Grade SB5, grade Sb5, tin-antimony alloy for service pressures up to 1034 kPa. Solder flux must be liquid or paste form, non-corrosive and conform to ASTM B813 CSA B701.

### 2.4.4 Brazing Filler Metal

Filler metal must conform to AWS A5.8/A5.8M, Type BAg-5 CSA W48 with AWS Type FB3-A CSA FB3-A Flux or Type FB3-C CSA FB3-C, except Type BCuP-3, BCuP-4, or BCuP-5 may be used for brazing copper-to-copper joints. BAlSi-4 with AWS Type FB1-A flux may be used when joining copper piping to



aluminum components.

#### 2.4.5 Brazing Flux

Brazing flux must conform to AWS A5.31/A5.31M, Type FB3-A Flux or Type FB3-C when using Type BAg-5 filler metal. Type FB1-A is to be used with Type BAlSi-4 filler metal.

#### 2.4.6 Press Fittings

Press fittings are not acceptable for use in refrigerant piping systems.

### 2.5 VALVES

Valves must be designed, manufactured, and tested specifically for refrigerant service. The valve material and all internal components must be compatible with the specific refrigerant and lubricant used. Valve bodies must be of brass, bronze, steel, or ductile iron construction. Valves 25 mm and smaller must have brazed or socket welded connections. Do not use threaded end connections, except in pilot pressure or gauge lines where maintenance disassembly is required and welded flanges cannot be used. Internal parts must be removable for inspection or replacement without applying heat or breaking pipe connections. Valve stems exposed to the atmosphere must be stainless steel or corrosion resistant metal plated carbon steel. Direction of flow must be legibly and permanently indicated on the valve body. Control valve inlets must be fitted with integral or adapted strainer or filter where recommended or required by the manufacturer. Purge, charge and receiver valves must be of manufacturer's standard configuration.

#### 2.5.1 Refrigerant Stop Valves

Valve must be the globe or full-port ball type with a back-seating stem especially packed for refrigerant service. Valve packing must be replaceable under line pressure. Provide valve with a wrench operator and a seal cap. Valve must be the straight or angle pattern design as indicated.

#### 2.5.2 Check Valves

Valve must be the swing or lift type as required to provide positive shutoff at the differential pressure indicated. Valve must be provided with resilient seat.

#### 2.5.3 Liquid Solenoid Valves

Provide valves that comply with AHRI 760 I-P CSA B52 and are suitable for continuous duty with applied voltages 15 percent under and 5 percent over nominal rated voltage at maximum and minimum encountered pressure and temperature service conditions. Valves must be direct-acting or pilot-operating type, packless, except that packed stem, seal capped, manual lifting provisions must be furnished. Provide solenoid coils that are moisture-proof, UL approved, totally encapsulated or encapsulated and metal jacketed as required. Valves must have safe working pressure of 4206 kPa and a maximum operating pressure differential of at least 1375 kPa at 85 percent rated voltage. Valves must have an operating pressure differential suitable for the refrigerant used.

#### 2.5.4 Expansion Valves

Provide valve conforming to AHRI 751 SI CSA B52 and ASHRAE 17. Valve must be the diaphragm and spring-loaded type with internal or external equalizers, and bulb and capillary tubing. Provide valve with an external superheat adjustment along with a seal cap. Internal equalizers may be utilized where flowing refrigerant pressure drop between outlet of the valve and inlet to the evaporator coil is negligible and pressure drop across the evaporator is less than the pressure difference corresponding to 1 degree C of saturated suction temperature at evaporator conditions. Bulb charge must be determined by the manufacturer for the application and such that liquid will remain in the bulb at all operating conditions. Do not use gas limited liquid charged valves and other valve devices for limiting evaporator pressure without a distributor or discharge tube or effective means to prevent loss of control when bulb becomes warmer than valve body. Pilot-operated valves must have a characterized plug to provide required modulating control. A de-energized solenoid valve may be used in the pilot line to close the main valve in lieu of a solenoid valve in the main liquid line. Provide an isolatable pressure gauge in the pilot line, at the main valve. Automatic pressure reducing or constant pressure regulating expansion valves may be used only where indicated or for constant evaporator loads.

#### 2.5.5 Electronic Expansion Valves

Valve must conform to AHRI 1371 SI CSA B52 and ASHRAE 17. The valve must prevent the return of liquid to the compressor in the event of power loss or low superheat.

#### 2.5.6 Safety Relief Valves

Valve must be the two-way type, unless indicated otherwise. Valve must bear the ASME code symbol. Valve capacity must be certified by the National Board of Boiler and Pressure Vessel Inspectors. Valve must be of an automatically reseating design after activation.

#### 2.5.7 Evaporator Pressure Regulators, Direct-Acting

Valve must include a diaphragm/spring assembly, external pressure adjustment with seal cap, and pressure gauge port. Valve must maintain a constant inlet pressure by balancing inlet pressure on diaphragm against an adjustable spring load. Pressure drop at system design load must not exceed the pressure difference corresponding to a 1 degree C change in saturated refrigerant temperature at evaporator operating suction temperature. Spring must be selected for indicated maximum allowable suction pressure range.

#### 2.5.8 Refrigerant Access Valves

Provide refrigerant access valves and hose connections in accordance with AHRI 720.

### 2.6 PIPING ACCESSORIES

#### 2.6.1 Filter Driers

Driers must conform to AHRI 711 CSA C743. Sizes 15 mm and larger must be the full flow, replaceable core type. Sizes 13 mm and smaller must be the sealed type. Cores must be of suitable desiccant that will not plug,

cake, dust, channel, or break down, and must remove water, acid, and foreign material from the refrigerant. Construct filter driers so that none of the desiccant will pass into the refrigerant lines. Minimum bursting pressure must be 10.3 MPa.

## 2.6.2 Sight Glass and Liquid Level Indicator

### 2.6.2.1 Assembly and Components

Assembly must be pressure- and temperature-rated and constructed of materials suitable for the service. Glass must be borosilicate type. Ferrous components subject to condensation must be electro-galvanized.

### 2.6.2.2 Gauge Glass

Gauge glass must include top and bottom isolation valves fitted with automatic checks, and packing followers; red-line or green-line gauge glass; elastomer or polymer packing to suit the service; and gauge glass guard.

### 2.6.2.3 Bull's-Eye and Inline Sight Glass Reflex Lens

Provide bull's-eye and inline sight glass reflex lens for dead-end liquid service. For pipe line mounting, provide two plain lenses in one body suitable for backlighting viewing.

### 2.6.2.4 Moisture Indicator

Indicator must be a self-reversible action, moisture reactive, color changing media. Indicator must be furnished with full-color-printing tag containing color, moisture, and temperature criteria. Unless otherwise indicated, the moisture indicator must be an integral part of each corresponding sight glass.

## 2.6.3 Vibration Dampeners

Dampeners must be of the all-metallic bellows and woven-wire type.

## 2.6.4 Flexible Pipe Connectors

Connector must be a composite of interior corrugated phosphor bronze or Type 300 Series stainless steel, as required for fluid service, with exterior reinforcement of bronze, stainless steel or monel wire braid. Assembly must be constructed with a safety factor of not less than 4 at 150 degrees C. Unless otherwise indicated, the length of a flexible connector must be as recommended by the manufacturer for the service intended.

## 2.6.5 Strainers

Strainers used in refrigerant service must have brass or cast-iron body, Y-or angle-pattern, cleanable, not less than 60-mesh noncorroding screen of an area to provide net free area not less than ten times the pipe diameter with pressure rating compatible with the refrigerant service. Screens must be stainless steel or monel and reinforced spring-loaded where necessary for bypass-proof construction.

### 2.6.6 Pressure and Vacuum Gauges

Provide gauges conforming to ASME B40.100 CSA B51/CAN/CSA B40.100 with throttling type needle valve or a pulsation dampener and shut-off valve. Gauge must be a minimum of 85 mm in diameter with a range from 0 kPa to approximately 1.5 times the maximum system working pressure. Select each gauge range so that at normal operating pressure, the needle is within the middle-third of the range.

### 2.6.7 Temperature Gauges

Provide industrial duty type temperature gauges for the required temperature range. Gauges must have Celsius scale in 1 degree graduations scale (black numbers) on a white face. The pointer must be adjustable. Provide rigid stem type temperature gauges in thermowells located within 1.5 m of the finished floor. Provide universal adjustable angle type or remote element type temperature gauges in thermowells located 1.5 to 2.1 m above the finished floor. Provide remote element type temperature gauges in thermowells located 2.1 m above the finished floor.

#### 2.6.7.1 Stem Cased-Glass

Provide stem cased-glass case composed of polished stainless steel or cast aluminum, 229 mm long, with clear acrylic lens, and non-mercury filled glass tube with indicating-fluid column.

#### 2.6.7.2 Bimetallic Dial

Provide bimetallic dial type case that is greater than 89 mm, stainless steel, and hermetically sealed with clear acrylic lens. Bimetallic element must be silicone dampened and unit fitted with external calibrator adjustment. Accuracy must be one percent of dial range.

#### 2.6.7.3 Liquid-, Solid-, and Vapor-Filled Dial

Provide liquid-, solid-, and vapor-filled dial type cases that are greater than 89 mm, stainless steel or cast aluminum with clear acrylic lens. Fill must be nonmercury, suitable for encountered cross-ambients, and connecting capillary tubing must be double-braided bronze.

#### 2.6.7.4 Thermowell

Thermowell must be identical size, 13 or 19 mm NPT connection, brass or stainless steel. Where test wells are indicated, provide captive plug-fitted type 13 mm NPT connection suitable for use with either engraved stem or standard separable socket thermometer or thermostat. Mercury must not be used in thermometers. Extended neck thermowells must be of sufficient length to clear insulation thickness by 25 mm.

### 2.6.8 Pipe Hangers, Inserts, and Supports

Provide pipe hangers, inserts, guides, and supports conforming to MSS SP-58.

### 2.6.9 Escutcheons

Escutcheons must be chromium-plated iron or chromium-plated brass, either one piece or split pattern, held in place by internal spring tension or set screws.

## 2.7 FABRICATION

### 2.7.1 Factory Coating

Unless otherwise specified, equipment and component items, when fabricated from ferrous metal, must be factory finished with the manufacturer's standard finish, except that items located outside of buildings must have weather resistant finishes that will withstand 500 hours exposure to the salt spray test specified in ASTM B117 using a 5 percent sodium chloride solution. Immediately after completion of the test, the specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond 3 mm on either side of the scratch mark. Cut edges of galvanized surfaces where hot-dip galvanized sheet steel is used must be coated with a zinc-rich coating conforming to ASTM D520, Type I CAN/CGSB-1.181.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, perform a verification of dimensions in the field. Submit a letter, at least 4 weeks prior to beginning construction, including the date the site was visited, conformation of existing conditions, and any discrepancies found before performing any work.

### 3.2 INSTALLATION

Pipe and fitting installation must conform to the requirements of ASME B31.1. Cut pipe accurately to measurements established at the jobsite, and work into place without springing or forcing, completely clearing all windows, doors, and other openings. Cutting or other weakening of the building structure to facilitate piping installation is not permitted without written approval. Cut pipe or tubing square, remove by reaming, and permit free expansion and contraction without causing damage to the building structure, pipe, joints, or hangers.

#### 3.2.1 Directional Changes

Make changes in direction with fittings, except that bending of pipe 100 mm and smaller is permitted, provided a pipe bender is used and wide weep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees is not permitted. The centerline radius of bends must not be less than 6 diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be accepted.

#### 3.2.2 Functional Requirements

Install piping 4 mm/m of pipe in the direction of flow to ensure adequate oil drainage. Properly cap or plug open ends of refrigerant lines or equipment during installation to keep moisture, dirt, or other foreign material out of the system. Piping must remain capped until installation. Equipment piping must be in accordance with the equipment manufacturer's recommendations and the contract drawings. Equipment and piping arrangements must fit into space allotted and allow adequate acceptable clearances for installation, replacement, entry, servicing, and maintenance.

### 3.2.3 Fittings and End Connections

#### 3.2.3.1 Threaded Connections

Make threaded connections with tapered threads and make tight with PTFE tape complying with ASTM D3308 CSA or CGSB or equivalent thread-joint compound applied to the male threads only. Do not show more than three threads after the joint is made.

#### 3.2.3.2 Brazed Connections

Perform brazing in accordance with AWS BRH CSA or CGSB, except as modified herein. During brazing, fill the pipe and fittings with a pressure regulated inert gas, such as nitrogen, to prevent the formation of scale. Before brazing copper joints, clean both the outside of the tube and the inside of the fitting with a wire fitting brush until the entire joint surface is bright and clean. Do not use brazing flux on copper-to-copper connections. Remove surplus brazing material at all joints. Make steel tubing joints in accordance with the manufacturer's recommendations. Paint joints in steel tubing with the same material as the baked-on coating within 8 hours after joints are made. Protect tubing against oxidation during brazing by continuous purging of the inside of the piping using nitrogen. Support piping prior to brazing and do not spring or force.

#### 3.2.3.3 Flared Connections

When flared connections are used, use a suitable lubricant between the back of the flare and the nut in order to avoid tearing the flare while tightening the nut.

### 3.2.4 Valves

#### 3.2.4.1 General

Install refrigerant stop valves on each side of each piece of equipment such as compressors condensers, evaporators, receivers, and other similar items in multiple-unit installation, to provide partial system isolation as required for maintenance or repair. Install stop valves with stems horizontal unless otherwise indicated. Install ball valves must be installed with stems positioned to facilitate operation and maintenance. Isolating valves for pressure gauges and switches must be external to thermal insulation. Safety switches must not be fitted with isolation valves. Filter dryers having access ports may be considered a point of isolation. Purge valves must be provided at all points of systems where accumulated non-condensable gases would prevent proper system operation. Valves must be furnished to match line size, unless otherwise indicated or approved.

#### 3.2.4.2 Expansion Valves

Install expansion valves with the thermostatic expansion valve bulb located on top of the suction line when the suction line is less than 54 mm in diameter and at the 4 o'clock or 8 o'clock position on lines larger than 54 mm. Fasten the bulb securely with two clamps. Insulate the bulb. Install the bulb in a horizontal portion of the suction line, if possible, with the pigtail on the bottom. If the bulb must be installed in a vertical line, the bulb tubing must be facing up.

#### 3.2.4.3 Valve Identification

Tag each system valve, including those which are part of a factory assembly. Tags must be in alphanumeric sequence, progressing in direction of fluid flow. Tags must be embossed, engraved, or stamped plastic or nonferrous metal of various shapes, sized approximately 34 mm diameter, or equivalent dimension, substantially attached to a component or immediately adjacent thereto. Attach tags with nonferrous, heavy duty, bead or link chain, 14 gauge annealed wire, nylon cable bands or as approved. Reference tag numbers in Operation and Maintenance Manuals and system diagrams.

#### 3.2.5 Vibration Dampers

Provide vibration damper in the suction and discharge lines on spring mounted compressors. Install vibration dampers parallel with the shaft of the compressor and anchor firmly at the upstream end on the suction line and the downstream end in the discharge line.

#### 3.2.6 Strainers

Provide strainers immediately ahead of solenoid valves and expansion devices. Strainers may be an integral part of an expansion valve.

#### 3.2.7 Filter Dryer

Provide a liquid line filter dryer on each refrigerant circuit located such that all liquid refrigerant passes through a filter dryer. Size dryers in accordance with the manufacturer's recommendations for the system in which it is installed. Install dryers such that it can be isolated from the system, the isolated portion of the system evacuated, and the filter dryer replaced. Install dryers in the horizontal position except replaceable core filter dryers may be installed in the vertical position with the access flange on the bottom.

#### 3.2.8 Sight Glass

Install a moisture indicating sight glass in all refrigerant circuits downstream of all filter dryers and where indicated. Provide full line size sight glasses.

#### 3.2.9 Accumulator

Provide accumulators in the suction line to each compressor.

#### 3.2.10 Flexible Pipe Connectors

Install connectors perpendicular to line of motion being isolated. Fit piping for equipment with bidirectional motion with two flexible connectors, in perpendicular planes. Install reinforced elastomer flexible connectors in accordance with manufacturer's instructions. Provide piping guides and restraints related to flexible connectors as required.

#### 3.2.11 Temperature Gauges

Thermowells for insertion thermometers and thermostats must extend beyond thermal insulation surface not less than 25 mm.

### 3.2.12 Pipe Hangers, Inserts, and Supports

Pipe hangers, inserts, and supports must conform to MSS SP-58 CSA Z259/ CSA W47.1, except as modified herein. Do not use pipe hanger types 5, 12, and 26. Fabricate hangers used to support piping 50 mm and larger to permit adequate adjustment after erection while still supporting the load. Support piping subjected to vertical movement, when operating temperatures exceed ambient temperatures, by variable spring hangers and supports or by constant support hangers.

#### 3.2.12.1 Hangers

Do not use Type 3 on insulated piping. Type 24 may be used only on trapeze hanger systems or on fabricated frames.

#### 3.2.12.2 Inserts

Secure Type 18 inserts to concrete forms before concrete is placed. Continuous inserts which allow more adjustments may be used if they otherwise meet the requirements for Type 18 inserts.

#### 3.2.12.3 C-Clamps

Torque Type 19 and 23 C-clamps in accordance with MSS SP-58 CSA Z259/ CSA W47.1 and have both locknuts and retaining devices, furnished by the manufacturer. Field-fabricated C-clamp bodies or retaining devices are not acceptable.

#### 3.2.12.4 Angle Attachments

Furnish Type 20 attachments used on angles and channels with an added malleable-iron heel plate or adapter.

#### 3.2.12.5 Saddles and Shields

Where Type 39 saddle or Type 40 shield are permitted for a particular pipe attachment application, the Type 39 saddle, connected to the pipe, must be used on all pipe 100 mm and larger when the temperature of the medium is 16 degrees C or higher. Use Type 40 shields on all piping less than 100 mm and all piping 100 mm and larger carrying medium less than 16 degrees C. Use a high-density insulation insert of cellular glass under the Type 40 shield for piping 50 mm and larger.

#### 3.2.12.6 Horizontal Pipe Supports

Space horizontal pipe supports as specified in MSS SP-58 CSA B2429/ CSA W47.1 and install a support no more than 300 mm from the pipe fitting joint at each change in direction of the piping. Space pipe supports no more than 1.5 m apart at valves.

#### 3.2.12.7 Vertical Pipe Supports

Support vertical pipe at each floor, except at slab-on-grade, and at intervals of not more than 4.5 m not more than 2.4 m from end of risers, and at vent terminations.

#### 3.2.12.8 High Temperature Guides with Cradles

Where there are high system temperatures and welding to piping is not



desirable, the Type 35 guide must include a pipe cradle, welded to the guide structure and strapped securely to the pipe. Separate the pipe from the slide material by at least 100 mm, or by an amount adequate for the insulation, whichever is greater.

#### 3.2.12.9 Multiple Pipe Runs

In the support of multiple pipe runs on a common base member, use a clip or clamp where each pipe crosses the base support member. Spacing of the base support members must not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run.

#### 3.2.12.10 Structural Attachments

Attachment to building structure concrete and masonry must be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors must be applied with a safety factor not less than 5. Do not attach supports to metal decking. Construct masonry anchors for overhead applications of ferrous materials only. Provide structural steel brackets required to support piping, headers, and equipment, but not shown, under this section. Specify material used for support under Section 05 12 00 STRUCTURAL STEEL.

#### 3.2.13 Building Surface Penetrations

Do not install sleeves in structural members except where indicated or approved. Provide galvanized sheet metal sleeves in non-load bearing surfaces conforming to ASTM A653/A653M, CSA G164, Coating Class G-90, 1.0 mm (20 gauge). Provide uncoated carbon steel pipe sleeves in load bearing surfaces conforming to ASTM A53/A53M, CSA Z245.1 Standard weight. Apply sealants to moisture and oil-free surfaces and elastomers to not less than 13 mm depth. Do not install sleeves in structural members.

##### 3.2.13.1 General Service Areas

Extend each sleeve through its respective wall, floor, or roof, and cut flush with each surface. Provide pipes passing through concrete or masonry wall or concrete floors or roofs with pipe sleeves fitted into place at the time of construction. Provide sleeves that allow a minimum of 6 mm all-around clearance between bare pipe and sleeves or between jacketed-insulation and sleeves. Except in pipe chases or interior walls, seal the annular space between pipe and sleeve or between jacket over-insulation and sleeve in accordance with Section 07 92 00 JOINT SEALANTS.

##### 3.2.13.2 Waterproof Penetrations

Install pipes passing through roof or floor waterproofing membrane through a 5.17 kg/sq. m copper sleeve, or a 0.81 mm thick aluminum sleeve, each within an integral skirt or flange. Form flashing sleeve, and extend skirt or flange greater than 200 mm from the pipe and set over the roof or floor membrane in a troweled coating of bituminous cement. Extend the flashing sleeve up the pipe a minimum of 50 mm above the roof or floor penetration. Seal the annular space between the flashing sleeve and the bare pipe or between the flashing sleeve and the metal-jacket-covered insulation as indicated. Seal penetrations by either one of the following methods.

### 3.2.13.2.1 Waterproofing Clamping Flange

Pipes up to and including 250 mm in diameter passing through roof or floor waterproofing membrane may be installed through a cast iron sleeve with caulking recess, anchor lugs, flashing clamp device, and pressure ring with brass bolts. Clamp waterproofing membrane into place and place sealant in the caulking recess.

### 3.2.13.2.2 Modular Mechanical Type Sealing Assembly

In lieu of a waterproofing clamping flange and caulking and sealing of annular space between pipe and sleeve or conduit and sleeve, a modular mechanical type sealing assembly may be installed. Provide seals consisting of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe/conduit and sleeve with corrosion protected carbon steel bolts, nuts, and pressure plates. Loosely assemble links with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut. After the seal assembly is properly positioned in the sleeve, tighten the bolt to cause the rubber sealing elements to expand and provide a watertight seal between the pipe/conduit and the sleeve. Size each seal assembly as recommended by the manufacturer to fit the pipe/conduit and sleeve involved. The Contractor electing to use the modular mechanical type seals must provide sleeves of the proper diameters.

### 3.2.13.3 Fire-Rated Penetrations

Seal penetration of fire-rated walls, partitions, and floors as specified in Section 07 84 00 FIRESTOPPING.

### 3.2.13.4 Escutcheons

Provide escutcheons for finished surfaces where exposed piping, bare or insulated, pass through floors, walls, or ceilings, except in boiler, utility, or equipment rooms. Where sleeves project slightly from floors, use special deep-type escutcheons. Secure escutcheon to pipe or pipe covering.

### 3.2.14 Access Panels

Provide access panels for all concealed valves, vents, controls, and items requiring inspection or maintenance. Provide access panels of sufficient size and locate so that the concealed items may be serviced and maintained or completely removed and replaced.

### 3.2.15 Field Applied Insulation

Field installed insulation is specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS, except as defined differently herein.

### 3.2.16 Field Painting

Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory are specified in Section 09 90 00 PAINTS AND COATINGS.

#### 3.2.16.1 Color Coding

Color coding for piping identification is specified in Section 09 90 00

## PAINTS AND COATINGS.

### 3.2.16.2 Color Coding Scheme

Provide a color coding scheme for locating hidden piping in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.

### 3.2.17 Identification Tags

Provide identification tags made of brass, engraved laminated plastic or engraved anodized aluminum indicating service and item number on all valves and dampers. Tags must be 35 mm minimum diameter and marking must be stamped or engraved. Indentations must be black for reading clarity. Attach tags to valves with No. 12 AWG copper wire, chrome-plated beaded chain or plastic straps designed for that purpose.

## 3.3 CLEANING AND ADJUSTING

Clean uncontaminated system(s) by evacuation and purging procedures currently recommended by refrigerant and refrigerant equipment manufacturers, and as specified herein, to remove small amounts of air and moisture. Systems containing moderate amounts of air, moisture, contaminated refrigerant, or any foreign matter are considered contaminated systems. Restore contaminated systems to clean condition including disassembly, component replacement, evacuation, flushing, purging, and re-charging, using currently approved refrigerant and refrigeration manufacturer's procedures. Restore contaminated systems at no additional cost to the Government as determined by the Contracting Officer. Do not use water in any procedure or test.

## 3.4 REFRIGERANT PIPING TESTS

After all components of the refrigerant system have been installed and connected, subject the entire refrigeration system to pneumatic, evacuation, and startup tests as described herein. Provide all material, equipment, instruments, and personnel required for the test. Provide the services of a qualified technician, as required, to perform all tests and procedures indicated herein. Coordinate field tests with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS. Submit an electronic copy of the tests report documenting all phases of the tests performed. Include initial test summaries, all repairs/adjustments made, and the final test results in the report.

### 3.4.1 Preliminary Procedures

Prior to pneumatic testing, isolate equipment which has been factory tested and refrigerant charged as well as equipment which could be damaged or cause personnel injury by imposed test pressure, positive or negative, from the test pressure, or remove from the system. Remove safety relief valves and rupture discs that are not part of factory sealed systems, and cap or plug openings.

### 3.4.2 Pneumatic Test

Provide pressure control and excess pressure protection at the source of test pressure. Valves must be wide open, except those leading to the atmosphere. Test gas must be dry nitrogen, with minus 55 degrees C dewpoint and less than 5 ppm oil. Apply test pressure in two stages before any refrigerant pipe is insulated or covered. In accordance with

ASME B31.5, CSA B52 a preliminary test not to exceed 170 kPa must be applied as a means of locating major leaks. Every joint being tested must be coated with a thick soap or color indicating solution. The second stage test pressure must be at least 110 percent of the design pressure, but cannot exceed 130 percent of the design pressure of any component in the system. For large systems that are not completely visible, the pressure in the system must be gradually increased to one-half of the test pressure after which the pressure must be increased in steps of one-tenth of the test pressure, until the required test pressure has been reached. The test pressure must be continuously maintained for at least 24 hours, after which it can be reduced to the leak test pressure. A correction factor of 2 kPa will be allowed for each degree C change between test space initial and final ambient temperature, plus for increase and minus for a decrease. The leak test pressure must be the design pressure, or a pressure specified in the engineering design. To repair leaks, the joint must be taken apart, thoroughly cleaned, and reconstructed as a new joint. Joints repaired by caulking, re-melting, or back-welding/brazing are not acceptable. Following repair, the entire system must be retested using the pneumatic tests described above. Reassemble the entire system once the pneumatic tests are satisfactorily completed.

#### 3.4.3 Evacuation Test

Following satisfactory completion of the pneumatic tests, relieve the pressure and evacuate the entire system to an absolute pressure of 300 micrometers. During evacuation of the system, the ambient temperature must be higher than 2 degrees C. Do not evacuate no more than one system at one time by one vacuum pump. Once the desired vacuum has been reached, close the vacuum line and allow the system to stand for 1 hour. If the pressure rises over 500 micrometers after the 1 hour period, evacuate the system again down to 300 micrometers and let set for another 1 hour period. Do not charge the system until a vacuum of at least 500 micrometers is maintained for a period of 1 hour without the assistance of a vacuum line. If during the testing the pressure rises above 500 micrometers, continue to repeat the evacuation procedures until all residual moisture has been removed. During evacuation, record pressures by a thermocouple-type, electronic-type, or a calibrated-micrometer type gauge.

#### 3.4.4 System Charging and Startup Test

Following satisfactory completion of the evacuation tests, charge the system with the required amount of refrigerant by raising pressure to normal operating pressure and in accordance with manufacturer's procedures. Following charging, the system must operate with high-side and low-side pressures and corresponding refrigerant temperatures, at design or improved values. Test the entire system tested for leaks. Test fluorocarbon systems with halide torch or electronic leak detectors.

#### 3.4.5 Refrigerant Leakage

If a refrigerant leak is discovered after the system has been charged, the leaking portion of the system must be immediately isolated from the remainder of the system and the refrigerant pumped into the system receiver or other suitable container. The refrigerant must not be discharged into the atmosphere.

### 3.4.6 Contractor's Responsibility

At all times during the installation and testing of the refrigeration system, take steps to prevent the release of refrigerants into the atmosphere. The steps must include, but not be limited to, procedures which will minimize the release of refrigerants to the atmosphere and the use of refrigerant recovery devices to remove refrigerant from the system and store the refrigerant for reuse or reclaim. At no time will the allowable leak rate exceed the leak rates allowed in Section 608 of the Clean Air Act: 30 percent of the full charge per year for industrial refrigeration, 20 percent of the full charge per year for commercial refrigeration, and 10 percent of the full charge per year for comfort cooling. Any system leaks within the first year must be repaired in accordance with the requirements herein at no cost to the Government including material, labor, and refrigerant if the leak is the result of defective equipment, material, or installation.

-- End of Section --

## SECTION 23 30 00

HVAC AIR DISTRIBUTION  
02/25

## PART 1 GENERAL

## 1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

## Canadian General Standards Board (CGSB)

|                        |                                                               |
|------------------------|---------------------------------------------------------------|
| CAN/CGSB-19.13         | Sealing Compound, One Component, Elastomeric, Chemical Curing |
| CAN/CGSB-37.54-95      | (2022) Adhesive and Sealant Applications                      |
| CGSB 51.34-M86         | Air duct systems                                              |
| CGSB 51.33-M89, Type 1 | Air duct sealants                                             |

## CSA GROUP (CSA)

|                       |                                                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CSA B52-13            | (2020) HVAC Duct Construction Standards Metal and Flexible, 4th Edition                                                                                                       |
| CSA B59               | Ball Bearings                                                                                                                                                                 |
| CSA B137.3            | (2020) Rigid polyvinylchloride (PVC) pipe and fittings for pressure applications                                                                                              |
| CAN/CSA ISO 16000-9   | CDPH SECTION 01350 (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
| CSA C22.1             | Canadian Electrical Code, Part I Safety Standard for Electrical Installations                                                                                                 |
| CAN/CSA C22.1-24      | Installation of Air Conditioning and Ventilating Systems in Residential Occupancies                                                                                           |
| CSA C22.2 No. 45.1-13 | Electrical rigid steel conduit                                                                                                                                                |
| CAN/CSA C22.2 No. 113 | (2016) Laboratory Methods of Testing Fans for Aerodynamic Performance Rating                                                                                                  |
| CSA C22.2 No. 100     | Motors and generators                                                                                                                                                         |
| CSA C22.2 No. 113     | Fans and ventilators                                                                                                                                                          |
| CSA C22.2 No. 236     | Safety Heating and Cooling Equipment                                                                                                                                          |

|                |                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------|
| CSA C390       | Test Methods, Marking Requirements, and Energy Efficiency Levels for Three-phase Induction Motors |
| CSA C260-16    | Performance of Centrifugal and Axial Fans and Blowers                                             |
| CSA G30.18     | Carbon Steel Bars for Concrete Reinforcement                                                      |
| CSA G40.20     | General Requirements for Rolled or Welded Structural Quality Steel/Structural Quality Steel       |
| CSA G40.21     | Grade 350W Class H                                                                                |
| CSA G164       | Hot dip galvanizing of irregularly shaped articles                                                |
| CAN/ULC-S701   | Standard for Calcium Silicate Block and Pipe Thermal Insulation                                   |
| CAN/ULC-S109   | Flame Tests of Flame-Resistant Fabrics and Films                                                  |
| CAN/ULC-S110   | (2003) Fibrous Glass Duct Construction                                                            |
| CAN/ULC-S111   | Standard Method of Fire Tests for Air Filter Units                                                |
| CAN/ULC-S112   | (2018) Laboratory Methods of Testing                                                              |
| CAN/ULC-S112.1 | (2014; Reprint Oct 2020) UL Standard for Safety Smoke Dampers                                     |
| CAN/ULC-S112.2 | (2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units     |
| CAN/ULC-S113   | Fans and ventilators                                                                              |
| CAN/ULC-S702   | Mineral fiber thermal insulation                                                                  |
| CSA Z246.1     | Scheme for the Identification of Piping Systems                                                   |
| CEC            | Canadian Electrical Code - Installation and safety requirements                                   |

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

|          |                                                                              |
|----------|------------------------------------------------------------------------------|
| AMCA 201 | (2002; R 2011) Fans and Systems                                              |
| AMCA 210 | (2016) Laboratory Methods of Testing Fans for Aerodynamic Performance Rating |
| AMCA 300 | (2014) Reverberant Room Method for Sound Testing of Fans                     |

AMCA 500-D (2018) Laboratory Methods of Testing  
Dampers for Rating

AMCA 511 (2010; R 2016) Certified Ratings Program  
for Air Control Devices

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 350 (2015; R 2021) Sound Rating of Non-Ducted  
Indoor Air-Conditioning Equipment

AHRI 410 (2001; Addendum 1 2002; Addendum 2 2005;  
Addendum 3 2011) Forced-Circulation  
Air-Cooling and Air-Heating Coils

AHRI 430 (2009) Central-Station Air-Handling Units

AHRI 440 (2008) Performance Rating of Room Fan-Coils

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

ABMA 9 (2015) Load Ratings and Fatigue Life for  
Ball Bearings

ABMA 11 (2014; R 2020) Load Ratings and Fatigue  
Life for Roller Bearings

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING  
ENGINEERS (ASHRAE)

ASHRAE 52.2 (2017) Method of Testing General  
Ventilation Air-Cleaning Devices for  
Removal Efficiency by Particle Size

ASHRAE 62.1 (2016) Ventilation for Acceptable Indoor  
Air Quality

ASHRAE 70 (2023) Method of Testing the Performance  
of Air Outlets and Inlets

ASHRAE 90.1 - SI (2019) Energy Standard for Buildings  
Except Low-Rise Residential Buildings

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A13.1 (2023) Scheme for the Identification of  
Piping Systems

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2024) Standard Specification for Zinc  
(Hot-Dip Galvanized) Coatings on Iron and  
Steel Products

ASTM A240/A240M (2025) Standard Specification for Chromium  
and Chromium-Nickel Stainless Steel Plate,  
Sheet, and Strip for Pressure Vessels and  
for General Applications



|                 |                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A924/A924M | (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process                                                                                                 |
| ASTM B117       | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus                                                                                                                                               |
| ASTM B152/B152M | (2019) Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar                                                                                                                                    |
| ASTM B209/B209M | (2021a) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                                                                                                                  |
| ASTM B766       | (2023) Standard Specification for Electrodeposited Coatings of Cadmium                                                                                                                                          |
| ASTM C534/C534M | (2024) Standard Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form                                                                                          |
| ASTM C553       | (2024) Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications                                                                                           |
| ASTM C1071      | (2019) Standard Specification for Fibrous Glass Duct Lining Insulation (Thermal and Sound Absorbing Material)                                                                                                   |
| ASTM D520       | (2000; R 2019) Zinc Dust Pigment                                                                                                                                                                                |
| ASTM D1654      | (2008; R 2016; E 2017) Standard Test Method for Evaluation of Painted or Coated Specimens Subjected to Corrosive Environments                                                                                   |
| ASTM D1785      | (2021) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120                                                                                                           |
| ASTM D2466      | (2023) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40                                                                                                                 |
| ASTM D2564      | (2020) Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems                                                                                                         |
| ASTM D2855      | (2020) Standard Practice for the Two-Step (Primer and Solvent Cement) Method of Joining Poly (Vinyl Chloride) (PVC) or Chlorinated Poly (Vinyl Chloride) (CPVC) Pipe and Piping Components with Tapered Sockets |
| ASTM D3359      | (2017) Standard Test Methods for Rating Adhesion by Tape Test                                                                                                                                                   |
| ASTM E2016      | (2022) Standard Specification for                                                                                                                                                                               |

## Industrial Woven Wire Cloth

## CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers

## INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60034-30-1 (2014) Rotating Electrical Machines - Part 30-1: Efficiency Classes of Line Operated AC Motors (IE Code)

## INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 21940-11 (2016) Mechanical vibration -- Rotor balancing -- Part 11: Procedures and Tolerances for Rotors with Rigid Behavior

## NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

NEMA MG 10 (2017) Energy Management Guide for Selection and Use of Fixed Frequency Medium AC Squirrel-Cage Polyphase Induction Motors

NEMA MG 10009 (2022) Energy Management Guide for Selection and Use of Single-Phase Motors

## NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NFPA 80 (2025) Standard for Fire Doors and Other Opening Protectives

NFPA 90A (2024) Standard for the Installation of Air Conditioning and Ventilating Systems

NFPA 90B (2024) Standard for the Installation of Warm Air Heating and Air Conditioning Systems

NFPA 701 (2023; ERTA 1 2023) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

## SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1819 (2022) Fire, Smoke and Radiation Damper Installation Guide for HVAC Systems, 5th Edition

|                                                                    |                                                                                               |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| SMACNA 1884                                                        | (2003) Fibrous Glass Duct Construction Standards, 7th Edition                                 |
| SMACNA 1966                                                        | (2020) HVAC Duct Construction Standards Metal and Flexible, 4th Edition                       |
| SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)               |                                                                                               |
| SCAQMD Rule 1168                                                   | (2022) Adhesive and Sealant Applications                                                      |
| U.S. DEPARTMENT OF ENERGY FEDERAL ENERGY MANAGEMENT PROGRAM (FEMP) |                                                                                               |
| PL-109-58                                                          | (1992; R 2005) Energy Efficient Procurement Requirements                                      |
| U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)           |                                                                                               |
| 40 CFR 82                                                          | Protection of Stratospheric Ozone                                                             |
| UNDERWRITERS LABORATORIES (UL)                                     |                                                                                               |
| UL 6                                                               | (2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel                            |
| UL 181                                                             | (2013; Reprint Dec 2021) UL Standard for Safety Factory-Made Air Ducts and Air Connectors     |
| UL 555                                                             | (2006; Reprint Aug 2016) UL Standard for Safety Fire Dampers                                  |
| UL 555S                                                            | (2014; Reprint Oct 2020) UL Standard for Safety Smoke Dampers                                 |
| UL 586                                                             | (2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units |
| UL 705                                                             | (2017; Reprint Sep 2024) UL Standard for Safety Power Ventilators                             |
| UL 900                                                             | (2015; Reprint Aug 2022) UL Standard for Safety Standard for Air Filter Units                 |
| UL 1995                                                            | (2015; Reprint Aug 2022) UL Standard for Safety Heating and Cooling Equipment                 |
| UL Bld Mat Dir                                                     | (updated continuously online) Building Materials Directory                                    |
| UL Fire Resistance                                                 | (2014) Fire Resistance Directory                                                              |

## 1.2 SYSTEM DESCRIPTION

Furnish ductwork, piping offsets, fittings, and accessories as required to provide a complete installation. Coordinate the work of the different trades to avoid interference between piping, equipment, structural, and electrical work. Provide complete, in place, all necessary offsets in piping and ductwork, and all fittings, and other components, required to

install the work as indicated and specified.

#### 1.2.1 Mechanical Equipment Identification

The number of charts and diagrams must be equal to or greater than the number of mechanical equipment rooms. Where more than one chart or diagram per space is required, mount these in edge pivoted, swinging leaf, extruded aluminum frame holders which open to 170 degrees.

##### 1.2.1.1 Charts

Provide chart listing of equipment by designation numbers and capacities such as flow rates, pressure and temperature differences, heating and cooling capacities, horsepower, pipe sizes, and voltage and current characteristics.

#### 1.2.2 Service Labeling

##### 1.2.2.1 Equipment Labeling

Label equipment as indicated below with labels made of self-sticking, plastic film designed for permanent installation. Provide labels in accordance with the typical examples below:

| SERVICE                      | LABEL AND TAG DESIGNATION |
|------------------------------|---------------------------|
| Dedicated Outdoor Air System | DOAS-1                    |
| Control and instrument air   | CONTROL AND INSTR.        |
| Exhaust Fan Number           | EF - 1                    |
|                              |                           |
| Fan Coil Unit Number         | FCU-1                     |
|                              |                           |
|                              |                           |

##### 1.2.2.2 Pipe Labeling

Identify similar services with different temperatures or pressures. Where pressures could exceed 860 kilopascal, include the maximum system pressure in the label. Color coding of all piping systems must be in accordance with ASME A13.1 CSA Z246.1. Label and arrow piping in accordance with the following:

- Each point of entry and exit of pipe passing through walls.
- Each change in direction, i.e., elbows, tees.
- In congested or hidden areas and at all access panels at each point required to clarify service or indicated hazard.
- In long straight runs, locate labels at distances within eyesight of each other not to exceed 22 meter. All labels must be visible and legible from the primary service and operating area.

e.

| For Bare or Insulated Pipes |           |
|-----------------------------|-----------|
| for Outside Diameters of    | Lettering |
| 13 thru 35 mm               | 13 mm     |
| 40 thru 60 mm               | 20 mm     |
| 65 mm and larger            | 30 mm     |

### 1.2.2.3 Ductwork Labeling

Label ductwork with 2 inch lettering indicating the type of ductwork, associated air handling device and arrow indicating direction of airflow. Coordinate ductwork labeling with any duct insulation and painting requirements to assure labeling is visible on final installation. Duct labels are to be black text on white background, produced by either paint/stenciling or made of self-sticking, plastic film designed for permanent installation. Locations of ductwork labels are to be in accordance with the following:

- a. Each point of entry and exit of ductwork passing through walls.
- b. In congested or hidden areas and at all access panels at each point required to clarify service.
- c. In long straight runs, locate labels at distances within eyesight of each other not to exceed 22 meter.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Detail Drawings; G

#### SD-03 Product Data

Insulated Nonmetallic Flexible Duct Runouts

Pre Manufactured Double Wall Ductwork

Duct Connectors

Duct Access Doors; G

Fire Dampers; G

Manual Balancing Dampers; G

Gravity Backdraft and Relief Dampers; G

Automatic Smoke-Fire Dampers; G

Automatic Smoke Dampers; G

Acoustical Duct Liner

Diffusers

Registers and Grilles

Louvers

Centrifugal Fans

In-Line Centrifugal Fans

Ceiling Exhaust Fans

PL-109-58 label for ceiling exhaust fan product; S

Air Handling Units; G

Room Fan-Coil Units; G

Energy Recovery Devices; G

Test Procedures

Indoor Air Quality for Duct Sealants; S

#### SD-06 Test Reports

Performance Tests; G

Damper Acceptance Test; G

#### SD-07 Certificates

Ozone Depleting Substances Technician Certification

#### SD-08 Manufacturer's Instructions

Manufacturer's Installation Instructions

Operation and Maintenance Training

#### SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

Fire Dampers; G

Manual Balancing Dampers; G

Automatic Smoke-Fire Dampers; G

Automatic Smoke Dampers; G

Centrifugal Fans; G

In-Line Centrifugal Fans; G

Ceiling Exhaust Fans; G

Air Handling Units; G

Room Fan-Coil Units; G

Energy Recovery Devices; G

#### SD-11 Closeout Submittals

Indoor Air Quality During Construction; S

### 1.4 QUALITY ASSURANCE

Except as otherwise specified, approval of materials and equipment is based on manufacturer's published data.

- a. Where materials and equipment are specified to conform to the standards of the Underwriters Laboratories, the label of or listing with reexamination in UL Bld Mat Dir SA Group Certification Listings UL Listed Products Database, and UL 6 CSA C22.2 No. 45.1-13 is acceptable as sufficient evidence that the items conform to Underwriters Laboratories requirements. In lieu of such label or listing, submit a written certificate from any nationally recognized testing agency, adequately equipped and competent to perform such services, stating that the items have been tested and that the units conform to the specified requirements. Outline methods of testing used by the specified agencies.
- b. Where materials or equipment are specified to be constructed or tested, or both, in accordance with the standards of the ASTM International (ASTM), CSA, the ASME International (ASME), or other standards, a manufacturer's certificate of compliance of each item is acceptable as proof of compliance.
- c. Conformance to such agency requirements does not relieve the item from compliance with other requirements of these specifications.
- d. Where products are specified to meet or exceed the specified energy efficiency requirement of FEMP-designated or ENERGY STAR covered product categories, equipment selected must have as a minimum the efficiency rating identified under "Energy-Efficient Products" at <http://femp.energy.gov/procurement>.

#### 1.4.1 Prevention of Corrosion

Protect metallic materials against corrosion. Provide rust-inhibiting treatment and standard finish for the equipment enclosures. Do not use aluminum in contact with earth, and where connected to dissimilar metal. Protect aluminum by approved fittings, barrier material, or treatment. Provide hot-dip galvanized ferrous parts such as anchors, bolts, braces, boxes, bodies, clamps, fittings, guards, nuts, pins, rods, shims, thimbles, washers, and miscellaneous parts not of corrosion-resistant steel or nonferrous materials in accordance with ASTM A123/A123M CSA G164 for exterior locations and cadmium-plated in conformance with ASTM B766 CSA G164 for interior locations.

#### 1.4.2 Asbestos Prohibition

Do not use asbestos and asbestos-containing products.

#### 1.4.3 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be certified as a Section 608 Technician to meet requirements in 40 CFR 82 Environmental Code of Practice for Elimination of Fluorocarbons Emissions from Refrigeration and Air Conditioning Systems (published by Environment and Climate Change Canada, Subpart F. Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

#### 1.4.4 Detail Drawings

Submit detail drawings showing equipment layout, including assembly and installation details and electrical connection diagrams; ductwork layout showing the location of all supports and hangers, typical hanger details, gauge reinforcement, reinforcement spacing rigidity classification, and static pressure and seal classifications. Include any information required to demonstrate that the system has been coordinated and functions properly as a unit on the drawings and show equipment relationship to other parts of the work, including clearances required for operation and maintenance. Submit drawings showing bolt-setting information, and foundation bolts prior to concrete foundation construction for all equipment indicated or required to have concrete foundations. Submit function designation of the equipment and any other requirements specified throughout this Section with the shop drawings.

#### 1.4.5 Test Procedures

Conduct performance tests as required in Section 23 05 93 Testing, Adjusting and Balancing for HVAC and Section 23 09 00 Instrumentation and Control for HVAC.

### 1.5 DELIVERY, STORAGE, AND HANDLING

Protect stored equipment at the jobsite from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Additionally, cap or plug all pipes until installed.

## PART 2 PRODUCTS

### 2.1 STANDARD PRODUCTS

Except for the fabricated duct, plenums and casings specified in paragraphs "Metal Ductwork" and "Plenums and Casings for Field-Fabricated Units", provide components and equipment that are standard products of manufacturers regularly engaged in the manufacturing of products that are of a similar material, design and workmanship. This requirement applies to all equipment, including diffusers, registers, fire dampers, and balancing dampers.

- a. Standard products are defined as components and equipment that have been in satisfactory commercial or industrial use in similar applications of similar size for at least two years before bid opening.



- b. Prior to this two year period, these standard products must have been sold on the commercial market using advertisements in manufacturers' catalogs or brochures. These manufacturers' catalogs, or brochures must have been copyrighted documents or have been identified with a manufacturer's document number.
- c. Provide equipment items that are supported by a service organization. In product categories covered by ENERGY STAR or the Federal Energy Management Program, provide equipment that is listed on the ENERGY STAR Qualified Products List or that meets or exceeds the FEMP-designated Efficiency Requirements.

## 2.2 IDENTIFICATION PLATES

In addition to standard manufacturer's identification plates, provide engraved laminated phenolic identification plates for each piece of mechanical equipment. Identification plates are to designate the function of the equipment. Submit designation with the shop drawings. Provide identification plates that are layers, black-white-black, engraved to show white letters on black background. Letters must be upper case. Identification plates 40 mm that are high and smaller must be 1.6 mm thick, with engraved lettering 3 mm high; identification plates larger than 40 mm high must be 3 mm thick, with engraved lettering of suitable height. Identification plates 40 mm high and larger must have beveled edges. Install identification plates using a compatible adhesive.

## 2.3 EQUIPMENT GUARDS AND ACCESS

Fully enclose or guard belts, pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts exposed to personnel contact according to OSHA requirements. Properly guard or cover with insulation of a type specified, high temperature equipment and piping exposed to contact by personnel or where it creates a potential fire hazard.

## 2.4 ELECTRICAL WORK

- a. Provide motors, controllers, integral disconnects, contactors, and controls with their respective pieces of equipment, except controllers indicated as part of motor control centers. Provide manual or automatic control and protective or signal devices required for the operation specified and control wiring required for controls and devices specified, but not shown. For packaged equipment, include manufacturer provided controllers with the required monitors and timed restart.
- b. For single-phase motors, provide high-efficiency type, including motors that are part of a system, in accordance with NEMA MG 10009 and NEMA MG 1 CSA C390. Provide premium efficiency type for alternating current motors in accordance with NEMA MG 1 CSA C390.
- c. For polyphase motors, provide either ECM or squirrel-cage medium induction motors. Utilize the best equipment for the given mechanical equipment characteristics, including motors that are part of a system, and that meet the efficiency ratings for premium efficiency motors in accordance with NEMA MG 1 CSA C390 or NEMA MG 10009 CSA C22.2 No. 100. Select premium efficiency polyphase motors in accordance with NEMA MG 10 CSA C390.

- d. Provide motors in accordance with NEMA MG 1 CSA C390 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor. Provide motors rated for continuous duty with the enclosure specified. Provide motor duty that allows for maximum frequency start-stop operation and minimum encountered interval between start and stop. Provide motor torque capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Provide motor starters complete with thermal overload protection and other necessary appurtenances. Fit motor bearings with grease supply fittings and grease relief to outside of the enclosure.
- e. Where two-speed or variable-speed motors are indicated, solid-state variable-speed controllers are allowed to accomplish the same function. Use solid-state variable-speed controllers for motors rated 7.45 kW or less and adjustable frequency drives for larger motors.

#### 2.4.1 Electronically Commutated Motors

Provide ECM Motors of a sufficient size for the duty to be performed while not exceeding the full-load rating when the driven equipment is operating at specified capacity under the most severe loading conditions.

##### 2.4.1.1 Design Requirements

- a. Provide Electronically Commutated Motors (ECM) designed as a DC brushless motor for variable speed or constant speed motor control. ECM design must allow for factory setting of the unit's fan flow. ECM must maintain respective constant fan speed at all set points regardless of external pressures that are within design ranges. Motor must be permanently lubricated with ball bearings with an L-10 rated life of not less than standards in accordance with ABMA 9 or ABMA 11 CSA B59.
- b. ECM will be complete with and be operated by a single-phase or three-phase integrated controller/inverter. AC power will be rectified, filtered and then converted to DC power. This will ensure variable speed operation at DC power.
- c. ECM controller must be capable of accepting a 0-10VDC or a 4-20mA signal for manual/automatic control.
- d. Ensure efficiency is a minimum of 70 percent over the motors entire operating range.

##### 2.4.1.2 Electrically Driven Equipment

When electrically driven equipment differs from that indicated, make adjustments to the motor size, wiring and conduit systems, disconnect devices, and circuit protection to accommodate the equipment installed, at no additional cost to the Government. Provide control and protective devices in accordance with NFPA 70, CEC, CSA C22.1 requirements.

##### 2.4.1.3 Voltage Ratings

Provide at a minimum, motors with the following voltage ratings:

| MOTOR SIZE                                |                       | FED FROM<br>POWER SOURCE                                              | MOTOR<br>VOLTAGE<br>RATING                                             |
|-------------------------------------------|-----------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------|
| MOTOR TYPE                                | HORSE<br>POWER        |                                                                       |                                                                        |
| Fractional<br>horsepower,<br>single-phase | 3/4<br>and<br>smaller | 120/208-volt,<br>120/240-volt,<br>480/277-volt,<br>3-phase,<br>4-wire | 115-volt,<br>200-volt,<br>230-volt,<br>277-volt 1-phase<br>50/60-hertz |
| Integer<br>horsepower,<br>single-phase    | 1                     | 120/208-volt,<br>120/240-volt,<br>277/480-volt,<br>3-phase,<br>4-wire | 115-volt,<br>200-volt,<br>230-volt,<br>277-volt 1-phase<br>50/60-hertz |
| Integer<br>horsepower,<br>single-phase    | 2 and 3               | 120/208-volt,<br>120/240-volt,<br>3-phase,<br>4-wire                  | 200-volt,<br>230-volt,<br>1-phase 50/60-hertz                          |
| Integer<br>horsepower,<br>three-phase     | 1 thru 5              | 120/208-volt,<br>120/240-volt,<br>277/480-volt,<br>3-phase,<br>4-wire | 200-volt,<br>230-volt,<br>460-volt, 3-phase<br>50/60-hertz             |
| Integer<br>horsepower,<br>three-phase     | 7.5 and above         | 277/480-volt,<br>3-phase,<br>4-wire                                   | 460-volt, 3-phase<br>50/60-hertz                                       |

#### 2.4.1.4 Temperature Rating and Insulation

Provide motors designed for continuous operation at the rated full load in an ambient temperature of 40 degrees C, 104 degrees F, with an insulation level of at least Class B.

#### 2.4.1.5 Motor Enclosures

##### 2.4.1.5.1 Indoor Type Enclosures

- a. For motors installed in indoor, clean, dry, non-hazardous locations, provide the following: Open-type drip-proof enclosures.
- b. For motors installed in indoor, wet, non-hazardous locations, provide the following: Open splash-proof enclosures.
- c. For motors installed in indoor, non-hazardous locations where it is necessary to protect the motor from dirt, moisture, chemical fumes, or other harmful ingredients in the surrounding atmosphere, provide either of the following type of enclosure: Totally enclosed fan-cooled enclosures for exterior cooling by means of a fan or fans integral with the machine.

#### 2.5 ANCHOR BOLTS

Provide anchor bolts for equipment placed on concrete equipment pads or on concrete slabs. Bolts to be of the size and number recommended by the

equipment manufacturer and located by means of suitable templates.  
Installation of anchor bolts must not degrade the surrounding concrete.

## 2.6 PAINTING

Paint equipment units in accordance with approved equipment manufacturer's standards unless specified otherwise. Field retouch only if approved. Otherwise, return equipment to the factory for refinishing. Paint in accordance with Section 09 96 00 HIGH-PERFORMANCE COATINGS.

## 2.7 FACTORY PAINTING

Factory paint new equipment, which are not of galvanized construction. Paint with a corrosion resisting paint finish according to ASTM A123/A123M CSA G164 or ASTM A924/A924M CSA G40.20/CSA G40.21. Clean, phosphatize and coat internal and external ferrous metal surfaces with a paint finish which has been tested according to ASTM B117, ASTM D1654 CAN/CGSB-1.189, and ASTM D3359 CAN/CGSB-1.53. Submit evidence of satisfactory paint performance for a minimum of 125 hours for units to be installed indoors and 500 hours for units to be installed outdoors. Provide rating of failure at the scribe mark that is not less than 6, average creepage not greater than 3 mm. Provide rating of the inscribed area that is not less than 10, no failure. On units constructed of galvanized steel that have been welded, provide a final shop docket of zinc-rich protective paint on exterior surfaces of welds or welds that have burned through from the interior according to ASTM D520 Type I CAN/CGSB-1.181.

Field paint factory painting that has been damaged prior to acceptance by the Contracting Officer in compliance with the requirements of paragraph FIELD PAINTING OF MECHANICAL EQUIPMENT.

## 2.8 CORROSION PROTECTION FOR COASTAL INSTALLATIONS

Provide air handling coils with aluminum fins on copper tubes or micro channel coils or otherwise custom treated for corrosion protection. Coils on all outdoor equipment of 35 kW capacity or greater must pass ASTM B117 3000 hour salt spray test as installed. For equipment installed within 300 meters of the ocean corrosion resistance must pass ASTM B117 6000 hour salt spray test as installed.

Heat transfer ratings of equipment must be indicated within the equipment submittals with consideration for the deratings associated with the corrosion protection systems as installed.

## 2.9 INDOOR AIR QUALITY

Provide equipment and components that comply with the requirements of ASHRAE 62.1 unless more stringent requirements are specified herein.

## 2.10 DUCT SYSTEMS

### 2.10.1 Metal Ductwork

Provide metal ductwork construction, including all fittings and components, that complies with SMACNA 1966 CSA B52-13, as supplemented and modified by this specification .

- a. Construct ductwork meeting the requirements for the duct system static pressure as indicated on the Ductwork Construction and Leakage Testing

Table provided on the drawings.

- b. Provide radius type elbows with a centerline radius of 1.5 times the width or diameter of the duct where space permits. Otherwise, elbows having a minimum radius equal to the width or diameter of the duct or square elbows with factory fabricated turning vanes are allowed.
- c. Provide branch duct supply take-offs and return connections to main ductwork using 45 degree branch connections or conical tap connections as indicated on the drawings.
- d. Provide ductwork that meets the requirements of Seal Class A.
- e. Provide sealants that conform to fire hazard classification specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS and are suitable for the range of air distribution and ambient temperatures to which it is exposed. Do not use pressure sensitive tape as a sealant. Provide duct sealant products that meet either emissions requirements of CDPH SECTION 01350 CAN/CSA ISO 16000-9 (limit requirements for either office or classroom spaces regardless of space type) or Volatile Organic Compounds (VOC) content requirements of SCAQMD Rule 1168 CAN/CGSB-37.54-95 and CAN/CGSB-19.13-M87 (HVAC duct sealants are classified as "Other" within the CAN/CGSB-37.54-95 and CAN/CGSB-19.13-M87 sealants table). The VOC emittance must be 420 g/l or less. Provide validation of indoor air quality for duct sealants.
- f. Make spiral lock seam duct, and flat oval with duct sealant and lock with not less than 3 equally spaced drive screws or other approved methods indicated in SMACNA 1966 CSA B52-13. Apply the sealant to the exposed male part of the fitting collar so that the sealer is on the inside of the joint and fully protected by the metal of the duct fitting. Apply one brush coat of the sealant over the outside of the joint to at least 50 mm band width covering all screw heads and joint gap. Dents in the male portion of the slip fitting collar are not acceptable.
- g. Fabricate outdoor air intake ducts and plenums with watertight soldered or brazed joints and seams. .

#### 2.10.1.1 Insulated Nonmetallic Flexible Duct Runouts

Use flexible duct runouts only where indicated. Runout length is indicated on the drawings, and is not to exceed 1.5 m. Provide runouts that are preinsulated, factory fabricated, and that comply with NFPA 90A CSA C22.1 and UL 181 CAN/ULC-S110. Provide either field or factory applied vapor barrier. Provide not less than 0.60 kg/square meter glass fabric duct connectors coated on both sides with neoprene. Where high velocity units are supplied with vertical air inlets, use a streamlined, vaned and mitered elbow transition piece for connection to the flexible duct or hose. Provide a die-stamped elbow and not a flexible connector as the last elbow to these units other than the vertical air inlet type. Insulated flexible connectors are allowed as runouts. Provide insulated material and vapor barrier that conform to the requirements of Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Do not expose the insulation material surface to the air stream.

#### 2.10.1.2 General Service Duct Connectors

Provide a flexible duct connector approximately 150 mm in width where sheet metal connections are made to fans or where ducts of dissimilar metals are connected. For round/oval ducts, secure the flexible material by stainless steel or zinc-coated, iron clinch-type draw bands. For rectangular ducts, install the flexible material locked to metal collars using normal duct construction methods. Provide a composite connector system that complies with NFPA 701 CAN/ULC-S109 and is classified as "flame-retardant fabrics" in UL Bld Mat Dir CSA Group Certification Listings UL Listed Products Database.

#### 2.10.1.3 High Temperature Service Duct Connections

Provide material that is approximately 2.38 mm thick, 1.2 to 1.36 kg per square meter weight, plain weave fibrous glass cloth with, nickel/chrome wire reinforcement for service in excess of 650 degrees C.

#### 2.10.1.4 Aluminum Ducts

ASTM B209/B209M CSA G40.20/CSA G40.21 3003 or CSA G40.20/CSA G40.21, alloy 3003-H14 for aluminum sheet and alloy 6061-T6 or equivalent strength for aluminum connectors and bar stock.

#### 2.10.1.5 Copper Sheets

ASTM B152/B152M, light cold rolled temper.

#### 2.10.1.6 Corrosion Resisting (Stainless) Steel Sheets

ASTM A240/A240M CSA G30.18, CSA G40.20/CSA G40.21

#### 2.10.2 Duct Access Doors

Provide hinged access doors conforming to SMACNA 1966 CSA B52-13 in ductwork and plenums where indicated and at all air flow measuring primaries, automatic dampers, fire dampers, coils, thermostats, and other apparatus requiring service and inspection in the duct system. Provide access doors upstream and downstream of air flow measuring primaries and heating and cooling coils. Provide doors that are a minimum 375 by 450 mm, unless otherwise shown. Where duct size does not accommodate this size door, make the doors as large as practicable. Equip doors 600 by 600 mm or larger with fasteners operable from inside and outside the duct. Use insulated type doors in insulated ducts.

#### 2.10.3 Fire Dampers

Use 2 hour rated fire dampers unless otherwise indicated. Provide fire dampers that conform to the requirements of NFPA 90A CAN/CSA C22.2 No. 113, NBCan and UL 555 CAN/ULC-S112. Perform the fire damper test as outlined in NFPA 90A, CAN/CSA C22.2 No. 113, NBC. Provide automatic operating fire dampers with a dynamic rating suitable for the maximum air velocity and pressure differential to which it is subjected. Provide fire dampers approved for the specific application, and install according to their listing. Equip fire dampers with a steel sleeve or adequately sized frame installed in such a manner that disruption of the attached ductwork, if any, does not impair the operation of the damper. Equip sleeves or frames with perimeter mounting angles attached on both sides of the wall or floor opening. Construct ductwork in fire-rated floor-ceiling or roof-ceiling

assembly systems with air ducts that pierce the ceiling of the assemblies in conformance with UL Fire Resistance. Provide curtain type with damper blades out of the air stream fire dampers. Install dampers that do not reduce the duct or the air transfer opening cross-sectional area. Install dampers so that the centerline of the damper depth or thickness is located in the centerline of the wall, partition or floor slab depth or thickness. Unless otherwise indicated, comply with the installation details given in SMACNA 1819 NFPA 80 and CAN/ULC-S112.1 and in manufacturer's instructions for fire dampers. Perform acceptance testing of fire dampers according to paragraph DAMPER ACCEPTANCE TEST and NFPA 80 CAN/ULC-S112.

#### 2.10.4 Manual Balancing Dampers

- a. Furnish factory manufactured and assembled manual balancing dampers with accessible locking-type quadrant operators. Manufacturer must publish damper data sheets and dampers must be sold on the open market. Use chromium plated operators (with all exposed edges rounded) in finished portions of the building where ductwork is exposed.
- b. Furnish stand-off mounting brackets, bases, or adapters, not less than the thickness of the insulation, to provide clearance between the duct surface and the operator. Stand-off mounting items are to be integral with the operator or standard accessory of the damper manufacturer. For duct widths greater than 750mm, provide locking regulator at each end of the damper shaft.
- c. Dampers installed in ductwork with an air velocity over 7.5 meters per second, or pressure class greater than 500 pascals must be rated for the design velocity and pressure class indicated.

##### 2.10.4.1 Single Blade Round Dampers

- a. May be furnished up to 500mm in diameter.
- b. Provide a minimum 10mm square continuous full length galvanized or plated steel axle shaft that supports the entire blade diameter and extends through the standoff bracket and quadrant operator. Firmly affix the axle shaft to the blade. Spring-loaded positioners are not permitted.
- c. Must be factory fabricated in a frame with a minimum length of 150mm made of galvanized steel with straightening ribs; minimum of 22 gage up to 200mm diameter, and 20 gage above 200mm diameter. Damper frames are to be mounted end to end with round duct. Supply the axle shaft on both ends with a pressed in elastomeric bearing/seal, for low infiltration/exfiltration air leakage.
- d. Provide galvanized steel damper blade; minimum of 22 gage up to 200mm diameter, and 20 gage above 200mm diameter.

##### 2.10.4.2 Single Blade Rectangular Dampers

- a. May be furnished up to 300mm in height for rectangular ducts.
- b. Provide square galvanized or plated steel axle shaft(s) that support the blade at both ends and extends through the standoff bracket and quadrant operator; minimum 10mm up to 475mm blade length, and 13mm over 475mm blade length. Firmly affix the axle shaft(s) to the blade.

Spring-loaded positioners are not permitted. Provide, at the duct wall shaft penetration, an elastomeric seal for low infiltration/exfiltration air leakage.

- c. Must be factory fabricated in a frame with a minimum length 75mm made of galvanized steel; minimum of 24 gage up to 475mm blade length, and 22 gage over 475mm blade length. Damper frames are to be mounted inside the duct. Support axle shaft on both ends with a bearing in the frame.
- d. Provide damper blades made of galvanized steel with roll formed or broke stiffener(s); minimum of 20 gage up to 475mm blade length, and 18 gage over 475mm blade length.

#### 2.10.4.3 Multi-blade Rectangular Dampers

- a. Furnish for round ducts larger than 500mm and rectangular duct greater than 300mm in height, except flanged duct system. Furnish square to round transitions for application in round duct.
- b. Provide 13mm galvanized or plated steel axle shafts that support each blade at both ends. Extend the operator shaft through the standoff bracket and quadrant operator. An elastomeric seal for low infiltration/exfiltration air leakage must be at the duct wall shaft penetration.
- c. Must be factory fabricated in a frame with a minimum length of 125mm made of minimum 16 gage galvanized steel. Damper frames are to be mounted inside the duct. Support axle shafts on both end with a bearing in the frame. Blade linkage must be galvanized or plated steel and concealed in the frame, out of the air stream.
- d. Provide opposed blades made of galvanized steel with roll formed or broke stiffeners(s); minimum of 16 gage.

#### 2.10.4.4 Rectangular Dampers for Flanged Duct

- a. Provide for flanged duct systems.
- b. Must be multi-blade rectangular.
- c. Provide 13mm galvanized or plated steel axle shafts that support each blade at both ends. Extend the operator shaft through the standoff bracket and quadrant operator.
- d. Must be factory fabricated in a frame with a minimum length of 75mm made galvanized steel with a thickness to match the duct and at least 16 gage, or 20 gage with additional reinforcing such as a C channel. Mount damper frames end to end with the duct. Support axle shafts on both ends with a pressed in elastomeric bearing/seal for low infiltration/exfiltration air leakage. Provide galvanized or plated steel blade linkage, concealed in a frame, out of the air stream.
- e. Provide opposed blades made of galvanized steel with roll formed or broker stiffeners(s); minimum of 16 gage.

#### 2.10.5 Automatic Balancing Dampers

Provide pressure independent automatic balancing dampers at locations as



shown on the plans or in accordance with drawing schedules. Automatic balancing dampers must operate without a required power supply and be field adjustable to design airflows without the use of tools using a slide setpoint indicator on the face of the damper. Automatic balancing dampers must be rated for operation with air temperatures ranging from -4 degrees to 60 degrees C 25 degrees to 140 degrees F. Automatic air balancing dampers must provide airflow within 10 percent of setpoint at air pressures from 50 to 300 pascal when tested in accordance with CAN/ULC-S112.

#### 2.10.6 Automatic Smoke-Fire Dampers

Multiple blade type with airfoil type blades, 82 degrees C fusible fire damper link; smoke damper assembly to include electric damper operator. UL 555 CAN/ULC-S112 as a 2 hour rated fire damper unless otherwise indicated; further qualified under UL 555S CAN/ULC-S112.2 as a leakage rated damper. Provide a leakage rating under UL 555S CAN/ULC-S112.2 that is no higher than Class II or III at an elevated temperature Category B ( 121 degrees C for 30 minutes).

#### 2.10.7 Automatic Smoke Dampers

UL listed multiple blade type with airfoil type blades, supplied by smoke damper manufacturer, with pneumatic electric damper operator as part of assembly. Qualified under UL 555S CAN/ULC-S112.2 with a leakage rating no higher than Class II or III at an elevated temperature Category B ( 121 degrees C for 30 minutes ).

#### 2.10.8 Gravity Backdraft and Relief Dampers

Provide gravity balanced backdraft dampers where indicated on plans or as required on equipment schedules. Backdraft dampers must be constructed for horizontal or vertical mounting as required for the application. Backdraft dampers must be testing and rated for the required application in accordance with AMCA 500-D CAN/ULC-S112.

Damper blades are to be equipped with neoprene blade seals mechanically locked into the blade edge.

Equip with adjustable counterbalancing weights to allow operation of the blades such that "start to open" occurs at 5 pascals and the damper achieves fully open position at 0.05" w.g.

Where backdraft dampers are provided as an accessory option to a fan, such backdraft dampers must be installed meeting all manufacturer's recommendations. The installer must demonstrate that the damper closes once the fan ceases operation.

#### 2.10.9 Plenums and Casings for Field-Fabricated Units

##### 2.10.9.1 Plenum and Casings

Fabricate and erect plenums and casings as shown in SMACNA 1966, CSA B52-13, as applicable. Construct system casing of not less than 1.6 mm galvanized sheet steel. Furnish cooling coil drain pans with 25 mm threaded outlet to collect condensation from the cooling coils. Fabricate drain pans from not lighter than 1.6 mm steel, galvanized after fabrication or of 1.3 mm corrosion-resisting sheet steel conforming to ASTM A240/A240M, Type 304, CSA G30.18 welded and stiffened. Thermally insulate drain pans exposed to the atmosphere to prevent condensation. Coat insulation with a flame

resistant waterproofing material. Provide separate drain pans for each vertical coil section, and a separate drain line for each pan. Size pans to ensure capture of entrained moisture on the downstream-air side of the coil. Seal openings in the casing, such as for piping connections, to prevent air leakage. Size the water seal for the drain to maintain a pressure of at least 500 Pa greater than the maximum negative pressure in the coil space.

#### 2.10.9.2 Casing

Terminate casings at the curb line and bolt each to the curb using galvanized angle, as indicated in SMACNA 1966 CSA B52-13.

#### 2.10.9.3 Access Doors

Provide access doors in each section of the casing. Weld doorframes in place, gasket each door with neoprene, hinge with minimum of two brass hinges, and fasten with a minimum of two brass tension fasteners operable from inside and outside of the casing. Where possible, make doors 900 by 450 mm and locate them 450 mm above the floor. Where the space available does not accommodate doors of this size, use doors as large as the space accommodates. Swing doors so that fan suction or pressure holds doors in closed position, airtight. Provide a push-button station, located inside the casing, to stop the supply.

#### 2.10.9.4 Factory-Fabricated Insulated Sheet Metal Panels

Factory-fabricated components are allowed for field-assembled units, provided all requirements specified for field-fabricated plenums and casings are met. Provide panels of modular design, pretested for structural strength, thermal control, condensation control, and acoustical control. Seal and insulate panel joints. Provide and gasket access doors to prevent air leakage. Provide panel construction that is not less than one mm galvanized sheet steel, assembled with fasteners treated against corrosion. Provide standard length panels that deflect not more than 13 mm under operation. Construct details, including joint sealing, not specifically covered, as indicated in SMACNA 1966 CSA B52-13. Construct the plenums and casings to withstand the specified internal pressure of the air systems.

#### 2.10.9.5 Duct Liner

Unless otherwise specified, duct liner is not permitted.

#### 2.10.10 Acoustical Duct Liner

Use fibrous glass designed or flexible elastomeric duct liner for lining ductwork and conforming to the requirements of ASTM C1071, Type I and II CAN/ULC-S702. Provide uniform density, graduated density, or dual density liner composition, as standard with the manufacturer. Provide not less than 25 mm thick coated lining. Where acoustical duct liner is used, provide the thermal equivalent of the insulation specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS for liner or combination of liner and insulation applied to the exterior of the ductwork. Increase duct sizes shown to compensate for the thickness of the lining used. In lieu of sheet metal duct with field-applied acoustical lining, provide acoustically equivalent lengths of fibrous glass duct, elastomeric duct liner or factory fabricated double-walled internally insulated duct with perforated liner .

### 2.10.11 Diffusers, Registers, and Grilles

Provide factory-fabricated units of aluminum that distribute the specified quantity of air evenly over space intended without causing noticeable drafts, air movement faster than 0.25 m/s in occupied zone, or dead spots anywhere in the conditioned area. Provide outlets for diffusion, spread, throw, and noise level as required for specified performance. Certify performance according to ASHRAE 70. Provide sound rated and certified inlets and outlets according to ASHRAE 70. Provide units with sound ratings at design conditions not exceeding the ASHRAE Applications Handbook, Noise and Vibration Control Chapter - table "Design Guidelines for HVAC-Related Background Sound in Rooms" refer to drawing air device schedule for maximum permissible sound rating by device. Provide diffusers and registers with volume damper with accessible operator, unless otherwise indicated; or if standard with the manufacturer, an automatically controlled device is acceptable. Where the inlet and outlet openings are located less than 2 m above the floor, protect them by a grille or screen according to NFPA 90A, CAN/CSA C22.2 No. 113, NBC.

#### 2.10.11.1 Diffusers

Provide diffuser types indicated. Furnish ceiling mounted units with anti-smudge devices, unless the diffuser unit minimizes ceiling smudging through design features. Provide diffusers with air deflectors of the type indicated. Install ceiling mounted units with rims tight against ceiling. Provide sponge rubber gaskets between ceiling and surface mounted diffusers for air leakage control. Provide suitable trim for flush mounted diffusers. For connecting the duct to diffuser, provide duct collar that is airtight and does not interfere with volume controller. Provide return or exhaust units that are similar to supply diffusers.

#### 2.10.11.2 Perforated Plate Diffusers

Provide adjustable four-way air pattern controls unless otherwise indicated. Provide diffuser faceplates that do not sag or deflect when operating under design conditions.

#### 2.10.11.3 Registers and Grilles

Provide registers and grilles of the types indicated on the drawings. Furnish registers with sponge-rubber gasket between flanges and wall or ceiling. Install wall supply registers at least 150 mm below the ceiling unless otherwise indicated. Locate return and exhaust registers 150 mm above the floor unless otherwise indicated. Provide grilles as specified for registers, without volume control damper.

#### 2.10.11.4 Registers

Double-deflection supply registers. Provide exhaust and return registers as specified for supply registers, except provide exhaust and return registers that have a single set of nondirectional face bars or vanes having the same appearance as the supply registers. Set face bars or vanes at 35 degrees.

### 2.10.12 Louvers

Provide louvers for installation in exterior walls that are associated

with the air supply and distribution system as specified in Section 07 60 00 FLASHING AND SHEET METAL.

#### 2.10.13 Bird Screens and Frames

Provide bird screens that conform to ASTM E2016, CAN/CSA C22.2 No. 113, NBC, No. 2 mesh, aluminum or stainless steel. Provide "medium-light" rated aluminum screens. Provide "light" rated stainless steel screens. Provide removable type frames fabricated from either stainless steel or extruded aluminum.

#### 2.10.14 Radon Exhaust Ductwork

Fabricate radon exhaust ductwork installed in or beneath slabs from Schedule 40 PVC pipe that conforms to ASTM D1785 CSA B137.3. Provide fittings that conform to ASTM D2466 CSA B137.3. Use solvent cement conforming to ASTM D2564 CSA B137.3 to make joints. Otherwise provide metal radon exhaust ductwork as specified herein.

#### 2.10.15 Pre Manufactured Double Wall Ductwork

For all interior exposed ductwork in finished/occupied spaces provide double wall pre-insulated ductwork with minimum 24 gauge solid galvanized inner wall. Double wall ductwork and fittings are to be from a single manufacturer. Provide ducts with clear internal dimensions of the inner duct equal to the duct dimensions indicated on the plans. Construct the outer duct of galvanized sheet steel, unless otherwise indicated. Double wall duct must comply with SMACNA 1966 CSA B52-13 based on static pressure class as indicated on the drawings, and be sealed in accordance to SMACNA Seal Class A CGSB 51.33-M89, Type 1 requirements.

##### 2.10.15.1 Double Wall Duct Insulation

Insulation thickness as installed must meet or exceed the thermal performance requirements of ASHRAE 90.1 - SI. Interstitial insulation may be one of the following:

- a) Fibrous glass duct liner complying with ASTM C1071 CAN/ULC-S702 and NFPA 90A, CAN/CSA C22.2 No. 113 NBC or NFPA 90B CAN/CSA C22.1-24. Fibrous glass duct liner insulation must be installed in the interstitial space with spacers that position the inner duct at a uniform distance from the outer duct without compressing the insulation. Provide with antimicrobial coating and cover with polyester film complying with UL 181 CGSB 51.33-M89, Type 1.
- b) Flexible elastomeric duct liner complying with ASTM C534/C534M, Type II CAN/ULC-S701 for sheet materials, and with NFPA 90A CAN/CSA C22.2 No. 113, NBC, CAN/CSA C22.1-24.

#### 2.11 AIR SYSTEMS EQUIPMENT

##### 2.11.1 Fans

Test and rate fans according to AMCA 210 CSA C260-16. Calculate system effect on air moving devices in accordance with AMCA 201 CSA or CGSB where installed ductwork differs from that indicated on drawings. Install air moving devices to minimize fan system effect. Where system effect is unavoidable, determine the most effective way to accommodate the inefficiencies caused by system effect on the installed air moving

device. The sound power level of the fans must not exceed 85 dBA when tested according to AMCA 300 CSA or CGSB and rated in accordance with AMCA 300 CSA or CGSB. Provide all fans with an AMCA seal. Provide direct drive fan motors. Provide fan and motor assemblies with vibration-isolation supports or mountings as required by ASHRAE Applications Handbook - Table "Selection Guide for Vibration Isolation. Use vibration-isolation units that are standard products with published loading ratings. Select each fan to produce the capacity required at the fan static pressure indicated. Provide units with sound ratings at design conditions that will not result in objectional background noise in accordance with the ASHRAE Applications Handbook, chapter Noise and Vibration Control, table "Design Guidelines for HVAC-Related Background Sound in Rooms" refer to drawing fan schedule for maximum permissible sound rating by device. Obtain the sound power level values according to AMCA 300 CSA or CGSB. Provide power ventilators that conform to UL 705 CSA C22.2 No. 113 and have a UL CSA label.

#### 2.11.1.1 Centrifugal Fans

Provide fully enclosed, single-width single-inlet, or double-width double-inlet centrifugal fans, with AMCA Pressure Class I, II, or III CGSB 51.34-M86as required or indicated for the design system pressure. Provide impeller wheels that are rigidly constructed and accurately balanced both statically and dynamically. Provide forward curved or backward-inclined airfoil design fan blades in wheel sizes up to 750 mm. Provide backward-inclined airfoil design fan blades for wheels over 750 mm in diameter. Provide fan wheels over 900 mm in diameter with overhung pulleys and a bearing on each side of the wheel. Provide fan wheels 900 mm or less in diameter that have one or more extra long bearings between the fan wheel and the drive. Provide sleeve type, self-aligning and self-oiling bearings with oil reservoirs, or precision self-aligning roller or ball-type with accessible grease fittings or permanently lubricated type. Connect grease fittings to tubing for serviceability from a single accessible point. Provide L50 rated bearing life at not less than 200,000 hours as defined by ABMA 9 and ABMA 11. Provide steel, accurately finished fan shafts, with key seats and keys for impeller hubs and fan pulleys. Provide fan outlets of ample proportions, designed for the attachment of angles and bolts for attaching flexible connections. Unless otherwise indicated, provide motors that do not exceed 1800 rpm and have open drip-proof. Provide magnetic reduced-voltage-start type motor starters with general-purpose enclosure.

#### 2.11.1.2 In-Line Centrifugal Fans

Provide in-line fans with centrifugal backward inclined blades, stationary discharge conversion vanes, internal and external belt guards, and adjustable motor mounts. Mount fans in a welded tubular casing. Provide a fan that axially flows the air in and out. Streamline inlets with conversion vanes to eliminate turbulence and provide smooth discharge air flow. Enclose and isolate fan bearings and drive shafts from the air stream. Provide precision, self aligning ball or roller type fan bearings that are sealed against dust and dirt and are permanently lubricated. Provide L50 rated bearing life at not less than 200,000 hours as defined by ABMA 9 CSA B59. Provide motors with opendrip-proof enclosure. Provide magnetic motor starters across-the-line with general-purpose enclosures.

#### 2.11.1.3 Ceiling Exhaust Fans

Provide centrifugal type, direct driven suspended cabinet-type ceiling

exhaust fans. Provide fans with acoustically insulated housing. Provide chatter-proof backdraft damper. Provide egg-crate design or louver design integral face grille. Mount fan motors on vibration isolators. Furnish unit with mounting flange for hanging unit from above. Provide U.L. listed fans. Provide PL-109-58 labeled ceiling exhaust fan product. Provide proof of PL-109-58 label for ceiling exhaust fan product.

#### 2.11.2 Coils

Provide fin-and-tube type coils constructed of seamless copper tubes and aluminum fins mechanically bonded or soldered to the tubes. Provide copper fins that are 0.114 mm minimum thickness. Provide casing and tube support sheets that are not lighter than 1.6 mm galvanized steel, formed to provide structural strength. When required, provide multiple tube supports to prevent tube sag. Mount coils for counterflow service. Rate and certify coils to meet the requirements of AHRI 410 CSA or CGSB. Provide factory applied phenolic, vinyl or epoxy/electrodeposition coating to all water coil within the DOAS and FCU units and to ductless split system condenser coils.

##### 2.11.2.1 Water Coils

Install water coils with a pitch of not less than 10 mm/m of the tube length toward the drain end. Use headers constructed of cast iron, welded steel or copper. Furnish each coil with a plugged vent and drain connection extending through the unit casing. Provide removable water coils with drain pans. Pressure test coils in accordance with UL 1995 CSA C22.2 No. 236.

##### 2.11.2.2 Electric Heating Coil

Provide an electric duct heater coil in accordance with UL 1995 CSA C22.2 No. 236 and NFPA 70 CEC. Provide duct- or unit-mounted coil. Provide nickel chromium resistor, single stage, strip or stainless steel, fin tubular type coil. Provide coil with a built-in or surface-mounted high-limit thermostat interlocked electrically so that the coil cannot be energized unless the fan is energized. Provide galvanized steel or aluminum coil casing and support brackets. Mount coil to eliminate noise from expansion and contraction and for complete accessibility for service.

##### 2.11.2.3 Eliminators

Equip each cooling coil having an air velocity of over 2 m/s through the net face area with moisture eliminators, unless the coil manufacturer guarantees, over the signature of a responsible company official, that no moisture can be carried beyond the drip pans under actual conditions of operation. Construct of minimum 24 gage zinc-coated steel, copper, copper nickel or stainless steel, removable through the nearest access door in the casing or ductwork. Provide eliminators that have not less than two bends at 45 degrees and are spaced not more than 63 mm center-to-center on face. Provide each bend with an integrally formed hook as indicated in the SMACNA 1884 NBCC 2020.

#### 2.11.3 Air Filters

List air filters according to requirements of UL 900, CAN/ULC-S111 except list high efficiency particulate air filters of 99.97 percent efficiency by the DOP Test method under the Label Service to meet the requirements of UL 586 CAN/ULC-S113.

### 2.11.3.1 Return Air Filter Grilles

Provide return air filter grilles as indicated on the drawing schedule. Return air filter grilles are to be pleated panel filters, 50 mm depth with a minimum MERV rating of 13 .

### 2.11.3.2 Holding Frames

Fabricate frames from not lighter than 1.6 mm sheet steel with rust-inhibitor coating. Equip each holding frame with suitable filter holding devices. Provide gasketed holding frame seats. Make all joints airtight.

### 2.11.3.3 Filter Gauges

Provide dial type filter gauges, diaphragm actuated draft for all filter stations, including those filters which are furnished as integral parts of factory fabricated air handling units. Provide gauges that are at least 98 mm in diameter, with white dials with black figures, and graduated in 0.0025 kPa, with a minimum range of 0.25 kPa beyond the specified final resistance for the filter bank on which each gauge is applied. Provide each gauge with a screw operated zero adjustment and two static pressure taps with integral compression fittings, two molded plastic vent valves, two 1.5 m minimum lengths of 6.35 mm diameter aluminum vinyl tubing, and all hardware and accessories for gauge mounting.

## 2.12 AIR HANDLING UNITS

### 2.12.1 Factory-Fabricated Air Handling Units

Provide draw through type constant volume dedicated outdoor air system (DOAS) units as indicated. Units must include direct drive plenum fans, coils, airtight insulated casing, , access sections where indicated, drain pan, vibration-isolators, and appurtenances required for specified operation. DOAS will be provided with a preheat coil, reheat coil, and fan inlet airflow measuring system with a piezometer ring. Provide additional features and accessories indicated on the drawings. Physical dimensions of each unit must be suitable to fit space allotted to the unit with the capacity indicated. Provide unit that is rated in accordance with AHRI 430 CSA or CGSB and AHRI certified for cooling.

#### 2.12.1.1 Casings

Provide the following:

- a. Casing sections 50 mm double wall type, constructed of a minimum 22 gauge galvanized steel, 0.040 inch aluminum, or 22 gauge stainless steel conforming to ASTM A240/A240M, Type 304 CSA G30.18. Make the inner casing of double wall units a minimum 20 gauge solid galvanized steel . Design and construct casing with an integral insulated structural galvanized steel or aluminum frame such that exterior panels are non-load bearing.

Incorporate a standing seam or thermoplastic polyolefin (TPO) on the exterior of outdoor unit roofs to ensure a rigid roof construction and prevent water infiltration. Roof assembly must be sloped not less than 1:100 for drainage and is to overhang all walls by 40mm minimum to prevent sheeting from roof to side panels. Rain gutters are to

also be provided over all doors shorter than total unit height to direct rain away from door assembly. Where outdoor units are shipped in multiple sections, provide standing-seam joiners at each split with adhesive, hardware, and cover strips for field joining by the installing contractor. All outdoor unit base frames must be welded construction.

Unit casing (wall/floor/roof panels and doors) must be able to withstand up to 1.5 times design static pressure, or 2000 pascals and not exceed 0.0042 per inch of panel span ( $L/240$ ) when measured at the panel midpoint. Air leakage to be guaranteed at no more than 1 percent of the design volume at 1.5 times the design operating pressure or 50 cubic feet per minute, whichever is greater. Specified air leakage accomplished without the use of caulk. Total estimated air leakage reported for each unit in liters per second or as a percentage of supply air.

- b. Casing panel inner liners must not extend to the exterior of the unit or contact the exterior frame. A mid-span, no-through-metal, internal thermal break is provided for all unit casing panels. Individually removable exterior panel sections are removable with standard tools. Removal must not affect the structural integrity of the unit. Furnish casings with access sections, according to paragraph AIR HANDLING UNITS, inspection doors, and access doors must be capable of opening a minimum of 90 degrees.
- c. Insulated, fully gasketed, double-wall type access doors, of a minimum 22 gauge galvanized steel, 0.040 inch aluminum, or 22 gauge stainless steel conforming to ASTM A240/A240M CSA G30.18, CSA G40.20/CSA G40.21. Provide rigid doors with heavy duty hinges and latches. Access doors must be a minimum 600 mm wide, the full height of the unit casing
- d. Double-wall insulated type drain pan (thickness equal to exterior casing) constructed of 0.95 mm stainless steel conforming to ASTM A240/A240M, Type 304, CSA G30.18 conforming to ASHRAE 62.1. Construct drain pans water tight, treated to prevent corrosion, and designed for positive condensate drainage. When 2 or more cooling coils are used, with one stacked above the other, condensate from the upper coils must not flow across the face of lower coils. Provide intermediate drain pans or condensate collection channels and downspouts, as required to carry condensate to the unit drain pan out of the air stream and without moisture carryover. Construct drain pan to allow for easy visual inspection, including underneath the coil without removal of the coil and to allow complete and easy physical cleaning of the pan underneath the coil without removal of the coil. Provide coils that are individually removable from the casing. Drain connections must be of the same material as the primary drain pan and extend a minimum of 60mm beyond the base to ensure adequate room for field piping of condensate traps.
- e. Casing insulation that conforms to CAN/CSA C22.2 No. 113, NBC. Insulate double-wall casing sections handling conditioned air with not less than 50 mm of the same insulation specified for single-wall casings. Foil-faced insulation is not an acceptable substitute for use with double wall casing. Seal double wall insulation completely by inner and outer panels.
- f. Factory applied fibrous glass insulation that conforms to ASTM C1071, CAN/ULC-S702 except that the minimum thickness and density



requirements do not apply, and that meets the requirements of NFPA 90A, CAN/CSA C22.2 No. 113, NBC. Make air handling unit casing insulation uniform over the entire casing. The performance of the air handler design is capable of meeting an ASHRAE 111 Class 6 leakage level..

- g. Duct liner material, coating, and adhesive that conforms to fire-hazard requirements specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Protect exposed insulation edges and joints where insulation panels are butted with a metal nosing strip or coat to meet erosion resistance requirements of ASTM C1071 CAN/ULC-S702 .
- h. A latched and hinged inspection door, in the fan and coil sections. Plus additional inspection doors, access doors and access sections where indicated.

#### 2.12.1.2 Heating and Cooling Coils

Provide coils as specified in paragraph AIR SYSTEMS EQUIPMENT.

#### 2.12.1.3 Air Filters

Provide air filters as specified in paragraph AIR SYSTEMS EQUIPMENT for types and thickness indicated.

#### 2.12.1.4 Fans

Provide the following:

- a. Direct drive plenum fans that are single width, single inlet. Fan blades are aluminum backward-inclined airfoil. Dynamically balance fans and shafts prior to installation into air handling unit, then after it has been installed in the air handling unit, statically and dynamically balance the entire fan assembly. Mount fans on steel shafts, accurately ground and finished.
- b. Fan bearings that are sealed against dust and dirt and are precision self-aligning ball or roller type, with L50 rated bearing life at not less than 200,000 hours as defined by ABMA 9 CSA B59 and ABMA 11. Provide bearings that are permanently lubricated or lubricated type with lubrication fittings readily accessible at the drive side of the unit. Support bearings by structural shapes, or die formed sheet structural members, or support plates securely attached to the unit casing. Do not fasten bearings directly to the unit sheet metal casing. Furnish fans and scrolls with coating indicated.
- c. Fans that are driven by a unit-mounted, or a floor-mounted motor connected to fans by V-belt drive complete with belt guard for externally mounted motors. Furnish belt guards that are the three-sided enclosed type with solid or expanded metal face. Design belt drives for not less than a 1.3 service factor based on motor nameplate rating.
- d. Where fixed sheaves are required, the use of variable pitch sheaves is allowed during air balance, but replace them with an appropriate fixed sheave after air balance is completed. Select variable pitch sheaves to drive the fan at a speed that produces the specified capacity when set at the approximate midpoint of the sheave adjustment. Furnish motors for V-belt drives with adjustable bases,

and with totally enclosed enclosures.

- e. Motor starters of magnetic cross-the-line type with general-purpose enclosure. Select unit fan or fans to produce the required capacity at the fan static pressure with sound power level as indicated. Obtain the sound power level values according to AMCA 300.

#### 2.12.1.5 Fan Arrays

Air handling units utilizing fan arrays are to comply with the following requirements:

- a. Fans require Class II (minimum) construction with single inlet, direct drive ACMA, CSA Arrangement 4 fans installed either in a modular fashion in the field or pre-assembled on a prefabricated frame. Fans must be dynamically balanced and internally isolated to minimize vibration. Fan performance rated in accordance with AMCA 210 CSA C260-16.
- b. Fan array plenum type fans driven by variable speed drives unless otherwise indicated; where indicated fans are driven by brushless DC electronically commutated motor (ECM) with external rotor and integrated maintenance free, permanently lubricated ball bearings and statically and dynamically balanced in accordance with ISO 21940-11. The ECM must be closed, protection level IP 54, thermal class 155; motor efficiency class complies with IEC 60034-30-1 IE3 Premium efficiency.
- c. Fan array controller provided by the air handling unit manufacturer to control the fan array modulating individual fan speed and number of active fans to provide the required airflows with minimum energy consumption.

#### 2.12.1.6 Access Sections and Filter/Mixing Boxes

Provide access sections where indicated and furnish with access doors. Construct access sections and filter/mixing boxes in a manner identical to the remainder of the unit casing and equip with access doors. Design mixing boxes to minimize air stratification and to promote thorough mixing of the air streams.

#### 2.12.1.7 Dampers

Provide dampers integral to factory fabricated air handling units constructed in accordance with 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC. Dampers connecting the air handling unit to the outdoors (outdoor air, exhaust air, relief Air) must be AMCA 511 leakage classification 1A.

### 2.13 TERMINAL UNITS

#### 2.13.1 Room Fan-Coil Units

Provide base units that include galvanized coil casing, coil assembly drain pan valve and piping package, outside air damper, wall intake box, air filter, fans, motor, fan drive, motor switch, an enclosure for cabinet models and casing for concealed models, leveling devices integral with the

unit for vertical type units, and sound power levels as indicated. Obtain sound power level data or values for these units according to test procedures based on AHRI 350 CSA or CGSB. Sound power values apply to units provided with factory fabricated cabinet enclosures and standard grilles. Values obtained for the standard cabinet models are acceptable for concealed models without separate test provided there is no variation between models as to the coil configuration, blowers, motor speeds, or relative arrangement of parts. Provide automatic valves and controls as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS. Fasten each unit securely to the building structure. Provide units with capacity indicated. Provide room fan-coil units that are certified as complying with AHRI 440, and meet the requirements of UL 1995 CSA or CGSB.

#### 2.13.1.1 Enclosures

Fabricate enclosures from not lighter than 1.3 mm steel, reinforced and braced. Provide enclosures with front panels that are removable and have 7 mm closed cell insulation or 13 mm thick dual density foil faced fibrous glass insulation. Make the exposed side of a high density, erosion-proof material suitable for use in air streams with velocities up to 23 m/s. Provide galvanized or factory finished ferrous metal surfaces with corrosion resistant enamel, and access doors or removable panels for piping and control compartments, plus easy access for filter replacement. Provide duct discharge collar and integral cabinet return grille for exposed view models. Provide supply duct collar and return duct collar for concealed view models.

#### 2.13.1.2 Fans

Provide steel or aluminum, multiblade, centrifugal type fans. In lieu of metal, fans and scrolls could be of non-metallic materials of suitably reinforced compounds with smooth surfaces. Dynamically and statically balance the fans. Provide accessible assemblies for maintenance. Disassemble and re-assemble by means of mechanical fastening devices and not by epoxies or cements.

#### 2.13.1.3 Coils

Fabricate coils from not less than 10 mm outside diameter seamless copper tubing, with copper or aluminum fins mechanically bonded or soldered to the tubes. Provide coils with not less than 13 mm outside diameter flare or sweat connectors, accessory piping package with thermal connections suitable for connection to the type of control valve supplied, and manual air vent. Test coils hydrostatically at 2000 kPa or under water at 1700 kPa air pressure. Provide coils suitable for 1400 kPa working pressure. Make provisions for coil removal.

#### 2.13.1.4 Drain Pans

Size and locate drain and drip pans to collect all water condensed on and dripping from any item within the unit enclosure or casing. Provide condensate drain pans designed for self-drainage to preclude the buildup of microbial slime and thermally insulated to prevent condensation and constructed of not lighter than 0.9 mm type 304 stainless steel or noncorrosive ABS plastic. Provide insulation with a flame spread rating not over 25 without evidence of continued progressive combustion, a smoke developed rating no higher than 50, and of a waterproof type or coated with a waterproofing material. Design drain pans so as to allow no

standing water and pitch to drain. Provide minimum 19 mm NPT or 15 mm OD drain connection in drain pan. Provide plastic or metal auxiliary drain pans to catch drips from control and piping packages, eliminating insulation of the packages; if metal, provide auxiliary pans that comply with the requirements specified above. Extend insulation at control and piping connections 25 mm minimum over the auxiliary drain pan.

#### 2.13.1.5 Manually Operated Outside Air Dampers

Provide manually operated outside air dampers according to the arrangement indicated, and parallel airfoil type dampers of galvanized construction. Provide blades that rotate on stainless steel or nylon sleeve bearings.

#### 2.13.1.6 Filters

Provide disposable type filter that complies with ASHRAE 52.2. Provide filters in each unit that are removable without the use of tools.

#### 2.13.1.7 Motors

Provide motors of either the permanent split-capacitor type with built-in thermal overload protection or ECM type, directly connected to unit fans. Provide motor switch with two or three speeds and off, manually operated, and mounted on an identified plate inside the unit below or behind an access door as indicated. In lieu of the above For multiple fan speed control, provide a solid-state variable-speed controller having a minimum speed reduction of 50 percent, if an ECM motor is provided, occupant interface to vary the airflow via the ECM may be utilized. Provide motors with permanently-lubricated or oilable sleeve-type or combination ball and sleeve-type bearings with vibration isolating mountings suitable for continuous duty. Provide a motor power consumption, shown in watts, at the fan operating speed selected to meet the specified capacity that does not exceed the following values:

| Free Discharge Motors |                                   |      |      |
|-----------------------|-----------------------------------|------|------|
| Unit Capacity (L/S)   | Maximum Power Consumption (Watts) |      |      |
|                       | 115V                              | 230V | 277V |
| 94                    | 70                                | 110  | 90   |
| 142                   | 100                               | 110  | 110  |
| 189                   | 170                               | 150  | 150  |
| 283                   | 180                               | 210  | 220  |
| 378                   | 240                               | 240  | 230  |
| 472                   | 310                               | 250  | 270  |
| 566                   | 440                               | 400  | 440  |

| High Static Motors  |                                   |
|---------------------|-----------------------------------|
| Unit Capacity (L/S) | Maximum Power Consumption (Watts) |
| 94                  | 145                               |
| 142                 | 145                               |
| 189                 | 210                               |
| 283                 | 320                               |
| 378                 | 320                               |
| 472                 | 530                               |
| 566                 | 530                               |

## 2.14 ENERGY RECOVERY DEVICES

### 2.14.1 Plate Heat Exchanger

Provide energy recovery ventilator unit that is factory-fabricated for indoor installation, consisting of a flat plate cross-flow heat exchanger, cooling coil, supply air fan and motor and exhaust air fan and motor. The casing must be 1 mm G90, galvanized steel, double wall construction with 25 mm insulation. Provide fibrous desiccant cross-flow type heat exchanger core capable of easy removal from the unit for sensible and latent heat exchangers. Heat exchanger must include condensate drain pan with means to connect to piping to drain condensate.

## 2.15 SUPPLEMENTAL COMPONENTS/SERVICES

### 2.15.1 Chilled, Condenser, or Dual Service Water Piping

The requirements for chilled, condenser, or dual service water piping and accessories are specified in Section 23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS

### 2.15.2 Condensate Drain Lines

Provide and install condensate drainage for each item of equipment that generates cooling coil condensate in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.

### 2.15.3 Backflow Preventers

The requirements for backflow preventers are specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE.

### 2.15.4 Insulation

The requirements for shop and field applied insulation are specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

### 2.15.5 Controls

The requirements for controls are specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS and Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

### 3.2 INSTALLATION

- a. Install materials and equipment in accordance with the requirements of the contract drawings and approved manufacturer's installation instructions. Accomplish installation by workers skilled in this type of work. Perform installation so that there is no degradation of the designed fire ratings of walls, partitions, ceilings, and floors.
- b. No installation is permitted to block or otherwise impede access to any existing machine or system. Install all hinged doors to swing open a minimum of 120 degrees. Provide an area in front of all access doors that clears a minimum of 910 mm or the minimum manufacturer's printed clearance recommendations, whichever is greater. In front of all access doors to electrical circuits, clear the area the minimum distance to energized circuits as specified in OSHA Standards, part 1910.333.
- c. Except as otherwise indicated, install emergency switches and alarms in conspicuous locations. Mount all indicators, to include gauges, meters, and alarms in order to be easily visible by people in the area.

#### 3.2.1 Condensate Drain Lines

Provide water seals in the condensate drain from all units except room fan-coil units. Provide a depth of each seal of 50 mm plus 0.1 mm for each Pa, of the total static pressure rating of the unit to which the drain is connected. Provide water seals that are constructed of 2 tees and an appropriate U-bend with the open end of each tee plugged. Provide pipe cap or plug cleanouts where indicated. Connect drains indicated to connect to the sanitary waste system using an indirect waste fitting. Insulate air conditioner drain lines as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

#### 3.2.2 Equipment and Installation

Provide frames and supports for tanks, compressors, pumps, valves, air handling units, fans, coils, dampers, and other similar items requiring supports. Floor mount or ceiling hang air handling units as indicated. Anchor and fasten as detailed. Set floor-mounted equipment on not less than 150 mm concrete pads or curbs doweled in place unless otherwise indicated assuring mounting height accommodates depth of condensate drain water seal (trap) where applicable. Make concrete foundations heavy enough to minimize the intensity of the vibrations transmitted to the piping, duct work and the surrounding structure, as recommended in writing by the equipment manufacturer. In lieu of a concrete pad foundation,

build a concrete pedestal block with isolators placed between the pedestal block and the floor. Make the concrete foundation or concrete pedestal block a mass not less than three times the weight of the components to be supported. Provide the lines connected to the pump mounted on pedestal blocks with flexible connectors. Submit foundation drawings as specified in paragraph DETAIL DRAWINGS. Provide concrete for foundations as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

### 3.2.3 Access Panels

Install access panels for concealed valves, vents, controls, dampers, and items requiring inspection or maintenance of sufficient size, and locate them so that the concealed items are easily serviced and maintained or completely removed and replaced.

### 3.2.4 Flexible Duct

Install pre-insulated flexible duct in accordance with the latest printed instructions of the manufacturer to ensure a vapor tight joint. Provide hangers, when required to suspend the duct, of the type recommended by the duct manufacturer and set at the intervals recommended.

### 3.2.5 Metal Ductwork

Install according to SMACNA 1966 CSA B52-13 unless otherwise indicated. Install duct supports for sheet metal ductwork according to SMACNA 1966, CSA B52-13 unless otherwise specified. Do not use friction beam clamps indicated in SMACNA 1966 CSA B52-13. Anchor risers on high velocity ducts in the center of the vertical run to allow ends of riser to move due to thermal expansion. Erect supports on the risers that allow free vertical movement of the duct. Attach supports only to structural framing members and concrete slabs. Do not anchor supports to metal decking unless a means is provided and approved for preventing the anchor from puncturing the metal decking. Where supports are required between structural framing members, provide suitable intermediate metal framing. Where C-clamps are used, provide retainer clips.

#### 3.2.5.1 Underground Ductwork

Provide Polyvinyl Duct Spiral Pipe (PVS) for underground systems. PVS must consist of prime, hot-dipped, G-90 galvanized steel, cleaned and fire treated with epoxy primer bonded to both sides of the metal, with final 4-mil polyvinyl coating heat fused to both sides (4x4). Install ductwork as indicated following all manufacturer's instructions. All underground ducts are to be PVS formed in spiral tubes. All couplings and fittings must be factory fabricated of PVS. All joints and connections must be screw fastened and taped in accordance with manufacturer's installation requirements to assure an air-tight system. All ducts and fittings are to be installed and back-filled according to manufacturer's recommendations. Maximum burial depth is 2 m.

#### 3.2.5.2 Radon Exhaust Ductwork

Perforate subslab suction piping where indicated. Install PVC joints as specified in ASTM D2855.

### 3.2.6 Dust Control

To prevent the accumulation of dust, debris and foreign material during

construction, perform temporary dust control protection. Protect the distribution system (supply and return) with temporary seal-offs at all inlets and outlets at the end of each day's work. Keep temporary protection in place until system is ready for startup.

### 3.2.7 Insulation

Provide thickness and application of insulation materials for ductwork, piping, and equipment according to Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Externally insulate outdoor air intake ducts and plenums up to the point where the outdoor air reaches the conditioning unit.

### 3.2.8 Duct Test Holes

Provide holes with closures or threaded holes with plugs in ducts and plenums as indicated or where necessary for the use of pitot tube in balancing the air system. Plug insulated duct at the duct surface, patched over with insulation and then marked to indicate location of test hole if needed for future use.

### 3.2.9 Power Transmission Components Adjustment

Test V-belts and sheaves for proper alignment and tension prior to operation and after 72 hours of operation at final speed. Uniformly load belts on drive side to prevent bouncing. Make alignment of direct driven couplings to within 50 percent of manufacturer's maximum allowable range of misalignment.

## 3.3 EQUIPMENT PADS

Provide equipment pads to the dimensions shown or, if not shown, to conform to the shape of each piece of equipment served with a minimum 75 mm margin around the equipment and supports. Allow equipment bases and foundations, when constructed of concrete or grout, to cure a minimum of 28 calendar days before being loaded.

## 3.4 CUTTING AND PATCHING

Install work in such a manner and at such time that a minimum of cutting and patching of the building structure is required. Make holes in exposed locations, in or through existing floors, by drilling and smooth by sanding. Use of a jackhammer is permitted only where specifically approved. Make holes through masonry walls to accommodate sleeves with an iron pipe masonry core saw.

## 3.5 CLEANING

Thoroughly clean surfaces of piping and equipment that have become covered with dirt, plaster, or other material during handling and construction before such surfaces are prepared for final finish painting or are enclosed within the building structure. Before final acceptance, clean mechanical equipment, including piping, ducting, and fixtures, and free from dirt, grease, and finger marks. When the work area is in an occupied space such as office protect all furniture and equipment from dirt and debris. Incorporate housekeeping for field construction work which leaves all furniture and equipment in the affected area free of construction generated dust and debris; and, all floor surfaces vacuum-swept clean.



### 3.6 PENETRATIONS

Provide sleeves and prepared openings for duct mains, branches, and other penetrating items, and install during the construction of the surface to be penetrated. Cut sleeves flush with each surface. Place sleeves for round duct 380 mm and smaller. Build framed, prepared openings for round duct larger than 380 mm and square, rectangular or oval ducts. Sleeves and framed openings are also required where grilles, registers, and diffusers are installed at the openings. Provide 25 mm clearance between penetrating and penetrated surfaces except at grilles, registers, and diffusers. Pack spaces between sleeve or opening and duct or duct insulation with mineral fiber conforming with ASTM C553, Type 1, Class B-2 CAN/ULC-S702.

#### 3.6.1 Sleeves

Fabricate sleeves, except as otherwise specified or indicated, from 1 mm thick mill galvanized sheet metal. Refer to design details on the drawings for sleeves in bearing walls or partitions.

#### 3.6.2 Framed Prepared Openings

Fabricate framed prepared openings from 1 mm galvanized steel, unless otherwise indicated.

#### 3.6.3 Insulation

Provide duct insulation in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS continuous through sleeves and prepared openings except firewall penetrations. Terminate duct insulation at fire dampers and flexible connections. For duct handling air at or below 16 degrees C, provide insulation continuous over the damper collar and retaining angle of fire dampers, which are exposed to unconditioned air.

#### 3.6.4 Closure Collars

Provide closure collars of a minimum 100 mm wide, unless otherwise indicated, for exposed ducts and items on each side of penetrated surface, except where equipment is installed. Install collar tight against the surface and fit snugly around the duct or insulation. Grind sharp edges smooth to prevent damage to penetrating surface. Fabricate collars for round ducts 380 mm in diameter or less from 1 mm galvanized steel. Fabricate collars for square and rectangular ducts, or round ducts with minimum dimension over 380 mm from 1.40 mm galvanized steel. Fabricate collars for square and rectangular ducts with a maximum side of 380 mm or less from 1 mm galvanized steel. Install collars with fasteners a maximum of 150 mm on center. Attach to collars a minimum of 4 fasteners where the opening is 300 mm in diameter or less, and a minimum of 8 fasteners where the opening is 500 mm in diameter or less.

#### 3.6.5 Firestopping

Where ducts pass through fire-rated walls, fire partitions, and fire rated chase walls, seal the penetration with fire stopping materials as specified in Section 07 84 00 FIRESTOPPING.

### 3.7 FIELD PAINTING OF MECHANICAL EQUIPMENT

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces

need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except clean to bare metal on metal surfaces subject to temperatures in excess of 50 degrees C. Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Provide aluminum or light gray finish coat.

#### 3.7.1 Temperatures less than 50 degrees C

Immediately after cleaning, apply one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm, one coat of primer applied to a minimum dry film thickness of 0.0255 mm; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm per coat to metal surfaces subject to temperatures less than 50 degrees C.

#### 3.7.2 Temperatures between 50 and 205 degrees C

Apply two coats of 205 degrees C heat-resisting enamel applied to a total minimum thickness of 0.05 mm to metal surfaces subject to temperatures between 50 and 205 degrees C.

#### 3.7.3 Finish Painting

The requirements for finish painting of items only primed at the factory, and surfaces not specifically noted otherwise, are specified in Section 09 90 00 PAINTS AND COATINGS.

### 3.8 IDENTIFICATION SYSTEMS

Provide identification tags made of brass, engraved laminated plastic, or engraved anodized aluminum, indicating service and item number on all valves and dampers. Provide tags that are 35 mm minimum diameter with stamped or engraved markings. Make indentations black for reading clarity. Attach tags to valves with No. 12 AWG 2 mm diameter corrosion-resistant steel wire, copper wire, chrome-plated beaded chain or plastic straps designed for that purpose.

#### 3.8.1 Identification of Hidden Utility Components

Provide labeling for equipment installed above the ceiling that requires periodic inspection or servicing. The labeling must be stick-on adhesive type with black lettering on white or clear background with lettering a minimum of 1/2" in height. For lay-in ceiling applications, locate stick-on on ceiling grid T-Bars closest to equipment such that recommended ceiling tile to be removed for servicing/access is direction below the label. For equipment located above hard ceilings, labeling is to be provided on the provided access door. The operable equipment requiring identification include, valves, dampers, fans.

### 3.9 DUCTWORK LEAK TESTS

The requirements for ductwork leak tests are specified in Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC.

#### 3.10 DAMPER ACCEPTANCE TEST

Submit the proposed schedule, at least 2 weeks prior to the start of

test. Perform the damper acceptance test in accordance with NFPA 80 CAN/ULC-S112. Operate all fire dampers and smoke dampers with the air handling and distribution system operating at maximum air flow conditions, prior to the occupancy of the building to verify that they function properly. Test each fire damper with a fusible link by manually releasing the damper. Visually inspect the mechanism. Reset all fire dampers after acceptance testing.

### 3.11 TESTING, ADJUSTING, AND BALANCING

The requirements for testing, adjusting, and balancing are specified in Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC. Begin testing, adjusting, and balancing only when the air supply and distribution, including controls, has been completed, with the exception of performance tests.

### 3.12 PERFORMANCE TESTS

Conduct performance tests as required in Section 23 05 93 Testing, Adjusting and Balancing for HVAC and Section 23 09 00 Instrumentation and Control for HVAC.

### 3.13 CLEANING AND ADJUSTING

Provide a temporary bypass for water coils to prevent flushing water from passing through coils. Inside of room fan-coil units, air terminal units, thoroughly clean ducts, plenums, and casing of debris and blow free of small particles of rubbish and dust and then vacuum clean before installing outlet faces. Wipe equipment clean, with no traces of oil, dust, dirt, or paint spots. Temporary filters must be provided and used when any air distribution system is in operation during construction. Provide new filters after all construction dirt has been removed from the building, and the ducts, plenums, casings, and other items specified have been vacuum cleaned. Perform and document that proper "Indoor Air Quality During Construction" procedures have been followed; provide documentation showing that after construction ends, and prior to occupancy, new filters were provided and installed. Maintain system in this clean condition until final acceptance. Properly lubricate bearings with oil or grease as recommended by the manufacturer. Tighten belts to proper tension. Adjust control valves and other miscellaneous equipment requiring adjustment to setting indicated or directed. Adjust fans to the speed indicated by the manufacturer to meet specified conditions. Maintain all equipment installed under the contract until close out documentation is received, the project is completed and the building has been documented as beneficially occupied.

### 3.14 OPERATION AND MAINTENANCE

#### 3.14.1 Operation and Maintenance Manuals

Submit an electronic copy and three hard copy manuals at least 2 weeks prior to field training. Submit data complying with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA. Submit Data Package 3 for the items/units listed under SD-10 Operation and Maintenance Data

#### 3.14.2 Operation And Maintenance Training

Conduct a training course for the members of the operating staff as

designated by the Contracting Officer. Make the training period consist of a total of 24 hours of normal working time and start it after all work specified herein is functionally completed and the Performance Tests have been approved. Conduct field instruction that covers all of the items contained in the Operation and Maintenance Manuals as well as demonstrations of routine maintenance operations. Submit the proposed On-site Training schedule concurrently with the Operation and Maintenance Manuals and at least 14 days prior to conducting the training course.

-- End of Section --